

Complex Transitivity Restrictions in Grammar

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Mike Berger

Gutachterinnen:

Prof. Dr. Barbara Stiebels

(Institut für Linguistik, Universität Leipzig)

Prof. Dr. Susanne Wurmbrand

(Institut für Sprachwissenschaft, Universität Wien)

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Abstract

This dissertation is about complex transitivity restrictions in grammar, where *transitivity* subsumes any grammatical property that relates to the number of a verb's arguments. The thesis consists of three case studies – serial verb constructions, transitivity matching, and non-finite clauses in Mayan languages – which focus on transitivity interactions in terms of such grammatical properties as they reside within the verb stem or syntactic argument positions. These constructions consist of at least a verb and one other, not necessarily verbal element that jointly give rise to a complex interaction in terms of transitivity. I derive a typology of complex transitivity restrictions couched within Minimalism, showing for each particular kind of restriction the necessary and sufficient conditions it is governed by.

I show that there three general types of restrictions exist. In *co-occurrence restrictions*, two or more verbal elements cannot co-occur with the same range of transitivity values as in isolation. These are found in serial verb constructions. *Matching restrictions* require two or more verbal elements to match with respect to some component of transitivity. These are investigated in the form of transitivity matching. And then there are *reductive restrictions*, which I define as cases in which a verb occurring in the context of another, non-verbal morpheme shows a range of transitivity values that is more restricted compared to its range available in isolation. This kind of restriction is found in non-finite clauses in Mayan.

First, I show how the co-occurrence restrictions in serial verb constructions follow from cross-cutting the *mode of structure-building* – i.e. adjunction or complementation – and the *level of structure-building* – i.e. V, VP or VoiceP. This first case study demonstrates how transitivity restrictions can arise due to principles of grammatical integration and projection, without the grammar making direct reference to a component of transitivity per se.

Second, regarding transitivity matching, I establish that it comes in three sub-types: semantic, syntactic and morphological. Each of these sub-types is the result of a distinct component of transitivity being tracked by different grammatical process. Semantic matching arises when the semantic types of verb stems are forced to match; syntactic matching reflects an agreement relationship with respect to structure-building features on Voice; and morphological matching is simply transitivity-conditioned allomorphy.

Third, I argue that transitive verb stems are often restricted in non-finite contexts

across Mayan because non-finite contexts involve nominalizations, and certain nominalizers impose a selectional requirement on their VoiceP complement that Voice lack an ergative case feature. This, when coupled with a high locus of absolutive case assignment, results in a situation where many languages must resort to an alternative Voice head, which in turn gives rise to an obligatory intransitive non-finite morpho-syntax.

The findings support a multi-parameter notion of transitivity that distributes its various components across VoiceP. In showcasing how transitivity components interact with one another, they also provide strong support for a derivational view of grammar, where the properties of larger structures follow from the properties of their parts, along with the processes by which they are brought together.

Acknowledgments

Finally! It's about time.

I remember my very first generative linguistics class – my very first proper linguistics class ever, in fact – which was about morpho-syntactic reflexes of A'-movement. Obviously, it was like looking at hieroglyphs since at the time I still had my fair bit of catching up to do on generative grammar (the book I used was on Government and Binding, a theory which I only realized after I had finished working through the book people no longer use). Strangely enough, among the movement reflexes discussed in that class was so-called *meN*-deletion in Indonesian, which I speak natively. This helped me understand what was going on at a very intuitive level after all: if you draw a line from the position of the questioned word to where it would be in the baseline, no verb on that line can bear the prefix *meN*-.

It completely blew my mind that in a language I had been speaking all my life, there was this funny little quirk I had never noticed. On top of that, the nature of this quirk made absolutely no sense to me: why on earth should Indonesian grammar feel the urge to do this? I have a better understanding of this phenomenon and other things linguistic now. Despite the fact that there remain countless things in language that I will never even remotely understand, and even if there are people younger than me with a greater output, I now feel confidence when telling people that I'm a linguist.

Most of all, I want to thank my friends in IGRA: Christine, Irene, Savio, Hyunjung and Luise. Without y'all, the time at IGRA would have been daunting and significantly less funny. Christine, I'm happy we could share those Baozi in Taipei. Irene, I'm still looking for that ugly hat I convinced us to buy in Times Square – do you have it? Savio, it's a shame you only discovered Agavendicksaft after you had already moved to the UK. Hyunjung, next to pig feet (which are delicious), I also think back fondly on our initiative in Mauerpark. And Luise, thanks for having the same weird taste in memes. Friendship was indispensable to me in IGRA. While I acknowledge that some people simply don't need it as much as others, I fail to understand those who take pride in shunning it for the sake of science.

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Thinking back on my time at IGRA, I have to say that the people who impressed me the most when I had the chance to meet them – and this is one of those rare moments where I'm quoting Chomsky and have actually read the source – are people whose names most other people won't ever know. I mean people who can act beyond the need to please others in order to be in favor with them, people with unconditional support for others, people like my friend Hyunjung, who are always willing to do something for somebody they've never met without expecting anything in return.

Thank you to my parents for their unconditional love and support. I'm sorry for all those times I was impatient with you because this thesis was bugging me from the back of my mind.

Christine, thanks for giving me many of my best memories – they'll forever hold a special place.

Finally, thank you Susa for your love, and for reminding me that there is more to language than structures and algorithms.

In dem Sinne: *chwaak chik!*

Untuk Ibu dan Ayah.

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List of symbols and abbreviations

MT	Magdalena Torres
PMP	Pedro Mateo Pedro
RP	Rosario Patzey
SR	Stirling Ramirez
C-SVC	contiguous serial verb construction
EC	Event Cumulation
EC _{TS}	Type-symmetric Event Cumulation
IAR	internal argument requirement
MNC	Morphological Non-head Constraint
NC-SVC	non-contiguous serial verb construction
RANT	Restriction Against Non-finite Transitivity
SVC	serial verb construction
TM	transitivity matching
A	Set A
ABS	absolutive
ACC	accusative
AF	agent focus
AGR	agreement
AGT	agentive
ANTIP	anti-passive
APPL	applicative
AUX	auxiliary
ART	article
ASP	aspect
ASSRT	assertive
AV	agent voice

B	Set B
CAUS	causative
CF	combining form
CLF	classifier
COND	conditional
CONJ	conjunctive
COMP	complementizer
COMPL	completive
DAT	dative
DECL	declarative
DEF	definite
DET	determiner
DISTR	distributive
DETR	detransitive
DIR	directional
DITR	ditransitive
DTRANS	derived transitive
DU	dual
DS	different subject
ENC	enclitic
EP	epenthetic
ERG	ergative
EXCL	exclusive
EVID	evidential
F	feminine
FACT	factitive
FOC	focus
FRUST	frustrative
FUT	future
GEN	genitive
GER	gerund
HSY	hearsay
IMPERS	impersonal
IMPF	imperfect
INCL	inclusive
INCOMPL	incompletive
INF	infinitive
INTR	intransitive
INTS	intensifier
IPFV	imperfective
LK	linking element
LIG	ligature

M	masculine
MID	middle
MOD	modal
NAR	narrative
NEG	negation
NF	non-feminine
NOM	nominative
NMLZ	nominalization
NPST	non-past
LOC	locative
O(BJ)	object
PASS	passive
PERF	perfect
PFV	perfective
PL	plural
PREP	preposition
PURP	purposive
PV	patient voice
POSS	possessive
POT	potential
POSTP	postposition
PROG	progressive
PROX	proximal
PRS	present
PST	past
REAL	realis
RED	reduplicant
RCNT	recent
REFL	reflexive
REL	relative
RN	relational noun
RTRANS	root transitive
S(UBJ)	subject
SEQ	sequential
SG	singular
SS	same subject
STAT	status suffix
SUB	subordination
TIMIT	timitive
TNS	tense
TOP	topic
TRANS	transitive

Chapter 1

Introduction

This dissertation is about complex transitivity restrictions in grammar. In this thesis, *transitivity* is understood as encompassing any grammatical property that relates to how many arguments a verb can project. The constructions forming the basis of this investigation consist of two morphemes which jointly give rise to some complex, syntactic interaction in terms of transitivity. Based on three case studies – serial verb constructions, transitivity matching and non-finite clauses in Mayan – I will try to answer the following two questions: first, what kinds of syntactic restrictions exist with respect to transitivity? Second, what are the necessary and sufficient conditions for these to arise – i.e. why is one restriction found in some configurations, but not others? The findings set up a typology of complex transitivity interactions, couched within a derivational theory of grammar in which the properties of larger structures follow from the properties of their parts, and the processes by which these are brought together.

1.1 Theoretical background and assumptions

I adopt the Minimalist Y-model of grammar (Chomsky 1995, Chomsky 2000, Chomsky 2001). Grammatical modules – the lexicon, the syntax, PF (Phonological Form) and LF (Logical Form) – are independent but connected via interfaces, with derivations proceeding step-wise and bottom-up, driven by the core operations of Merge, Move and Agree. Merge builds structure by combining two separate syntactic objects into a new unit; Move does the same, except that one of these objects originates in the other.¹ Agree creates syntactic dependencies between a probe and a goal in the former's search space, with respect to some feature(s). These operations are subject to various conditions defined over phrase markers, most notably c-command and Minimality. I also assume Distributed Morphology (Halle & Marantz 1993, Harley & Noyer 1999), where syntax manipulates abstract morpho-syntactic feature bundles that are only later mapped onto vocabulary items in a process called Spell-Out. Note

¹Merge and Move are also known as *external* and *internal Merge*, respectively (Chomsky 2001).

that the adoption of Distributed Morphology is not crucial to this thesis, and it will figure only briefly in the context of allomorphy.

Turning to argument structure, I adopt a neo-Davidsonian architecture in which internal arguments are anchored within a verb's lexical semantics, but external arguments are introduced by Voice (Kratzer 1996). This view is hence projectionist with respect to internal arguments, and constructionist with respect to external arguments.^{2,3} By default, I assume that languages obey an unaccusative-unergative distinction, in which unaccusatives merge internal arguments, while unergatives merge external arguments (Hale & Keyser 1993, Hale & Keyser 2002).⁴ I also assume a division between Voice (which introduces external arguments and may assign case) and *v* (which verbalizes and introduces agentive-causative semantics), both of which may be bundled into a single head (Legate 2014, Harley 2017); this division will be left implicit throughout.

1.2 Transitivity

While typically used in a ternary morpho-syntactic opposition to distinguish verbs based on how many arguments they take – i.e. *intransitive* vs *transitive* vs *ditransitive* – the notion of *transitivity* actually decomposes into a multi-parameter and partially scalar concept that subsumes a whole range of additional properties relating to argument structure, event structure, aspect and participant individuation, along with how these are mapped onto the morpho-syntax; important works in this regard include Hopper & Thompson (1980) and Næss (2007).

On one end of the spectrum, we find high-transitive VPs which denote telic events with a volitional agent and a highly individuated and affected patient; on the other end, we find low-transitive VPs which denote atelic events with a non-volitional agent and a patient with low individuation and affectedness (Hopper &

²Theories of argument structure seek to uncover the argument-taking properties of verbs and how these are to be represented, and all of them agree that the grammar plays some role or other in this regard. As laid out in Rappaport Hovav & Levin (1998), one can distinguish two broad approaches to argument structure: *projectionism* and *constructionism*. According to projectionist views (Williams 1981a, Grimshaw 1990, Hale & Keyser 1993, Levin & Rappaport Hovav 1995, Wunderlich 1997, Bresnan 2001), a verb's argument structure is anchored within its lexical semantics. In other words, the number and types of arguments characterized by a given verb are determined on the basis of its meaning, and hence the morpho-syntactic frame a verb and its arguments occur in likewise follow from its meaning. On the other hand, constructionist views (Marantz 1997, von Stechow 1998, Ritter & Rosen 1998, Embick 2004, Borer 2005, Ramchand 2008) hold that lexical verb meanings are underspecified with respect to the number and types of their arguments. Instead, verbs appear within functional syntactic configurations which themselves add event-structural meanings and introduce arguments. Of course, hybrid approaches exist as well, under which some arguments are lexicalized while others are "syntacticized", as in Marantz (1984), Kratzer (1996) and Ramchand (1997).

³Note that despite its wide adoption, the neo-Davidsonian view is not unchallenged. See Kiparsky (1997), Krifka (1999), Horvath & Siloni (2003), Wechsler (2005), Wechsler (2020), Bale (2007), Ausensi et al. (2020) and Ausensi et al. (2021) for criticisms, and Harley & Stone (2013) for a reply to (some of) these.

⁴Ch'ol is one of the languages that has been argued to lack such a distinction (Coon 2010).

Thompson 1980: 252). Numerous transitivity-based distinctions may occupy any zone between these poles across languages, often characterized by language-specific property clusters. Depending on the type of verb, the construction and the language, event- and argument-structural factors may carry over into the morpho-syntax in the form of differential argument marking, argument linking, case, ϕ -agreement, and so on. Without specification, then, the notion of transitivity by itself is partially vague from a technical standpoint, and what counts as transitive in one language often does not map clearly onto another.⁵

Transitivity manifests differently across grammatical modules. Semantically, it takes the form of distinct eventualities with varying numbers of participants, which in turn characterize certain semantic types, e.g. $\langle v, t \rangle$ for unergatives, and $\langle e, vt \rangle$ for unaccusatives and transitives (with only the latter intrinsically compatible with external arguments). Morphologically, transitivity relates to formal properties which encode and reflect semantic and syntactic transitivity, such as voice operations and ϕ -agreement. Syntactically, transitivity is brought out in various phrase-structural properties, including how arguments project and verb-argument relations such as differential argument marking. Numerous phenomena lie at the interfaces, i.e. there is not always a clear division of labor across modules with respect to a given transitivity-related parameter.

Moreover, transitivity means different things depending on the particular structural element at hand: when talking about verb stems without arguments, an *intransitive verb* can refer to a verb that is morphologically intransitive (e.g. is formally anti-passive), or one that gives rise to an intransitive clause based on its semantic type (e.g. $\langle v, t \rangle$), while a *transitive verb* would be formally transitive (e.g. bears causative morphology) or have a semantic type that gives rise to a canonical transitive clause (e.g. $\langle e, vt \rangle$); at the phrasal level a canonical *transitive verb phrase* might involve both a full DP object and object agreement, while an *intransitive verb phrase* would lack object agreement, and either lack an object entirely or occur with one that is incorporated or oblique. What this element-dependence entails is that depending on where we are in the derivation, a verbal element can change with respect to some transitivity-related property. A verb meaning ‘eat’, for example, may start out as a lexically transitive root of type $\langle e, vt \rangle$, but become intransitive at the stem- and phrase-level via anti-passivization, which also changes its type to $\langle v, t \rangle$.

Transitivity is manipulated by *voice*, i.e. any grammatical category that manipulates a verb’s argument structure, either semantically and / or syntactically (Zuñiga & Kittilä 2019), and may interact with argument linking.

It is now common in the Minimalist tradition to have some components of tran-

⁵Language families for which a binary intransitive-transitive distinction proves especially inadequate include the Algonquian, Salishan and Oceanic languages. As a result, scholars have introduced various additional concepts intended to capture the more fine-grained distinctions found in these languages, such as *semi-transitive*, *non-transitive* and *pseudo-transitive*. For an overview of the relevant phenomena, see Zuñiga & Kittilä (2019: Chapter 7), and references therein.

sitivity – such as agentivity, causation and dependent case – reside in dedicated functional heads. What used to be a bare VP is now generally taken to be a layered complex containing functional items like Voice and *v* on top of the lexical verb; see Marantz (1984), Chomsky (1995), Kratzer (1996), Folli & Harley (2004), Pylkkänen (2008), Ramchand (2008) and Alexiadou (2010) for important works in this line of research.⁶

As mentioned at the outset, *transitivity* is here understood as encompassing any grammatical property that relates to the number of a verb's arguments.⁷ The case studies in this thesis will be concerned with transitivity in terms of such grammatical properties as they reside within VoiceP, specifically in the verb stem and on Voice on the one hand, and syntactic argument positions on the other: in serial verb constructions, transitivity is investigated indirectly by way of syntactic argument positions, which – depending on the structural level – characterize distinct sets of possible transitivity values; in transitivity matching, we will discuss transitivity mainly in terms of the semantic types of verb stems, and structure-building features on Voice which correlate with different transitivity values; finally, non-finite clauses in Mayan will again characterize restrictions on transitivity in terms of features on Voice. VoiceP delimits the clausal locus of transitivity alternation, though factors such as aspect – typically located by scholars in a projection directly above VoiceP – are also known to condition transitivity. These kinds of effects are not considered here.

A *transitive verb* will refer to a verb stem that is semantically dyadic and licenses an object, even though it does not necessarily have to express this object in the syntax. A *transitive verb phrase* will refer to a phrase consisting of a transitive verb and an object. An *intransitive verb* is a semantically monadic verb which does not license an object, and an *intransitive verb phrase* is a phrase consisting of an intransitive verb without any objects. The terms *ditransitive verb* and *ditransitive verb phrase* will be used analogously. Also, I adopt from Dixon (1994: 122ff) the term *extended intransitive* to refer to morpho-syntactically intransitive verbs occurring with oblique objects. When speaking in general terms, I will use the term *transitivity* to refer to the overall notion itself, regardless of specific parameters and values.

Next to this syntactic sense, transitivity will also figure prominently in a type-theoretic sense. In the context of transitivity matching, verbs will mostly be referred

⁶The situation is complicated by the fact that scholars assume distinct functional heads with varying labels, and that these may come in different "flavors" with different semantic and syntactic properties; cf. Folli & Harley (2004) and Alexiadou et al. (2015), among others. So while there is general consensus in the constructionist tradition on the need for the syntacticization of certain aspects of argument structure, there is as yet no agreement on the number of types of functional verbal heads, or even the role of VoiceP as separate from these; see Ramchand (2013) for a recent review of argument structure and its treatment in the generative tradition.

⁷This definition is intended to rule out cases in which e.g. the verb 'eat' is lexically an unergative verb that cannot take an object without further derivation (as is the case in some Mayan languages), although it is clear that semantically (or rather encyclopedically), the activity of eating necessarily entails something that is eaten.

to in terms of their *semantic types*. Given the verbal architecture adopted in this thesis, this will mean the types $\langle e, vt \rangle$ for unaccusatives, $\langle v, t \rangle$ for unergatives and $\langle e, vt \rangle$ for transitives (unless otherwise noted).

Finally, *morphological* or *formal transitivity* will play a marginal role in cases where verbs will need to be referred to based on their formal properties, independently of any associated semantic and / or syntactic transitivity-related features they may have.

1.3 Complex verb constructions

The constructions forming the baseline objects of each case study in this thesis are what I call *complex verb constructions*, which I define as having the following three properties: first, they consists of at least one verb; second, they consist of at least one other, not necessarily verbal morpheme; third, these two elements jointly give rise to some kind of complex interaction with respect to some grammatical property. In our case, then, we are interested in complex verb constructions with respect to transitivity. This notion is not meant to pick out a natural class of constructions per se, but simply any construction that brings together two or more transitivity-importing elements which (i) need not all be verbal, and (ii) end up giving rise to some kind of complex transitivity interaction.⁸ Given the multi-parameter nature of transitivity, distributed across distinct parts of the verbal projection, we have an a priori expectation that such interactions should sometimes arise.

Complex verb constructions thus subsume the more familiar *complex predicates* in the sense of Butt (1993) and Butt (2010), defined as combinations of multiple verbal heads that contribute to argument structure (which conditions transitivity), jointly expressed in a single inflectional domain. Complex predicates in this sense include serial verb constructions, light verb constructions, causative and applicative constructions and restructuring configurations; but they exclude, for example, deverbal nominalizations on grounds that the nominalizer itself is taken to be devoid of any contribution to argument structure itself, serving a purely re-categorizing function.

By contrast, certain kinds of nominalizations do fall under complex verb constructions with respect to transitivity, provided that the nominalizer interacts with the verb in a way that ends up restricting its transitivity. We will see that this definition adequately places restrictions on transitivity in Mayan non-finite clauses within the broader context of this dissertation. It also allows us to discuss phenomena like transitivity-conditioned allomorphy, which do not obviously instantiate complex predication, alongside Mayan non-finite clauses in a way that is coherent but also does justice to their differences.

⁸I take a *transitivity-importing* element to be an element which either itself gives rise to transitivity due to having argument structure, i.e. is verbal, or which has a property that interacts with some component of transitivity.

1.4 Restrictions on transitivity

I take a *restriction on transitivity* to be any restriction on some transitivity-related property of a verbal element. Within a complex verb construction, this will mean any restriction as compared to at least one of the elements' behavior in isolation.⁹ This dissertation establishes three basic kinds of transitivity restrictions in complex verb constructions, based on the general kind of phenomenon involved: *co-occurrence*, *matching* and *reductive restrictions*. These can further be distinguished in terms of the sources from which they arise.

1.4.1 General types of restrictions

1.4.1.1 Co-occurrence restrictions

In co-occurrence restrictions, two or more verbal elements cannot co-occur with the same range of values as in isolation. We will encounter these in our first case study on serial verb constructions, where depending on the language and construction, only verbs of certain transitivity values can co-occur with one another in the context of serialization.

1.4.1.2 Matching restrictions

Matching restrictions require two verbal elements to match with respect to some property. In our second case study, we will see that transitivity matching requires verbal elements to match with respect to one of various transitivity-related properties.

1.4.1.3 Reductive restrictions

A reductive restriction is one in which a verbal element occurring in the context of another, non-verbal morpheme cannot show the same range of values as in isolation. Reductive restrictions characterize non-finite clauses in Mayan, our third case study, where transitive verbs are often restricted in obligatorily requiring a detransitivizing workaround.

1.4.2 Restrictional sources

1.4.2.1 Configurational sources

Turning to the sources from which these restrictions may arise, we can first distinguish what I will call *configurational sources*. A restriction has a configurational source whenever it is conditioned specifically by the way in which elements are syntactically brought together into certain structural dispositions to one another, i.e. configurations. Configurational sources will usually amount to the distinct structures that are

⁹Because of its use in Optimality Theory, I will do my best to avoid the term *constraint*.

output by cross-cutting the *mode* and the *level of structure-building*. The former refers to how structure is built, and the latter to the height at which structure is built.

For example, in contiguous serial verb constructions involving adjunction at the V-level, transitive verbs which are non-heads systematically cannot project their internal arguments – a restriction that is conditioned by the configuration of V-level adjunction itself.

1.4.2.2 Natural vs ad hoc sources

A restriction with a *natural source* is one that follows automatically from what we know about its context; conversely, an *ad hoc source* does not follow from the context per se, such that it must be stated in addition to what know about the context – as a kind of "grammatical stipulation".

For example, the inability of non-head transitive verbs to project in V-level adjunction configurations follows naturally from what we know about adjunction and projection, with nothing more needing to be said. Transitivity matching, on the other hand – despite clearly being conditioned by certain configurations – does not follow from these per se, and must therefore be formulated as an ad hoc requirement that is arbitrarily enforced by the grammar.

1.4.2.3 Selectional sources

A restriction can have a *selectional source*, in which case it is simply a selectional requirement internal to certain lexical items, imposed on their complement.

For example, a subset of nominalizers in Mayan require their verbal complement to lack a feature that assigns ergative case, which has the effect of restricting the range of possible verb stems under nominalization.

1.4.2.4 Other sources

Another recurring restriction will be what I call a verb's *Internal Argument Requirement*, or *IAR*, which refers to the general need of transitive verbs (and unaccusatives) to have their internal argument slot closed, e.g. via Function Application or existential binding. Many languages considered here impose a strong Internal Argument Requirement on verbs, to the effect that e.g. transitive verbs must obligatorily project some kind of object, or else undergo an operation like anti-passivization.

An important role will also be played by the *Principle of Full Interpretation* (Chomsky 1986), which requires every element to be interpretable by the time it reaches the interfaces, LF and PF. It amounts to a fundamental grammatical licensing condition which forces every element to be integrated in accordance with one of various clear-defined grammatical principles. Full Interpretation plays an important role in certain serial verb constructions, where it regulates which kinds of arguments can project as

regular nominals and which cannot, a restriction which ends up indirectly affecting transitivity.

Finally, we will briefly invoke a particular *thematic source* in order to rule out a handful of cases that would otherwise incorrectly be predicted to be well-formed. Specifically, I will make the assumption that within certain domains, one and the same argument cannot be assigned contradictory thematic roles. This idea is an old one, going back to e.g. Baker (1996) in his work on causatives in poly-synthetic languages.

1.5 Outline of the dissertation

The dissertation consists of three case studies. These were chosen specifically not only because each involves a systematically recurring, broad kind of restriction, but also because each restriction also shows some degree of cross-linguistic variation – allowing us to better tease apart the more general factors from the more specific ones. Also, the fact that numerous descriptive and theoretical works exist on these phenomena to varying degrees informs my analyses.

On the one hand, each case study accounts for a particular kind of complex transitivity interaction in its own right. On the other, this dissertation aims to bring these case studies together using a common formal vocabulary, in order to set up an initial typology of the kinds of syntactic transitivity restrictions that can arise in grammar, along with their governing conditions. A welcome fact about the three phenomena chosen is that while two of them involve interactions in which the grammar tracks some component directly pertaining to transitivity, one of them shows how restrictions on transitivity can also arise indirectly without the grammar tracking any such property *per se*.

1.5.1 Part I: serial verb constructions

We begin with Part I on serial verb constructions (SVCs), which I define as monoclausal constructions consisting of multiple verbs, built without any marker of coordination within VoiceP. Despite seeming remarkably unconstrained in terms of transitivity, we will see in Chapter 2 that SVCs characterize a number of robust profiles in terms of which transitivity values verbs can have once they enter into serialization. Chapter 3 presents the basic analysis, which is that the transitivity profiles we find follow from cross-cutting the level and mode of structure-building, along with independent principles on how elements must be integrated, and the fact that transitive verbs must generally have their internal argument slot closed off. Chapter 4 presents derivations of the patterns found in non-contiguous SVCs, and Chapter 5 of those in contiguous SVCs.

The co-occurrence restrictions we find in SVCs have a largely configurational

source, alongside fundamental well-formedness conditions on syntactic and semantic integration. The restrictions follow naturally from what we know about the structures involved, without the need for ad hoc grammatical statements.

1.5.2 Part II: transitivity matching

Part II looks into transitivity matching, a phenomenon in which two verbal elements co-vary with respect to some component of transitivity. Chapter 6 starts with semantic transitivity matching as found in the Oceanic language Saliba, where all matching reflexes come with a semantic effect. The analysis of semantic transitivity matching, which treats it as the result of the grammar tracking the semantic types of verb stems in V-level adjunction configurations, is presented in Chapter 7, with derivations following in Chapter 8; this chapter also looks into a phenomenon where certain verbs that are generally intransitive exceptionally become transitive in the context of serialization, in order to indirectly satisfy the internal argument requirement of their co-verb. We venture outside Saliba in Chapter 9, where I show for the Arawakan language Tariana that the predictions of my analysis of semantic transitivity matching are borne out. Finally, Chapter 10 turns to syntactic and morphological transitivity matching, which I will argue are the result of agreement with structure-building features, and transitivity-conditioned allomorphy, respectively.

The restrictions at play in transitivity matching are all conditioned configurationally, with the mode and level of structure-building again playing a crucial role: semantic matching can only arise in V-level adjunction configurations, while syntactic and morphological matching can only arise in larger, phrasal configurations built via complementation. These configurational sources, however, only make up the necessary conditions for a particular type of matching to arise. Therefore, we must in each formulate ad hoc grammatical requirements in order to fully derive matching, which occurs either via type-matching (semantic), agreement (syntactic) or allomorphy (morphological) – in each case with respect to a distinct component of transitivity.

1.5.3 Part III: non-finite clauses in Mayan

Finally, Part III turns to non-finite clauses in Mayan languages, where transitive verbs must often occur with some detransitivizing workaround in order to be well-formed. The general restriction is introduced in Chapter 11, along with the fact that non-finite clauses in Mayan are deverbal nominalizations. Chapter 12 discusses existing approaches to this restriction, which lead me to conclude that – while it is directly conditioned by nominalization – something more needs to be said about nominalization in order to fully explain it. Based on the internal properties of non-finite clauses, discussed in detail in Chapter 13, I present the analysis in Chapter 14, arguing that Mayan nominalizers taking VoiceP complements systematically select

against an ergative-assigning feature on Voice, which – depending on whether a given language provides a high or low source of absolutive case – may ban transitive verbs from occurring under nominalization as they are.

This restriction is the only one discussed here that has a selectional source. While the ability of certain nominalizers to restrict their verbal complement in this way is conditioned configurationally – i.e. the two must be in a local selectional relationship – it does not clearly follow from nominalization and selection per se, meaning that the sufficient condition on this restriction must ultimately be ad hoc.

1.5.4 Part IV: concluding remarks

This dissertation ends with Part IV, which consists of Chapter 15. Next to summarizing the findings, I extend them to one other construction that shows a peculiar pattern with effects on transitivity, discuss the application of Optimality Theory, and bring up further avenues into which the general logic and results of this thesis may take us in future work.

Part I

Serial verb constructions

Chapter 2

Co-occurrence restrictions in serial verb constructions

2.1 Introduction

When one starts looking into SVCs, what stands out is how remarkably productive and seemingly unconstrained they are. Regarding licit transitivity pairings, any pairing of transitivity values that is logically possible is attested cross-linguistically, in any order. Non-contiguous serialization in Korean, for example, places no restrictions whatsoever on the stem-level transivities of constituent verbs:

(1) **Korean**

a. *intr + intr*

Say-ka [nal-a] [ka-ss-ta]
bird-NOM fly-LK go-PST-DECL
'The bird flew away'

(Kang 1997: 6)

b. *trans + trans*

John-i thokki-lul [cap-a] [mek-ess-ta]
J.-NOM rabbit-ACC catch-LK eat-PST-DECL
'John caught and ate a rabbit'

c. *intr + trans*

John-i wulthali-lul [ttwi-e] [nem-ess-ta]
J.-NOM fence-ACC jump-LK go.over-PST-DECL
'John jumped over a fence'

(Ko & Sohn 2015: 100, 108)

d. *trans + intr*

Say-ka thokki-lul [cap-a] [nal-ass-ta]
bird-NOM rabbit-ACC catch-LK fly-PST-DECL
'The bird caught a rabbit and flew away'

(Hyunjung Lee p.c.)

Compare this to Edo (Niger-Congo, Nigeria), which allows non-contiguous SVCs to

consist of two transitive stems (2a), but not e.g. two unergatives (15):

(2) **Edo**

a. *trans + trans*

Òzó ghá [gbè] èwé [khièn]

O. FUT hit goat sell

'Ozo will kill the goat and sell it'

b. *unerg + unerg*

*Òzó gié!gié rhú!lé sálán

O. quickly run jump

intended: 'Ozo quickly ran and jumped' (Baker & Stewart 2002: 2, 49)

I will show that the major transitivity patterns we find in SVCs across languages are derivable from cross-cutting two parameters: the level at which structure is built, and the mode by which it is built.

This chapter lays the groundwork for our first case study. After defining SVCs in Section 2.2, I establish the role of contiguity in Section 2.3, and introduce important concepts and conventions to be used here in Section 2.4. The major transitivity profiles found in SVCs are presented in Section 2.5. This chapter ends with Section 2.6, which explains why SVCs with resultative meanings are not considered as falling under serialization proper.

2.2 Defining serial verb constructions

There is no generally agreed upon definition of a serial verb construction. Every definition I am aware of includes the properties of mono-clausality and the absence of any kind of linking element marking subordination or coordination; yet despite this convergence, there remain varying criteria which inadvertently rule out constructions that are typically considered instances of serialization, and conversely others which rule in constructions that are typically not seen as serial verbs.

To take just two of the various definitions out there, consider first the one in (3), from Veenstra & Muysken (2017):

(3) ***Serial Verb Construction I***

a. mono-clausal

b. no linking element

c. only one grammatical subject

d. at most one shared grammatical object

(Veenstra & Muysken 2017: 6)

Even though (3) covers many uncontroversial cases of serialization, it also rules in cases which would just as uncontroversially not be seen as falling under serialization,

such as the following:

- (4) a. *He made her cry* (Haspelmath 2016: 15)
 b. *Alma will read Dune*

Assuming (4a) is mono-clausal, (4a) and (4b) both satisfy the definition (3), yet hardly anybody would claim that these constructions are SVCs.

In an attempt to establish serialization as a comparative concept, Haspelmath (2016) offers the definition in (5):

- (5) ***Serial Verb Construction II***
 a. mono-clausal
 b. no linking element
 c. consists of multiple verbs
 d. the verbs may occur independently
 e. no complementation relation between verbs
 (Haspelmath 2016: 6)

(5) seems more promising than (3) because it rules out the cases in (4). Yet (5) also rules out instrumental SVCs like (6) under the analysis that is going to be developed here, which treats instrumental SVCs as cases of complementation, whereby the V_1 serves an applicative-like role with respect to V_2 in introducing the non-core role of Instrument:

- (6) **Gungbe: Instrumental NC-SVC**

S'etù [zé] kpò ló [xò] Kòjò
 S. take stick DET hit K.

'Setu took the stick hit Kojo'

i.e. 'Setu hit Kojo with the stick' (Aboh 2009: 16)

The problem is that cases like (6) are unanimously considered bona fide instances of serialization. More generally, I submit that the problem with definitions like (3) and (5) is that they do not sufficiently rely on clear grammar-theoretical criteria.

While I did not originally intend to define SVCs in this thesis, the present case study has led to the welcome side effect of leading us to set up what I believe is a viable grammar-theoretical diagnostic of SVCs and their sub-types.

However, instead of a clear-cut definition of what SVCs are and what they are not, I provide a set of grammar-theoretical conditions that will enable us to clearly distinguish and discuss different kinds of constructions, some of which are more or less universally seen as SVCs, while others are only seen as SVCs by some authors, and yet others are generally not seen as instances of serialization by anybody. This may seem unsatisfactory, but I contend that it is actually an expected result given

that definitions of SVCs – despite a clear intuitive understanding of what makes a SVC – generally remain as descriptions of recurring clusters of properties, informed both by the language(s) investigated and authors’ varying criteria, which do not make reference to grammar-theoretical principles such as the mode or the level of structure-building.

I thus propose the minimally necessary set of conditions for a SVC in (7), which I adopt in this thesis:

(7) *Necessary conditions for a Serial Verb Construction*

- a. mono-clausal
- b. consists of multiple verbs
- c. the verbs may occur independently
- d. no linking element
- e. structure-building within VoiceP

The idea is that in order for a construction to count as a SVC, it must exhibit all the properties in (7); however, the joint presence of all of these properties does not guarantee SVC status. In other words, every SVC has the properties in (7), but not every construction with the properties in (7) will count as a SVC.

I take mono-clausality to be characterized by a single inflectional domain. While what counts as a single inflectional domain may vary from language to language (Butt 2010: 57), it generally amounts to a single domain with respect to the marking and interpretation of TAM, inflection and polarity, but also with respect to certain domain-sensitive dependencies like negative polarity and clitic placement.¹⁰ Bohnemeyer et al. (2007: 501) propose to rely on the criterion of single negatability, citing it as a cross-linguistically applicable test for clausehood – and indeed, this property is practically always referenced in definitions of SVCs; see also Haspelmath (2016: 9ff).^{11,12} Prosodically, SVCs have the same intonational properties as simplex clauses,

¹⁰While negation in SVCs usually scopes over the entire string, this is not always the case. For example, negation of SVCs in Alamblak (Sepik, Papua New Guinea) can scope over any number of verbs to the exclusion of all others, thus giving rise to multiple negative interpretations (Bruce 1988: 27f). Crucially, even though more than one negative reading is available, Alamblak SVCs allow only a single negator.

¹¹I follow Bohnemeyer et al. (2007) in assuming that a structure is semantically mono-clausal if it has the *Macro-event property*, i.e. if any time-positional or durational operators it combines with necessarily have scope over all of the sub-events it entails.

¹²Note that in some SVCs, inflectional uniqueness may surface as concordant marking, whereby certain categories are marked on every verb as a kind of agreement. In (i) from Tariana, each verb shows 1PL subject agreement despite the presence of only a single subject (and there is no coordination); compare (ii) from Igbo, where only V₁ is marked for tense:

(i) **Tariana:** *Concordant marking*

Mawina-nuku wasā [wheta] [wa-hña]
 pineapple-TOP let’s 1PL.take 1PL-eat

‘Let’s take and eat the pineapple!’

(Aikhenvald 2006a: 183)

and speakers systematically conceptualize SVCs as single yet complex events (Durie 1997, Stewart 2001, Bohnemeyer & van Valin 2017).

The crucial criterion in (7) is , i.e. that structure-building must proceed within VoiceP. This criterion makes reference to a clear-defined grammatical property, namely a structural delimitation by VoiceP, the domain in which arguments are introduced, their thematic roles established and the overall argument structure fixed. As we will see, this set of necessary conditions makes enables us to take about constructions from the perspective of "SVC-hood" not only in a way that does justice to the long-standing intuition of what it means to be a SVC, but is also from a grammar-theoretical perspective. We thus have a viable comparative concept that is rooted in grammatical, and not merely formal criteria.

Note that the final two criteria in (7) entail that all SVC must be built via adjunction or complementation, given my assumption that even covert coordination structures involve coordinating heads, i.e. linking elements; this is discussed in Section 3.4 of the next chapter.

2.3 Contiguity

A common dimension along which SVCs are distinguished is whether or not their constituent verbs may be separated by any material that does not belong to the domains of derivation or inflection. Throughout, I will use the terms *non-contiguous SVC* (NC-SVC) and *contiguous SVC* (C-SVC) for this distinction, based on Durie (1997): non-contiguous SVCs allow the occurrence of inter-verbal material, most notably objects, and verbs in contiguous SVCs exhibit strict adjacency, only ever allowing derivational or inflectional material to occur between them, but never arguments or adverbs. The ditransitive SVC in (8) from Cantonese is an example of a non-contiguous SVC, while its counterpart in Jeh is contiguous (9):

(ii) **Igbo:** *Single marking*

Úchè [gbù-rù] òkúkò [sí-é]
U. kill-PST chicken cook

‘Uche killed and cooked the chicken’

(Zimmermann & Amaechi 2020: 2)

(iii) further shows the purely formal nature of concordant marking, where the subject of V_2 is also φ -marked on V_1 , whose thematic subject, however, corresponds to the overall direct object:

(iii) **Tariana:** *Concordant marking*

Emite-tiki nu-na [dihpani] [di-adeta-naka]
child-DIM 1SG-OBJ 3SG.NF.work 3SG.NF-prevent-PRES

‘The little boy is preventing me from working’

(Aikhenvald 2006b: 40)

(8) **Cantonese:** NC-SVC

ngo₅ [lo₂-zo₂] bun₂ syu₁ [bei₂] keoi₂
 1SG take-PERF CLF book give 3SG

'I have taken a book for him/her'

(Bodomo et al. 2004: 71)

(9) **Jeh:** C-SVC

Mi [ruat] [dóh] au pheí
 you buy give me rice

'You buy rice for me'

(Kroeger 2004: 240)

Unlike Durie's use of the contiguous vs non-contiguous distinction, my use of this notion is meant to be able to be compatible with an analysis that derives the contiguity of a subset of superficially contiguous SVCs from an underlyingly non-contiguous structure. Even though overwhelmingly, surface contiguity will reflect underlying contiguity, we will encounter a few rare cases for which it will be argued that what appears as a contiguous V-to-V structure is actually derived from an underlying phrasal structure via verb incorporation, as proposed in e.g. Déchaine (1993) and Collins (2002). So by default, I will use C-SVC and NC-SVC to mean serial verbs which are also underlyingly contiguous and non-contiguous, respectively; when called for, I will explicitly distinguish *derived* C-SVCs and *true* C-SVCs in order to express whether the contiguity we see is a derived surface property, or truly underlying.¹³

Unless otherwise noted, all of the C-SVCs discussed here are obligatorily contiguous. Moreover, all the C-SVCs discussed here are classified as contiguous based both on authors' explicit claim of contiguity, and also the fact that in SVCs comprising at least one transitive verb, no object can ever occur inter-verbally (i.e. contiguity cannot be inferred from SVCs made up of only intransitives, given that these lack objects for diagnosing adjacency). Whenever a SVC appeared to be contiguous, but for which the relevant classification and crucial transitivity variants were missing, I have excluded it from discussion.

2.3.1 C-SVCs and V-V compounds

While NC-SVCs have received ample attention in the literature, there has been a tendency to sideline C-SVCs as bona fide serialization and instead treat them as V-V compounds, typically understood as being made up of multiple verbal lexemes

¹³The notion of contiguity is related to, but different from the *nucleus* vs *core* distinction from Role and Reference Grammar (Foley & Van Valin 1984, Van Valin & LaPolla 1997). In this framework, the *nucleus* refers to the predicate without any arguments, while the *core* refers to the predicate with its arguments. Serial verbs hence characterize so-called *nuclear* and *core junctures*, depending on the level at which their verbal projections are merged with one another. In phrase-structural terms, nuclear and core junctures roughly correspond to V-level (before object introduction) and phrase-level structure-building (after object introduction), respectively.

with the same properties as simplex verbs due to being drawn from the lexicon as prefabricated units; for works with this disposition, see Baker (1989), Déchaine (1993) and Collins (2002), among others.¹⁴¹⁵ Yet relabeling C-SVCs as V-V compounds does not add to our understanding of their systematically recurring properties, and as pointed out in Owens (2011: 25), the way this classification is handled by some authors is oblivious to the morpho-syntactic properties that a compounding analysis would actually entail. Still, authors have come to point out numerous parallels between NC-SVCs (in our terms) on the one hand and what they take to be V-V compounds on the other, accordingly seeking to derive the latter from the former transformationally (Déchaine 1993, Gruber & Collins 1997, Nishiyama 1998, Cummings 2001, Collins 2002).

In this thesis, C-SVCs are understood as constructions that are built in the syntax via a productive structure-building mechanism at the V-level, i.e. the level of the verb stem prior to the introduction of any arguments. This verb stem may be simplex or complex, i.e. it may contain valence-changing operations, material relating to transitivity and inflection; crucially, this material may target one verb stem individually to the exclusion of the other.

In opposition to C-SVCs, I will take V-V compounds to be constructions that are drawn from the lexicon as prefabricated units. This calls for a set of diagnostics to tell the two apart. Even though this may be tricky at times, there exist a set of principled differences between C-SVCs and V-V compounds, which in turn justifies the treatment of both NC-SVCs and C-SVCs as instances of bona fide serialization, differing only in the level at which serialization takes place. It follows that C-SVCs cannot be reduced to compounding.

First, as prefabricated lexical units, V-V compounds morpho-syntactically behave just like simple verbs, meaning that they generally show utter impermeability not only to arguments, but also derivation and inflection.¹⁶ Any derivation or inflection must necessarily attach outside of the verbal complex in a V-V compound, whereas C-SVCs in principle allow such material to occur inter-verbally, including the occasional particle, incorporate or clitic.

Consider the following pair from Northern Paiute: while the applicative suffix occurs on V_1 in C-SVCs, in between V_1 and V_2 (10a), it appears outside the verbal complex in V-V compounds, with both verbs falling into its scope (10b):^{17,18}

¹⁴Compounds often behave as single phonological words. That said, their constituents receive independent stress in languages like German and English, such that they may be seen as consisting of two phonological words (Susanne Wurmbrand p.c.).

¹⁵Owens (2011) and Aikhenvald (2018) attribute the widespread tendency to class V-V compounds and C-SVCs as one and the same to the heavy initial focus on SVCs in languages with highly analytic isolating profiles, most notably the West African languages. See also Haspelmath (2016).

¹⁶Note that this is not always the case – see e.g. Harðarson (2016) on Icelandic compounds containing inflectional material.

¹⁷The suffix *-ki*, glossed as an applicative, may yield causative meanings when attached to e.g. stative verbs (Thornes 2003: 284f), as appears to be the case in (10a).

¹⁸Note that I am relying largely on the author's word that the applicative in (10a) scopes over both

(10) **Northern Paiute**a. *C-SVC*

ni i=giki [ʔyui-**ki**]-[kati]
 I my=feet warm-APPL-sit.SG
 'I sit warming my feet'

b. *V-V compound*

i=sakwa ka=toissapui i=[sami]-[tiki]-**ki**
 you=MODAL OBL=chokecherry me=soak-put.SG-APPL
 'You should put the chokecherries in to soak for me'

(Thornes 2011: 32)

Thornes (2003: 347ff) lists further differences between C-SVCs and V-V compounds, e.g. the V₂ slot in C-SVCs but not V-V compounds is restricted to motion and posture verbs, and stress assignment takes into account the entire complex stem in V-V compounds but not C-SVCs.

Second, under a uniform V-V compounding analysis, all contiguous multi-verb constructions in a given language are expected to pattern morpho-phonologically like other compounds in that language. That this is not generally true is demonstrated by languages such as Paamese, where cliticization diagnoses differences between the two in systematically attaching to V₁ in C-SVCs, but the entire complex in V-V compounds (Crowley 1982: 20ff, 155f).¹⁹

Korean distinguishes C-SVCs and V-V compounds along similarly clear morpho-syntactic lines: C-SVCs contain a linking vowel, and when they undergo predicate doubling, only one of the constituent verbs is doubled (11a); in contrast, V-V compounds have no linking vowels, and disallow predicate doubling if it does not target the entire verbal complex (11b):

(11) **Korean**a. *C-SVC*

- (i) John-i kom-eykey [cap-a]-[mek]-hi-ess-ta
 J.-NOM bear-by catch-LK-eat-PASS-PST-DECL
 'John was caught and eaten by a bear'
- (ii) John-i kom-eykey [cap-a]-[mek-hi-**ki-nun**]
 J.-NOM bear-by catch-LK-eat-PASS-NMLZ-TOP
 [**mek-hi-ess-ta**]
 eat-PASS-PST-DECL
 'It is true that John was caught and eaten by a bear'

b. *V-V compound*

- (i) John-i cip-kwa hakkyo-lul mayil [o]-[ka]-ss-ta
 J.-NOM house-and school-ACC everyday come-go-PST-DECL

verbs, rather than V₂ alone.

¹⁹C-SVCs and V-V compounds are further distinguished by abundant morpho-phonological processes (Crowley 1987: 60).

- 'John went back and forth between (his) house and (his) school'
- (ii) *John-i cip-kwa hakkyo-lul mayil [o]-[ka]-**ki-nun**
 J.-NOM house-and school-ACC everyday come-go-NMLZ-TOP
ka-ss-ta
 go-PST-DECL
 intended: 'It is true that John went back and forth between (his)
 house and (his) school'

(Ko & Sohn 2015: 84, 86)

A third difference concerns the productivity and transparency of each construction. As a rule of thumb, serialization is productive and yields transparent meanings, while compounding is more prone to non-compositionality, lexicalization and accidental gaps (Durie 1997, Crowley 2002, Fukushima 2005, Paschen 2014, Haspelmath 2016). This is not to say, however, that one is free to serialize just any two verbs; rather, the two must form what Durie (1997) terms a "construable (super)event" (whatever this means in a given language). In Alambak, for instance, one can serialize 'climb tree, find insects' but not 'climb tree, look at stars' because only the former is conceptualized as a sufficiently cohesive sequence of actions by Alambak speakers (Bruce 1988). Conversely, C-SVCs may also give rise to non-compositional meanings, e.g. 'pass-stand' in Tariana, which translates to 'be arrogant' (Aikhenvald 2003). Relatedly, meanings are usually symmetrical in V-V compounds, and may be symmetrical or asymmetrical in C-SVCs, in the sense that V-V compounds tend to consist of verbs with lexical meanings only, while C-SVCs may contain a verb with a more specialized or functional meaning, most often relating to aspect / aktionsart, causation, motion / directionality (Aikhenvald 2006b, Aikhenvald & Dixon 2006), and also – as we will see – applicative-like argument introduction.

Fourth, there are numerous parallels between NC-SVCs and C-SVCs which are absent in V-V compounds. One of these is the existence of highly robust, and what at times appear to be universal constraints on the ordering of verbs in both C-SVCs and NC-SVCs (which are often blind to the general headedness of a language). While exceptions exist, the generalization is that verb order in serial verbs is iconic to event structure (Muysken 1988, Li 1993, Carstens 2002, Aikhenvald 2006b). For example, in SVCs expressing cause and effect, V_1 is always the manner verb and V_2 always the verb entailing the effect or result of the action denoted by V_1 . This ordering replicates in C-SVCs (12) and NC-SVCs (7a):

(12) **Igbo: C-SVC**

ó [tì]-[gbù]-rù nwóké áhù
 he hit-kill-TNS man that

'He beat that man to death'

(Lord 1975: 28)

(13) **Sranan: NC-SVC**

Kofi [naki] amba [kiri]
 K. hit A. kill

‘Kofi killed Amba’

(Sebba 1987)

This generalization extends to many other kinds of SVCs; see Durie (1997: 304ff). In the realm of compounds, despite the existence of clear cross-linguistic tendencies, we find significantly more variation in ordering (Bauer 2001, Gaeta 2008).

Turning to argument-structural diagnostics, these are not always as informative as one might hope in distinguishing C-SVCs from V-V compounds, the reasons being that (i) argument structure may be accessible to pre-syntactic lexical processes, and (ii) event-structural restrictions may also determine whether a pair of verbs is combinable, both in serialization and compounding (Li 1990, Fukushima 2005, Paschen 2014). We will nevertheless see that C-SVCs follow the same abstract argument sharing and linking principles as NC-SVCs. Specifically, the verbal constituents which make up what I am treating as C-SVCs are all syntactically active, in that their argument-structural properties are not only legible to the syntax, but also factor into determining the syntactic properties of the overall construction. The most telling example in this regard is – as we will see – the fact that many languages block the formation of V-level configurations precisely when the syntactic non-head verb is transitive, but the head verb is transitive (a restriction that is due to the non-head verb’s inability to project in such configurations). A compounding analysis, which grants no proper syntactic status to the non-head verb, leaves this effect unexplained. In short, lexical autonomy feeds syntactic autonomy in C-SVCs, whereas lexical unity bleeds syntactic autonomy in V-V compounds.

It is based mainly on these criteria – (i) productivity, (ii) the syntactic independence and activity of all constituent verbs, and (iii) the argument-structural parallelism to NC-SVCs – that the constructions which are here classified as C-SVCs were identified and included as instances of serialization proper, alongside NC-SVCs but in opposition to V-V compounds; the latter are not investigated here any further.

2.4 Concepts and conventions

This section introduces further major concepts and conventions used throughout this case study.

2.4.1 SVC sub-types

A SVC with *object sharing* will be any SVC containing an object that is understood as being shared among two or more transitive verbs. *Single-object* and *multiple-object*

SVCs are SVCs which express only one or multiple objects, respectively. I will use *intransitive* and *transitive* SVCs to mean SVCs in which all verbs are intransitive and transitive, respectively (rather than SVCs in terms of the highest transitivity value among their components).

For example, (2a) from Edo is a transitive SVC with object sharing, (18a) from Korean is an intransitive SVC; while (2a) from Edo is a single-object SVC, the SVCs in (8) from Cantonese and (9) from Jeh are multiple-object SVCs.

2.4.2 SVC-internal syntactic properties

In order to distinguish verbal projections in terms of their morpho-syntactic status within a major subset of SVCs, I will use the terms *main* and *adjoined*; this distinction is reserved for SVCs which we will see involve adjunction. Among these, in SVCs with object sharing, main projections are characterized by hosting overt objects, and adjoined projections by hosting at least one silent object, understood as identifying an overt object in the main projection. Adjoined projections are hence dependent on main projections for full interpretation, a dependence which we will see may have formal reflexes.

For example, in the Edo NC-SVC in (2a), ‘hit goat’ is the main projection, and ‘sell’ the adjoined one.

For a particular subset of SVCs involving complementation, the main vs adjunct distinction is inadequate; I will introduce the proper terminology for these when discussion is due.

In this connection, we will also need terms to refer to verbs by way of their various dispositions within SVCs. To distinguish verbs in terms of their SVC-internal linear position, the shorthand V_n will be used: V_1 is thus the linearly first verb in a string, V_2 the second one, and so on. V_F will refer to the final verb in a string, and V_H to the syntactic head. At times, I will equate two properties, e.g. $V_{2=H}$ denotes the second verb which is also the syntactic head, and $V_{F=H}$ the string-final verb that is also the head. Whenever helpful for illustration, material belonging to main projections is colored red, and material belonging to dependent projections is colored blue.

Finally, I will at times use *VerbP* to abstract over verbal projections beyond the V-level, i.e. the phrasal projections VP or VoiceP; this label goes back to (Baker & Stewart 2002).

2.4.3 Transitivity and internal arguments

In this case study on SVCs, the notion of transitivity will by default pick out transitivity at the level of the verb stem, regardless of whether the respective verb stem occurs with an overt or projected object or not. E.g. in the Edo and Sranan transitive NC-SVCs in (2a) and (7a), both the main V_1 and the adjoined V_2 are transitive even though only the main V_1 occurs with an overt object. As we will see, the adjoined

transitive V_2 also projects an object, albeit a silent one. As both V_1 and V_2 are transitive at the stem-level, it follows that both V_1 and V_2 are also semantically transitive. Whenever discussions move to other parameters of transitivity, e.g. at the phrasal level in terms object agreement, I will make this explicit.

The terms *intransitive*, *transitive*, *ditransitive* and *ambi-transitive* will be abbreviated as *intr*, *trans*, *ditr* and *ambi-trans*, respectively, mostly in the context of pairings of transitivity values. E.g. [*intr* + *trans*] designates a pairing of an intransitive V_1 and a transitive V_2 .

I take the *Internal argument Requirement* of a verb, or its *IAR*, to be a consequence of the general assumption that all roles in a verb's argument structure must somehow be discharged – as expressed by the *Theta-Criterion* in generative grammar (Chomsky 1981), or the *Completeness Constraint* in Role and Reference Grammar (Van Valin 2005). Along the lines of Coon (2019), I assume that IARs are handled equally strictly across languages, with any differences reducing to the availability of different means of satisfaction, and how the effects of these are reflected morphologically.

2.4.4 Null arguments vs implicit arguments

One of the hallmark properties of prototypical transitive SVCs is that they show a many-to-one mapping from the number of transitive verbs to the number of overt objects, i.e. more objects are interpreted than there are pronounced. As a result, the issue of how non-overt objects come to be interpreted as co-referent with overt ones has been at the forefront of research into SVCs.

Slightly adopting the terminology of Bhatt & Pancheva (2017) and Williams (2015: 96), I define a *silent argument* as any argument that is non-overt, but which nevertheless participates in some grammatical dependency into which otherwise only overt ones can enter.²⁰ Because this by itself leaves open how exactly an argument is non-overt, let us further establish that a silent argument is an *null argument* if it is projected syntactically but remains covert, and that it is an *implicit argument* if it is not projected syntactically at all. Null arguments thus canonically include PRO, *pro*, traces and operators, while implicit arguments have been argued to be present in e.g. nominals, passives and modals; note that the question of which arguments syntactically project or not is far from settled.²¹ We can generalize for our purposes that implicit arguments do not occupy a syntactically projected position, but remain

²⁰Silent arguments thus construed are different from Williams (2015: 95)' *unrealized roles*, which broadly refer to any entailed relation that does not correspond to any obvious part of the derivation. This by itself, however, is too permissive, given that the lexical semantics of a verb may entail quite a range of unrealized roles, which may or may not be referable to by the grammar.

²¹PRO and *pro* are conventionally often grouped together under the separate label of *null subjects* (Bhatt & Pancheva 2017). For differing views on the projection status of silent arguments in various constructions, typically centering around the elements referred to as *pro* / PRO, see Bach & Partee (1980), Epstein (1984), Rizzi (1986), Jackendoff (1987), Roeper (1987), Partee (1989), Alexiadou & Anagnostopoulou (1998), Sichel (2009), Solstad (2009), Landau (2010), Borer (2021) and Michelioudakis (2021), among others.

grammatically active. In terms of linking, then, implicit arguments correspond to those slots in the argument structure of a verb which remain unlinked (Williams 1985: 314). We will see that NC-SVCs involve null arguments in the form of minimal pronouns (Kratzer 2009), while C-SVCs involve implicit arguments.

2.5 Transitivity profiles

SVCs characterize three major profiles in terms of which transitivity values their constituent verb stems may have. One important dimension along which SVCs turn out to pattern in this regard is contiguity. Among NC-SVCs, we find two types of languages: *Type A* languages place no restrictions whatsoever on licit transitivity pairings, whereas *Type B* languages require each verb to be at least transitive. Among C-SVCs, we find one basic type, *Type C*: these languages characterize a 3/4 pattern in which pairings of a transitive V_1 with an intransitive V_2 are ill-formed when V_2 is the syntactic head:

		<i>intr + intr</i>	<i>trans + trans</i>	<i>intr + trans</i>	<i>trans + intr</i>
NC-SVCs:	Type A	✓	✓	✓	✓
	Type B	*	✓ ^a	*	*
C-SVCs:	Type C	✓	✓	✓	* ^b

a. Either V_1 or V_2 may be ditransitive

b. Where V_2 is the syntactic head

Table 2.1: Major transitivity profiles in SVCs

Note that I am factoring out SVCs with resultative meanings because these involve a dedicated structure-building mechanism that is independent of serialization; see Section 2.6.

2.5.1 NC-SVCs

2.5.1.1 Type A: no restrictions

In profile Type A, found among NC-SVCs, there are no restrictions whatsoever, i.e. verb stems of any transitivity values can be paired. This was illustrated for Korean at the outset, with the relevant data repeated here. Next to allowing verb stems of like transitivity (19)-(20), Korean allows disparate pairings in either order, i.e. [*intr + trans*] (14c) and [*trans + intr*] (22):

(14) **Korean**

- a. *intr + intr*
 Say-ka [nal-a] [ka-ss-ta]
 bird-NOM fly-LK go-PST-DECL
 ‘The bird flew away’ (Kang 1997: 6)
- b. *trans + trans*
 John-i thokki-lul [cap-a] [mek-ess-ta]
 J.-NOM rabbit-ACC catch-LK eat-PST-DECL
 ‘John caught and ate a rabbit’
- c. *intr + trans*
 John-i wulthali-lul [ttwi-e] [nem-ess-ta]
 J.-NOM fence-ACC jump-LK go.over-PST-DECL
 ‘John jumped over a fence’²² (Ko & Sohn 2015: 100, 108)
- d. *trans + intr*
 Say-ka thokki-lul [cap-a] [nal-ass-ta]
 bird-NOM rabbit-ACC catch-LK fly-PST-DECL
 ‘The bird caught a rabbit and flew away’²³ (Hyunjung Lee p.c.)

Profile Type A is very widespread, and from what I gather impressionistically, it may be the most frequent one cross-linguistically. Other languages with this profile include Goemai (Chadic) (Hellwig 2006), Khwe (Khoisan) (Kilian-Hatz 2006), Malayalam (Dravidian) (Jayaseelan 2004, Son & Svenonius 2008), Sranan (English creole) (Sebba 1987) and White Hmong (Hmong-Mien) (Jarkey 1991, Jarkey 2015).²⁴

2.5.1.2 Type B: $\geq$ [*trans + trans*] pairings

Unlike Korean-type languages, other languages show a more restricted transitivity profile in NC-SVCs. The most well-known of these are perhaps the Niger-Congo languages Edo, Nupe and Yoruba, discussed extensively in Stewart (2001) and Baker & Stewart (2002).

In Edo, for example, we find NC-SVCs with object sharing, where both V_1 and V_2 are transitive; only the main V_1 appears with an overt object, while the dependent V_2 ’s object is silent:

- (15) **Edo:** *trans + trans*
- a. Òzó ghá [gbè] èwé [khièn]
 O. FUT hit goat sell

²²Note that the V_1 *ttwi-e* ‘jump’ is unergative and cannot take objects, i.e. *wulthali-lul ttwi-ta* ‘jump a fence’ is ungrammatical (Ko & Sohn 2015: 96). The object *wulthali* ‘fence’ belongs only to the transitive V_2 *nem-ess-e* ‘go over’.

²³(22) requires a single-interpretation, in which the bird catches the rabbit and then flies away with the rabbit; a multiple-event interpretation, as afforded or forced by the addition of distinct temporal adverbs, is degraded (Hyunjung Lee p.c.).

²⁴Regarding Goemai, the only [*intr + trans*] SVCs given in Hellwig (2006) involve an intransitive motion verb as V_1 . Hellwig (2006: 96) states explicitly that there are (i) no restrictions on the transitivity values on constituent verbs, and that (ii) these may occur in any order.

- 'Ozo will kill the goat and sell it'
- b. Òzó ghá [mièn] iyán [èvá] lé
 O. FUT find yam two cook
 'Ozo will find two yams to cook (and do so)'

(Baker & Stewart 2002: 2, 3)

Edo also allows [*trans* + *intr*] pairings with resultative meanings:

- (16) **Edo:** *trans* + *intr* (resultative)

- Òzó ghá [gbè] èwé [wù]
 O. FUT hit goat die

'Ozo will strike the goat dead'

(Baker & Stewart 2002: 2)

These are discussed separately below, where they will be factored out due to involving a mechanism that is distinct from serialization proper. Once resultatives NC-SVCs are controlled for, then, it becomes clear that Edo allows all verb in NC-SVCs to be at least transitive, which I notate as $\geq[trans + trans]$. Neither two unergative verb stems (17a), nor an unergative and a transitive verb stem in any order (17b)-(17c), are possible:

- (17) **Edo**

- a. *unerg* + *unerg*
 *Òzó gié!gié rhú!lé [sá!án]
 O. quickly run jump
 intended: 'Ozo quickly ran and jumped'
- b. *unerg* + *trans*
 *Ékítà gié!gié [gbó!ó] [khú] áhiánmwèn
 dog quickly bark chase bird
 intended: 'The dog quickly barked and chased a bird'
- c. *trans* + *unerg*
 *Ékítà [khú] áhiánmwèn [gbò!ó]
 dog chase bird bark
 intended: 'The dog chased a bird and barked'

(Baker & Stewart 2002: 49, 52)

The same restriction holds for Ewe (Collins 1997):²⁵

²⁵The claim that Ewe allows only $\geq[trans + trans]$ NC-SVCs seems problematic in face of Ameka (2006), who lists a number of apparent counter-examples as well-formed NCSVCs:

- (i) **Ewe**

- a. *intr* + *trans* + *trans* + *trans*
 éyata [é-dzo] [hé-vá-dze] anyígbá [hé-tso] atukpá lá
 therefore 3SG-move.above.ground ITIVE-VENT-contact ground ITIVE-take bottle DEF

(18) **Ewe:** *trans + trans*

Me [ɖa] nu [ɖu]
I cook thing eat

‘I cooked something and ate it’

(Collins 1997: 467)

Like in Edo, the restriction to $\geq[trans + trans]$ NC-SVCs in Ewe only becomes fully apparent once resultative NC-SVCs are controlled for.

The well-formed $[trans + trans]$ pairings above all show object sharing, where V_2 ’s object is silent; yet Edo, Nupe and Ewe also allow instrumental NC-SVCs in which both transitive verbs appear with overt objects:

(19) **Ewe:** *trans + trans*

Kofi [tsɔ] ati-ε fo Yao
K. take stick-DEF hit Y.

‘Kofi took the stick and hit Yao with it’

(Collins 1997: 461)

These instrumental NC-SVCs will be shown to not contradict the proposed ban against pairings other than $[trans + trans]$, because they involve complementation, as opposed to object-sharing NC-SVCs like (15) and (18), which involve adjunction.

In profile Type B, either verb may also be ditransitive; such pairings are discussed later.

2.5.2 C-SVCs

2.5.2.1 Type C: no $[trans + intr_H]$ pairings

Turning to C-SVCs, we find one illicit transitivity pairing which recurs systematically in numerous languages, namely those in which V_1 is transitive and V_2 is intransitive, and the meaning is non-resultative. This restriction is tied to headedness: when the transitive V_1 is the non-head and the intransitive V_2 is the syntactic head – and the meaning is non-resultative – C-SVC formation fails. I therefore notate this transitivity

-
- [zi ɖi]...
press down
‘Therefore he jumped onto the ground and smashed the bottle on the ground...’
b. [Mié-le] klo dzi [le] kúkú ɖe-m
1PL-be.at.PRES knee upper.surface be.at.PRES hat remove-PROG
‘We were on our knees begging’ (Ameka 2006: 129, 136)

However, Ameka’s definition of SVCs subsumes constructions which include independent marking for aspect and modality (Ameka 2006: 128, 137). E.g. in (ib), V_1 is specified for aorist aspect (which is zero-marked), and V_2 for progressive aspect. The definition of SVCs adopted in this thesis (and Collins (1997)) is more restrictive in not allowing constituent verbs to be independently marked for TAM. Constructions like (ia) and (ib) do not involve serialization as understood here, and therefore pose no problem for the generalization that Ewe restricts NC-SVCs to $[trans + trans]$ pairings (perhaps these are instances of covert coordination, or what Kalin & Keenan (2011) call *TP-serialization*.)

profile as $*[trans + intr_H]$.

Consider first Tlachihilco Tepehua, a Totonacan language spoken in Mexico, where C-SVCs generally consist of two verbs. These characterize a 3/4 pattern in terms of licit transitivity pairings: $[intr + intr]$ (20a), $[trans + trans]$ (20b), and $[intr + trans]$ pairings can form a C-SVC:²⁶

(20) **Tlachihilco Tepehua**

a. *intr + intr*

[talaqsta]-[laka:wa:nan]
turn.face.up-look.IPFV

‘(S)he looks, facing upward’

b. *trans + trans*

[ʔap’i]-[ʔu-y]
peel-eat-IPFV

‘(S)he eats it, peeling it’

c. *intr + trans*

[tapasa]-[ʔi:-ɬ]
pass.by-get-PFV

‘(S)h got it passing by’

(Watters 2017: 288)

The only illicit pairing involves a transitive V_1 and an intransitive V_2 (21a), provided that the meaning yielded is non-resultative. In order to get around this restriction, two adjustments are available: either the transitive V_1 can be detransitivized (21b), or the intransitive V_2 can be transitivized (21c):²⁷

²⁶In Watters (2017), these constructions are classified as V-V compounds, though Watters himself notes similarities to SVCs. Given the distinction between C-SVCs and V-V compounds adopted here, these constructions qualify as C-SVCs on grounds that their constituents exhibit lexical autonomy: phonological processes may make reference to complex-internal boundaries (Watters 2017: 285), certain inflectional markers appear between the two members (Watters 2017: 286), and valence-changing operations may target individual verbs (33d).

²⁷In light of the fact that motion verbs may give rise to restructuring, note that we can rule out a complementation analysis of the C-SVCs in (33d) given that – if the V_2 *ʔan* ‘go’ took a complement – it would not need to undergo a valence increase in (21c). In foreshadowing of the analysis to come, the C-SVCs in (33d) involve V-level adjunction.

(21) **Tlachihilco Tepehua**a. *trans + intr*

*[tʃaqʃ]-[ʔan]

chop-go.IPFV

intended: '(S)he goes chopping'

b. *intr + intr*[tʃʼaqʃ-**nah**]-[ʔan]

chop-ANTIP-go.IPFV

'(S)he goes chopping'

c. *trans + trans*

[tʃaqʃ]-[ʔa:-ʔan]

chop-APPL-go.IPFV

'(S)he goes chopping it'

(Watters 2017: 289f)

According to Watters (2017: 289), if V_1 is transitive, V_2 must also be transitive, a condition which strictly limits the set of possible C-SVC pairings.²⁸

Another language which bans non-resultative [*trans + intr_H*] pairings in C-SVCs is the Oceanic language Paamese, spoken in Vanuatu. V_1 and V_2 are allowed to have the same transitivity ((22a),(22b)), and [*intr + trans*] pairings are also fine (22c):

²⁸In Watters (1988: 94), this condition is stated as follows: "The modifying verb cannot be transitive if the head verb is intransitive". To further see the systematicity of this condition, consider the following pair, in which the (b) example is ruled out because V_1 is transitive and V_2 intransitive:

(i) **Tlachihilco Tepehua**a. *trans + trans*

[tʃʼan]-[lhi:-min]-ta

sow-APPL-come-IMPF

'(S)he is coming sowing / planting it'

b. *trans + intr*

*[tʃʼan]-[min]-ta

sow-come-PERF

intended: 'She is coming sowing / planting (it)'

(Watters 2017: 290)

(22) **Paamese**a. *intr + intr*

Kaie [muali] [suai]
 3SG 3SG.REAL.walk disappear
 'He was walking without being seen' (Crowley 1987: 38)

b. *trans + trans*

Samsene [mungali] [vaasi] velaase-nV laiane
 S. 3SG.REAL-rip.open split jaw-CONST lion
 'Samson ripped apart the lion's jaw'

c. *intr + trans*

Kaie [matilu] [vaasi-e]
 3SG 3SG.REAL.sleep split-SG
 'He slept (on it), thereby splitting it' (Crowley 1987: 62)

However, if V_1 is transitive, V_2 must also be transitive (23). According to Crowley (1987: 62f), because transitive verbs internal to the verbal complex in C-SVCs cannot independently express their objects, any object must be expressed as an object to the entire construction, so that when V_1 is transitive, so must V_2 .²⁹

(23) **Paamese:** *trans + trans*

Inau [na-lahi] [pilu-ni] puuke ona-ku
 1SG 1SG.REAL-carry stick.together-TRANS book POSS-1SG
 'I carried my books in one hand' (Crowley 1987: 63)

Just like in Tlachihilco Tepehua, non-resultative [*trans + intr*] pairings are unavailable, such that either V_1 must be detransitivized or V_2 transitivized.³⁰

While languages like Paamese lack [*trans + intr*] pairings entirely (Crowley 1982, Crowley 1987), other languages systematically restrict [*trans + intr*] pairings to only those expressing resultative meaning. One such language is #Hoan, which allows [*intr + intr*] (24a), [*trans + trans*] (24b) and [*intr + trans*] pairings (24c).^{31,32}

²⁹Alternatively, V_2 can be made into an extended intransitive via the formally related preposition *eni*, which renders the object oblique rather than structural (Crowley 1987: 63).

³⁰As is typical of Oceanic languages, Paamese lacks productive valence-decreasing operations (Crowley 1982).

³¹Regarding (24a), Collins glosses the V_1 | | *hoam*- | | *hoam* 'jog' as a non-reduplicated form. In (24c), we can think of the V_2 *ka* 'go in' as containing a semantically incorporated adposition, similar to the transitive verb 'enter'.

³²Collins (2002: 15) notes that not just any [*intr + trans*] pairing is available, citing ungrammatical examples like the following:

(i) *intr + trans*

- a. Ma a- [ki'n'o] [kisile] n!ui
 1SG PROG run.SG imitate ostrich
 intended: 'I am running imitating an ostrich'

(24) #Hoan

a. *intr + intr*

Ma | | oe na ka [| | hoam- | | hoam] [tca]
 1SG still AUX SUB jog come
 ‘while I was still coming jogging’

b. *trans + trans*

Ma qo kí- [tsaxo] [‘am] | | a’’e
 1SG FUT PLURACT cook eat meat
 ‘I will cook and eat meat (repeatedly)’

c. *intr + trans*

#i-si a- [kala] [ka] | ’on ci q | | ’am
 bird-DIM PROG fly go.to tree POSS top
 ‘The bird is flying to the top of the tree’

(Collins 2002: 4, 15)

[*trans + intr*] pairings are limited to resultatives, as in (25a); (25b) shows that as an independent verb, the V₂ in (25a) is an intransitive stative:³³

(25) #Hoan

a. *trans + intr*

Ma i [q | | a] [| | hon- | | hon] bele-qa
 1SG PST pound ground.up sorghum
 ‘I ground up the sorghum by pounding it’

b. Bele-qa | ho’on i | | hon- | | hon
 sorghum not PST ground.up
 ‘The sorghum is not ground up well’

(Collins 2002: 23)

Another language that restricts [*trans + intr*] C-SVCs to resultatives is Igbo (Mary Amaechi p.c.).

To conclude, the ban against non-resultative [*trans+intr_H*] pairings in C-SVCs is

- b. *Ma a- [n | obo] [‘am] | | a’’e
 1SG PROG speak eat meat
 intended: ‘I am eating meat while speaking’

(Collins 2002: 15f)

Collins proposes that the [*intr + trans*] pairings in (i) are ruled out not for transitivity, but rather event-structural reasons, arguing that #Hoan requires SVCs to be non-simultaneous in their event structure (cf. (24b), which only has a reading of repeated alternation), and that the intended meanings of (i) express complex events with simultaneous sub-events.

³³Note that the V₂ | *hon- | hon* in (25a) does not refer to the semantically dyadic activity of grinding, but to the monadic state of being ground up; cf. (25b). It is in fact impossible to transitivize | *hon- | hon* ‘ground up’:

(i) #Hoan

*Ma i (ki-) | | hon- | | hon bele-qa
 1SG PST CAUS-ground.up sorghum

intended: ‘I caused the sorghum to be ground up’

(Collins 2002: 23)

systematic. We will see that precisely these pairings are blocked in C-SVCs not simply as a matter of linear order, but because of how their underlying V-level configurations interact with headedness: when the transitive V_1 is the syntactic non-head and the intransitive V_2 is the syntactic head, C-SVC formation fails. The head-finality of these C-SVCs is diagnosed by various morpho-syntactic, such as inflection and allomorphy, and argument blocking effects; these are discussed in detail in the next chapters, and also in Part 2 on transitivity matching.

Other languages which ban non-resultative [*trans+intr_H*] pairings in C-SVCs are Khwe (Central Khoisan) (Kilian-Hatz 2006), as well as numerous Oceanic languages, including East Uvean (Moyse-Faurie 2004), Mwotlap (François 2004, François 2006), Nêlêmwa (Bril 2004), Pileni (Næss 2004), Teop (Reinig 2004, Mosel 2006) and Toqabaqita (Lichtenberk 2006, Lichtenberk 2008).³⁴

2.6 Resultative and directional SVCs

In this section, we will see that constructions expressing resultative and (certain kinds of) directional meanings involve a dedicated structure-building mechanism that is independent of serialization; for this reason, they will be factored out and not considered instances of serialization proper.

Consider the following Edo resultative NC-SVC, where V_1 is transitive and V_2 intransitive:

(26) **Edo:** *trans + intr (resultative)*

Òzó ghá [gbè] èwé [wù]
O. FUT hit goat die

‘Ozo will strike the goat dead’

(Baker & Stewart 2002: 2)

As argued by Baker & Stewart themselves, NC-SVCs like (26) are built by a mechanism that is distinct from the one involved in NC-SVCs with object-sharing like (27):

(27) **Edo:** *trans + trans (non-resultative)*

Òzó ghá [gbè] èwé [khièn]
O. FUT hit goat sell

‘Ozo will kill the goat and sell it’

(Baker & Stewart 2002: 3)

The authors back up this view using Edo-specific evidence. In Edo, the emphatic

³⁴Note that for some of these languages, the respective works do not explicitly state that such pairings are unavailable, but rather such pairings are systematically absent in the descriptions. There are other languages whose C-SVCs also appear to ban [*trans + intr*] pairings, yet for which I was unable to clear up all reasonable doubts due to the absence of certain data points. These include Imonda (Upper Tami) (Seiler 1986), Kalam (Trans-New Guinea) (Lane 1991) and Tetun Dili (Hajek 2006).

reflexive-like element *tobore*, which can right-adjoin to any overt or empty category, diagnoses the presence of a trace in simple unaccusative clauses (28a), and conversely the absence of a trace in simple unergative clauses (28b):

(28) Edo

a. *Unaccusative* + *tobore*

Úyì_i dé **tòbórè_i**

U. fall by.self

‘Uyi fell by himself’

b. *Unergative* + *tobore*

*Úyì_i só **tòbórè_i**

U. shout by.self

‘Uyi shouted by himself’

(Baker & Stewart 2002: 21f)

tobore can follow the V_2 in non-resultative NC-SVCs (29a), but not the V_2 in what the authors classify as resultative NC-SVCs (29b):

(29) Edo

a. *Consecutive NC-SVC*

Òzó [dé] ìyán_i [dùnmwún] **tòbórè_i**

O. buy yam pound by.self

‘Ozo bought the yam and pounded it by itself’

b. *Resultative NC-SVC*

*Òzó [sùá] ògò_i [dé] **tòbórè_i**

O. push bottle fall by.self

intended: ‘Ozo pushed the bottle down by itself’

(Baker & Stewart 2002: 22f)

Baker & Stewart conclude from this that only non-resultative NC-SVCs involve an empty category after V_2 .

The authors propose that what they call resultative NC-SVCs are formed via complementation, rather than adjunction as in the other NC-SVCs, and invoke the notion of *LF head movement* (Saito & Hoshi 2000, Saito 2001), which enables the transitive V_1 and unaccusative V_2 to covertly form a complex predicate.³⁵

Recognizing that resultatives involve a distinct compositional mechanism from other kinds of serialization is important, not only in Edo but in general. We should therefore expect the two to be independent, which is correct. Saramaccan, for example, has NC-SVCs with object sharing (like Edo), but cannot use serialization to express resultative meaning (unlike Edo) (30a). The so-called *means construction* must be used instead (Sæbø 2016, Hopperdietzel 2020), in which the result state is not

³⁵Collins (1997) uses LF movement for a similar problem in Ewe. See Kamiya (2006) for problems with this notion.

expressed directly by an intransitive V_2 , but rather entailed by a transitive V_2 (30b):

(30) **Saramaccan**

a. *trans + unacc (resultative)*

*De [faa] dí páu [kaí]
3PL chop DET tree fall
intended: ‘They felled the tree’

b. *trans + trans (means construction)*

De [fáa] dí páu [túe]
3PL chop DET tree throw
‘They felled the tree’

(Veenstra 2004: 275, 274)

The same holds for Korean (Ko & Sohn 2015), Malayalam (Jayaseelan 2004, Son & Svenonius 2008), Paiwan (Crowley 1987) and Saliba (Margetts 1999, Margetts 2005), all of which have serialization but no resultatives; conversely, many languages have resultatives but not serialization, such as English and German. See Kaufmann & Wunderlich (1998) for cross-linguistic patterns of resultatives, and Hopperdietzel (2022) for how serializing languages vary in the expression of resultative meaning.

The distribution of resultative SVCs hence cross-cuts the contiguity dimension: just like C-SVCs may allow (e.g. #Hoan) or ban resultatives (e.g. Paamese), so, too, NC-SVCs may allow (e.g. Edo) or ban resultatives (e.g. Saramaccan), subject to language-specific conditions on resultative formation.

This is expected in view of the fact that a lot of work has shown the kind of complex predication underlying resultatives to not correspond to a mere conjunction of the manner and the result component – as will be argued here to underlie serialization proper – but that resultatives are instead either (i) the result of a dedicated operation of resultative formation (Kaufmann & Wunderlich 1998), or that (ii) causative / resultative meaning is read off of structural configurations meeting certain conditions (Alexiadou et al. 2015, Hopperdietzel 2020). Moreover, works like Kaufmann & Wunderlich (1998) and Williams (2012) have argued that objects of resultatives are not arguments of either the manner or the result component, but introduced as arguments of the entire resulting complex predicate.³⁶

In terms of the mode of structure-building, resultatives show properties of both complementation (e.g. due to tight syntactico-semantic restrictions between the manner and result component) and adjunction (both the manner and the result component are free to occur alone, and resultative formation is optional) (Zhang 2007, Beavers 2012). Resultatives hence correspond to what we can think of as an optional, yet natural extension of a verb’s argument structure, a fact which is reflected in (i) the sensitivity of resultatives to their constituents’ lexical semantics (Kaufmann &

³⁶This seems especially evident in so-called *strong resultatives*, a term going back to Washio (1997), which project objects that are not sub-categorized for by the manner verb, e.g. *She sang her throat hoarse* (cf. **She sang her throat*).

Wunderlich 1998, Williams 2012, Williams 2014, Iwata 2020), and (ii) their uniformly belonging to the most highly integrated class of complex predicates across languages (Beavers 2012, Williams 2012, Zimmermann & Amaechi 2018); by contrast, adjuncts constitute truly independent optional additions, with degrees of integration that are much more spread out (Williams 2015).

By granting special status to a subset of [*trans* + *intr*] pairings in C-SVCs, namely those which express resultative meaning, we can factor these out from the broader landscape of what is otherwise a robust ban against contiguous [*trans* + *intr*] pairings. In other words, the availability of resultative [*trans* + *intr*] pairings obscures what is otherwise a robust generalization. Likewise, recognizing that in a language like Edo, the only [*trans* + *intr*] pairings allowed in NC-SVCs are resultative is what allows the generalization that Edo NC-SVCs only accept [*trans* + *trans*] pairings to fully emerge.

Finally, there is another class of C-SVCs which at first glance appear to falsify the ban against contiguous [*trans* + *intr*] pairings, namely those expressing directional meanings. In e.g. Teop (Oceanic, Bougainville), C-SVCs in (31) consist of verbs of like transitivity, and also of disparate transitivity with an intransitive *V*₁ and a transitive *V*₂:

(31) **Teop**

a. *intr* + *intr*

Me-nam paa [rosin] [tagihu]
and-1EXCL PST flee go.to.the.bush
'And we fled into the bush' (Reinig 2004: 98)

b. *trans* + *trans*

Nomana to kahi [asun] [va-mate=mate] ri nom naa
today REL TAM hit CAUS-RED=dead 3PL 1SG.IMPF 1SG.SUBJ
ori
3PL.O
'It's today that I'll kill them!' (Mosel 2006: 3)

c. *intr* + *trans*

E bubuu na [hio] [ihuana] nana bona nahu kaukau
ART granny REAL sit wait.for IMPF.3SG ART pot sweet.potato
'Granny is sitting and waiting for the pot of sweet potatoes' (Reinig 2004: 99)

[*trans* + *intr*] pairings are limited to pairings in which *V*₂ is a directional verb:

(32) **Teop: Directional C-SVC**

a. *trans* + *intr*

A anohi na [hiri] [hiava] bata-na moo-naa
ART black ant REAL crawl go.up SIMUL-IMPF.3SG leg-1SG
'The black ant is crawling up my leg'

b. *trans + intr*

Ausu e tama-man tea [deedee] [hovo] a maa ta naono
 help ART father-2SG COMP carry go.in ART PL timber
 ‘Help your father carry the pieces of timber inside (the house)’

(Reinig 2004: 96)

That directional and resultative C-SVCs should pattern alike in this regard is not accidental, as authors have proposed a tight correlation between resultative and directional meanings (Kaufmann & Wunderlich 1998, Beck & Snyder 2001, McIntyre 2004). The directional V_2 expresses a directional change of state which the object referent enters into as a result of the action denoted by V_1 . Given that directional verbs may be agentive, i.e. are unergatives, the question arises as to how they can end up in result functions, which are otherwise expressed by change-of-state or stative verbs in resultatives, i.e. typically unaccusatives. It is in fact a cross-linguistically common process for unergative verbs to become unaccusative when telic Levin & Rappaport Hovav (1995: 185).³⁷ As with their resultative siblings, then, the apparent exceptionality of directional C-SVCs to the ban against [*trans + intr*] pairings finds a principled explanation once they are treated apart from other cases of serialization.

2.7 Summary

To summarize, we find the following transitivity profiles in NC-SVCs: Type A is completely unrestricted; in Type B, each verb must be at least transitive. C-SVCs characterize a 3/4 pattern, with a systematically recurring ban against non-resultative [*trans + intr_H*] pairings, with V_2 as the syntactic head:

		<i>intr + intr</i>	<i>trans + trans</i>	<i>intr + trans</i>	<i>trans + intr</i>
NC-SVCs:	Type A	✓	✓	✓	✓
	Type B	*	✓ ^a	*	*
C-SVCs:	Type C	✓	✓	✓	* ^b

a. Either V_1 or V_2 may be ditransitiveb. Where V_2 is the syntactic head

Table 2.2: Major transitivity profiles in SVCs

³⁷The same phenomenon is noticed by Collins (1997) in NC-SVCs in Edo.

Chapter 3

Analysis

3.1 Introduction

The previous chapter established the major transitivity profiles found in SVCs. In this chapter, we will see that these patterns can be derived if we cross-cut the theory of serialization to be introduced for a major subset of SVCs with the level of syntactic structure-building.

This chapter is structured as follows: we begin in Section 3.2 with an overview of the modes of structure-building. Section 3.3 outlines existing generative approaches to the syntax of SVCs, after which Section 3.4 argues why covert coordination is here not considered as falling under serialization proper. Section 3.5 goes into diagnostics for telling apart main and dependent projections in SVCs, and discusses important cross-linguistic generalizations about headedness and verb order in serialization that will inform the analysis. The general analysis presented at the end in Section 3.6.

3.2 Modes of structure-building

This section outlines the core properties of the major modes of structure-building: complementation, adjunction and coordination.

The main difference between complementation and adjunction is that in complementation, a head carries an intrinsic selectional requirement that must be satisfied by an appropriate argument. Merging this argument saturates or "checks off" this intrinsic need. No such relationship holds in adjunction: adjuncts do not satisfy any intrinsic need of the head it is a dependent of, i.e. they are not selected. More technically, complements and adjuncts differ in their effects on the features of the phrase resulting after their merger: Y is a complement to X if the phrase resulting from merging X and Y differs in some feature from X; Y is an adjunct to X if this is not the case (Williams 2015: 51). Arguments are generally obligatory, while adjuncts are always optional.³⁸

³⁸Note that it does not suffice for an element to be a complement-taking head if it is a functor. If

While the question of whether coordination actually involves a kind of complementation (between the coordinator and one or more of the coordinated elements) or adjunction is a matter of ongoing debate, it is undeniably closer to adjunction in that coordination per se is optional, not required by some property that is intrinsic to any of the elements coordinated; see e.g. Neeleman et al. (2022), and references therein.

Given their nature, adjunction and coordination configurations can express a broad variety of meanings that are relatively unconstrained, many of which would be highly unusual or unattested as the result of complementation. Complementation, by contrast, is known to yield a more restricted set of meanings, which – in the verbal domain – include temporal, aspectual / phasal, desiderative, conative, and manipulative / causative meanings; many of these involve a more abstract semantics, and are hence more prone to be realized as functional items in a given language; see e.g. Grano (2015). Canonical instances of complementing verbs are – roughly in order of an increasingly functional (i.e. decreasingly lexical) semantics – control, light verb, raising and auxiliary constructions.

These systematically map onto an abstract interpretive core, which is that the meaning of the complement Y is an integral part of the meaning of the phrase resulting from the very operation of complementing Y with X. Complementation thus involves mereological meaning relationships, and it is monotonic in the sense that the meaning contributed by the head elaborates on the meaning of the complement. Adjunction and coordination, by contrast, add optional information, and express conjunctive meanings (Svenonius 2016). Despite both yielding conjunctive meanings, adjunction generally lacks a linking element, unlike coordination.

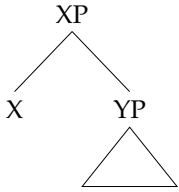
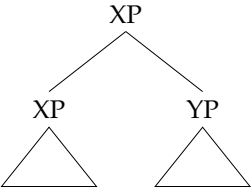
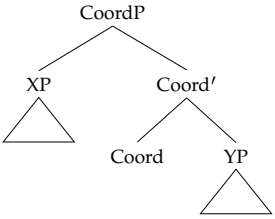
In terms of extraction, complements generally allow (sub-)extraction, while adjunction and coordination do not, except in principled cases; see Ross (1967), Cattell (1976) and Huang (1982).³⁹

See Baker & Harvey (2010), Ramchand & Svenonius (2014) and Svenonius (2016) for general discussion of the differences between these modes of structure-building, among others; for discussion specifically with respect to serialization, see Veenstra & Muysken (2017) and Zimmermann & Amaechi (2018).

These major properties are summarized in Table 3.1:

this were the case, adjuncts would – under a Montagovian account that treats adjuncts as dependent elements – also count as complement-taking heads, namely derived ones. An element thus counts as complementing if its need for an argument is intrinsic.

³⁹Huang's (1982) *Condition on Extraction Domains* (CED) has come to face numerous challenges over the years, both empirically and theoretically; see Stepanov (2007) for a summary of its shortcomings. For recent works that either attempt to resuscitate the CED and / or derive its exceptions, see Truswell (2007), Müller (2010), Sheehan (2013) and Ernst (2022).

Property	Complementation	Adjunction	Coordination
			
Interpretation	mereological	conjunctive	conjunctive
Selection	✓	*	*
Linking element	*	*	✓
Extraction	YP: ✓	XP: ✓; YP: * ^a	XP: *; YP: * ^b

a. Except in rare cases, cf. Truswell (2007).

b. Except in the case of ATB extraction.

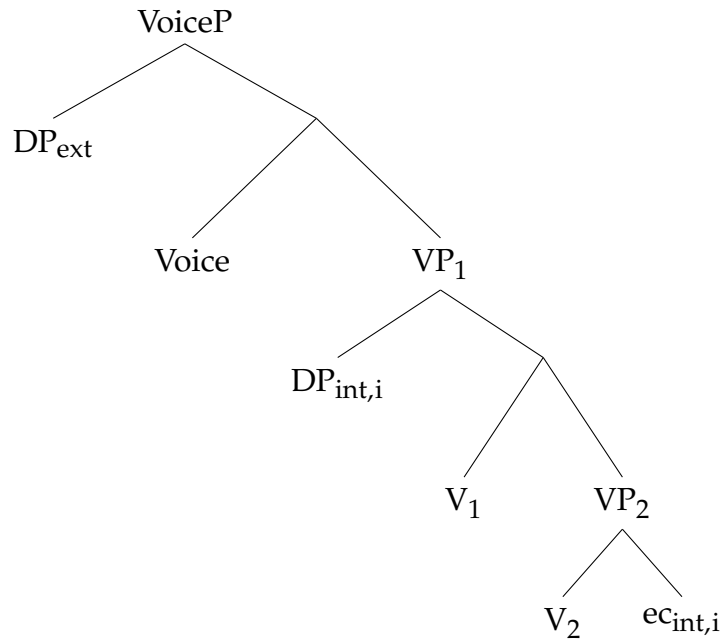
Table 3.1: Modes of structure-building

3.3 Generative approaches to the syntax of SVCs

SVCs in the generative tradition have been claimed to involve any of the basic modes of syntactic structure-building. In the following, each mode is illustrated for NC-SVCs with object sharing serialized at the VP-level, because it is this sub-type which most clearly brings out the syntax underlying serialization. Note that coordination is considered here for the purposes of this overview, though we will later discard it.

3.3.1 Serialization as complementation

Under a complementation analysis, V_1 takes VP_2 as its complement. Arguments for this kind of analysis have included the extractability of arguments, syntactico-semantic restrictions between V_1 and V_2 , and the obligatoriness of VP_2 (conversely, the inability of V_1 to occur on its own). Analyses vary with respect to the nature of V_1 , with some treating it as a functional or light verb, and others taking it to be a normal lexical verb. Object sharing is generally analyzed in terms of syntactic binding: V_1 's object is projected overtly while V_2 's object is projected as an empty argument, with the former determining the reference of the latter by virtue of c-command. See Tree 3.1:

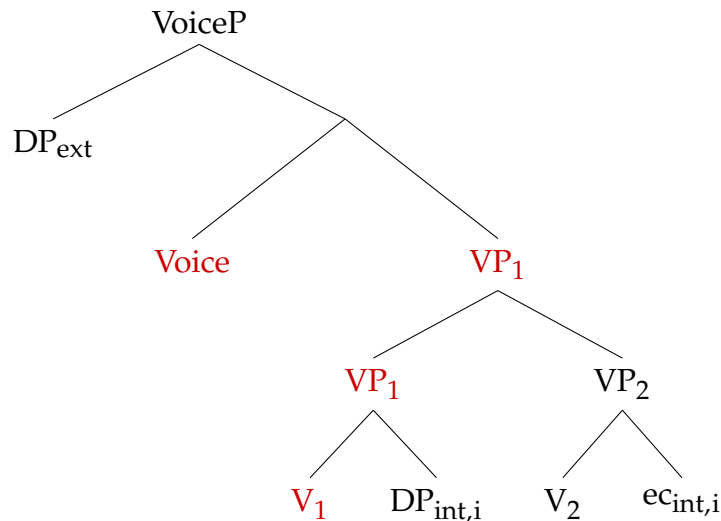


Tree 3.1: NC-SVC via complementation

Works arguing that (some or all) SVCs are built via complementation include Jansen et al. (1978), Sebba (1987), Larson (1991), Lefebvre (1991), Collins (1997), Collins (2002), Hagemeijer (2001), Aboh (2009) and Cleary-Kemp (2015).

3.3.2 Serialization as adjunction

Under the second kind of analysis, adjunction, a VP_2 is adjoined to a main VP_1 . Evidence for the adjunct status of VP_2 comes from the non-extractability of it or its proper constituents, the lack of a tight syntactico-semantic bond between V_1 and V_2 , and the optionality of VP_2 (conversely, the ability of VP_1 to occur on its own). Co-reference between V_1 's and V_2 's objects is achieved not necessarily via c-command, but may be brought about by e.g. co-indexation and / or m-command. See Tree 3.2:

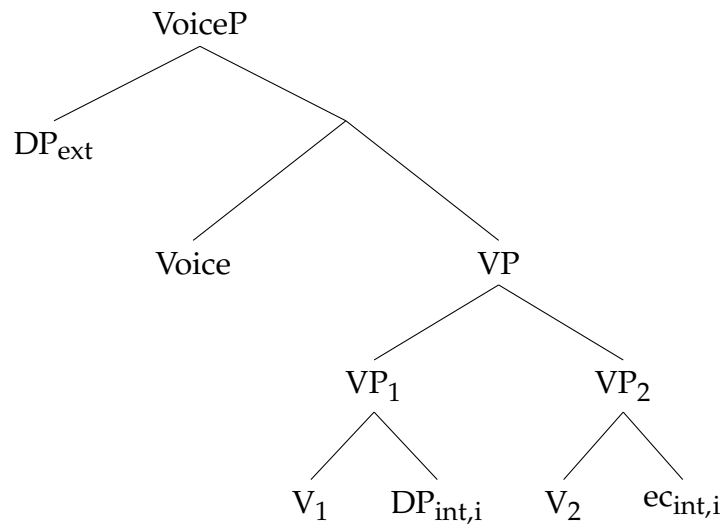


Tree 3.2: SVC via adjunction

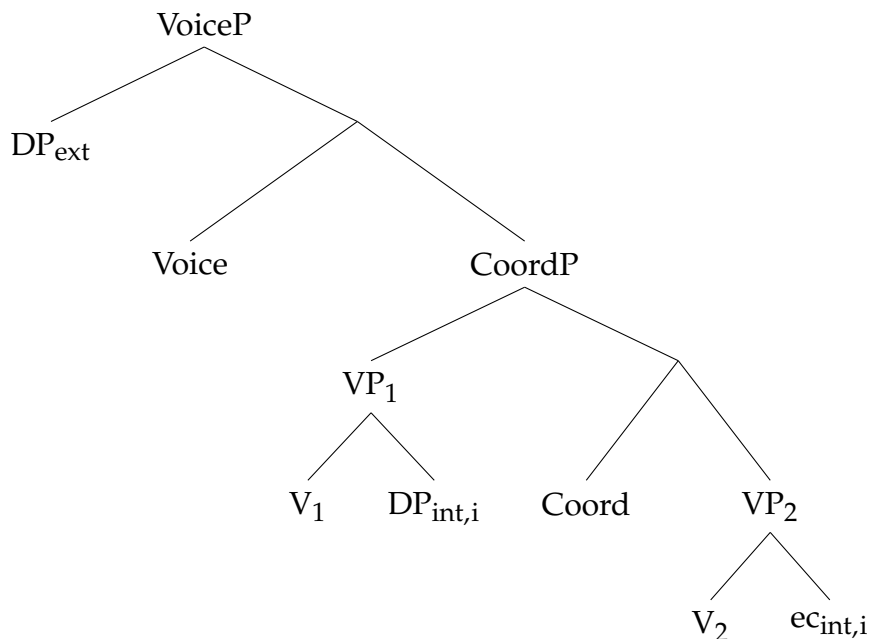
Adjunction analyses for (certain kinds of) SVCs have been proposed in Bickerton & Iatridou (1987), Larson (1991), Law & Veenstra (1992), Lawal (1993), Baker & Stewart (2002), Ko & Sohn (2015), Veenstra (2018) and Hopperdietzel (2020), among others.

3.3.3 Serialization as covert coordination

Finally, SVCs have been analyzed as involving covert coordination, whereby VP_1 and VP_2 are combined in a way that is similar to adjunction, yet without one of the VPs constituting the main projection; the arguments for this kind of analysis are similar to those for an adjunction analysis. Serialization via covert coordination have been argued to involve either no coordinating projection, as in Tree 3.3, or one that remains silent, as in Tree 3.4:



Tree 3.3: SVC via coordination without CoordP



Tree 3.4: SVC via coordination with silent CoordP

Hale (1991) is one of the very few analyses employing covert coordination; see Sebba (1987) and Baker (1989) for discussion of covert coordination analyses of serialization.

3.3.4 Taking stock

Various counter-arguments exist against each kind of analysis. The major problem faced by complementation analyses is that in many cases, there is no plausible selectional relationship holding between V_1 and V_2 , as reflected in the fact that the verbs making up SVCs are often able to occur on their own as independent verbs (a property that is often built into definitions of SVCs, including the one used here). A complementation analysis would, among other things, require all serializable verbs to be able to select verbal in addition to nominal arguments, an unwarranted complication of the lexicon and grammar of these languages in the absence of independent empirical and conceptual support (e.g. in type-theoretic terms, given the type-shifting that would be required).⁴⁰

One problem faced by adjunction analyses concerns extractability: constituents that are argued to be adjoined unexpectedly allow sub-extraction. This has led e.g. Veenstra (1993) to posit weak islandhood for such projections only in the context of serialization. Conversely to the counter-argument against rampant multiple lexical entries faced by complementation, a further argument against adjunction across the board in serialization is that there exists a class of systematically recurring verbs within SVCs which require a complementation analysis, namely those serving an applicative-like function, e.g. ‘take’ and ‘give’, which add non-core arguments to yield instrumental and benefactive meanings, respectively.

Finally, analyses according to which SVCs involve covert coordination are both least favored and most problematic. Like adjunction analyses, coordination analyses cannot explain that VPs in SVCs generally lack the island effects that have come to be associated with coordinate structures since Ross (1967), most notably the *Coordinate Structure Constraint* (CSC) (modulo across-the-board extraction). Moreover, in languages that have been shown to have covert coordination alongside serialization, covert coordination constructions differ from SVCs in a number of systematic respects, including include the possibility of multiple subjects and independent TAM marking across components, intonational breaks between components, movement operations, and overall a much more unconstrained set of possible verb combinations; what is more, speakers conceptualize SVCs as single complex events, but covert coordinations as sequences of events. Some of these contrasts are illustrated in the next section. See Sebba (1987: 107ff) and Baker (1989: 547ff) for extensive discussion of covert coordination analyses of serialization, and why they should be rejected.

⁴⁰Complementation analyses of SVCs involving verbs lacking a plausible corresponding semantic selectional requirement are thus *pseudo-complements* in the sense of Seuren (1991), defined as instances of syntactic complementation whereby the complement does not satisfy any semantic requirement of the head (i.e. syntactic without semantic complementation).

Notwithstanding the problems that come with a claim to uniform syntactic composition across all kinds of SVCs, many works have taken precisely this stance, i.e. that all serialization involves either adjunction or complementation, but not both. As we will see, such a view is inadequate: while most of the SVCs discussed here will be shown to involve adjunction, there are some SVCs which require a complementation analysis; adopting only adjunction or complementation for all kinds of SVCs fails to account for certain clusters of systematically recurring properties.⁴¹

That said, the idea that different sub-types of SVCs are built by distinct modes of structure-building is an old one, espoused in e.g. Li & Thompson (1973), Finney (1991), den Dikken (1991), Gamerschlag (2005) and Nishiyama & Ogawa (2014).

3.4 Serialization and covert coordination

Following Sebba (1987), Baker (1989) and Collins (1997), I do not take covert coordination to fall under serialization. The reason is that although serialization and covert coordination are superficially similar, the two actually differ in a number of respects which become clear upon closer inspection, including object sharing, the kinds of verbs that can be paired, and extractability. These differences are especially telling in languages which have covert coordination alongside serialization.

Regarding objects, SVCs with object sharing require an obligatorily silent object in the adjoined VerbP, which additionally must refer back to the overt object in the main VerbP. In e.g. the NC-SVC in (1a) from Sranan, the silent object in VP₂ can only identify the object of VP₁, Amba, while the overt pronoun following VP₂ in the covert coordination in (7b) is free to denote either Amba or someone else:

(1) **Sranan:**

a. *NC-SVC*

Kofi [naki] Amba_i [kiri] ec_i/*k
 K. hit A. kill
 ‘Kofi struck Amba dead’

b. *Covert coordination*

Kofi [naki] Amba_i [kiri] en_i/k
 K. hit A. kill 3SG
 ‘Kofi struck Amba_i and killed her_i/k’ (Sebba 1987: 109)

The same restriction is found in NC-SVCs from Ewe (Ameka 2006, Collins 1997) and Korean (Ko & Sohn 2015).

Second, if both verbs project overt arguments, serialization systematically restricts one of the verbs to a particular kind of verb. Take the instrumental NC-SVC in (32),

⁴¹I note in passing how this is at odds with the fact that the definitions which authors set up for SVCs are generally based on formal criteria, and therefore neutral with respect to the syntactic structures involved.

again from Sranan, which has the same surface sequence of constituents as the covert coordination in (40b):

(2) **Sranan**

a. *Instrumental NC-SVC*

Kofi [teki] a ne fi [koti] a bre de
 K. take the knife cut the break
 ‘Kofi cut the bread with the knife’
 cf. ‘Kofi took the knife and cut the bread’

b. *Covert coordination*

Kofi [sutu] Amba [kiri] Kwaku
 K. shoot A. kill K.
 ‘Kofi shot Amba and killed Kwaku’ (Sebba 1987)

Both can be translated as coordinations in English, yet the two are argument-structurally and syntactically distinct. In terms of argument structure, the object of V_1 in (40a) bears the ϑ -role of Instrument with respect to the event denoted by V_2 , and correlated with this property is that both objects can extract, cf. (32b) and (32c):

(3) **Sranan: Instrumental NC-SVC**

a. *Extraction of Instrument*

San_i Kofi [teki] —_i [koti] a-bre de?
 what K. take cut DET-bread
 ‘What did Kofi cut the bread with?’

b. *Extraction of Patient*

San_i Kofi [teki] a-ne fi [koti] —_i?
 what K. take DET-knife cut
 ‘What did Kofi cut with the knife?’ (Sebba 1987)

In the covert coordination in (40b), by contrast, no designated thematic relationship holds between V_1 ’s object and the event denoted by V_2 , and neither object can extract:

(4) **Sranan: Covert coordination**

a. *Extraction of V_1 ’s object*

*Suma_i Kofi sutu —_i [kiri] Kwaku?
 who K. shoot kill K.
 intended: ‘Who did Kofi shoot and kill Kwaku?’

b. *Extraction of V_2 ’s object*

*Suma_i Kofi [sutu] Amba [kiri] —_i?
 who K. shoot A. kill
 intended: ‘Who did Kofi shoot Amba and kill?’ (Sebba 1987)

We will see that – rather than constituting an exception to the generalization on

object sharing in serialization – instrumental NC-SVCs are similar to applicatives and involve complementation.

In Ewe, the differences between SVCs and coordination are brought out clearly once a clause is put into the future tense: while SVCs (which have single objects) mark future tense once for the entire construction (5a-ii), covert coordinations (which have multiple objects) requires future tense to be marked on every verb (5b-ii):

(5) **Ewe**

a. *NC-SVC*

- (i) Me [fo] kaɖɛgbɛ [gba]
I hit lamp break
'I hit the lamp and broke it'
- (ii) Me a [fo] kaɖɛgbɛ [gba]
I FUT hit lamp break
'I will hit the lamp and break it'

b. *Covert coordination*

- (i) Me [fo] kaɖɛgbɛ [gba] (yɛme) tsimini
I hit lamp break its glass
'I hit the lamp and broke its glass'
- (ii) Me a [fo] kaɖɛgbɛ *(a) [gba] (yɛme) tsimini
I FUT hit lamp FUT break its glass
'I hit the lamp and broke its glass' (Collins 1997: 463)

Instrumental NC-SVCs show the exact same behavior as object-sharing NC-SVCs, allowing only a single tense marker:

(6) **Ewe**

a. *Instrumental NC-SVC*

- (i) Kofi [tsɔ] ati-ɛ [fo] Yao
K. take stick-DEF hit Y.
'Kofi took the stick and hit Yao with it'
- (ii) Kofi a [tsɔ] ati-ɛ [fo] Yao
K. FUT take stick-DEF hit Y.
'Kofi will take the stick and hit Yao with it'

b. *Covert coordination*

- (i) Kofi [liɛ] ati [gbe] ne
K. climb tree pick coconut
'Kofi climbed a tree and picked a coconut'
- (ii) Kofi a [liɛ] ati *(a) [gbe] ne
K. FUT climb tree FUT pick coconut
'Kofi will climb a tree and pick a coconut' (Collins 1997: 466)

Following Collins (1997), NC-SVCs involve VP-level serialization, while covert coordination involves coordination at the level of TP (at least in Ewe).⁴²

⁴²As noted by Collins (1997: 464), there remain cases in which SVCs and covert coordination may

SVCs and covert coordination thus have distinct underlying structures. These structural differences correlate with differences in how speakers perceive the respective kinds of events denoted by these constructions. For example, while SVCs describe single yet complex actions, covert coordinations are understood as describing less cohesive series of independent actions (Sebba 1987, Baker 1989, Durie 1997). Covertly coordinated structures in Sranan also differ from serialized ones in terms of prosody and various syntactic processes such as clefting (Sebba 1987: 109ff).⁴³

The attractiveness of a coordination analysis lies in the fact that, cross-linguistically, some SVCs appear to adhere to the Law of Coordination of Likes, in that the serialized elements are generally of the same category and size (Veenstra & Muysken 2017). While a general categorial matching effect is indeed active in verbal adjunction, which in turn correlates with size effects – cf. Endo & Haegeman (2019) – the proposal that SVCs generally involves coordination due to the Law of Coordination of Likes does not go through given that countless SVCs cross-linguistically lack such matching effects. That is, if serialization involves coordination, and the Law of Coordination of Likes is treated as a property intrinsic to coordination, a coordination analyses cannot explain those languages without absence of matching, such as Korean.

A last argument against a coordination analysis of serialization is that SVCs tend to exhibit rigid ordering, regardless of the headedness of a language (Durie 1997, Veenstra & Muysken 2017).⁴⁴ This fact is unexpected from a coordination perspective, but follows from adjunction or complementation, both of which may show ordering effects.

Despite the fact that there is little in-depth discussion dedicated to covert coordination, I adopt the null hypothesis that covert coordination is just like overt coordination, with the only difference that the coordinating head is not pronounced.⁴⁵ We thus come full circle back to our set of necessary criteria set up for SVC-hood in the preceding chapter (Section 2.2), which requires that SVCs involve no linking element.^{46,47}

be hard to distinguish, such as when (i) structure-building takes place at the VP-level in both, and (ii) any other morphological indicators of distinct underlying structures are absent.

⁴³See Jansen et al. (1978), Collins (1997), Syea (2013) for further arguments against coordinate analyses of serialization, and Veenstra & Muysken (2017) for detailed discussion of this kind of analysis.

⁴⁴This has been termed *Muysken's anomaly* in Carstens (2002). Note that this is not a universal property of SVCs; see e.g. Zimmermann & Amaechi (2020) on Igbo.

⁴⁵This view goes contra Baker (1989) and Collins (1997), who take covert coordination to lack mediating coordinating heads entirely.

⁴⁶See e.g. Mitrovic & Sauerland (2014), who argue that languages differ in terms of which parts of an underlying universal structure for individual coordination are pronounced. For an opposing view, see Neeleman et al. (2022), who claim that coordination reduces to a special kind of adjunction, in which the two conjuncts are mutually adjoined to one another. I thank Philipp Weisser for these references, and for discussion on this topic.

⁴⁷See Jansen et al. (1978) and Syea (2013) for further arguments against coordinate analyses of serialization, and Veenstra & Muysken (2017) for detailed discussion of this kind of analysis.

3.5 On SVC-internal relationships

3.5.1 Main vs adjoined vs complemented projections

The aim of this section is to outline some of the diagnostics that inform us in distinguishing main, adjoined and complemented projections.

First, differences between main and dependent projections may be reflected formally in derivation or inflection. In Korean, for instance, only V_2 is finite, while V_1 is limited to being suffixed with *-e ~ -a*, a marker indicating non-finiteness and morphological closure (Kang 1997, Ko & Sohn 2015):

- (7) **Korean:** *Only V_2 is finite*

Say-ka [nal-a] [ka-ss-ta]
bird-NOM fly-LK go-PST-DECL

‘The bird flew away’

(Kang 1997: 6)

Headedness asymmetries in serialization are brought out especially clearly in Oceanic languages, where transitive verbs often carry special morphology indicating semantic and / or syntactic transitivity, traditionally as TRANS or APPL. Take Toqabaqita, where the transitive suffix borne by transitive verbs in isolation (8a) is lost on any transitive verb within C-SVCs that is not the string-final verb (8b) (note that TRANS has various allomorphs):

- (8) **Toqabaqita**

- a. ... ada keka thau-ngi qoe
TIMIT 3PL.SEQ hit-TRANS 2SG
‘...they might beat you up / kill you’

- b. *Only V_2 bears TRANS*

Ada keka [thau] [mae-li] qoe
TIMIT 3PL.SEQ hit.DF die-TRANS 2SG

‘(Be careful,) they might kill you by beating you up / they might beat you to death’

(Lichtenberk 2006: 267)

This helps diagnose V_2 as the head verb in both Korean and Toqabaqita, and V_1 as the adjoined verb.⁴⁸

Another property which helps us tell main from adjoined projections comes from argument linking. Recall that in two-verb NC-SVCs with object sharing, the object of V_1 is overt while that of V_2 is silent (9a); this behavior iterates when more transitive verbs are added, i.e. every transitive verb except V_1 occurs with a silent object (9b):

⁴⁸See Section 6.5.2.3 for an in-depth discussion of headedness asymmetries in Oceanic C-SVCs.

(9) **Saramaccan**a. *NC-SVC with two transitive VerbPs*

De [sikópu] hen [kíi]

3PL kick 3SG kill

‘They kicked him dead’

(Veenstra 2018: 12)

b. *NC-SVC with six transitive VerbPs*

A [kísi] dí fou [náki] [kíi] [limbó] [bói] [njan]

3SG catch DET bird hit kill clean cook eat

‘He caught the bird, struck it dead, cleaned, cooked and ate it’

(Bickerton & Iatridou 1987)

Regarding (9), we know that the object belongs to V_1 because the overall headedness of a language is generally reflected within VerbPs in serialization (though not necessarily between VerbPs). The simple transitive in (10) shows that Saramaccan is head-initial, a fact which is preserved within VerbP₁ in (9). Compare Ijo, which is head-final (11a), and where the VerbP-internal order of verb and object is accordingly reversed in serialization (11b):

(10) **Saramaccan: Simple transitive clause**

A naki di tatai

3SG hit DEF rope

‘He hits the rope’

(McWhorter & Good 2012: 51)

(11) **Ijo**a. *Simple transitive clause*

keni bila eri mi

an elephant see TNS

‘saw an elephant’

(Veenstra & Muysken 2017: 24)

b. *NC-SVC with object sharing*

Arau ingo [deri] [pite-mi]

A. trap weave set-PST

‘She wove a trap and set it’

(Williamson 1965: 52)

I am going to make the assumption that for any two non-contiguously serialized transitive verbs with object sharing, one of which appears with an overt object, the silent object of the other verb is always an null argument. This will fall out from (i) the uncontentious premise that argument sharing must somehow be achieved in SVCs, and (ii) the theory on non-contiguous serialization that will be introduced in the next chapter, which dictates that a VerbP can only be adjoined to a main VerbP if there is an overt category in the adjoined VerbP and a null argument in the main VerbP. We can thus leverage this theory-internal well-formedness condition to infer

that in NC-SVCs, transitive VerbPs lacking visible objects must have null objects, and therefore constitute dependent projections; conversely, transitive VerbPs with overt objects must form the main projection.

Turning to complementation, a prime diagnostic for telling main and complemented projections is extraction, which is generally impossible out of adjuncts and coordinates. In instrumental NC-SVCs like (12a), for example, the fact that *wh*-extraction may target both the Instrument (12b) and the Patient (12c) indicates that neither VerbP is adjoined:

(12) **Sranan:** *Instrumental NC-SVC*

- a. Kofi [teki] a-nefi [koti] a-brede
K. take DET-knife cut DEF-bread
'Kofi cut the bread with the knife'

b. *Extraction of Instrument*

San_i Kofi [teki] —_i [koti] a-brede?
what K. take cut DEF-bread
'What did Kofi cut the bread with?'

c. *Extraction of Patient*

San_i Kofi [teki] a-nefi [koti] —_i?
what K. take DET-knife cut
'What did Kofi cut with the knife?'

(Sebba 1987)

This extraction potential contrasts with that of the superficially parallel covert coordination in (13a), where neither object can undergo *wh*-extraction ((13b),(13c)):

(13) **Sranan:** *Covert coordination*

- a. Kofi [sutu] Amba [kiri] Kwaku
K. shoot A. kill K.
'Kofi shot Amba and killed Kwaku'

- b. *Suma_i Kofi [sutu] Amba [kiri] —_i?
who K. shoot A. kill
'Who did Kofi shoot Amba and kill?'

- c. *Suma_i Kofi sutu —_i [kiri] Kwaku?
who K. shoot kill K.
'Who did Kofi shoot and kill Kwaku?'

(Sebba 1987)

Additionally, we can again draw on a theory-internal reason for why neither (12a) nor (13a) can involve adjunction: because the theory of non-contiguous serialization via adjunction to be adopted entails null arguments in adjoined VerbPs, yet given that that each VerbP in both (12a) and (13a) has an overt object, it follows that neither (12a) nor (13a) can involve adjunction.

More generally, we can also leverage the fact that complementation involves selection and adjunction does not. Because it is overwhelmingly the case across languages that verbs in SVCs lead independent lives outside serialization, i.e. that

serialization is practically always optional (Aikhenvald & Dixon 2006, Veenstra & Muysken 2017, Veenstra 2018), our null hypothesis about any given SVC should be that it is composed via adjunction – an assumption that I will adopt.

That said, there is a subset of verbs that systematically recur cross-linguistically in specific functions within SVCs, for which a complementation analysis is called for. These include verbs giving rise to instrumental (mostly ‘take’) and benefactive (mostly ‘give’) meanings, and which we will see are similar to applicatives.

3.5.2 Ordering and headedness

There will be no attempt here to derive properties of ordering and headedness in serialization. Nevertheless, our investigation will be informed by certain established generalizations about ordering and headedness.

Generally speaking, we first have the fact that verb order in both NC-SVCs and C-SVCs tends to be iconic to event structure, as mentioned at the outset; see e.g. Durie (1997), Li (1993) and Aikhenvald (2006b).

Second, there is *Muysken’s anomaly*, so coined in Carstens (2002) and first pointed out in Muysken (1988). Muysken’s anomaly refers to the surprising fact that the order of verbs in SVCs is overwhelmingly blind to a language’s headedness parameter. To see this, consider the following simple transitive clauses from Ijò (14a) and Sranantongo (14b), which show that the former is head-final and the latter head-initial:

(14) *Simple transitive clause*

a. **Ijò**

keni bila eri mi
an elephant see TNS
‘saw an elephant’

b. **Sranantongo**

koti a brede
cut DET bread
‘cut the bread’

(Veenstra & Muysken 2017: 24)

In NC-SVCs, however, both languages show the exact same order of verbs, as illustrated by the following benefactive NC-SVCs: while the VerbP-internal orders reflect the overall headedness of the respective language, the main V_1 always precedes the adjoined V_2 ‘give’:

(15) *Benefactive NC-SVC*a. **Ijo**

Dúma [tun-nì] a [píri]
 song sing-PRT 3SG.F give
 ‘Sing a song for her’

b. **Sranantongo**

[prani] a karu [gi] yu
 plant DET corn give 2SG
 ‘plant the corn for you’

(Veenstra & Muysken 2017: 25)

Muysken’s anomaly is given in (16):

(16) *Muysken’s anomaly*

(Carstens 2002: 5)

Head-finality has no correlates in the relative orderings of verbs [$V_1 \dots V_n$]
 and arguments [$O_1 \dots O_n$] in SVCs.

Note that despite its robustness, Muysken’s anomaly is merely a tendency, as illustrated by e.g. Korean, where the language’s general head-finality is replicated in NC-SVCs at the inter-VerbP level (Ko & Sohn 2015). Carstens (2002), building on Muysken (1988) and Kayne (1994)’s theory of *Anti-symmetry* – which states that all languages are underlyingly head-initial, with left-branching specifiers and right-branching complements – proposes that NC-SVCs are universally right-branching, and that all non-head-initial orders are the result of movement operations.

Now, it is specifically in the domain of C-SVCs where Muysken’s anomaly appears to more robust, and perhaps universal: recall the $*[trans + intr_H]$ restriction, which constitutes transitivity profile Type C. In fact, all the C-SVCs considered here are head-final, and so far I am not aware of any exception to this property. We will therefore leverage this property, especially in those cases where formal diagnostics for syntactic headedness within C-SVCs are sparse or lacking. Kayne’s theory of *Anti-symmetry* offers a potential explanation for this generalization because it rules out right-adjunction of one terminal node to another (Kayne 1994: 17ff); see also Williams (1981b)’s *Righthand head rule*, which states that the righthand member of a morphologically complex word is the head of that word.

In what follows, I will take the head-finality of C-SVCs as a given, and set aside the derivation of ordering and headedness in serialization for future research, referring the reader to works like Williams (1981b), Muysken (1988), Kayne (1994) and Carstens (2002).

3.6 Analysis

This section outlines the general analysis that will be adopted in these case study, with additional details filled in as necessary along the way.

3.6.1 Proposal

Recall that SVCs show the following profiles of licit transitivity pairings: among NC-SVCs, Type A is completely unrestricted, and Type B allows only pairings of two transitive verbs; C-SVCs characterize a 3/4 pattern, with a systematically recurring ban against combinations of a transitive V_1 and an intransitive V_2 , where V_2 is the syntactic head. As argued in the previous chapter, these generalizations factor out SVCs with resultative and directional meanings. See Table 3.2:

		<i>intr + intr</i>	<i>trans + trans</i>	<i>intr + trans</i>	<i>trans + intr</i>
NC-SVCs:	Type A	✓	✓	✓	✓
	Type B	*	✓ ^a	*	*
C-SVCs:	Type C	✓	✓	✓	* ^b

a. Either V_1 or V_2 may be ditransitive

b. Where V_2 is the syntactic head

Table 3.2: Major transitivity profiles in SVCs

I propose that these major profiles fall out from cross-cutting the mode of structure-building with the level of structure-building. By cross-cutting adjunction on the one hand, and the levels of V , VP and $VoiceP$ on the other, we end up with all and only those profiles in Table 3.2. The labels V , VP and $VoiceP$ are meant to abstract over more fine-grained articulations of the verbal projection; they are used here to refer to levels of structure-building prior to the introduction of any arguments (V), prior to the introduction of external arguments (VP), and after the introduction of external arguments ($VoiceP$).

We will see that in NC-SVCs, (i) the completely unrestricted Type A profile follows from $VoiceP$ -level adjunction, while (ii) Type B, which allows only [*trans + trans*] pairings, follows from VP -level adjunction; (iii) Type C in C-SVCs, which bans [*trans + intr_H*] pairings, follows from V -level adjunction whenever the language shows a strict IAR that does not allow the internal argument slots of transitive verbs to remain unsatisfied.

		<i>intr + intr</i>	<i>trans + trans</i>	<i>intr + trans</i>	<i>trans + intr</i>	Follows from
NC-SVCs:	Type A	✓	✓	✓	✓	VoiceP-level adjunction
	Type B	*	✓	*	*	VP-level adjunction
C-SVCs:	Type C	✓	✓	✓	* a	V-level adjunction + strict IAR

a. Either V_1 or V_2 may be ditransitive

b. Where V_2 is the syntactic head

Table 3.3: Major transitivity profiles in SVCs

This analysis presupposes that adjunction at the levels of V, VP and VoiceP always involves the adjunction of like categories, i.e. V-to-V, VP-to-VP and VoiceP-to-VoiceP adjunction. The adjunction of like categories seems to be a general property not only of serialization (Baker & Stewart 2002, Kalin & Keenan 2011, Ko & Sohn 2015, Hopperdietzel 2020), but also more generally of certain types of adjunction in the verbal domain (Endo & Haegeman 2019), as discussed in the next section.

The theory of serialization via adjunction to be adopted here, namely Baker & Stewart (2002), requires each adjoined VerbP in a SVC to be linked to the main VerbP by way of syntactic argument positions: there must be an overt argument in the main VerbP and a null argument in the adjoined VerbP for the two to be able to serialize. One of the empirical facts this captures is that in transitive NC-SVCs with object sharing, the object in the adjoined VerbP is always silent. The presence of multiple objects, then, implies the lack of adjunction, which I show is correct for subset of multiple-object SVCs: instrumental NC-SVCs like (40a), which introduce non-core roles, do not involve adjunction but complementation, thus patterning like applicatives.

In sum, cross-cutting adjunction and complementation with the level of structure-building yields a set of structures that is a superset of the structures we see, which in turn characterizes the attested transitivity profiles.

3.6.2 On the adjunction of like categories

As mentioned, I take serialization via adjunction to always involve the merger of like categories, meaning that any projection of category XP must be adjoined to a projection that is also of category XP. Concretely in our case, this means that only V-to-V, VP-to-VP and VoiceP-to-VoiceP adjunction is predicted to be well-formed, but not e.g. V-to-VP or VP-to-VoiceP adjunction. We can rule out X-to-XP adjunction given the uncontroversial assumption that heads can only be adjoined to heads and phrases to phrases, but more needs to be said on VP-to-VP and VoiceP-to-VoiceP

adjunction.

Even though previous work on serialization via adjunction has implicitly both assumed and confirmed the adjunction of like categories – see Baker & Stewart (2002) for VP-level serialization, and Ko & Sohn (2015) and Hopperdietzel (2020) for VoiceP-level serialization – I am not aware of any work on SVCs that attempts to explicitly derive this generalization, which appears to hold beyond the domain of VoiceP, as shown by the adoption of TP-level serialization in Kalin & Keenan (2011). I claim that the adjunction of like categories in serialization is an instance of the conjecture in Endo & Haegeman (2019), which states that the more internal structure a clause manifests, the higher its merge site.⁴⁹ Based on data from coordination, movement, scope and the distribution of information-structural markers in adverbial clauses across various languages, Endo & Haegeman propose that the merge site of an adjunct clause is determined by its derivational history.

Take Japanese, which has the clausal hierarchy in (17). Japanese adverbial clauses give rise to a cline in terms of the availability of various post-verbal functional elements, with clauses headed by *nagara* ‘while’ located at one end of the spectrum in allowing only voice (18a), and clauses headed by *ga* ‘though’ at the other end in allowing the expression of everything up to speaker-oriented mood (18c) (Minami 1974, Noda 1989, Endo 2012):

(17) *Japanese clause structure*

Voice < Asp < Neg < T < Mood_{Speaker} < Mood_{Addressee}

(Endo & Haegeman 2019: 15)

(18) **Japanese**

a. Voice < Asp < Neg < T < Mood_{Speaker} < Mood_{Addressee}

[Neko-wa atama-o nade-rare-(***tei**) nagara] [**zitto si-tei-ru**]
cat-TOP head-ACC pat-PASS-(***ASP**) while still stay-ASP-NPST
‘While its head is patted, the cat stays still’

b. Voice < Asp < Neg < T < Mood_{Speaker} < Mood_{Addressee}

[Neko-wa atama-o nade-rare-tei-na-i-(***daroo**) node]
cat-TOP head-ACC pat-PASS-ASP-NEG-PRES-(***MOOD_S**) because
[**zitto si-tei-ru**]
still stay-ASP-NPST
‘Because its head is not patted, the cat stays still’

c. Voice < Asp < Neg < T < Mood_{Speaker} < Mood_{Addressee}

⁴⁹Thanks to Susi Wurmbrand (p.c.) for bringing this paper to my attention.

[Neko-wa atama-o nade-rare-tei-na-i-daroo-(*ne)
 cat-TOP head-ACC pat-PASS-ASP-NEG-PRES-MOOD_S-(*MOOD_A)
 ga] [zitto si-tei-ru]
 though still stay-ASP-NPST

‘Though its head does not seem to be patted, the cat stays still’

(Endo & Haegeman 2019: 16)

Simplifying somewhat, Endo & Haegeman (2019: 17) argue that this cline follows because the highest head in an adverbial clause determines a clause-typing feature on that clause, which in turn must be licensed by a projection in the matrix clause bearing a matching feature. So underlyingly, the adverbial clause in (18c) exceeds the one in (18a) in both its internal size and the height at which it is merged, with the one in (18b) falling somewhere in between.

The syntactic nature of this *concord* relation between an adverbial clause and its merge site – so termed by Noda (1989) – is brought out more clearly in cases involving two adverbial clauses.⁵⁰ Clauses headed by *ba* ‘if’ project up to negation, whereas clauses headed by *zuni* ‘without’ only project up to aspect, meaning that *ba*-clauses are both smaller and merged lower than *zuni*-clauses. The highly preferred order when these clauses co-occur is the one predicted by the adjunction of like categories, i.e. where the *ba*-clause precedes the *zuni*-clause due to the latter being merged before the former:

(19) **Japanese**

- a. *ba*-clause < *zuni*-clause

[Ame-ga fure-ba] [soto-ni ika zuni] [ie-ni iru]
 rain-NOM fall-if outside go without home-at stay
 ‘If it rains, without going out, I stay home’

- b. *zuni*-clause < *ba*-clause

??[soto-ni ika zuni] [Ame-ga fure-ba] [ie-ni iru]
 outside go without rain-NOM fall-if home-at stay
 intended: ‘If it rains, without going out, I stay home’

(Endo & Haegeman 2019: 22)

Put simply, an adverbial clause’s internal syntax determines its merge site in direct proportion. Endo & Haegeman implement this idea in terms of a feature-matching requirement between the main and the adjoined clause, such that – roughly – for any clause-typing feature *F* on the highest head of a modifying clause *XP*, there must be a matching feature *F* on the highest head of the main clause *YP* (Endo & Haegeman

⁵⁰This is not to say that the adjunction of like categories does not also have a semantic basis, which I assume it does.

2019: 24). In relative terms, (i) this rules out the merger of larger adjunct clauses with smaller matrix clauses because the required licensing feature is not yet present in the matrix clause; and conversely (ii) it rules out the merger of smaller adjunct clauses with larger matrix clauses because the required licensing feature is too deeply embedded within the matrix clause.

I adopt Endo & Haegeman's proposal, extending it to adjunction in the domain of serialization. From here on, I will take this likeness condition as a given, neither attempting to derive it from more general principles nor representing it explicitly in syntactic derivations, submitting only that (i) the merger of like categories must be a deeper grammatical condition on (some kinds of) adjunction, given that it is observed by both adverbial and serial clauses; and (ii) if it is indeed correct that these likeness conditions exist for the same reason, we have a welcome, reductionist result.⁵¹

3.7 Summary

This chapter presented the general analysis of co-occurrence restrictions on transitivity in SVCs: depending on whether occurs via serialization or complementation, and depending on the structural level at which it occurs, distinct transitivity profiles arise. This general analysis will be expanded in detail in the next two chapters for NC-SVCs on C-SVCs, respectively.

⁵¹Non-like categories are regularly adjoined in head movement, like when V moves through Voice to T yielding a complex T-Voice-V head. However, these cases are distinct because the moving element originates within the complement domain of the element that it moves to.

Chapter 4

Non-contiguous serial verb constructions

4.1 Introduction

This chapter presents a detailed analyses of the co-occurrence restrictions on transitivity found in NC-SVCs. Section 4.2 is dedicated to NC-SVCs that are built via adjunction. Here, I establish that these kinds of NC-SVCs involve null arguments, and present the theory of serialization via adjunction that I will adopt. We then move on to derivations of serialization via adjunction at the levels of VP and VoiceP. In Section 4.3, I present my analysis of of serialization via complementation at the level of VP, briefly discussing VoiceP-level serialization via complementation, and also why V-level constitutes a potential gap in the typology.

4.2 Non-contiguous serialization via adjunction

4.2.1 NC-SVCs involve null arguments

Recall that for our purposes, two kinds of silent arguments are distinguished: *null arguments* and *implicit arguments*. Slightly adopting the notions in Bhatt & Pancheva (2017) and Williams (2015), I take null arguments to be categories that are projected in the syntax but remain silent, and implicit arguments to be arguments that are not projected syntactically, but which are nevertheless active in the syntax by virtue of participating in certain grammatical dependencies.

For transitive NC-SVCs with object sharing, we will see that the silent argument in VerbP₂ is a null argument; it is this null argument which induces the necessary status for VerbP₂ to be stitched together with VerbP₁. Diagnosing the presence of this null argument will here be illustrated for Edo, based on Baker & Stewart (2002).

Edo has an emphatic, reflexive-like element *tobore* 'by him-/her-/itself', which can be adjoined to the right of an overt or covert nominal in any syntactic position.

(1a) shows *tobore* adjoined to the subject, and (1b) to the direct object of a simple transitive:

(1) **Edo**

a. *Subject + tobore*

Òzó_i **tòbórè_i** ghá là èvbàré

O. by.self FUT cook food

‘Ozo by himself will quickly cook the food’

b. *Object + tobore*

Òzó kpàán àlímó_i **tòbórè_i**

O. pluck orange by.self

‘Ozo plucked the orange by itself’

(Baker & Stewart 2002: 20)

A contrast emerges in the behavior of *tobore* with respect to intransitives: *tobore* can follow unaccusatives (2a), but not unergatives (2b):

(2) **Edo**

a. *Unaccusative subject + tobore*

Úyì_i dé **tòbórè_i**

U. fall by.self

‘Uyi fell by himself’

b. *Unergative subject + tobore*

*Úyì_i só **tòbórè_i**

U. shout by.self

‘Uyi shouted by himself’⁵²

(Baker & Stewart 2002: 21f)

This contrast follows because the unaccusative subject in (2a) is base-merged as an internal argument, unlike the unergative subject in (2b), which is base-merged as an external argument Baker & Stewart (2002: 22); for this reason, only the unaccusative subject leaves a trace to which *tobore* can right-adjoin after it moves to subject position.

Turning to the behavior of *tobore* in the context of NC-SVCs, we find that it can be added to non-resultative NC-SVCs, following the transitive V_2 (3a), but not to resultative NC-SVCs, where V_2 is intransitive (3b):

(3) **Edo**

a. *Consequential NC-SVC: V_2 + tobore*

Òzó [dé] ìyán_i [dùnmwún] **tòbórè_i**

O. buy yam pound by.self

‘Ozo bought the yam and pounded it by itself’

b. *Resultative NC-SVC: V_2 + tobore*

⁵²We expect *tobore* to be well-formed in an unergative clause if it were to directly follow the subject; unfortunately, Baker & Stewart (2002) do not provide the relevant example.

*Òzó [sùá] ògó_i [dé] tòbóre_i
 O. push bottle fall by.self

‘Ozo pushed the bottle down by itself’ (Baker & Stewart 2002: 22f)

This shows that VerbP₂ in non-resultative NC-SVCs contains a null object, but that there is no null object associated with the intransitive V₂ in resultative NC-SVCs. Baker & Stewart therefore conclude that resultative NC-SVCs like (3b) involve a structure-building mechanism that is distinct from the one in non-resultative NC-SVCs like (3a), one that is less like adjunction but closer to complementation. As elaborated on in Section 2.6 of the previous chapter, this is in line with generalizations in the literature, according to which result components are not canonical adjuncts, but more like derived complements. A similar diagnostic for the presence of null objects in (a subset of) NC-SVCs is used in Collins (1997) for Ewe, based on the adposition-like element *yi*.

Null objects in Ewe transitive NC-SVCs are always obligatorily null, i.e. they can never alternate with any kind of overt nominal. This is illustrated in (4), where an overt object following V₂ is ungrammatical:⁵³

(4) *VerbP₂ cannot contain an overt object*

a. **Nupe**

*Musa [si] èbi [ba] nakan

M. buy knife cut meat

intended: ‘Musa bought a knife and cut the meat’

b. **Edo**

*Òzó [mién] èrhán [guó!ghó] àkhé

O. find stick break pot

intended: ‘Ozo found a stick and (then) broke the pot’

(Baker & Stewart 2002: 39)

For Ewe, a language which otherwise heavily restricts the possibility of omitting core arguments even if these are recoverable from context (Ameka 2006: 127), Collins (1997) shows that overt pronouns following V₂ and coreferring with V₁’s overt object are impossible in NC-SVCs:

⁵³In Edo, an overt object following V₂ is grammatical in the superficially similar purposive construction in (i), which, however, is not mono-clausal and hence not a SVC, as evinced by the presence of the marker *ya*, which marks non-finite clauses:

(i) **Edo**

*Òzó [mién] èrhán yá [guó!ghó] àkhé

O. find stick INF break pot

‘Ozo found a stick in order to break the pot’

(Baker & Stewart 2002: 39)

(5) **Ewe**

Wo-a [d̥a] fufu [d̥u] (*i)
 they-FUT cook fufu eat it

‘They will cook fufu and eat it’ (Collins 1997: 480)

Similarly, Korean NC-SVCs – which we will see are serialized at a higher structural level than their Edo and Ewe counterparts – disallow the occurrence of an overt subject in both VerbPs:

(6) **Korean**

*John-i kaymi-lul_i [kwulm-e] John-i *pro*_i [cwuk-i-ess-ta]
 J.-NOM ant-ACC starve-LK J.-NOM it die-CAUS-PST-DECL

intended: ‘John starved an ant to death’ (Ko & Sohn 2015: 112)

Second, null arguments in VerbP₂s can only refer back to the overt argument in VerbP₁s. This is best illustrated by a comparison to languages like Sranan, where serialization exists alongside covert coordination. In the NC-SVC in (7a), the understood object of VerbP₂ can only identify the overt object of VerbP₁, Amba, while the overt object pronoun following V₂ in the covert coordination structure in (7b) is free to denote either Amba or someone else:

(7) **Sranan:**a. *NC-SVC*

Kofi [naki] Amba_i [kiri]
 K. hit A. kill
 ‘Kofi struck Amba dead’

b. *Covert coordination*

Kofi [naki] Amba_i [kiri] en_{i/k}
 K. hit A. kill 3SG
 ‘Kofi struck Amba_i and killed her_{i/k}’ (Sebba 1987: 109)

Finally, the obligatorily null argument status of the argument in VerbP₂s is recursive. Some languages permit the serialization of more than two transitive VerbPs, in which case – just like in cases with a single VerbP₂ – only VerbP₁ hosts an overt object:

(8) *Multiple dependent transitive VPs*a. **Igbo**

Ógù [gòrò] ọkụkọ [gbúo] [síce] [ríe]
 O. buy chicken kill cook eat

‘Ogu bought a chicken, killed [it], cooked [it] and ate [it]’

(Zimmermann & Amaechi 2018: 2)

b. **Saramaccan**

A [kísi] dí fou [náki] [kíi] [limbó] [bói] [njan]
 3SG catch DET bird hit kill clean cook eat

‘He caught the bird, struck it dead, cleaned, cooked and ate it’

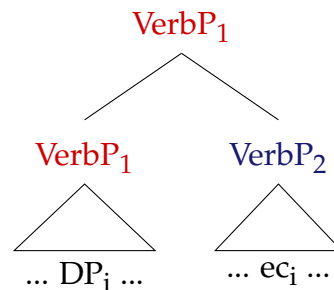
(Bickerton & Iatridou 1987)

In sum, that silent arguments in NC-SVCs are null arguments is tied specifically to the fact that they refer back to the overt argument. Moreover, these null arguments are obligatorily null.

Numerous works have argued for the presence of null arguments in NC-SVCs. See e.g. Bickerton & Iatridou (1987), Carstens (1988), Larson (1991), Law & Veenstra (1992), DeGraff (1993), Collins (1993), Collins (1995), Veenstra (1993), Veenstra (1996), da Cruz (1994), Campbell (1996), Hagemeijer (2001) and Stewart (2001).

4.2.2 Non-contiguous adjunction via predication

I will now present the theory of serialization that I will adopt for NC-SVCs involving adjunction. This theory is developed in Baker & Stewart (2002), who base it on the theory of predication initiated by Williams (1980). Basically, NC-SVCs involve adjunction structures in which a VerbP₂ is adjoined to a main VerbP₁ by virtue of a null argument in VerbP₂ that acts as a predicate variable and is co-indexed with an overt argument in VerbP₁. The relationship by which the overt nominal in VerbP₁ and the null argument in VerbP₂ are linked is similar to the one that has been developed for relative clauses. This general configuration, which we will refine, is given in Tree 4.1:



Tree 4.1: Non-contiguous serialization via adjunction (to be refined)

The conditions on VerbP-level serialization via adjunction are listed in (9):

- (9) *Conditions on VerbP-level serialization via adjunction* (Baker & Stewart 2002)
- i. VerbP₁ must contain an overt argument
 - ii. VerbP₂ must contain a null argument
 - iii. VerbP₁'s overt argument and VerbP₂'s null argument must be co-indexed
 - iv. Each argument must be the highest nominal in its projection

Conditions (i) and (ii) are needed because VerbP₂ must have an open position in order to be able to function as a predicate, and condition (iii) is needed in order for this predicate variable in VerbP₂ to have something to refer back to. The null argument in VerbP₂ must bear the same index as the overt argument in VerbP₁, or else the former cannot be identified with the latter.⁵⁴ Finally, condition (iv) ensures that VerbP₁ and VerbP₂ will each inherit the index of their respective highest nominal category – a point which requires explanation.

Baker & Stewart's reasoning is that, first, all and only lexical nominal categories have inherent indices, i.e. NPs and DPs. Second, non-nominal phrases acquire an index from the structurally highest NP / DP they contain. Third, predication in their account does not technically hold between the adjoined VerbP₂ and the overt argument in VerbP₁, but instead between VerbP₂ and VerbP₁ because the overt and the null argument stand in no relationship of c-command. Finally, they assume that null arguments must bear the same index as some other, inherently index-bearing nominal in the structure due to lacking lexical content. This way, the null argument in VerbP₂ comes to be linked to overt one in VerbP₁ via a relation similar to the kind that has been developed for relative clauses (10a) and control into non-finite clauses (10b):

- (10) a. *Relative clause*
 the book_i [CP_i Op_i that Alma read —_i]
 b. *Non-finite control*
 the woman_i [TP_i PRO_i reading *Dune*]

(based on Baker & Stewart 2002: 43)

Now, even though both the serialized VerbP₂ in Tree 4.1 and the clauses in (10) are adjoined, there is an important difference between the two, which is that in the latter, the DPs to which the relative clause and the non-finite clause are adjoined c-command the null arguments inside these clauses; this is not the case in Tree 4.1, where the overt argument in VerbP₁ does not c-command the null argument in VerbP₂. It is for this reason that the true relationship of syntactic predication is taken to hold between VerbP₁ and VerbP₂ (Baker & Stewart 2002: 50). This way, Baker & Stewart draw a parallel between serialization via adjunction (11) to purposive clauses like (12a), which are likewise assumed to involve adjunction of a non-finite clause to VP, rather

⁵⁴In Baker & Stewart's original account, it must be assumed that the null argument in VerbP₂ bears the same index as DP in VerbP₁ prior to its adjunction to VerbP₁. If serialization via adjunction merely required there to be an null argument in VerbP₂, we would falsely predict well-formed NC-SVCs in which the null argument in the adjoined VerbP₂ refers arbitrarily, rather than necessarily to the overt argument in VerbP₁. It is tempting to have the null argument be without an index prior to serialization, and acquire the same one as the overt argument as the result of adjunction of VerbP₂ to VerbP₁. The problem with this view is that in the configuration yielded by adjoining VerbP₂ to VerbP₁, no relationship of (asymmetrical) c-command holds between the overt and the null argument, which makes it non-trivial to force their co-indexation as a consequence of adjunction.

than the object contained therein:⁵⁵

(11) *Serialization via adjunction*

[VerbP_{1i} ... DP_i ...] [VerbP_{2i} ... ec_i ...]

(12) a. *Purposive clause*

Alma [VP_i bought a book_i] [VP_i Op_i to read —_i]

(based on Williams 1980: 231, 238)

One shortcoming of Baker & Stewart's account concerns their proposal that in a subset of NC-SVCs, the null argument in the adjoined VerbP₁ is *pro*: because *pro* saturates the internal argument slot of a transitive V₂ via Function Application, the resulting VerbP₂ no longer has an open position, i.e. cannot function as a predicate. Moreover, the fact that the relationship of predication is taken to hold between VerbP₁ and VerbP₂ – and not between VerbP₂ and the overt argument in VerbP₁ – characterizes a mismatch given that the subject of predication is clearly an individual argument.

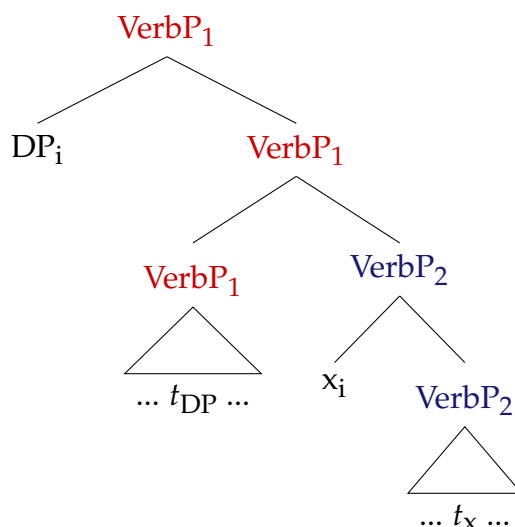
These shortcomings will here be remedied by invoking (i) movement, and (ii) minimal pronouns in the sense of Kratzer (2009). I propose that the null argument in VerbP₂ is a minimal pronoun, i.e. a non-referential pronominal with unvalued φ -features that must come to inherit the φ -values of a binder in the derivation. It will be represented here as *x*. Even though this minimal pronoun saturates the internal argument slot of V₂, it subsequently moves to the edge of VerbP₂, a step which translates to a λ -abstractor and hence induces the desired predicate status for VerbP₂.⁵⁶ VerbP₂ then adjoins to VerbP₁, after which the overt argument of VerbP₁ also moves, namely to the edge of the complex VerbP₁; from here, it c-commands both VerbP₁ and VerbP₂, including the minimal pronoun in VerbP₂.⁵⁷ In this configuration, (i) the minimal pronoun comes to be bound by and thus interpreted as co-referent with the overt argument; and (ii) VerbP₂ is part of a complex predicate that predicates over the overt argument, which denotes an individual.⁵⁸ This configuration is given in Tree 4.2:

⁵⁵See Baker & Stewart (2002: 50) for phrase-structural arguments that purposive clauses like (12a) are not adjuncts to the object, but to its containing VP. It seems to me, however, that examples like *an easy [book to read]*, which suggest that the clause does directly attach to the object, may be problematic for this view.

⁵⁶ λ -abstraction is an automatic outcome of movement (Heim & Kratzer 1998). Following Landau (2015), the fact that the moved minimal pronoun does not saturate the λ -predicate formed follows naturally from its being a non-referential numerical index.

⁵⁷I assume that movement to the edge of VerbP₁ and VerbP₂ is triggered by an EPP feature on V / v / Voice, not shown here.

⁵⁸Baker & Stewart follow Williams (1980) in assuming that A and B must c-command each other. This is satisfied in Tree 4.2, where the overt argument at the edge and the complex predicate consisting of the complex VerbP₁ and VerbP₂ are in a relationship of mutual c-command.



Tree 4.2: Non-contiguous serialization via adjunction

This refinement reconciles the insights and general logic of Baker & Stewart’s original account with the above-mentioned shortcomings. It also does away with a major problem entailed by a *pro* analysis of the null argument – also proposed in e.g. (Collins 1997) – namely the incorrect prediction that it should be able to alternate with an overt pronoun (cf. Section 4.2.1).

I follow Baker & Stewart in drawing on Chomsky (1986)’s *Principle of Full Interpretation*, according to which the integration of any element into a clause must be sanctioned by certain principles in order for it to be interpretable at PF and LF. For adjoined phrases, which are neither selected nor eligible for complex predicate formation via incorporation derived by head movement, Full Interpretation entails that they must be integrated via predication. This ensures that VerbPs must be stitched together via an open argument position, and that it is specifically the adjoined VerbP which must contain this position.

Let us recap the logic of Baker & Stewart’s account, now augmented with minimal pronouns: every element is subject to well-formedness conditions on integration; VerbP₂ is an adjunct to VerbP₁; because VerbP₂ is an adjunct, it must be integrated via predication; in order to become a predicate, the minimal pronoun merged in VerbP₂ moves to its edge, which translates to a λ -abstractor; the overt argument in VerbP₁ likewise moves to the edge of the complex VerbP₁, thereby both binding the minimal pronoun and becoming the subject of the complex predicate; serialization is successful.⁵⁹

In Baker & Stewart (2002), who focus on NC-SVCs in Edo and Nupe, this theory is applied to projections containing internal arguments but lacking external arguments, i.e. VP for our purposes. Even though licit transitivity pairings are not of concern to Baker & Stewart, their approach derives the fact that non-resultative NC-SVCs in these languages can only consist of two transitive verbs, i.e. Type B profiles. Ko

⁵⁹This refined approach involving minimal pronouns is similar to Landau (2015)’s analysis of predicative control.

& Sohn (2015) extend this theory to the level of VoiceP, thereby freeing up all other remaining logically possible transitivity pairings.

4.2.3 VP-level adjunction

Cross-cutting adjunction with the level of VP derives profile Type B, which is restricted to [*trans* + *trans*] pairings, found in e.g. Edo::

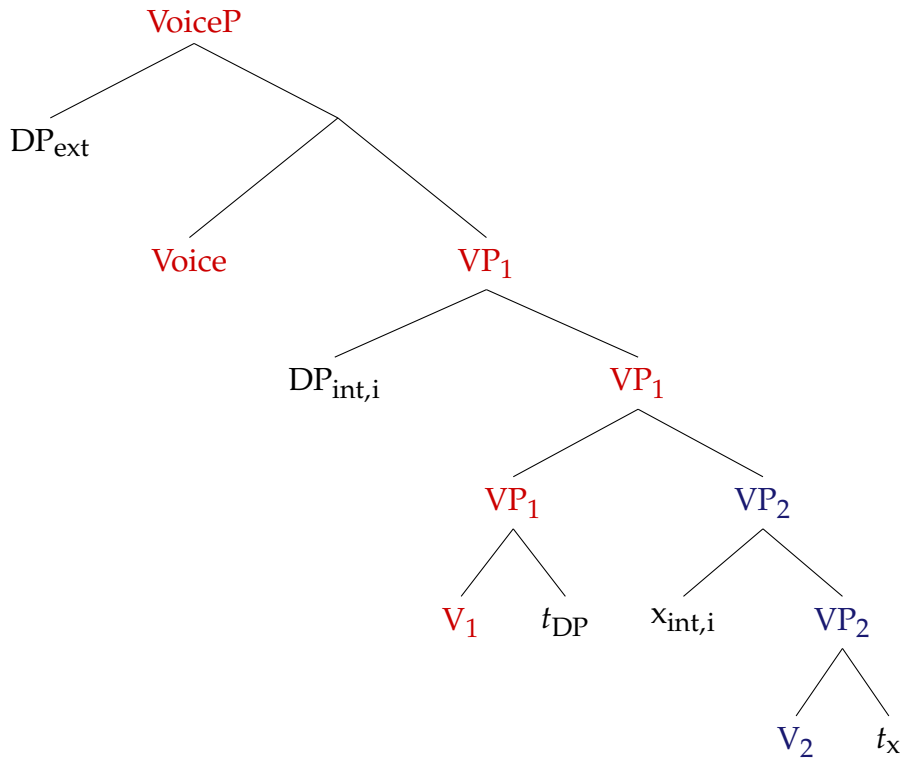
(13) **Edo:** *trans* + *trans*

Òzó ghá [gbè] èwé [khièn]
O. FUT hit goat sell

‘Ozo will kill the goat and sell it’

(Baker & Stewart 2002: 2)

The general syntax of VP-level adjunction is shown in Tree 4.3:



Tree 4.3: VP-level serialization via adjunction

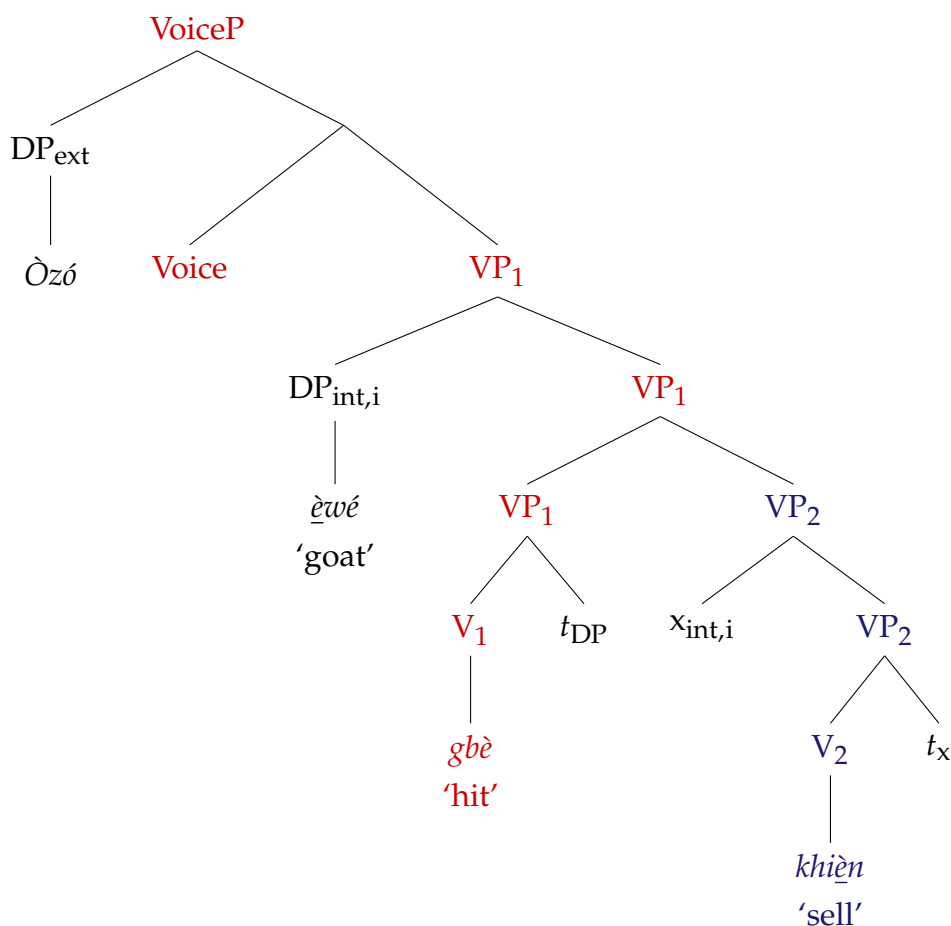
This analysis is taken directly from Baker & Stewart (2002), who apply it to non-resultative NC-SVCs with object sharing in Edo, Nupe and Yoruba. A NC-SVC like (14) thus has the structure in Tree 4.4:

(14) **Edo:** *trans* + *trans*

Òzó ghá [gbè] èwé [khièn]
O. FUT hit goat sell

‘Ozo will kill the goat and sell it’

(Baker & Stewart 2002: 2)

Tree 4.4: Edo NC-SVC: *trans* + *trans*

After the main, transitive V_1 merges with its object, VP_2 is adjoined. Because VP_2 is an adjunct, which must be integrated via predication, its minimal pronoun object has undergone movement to induce predicate status. Finally, the overt object of VP_1 moves to the edge of the complex VP_1 , binding the minimal pronoun object. Because the level at which serialization occurs is VP, argument sharing must occur via internal arguments. It follows that both VPs must be transitive, and hence we derive profile Type A.

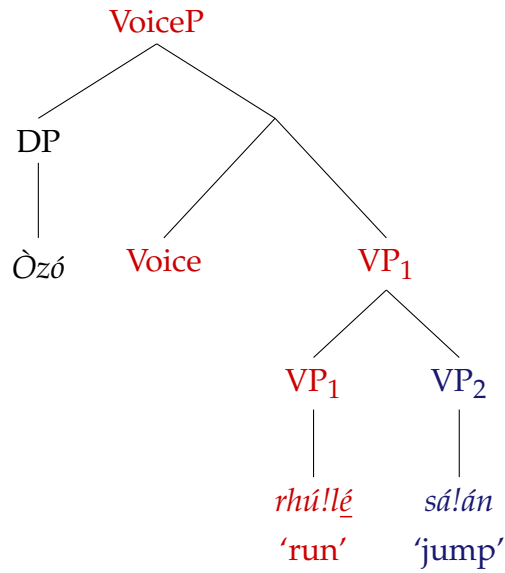
Whenever either the main VP lacks an overt object, or the adjoined VP lacks a null object, or both, serialization cannot occur. This is illustrated below for a NC-SVC consisting of two unergatives (15) / Tree 4.5 and a NC-SVC where V_1 is unergative and V_2 transitive with an overt object (16) / Tree 4.6:

(15) **Edo:** *unerg* + *unerg*

*Òzò gié!gié [rhú!lé] [sá!án]
O. quickly run jump

intended: 'Ozo quickly ran and jumped'

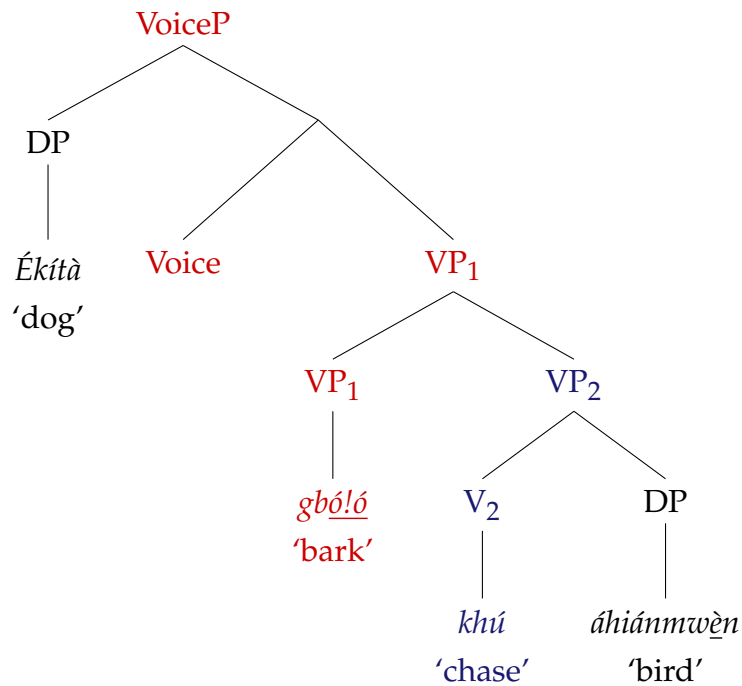
(Baker & Stewart 2002: 49)

Tree 4.5: Edo NC-SVC: *unerg* + *unerg*(16) **Edo:** *unerg* + *trans*

*Ékítà gié!gié [gbó!ó] [khú] áhiánmwèn
 dog quickly bark chase bird

intended: 'The dog quickly barked and chased a bird'

(Baker & Stewart 2002: 2)

Tree 4.6: Edo NC-SVC: *unerg* + *trans*

The derivation crashes in Tree 4.5 because an unergative main V_1 does not provide an internal argument position to be bound by, and conversely an unergative adjoined V_2 does not provide a null argument position to be bound. Similarly, Tree 4.6 cannot give

rise to a well-formed NC-SVC in Edo because VP₂'s overt object bleeds integration by not giving rise to predicate status.

Turning to non-resultative double unaccusatives, these are predicted to be well-formed because unaccusatives have internal arguments.⁶⁰ A double unaccusative NC-SVC would have a syntax similar to Tree 4.4, without external argument-introducing Voice. However, no examples of Edo NC-SVCs consisting of two unaccusatives are discussed in Baker & Stewart (2002), and there is no mention of them being well-formed or not. I leave open why this should be the case, leaning towards the view that it is due to properties idiosyncratic to Edo, such as the kinds of resultatives available and which elements may enter into resultative formation; cf. e.g. Kaufmann & Wunderlich (1998).⁶¹

Finally, multiple-object NC-SVCs like (17) are impossible because there is no argument sharing, i.e. VP₁ and VP₂ cannot be stitched together; cf Tree 4.7:

(17) **Edo:** *trans + trans*

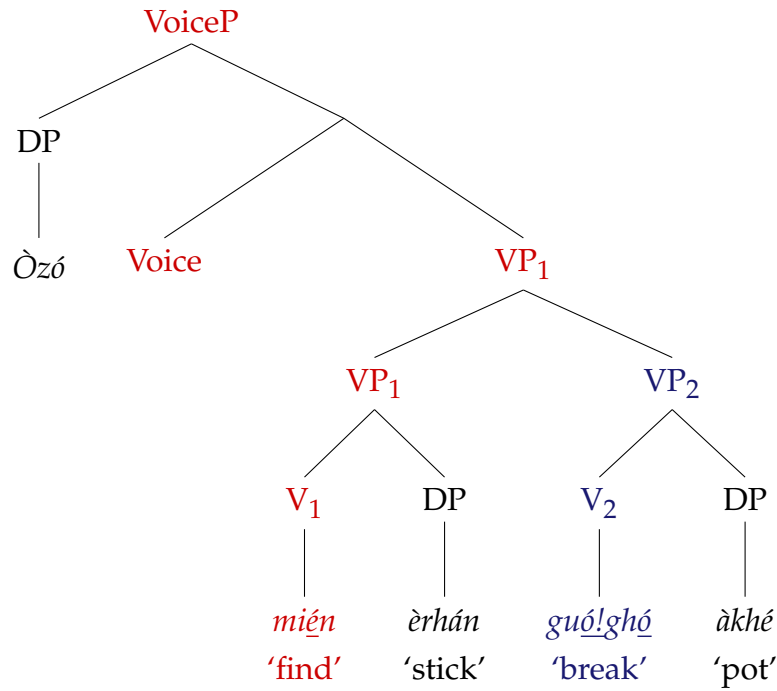
*Òzó [**mién**] èrhán [**guó!ghó**] àkhé
O. **find** stick **break** pot

intended: 'Ozo found a stick and (then) broke the pot'

(Baker & Stewart 2002: 39)

⁶⁰We are interested in [*unacc + unacc*] pairings derived by adjunction, rather than the complementation-like mechanism underlying resultative formation. These may be hard to distinguish without testing, depending on how resultatives are formed in a given language, cf. *She fell and died* vs *She fell such that she died*.

⁶¹This view is supported by the fact that non-resultative [*unacc + unacc*] pairings are attested in both VoiceP-level serialization and V-level serialization, which suggests that they should in principle be available at the intermediate level of VP-level serialization.

Tree 4.7: Edo NC-SVC: *trans* + *trans*

In sum, profile Type B falls out from adjunction at the VP-level, which lacks external arguments. This, coupled with the fact that integration proceeds via predication, ensures that argument sharing must occur via non-external arguments, meaning that each VP must be at least transitive. Ditransitives are discussed in Section 4.2.5.

4.2.4 VoiceP-level adjunction

Remaining with adjunction but moving upwards one level gives us VoiceP-level adjunction. This combination gives rise to profile Type A, which permits all logically possible transitivity pairings, illustrated here for Korean (18):

(18) **Korean:** *Type A transitivity profile*

a. *intr + intr*

Say-ka [nal-a] [ka-ss-ta]
bird-NOM fly-LK go-PST-DECL

‘The bird flew away’

(Kang 1997: 6)

b. *trans + trans*

John-i thokki-lul [cap-a] [mek-ess-ta]
J.-NOM rabbit-ACC catch-LK eat-PST-DECL

‘John caught and ate a rabbit’

c. *intr + trans*

John-i wulthali-lul [ttwi-e] [nem-ess-ta]
J.-NOM fence-ACC jump-LK go.over-PST-DECL

‘John jumped over a fence’

(Ko & Sohn 2015: 100, 108)

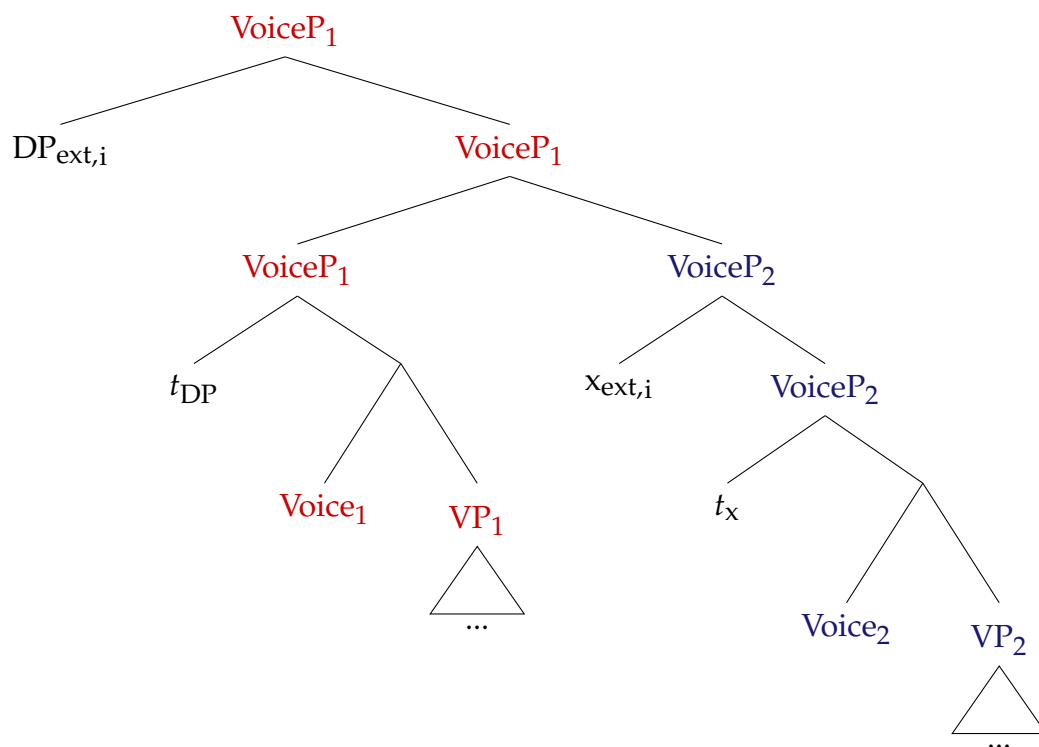
d. *trans + intr*

Say-ka thokki-lul [cap-a] [nal-ass-ta]
bird-NOM rabbit-ACC catch-LK fly-PST-DECL

‘The bird caught a rabbit and flew away’

(Hyunjung Lee p.c.)

Ascending from VP to VoiceP frees up the external argument position in Spec,VoiceP as an additional locus of argument sharing, to the effect that the internal argument position no longer needs to be always be present for serialization to occur. The general syntax of VoiceP-level adjunction is shown in Tree 4.8:



Tree 4.8: VoiceP-level serialization via adjunction

This analysis of VoiceP-level adjunction for certain NC-SVCs is proposed in Ko &

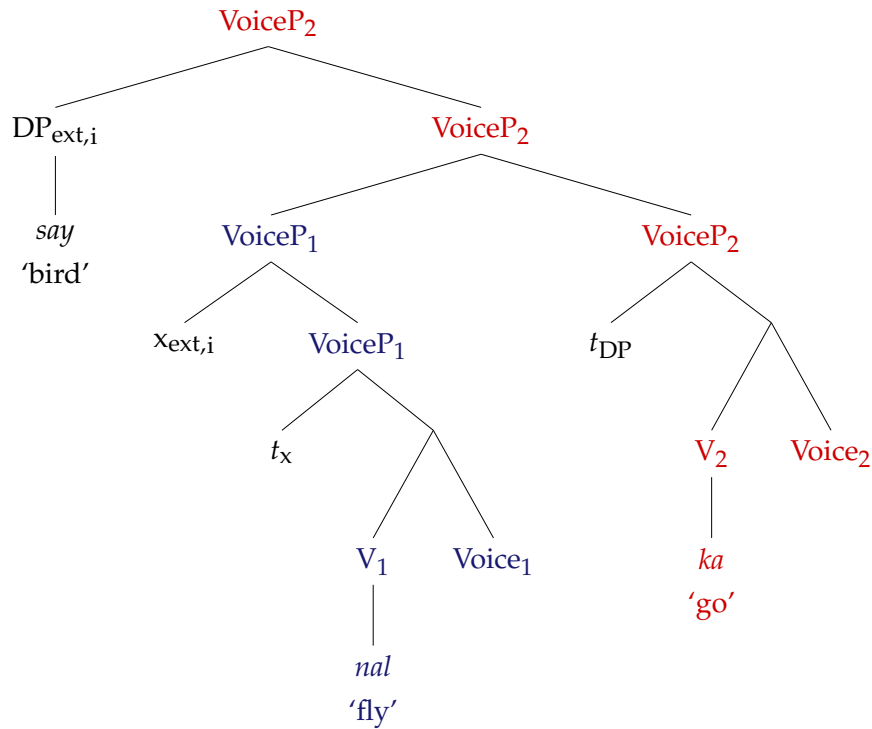
Sohn (2015) for Korean. Their approach continues the logic of Baker & Stewart (2002)'s theory of serialization via predication by extending the level of serialization from VP to VoiceP.

A NC-SVC like (19), consisting of two unergatives, thus has the structure in Tree 4.9, where the two verbs are serialized by way of the external argument position in Spec, VoiceP:⁶²

(19) **Korean:** *unerg + unerg*

Say-ka [nal-a] [ka-ss-ta]
bird-NOM fly-LK go-PST-DECL

'The bird flew away'



Tree 4.9: Korean NC-SVC: *unerg + unerg*

The derivation proceeds just as in VP-level serialization, except that argument sharing now proceeds via the external argument; consequently, it is this argument which undergoes movement in the form of an overt DP in the main VoiceP, and in the form of a minimal pronoun in the adjoined VoiceP, respectively.

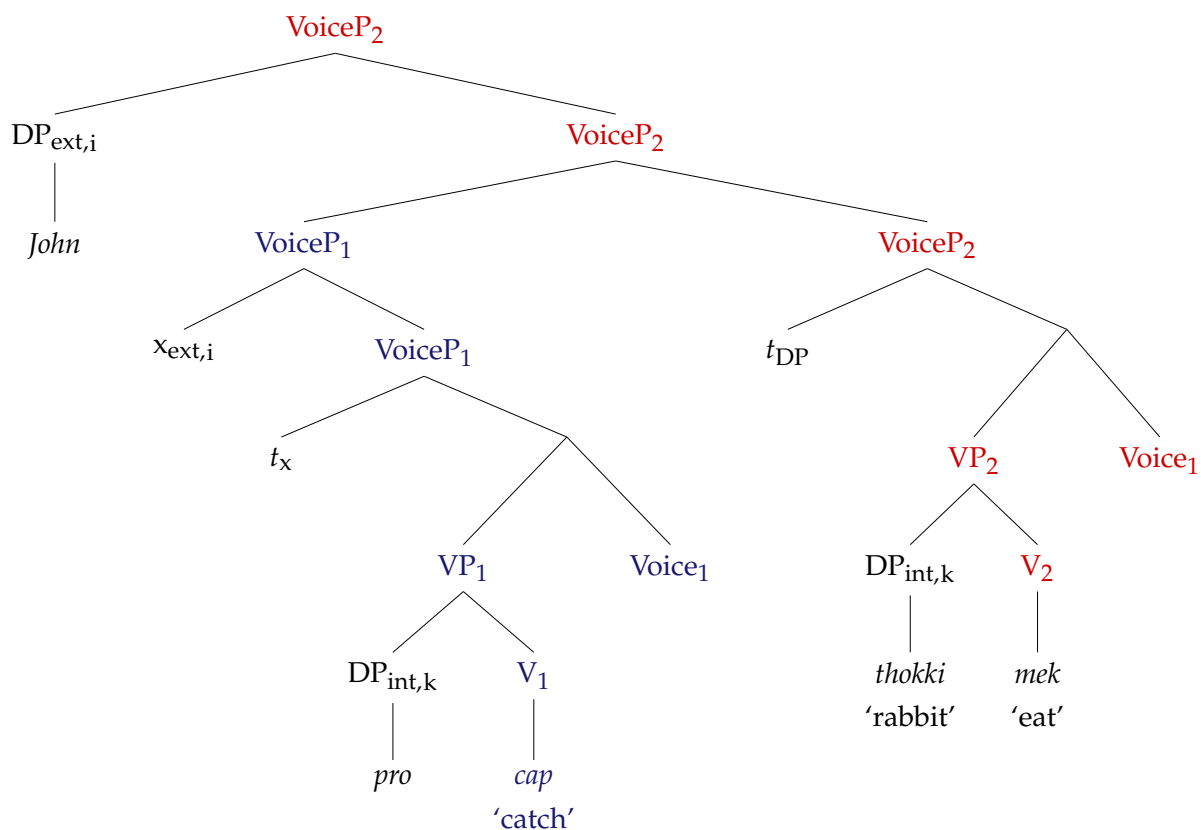
Likewise, two transitive VoicePs can be stitched together to NC-SVCs like (20) by virtue of the fact that the main VoiceP's subject in Spec, VoiceP projects overtly, while the dependent one has an null argument in that same position; cf. Tree 4.10:

⁶²Note that Tree 4.9 does not reflect the final word order of (19). I assume, following e.g. Lee (2005) and Koopman (2005), that subjects move from Spec, VoiceP to Spec, TP in Korean. Thanks to Hyunjung Lee for discussion and bringing these works to my attention.

(20) **Korean:** *trans + trans*

John-i thokki-lul [cap-a] [mek-ess-ta]
 J.-NOM rabbit-ACC catch-LK eat-PST-DECL

‘John caught and ate a rabbit’



Tree 4.10: Korean NC-SVC: *trans + trans*

One consequence of the fact that argument sharing need no longer exclusively proceed via internal arguments is that verbs of disparate transitivity can now enter serialization, in either order. Next to well-formed pairings in which an adjoined unergative is serialized with a main transitive (21) / Tree 4.11, we hence find pairings of an adjoined transitive and a main unergative (22) / Tree 4.12.⁶³

⁶³Hyunjung Lee (p.c.) informs me that (i), where the object of the transitive V_2 follows the unergative V_1 (rather than preceding it, as in (21)), is ungrammatical:

(i) **Korean:** *unerg + trans*

*John-i [ttwi-e] wulthali-lul [nem-ess-ta]
 J.-NOM jump-LK fence-ACC go.over-PST-DECL

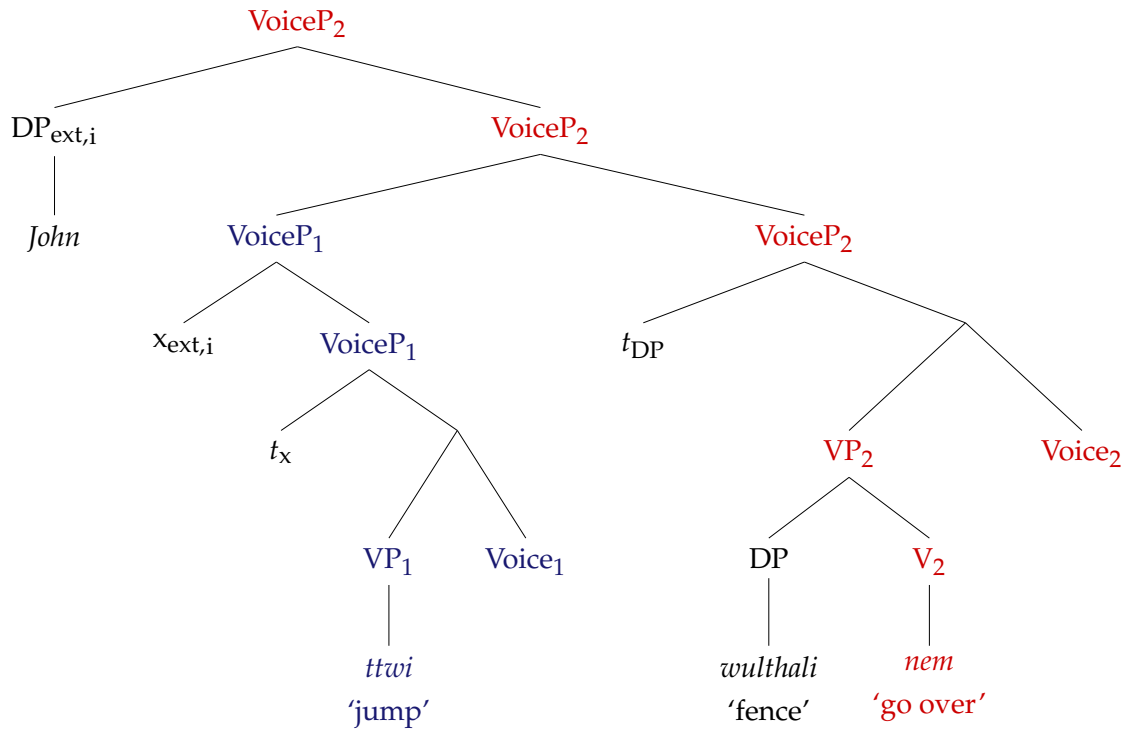
intended: ‘John jumped over a fence’

I leave this issue aside, tentatively suggesting that – even though there does not seem to be independent work arguing for the necessity of object movement in Korean – at least in some cases, object movement to a position at the edge of or external to the complex VoiceP_2 is obligatory, in any case to a position that ends up linearly preceding the adjoined VoiceP_1 . Alternatively, one might assume that Korean’s base

(21) **Korean:** *unerg + trans*

John-i wulthali-lul [ttwi-e] [nem-ess-ta]
 J.-NOM fence-ACC jump-LK go.over-PST-DECL

'John jumped over a fence'



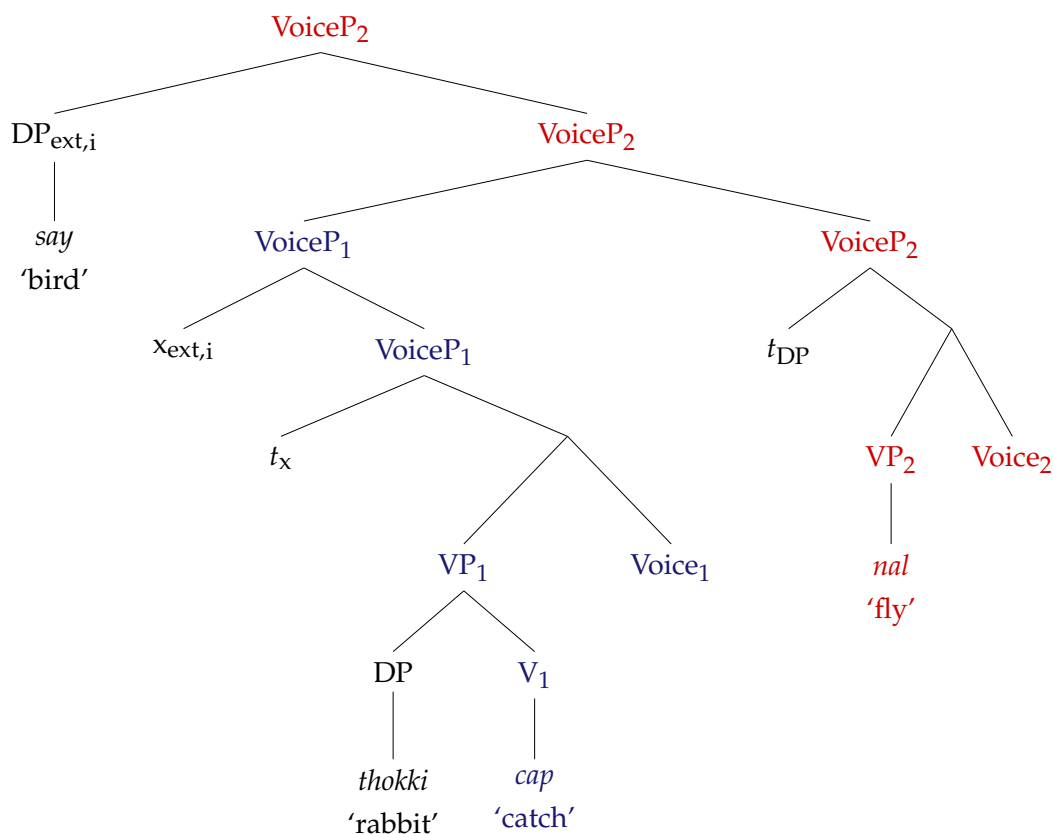
Tree 4.11: Korean NC-SVC: *unerg + trans*

(22) **Korean:** *trans + unerg*

Say-ka thokki-lul [cap-a] [nal-ass-ta]
 bird-NOM rabbit-ACC catch-LK fly-PST-DECL

'The bird caught a rabbit and flew away'

order is SVO and that SOV order is derived by verb and / or object movement, as per Anti-symmetry (Kayne 1994); see e.g. Lee (2007).

Tree 4.12: Korean NC-SVC: *trans* + *unerg*

So far, the above examples have all involved external argument sharing. But serialization can also proceed via internal arguments – this happens when e.g. both verbs are unaccusative (23a), or one is unaccusative and the other passive (23b):

(23) **Korean**a. *unacc* + *unacc*

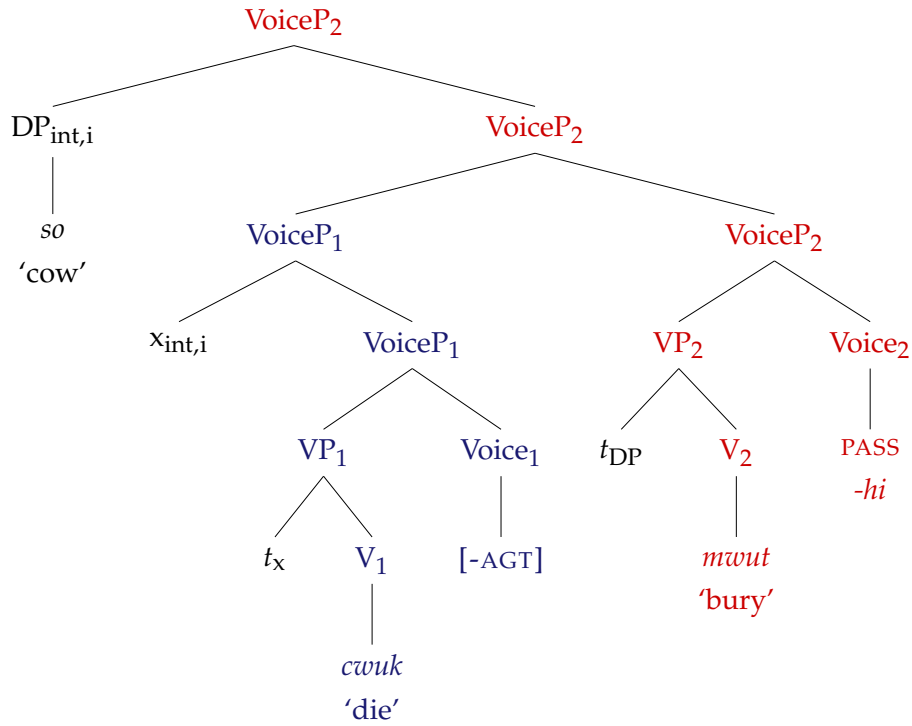
Ceylli-ka [el-e] [kwut-ess-ta]
 jelly-NOM freeze-LK solidify-PST-DECL
 'Jelly got frozen and became solid'

b. *unacc* + *pass*

So-ka [cwuk-e] [mwut-hi-ess-ta]
 cow-NOM die-LK bury-PASS-PST-DECL
 'A cow died and was buried'

(Ko & Sohn 2015: 101)

Following Ko & Sohn (2015: 89), who assume that unaccusatives project a non-agentive Voice head, (23b) has the structure in Tree 4.13:

Tree 4.13: Korean NC-SVC: *unacc* + *pass*

In Tree 4.13, the sole and overt internal argument of the main, passive VoiceP is shared with the sole and null internal argument of the dependent, unaccusative VoiceP; each internal argument is still the structurally highest one within its own VoiceP.^{64,65} The only combinations that are ungrammatical in NC-SVCs (at least in Korean) are those which involve co-predication over thematically distinct subjects, as when an unaccusative is paired with a transitive (24a), or a passive is paired with a transitive (24b):

(24) **Korean**a. *unacc* + *trans*

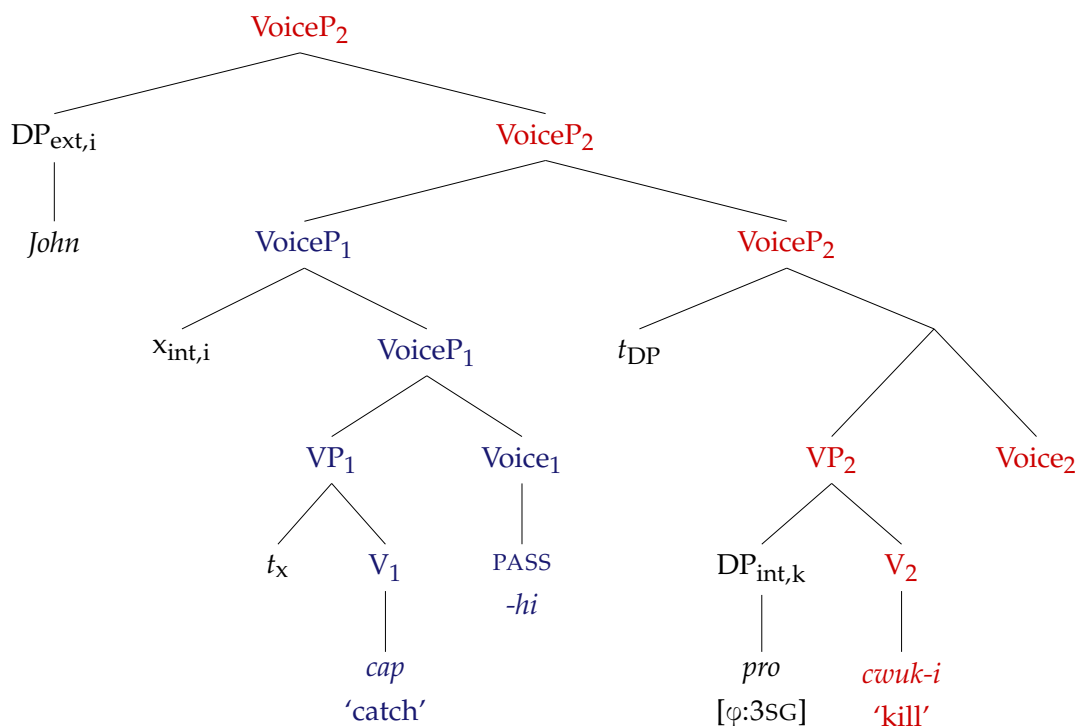
*John-i mwul-ul [kkulh-e] [masi-ess-ta]
 J.-NOM water-ACC boil-LK drink-PST-DECL
 intended: 'John boiled and drank water'

b. *pass* + *trans*

*John-i kom-eykey [cap-hi-e] [cwuk-i-ess-ta]
 J.-NOM bear-by catch-PASS-LK die-CAUS-PST-DECL
 intended: 'John was caught by a bear and killed [the bear]'

⁶⁴A possible alternative to the analysis in Tree 4.13, which is more in line with standard assumptions, would be to assume that unaccusatives lack Voice entirely, and are simply VPs / vPs. However, this would run us into a problem in light of the assumption adopted here that only like categories can be adjoined in serialization, from which it follows that adjoining VP / vP to VoiceP is ruled out. On the other hand, the availability of [*unacc* + *pass*] pairings like (23b) provides potentially strong evidence for the view that (some) unaccusatives contain Voice; I leave this issue open. See Alexiadou et al. (2015) for arguments that Voice may be present morpho-syntactically, but expletive semantically.

⁶⁵Note that the grammaticality of (23b), which pairs an unaccusative and a passive verb, shows that the implicit semantic presence of an agent in the latter does not block serialization.

Tree 4.14: Korean NC-SVC: *pass + trans*

Ko & Sohn (2015) attribute this effect to distinct selectional properties of the voice heads involved and a generalized matching condition on serialization: because unaccusatives and passive do not require the merger of an external argument in Spec,VoiceP (i.e. they lack the feature [$\bullet$ DP $\bullet$]), but unergatives and transitives do (i.e. they bear the feature [$\bullet$ DP $\bullet$]), it follows that the former cannot be serialized with the latter.

It seems equally plausible to me, however, that this effect is attributable to the following event-structural explanation: because serializing unaccusatives or passives with unergatives or transitives via predication over one and the same individual entails ascribing contradictory thematic roles to one and the same argument, a thematic contradiction arises and causes the derivation to crash.⁶⁶

In sum, profile Type A falls out from adjunction at the VoiceP-level. Because argument sharing can now occur via external arguments, in addition to internal ones, VoicePs of any transitivity can be serialized.

4.2.4.1 Multiple-object NC-SVCs

Because argument sharing is now no longer restricted to internal arguments, an adjoined VoiceP headed by a transitive verb should no longer be subject to the

⁶⁶Note that regarding cases where the subject and object denote the same participant, e.g. in resultatives like *She sang herself tired*, I do not take the restriction against thematic contradiction to be violated because I assume that such cases involve two arguments which happen to be co-referent.

condition that its object be null. As it turns out, however, languages appear to differ in this regard: while some allow for multiple lexical objects that are not shared, others place a ban on referentially disjoint objects in serialization.

In Korean for example, it is ungrammatical for VoicePs to host referentially distinct objects. (25) shows that it is impossible for the adjoined VoiceP₁ and the main VoiceP₂ to each host a referentially distinct, overt object:

(25) **Korean:** *trans* + *trans*

*Susi-nun maykcwu-lul [masi-e] kokil-ul [mek-ess-ta]
S.-NOM beer-ADD drink-LK meat-ACC eat-PST-DECL

intended: ‘Susi drank beer and ate meat’ (Hyunjung Lee p.c.)

The only option with two transitive VoicePs is for both to share the same object, as in (20). Compare this to (26) from Khwe, where the first two VoicePs have their own, distinct object:⁶⁷

(26) **Khwe:** *trans*₁ + *trans*₂ + *unerg*₃

tí tĩyá cáca à [dx’áa-a] kx’óxò à [kx’ó-ro] [txóro-è-tè]
1SG standing beer OBJ drink-II meat OBJ eat-II dance-I-PRES

‘In standing position I am drinking beer, eating meat, and dancing’

(Kilian-Hatz 2006: 113)

Judging by the criteria employed here and in Kilian-Hatz (2006), (26) instantiates bona fide serialization. This implies that it must be stated on a language- and perhaps also SVC-specific basis, then, whether or not object sharing is obligatory in addition to subject sharing. This by itself seems unproblematic in light of the fact languages place what at times appear to be arbitrary restrictions on what kinds of verbs may enter serialization. That said, multiple-object NC-SVCs like (26) are the exception rather than the norm, leading me to speculate that (26) might involve covert coordination after all. I set this issue aside for future work.

⁶⁷Khwe NC-SVCs permit all logically possible transitivity pairings, and hence instantiate profile Type A. Next to lexical object adding (26), Khwe NC-SVCs also come in the object sharing type:

(i) **Khwe:** *trans* + *trans*

Táténò xàm-tcà à [c’é-ε] [≠’áka-ra-xu-a-tè]
then lion-2DU.M OBJ take-II hide-II-COMPL-I-PRES

‘Then she takes and hides the two lions’

(Kilian-Hatz 2006: 112)

4.2.5 Ditransitives

The theory of serialization adopted here easily accommodates ditransitive projections, as Baker & Stewart (2002) show.

We begin with the ditransitive NC-SVC in (27a) from Edo, in which the main V_1 is a ditransitive, and the adjoined V_2 is transitive. The main V_1 licenses an indirect and a direct object outside serialization, corresponding thematically to source and patient, respectively (27b):

(27) **Edo**

a. *Main ditransitive V_1*

Òzó [**vbó**] òkhókhò ìgàn [khien]

O. **pluck** chicken feathers **sell**

‘Ozo plucked the chicken_i of feathers_k and sold them_{*i/k}’

(Baker & Stewart 2002: 48)

b. *Ditransitive V_1 takes double objects in isolation*

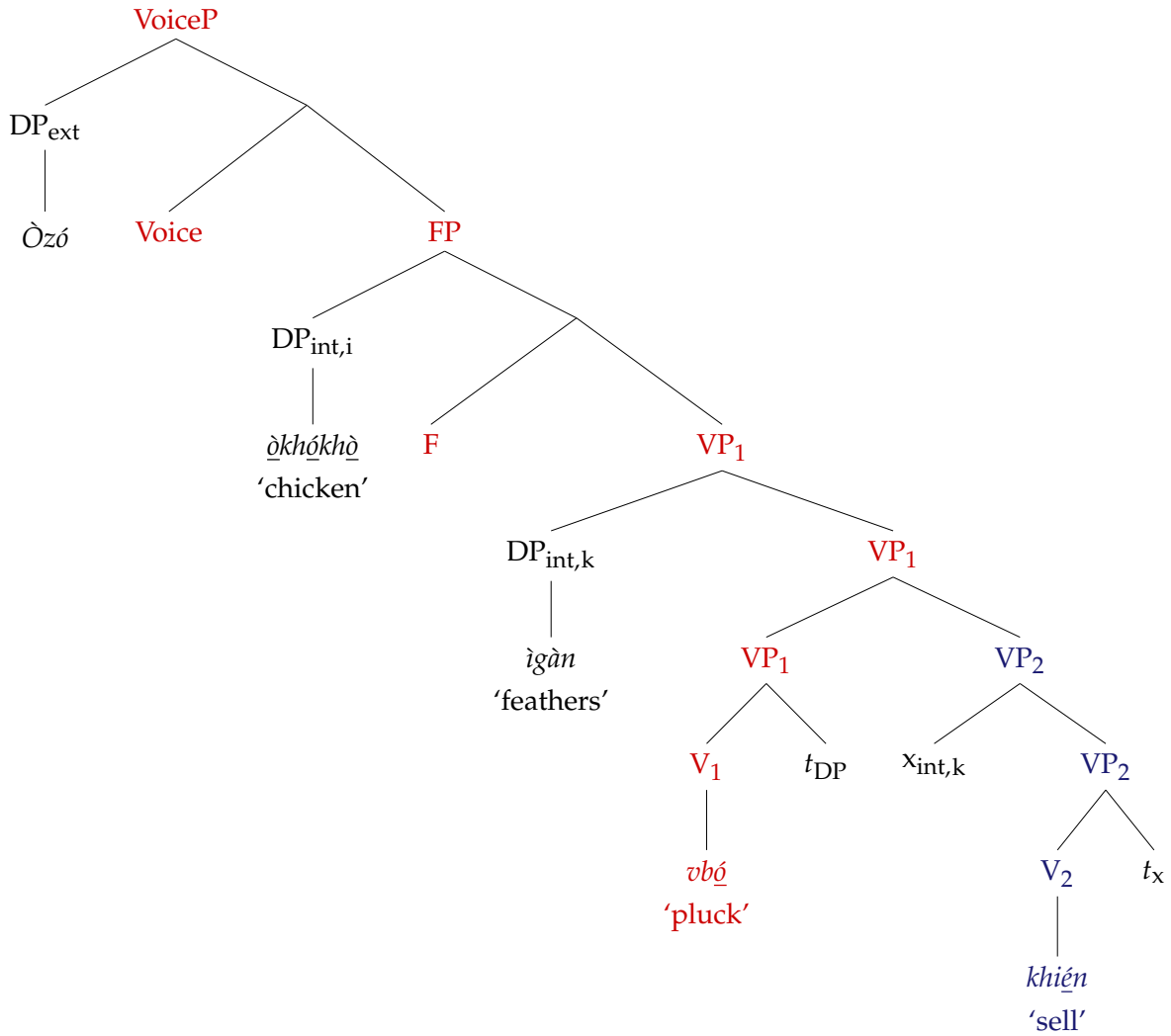
Úyì vbó òkhókhò ìgàn

U. pluck chicken feather

‘Uyi plucked the chicken’s feathers’

(Stewart 2001: 123)

The adjoined transitive V_2 ’s null object can only be understood as identifying V_1 ’s patient, not its source, even though the latter would be pragmatically more plausible (cf. (27a)). I follow Baker & Stewart (2002) in assigning to (27a) the structure in Tree 4.15:



Tree 4.15: Edo ditransitive NC-SVC with main ditransitive

In this structure, the null object in the adjoined VP₂ necessarily identifies the patient in VP₁ because the source is located in a position above the point of serialization, i.e. it is too high to be the subject of predication.⁶⁸ That the source should be located in a VP-external position goes back to observations since Larson (1988), according to which (non-oblique) Goals (and Benefactors, Sources, etc.) are structurally higher than Patients. In terms of timing, then, the source comes in too late for it to be able to enter object sharing.⁶⁹

⁶⁸I assume, for clarity, that indirect objects are merged in a functional projection FP, located between VP and VoiceP. Nothing hinges on this assumption, and we could just as well place them in Spec,VP.

⁶⁹Baker & Stewart (2002: 47) draw attention to similar restrictions that are known to hold in more familiar cases of predication, noted in Williams (1980). Specifically, there are tight constraints on what can be the object of a predicative adjunct that is attached to VP, which is that the predicate must be related to a non-oblique overt argument that receives a Patient role from the verb. (i) illustrates this characteristic paradigm for the English purposive construction:

- (i) a. I gave Pete the book_i [to read ____i]
 b. *I gave Pete_i the book [to impress ____i]
 c. I presented it_i to Bill [to read ____i]
 d. *I presented Bill [with it_i] [to read ____i]
- (Baker & Stewart 2002: 47)

The understood object of the adjoined purpose clause can link to the direct object / Patient in the

I will now show that this extends readily to cases in which it is the adjoined verb that licenses more than one internal argument. While I was unable to find NC-SVCs with object sharing in which the adjoined verb is a true ditransitive with two structural arguments, consider nevertheless (28a) from Sranan, where the adjoined V_2 takes two arguments in isolation, a patient and a source, the latter of which is oblique (28b):

(28) **Sranan**

a. *Dependent ditransitive V_2*

Kofi [hari] a ston [puru] na ini a olo
 K. pull the stone remove LOC in the hole
 ‘Kofi pulled the stone out of the hole’

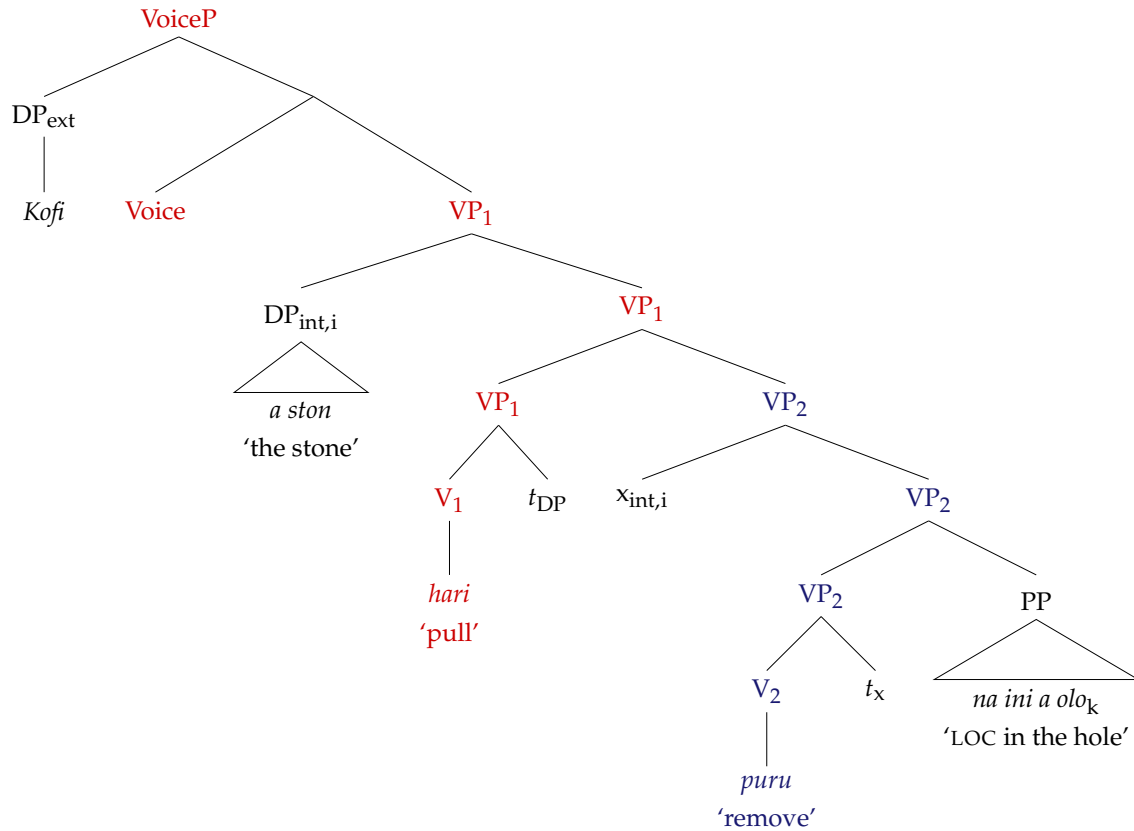
b. *Ditransitive V_2 takes double objects in isolation*

Kofi puru a ston na ini a olo
 K. remove the stone LOC in the hole
 ‘Kofi removed the stone from the hole’

(Sebba 1987)

Just like in (27a), argument sharing holds between patients. For (28a), I propose the structure in Tree 4.16, which follows the logic of Tree 4.15, except that now, the oblique argument of the adjoined V_2 is too deeply embedded to partake in argument sharing: because the minimal pronoun merges in *Comp,V*, which corresponds to the patient role, this is the argument position that gets abstracted over upon movement, and subsequently bound by the patient of the main VP_1 :

main clause (ia), but not the indirect object / Goal (ib); moreover, it can only link to a Patient that is realized structurally as a direct object (ic), but not obliquely in a PP (id). Baker & Stewart capture the fact that purposive constructions obey restrictions analogous to those found in SVCs as generalized consequences of predication: because the Goal is structurally higher than the point of the adjoined VP, the only argument that can be predicated of within the VP_H is the Patient.



Tree 4.16: Sranan ditransitive NC-SVC with dependent ditransitive

4.2.6 Transitivity restrictions

Profiles Type A and B fall out from cross-cutting adjunction with either the level of VP or VoiceP. In this way, distinct sets of possible transitivity co-occurrences fall from the distinct argument positions that are available as a function of the structural level: because external arguments are unavailable at the VP-level, syntactic argument sharing must proceed internal arguments, which gives rise to profile Type B. The remaining, logically possible pairings become unlocked at the VoiceP-level, which additionally makes external arguments available as loci for argument sharing; this gives us profile Type A.

4.3 Non-contiguous serialization via complementation

In the NC-SVCs we have considered so far, the adjoined VerbP must always project a null argument due to how it must be integrated. This robust linking effect appears to be falsified by a particular type of construction in which each verb projects an overt internal argument. These properties follow because these constructions are built not via adjunction, but rather complementation.

4.3.1 VP-level complementation

Cross-cutting complementation with the VP-level gives us what I classify as *applicative* NC-SVCs, which involve the addition of non-core roles. This kind of NC-SVCs is found in e.g. Sranan:

(29) **Sranan:** *Instrumental* NC-SVC

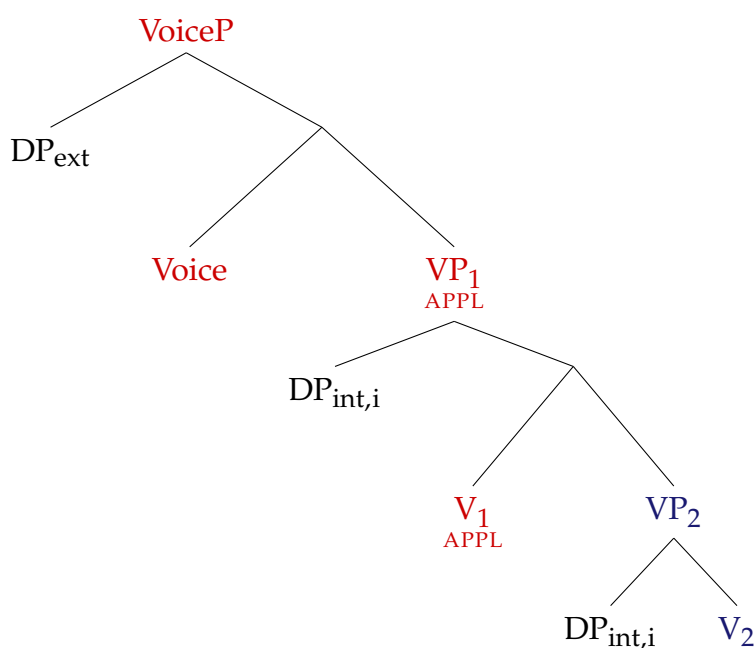
Kofi [teki] a nefi [koti] a brede
K. take the knife cute the break

‘Kofi cut the bread with the knife’

(Sebba 1987)

Because no relationship of argument sharing holds between V_1 and V_2 , but one verb introduces an argument such as an instrument or a beneficiary to the overall argument structure, serialization proceeds via complementation instead of adjunction (which would require a null argument in the adjoined projection).

The general syntax of VP-level complementation is given in Tree 4.17:



Tree 4.17: VP-level serialization via complementation

Unlike the other structure-building combinations discussed here, VP-level complementation does not so much have a direct bearing on the transitivity of verbs as it does on the sharing and linking of arguments; this way, the mode of structure-building is intimately connected to argument structure. We return to this issue below.

Given their functional similarity to morphological applicatives in languages like that have them, I propose to classify NC-SVCs in which one of the verbs adds a non-core role as *applicative* SVCs or *verbal periphrastic applicatives*; accordingly, I refer to the complementing verb that introduces the non-core role as the *functional* or

applicative verb in the context of serialization, and the other, complemented verb as the *lexical verb*.

Although a special interpretive and link between one of the verb's objects and the action denoted by the other verb has been pointed out in other generative work on SVCs (Baker 1989, da Cruz 1997, Veenstra & Muysken 2017), and works like Aboh (2009) and Paul & Whitman (2010) employ complementation analyses which attribute a clear functional role to one of the verbs, the classification of these kinds of NC-SVCs as verbal applicatives has so far received little attention. Heine & König (2010) is an important work in this regard due to its recognition that what I am calling applicative NC-SVCs are functional equivalents to canonical ditransitive constructions in non-serializing languages with respect to argument linking.

In Tree 4.17, an applicative V_1 takes a lexical VP_2 as its complement and introduces a non-core argument in its specifier – e.g. 'take' adds an Instrument (31a), 'give' a Beneficiary (30) and 'accompany' a Comitative (31b):

(30) **Yoruba:** *'give' introduces Beneficiary*

Ajé [wɔ] aṣo [fún] Olú
A. wear clothing **give** O.

'Aje wore an outfit for Olu'

(Awoyale 1988: 5)

(31) **Gungbe:**

a. *'take' introduces Instrument*

Sétù [zɛ] kpò ló [xò] Kòjò
S. **take** stick DET hit K.

'Setu took the stick hit Kojo'

i.e. 'Setu hit Kojo with the stick'

b. *'accompany' introduces Comitative*

Sétù [kplán] Kòjò yì Kútònú
S. **accompany** K. go C.

'Setu accompanied Kojo to Cotonou

i.e. 'Setu went to Cotonou with Kojo'

(Aboh 2009: 16)

As complementation structures, applicative NC-SVCs should allow extraction of both objects; where the relevant facts have been documented, this is borne out. In the instrumental NC-SVC from Sranan in (32a), an English-based creole spoken in Suriname, *wh*-extraction can target both the Instrument object of the applicative V_1 (32b), and the Patient object of the lexical V_2 (32c):⁷⁰

⁷⁰Even though Aboh (2009) explicitly assumes a complementation analysis of instrumental SVCs, he does not give extraction data specifically for the Gungbe instrumental SVCs in (31b). He does claim that the Kwa NC-SVCs he considers generally show no extraction asymmetries, and also shows that Instruments are extractable in closely related Fongbe:

(i) **Fongbe:** *Instrumental NC-SVC*

(32) **Sranan:** *Instrumental NC-SVC*

- a. Kofi [teki] a-nefi [koti] a-brede
K. take DET-knife cut DEF-bread
'Kofi cut the bread with the knife'

b. *Extraction of Instrument*

San_i Kofi [teki] —_i [koti] a-brede?
what K. take cut DEF-bread
'What did Kofi cut the bread with?'

c. *Extraction of Patient*

San_i Kofi [teki] a-nefi [koti] —_i?
what K. take DET-knife cut
'What did Kofi cut with the knife?'

(Sebba 1987)

Similarly, benefactive NC-SVCs in Yoruba (33a) allow both the beneficiary (33b) and the patient (33c) to extract by clefting:

(33) **Yoruba:** *Benefactive NC-SVC*

- a. Jímò ọ [ra] èwù [bùn] mi
J. AGR buy garment present 1SG
'Jimo brought me (a) garment'

b. *Extraction of Beneficiary*

Èmì_i ni Jímò ọ [ra] èwù [bùn] —_i
1SG FOC J. AGR buy garment present
'It's me that Jimo bought a garment for'

c. *Extraction of Patient*

Èwù_i ni Jímò ọ [ra] —_i [bùn] mi
garment FOC J. AGR buy present 1SG
'It's a garment that Jimo bought me'

(Déchaine 1993: 799, 800)

This follows from a configuration in which neither verbal element is adjoined to the other, but instead the applicative V_1 takes the lexical VP_2 as its complement. Because neither object is in an adjunct, but is a specifier and complement, respectively, both can extract. Note that even though the above NC-SVCs superficially appear to consist of two transitive VPs, they actually do not on this analysis, with only one of the VPs

-
- a. Kòkú [sò] {jìvì ọ} [sén] wóxúxú ọná
K. take knife DET cut bread DET with
'Koku cut the bread with the knife'
- b. Ètè_i wèKòkú [sò] sèn wóxúxú ọná
what FOC K. take cut bread DET with
'What did Koku cut the bread with?'

(Aboh 2009: 7)

Unfortunately, the Fongbe instrumental SVCs do not map perfectly onto the Gungbe ones, because the former contain the preposition *ná* 'with' in addition to the applicative V_1 'take'. Aboh (2009: 7) assumes that the Instrument is introduced as a complement to *ná* 'with', and subsequently shifts to a position between the V_1 and the object of V_2 .

(the lexical one) forming a proper constituent with its object.

I adopt the analysis of high applicatives in Pylkkänen (2008) for the applicative uses of verbs like ‘take’ and ‘give’ (under a slight notational change): like Voice, high Appl is a head that (i) takes a VP complement, (ii) introduces an external argument in its specifier, and (iii) expresses a relation between an individual (the referent of the external argument) and an event (denoted by the VP complement). The denotation and syntactic selectional requirements of high applicatives are given in (34):^{71,72}

(34) *High applicative*

- a. $\llbracket \text{Appl} \rrbracket = \lambda x \lambda e. [\text{R}(e, x)],$
 where $\text{R} \in \{\text{BENEFICIARY, INSTRUMENT, COMITATIVE, LOCATIVE, ...}\}$
 (Pylkkänen 2008: 75)
- b. $\text{Appl: } [\bullet \text{V} \bullet < \bullet \text{D} \bullet]$

By full analogy, I propose the following semantic and syntactic specifications for verbal applicatives, here illustrated for instrumental ‘take’ and benefactive ‘give’:

(35) *‘take’ verbal applicative*

- a. $\llbracket \text{TAKE} \rrbracket = \lambda x \lambda e. [\text{INSTRUMENT}(e, x)]$
- b. $\text{TAKE: } [\bullet \text{V} \bullet < \bullet \text{D} \bullet]$

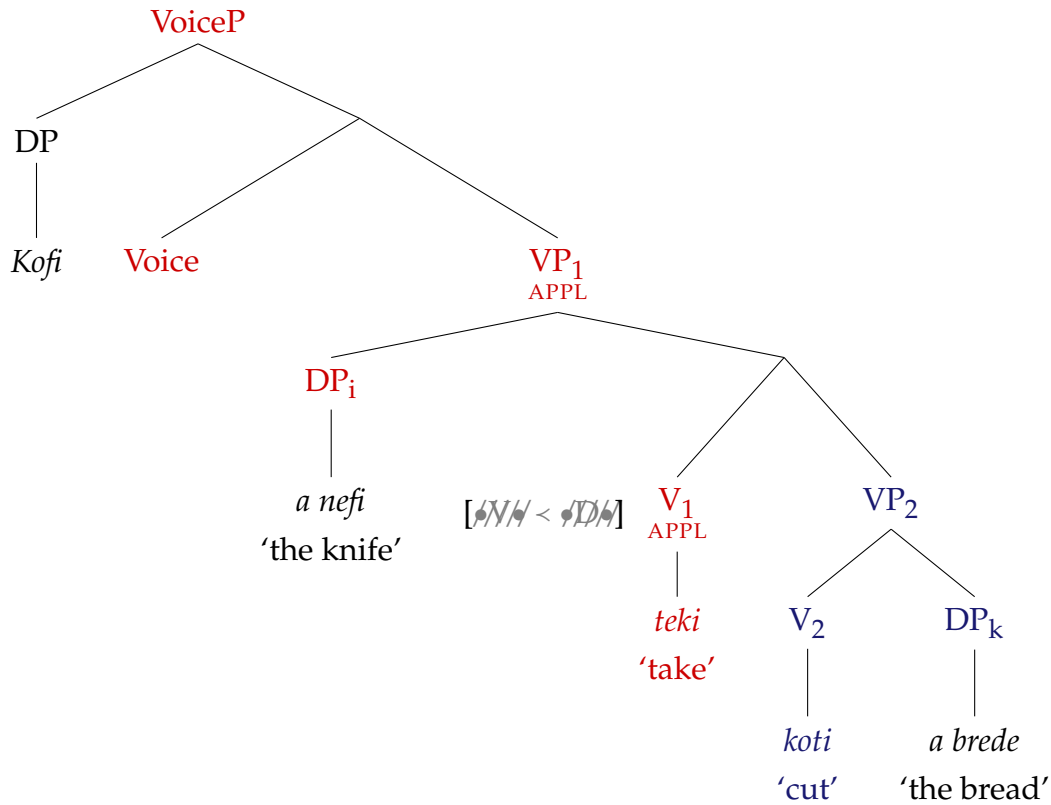
(36) *‘give’ verbal applicative*

- a. $\llbracket \text{GIVE} \rrbracket = \lambda x \lambda e. [\text{BENEFICIARY}(e, x)]$
- b. $\text{TAKE: } [\bullet \text{V} \bullet < \bullet \text{D} \bullet]$

The instrumental NC-SVC in (29) has the structure in Tree 4.18:

⁷¹As with Voice, I assume that high Appl composes with its syntactic VP complement via Event Identification (Kratzer 1996), though nothing hinges on this. In works like Bruening (2013), for instance, Voice heads are higher-order predicates which compose with their complements via Function Application.

⁷²At times, authors have hypothesized opposite syntactic positions for a particular kind of applied object. Regarding Instruments, for example, Pylkkänen (2008) takes them to be high applicatives, while Marantz (1984) assumes that they reside within VP, i.e. are low applicatives.



Tree 4.18: Instrumental NC-SVC via VP-level complementation

- (37)
- a. $[[VP_2]] = \lambda e. [\text{cut}(e, \text{the bread})]$
 - b. $[[V_1_{APPL}]] = \lambda x \lambda e. [\text{INSTR}(e, x)]$
 - c. $[[VP_1_{APPL}]] = \lambda e. [\text{cut}(e, \text{the bread}) \ \& \ \text{INSTRUMENT}(e, \text{the knife})]$
 - d. $[[VoiceP]] = \lambda e. [\text{cut}(e, \text{the bread}) \ \& \ \text{INSTRUMENT}(e, \text{the knife}) \ \& \ \text{AGENT}(e, \text{Kofi})]$

Note that this analysis requires double lexical entries for the applicative verb since it generally patterns as a regular transitive verb outside serialization. In Gungbe, for instance, the verb *zé* ‘take’ is applicative within serialization (38a), but takes an individual-denoting direct object in isolation (38b):

(38) **Gungbe**

- a. *Applicative use of ‘take’*

Sétù [zé] kpò ló [xò] Kòjò
 S. take stick DET hit K.
 ‘Setu took the stick hit Kojo’

- b. *Lexical use of ‘take’*

Sétù zé gbó ðàkpó
 S. take goat one
 ‘Setu took one goat’

(Aboh 2009: 16, 17)

I assume double lexical entries along the lines of (39) (leaving aside how the applicative entry should be derived from the lexical one):⁷³

(39) *Double lexical entries for ‘take’*

a. *Lexical ‘take’*

(i) $\llbracket take \rrbracket = \lambda x \lambda e. [take(e, x)]$

(ii) *take*: [$\bullet D \bullet$]

b. *Applicative ‘take’*

(i) $\llbracket TAKE \rrbracket = \lambda x \lambda e. [INSTRUMENT(e, x)]$

(ii) *TAKE*: [$\bullet V \bullet < \bullet D \bullet$]

It is useful at this point to remind ourselves of the differences between applicative serialization and covert coordination. Take again the instrumental NC-SVC in (32) from Sranan, repeated in (40a), which has the same surface sequence of constituents as the covert coordination in (40b):

(40) **Sranan**

a. *Instrumental NC-SVC*

Kofi [teki] a ne fi [koti] a bre de

K. take the knife cut the break

‘Kofi cut the bread with the knife’

cf. ‘Kofi took the knife and cut the bread’

b. *Covert coordination*

Kofi [sutu] Amba [kiri] Kwaku

K. shoot A. kill K.

‘Kofi shot Amba and killed Kwaku’

(Sebba 1987)

Both can be translated as coordinations in English, yet the two are argument-structurally and syntactically distinct. In terms of argument structure, the object of V_1 in (40a) bears the ϑ -role of instrument with respect to the event denoted by V_2 , and correlated with this property is that syntactically – as we know – both objects can extract (cf. (32b) and (32c)). In the covert coordination in (40b), by contrast, no designated

⁷³This contrasts with e.g. the verb *de* ‘take’ in Akan, which occurs in isolation, but disallows objects (ib):

(i) **Akan**

a. Kofi [de] Yaw kɔɔ Kumase

K. take Y. go K.

‘Kofi took Yaw to Kumase’

b. *Kofi de Yaw

K. take Y.

‘Kofi took Yaw’

(Aboh 2009: 17f)

thematic relationship holds between V_1 's object and the event denoted by V_2 , and neither object is able to extract:

(41) **Sranan:** *Covert coordination*

a. *Extraction of V_1 's object*

*Suma_i Kofi sutu —_i [kiri] Kwaku?

who K. shoot kill K.

intended: 'Who did Kofi shoot and kill Kwaku?'

b. *Extraction of V_2 's object*

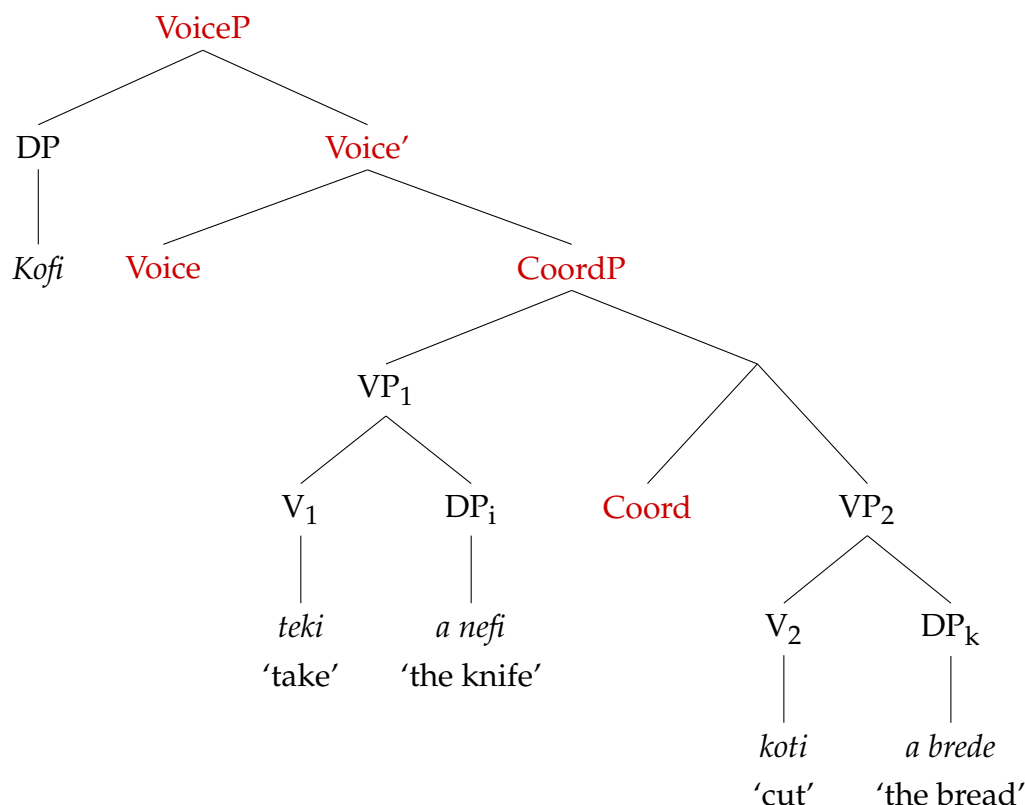
*Suma_i Kofi [sutu] Amba [kiri] —_i?

who K. shoot A. kill

intended: 'Who did Kofi shoot Amba and kill?'

(Sebba 1987)

This contrast follows because despite their surface similarities, instrumental NC-SVCs and covert coordinations are distinct: while the former involve the complementation structure in Tree 4.18, the latter involve the coordination structure in Tree 4.19, with a silent coordinating projection:



Tree 4.19: Covert coordination with silent CoordP

Extraction out of coordination is ruled out by the Coordinate Structure Constraint (Ross 1967).⁷⁴ See Sebba (1987: 109ff), Baker (1989: 546ff) and Aboh (2009: 4ff) for discussions of these and further differences between serialization and covert

⁷⁴Crucially, the objects of the coordinated VPs in (40b) are non-identical, thus controlling for ATB extraction.

coordination, and also Section 3.4 in the preceding chapter.

Turning to semantic and syntactic integration, recall how our theory of serialization via adjunction requires adjoined elements to be integrated via predication, in accordance with Full Interpretation. Now, when serialization does not occur via serialization, but complementation, integration proceeds by selection. With respect to our applicative NC-SVCs, this means that the lexical VP is simply merged with the applicative verb by virtue of being selected by it. In this way, the mode of structure-building is intimately connected to argument sharing and linking: where VerbP-level serialization via adjunction forces the adjoined projection to contain a null argument to be linked with an overt category in the main projection, no such restriction holds in VerbP-level complementation. Consequently, there is no argument sharing in applicative NC-SVCs, but the lexical verb's argument structure is extended by a thematically distinct, non-core argument.

From a conceptual point of view, an analysis of (a subset of) multiple-object NC-SVCs as applicatives involving complementation comes with several merits. First, it recognizes and adequately models the systematic recurrence of verbs like 'take' and 'give' across languages precisely in connection with the introduction of non-core roles like Instruments and Beneficiaries, respectively.

Second, it explains the availability of only a very particular kind of multiple-object NC-SVC among what is otherwise a landscape that is practically bare of multiple-object NC-SVCs in numerous languages.

Third, it explains why instrumental meanings are robustly absent among C-SVCs, the reason being that only VerbP-level serialization provides the morpho-syntactic configuration necessary for projecting two structural arguments. See Section A.1 of the Appendix.

And finally, it does away with the need of invoking construction-specific properties. In Veenstra (1993), for example, the lexical V_2 in instrumental NC-SVCs from Sranan and Saramaccan is analyzed as an adjunct to the V_1 'take' rather than a complement; because V_2 's object can extract (just like V_1 's), Veenstra (1993: 161) proposes that V_2 constitutes a weak island specifically in serialization – crucially, despite the fact that extraction out of adjuncts is generally bad in these languages. Similarly, Collins (2002: 21) assumes that NC-SVCs in #Hoan are exempt from the Coordinate Structure Constraint. These stipulations simply fall away under an applicative analysis.

In sum, applicative NC-SVCs hold a special place among the various NC-SVCs discussed here in that they allow integration without syntactic argument sharing between an overt and a null argument, by virtue of selection between the applicative verb and the lexical VP.⁷⁵

⁷⁵Note that the presence of a 'take'-type verb alone does not entail that the SVC is applicative because the verbs which typically occur in this function may also (i) be serialized as regular lexical verbs via adjunction, or (ii) occur in what similarly appear to be complementation configurations, yet with distinct grammaticalized meanings (e.g. aspect); see Lee (2019) for a cross-linguistic survey.

4.3.1.1 Raising applicatives

So far, the applicative NC-SVCs considered here have all involved an applicative verb that is thematic, meaning that the applicative verb itself introduces a non-core argument that is not lexically specified by its non-applicative co-verb. In opposition to *thematic applicatives*, a growing body of work going back to Georgala et al. (2008), Paul & Whitman (2010) and Georgala (2012) has argued for so-called *raising applicatives*, which do not themselves contribute any thematic role, but instead attract an argument that is licensed thematically by a lower verb to their specifier for Case reasons.⁷⁶ While differing in the specifics, these accounts share the idea that raising applicatives have no thematic import of their own, but serve a purely syntactic purpose in ditransitive constructions by attracting the highest DP introduced by the lexical verb for Case reasons. After this DP raises from its VP-internal position to Spec,ApplP, it receives Case either from a higher functional head associated with transitivity, e.g. Voice, or from Appl itself. Thematic and raising applicatives are schematized in (42) and 8.6, respectively:

(42) *Thematic applicative*

$$[_{\text{ApplP}} \text{DP}_{\text{BEN}} [_{\text{Appl}'} \text{Appl} [_{\text{VP}} \text{V} \text{DP}_{\text{PAT}}]]]$$

(43) *Raising applicative*

$$[_{\text{ApplP}} \text{DP}_{\text{GOAL}} [_{\text{Appl}'} \text{Appl} [_{\text{VP}} t_i [_{\text{V}'} \text{V} \text{DP}_{\text{PAT}}]]]]$$

For a subset of NC-SVCs consisting of an applicative verb like ‘take’ or ‘give’, the applicative verb qualifies as a raising applicative verb on grounds that it (i) appears to not itself introduce a non-core role such as Instrument or Beneficiary, and (ii) systematically appears with Goal or Source arguments and, correlated with this fact, is restricted to ditransitive co-verbs like ‘sell’, ‘show’ and ‘send’. A raising applicative analysis is proposed by Paul & Whitman (2010) for Mandarin ditransitive constructions involving the verb *gěi* ‘give’, which is argued to Case-license Goals that are thematically licensed by lexically ditransitive verbs.

A strong case for a raising applicative analysis is made by the following NC-SVC from Haitian Creole, where the argument occurring to the left of the V_1 *pran* ‘take’, Mary, is hardly construable as an Instrument in the event denoted by the semantically triadic V_2 ‘show’:

⁷⁶In Georgala et al. (2008), markers passing low-applicative diagnostics in Mandarin and Chi-Mwi:ni are argued to also function as Case-assigners in the presence of VP-internal non-core arguments; for Niuean, Massam (2015) shows that the marker *aki* - next to introducing Instruments - doubles as a case marker for causees in causatives of transitives; in Nie (2019), Tagalog external possession is taken to involve raising of the possessor to Spec,ApplP in order to become licensed for Case assignment. The idea that applicative morphology may contribute to syntactic licensing goes back to at least Baker (1988), who treats applicatives as incorporated prepositions.

(44) **Haitian Creole**

M [pran] Mari_i [montre] tet-li_i / li_i nan glas la
 1SG take M. show HEAD.3SG / 3SG LOC mirror DET

‘I showed Mary herself in the mirror’ (Veenstra & Muysken 2017: 22)

Relatedly, Collins (1997: 489) reports for Akan that bare Instruments are possible with neither an instrumental preposition nor a default Case-assigner (45a), in addition to being possible in connection with the verb ‘take’ (45b) (though he does not specify any additional differences between the two options):

(45) **Akan**a. *Bare Instrument*

Otwaa nIyir sikan
 he-cut his-wife knife
 ‘He stabbed his wife with a knife’

b. *Instrument via ‘take’*

[Odzi] sIkan [twaa] nIyir
 he-take knife cut his-wife
 ‘He stabbed his wife with a knife’

(Collins 1997: 489)

On the assumption that some Case-theoretic difference exists between the bare (45a) and the non-bare Instrument (45b), the use of ‘take’ in (45b) may turn out to instantiate a raising applicative.

Finally, another prediction of the structure in Tree 4.17 – regardless of an analysis in terms of thematic or raising applicativization – is that because the applied object asymmetrically c-commands the lexical object, the former should be able to bind the latter.⁷⁷ While the hierarchical relations between arguments in SVCs have so far received little attention, this is correct where the facts have been documented (all the relevant examples I am aware of seem to be raising applicative NC-SVCs):⁷⁸

(46) *DP₁ asymmetrically c-commands DP₂*a. **Haitian Creole: Anaphoric binding**

M [pran] ⟨Mari⟩_i [montre] ⟨tet-li⟩_i / ⟨li⟩_i nan glas la
 1SG take M. show HEAD.3SG / 3SG LOC mirror DET
 ‘I showed Mary herself in the mirror’

b. **Saramaccan: Quantifier binding**

mi [téi] ⟨híniwan búku⟩_i [náki] ⟨dí sikífi-ma f’én⟩_i
 1SG take each book hit DET write-MA FOR.3SG

⁷⁷This c-command asymmetry is one of the defining properties of double object and applicative constructions cross-linguistically (Barss & Lasnik 1986, Marantz 1993, Pytkäinen 2008).

⁷⁸Haitian Creole possesses multiple anaphoric elements, with both the complex *tet-li* ‘HEAD-3SG’ and the simplex 3SG pronoun *li* able to function as anaphors (Veenstra & Muysken 2017: 45).

‘I took each book and hit its writer with it’

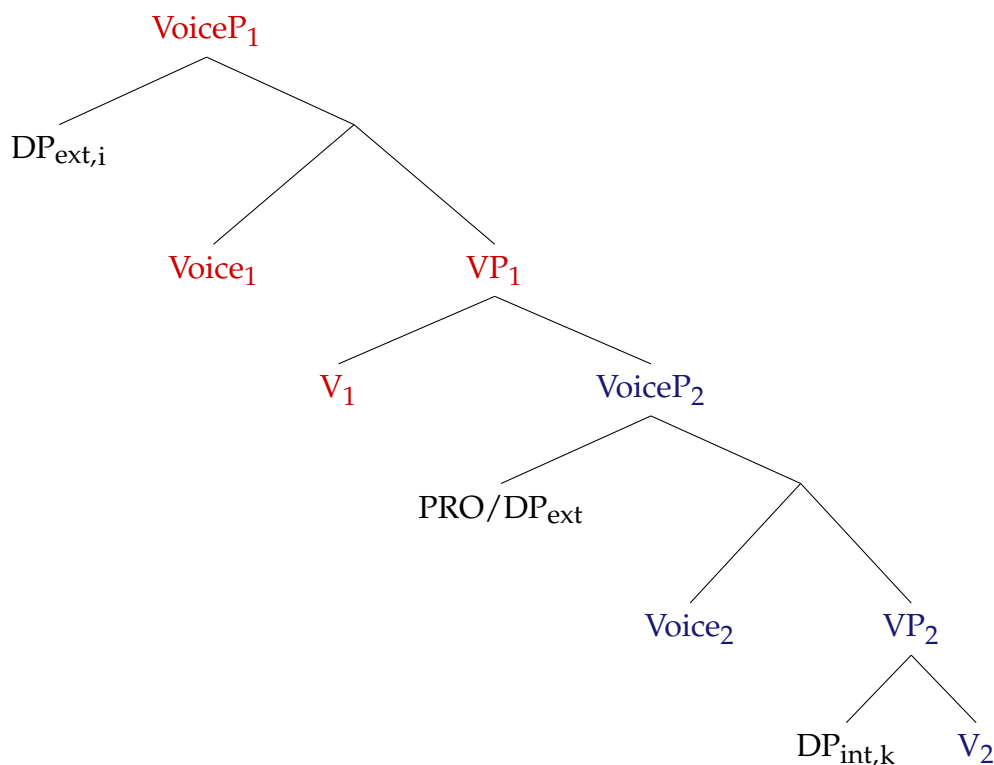
(Veenstra & Muysken 2017: 22, 23)

Aboh (2009) proposes an account that comes conceptually close to a raising applicative analysis. The instrument or comitative argument is introduced lower, either by an overt adposition or a silent Appl head, and subsequently raises to a position that linearly follows the ‘take’-type V_1 . Simplifying somewhat, V_1 itself is argued to be a light verb lacking a thematic import, serving a purely morpho-syntactic purpose.

I will not pursue in anymore detail the differences between thematic and raising verbal applicatives. What is important for our purposes is that both involve complementation, whereby an applicative verb takes a lexical VP as its complement without syntactic argument sharing.

4.3.2 VoiceP-level complementation

In NC-SVCs, we are left with the structural level of VoiceP to cross-cut with complementation. This combination yields the general configuration in Tree 4.20, illustrated here with an embedded transitive VoiceP:



Tree 4.20: VoiceP-level serialization via complementation

Even though the structure in Tree 4.20 meets the necessary conditions on serialization, it turns out that many authors exclude it from serialization. Haspelmath (2016), for example, explicitly defines SVCs as constructions in which there is no relationship of complementation between constituent verbs. His choice is partly arbitrary, but

also motivated by the simple fact that historically, complement relations have normally been treated as distinct from serialization. Others do take (certain kinds of) complementation to fall under serialization, such as Aikhenvald (2018).⁷⁹

Structures like in Tree 4.20 have perhaps most famously been proposed for analytical causatives, as in Pylkkänen (2008) and Guasti (2017). These works investigate non-serializing languages, though explicit links between analytic causatives from non-serializing languages and serialization have been drawn in e.g. Kroeger (2004).⁸⁰

For the sake of illustration, we can assume that the causative NC-SVC from Ambae in (47), with a causative V_1 and a lexical V_2 , has a structure along the lines of Tree 4.20 – an analysis which parallels the one proposed for Italian *fare* causatives by (Guasti 2017):

(47) **Ambae:** *Causative NC-SVC*

Mo [vai] ngure dolegi ra=mo [inu=e]
 REAL make 3NSG all 3NSG.A=REALIS drink=3SG.O

‘He made all of them drink it’ (?: 303)

As with applicative NC-SVCs involving VP-level complementation, we do not expect VoiceP-level complementation configurations per se to give rise to restrictions on transitivity. Even though e.g. causatives may restrict the transitivity of their complement, this is correct to my knowledge, i.e. there is nothing intrinsic to structures like Tree 4.20 that restricts transitivity.

4.3.3 Transitivity restrictions

In the case of adjoined NC-SVCs, transitivity restrictions may arise depending on the level of serialization, the generalization being that syntactic argument sharing requires a one-to-one mapping between the highest arguments of the main and the adjoined VerbP. With complementation, all else being equal, no such restrictions are predicted to arise because integration occurs via selection.

That said, at least with regard to the instrumental NC-SVCs I am aware of, these always involve a transitive lexical VP containing an object:

(48) *Instrumental NC-SVCs*

a. **Gungbe:** ‘take’

⁷⁹This view does come with its problems, such as the fact that it includes countless cases that would unanimously be considered as standard non-finite control. It seems that for Aikhenvald, the overall profile of a language also factors into deciding whether a given construction is a SVC.

⁸⁰The distinction between serializing and non-serializing languages seems to be another reason for why some constructions have traditionally been labeled SVCs to the exclusion of causatives. Serializing languages often have poor inflection and lack productive morphological valence-decreasing operations, a fact which constitutes a potential bias upon which classifications of (non-)SVCs are formed.

Sétù [zé] kpò ló [xò] Kòjó

S. take stick DET hit K.

‘Setu took the stick hit Kojo’

i.e. ‘Setu hit Kojo with the stick’

(Aboh 2009: 16)

b. **Mandarin:** ‘use’

Ta [yòng] zhèbâ yaòshi [kai-de] mén

3SG use key this open-DE door

‘He opened the door with his key’

(Déchaine 1993: 253)

c. **Numbami:** ‘hold’

[Na.ambi] wosa [na.wosa] wanga

1SG.hold oar 1SG.IRREAL.paddle canoe

‘I will paddle the canoe with the oar’

(Bradshaw 1993: 148)

Of course, it is entirely possible that this is simply an accidental gap in the data. Assuming it is not, I briefly consider some potential explanations.

On the one hand, that instrumental NC-SVCs should involve multiple objects is unsurprising because event-structurally, the presence of an instrument entails an agent using this instrument, typically in order to act on a patient (Jackendoff 1990, Croft 1991, Koenig et al. 2007).⁸¹

On the other hand, it is not clear how this event-structural restriction would translate to a grammatically reified restriction which would ensure only transitive VPs to be combinable with ‘take’-type verbs. It is also unclear how the following cases would be handled, which involve instruments but intransitive verbs:

(49) *Instruments with intransitive verbs*

a. Dennis walked with a cane

b. Maddie breathed with a ventilator

(Owens 2011: 176)

Barbara Stiebels (p.c.) suggests that the role of the instruments in (49) differs substantially from those in (48): roughly, the transitive action (including its result) can be seen as a causative-like consequence of the use of the instrument in (48); in (49), by contrast, the use of the instrument is more of an elaboration of the overall intransitive action.⁸²

Alternatively, it is possible that the instrumental verb in instrumental NC-SVCs is actually a low applicative. In contrast to high applicatives, low applicatives are higher-order predicates that attach below the verb, and they do not express a relation

⁸¹This contrasts with the modal analysis in Rissman & Rawlins (2017), who take instrumental events to be characterized by a dyadic relationship between an agent and an instrument. This is reflected in their denotations for *use* and *with*, which do not make reference to patients.

⁸²This idea is consistent with the conclusions reached in Koenig et al. (2007: 213), who distinguish five types of relationships between instruments and patients, ranked according to the nature of the causal force involved. These range from cases where instruments are true causes, to cases where they are construed as merely helping in achieving an overall better performance. In thematic-role parlance, we can think of the participants in (48) as “true” instruments, and those in (49) as mere “aids”.

between an individual and an event, but a kind of transfer-of-possession relation between two individuals, namely a patient and a recipient / source (Pylkkänen 2008: 18, 75). In other words, low applicatives require transitive verbs as their input.⁸³

A final, grammar-theoretical explanation may be found in the fact that serializing languages (i) tend to have poor morphological Case-licensing means, resorting instead to analytic strategies such as verbs and adpositions in order to license non-core arguments; and, correlated with this, (ii) tend to have few to no ditransitive verbs; see e.g. Heine & König (2010) and Aikhenvald (2018: 189). This is basically a restatement of the applicative NC-SVC analysis given above, from a purely licensing perspective.

Similar facts and arguments apply to benefactive NC-SVCs. I leave this issue to future work.

4.4 Summary

This chapter presented the analyses for NC-SVCs by cross-cutting adjunction and complementation with the structural levels of VP and VoiceP. By using adjunction, we derived Type A and B transitivity profiles; by switching to complementation, we explained an apparent exception to the robust linking effect observed in NC-SVCs derived via adjunction. These results are summarized below in Table 5.1:

⁸³Even though Pylkkänen's original work rules out low instrumental applicatives, Jerro (2020) elaborates on her typology by extending the range of the first individual argument of the low-applicative relation from merely recipients / sources to all non-core roles.

VoiceP	⇒ Type A (no restrictions)	
VP	⇒ Type B ($\geq[trans + trans]$)	
	Adjunction	Complementmentation
Level of structure-building	Mode of structure-building	

Table 4.1: Typology of NC-SVCs via the mode and level of structure-building

Chapter 5

Contiguous serial verb constructions

5.1 Introduction

This chapter turns to the analysis of C-SVCs. Section 5.2 establishes that the kinds of silent arguments found in C-SVCs are implicit arguments. In Section 5.3, I propose the general mechanism by which C-SVCs built via adjunction are integrated in contrast to NC-SVC, which helps explain that C-SVCs lack the robust linking effect found in NC-SVCs. This section also introduces a basic event structural semantics that will be worked with from here on, so that we can make reference to internal arguments that remain unlinked but grammatically active. Next, Section 5.4 discusses the Morphological Non-head Constraint, which entails that non-head verbs cannot project in V-level configurations – a property crucial to C-SVCs. While derivations of C-SVCs involving adjunction are given in Section 5.5, derivations cannot be given for C-SVCs involving complementation on grounds that I have not come across any such clear cases, as touched on in 5.6. Section 5.8 takes up various grammar-theoretical issues, and Section 5.9 summarizes the findings of transitivity restrictions not only with respect to C-SVCs, but also NC-SVCs.

5.2 C-SVCs involve implicit arguments

Unlike NC-SVCs, whose silent arguments – if present – project syntactically as null arguments (modeled here as minimal pronouns), C-SVCs involve implicit arguments, which remain unlinked but accessible for the purposes of argument sharing. Let us run through the evidence for this.

First, there is the simple fact that C-SVCs are strictly contiguous, i.e. do not permit any non-derivational or non-inflectional material to occur between verbs, only ever outside of the verbal complex. The material that cannot occur internal to the

verbal complex hence subsumes any kind of argument (apart from the occasional incorporate or clitic).

For Daakaka, Hopperdietzel (2020) gives an explicit example showing that it is impossible for the internal argument of a non-final transitive V_1 in C-SVCs to be realized overtly directly after V_1 :

(1) **Daakaka:** *trans + trans*

- a. Bong ma [ta] [sengave] seli
B. REAL cut open.TRANS road
'Bong opened/cleared the road by cutting [the grass]'
 - b. *Bong ma [ta] **pwesye** [sengave] seli
B. REAL cut **branch** open.TRANS road
intended: 'Bong opened / cleared the road by cutting (the) branches'
- (Hopperdietzel 2020: 304)

Second, recall that a certain licit transitivity and linking profile is connected to contiguity: C-SVCs (i) often disallow [*trans + intr*] pairings, and (ii) they always show a many-to-one mapping from the number of verbs to argument projection lines – i.e. only one verb can project.

A case in point was Tlachihilco Tepehua, which shows a 3/4 pattern in terms of licit transitivity pairings, where only the [*trans + intr*] pairing is ungrammatical (2); in order to get around this restriction, either V_1 must be made intransitive (2b-i) or V_2 transitive (2b-ii):

(2) **Tlachihilco Tepehua**

- a. *trans + intr*
*[tʃaqʃ]-[ʔan]
chop-go.IPFV
intended: '(S)he goes chopping'
 - b. (i) *intr + intr*
[tʃ'aqʃ-**nah**]-[ʔan]
chop-ANTIP-go.IPFV
'(S)he goes chopping'
 - (ii) *trans + trans*
[tʃaqʃ]-[ɬa:-ʔan]
chop-APPL-go.IPFV
'(S)he goes chopping it'
- (Watters 2017: 289f)

As we will see shortly, these two facts are intertwined: first, given V-level configurations, only one verb – the syntactic head – can project its internal argument as object, but whenever this head verb does not itself project an object, it cannot satisfy the IAR of the non-head, transitive co-verb. Second, the string-final verb is uniformly the syntactic head in the C-SVCs discussed here. Jointly, these facts reveal that the non-head transitive verb's internal argument remains argument-structurally active

with potential syntactic effects, yet crucially unprojected.⁸⁴

5.3 Contiguous serialization via adjunction

Let us now turn to the issue of how constituent verbs in C-SVCs are integrated syntactically. For NC-SVCs involving adjunction, we used Baker & Stewart's theory of serialization via predication, which forces dependent VerbPs in NC-SVCs to contain null arguments in order to be integrated. The fact that we have just established the presence of implicit arguments in C-SVCs, however, suggests that C-SVC should involve a fundamentally distinct mechanism.

Another fact pointing to this conclusion is that – unlike VerbP-level serialization – V-level serialization does not require constituent verbs to be associated with argument positions that can be stitched together. This is brought out clearly in well-formed pairings of e.g. an unergative V_1 and a transitive V_2 , as in (7) from Daakaka:

(3) **Daakaka:** *unerg + trans*

Bong ma [tas] [tiwiye] etastas
B. REAL sit break bench

'Bong broke the bench by sitting (down)' (Hopperdietzel 2020: 302)

Because the unergative V_1 *tas* 'sit' has no IAR, but the transitive V_2 *tiwiye* 'break' does, these verbs have the denotations in (4):⁸⁵

- (4) a. $\llbracket tas \rrbracket = \lambda e_1. [\text{sit}(e_1)]$
b. $\llbracket tiwiye \rrbracket = \lambda y \lambda e_2. [\text{break}(e_2, y)]$

This means that at the point of serialization, where V_1 is adjoined to V_2 , only the V_2 *tiwiye* 'break' (4b) has an open, unlinked individual argument (4b), but not the V_1 *tas* 'sit'. As the derivation goes on, no relationship of internal argument sharing ever comes to hold between V_1 and V_2 – or any relationship of individual argument sharing at all, for that matter. Yet (4) is well-formed, unlike its non-contiguous counterpart from e.g. Edo, repeated here in (5):⁸⁶

(5) **Edo:** *unerg + trans*

⁸⁴In light of the fact that cross-linguistically, motion verbs sometimes give rise to restructuring, note that the fact the motion V_2 must change its transitivity depending on the V_1 in (2b) already shows us that complementation cannot be involved. If complementation were actually involved in, the V_2 should remain invariant. In fact, V-level complementation may turn out to be absent in the world's languages, cf. Section 5.6.

⁸⁵See von Prince (2015: 50) and Hopperdietzel (2020: 271, 277ff, 297, 302) for differences between unaccusative and unergative verbs in Daakaka.

⁸⁶Recall that the reverse pairing, *trans + unerg*, is also impossible in Edo.

*Ékítà gié!gié [gbó!ó] [khú] áhiánmwèn
 dog quickly bark chase bird

intended: ‘The dog quickly barked and chased a bird’ (Baker & Stewart 2002: 52)

Where the ungrammaticality of (5) is explained in terms of Baker & Stewart’s theory by the lack of a minimal pronoun object in the unergative VP₁, such that argument sharing with VP₂ cannot proceed and integration fails, the grammaticality of (7) is surprising. It would seem, then, that serialization via adjunction at the levels of V and VP are fundamentally distinct.

I propose, however, that serialization via adjunction at the levels of V and VP follow the same principles, except that integration at the V-level does not necessarily proceed through argument sharing. Recall how Baker & Stewart’s account relies on Full Interpretation (Chomsky 1986) in order to derive the fact that adjoined VerbPs must contain open argument positions: adjoined phrases are not selected, and they cannot give rise to complex heads through head movement; consequently, they must be integrated via predication, which entails that they must have an open argument position (modeled here as a minimal pronoun that moves to the edge of its VerbP).

It follows from this that if adjunction occurs at the V-level, integration goes through because adjunction yields a base-generated complex head. That is, integration comes for free at the V-level, without the need for syntactic argument sharing or head movement, because adjunction automatically yields a well-formed complex predicate. There are hence two routes to complex heads in the syntax: VerbP-level complementation plus head movement, or simply V-level adjunction.⁸⁷

We thus have an explanation for why C-SVCs like (7) are well-formed in spite of the fact that at the point of serialization, no individual arguments are shared. This way, we can accommodate V-level serialization alongside VerbP-level serialization via adjunction, with differences between the two following from Full Interpretation on the one hand, and the level of structure-building level on the other.

5.3.1 Event Cumulation

I will now introduce a basic event-structural semantics for C-SVCs. This move is necessary given that the core effects we are interested in with respect to transitivity restrictions in C-SVCs – though conditioned by syntactic configurations and principles of integration – will play out non-syntactically at the level of a verb’s lexical argument structure. Where the internal arguments we were concerned with in NC-SVCs with object sharing were always projected (some overtly, others not), those in C-SVCs with object sharing will now take the form of either projected or

⁸⁷This effectively recapitulates the idea of *LF head movement*, which states that with phrasal constituents, covert verb incorporation can feed theta-role assignment in order to give rise to well-formed complex predicates (Saito & Hoshi 2000, Saito 2001).

non-projected arguments. We therefore need a level of representation that makes visible not only the relationships between projected and non-projected arguments, but also how morpho-syntactic factors such as headedness and argument linking factor in.

As discussed in Stewart (1998), Zimmermann & Amaechi (2018) and Zimmermann & Amaechi (2020), SVCs cross-linguistically come with various semantic interpretations. Of the various kinds of event composition discussed in these works, I will handle the C-SVCs investigated here in terms of *Event Cumulation* (recall that material belonging to head projections is red, while material belonging to adjoined projections is blue):

- (6) *Event Cumulation:* (Zimmermann & Amaechi 2018: 6)
 $\exists E \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ P(e_1) \ \& \ Q(e_2)]$

For a two-verb C-SVC, Event Cumulation expresses a *macro-event* E consisting of two, symmetrically related *sub-events* e_1 and e_2 . The cumulation, or unification, of the sub-events e_1 and e_2 into the macro-event E derives the fact that, for the purposes of inflection and operators like TAM and negation, C-SVCs behave as single verbs. Any operators will necessarily scope over the complex macro-event E , but not be able to scope into E and modify any of its sub-events e_1 or e_2 individually. The sum operator $\oplus$ is Boolean in the sense that event composition yields a complex macro-event consisting of independent sub-events. Event Cumulation applies to a heterogeneous set of atomic events, and by itself does not impose any constraints on causal, temporal or spatial cohesion. This is welcome for our purposes because it means that the use of Event Cumulation is vague with respect to any more fine-grained semantic components a given construction may turn out to have upon closer inspection. Also, note that Event Cumulation is not restricted to V-level configurations – as discussed in Zimmermann & Amaechi (2018), the internal event structure of an SVC is, to a certain extent, independent of its internal syntax.

In the case of C-SVCs consisting of multiple transitive verbs with object sharing, I assume that the internal argument slot of any non-head transitive verb is co-indexed with the internal argument slot of the head transitive verb, i.e. the only verb that can project its internal argument. While I take the operation of co-indexation to be achieved by the rule of Event Cumulation, I take the need for co-indexation itself to follow from a transitive verb's IAR, regardless of whether it is a head or non-head in a given configuration, rather than from a property specific to C-SVCs. That is, all transitive verbs have an IAR, but only those that project can have it saturated via Function Application, while non-projecting ones must have it satisfied differently via co-indexation. When co-indexation fails due to sortal distinctness, existential binding may apply as a kind of Last Resort (depending on the language and construction).

For a C-SVC consisting of two unergatives like (7a), Event Cumulation thus gives us the denotation in (7b) with the paraphrase in (8b):

- (7) a. **Tlachihilco Tepehua:** *intr + intr*
 [talaqsta]-[laka:wa:snan]
 turn.face.up-look.IPFV
 '(S)he looks, facing upward' (Watters 2017: 288)
- b. $\exists E \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ \text{turn face up}(e_1) \ \& \ \text{look}(e_2) \ \& \ \text{AGENT}(E) = (s)\text{he}]$
- c. 'There is a complex macro-event E comprising the two sub-events e_1 and e_2 , where e_1 is an event of turning one's face up, and e_2 is a looking event, and the Agent of E is him / her'

Because there is no object sharing in (7a), there is no co-indexation in (7b). This contrasts with the C-SVC in (8), made up of two transitives, where the interpretation of object sharing results from co-indexing V_1 's and V_2 's internal arguments (8b):

- (8) **Tlachihilco Tepehua:** *trans + trans*
- a. [ʔap'i]-[ʔu-y]
 peel-eat-IPFV
 '(S)he eats it, peeling it' (Watters 2017: 288)
- b. $\exists E \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ \text{peel}(e_1, x_i) \ \& \ \text{eat}(e_2, y_i) \ \& \ \text{AGENT}(E) = (s)\text{he}]$
 $\begin{array}{c} \circ \quad \quad \quad \circ \\ \text{-----} \\ \text{Co-indexation} \end{array}$
- c. 'There is a complex macro-event E comprising the two sub-events e_1 and e_2 , with e_1 consisting in peeling and e_2 in eating it, and the Patient of e_1 is identical to the Patient of e_2 , and the Agent of E is him / her'

In (9) there is again no co-indexation because V_1 is an unergative and V_2 transitive, i.e. only the latter has an internal argument:

- (9) **Tlachihilco Tepehua:** *intr + trans*
- a. [tapasa]-[ʔi:-ɬ]
 pass.by-get-PFV
 '(S)h got it passing by' (Watters 2017: 288)
- b. $\exists E \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ \text{pass by}(e_1) \ \& \ \text{get}(e_2, \text{it}) \ \& \ \text{AGENT}(E) = (s)\text{he}]$
- c. 'There is a complex macro-event E comprising the two sub-events e_1 and e_2 , with e_1 consisting in passing by and e_2 in getting it, and the Agent of E is him / her'

One of various alternative construals to Event Cumulation is Predicate Modification, which expresses the existence of a single event with multiple event predicates:

- (10) *Predicate Modification:*
 $\exists e [P(e) \ \& \ Q(e)]$ (Zimmermann & Amaechi 2018: 6)

In (10), the predicates P and Q predicate of one and the same event e – a property which makes it impossible to distinguish arguments as functions of distinct sub-

events. To see why this is important for our purposes, recall that some languages have object-sharing C-SVCs whose transitive verbs characterize sortally distinct internal arguments. The Saliba C-SVC in (11), for example, has ‘pour’ as V_1 and ‘fill’ as V_2 , which sortally characterize a thing that is poured and a thing that is filled, respectively; yet it is only the latter (the cup corresponding to the thing filled) that is projected as object, while the former (the thing poured) is blocked from realization:

(11) **Saliba:** *trans + trans*

Kaputi ku-[ini]-[he-mwayau]- $\emptyset$
cup 2SG.S-pour-CAUS-fill-3SG.O

‘Fill the cup’

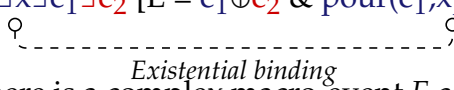
(Margetts 2005: 81)

PATIENT(pour) = liquid $\neq$ the cup = PATIENT(fill)

A Predicate Modification analysis of (11) is given in (12a) (which ignores the imperative mood of (11)). Notice how (12a) cannot accommodate the non-identity of the Patient roles of the respective pouring and filling events. Because there is only one event argument shared by two event predicates, a Predicate Modification construal entails the contradictory statement in (12b), which says that the Patients of the pouring and filling events are identical:

- (12) a. $\exists x \exists e [\text{pour}(e, x) \ \& \ \text{fill}(e, \text{cup}) \ \& \ \text{AGENT}(e) = \text{you}]$
b. $\text{PATIENT}(\text{pour}) = \text{PATIENT}(\text{fill})$ $\nexists$

Now compare the alternative in (13a) with Event Cumulation. Because (13a) has two event arguments that can be made reference to, i.e. P and Q are not neutralized with respect to the identity of their event argument but instead retain a distinction in terms of e_1 and e_2 , we can integrate sortally distinct Patient roles under a single macro-event as functions of distinct sub-events without running into a contradiction (13c):

- (13) a. $\exists E \exists x \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ \text{pour}(e_1, x_k) \ \& \ \text{fill}(e_2, \text{cup}_i) \ \& \ \text{AGENT}(E) = \text{you}]$


$$\begin{array}{c} \text{ } \\ \text{ } \end{array}$$

Existential binding

b. ‘There is a complex macro-event E comprising the two sub-events e_1 and e_2 , with e_1 consisting in pouring (something) and e_2 in filling the cup, and the Agent of E is the addressee’

c. $\text{PATIENT}(\text{pour}) \neq \text{PATIENT}(\text{fill})$

At a more abstract level, the difference between Event Cumulation and Predicate Modification is minimal: at their cores, both construals express eventive conjunction. The difference between the two lies in whether or not the event predicates denoted by the constituent verbs end up predicating over the same event argument. This is not the case in Predicate Modification, which collapses these into one and the

same predicate without introducing a new event argument, i.e. they end up co-predicating over a single event. This neutralizes the identities of prior individual sub-events, such that sortally distinct internal arguments become impossible to distinguish. By contrast, Event Cumulation does not yield such a fusion, but instead brings the individual sub-events together under a new, higher event argument, i.e. the macro-event E . Because the individual sub-events retain their identities, sortally distinct internal arguments are automatically distinguished as a function of distinct sub-events – a desirable result in light of cases like (11).

C-SVCs will here be derived by the catagorematic rule of *generalized Event Cumulation*, henceforth *EC*. EC takes a predicate and extends its predicate structure by an additional predicate variable, thereby requiring it to combine with another predicate. EC ends up yielding a complex predicate which, internally, expresses a symmetric cumulation of the two sub-events denoted by these two predicates under a macro-event E ; externally, in terms of the individual and event argument slots in need of satisfaction, it behaves just like the predicate corresponding to the syntactic head:

$$(14) \quad \text{Generalized EC:}$$

$$\lambda Q \lambda P \{ \lambda y \} \lambda E. \underbrace{\{ \exists x \} \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ P(e_1, \{x_i/k\}) \ \& \ Q(e_2, \{y_i\})]}_{\text{Complex Predicate}}$$

The predicate whose predicate structure is extended (Q) is the first input fed into EC, while the predicate that fills the added predicate slot (P) is the second input fed into EC. Note that the first predicate corresponds to the syntactic head verb, while the second one corresponds to the adjunct. EC thus instantiates a kind of head extension, whereby the head accommodates to the adjoined verb, rather than vice versa. Similar head-extending operations on verbs have been proposed and discussed by e.g. (Stiebels 1996) for verbal particles and prefixes, Wunderlich (1997), van Geenhoven (1998) for noun incorporation and Chung & Ladusaw (2004) for pseudo noun incorporation.

The curly braces around certain elements in (14) indicate that these elements are not always present in the complex predicate yielded by EC; the presence of these elements depends on the transitivity of the main and the adjoined verb, and whether or not – if both verbs are transitive – their internal arguments are sortally identical and hence shared, or sortally distinct and hence not shared.

The denotation of the complex predicative cores underlying C-SVCs derived by EC are given in the following. (13) gives the result of pairing two unergatives. C-SVCs made up of two transitives characterize two sub-types, depending on whether the objects of the constituent transitive verbs are shared or not (13b): in object sharing, EC co-indexes the internal arguments of P and Q , and yields a complex predicate requiring only the internal argument of Q as its next input (13b-i); in non-object sharing, EC also yields a complex predicate requiring the internal argument of Q as

its next input, but instead of co-indexing the internal argument of P with that of Q , it existentially binds P 's internal argument (13b-ii). Finally, (15c) shows the pairing of an unergative V_1 and a transitive V_2 :

- (15) a. *unerg + unerg*:

$$\underbrace{\lambda E. \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ P(e_1) \ \& \ Q(e_2)]}_{\text{Complex Predicate: } \langle v, t \rangle}$$
- b. *trans + trans*:
- (i) *Object sharing*

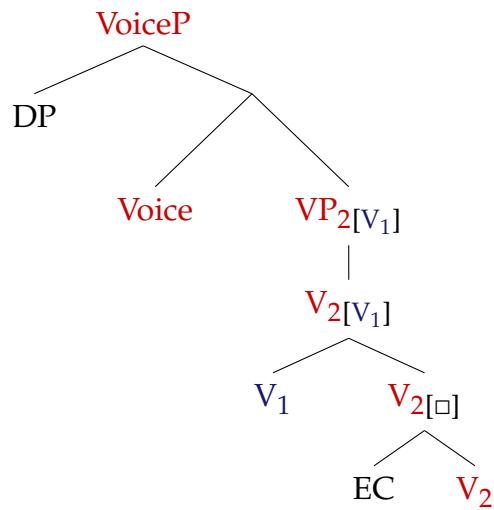
$$\underbrace{\lambda y \lambda E. \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ P(e_1, x_i) \ \& \ Q(e_2, y_i)]}_{\text{Complex Predicate: } \langle e, vt \rangle}$$
- (ii) *No object sharing (existential binding)*

$$\underbrace{\lambda y \lambda E. \exists x \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ P(e_1, x_k) \ \& \ Q(e_2, y_i)]}_{\text{Complex Predicate: } \langle e, vt \rangle}$$
- c. *unerg + trans*:

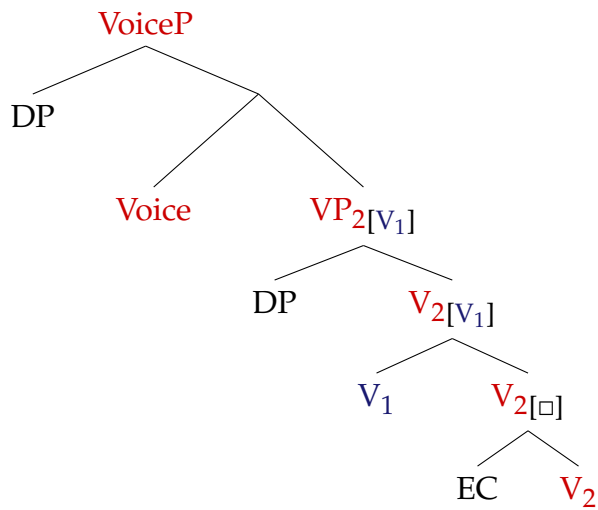
$$\underbrace{\lambda y \lambda E. \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ P(e_1) \ \& \ Q(e_2, y)]}_{\text{Complex Predicate: } \langle e, vt \rangle}$$

Notice that the semantic type of a complex predicate derived by EC always corresponds to the semantic type of the predicate associated with the syntactic head, i.e. it is always the syntactic head verb that projects its argument structure into the syntax.

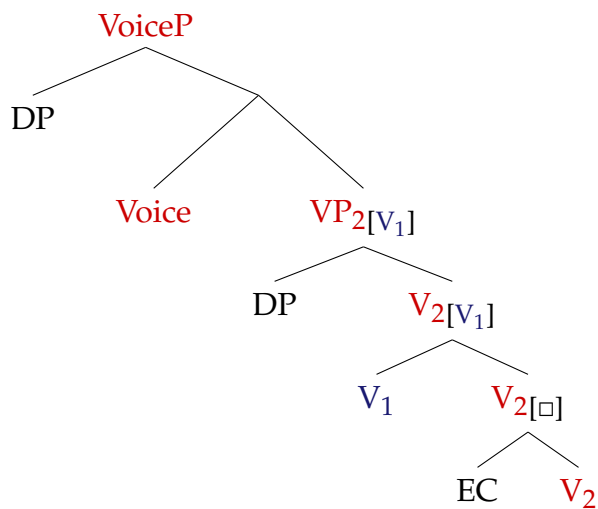
The basic tree structures associated with these denotations are given in Trees 7.3, 7.4 and 5.3, respectively (even though EC is a categorematic rule, not a covert syntactic head, it will be woven into derivations throughout for compositional clarity). I adopt the following notation, with linear order indicated by numerical indices: " $V_{2[\square]}$ " denotes a head V_2 that has undergone head extension via EC, and which therefore must combine with an adjunct verb; " $V_{2[V_2]}$ " denotes a head V_2 that has combined with an adjunct V_1 ; step-by-step derivations in which syntax and semantics go hand in hand are provided in the next section.



Tree 5.1: *unerg + unerg*



Tree 5.2: *trans + trans*



Tree 5.3: *unerg + trans*

Next to giving us the desired complex event and argument structure for C-SVCs, EC needs to fulfill two other things: first, it must be compatible with head-finality; and

second, we need to derive a complex predicate that is mapped onto the format of a single verb, so that the cluster can morpho-syntactically be dealt with like a single verb. Let us see how these requirements are managed by EC.

First, the headedness of the string-final verb follows from the simple fact that whichever verb is fed first into EC is necessarily the syntactic head to which further, non-head verbs are subsequently attached. This is part of what it means for an operation to instantiate head extension. Any non-first verb fed into EC consequently becomes a non-head, as per the standard view that adjuncts are added to main projections after the main projection has entered the derivation. This gives us exactly the configuration we need, wherein any non-final verbs preceding the head V_H are adjuncts to V_H .

Crucially, the head V_H does not inherently require merger with another verb, a property consistent with the fact adjunction does not involve selection. The necessity of the head V_H to subsequently merge with another verb is a derived property that follows from the application of EC; there is generally nothing intrinsic to the verbs making up C-SVCs that requires them to serialize. Because this requirement is introduced by the rule EC, we correctly capture the fact that the verbs making up C-SVCs can equally occur as independent verbs. If we choose to iterate EC by applying it to its own output, V_H will remain the syntactic head given that it has been established as the syntactic head by the first application of EC.

Second, complex event- and argument-structure formation follows from the cumulation of the sub-events e_1 and e_2 , associated respectively with P and Q . This is achieved by first symmetrically relating them via the sum operator $\oplus$, then equating their cumulation with the macro-event E – itself introduced by EC – and lastly existentially quantifying over them.⁸⁸ The symmetry of their cumulation reflects the fact that the basic complex event structure of C-SVCs is conjunctive, in consistency with their syntactic lives as adjunction configurations (Baker & Harvey 2010, Svenonius 2016). The cumulation of the sub-events e_1 and e_2 into a macro-event E and their subsequent existential binding ensures that only the highest λ -bound eventuality argument, i.e. the macro-event E , will be accessible to subsequent operators such as TAM and negation, which in turn derives the single-verb inflectional behavior of C-SVCs. Alternatively, the resulting complex predicate can enter into another cycle of EC, in which case the λ -bound macro-event E will likewise be the only eventuality argument accessible to EC.

To be clear: I am not claiming that all the C-SVCs discussed here involve Event Cumulation – this would require testing with more fine-grained diagnostics. Rather, I am using Event Cumulation here to investigate the basic event- and argument-structural properties of C-SVCs because it is both fine-grained enough to distinguish

⁸⁸While these steps are left implicit in (13) and the following, I do take them to actually occur in this particular order. The sub-events e_1 and e_2 must remain open until after they have been cumulated; if they were to be closed earlier, they would be inaccessible to cumulation.

internal arguments as a function of distinct sub-events, but also relatively compatible with any additional event-structural relationships and properties that may turn out to be involved, depending on the language and construction. It is therefore as fine-grained as we need the semantics of C-SVCs to be, but not more so than we can know it to be.

5.4 The Morphological Non-head Constraint

We now turn to a general constraint on projection that arises in base-generated V-level complexes, namely the inability of any non-head transitive verb to project structural arguments. This reductionist move will allow us to explain the recurring ban against [*trans* + *intr*] pairings in C-SVCs without appealing to any construction-specific property.

Taking stock of a range of observations made in preceding literature, McIntyre (2014) proposes a solution to the problem of why some cases of English *-er* nominalizations block the phrasal realization of the non-head verb's arguments (16a), while others do not (16b); in (16c), the understood thematic object of 'tune' surfaces as a non-phrasal, i.e. structurally reduced nominal in pre-verbal position, such that its status as a true argument is questionable:

- (16) *-er nominalizations*
- a. *non-eventive -er*: ✗ *phrasal internal argument*
The tuner (*of guitars) in the drawer
 - b. *eventive -er*: ✓ *phrasal internal argument*
The tuner of the guitar was Mary
 - c. *non-eventive -er*: ✓ *non-phrasal internal argument*
The guitar tuner in the drawer
- (McIntyre 2014: 121)

McIntyre's proposal builds on the insight that the behavior of those *-er* nominalizations which block phrasal argument linking – i.e. non-eventive *-er* nominalizations – parallels that of English *Ving-N* compounds, where the non-head verb is equally unable to realize its arguments:

- (17) *[V-ing + N] compounds*
- a. washing machine (*of clothes)
 - b. washer (*of clothes)
 - c. kneeling cushion (*of/by worshipers)
 - d. kneeler (*of/by worshipers)
- (McIntyre 2014: 127)

This leads to the descriptive generalization in (18), which is schematized in Tree 5.4:

- (18) *The Morphological Non-head Constraint (MNC):*

In a base-generated complex head [_Y X Y], X's arguments cannot be realized outside [_Y X Y], unless Y realizes an argument of X or is otherwise sensitive to X's argument structure. (McIntyre 2014: 130)

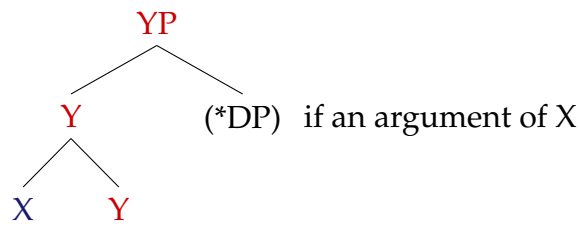


Figure 5.4: MNC configurations

The MNC has its precursors in works on compounding like Di Sciullo & Williams (1988: 30) and Ackema & Neeleman (2004: 26). It entails that "there is no generally available mechanism of percolation which allows arguments of non-heads to become arguments of the whole complex head, or otherwise to be realized at a distance from the non-head" (McIntyre 2014: 130). In order to capture cases like (16b), where the internal argument of the non-head verb can indeed be expressed phrasally via nominal linking – in apparent violation of the MNC – McIntyre proposes that only if *-er* is *sensitive to argument structure* can it realize an argument of its verbal complement. This sensitivity is in turn argued to be connected to eventiveness: the use of *-er* is eventive in (16b) but non-eventive in (16a), in that (16b) entails an actual, episodic tuning event whose Agent is Mary and Patient the guitar, whereas (16a) involves a noun denoting an entity with a function or disposition to tune in generic, non-episodic situations; hence, only eventive (16b) is able to realize the internal argument of *tune*.⁸⁹

The MNC establishes an important parallel between certain kinds of nominalizations and certain kinds of compounds. To these, I propose to add a subset of C-SVCs, namely those discussed here with non-head transitive verbs. *-er* nominalizations, *Ving*-N compounds and C-SVCs with a transitive non-head verb all share the property of being base-generated complexes containing a non-head transitive verb that cannot have its IAR fulfilled by projection. Except in the case of *Ving*-N compounds, this non-head verb's IA appears to be able to be realized indirectly by the head element under principled conditions, via what can informally be characterized as a kind of argument inheritance. That the non-head verb cannot realize its internal argument must be due to its status as a non-head terminal node in immediate configuration with another terminal node, as opposed to its status as a complement (*-er* nominals) or an adjunct (verb-noun compounds, C-SVCs), because this restriction follows neither from complementhood or adjuncthood per se. This is reflected in the MNC in (18), which does not make reference to how the non-head verb is integrated into a complex, merely to its status as the non-head within the complex.

⁸⁹For more on the properties of *-er*, see Alexiadou & Schäfer (2010) and Roy & Soare (2014).

Given that there is nothing about adjunction or complementation in and of themselves that would enforce such an effect, it follows that the inability of non-head transitive verbs in non-eventive *-er* nominalizations, *Ving*-N compounds and C-SVCs to syntactically realize their internal argument results from the very fact that they are non-heads at the V-level.

Specifically with respect to C-SVCs, this restriction is termed the *Atransitivity Generalization* by Nishiyama & Ogawa (2014), given in (19):

- (19) *Atransitivity Generalization* (Nishiyama & Ogawa 2014: 76)
When a verb is adjoined to another head, it cannot realize its internal argument.

In their analysis of Japanese C-SVCs whose non-head verbs are similarly unable to project their internal arguments, Nishiyama and Ogawa assume a full-blown neo-Davidsonian architecture in which not only external, but also internal arguments are severed from the verb and introduced by a functional head F (Jelinek 1998, Borer 2005). They go on to declare (19) as a consequence of this architectural property: because the adjoined verb is not selected by F, only the head verb selected by F can have its internal argument projected in Spec,FP (Nishiyama & Ogawa 2014: 76).

I believe that we can do without this additional assumption by simply invoking the MNC (18), which itself follows naturally from X-level configurations.⁹⁰

5.4.1 Co-indexation vs existential binding of internal arguments

Given the MNC, the IAR of a non-head transitive verb in a C-SVC must be satisfied via a strategy other than Function Application. I propose that there are two ways of doing so: co-indexation with the internal argument of the head transitive verb (object sharing), and existential binding (non-object sharing due to sortal distinctness).

Consider again (20a) from Saliba, where the single syntactic object is understood as undergoing both the hitting and the breaking. The single syntactic object of (20b), by contrast, is not understood as undergoing both the pouring and the filling; rather, these two participants are sortally distinct, and the cup specifies only the internal argument of the $V_{2=H}$ 'fill', while the internal argument of the non-head V_1 'pour' remains implicit:

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There are principled exceptions to the blocking effect of the MNC, such as certain complex NPs headed by N-N compounds. In e.g. *Härtegrad des Wassers* 'degree of hardness of water' (literally 'hardness-degree of water'), the modifier *Härte* of the N-N compound *Härtegrad* corresponds to what is otherwise associated with the dependent *Wasser* in the complex nominal *Härte des Wassers*; that is, in *Härtegrad des Wassers*, an element of the non-head is "inherited" and expressed by the head (cf. *Grad der Härte des Wassers* and *Grad des Wassers*) (Siebert 1999). Cases like this involve Function Composition, and unfortunately, exploring the relationship between Function Composition on the one hand, and both MNC and non-MNC configurations on the other – including the conditions under which it may be obviated – is beyond the scope of this thesis. I leave this issue open, referring the reader to works like Steedman (1987), Stiebels (1996), Siebert (1999) and Wunderlich (2000).

(20) **Saliba**a. *Object sharing*

Ye-[koi]-[kesi]-di
 3SG.S-hit-break-3PL.O
 'He broke them'

b. *Non-object sharing*

Kaputi ku-[ini]-[he-mwayau]-∅
 cup 2SG.S-pour-CAUS-full-3SG.O
 'Fill the cup'

(Margetts 2005: 66, 81)

(20a) and (20b) thus have the denotations in (21a) and (21b), respectively (full derivations are given later):

(21) a. $\llbracket (20a) \rrbracket =$

$\exists E \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ \text{hit}(e_1, x_i) \ \& \ \text{break}(e_2, \text{they}_i) \ \& \ \text{AGENT}(E) = \text{he}]$
 = 'He hit them and broke them' $\approx$ 'He hit-broke them'

b. $\llbracket (20b) \rrbracket =$

$\exists E \exists x \exists e_1 \exists e_2 \exists e_2' [E = e_1 \oplus e_2 \ \& \ \text{pour}(e_1, x_k) \ \& \ \text{CAUSE}(e_2, e_2') \ \& \ \text{full}(e_2', \text{cup}_i)$
 $\& \ \text{AGENT}(E) = \text{you}]$
 = 'You pour something and fill the cup' $\approx$ 'Fill the cup'

This blocking effect with respect to non-head internal arguments is not specific to Saliba, but in fact a more general property of C-SVCs. In (22) from Mwotlap, the semantic object of the non-head V_1 in 'drink' is the beer, while that of the head $V_{2=H}$ *biyiñ* 'accompany' is the kava, but only the kava projects as object:

(22) **Mwotlap:** *trans + trans*

Nēk ta-kalē vēh mi ni-bia... a so [in] [biyiñ] ēwē
 2SG POT₁-chock POT₂ with ART-beer that is drink accompany just
 na-ga
 ART-kava

'[when drinking kava] one may force it down with beer... that means nothing
 but drink together with kava' (François 2004: 128)

Recall also (1) from Daakaka, repeated here in (23), where it is impossible for the internal argument of a non-head transitive V_1 in C-SVCs to project:

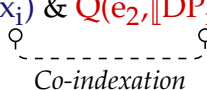
(23) **Daakaka:** *trans + trans*

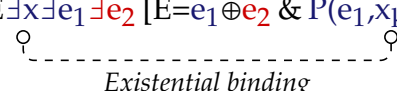
- a. Bong ma [ta] [sengave] seli
 B. REAL cut open.TRANS road
 ‘Bong opened/cleared the road by cutting [the grass]’
- b. *Bong ma [ta] pwesye [sengave] seli
 B. REAL cut branch open.TRANS road
 intended: ‘Bong opened/cleared the road by cutting (the) branches’
 (Hopperdietzel 2020: 304)

We can thus generalize the argument linking profiles of transitive C-SVCs with respect to their internal arguments as follows: in C-SVCs with object sharing, the internal arguments of the head and non-head verbs are co-indexed (24a); in C-SVCs without object sharing, the internal arguments of the head and non-head transitive verb cannot be co-indexed (due to sortal distinctness), so that the non-head internal argument is rendered implicit by existential binding (24b). In both object-sharing and non-object sharing C-SVCs, only the head internal argument is realized syntactically; where they differ is the way in which the non-head verb’s IAR is satisfied, viz. co-indexation (24a) or existential binding (24b):

 (24) *Internal argument relations in [trans + trans] C-SVCs*

- a. *Object sharing via co-indexation*

$$\exists E \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ P(e_1, x_i) \ \& \ Q(e_2, \llbracket DP_{int} \rrbracket_i) \ \& \ AGENT(E) = \llbracket DP_{ext} \rrbracket]$$

- b. *Existential binding of non-head internal argument*

$$\exists E \exists x \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ P(e_1, x_k) \ \& \ Q(e_2, \llbracket DP_{int} \rrbracket_i) \ \& \ AGENT(E) = \llbracket DP_{ext} \rrbracket]$$


(24a) and (24b) represent two-verb C-SVCs, but this behavior generalizes to C-SVCs with n number of verbs (subject to language-specific upper bounds on the number of verbs in C-SVCs).

To further see that co-indexing a non-head transitive verb’s internal argument with the projected internal argument of the head transitive verb V_H indeed satisfies the former’s IAR, let us consider cases where such co-indexation is not possible. Specifically, there are languages that simply do not allow the pairing of a non-head transitive verb with a head transitive verb, instead forcing speakers to either utilize some workaround, or – if no workaround is available – outright banning [*trans + intr*] pairings.

First, consider Nêlêmwa, which prohibits the pairing of a transitive V_1 with an intransitive $V_{2=H}$, except if $V_{2=H}$ comes from a particular class of verb which (i) express manner-adverbial meanings and (ii) can exceptionally transitivize within C-SVCs (this phenomenon is discussed in detail in Part 2 on transitivity matching)

(25):

(25) **Nêlêmwa:** *trans* + *trans*

I [hnawe] [mwang-ilî]
 3SG drop be.bad-TRANS

'He dropped it awkwardly'

(Bril 2004: 176)

Yet Nêlêmwa also has at its disposal a workaround that does allow the pairing of a transitive verb with a regular intransitive verb, which is that transitive verbs to come last in the string (26):

(26) **Nêlêmwa:** *intr* + *intr* + *trans*

a. Ma u [oda] [uya] [tu] axaleny Boovalan
 1DU.EXCL PERF go.up arrive find this man B.
 'We went up, arrived, found this Boovalan'

b. Na [tu] [pwe] [yaagi] caaxi caa-hî
 1SG go.down fish look.for side-dish starch-food-POSS.1DU.INCL
 'I went down (to) fish (and) look for food as a side-dish to our starch-food'

(Bril 2004: 172f)

Crucially, there is no absolute ban against transitive verbs in non-final position in C-SVCs, as shown by the all-transitive (27) below, and (25) above. It is only whenever the constituent verb stems have disparate transitivities that the transitive verb must come last in the string:

(27) **Nêlêmwa:** *trans*₁ + *trans*₂ + *trans*₃

Kio i [tâlâ] [mwemwelî] [yoo-lî] vhaa Nêlêmwa
 NEG 3SG hear know.TRANS be.good-TRANS talk N.

'He doesn't understand the Nêlêmwa language very well' (Bril 2004: 172)

This state of affairs finds a natural explanation if the string-final verb is the syntactic head, such that the obligatory reordering of a transitive verb from non-final to final position in these cases amounts to a rearrangement from non-head to head status.⁹¹ This, I claim, happens in order for the transitive verb to be able to satisfy its IAR, absent a string-final, head transitive verb with a direct object to satisfy its IAR via co-indexation, and given the configurational restrictions of the MNC.

Second, we know that there are languages like Tlachihiilco Tepehua, where the only illicit pairing involves a transitive V_1 and an intransitive $V_{2=H}$. In order to get around this restriction, two adjustments are available: either the transitive V_1

⁹¹Not transformationally, but in terms of which verb is established as the syntactic head in the derivation.

is detransitivized, or the intransitive $V_{2=H}$ is transitivized. As Watters (2017: 281) notes, the "Actor and Undergoer of the non-head verb must both be licensed by the transitivity of the head verb". In our terms, this means that Tlachihilco Tepehua bans only [*trans* + *intr*] pairings in syntactically right-headed C-SVCs because these constitute the only case in which a non-head transitive verb's IAR cannot be satisfied via co-indexation. Once the intransitive $V_{2=H}$ is transitivized, co-indexation becomes available, due to the $V_{2=H}$ now projecting an internal argument, and hence V_1 's IAR is satisfiable in non-head position.

Like Nêlêmwa, then, Tlachihilco Tepehua bans [*trans* + *intr*] pairings, but unlike Nêlêmwa, it does not have the option of rearranging the non-head transitive verb to head status (for whatever reason). These languages show two things: first, the ill-formedness of [*trans* + *intr_H*] pairings further illustrates the effects of the MNC; second, they provide further evidence that co-indexing a non-head transitive verb's internal argument with a projected internal argument indeed satisfies that non-head transitive verb's IAR.

Now, recall that according to the MNC, repeated in (28), the non-head X 's arguments can only be realized outside of [${}_Y X Y$] if (i) the head Y realizes an argument of X , or (ii) Y is otherwise sensitive to X 's argument structure:

(28) *The Morphological Non-head Constraint (MNC):*

In a base-generated complex head [${}_Y X Y$], X 's arguments cannot be realized outside [${}_Y X Y$], unless Y realizes an argument of X or is otherwise sensitive to X 's argument structure. (McIntyre 2014: 130)

Insofar as "argument realization" in (28) only refers to the *syntactic realization* of an argument, sanctioning condition (i) ("unless Y realizes an argument of X ") is not fulfilled in transitive C-SVCs because the internal argument of the non-head V_1 is not realized by the head $V_{2=H}$. $V_{2=H}$ does not syntactically realize V_1 's internal argument, but its own internal argument, yet which is co-indexed with V_1 's internal argument. As for sanctioning condition (ii) (" Y is otherwise sensitive to X 's argument structure"), I propose that while the head $V_{2=H}$ is indeed sensitive to V_1 's argument structure (in a sense to be made precise shortly), it is again not the case that this sensitivity to argument structure allows V_1 's internal argument to be realized syntactically because it is only ever V_2 's internal argument that is so realized. This means that given the mechanism of argument linking in C-SVCs proposed here, circumvention of the MNC with respect to the non-head V_1 is never achieved by syntactically realizing its internal argument outside of the verbal complex.

If we move away from syntactic projection only and consider other ways in which the non-head V_1 's IAR can be satisfied, however, then sanctioning conditions (i) and (ii) are clearly fulfilled with respect to $V_{2=H}$: condition (i) is fulfilled because $V_{2=H}$'s syntactically realized internal argument is co-indexed with V_1 's, and condition (ii) is fulfilled because $V_{2=H}$ is sensitive to V_1 's argument structure. In what sense,

though, is $V_{2=H}$ sensitive to V_1 's argument structure? In McIntyre (2014), a head's sensitivity to argument structure is understood as the property of not rendering inactive or suppressing the argument-structural requirements of the non-head, but instead retain and somehow be able to operate on them. I claim that $V_{2=H}$ fulfills this condition because its combination with V_1 does not suppress V_1 's argument-structural requirements, but instead operates on these to yield a complex argument structure. Crucially though, this sensitivity to argument structure is not intrinsic to $V_{2=H}$, but rather a derived property which is brought about by Event Cumulation, which extends the predicate structure of the head $V_{2=H}$, ultimately yielding a complex predicate.

5.4.1.1 Conditions on existential binding

We have yet to discuss the conditions under which the internal arguments of adjoined transitive verbs in C-SVCs may be existentially bound. Even though we might a priori take existential binding and Function Application to be freely available equivalents with respect to satisfying V_1 's IAR (Chung & Ladusaw 2004: 11), existential binding of internal arguments generally does not come for free in the languages at hand. In Saliba, for instance, all transitive verbs occurring outside C-SVCs without valence-reducing morphology must obligatorily φ -mark a referentially interpreted object (Anna Margetts p.c.). Similarly, internal arguments of transitive verbs without a lexical object in Mwotlap are generally interpreted as zero anaphora referring back to a previously established referent (Alexandre François p.c.).⁹²

I propose a two-part answer to this question, the first of which concerns the syntactic conditions, and the second one the event-structural conditions under which existential binding of adjoined internal arguments may occur.

In terms of syntax, the availability of existential binding within C-SVCs seems to follow from a *failed attempt* by the grammar to relate the internal argument positions of transitive verbs via co-indexation C-SVCs, perhaps analogous to how Agree may fail in the sense of (Preminger 2014). That is, given the adjoined V_1 's IAR, the rule of EC attempts to co-index the two internal argument positions, but fails once confronted with their sortal distinctness. This, in conjunction with the fact that V_1 cannot project, sanctions a Last Resort-type application of existential binding of V_1 's

⁹²In order to express a generic transitive activity with unspecified object deletion in Mwotlap – i.e. to derive a kind of anti-passive function – the verb must obligatorily be made intransitive via reduplication:

- (i) **Mwotlap:**
 - a. Kē ni-in
3SG AOR-drink
'He drank it'
 - b. Kē ni-in=in
3SG AOR-RED=drink
'He went to drink'

(Alexandre François p.c.)

internal argument. I suggest that this situation qualifies as a repair in Optimality-Theoretic terms, referring the reader to Section 15.2.2 of the last chapter.

Second, existential binding of internal arguments must additionally be sanctioned by event-structural conditions, some of which are language-specific, while others are universal. The idea is that because we know independently that constructions expressing causation and resultativity are governed by systematic principles, the choice of what kinds of arguments may remain implicit in cases like (29) is predictable.

(29) **Saliba:** *Non-object sharing*

Kaputi ku-[ini]-[he-mwayau]-Ø
cup 2SG.S-pour-CAUS-full-3SG.O

‘Fill the cup’

(Margetts 2005: 81)

Wolff (2003) and Kratzer (2005) show that resultatives are subject to the requirement that their causation component correspond to that causal sub-event immediately leading up to the result or transition state. For this reason, (29) can only mean that the cup becomes full as a direct result of something being poured into it, as opposed to, say, it becoming full as an indirect result of someone pouring tea into their hands and then emptying their hands into the cup. Similarly, the English resultative in (30) can only describe an event where the teapot becomes empty due to being directly affected by Peter’s drinking from it, but not an event where Peter’s drinking of all the water in a well causes the teapot to remain empty because there is no water left to make tea from (Hopperdietzel 2020: 80):

(30) a. Peter drank the teapot empty

b. $\lambda e\exists s. [\text{drink}(e) \ \& \ \text{CAUSE}(e,s) \ \& \ \text{empty}(s) \ \& \ \text{HOLDER}(s, \text{teapot})]$

(Hopperdietzel 2020: 80)

Informally, the causal chain of events running from the manner event to the result state cannot be broken up by intermediate events not standing in a contactive relationship to one another, i.e. the causal chain must be contiguous in terms of contact and the affectedness of participants. Resultatives thus show a *tightness* property which requires the object of the result state to directly be affected by the action expressed by the manner verb, even if the object of the result state is distinct from that the manner verb (Tomioka 2006, Levin 2020).

This explanation fares well for C-SVCs expressing causation or resultativity, but it does not explain why in the Mwotlap case in (22), repeated here in (31), the internal argument of the V_1 in ‘drink’ is blocked from realization despite the construction not obviously expressing causation or resultativity:

(31) **Mwotlap:** *trans + trans*

Nēk ta-kalē vēh mi ni-bia... a so [in] [biyiñ] ēwē
 2SG POT₁-chock POT₂ with ART-beer that is drink accompany just
 na-ga
 ART-kava

‘[when drinking kava] one may force it down with beer... that means nothing
 but drink together with kava’ (François 2004: 128)

For this issue, I draw on an explanation that is provided by a more fundamental line of reasoning in the framework of *Lexical Decomposition Grammar* (LDG) (Wunderlich 1997a, Wunderlich 1997b, Kaufmann & Wunderlich 1998, Wunderlich 2000). This reasoning characterizes a set of generalizations which, among others, entails the just-mentioned tightness constraint on causation. In LDG, there are four levels of representation: *Conceptual Structure* (CS), which represents extra-linguistic conceptual information; *Semantic Form* (SF), which explicitly decomposes that information from CS which is grammatically relevant; *Thematic Structure* (TS), which hosts λ -abstracted individual and situation arguments that must be realized morpho-syntactically (or be existentially bound), on the basis of SF; and *Morphology / Syntax* (MS), which regulates the linking of arguments from TS into the morpho-syntax. Mapping relations thus hold between CS and SF, SF and TS, and TS and MS.

SF can be considered a streamlined version of CS that has been stripped away of all information that is not directly managed by the grammar, such as certain event-structural or thematic properties that can be inferred. TS corresponds to a predicate’s argumental requirements in terms of participants and events, while MS does not directly correspond to any designated level of representation in Minimalism, where core argument linking is taken to follow from structural mechanisms such as case-licensing functional heads or dependent case.

Within LDG, two constraints have been proposed that are meant to regulate the mapping between CS to SF in complex predication, and which are relevant to the present context. These are CONNEXION (32a) and COHERENCE (32b):

- (32) a. CONNEXION:
 In a decomposed SF structure, each predicate must share at least one argument with another predicate, either explicitly or implicitly.
 b. COHERENCE:
 Sub-events encoded by the predicates of a decomposed SF structure must be contemporaneously or causally connected.

(Kaufmann 1995a, Kaufmann 1995b)

CONNEXION requires that there be some form of (direct or indirect) argument sharing between the constituent predicates of any complex predicate, while COHERENCE regulates its event-structural profile in terms of temporal and causal coherence.

Coming back to our Mwotlap example (31), we can now see that it respects both

CONNEXION and COHERENCE, despite lacking resultativity: CONNEXION is fulfilled because even though the object of drinking (beer) does not correspond to that of accompaniment (kava), the two are nevertheless tightly connected at a conceptual level due to generally being drunk together in Mwotlap culture, and COHERENCE is fulfilled because the beer-drinking and its accompaniment with kava are construed as a temporally coherent, complex beer-and-kava-drinking event. One might therefore also argue that the kava qualifies as a further object of drinking, at least at the level of CS.⁹³

As Owens (2011: 57) notes, there is some amount of vagueness built into these two constraints, in that (i) argument sharing has the option of being implicit (CONNEXION), and that (ii) predicates must be contemporaneously or causally connected (COHERENCE). I propose that even though CONNEXION and COHERENCE heavily constrain the set of possible internal argument relations in complex predication, this vagueness nevertheless allows for a certain degree of language-specific variation in terms of which kinds of arguments may ultimately remain implicit in C-SVCs, and which may not.⁹⁴

Some languages place an outright general ban on the existential binding of internal arguments, such as Toqabaqita, where non-head verbs can only be transitive if the head itself is also transitive (Lichtenberk 2006: 258), which I take to mean that IAR satisfaction via co-indexation is an available option, but not existential binding.

Hopperdietzel (2020) shows that while internal arguments of non-head transitive verbs in SVCs may rather freely undergo existential binding in Daakaka, this option is never available in Samoan.

While more fine-grained data are needed in order to obtain a full picture, it appears that Daakaka constitutes one end of the continuum in possessing comparatively liberal conditions on existentially binding non-head internal arguments, followed by Mwotlap (where it is not restricted to causative / resultative C-SVCs), then Saliba (where it is restricted to causative / resultative C-SVCs), and finally Samoan and Toqabaqita (which disallow it entirely) constituting the other end. For an in-depth discussion of the directness of causation in Oceanic resultative and causative SVCs, see Hopperdietzel (2020: 76).

Finally, if my claim is on the right track that NC-SVCs and C-SVCs built via adjunction differ in that only the former necessarily involve argument sharing while the latter do not – in turn because complex V-level adjunction obviates the need for

⁹³An alternative to treating the internal argument of *ni* ‘drink’ in (31) as existentially bound would be to treat it as referring back to the noun *ni-bia* ‘the beer’ in the preceding clause. In Mwotlap, omitted objects of transitive verbs are always interpreted as zero anaphora (Alexandre François p.c.). I will not go further into how such discourse-structural conditions would need to interact with the binding status of internal arguments in C-SVCs, but submit that – regardless of whether the non-final transitive verb’s internal argument is bound existentially or anaphorically – the exact same general line of reasoning applies. In either scenario, the grammar would attempt to co-index the non-head’s internal argument with that of the head, but fail under sortal distinctness and then resort to an alternative binding strategy.

⁹⁴For a thorough discussion of this issue with respect to serial verbs in the framework of Lexical Decomposition Grammar, see Owens (2011).

integration via syntactic predication – we make a major prediction with respect to the interpretation of internal arguments in adjoined projections: given the necessity of argument sharing in NC-SVCs, we predict that internal arguments in adjoined VerbPs must always be co-referent with an overt argument in the main VerbP; conversely, we know that internal arguments belonging to adjoined verbs may be interpreted existentially due to sortal distinctness (provided that the language allows it). This appears to be correct: I have yet to come across a NC-SVC consisting of transitive verbs in which the internal arguments are not understood as co-referent.

5.5 V-level adjunction

We are ready to run through some derivations. V-level adjunction derives Type C transitivity profiles, which ban [*trans* + *intr_H*] pairings in the face of a strict IAR, as in Tlachihilco Tepehua, where every acquisition or omission of an object must be sanctioned by some overt derivation (Watters 2017):

(33) Tlachihilco Tepehua

a. *intr* + *intr*

[talaqsta]-[laka:wa:nan]
turn.face.up-look.IPFV

‘(S)he looks, facing upward’

b. *trans* + *trans*

[ʔap’i]-[ʔu-y]
peel-eat-IPFV

‘(S)he eats it, peeling it’

c. *intr* + *trans*

[tapasa]-[ʔi:-t]
pass.by-get-PFV

‘(S)h got it passing by’

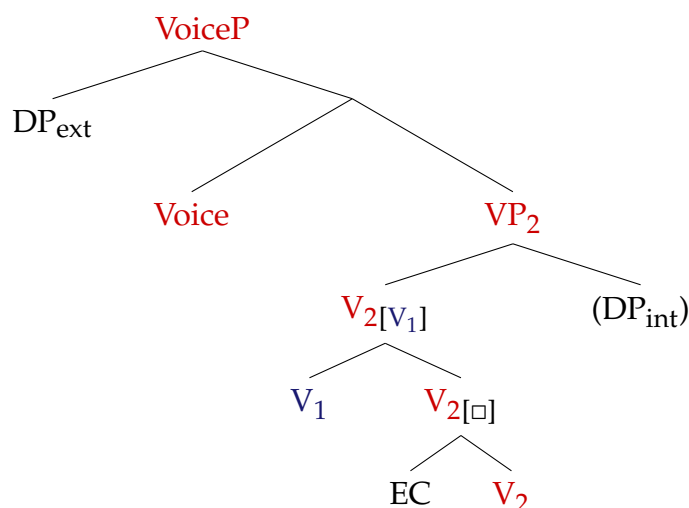
d. *trans* + *intr*

*[tʃaqʃ]-[ʔan]
chop-go.IPFV

intended: ‘(S)he goes chopping’

(Watters 2017: 288, 289)

The ability of only the head verb to project follows from the MNC. V-level adjunction configurations are schematized in Tree 5.5:



Tree 5.5: V-level serialization via adjunction

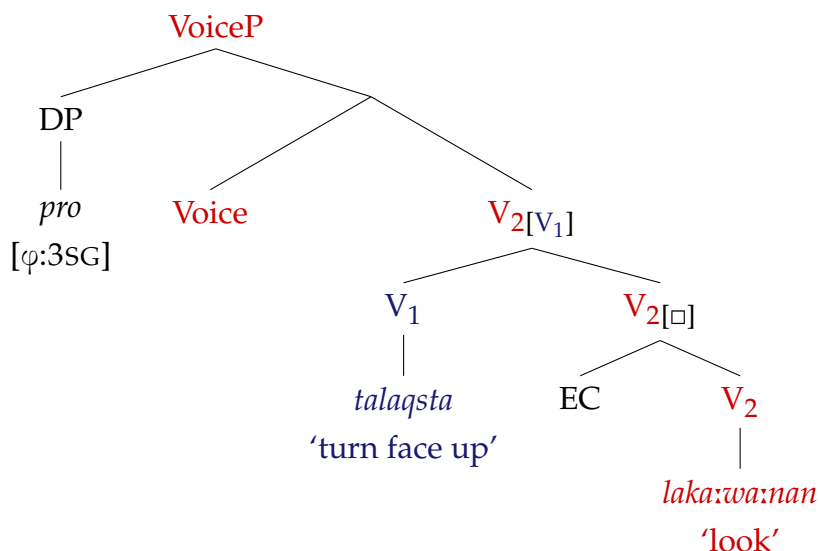
A C-SVC like (34), which I assume consists of two unergatives, thus has the structure in Tree 5.6:

(34) **Tlachihilco Tepehua:** *intr + intr*

[*talaqsta*]-[*laka:wa:nan*]
 turn.face.up-look.IPFV

‘(S)he looks, facing upward’

(Watters 2017: 288)


 Tree 5.6: Tlachihilco Tepehua C-SVC: *intr + intr*

The two unergatives making up (34) have the denotations in (35). After EC applies to V_2 and extends its predicate structure (36a), V_2 combines with V_1 to yield the complex predicate in (36b). Note how at this step in the derivation, which corresponds to the point of serialization, there are no individual arguments present at the complex argument structure in (36b) because V_1 and V_2 are both unergatives; consequently,

there is no argument sharing between V_1 and V_2 . After the only individual argument, the Agent, is introduced by Voice, Event Closure applies and we end up with (37):

- (35) a. $\llbracket V_1 \rrbracket = \llbracket \textit{talaqsta} \rrbracket = \lambda e_1. [\text{turn face up}(e_1)]$
 b. $\llbracket V_2 \rrbracket = \llbracket \textit{laka:wa:nan} \rrbracket = \lambda e_2. [\text{look}(e_2)]$
- (36) a. $\llbracket V_2[\square] \rrbracket = \lambda P \lambda E. \{\exists x\} \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ P(e_1, \{x_i/k\}) \ \& \ \text{look}(e_2)]$
 b. $\llbracket V_2[V_1] \rrbracket = \lambda E. \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ \text{turn face up}(e_1) \ \& \ \text{come}(e_2)]$
- (37) $\llbracket (34) \rrbracket = \exists E \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ \text{turn face up}(e_1) \ \& \ \text{look}(e_2) \ \& \ \text{AGENT}(E) = (s)\text{he}]$

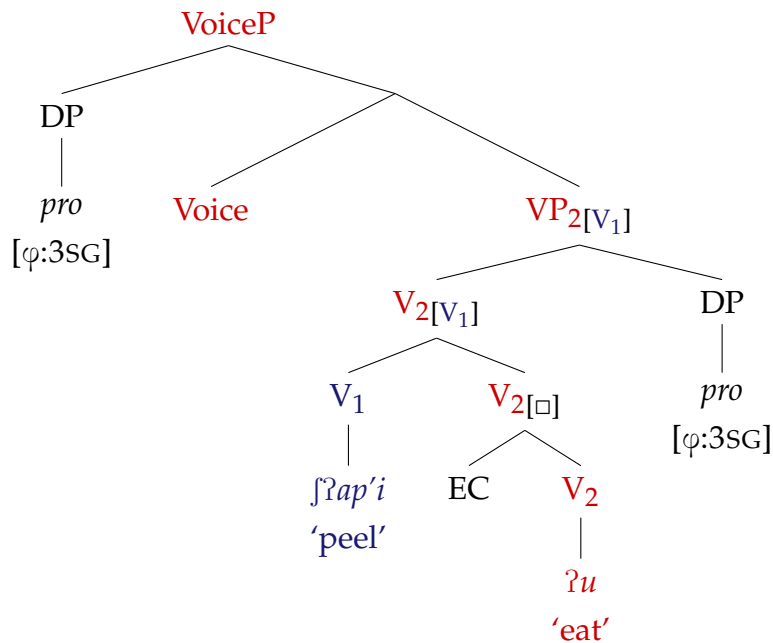
Next, let us derive the C-SVCs with object sharing in (38), which has the structure in Tree 5.7:

- (38) **Tlachihilco Tepehua:** *trans + trans*

$\llbracket \textit{ʃʔap'i} \rrbracket$ - $\llbracket \textit{ʔu-y} \rrbracket$
 peel-eat-IPFV

‘(S)he eats it, peeling it’

(Watters 2017: 288)



Tree 5.7: Tlachihilco Tepehua C-SVC: *trans + trans*

The two verbs making up (34) have the denotations in (39). As with the [*unerg* + *unerg*] pairing above, after $V_{2=H}$ undergoes head extension via EC (40a), it can serialize with V_1 . Because V_1 and V_2 are both transitives, the resulting complex predicate in (40b) has two internal arguments. As these arguments are sortally non-distinct they can be co-indexed, such that upon merger of a DP object the resulting VP_2 denotes a complex event with object sharing (40c): even though the internal argument position that is saturated syntactically via Function Application is only

that of the head transitive verb $V_{2=H}$, the adjoined internal argument of V_1 comes to co-refer with that of $V_{2=H}$ by virtue of co-indexation, which satisfies its IAR. The rest of the derivation proceeds normally, ultimately giving us (41):

- (39) a. $\llbracket V_1 \rrbracket = \llbracket \text{?ap'i} \rrbracket = \lambda x \lambda e_1. [\text{peel}(e_1, x)]$
 b. $\llbracket V_2 \rrbracket = \llbracket \text{?u-y} \rrbracket = \lambda y \lambda e_2. [\text{eat}(e_2, y)]$
- (40) a. $\llbracket V_{2[\square]} \rrbracket = \lambda P \lambda y \lambda E. \{\exists x\} \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ P(e_1, \{x_i/k\}) \ \& \ \text{eat}(e_2, y_i)]$
 b. $\llbracket V_{2[V_1]} \rrbracket = \lambda y \lambda E \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ \text{peel}(e_1, x_i) \ \& \ \text{eat}(e_2, y_i)]$
 c. $\llbracket VP_{2[V_1]} \rrbracket = \lambda E \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ \text{peel}(e_1, x_i) \ \& \ \text{eat}(e_2, it_i)]$
- (41) $\llbracket (38) \rrbracket = \exists E \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ \text{peel}(e_1, x_i) \ \& \ \text{eat}(e_2, it_i) \ \& \ \text{AGENT}(E) = (s)\text{he}]$

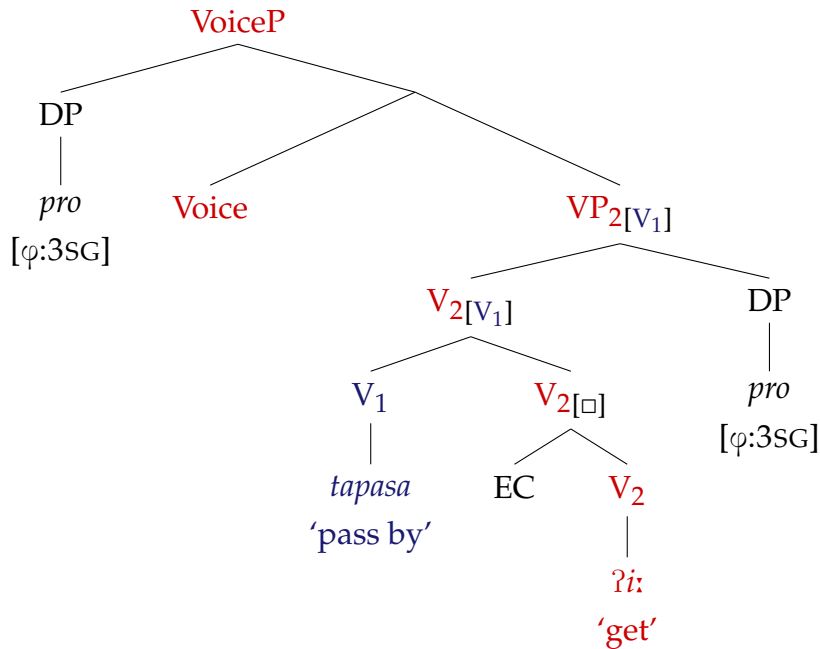
The [*unerg* + *unerg*] and [*trans* + *trans*] C-SVCs above are pairings of like transitivity. By contrast, the C-SVC in (42) involves verbs of disparate transitivity, with an unergative V_1 and a transitive $V_{2=H}$, and the structure in Tree 5.8:

- (42) **Tlachihienco Tepehua:** *intr* + *trans*

[tapasa]-[?i:-4]
 pass.by-get-PFV

‘(S)h got it passing by’

(Watters 2017: 288)



Tree 5.8: Tlachihienco Tepehua C-SVC: *intr* + *trans*

Given the denotations in (43), EC first carries out head extension of the transitive $V_{2=H}$, thereafter making it require another predicate (44); this open slot is subsequently filled by the unergative V_1 , which gives us the complex predicate in (44b). Next, the transitive $V_{2=H}$'s IAR is satisfied by merger of a DP, and because V_1 is

unergative and lacks an IAR, neither co-indexation nor existential binding takes place. We end up with (45):

- (43) a. $\llbracket V_1 \rrbracket = \llbracket \textit{tapasa} \rrbracket = \lambda e_1. [\textit{pass by}(e_1)]$
 b. $\llbracket V_2 \rrbracket = \llbracket \textcolor{red}{?i:-\textcolor{red}{t}} \rrbracket = \lambda y \lambda e_2. [\textit{get}(e_2, y)]$
- (44) a. $\llbracket V_{2[\square]} \rrbracket = \lambda P \lambda y \lambda E. \{ \exists x \} \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ P(e_1, \{x_i/k\}) \ \& \ \textit{get}(e_2, y_i)]$
 b. $\llbracket V_{2[V_1]} \rrbracket = \lambda y \lambda E \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ \textit{pass by}(e_1) \ \& \ \textit{get}(e_2, y_i)]$
 c. $\llbracket VP_{2[V_1]} \rrbracket = \lambda E \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ \textit{pass by}(e_1) \ \& \ \textit{get}(e_2, it_i)]$
- (45) $\llbracket (42) \rrbracket = \exists E \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ \textit{pass by}(e_1) \ \& \ \textit{get}(e_2, it_i) \ \& \ \text{AGENT}(E) = (s)\text{he}]$

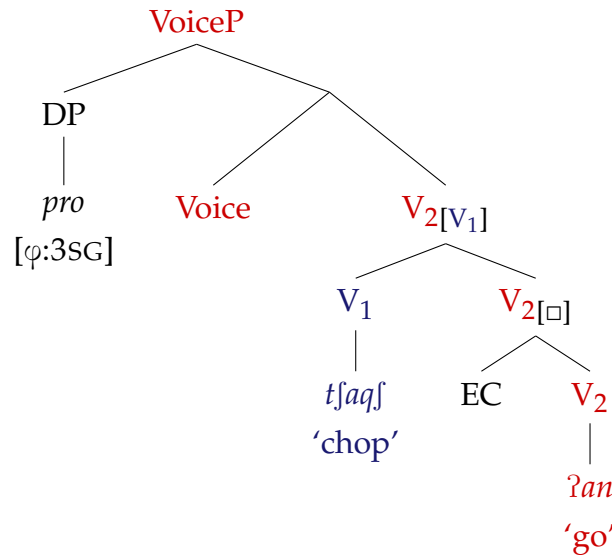
Having derived the three well-formed pairings – $[\textit{intr} + \textit{trans}]$, $[\textit{trans} + \textit{trans}]$ and $[\textit{intr} + \textit{trans}]$ – we lastly derive why $[\textit{trans} + \textit{intr}]$ pairings are ungrammatical. Consider the ungrammatical C-SVC in (46), which consists of a transitive V_1 and an unergative $V_{2=H}$. Tree 5.9 shows the hypothetical structure of (46):

- (46) **Tlachihilco Tepehua:** **trans + intr*

$\textcolor{blue}{*[tʃaɟʃ]} - \textcolor{red}{[ʔan]}$
 $\textcolor{blue}{\textit{chop}} - \textcolor{red}{\textit{go.IPFV}}$

intended: ‘(S)he goes chopping’

(Watters 2017: 289)



Tree 5.9: Tlachihilco Tepehua C-SVC: *trans + intr*

Note that in purely syntactic terms, nothing would go wrong in the derivation of (46), as the syntactic conditions on V-level serialization via adjunction are met at all times. The derivation of (46) fails for a non-syntactic reason, which is that the IAR of the adjoined transitive V_1 remains unsatisfied. To see this, given the denotations in (47), let us review that step in the derivation after the unergative $V_{2=H}$ has undergone head extension by EC (48a), and subsequently combined with the transitive V_1 into a

complex predicate prior to the introduction of any arguments, namely (48b):

- (47) a. $\llbracket V_1 \rrbracket = \llbracket tjaqf \rrbracket = \lambda x \lambda e_1. [\text{chop}(e_1, x)]$
 b. $\llbracket V_2 \rrbracket = \llbracket ?an \rrbracket = \lambda e_2. [\text{go}(e_2)]$
- (48) a. $\llbracket V_{2[\square]} \rrbracket = \lambda P \lambda E. \{\exists x\} \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ P(e_1, \{x_i/k\}) \ \& \ \text{go}(e_2)]$
 b. $\llbracket V_{2[V_1]} \rrbracket = \lambda E \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ \text{chop}(e_1, \boxed{x}) \ \& \ \text{go}(e_2)]$

The boxed internal argument of V_1 in (48b) represents the fact that it remains open, thereby leading to a crash. This happens because the head $V_{2=H}$ is unergative, meaning that it has no way of indirectly satisfying the adjoined V_1 's IAR. (49) shows that even as the derivation goes on, the internal argument position of the non-head transitive V_1 continues to remain open:

- (49) $\llbracket (46) \rrbracket = \exists E \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ \text{chop}(e_1, \boxed{x}) \ \& \ \text{go}(e_2) \ \& \ \text{AGENT}(E, (s)he)]$

In sum, the ban against *trans* + *intr* pairings (Type C) and the inability of non-head transitive verbs to project and hence realize their internal arguments are intimately connected: because serialization occurs at the V-level prior to the introduction of any arguments, and given the MNC, only the head verb can uniquely project. Consequently, if (i) any non-head verb is transitive, (ii) the language strictly requires all transitive verbs to have their IAR satisfied, and (iii) the head verb is unable to satisfy the IAR of its transitive co-verb(s), the derivation crashes, and hence *trans* + *intr* pairings are unavailable. As in all the other cases so far, at no point in the derivation does the grammar directly make reference to some parameter of transitivity; rather, the restrictions we find reduce to those arising in V-level adjunction configurations, coupled with a language's strict IAR. So while the Type IV linking profile is a necessary property of C-SVCs, the Type C transitivity profile is not, arising only in languages placing a strict IAR on transitive verbs; this point of variation is the topic of the next subsection.

5.5.1 Ditransitives

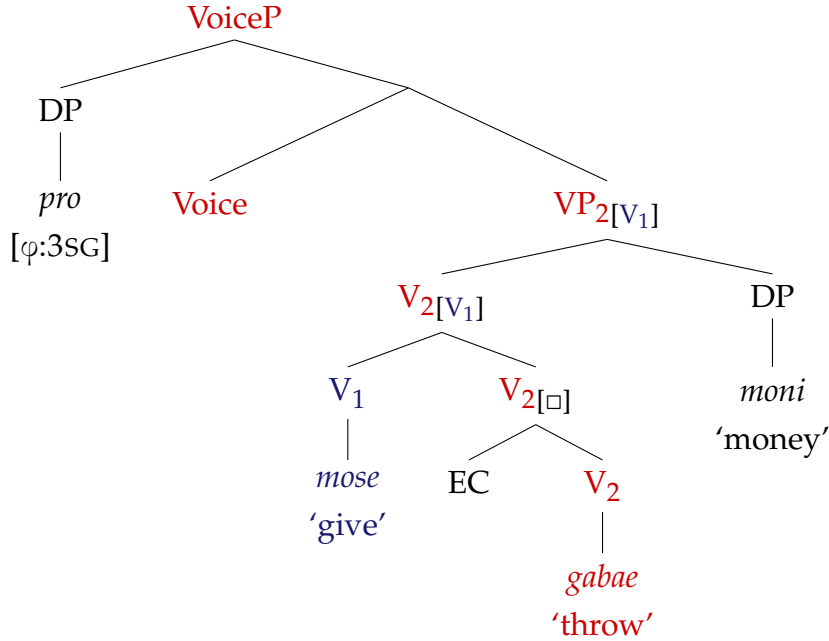
Lastly, as with NC-SVCs, we turn to cases involving ditransitives. In (50a) from Saliba, the adjoined V_1 is ditransitive and the main $V_{2=H}$ transitive; (50b) shows the double object structure that is licensed by the ditransitive V_1 outside serialization:

- (50) **Saliba**
- a. *Adjoined ditransitive V_1*
 Moni ye-[mose]-[gabae]- $\emptyset$
 money 3SG.S-give-throw-3SG.O
 'He gave the money away'
- b. *Ditransitive V_1 takes double objects in isolation*

Bosa kesega ye-mose-i-di
 basket one 3SG.S-give-APPL-3PL.O
 ‘He gave them one basket’

(Margetts 1999: 300)

The most important thing to note about (50a) is that – just like in NC-SVCs – the argument shared between V_1 and $V_{2=H}$ is the Patient. I assume the structure in Tree 5.10 for (50a):



Tree 5.10: C-SVC with adjoined ditransitive V_1

The derivation of (50a) proceeds similarly to the NC-SVC with an adjoined ditransitive V_2 in Edo, cf. Section 4.2.5. Departing from the denotations in (51), the transitive $V_{2=H}$ undergoes head extension via EC (52a), and then combines with the ditransitive V_1 to yield the complex predicate in (52b). At this point in the derivation, there are no individual arguments but Patients present in (52b) that could be co-indexed with one another, such that object sharing must necessarily come to hold between these. Because the ditransitive verb is the non-projecting adjoined verb, not only its Patient internal argument must remain implicit, but also its Goal internal argument.⁹⁵ The derivation ends with (53):

- (51) a. $\llbracket V_1 \rrbracket = \llbracket mose \rrbracket = \lambda x \lambda e_1. [give(e_1, x)]$
 b. $\llbracket V_2 \rrbracket = \llbracket gabae \rrbracket = \lambda y \lambda e_2. [throw(e_2, y)]$
- (52) a. $\llbracket V_{2[\square]} \rrbracket = \lambda P \lambda y \lambda E. \{ \exists x \} \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ P(e_1, \{x_i/k\}) \ \& \ throw(e_2, y_i)]$
 b. $\llbracket V_{2[V_1]} \rrbracket = \lambda y \lambda E. \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ give(e_1, x_i) \ \& \ throw(e_2, y_i)]$
 c. $\llbracket VP_{2[V_1]} \rrbracket = \lambda E. \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ give(e_1, x_i) \ \& \ throw(e_2, money_i)]$
- (53) $\llbracket (50a) \rrbracket = \exists E \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ give(e_1, x_i) \ \& \ throw(e_2, money_i) \ \& \ AGENT(E, he)]$

⁹⁵Here, I am assuming that the Goal is simply absent entirely. Alternatively, we can assume that it is present thematically as a lexical argument of the V_1 ‘give’, but existentially bound.

This leaves us with C-SVCs where the main verb is ditransitive and the adjoined one transitive. Unfortunately, I have not yet come across such cases which are also conclusively non-applicative, i.e. where the ditransitive verb does not have a more abstract meaning and does not serve an argument-introducing function with respect to the event denoted by V_1 ; moreover, the SVCs that are candidates in this regard do not appear to be robustly contiguous, potentially allowing inter-verbal markers of coordination, despite being clearly contiguous in other cases; see e.g. Jeh (Cohen 1976, Gradin 1976).⁹⁶ I leave this issue aside for future work, but submit that (i) they surely exist, and that (ii) object sharing should again hold between Patients under the assumption that Goals / Sources are introduced higher in the structure (Larson 1988), such that only Patients can come to be shared with one another.

5.5.2 On the strictness of the internal argument requirement

So far, I have not found a language with a clear example of a well-formed non-resultative [*trans* + *intr*] C-SVC, consisting of a transitive non-head and an intransitive head. Such a language would need to have a lax IAR, allowing transitive verbs to omit their objects without the need of an overt valence-changing derivation. Ideally, it should not be a language lacking productive morphological valence changes in general, so that we can rule such operations applying covertly. I am sure, however, that this kind of language exists.

5.6 V-level complementation

So far, I have not come across any clear examples of V-level complementation, in which one verb takes another verb as its complement. To be clear: I am referring to configurations in which the complement to one verb is itself a verbal head, not a phrase. I leave this issue open, submitting that if V-level complementation configurations are indeed unattested, they may again be explained by Kayne (1994)'s Anti-symmetry, which rules out head-to-head complementation.

5.7 Transitivity restrictions

The transitivity profile we find in C-SVCs, i.e. Type C ($*[trans + intr_H]$), follows from the interaction of (i) the configuration yielded by V-level adjunction, (ii) the MNC, in turn a consequence of V-level adjunction, and (iii) a transitive verb's IAR. Once these are accounted for, profile Type C falls into place naturally given what we know about principles of integration and projection.

⁹⁶Should this impression turns out to be correct, the facts would be consistent with the absence of V-level complementation.

5.8 Discussion

5.8.1 The role of the grammar in implicit argument identification

Now that we have also discussed C-SVCs with ditransitives, let us briefly reconsider the role of the grammar in the identification of implicit arguments.

Given the MNC, nothing more needs to be said on why an adjoined transitive verb cannot project in V-level adjunction configurations. It is also clear that an adjoined verb's IAR must be satisfied in some way. This raises the question of whether the mechanisms of co-indexation and existential binding are actually enforced grammatically, or achieved pragmatically.

In the domain of resultatives, for instance, it has been argued that implicit objects may be vague and identified pragmatically, e.g. in Kaufmann & Wunderlich (1998). Evidence for this analysis comes from examples like (54), which has two readings:

(54) **German:**

Er fuhr das Auto seines Nachbarn kaputt
he drove the car his neighbor.GEN broken

- i. 'He drove his neighbor's car and demolished it'
- ii. 'He drove someone else's car, by means of which he caused his neighbor's car to become demolished' (Kaufmann & Wunderlich 1998: 19)

On the first reading of (54), the direct object of the construction is identified with the neighbor's car, whereas on the second one, it is identified with someone else's car. Note that under Kaufmann & Wunderlich's analysis, the direct object realizes an argument introduced by resultative formation in both readings; what differs is whether or not this added argument is also pragmatically identified with the internal argument sub-categorized by the verb.

By contrast, work like Williams (2005) *et seq.* argues that all instances of argument (non-)sharing in resultatives are established grammatically. All the variation we find is argued to fall out systematically from language-specific grammatical properties, such as the kinds of strategies available for resultative formation, or whether or not internal arguments are lexical arguments of verbs. Consequently, the two readings of the resultative in (54) would, under Williams' account, result from either two distinct argument structures, or two distinct syntactic structures, rather than pragmatic differences; in other words, (54) is underlyingly ambiguous in his system, not vague.

In this chapter, we have seen that the sharing and linking principles found in NC-SVCs are abstractly reproduced in C-SVCs: in both cases, an internal argument of an adjoined verb remains silent and is identified with an overtly projected internal argument of the main verb. We have also seen that object sharing systematically targets patients because these are the only arguments present in the derivation at the

point where serialization occurs. By extension, existential binding of non-projected internal arguments must also proceed grammatically, given that it results from a failed attempt to co-index two internal arguments due to their sortal distinctness.

I therefore conclude that implicit argument identification in both NC-SVCs and C-SVCs is achieved grammatically, and subject to the same grammatical principles. The observed differences relate to the structural size of the adjoined verbal constituent, i.e. V vs VP, and to how exactly adjoined internal arguments are rendered silent, i.e. as null arguments in the form of minimal pronouns in NC-SVCs vs as implicit arguments in C-SVCs.

5.8.2 On the complement-like behavior of adjoined material

This section addresses the issue of how within serialization via adjunction, the adjoined material can feed into the overall complex event, argument and syntactic structure of the resulting serial verb in a way that is reminiscent of complementation configurations. That is, in light of the fact that adjuncts are often invisible to certain semantic and syntactic processes (cf. Chomsky (2004)'s metaphor that adjuncts are merged on a "separate plane"), how can the adjoined verb and the head verb jointly give rise to complex-predicate behavior, as if the adjoined verb were a complement of the head verb? This behavior is especially evident in C-SVCs, where co-reference between arguments is not established via syntactic predication, but co-indexation at the level of complex argument structure. It is even more evident with so-called *contextually bivalent* verbs, discussed in upcoming chapters, where the adjoined verb also determines the thematic reference of the head verb, i.e. the head verb's internal argument depends thematically on the non-head verb.

I propose that this behavior has a semantic basis, following Truswell (2007)'s analysis of exceptionally well-formed extraction from adjuncts. Even though adjuncts have been taken to be islands since Ross (1967) – as illustrated by the contrast in (55) – there exist constructions where $\bar{A}$ -extraction from adjuncts is grammatical, cf. (56):

- (55) a. Who_i —_i cried [after John hit Mary]?
 b. *Why_i did Mary cry [after John hit —_i]
 (Huang 1982: 503)

- (56) a. What_i did John arrive [whistling —_i]?
 (Borgonovo & Neeleman 2000: 200)

- b. What_i did John drive Mary crazy [trying to fix —_i]?
 (Truswell 2007: 2)

According to Truswell, adjuncts are transparent for extraction only if the event they denote stands in a particular event-structural relation to the event denoted in the matrix projection. Specifically, the adjoined event must express a depictive relation

to the matrix event, as in (56a), or express a cause immediately leading up to the matrix event, as in (56b). Adjuncts must be related semantically to main verbal projections via one of these two interpretations (57), which in turn characterizes the generalization in (58):

(57) *Interpretation of adjoined predicates*

- a. THEN(e_1, e_2)
 $\Rightarrow$ in (56a), John's whistling leads up to (and potentially overlaps with) his arrival
- b. CAUSE(e_1, e_2)
 $\Rightarrow$ in (56b), John's attempt at fixing causes Mary to become crazy
 (Truswell 2007: 26)

(58) *Extraction from adjoined predicates:*

Extraction of a complement from a secondary predicate is permitted only if the event it denotes is identified with an event position in the matrix predicate

(Truswell 2007: 39)

Roughly, the generalization is that the adjoined event must express an immediate event-structural antecedent with respect to the change of state expressed by the matrix event, which in some cases is interpreted as causation (cf. (57b)). Because the event-structural function of the adjoined event must naturally be induced by and complement the semantics of the matrix verb, thereby giving rise to complex event structures similar to Vendlerian accomplishments and achievements, adjuncts expressing simple relations like temporal succession (55b) fail to be transparent.⁹⁷

Coming back to SVCs, it is well known that these, too, exhibit a kind of event-structural complementarity, subject to language-specific differences; see e.g. Durie (1997) and Aikhenvald (2006b). I suggest that the exceptional transparency of certain adjuncts for extraction on the one hand, and the complement-like behavior found in SVCs built via adjunction on the other are afforded by one and the same semantic principle, namely the fact that both express complex event structures which parallel those of simple verbs. Put differently, both give rise to a complex-predicate behavior because their denoted events belong to the same macro-event.

⁹⁷That is, "verbs do not express relations of simultaneity among events, instead expressing either simplex events or states, or relations of temporal precedence or causation among events and states" Truswell (2007: 9). Consider the following example:

- (i) *What_i did John arrive [while whistling —_i]?

(Truswell 2007: 9)

5.8.3 Serialization and restructuring

Recall from the outset the set of necessary conditions that we set up in order for a construction to count as a SVC, repeated here:

(59) *Necessary conditions for a Serial Verb Construction*

- a. mono-clausal
- b. consists of multiple verbs
- c. the verbs may occur independently
- d. no linking element
- e. structure-building within VoiceP

Rather than telling us in absolute terms for any given construction whether or not it is a SVC, (59) provides us with comparative diagnostics which make reference to clearly defined grammatical principles, based upon which we can decide what a SVC must have in any case. The main reason for doing so is that – while there is a clear intuition shared by scholars as to what a SVC is – there remain numerous cases which are seen as falling under serialization by some, but not others. The reasons for this, in turn, are partly due to the distinct sets of criteria employed by different authors, partly due to properties specific to the languages investigated, and partly due to historical reasons. The criteria in (59), then, clearly constitute progress in this regard because it reconciles the long-standing intuition that scholars have had regarding SVC-hood on the one hand, with robust grammatical principles on the other.

In the strictest sense, then, prototypical serialization involves V-, VP- or VoiceP-level adjunction – something which empirically captures what authors more or less universally understand as falling under serialization. To these, we can add applicative SVCxs, derived by VP-level complementation, which have similarly been unanimously regarded as instances of serialization. VoiceP-level complementation, on the other hand, has traditionally not been viewed as serialization by many, presumably because of the obvious relationship of complementation holding between the kinds of verbs typically found in this configuration, most notably causatives. Conversely, I propose that what has here been termed applicative SVCs, despite having been uncovered here as also involving complementation, did not obviously bear the symptoms of complementation to most scholars, hence their frequent traditional classification as SVCs.

I would like to point out another advantage of the present account in this connection, namely the relationship it turns out to bear to the phenomenon of *restructuring* (Wurmbrand 2001, Wurmbrand 2016). A restructuring configuration is an infinitival construction that lacks the clause-boundedness effects which are otherwise associated with infinitives in a given language. While certain diagnostics for restructuring systematically recur (e.g. clitic climbing), whether or not a configuration involves restructuring is to a certain extent language-specific.

Based on Wurmbrand (2016), I adopt a mono-clausal view of restructuring, and assume that it falls into two broad types: *lexical* and *functional restructuring*. Lexical restructuring verbs optionally induce restructuring, they are thematic in obligatorily introducing arguments, and they lack co-occurrence restrictions. In functional restructuring, by contrast, restructuring is obligatory, the verb is athematic, and it shows co-occurrence restrictions; functional restructuring thus patterns like raising. Typical restructuring verbs are modal, motion, causative and aspectual verbs.

Without going too much into detail, the necessary conditions on serialization in (59) significantly relieve us from the pressure to always have to decide between restructuring and serialization. The difference has generally been somewhat unclear, but – labels aside – we can now see that certain constructions actually fall into both categories. VP-level complementation, for instance, which derives applicative NC-SVCs, would instantiate semi-functional restructuring because (i) the applicative ‘take’-type verb occurs independently and need not obligatorily serialize, but (ii) when it does, it introduces a thematic argument in its specifier, i.e. the applied role. Similarly, VoiceP-level complementation, which may be instantiated by causation, would count as (semi-)functional restructuring, or perhaps Voice restructuring (Wurmbrand & Shimamura 2017) (despite not always being viewed as serialization).

Finally, Susi Wurmbrand (p.c.) points out another parallel to restructuring in a more abstract sense: just like restructuring falls into two general types, lexical and functional, so does the set of conditions on serialization in (59) characterize two broad types of serialization: in “lexical” serialization, verbs are serialized via adjunction, while in “(semi-)functional” serialization, verbs are serialized via complementation. On this view, serialization partly recapitulates the lexical-functional distinction found in restructuring, despite restructuring always involving complementation.⁹⁸

5.9 Summary

Let us now come to a close in our first case study and summarize our findings about transitivity restrictions in SVCs. NC-SVCs show two transitivity profiles: Type A, which shows no restrictions, and Type B, which requires each VerbP to minimally contain an internal argument. C-SVCs show profile Type C, which disallows adjoined transitives unless the head verb is itself transitive. These transitivity profiles fall out from cross-cutting the mode of structure-building with the level of structure-building.

In NC-SVCs built via adjunction, VP-level adjunction yields profile Type B, while VoiceP-level adjunction yields profile Type A. Type B follows because only internal arguments are available at the VP-level, and because serialization via adjunction requires integration to proceed via predication, which entails syntactic argument sharing at the phrasal level. The same conditions apply to the VoiceP-level, with the

⁹⁸Note that serialization cannot be reduced to restructuring, contra Aboh (2009); see e.g. Veenstra & Muysken (2017), who show that the two are to a certain extent independent.

difference that the external argument position is freed up as an additional locus for argument sharing; consequently, there are no transitivity restrictions at this level, which gives us Type A.

In C-SVCs, V-level adjunction yields profile Type C. Because V-level adjunction configurations yield base-generated complexes of terminal nodes, C-SVCs sidestep the need for integration to proceed via syntactic argument sharing. Consequently, the only transitivity restriction that arises at this level reduces to the IAR of adjoined transitive verbs and the MNC: whenever an adjoined transitive cannot have its IAR satisfied indirectly via co-indexation or existential binding through a head transitive verb, the head verb cannot itself be intransitive.

In SVCs built via adjunction, these transitivity restrictions are intimately connected to the restriction that arguments in adjoined verbal elements cannot canonically project into the syntax: they must either project as null arguments, modeled here as minimal pronouns (NC-SVCs), or remain entirely implicit (C-SVCs). In NC-SVCs, this follows from our theory of serialization via adjunction, while in C-SVCs, this follows from the MNC.

This effect is absent in a subset of NC-SVCs involving systematically recurring verbs with an argument-adding function, here analyzed as verbal periphrastic counterparts of morphological applicatives. As applicatives, these NC-SVCs involve complementation, and because integration in complementation proceeds via selection, there is no requirement that any argument in either the complement or the main projection be linked in some special way. That said, the applicative NC-SVCs discussed here – derived via VP-level complementation – robustly involved lexical verbs with direct objects, though this was surmised to likely be due to a coincidence and / or event-structural conditions.

Likewise, we do not expect any transitivity restrictions to arise from VoiceP-level complementation *per se*, a level which is often not regarded as falling under serialization due to the obviously complementing nature of the kinds of verbs that are typically involved, e.g. causatives.

Finally, the status of V-level complementation remains unclear; it may turn out to be a configuration that is unattested.

These results are summarized in Table 5.1:

VoiceP	$\Rightarrow$ Type B ($\geq[trans + trans]$)	
VP	$\Rightarrow$ Type B ($\geq[trans + trans]$)	
		status unclear
V	$\Rightarrow$ Type C ($*[trans + intr_H]$)	
	Adjunction	Complementation
Level of structure-building	Mode of structure-building	

Table 5.1: Typology of serialization via the mode and level of structure-building

The restrictions on transitivity we find in SVCs are thus largely configurational in that for the most part, they are conditioned by the way in which things are brought together in the syntax, in conjunction with the fundamental grammatical principles of the MNC and Full Interpretation. This also means that these restrictions are natural because nothing more needs to be stated in order to derive them. One exception to this is the strictness of a language's IAR, which – even though all languages discussed here show a strict IAR – can be considered partially ad hoc in the sense that it is simply an arbitrary fact that it is stronger in some languages and weaker in others. Still, once we allow for ad hoc restrictions such as the strictness of a language' IAR, we arrive at a fully derivational account of transitivity restrictions in SVCs, with all the restrictions we find following from basic grammatical principles, the properties of the parts involved, and how these are brought together.

Part II

Transitivity matching

Chapter 6

Semantic transitivity matching in Saliba

6.1 Introduction

This chapter presents the phenomenon of semantic transitivity matching, as found in Saliba. Section 6.2 outlines the basic properties of Saliba. Section 6.3 introduces C-SVCs; these show semantic transitivity matching, the topic of Section 6.4. Finally, Section 6.5 presents the syntax and semantics of Saliba C-SVCs.

6.2 Saliba

Saliba is an Oceanic language of southeast Papua New Guinea (Margetts 1999, Margetts 2005, Margetts 2008, Margetts 2013). It is largely head-final, agglutinating and head-marking. Subjects and objects are indexed on the verb via prefixes and suffixes, respectively, and the dominant constituent order is SOV; pronouns and lexical nominals are often omitted. A simple intransitive clause is given in (1a), a transitive one in (1b):

- (1) a. Se-lao-ko
 3PL.S-go-PERF
 ‘They left already’
 b. Kaleko ya-deuli-di
 clothes 1SG.S-wash-3PL.O
 ‘I washed the clothes’

(Margetts 1999: 13,93)

Margetts (1999) classifies Saliba as a *fundamentally intransitive* language with *rigid transitivity*: most verbal roots are intransitive, and most verbal stems can occur in either only intransitive or transitive frames, but not both. At the VP-level, this means that most simplex verbs generally cannot occur with objects without a valence-increasing derivation, and that verbs in general cannot occur with objects when intransitive,

and, conversely, cannot occur without objects when transitive.⁹⁹ Consequently, (i) most transitive verbs exhibit valence-increasing morphology, and (ii) most verbs require valence-increasing or valence-decreasing morphology in order to function as transitives and intransitives, respectively. Correlated with this is the fact that while Saliba has two productive valence-increasing operations, it lacks productive valence-decreasing operations (Margetts 1999: 145ff, 181ff). Ambi-transitives exist, i.e. verbs which can be used intransitively or transitively without valence-changing operations, but they are rare.

6.3 Contiguous serial verb constructions

Saliba serial verbs are contiguous combinations of two to four verbs behaving as single morpho-syntactic units (Margetts 1999, Margetts 2005), i.e. they are C-SVCs. Most of the verbs occurring in C-SVCs also occur as full verbs in isolation, and the meaning of a serialized verb always corresponds to its independent meaning.¹⁰⁰ A C-SVC consisting of two intransitive verbs is given in (2a), and (2b) shows a C-SVC consisting of two transitive verbs:

- (2) a. *intr + intr*

Ye-[kamposi]-[dobi]
3SG.S-jump-go.down
'He jumped down'

- b. *trans + trans*

Ye-[koi]-[kesi]-di
3SG.S-hit-break-3PL.O
'He broke them'
(literally: 'He hit-break them')

(Margetts 2005: 66f)

The subject of (2a) is understood as the sole participant of a single jumping-and-going-down event. Likewise in (2b), the subject denotes both the hitter and the breaker, and the thing that is broken is understood as identifying the thing that is hit. This kind of argument sharing always occurs with subjects, and also with objects as long as they are sortally identical (Margetts 1999). Despite the occurrence of more than one verb, (2a) and (2b) contain a single set of subject and object agreement affixes, enclosing the verbal complex, just like the simple verbs in (1).

The general template for Saliba C-SVCs is given in (3):

⁹⁹Transitive verbs need not always occur with a lexical object, but they can never occur without an object agreement suffix, except in the case of so-called discord verbs, discussed in the next chapter.

¹⁰⁰Certain verbs may acquire a more abstract meaning within C-SVCs, yet even here the particular kind of abstraction is predictable and systematically derivable from its independent lexical meaning.

(3) *Saliba C-SVCs*:

(Margetts 1999: 99)

(Subject) (Object) AGR_S-[V₁-...-V_F]-AGR_O

Morpho-syntactically, C-SVCs behave just like simple verbs. Their transitivity is not cumulative, i.e. a C-SVC consisting of two transitive verbs never yields a ditransitive construction with two objects, but a transitive construction with a single object. C-SVCs always have exactly one subject, and – if present – exactly one object.

6.3.1 Positional slots

Saliba C-SVCs characterize four positional slots (Margetts 2005), which I label *A*, *B*, *C* and *D*. Each slot contributes a typical meaning component to the overall meaning of the C-SVC. Interpretation-wise, Slot A houses the verb most readily construed as denoting the main event; verbs in Slot B typically express an event which expresses or entails what can be construed as the result of the event in Slot A; Slot C expresses the direction along which an action is carried out; and Slot D expresses manner or phasal modification. Depending on the construal of the overall macro-event, a reversed event-structural relationship between Slots A and B is at times equally plausible: Slot A may thus also be taken to provide the means event (rather than the main event), and Slot B be taken to provide the main event (rather than the result event). This is summarized below:

	A	B	C	D
Meaning	main event	cause + result	directionality	manner
	means	main event		phasal
# of stems	open	≥ 39	closed: 7	closed: 12

Table 6.1: Positional slots in Saliba C-SVCs (Margetts 2005)

These slots correspond to fixed relative positions that need not all be filled, except for Slot A. For example, (4b) contains Slots A and C; (4c) contains Slots A and D; and (4d) contains Slots A, C and D:

- (4) a. $trans_A + \underline{trans}_B$
 Ye-[koi]-[*(**he**)-beku]-Ø
 3SG.S-hit-CAUS-fall-3SG.O
 ‘He made it fall down’ (Margetts 1999: 99)
- b. $intr_A + intr_C$
 Ye-[kamposi]-[dobi]
 3SG.S-jump-go.down
 ‘She jumped down’ (Margetts 2005: 67)
- c. $intr_A + intr_D$
 Se-[paisowa]-[gehe]
 3PL.S-work-finished
 ‘They finish working’ (Margetts 2005: 67)
- d. $trans_A + ambi-trans_C + \underline{trans}_D$
 kwa-[tole]-[sae]-[uyo-i]-Ø
 2PL.S-put-raise-return-APPL-3SG.O
 ‘Put it back up’ (Margetts 2005: 67)

A C-SVC most commonly consists of two verbs, but examples with three (4d) and even four (5) verbs are possible:

- (5) $trans_A + trans_B + ambi-trans_C + \underline{trans}_D$
 Ye-[tu]-[isini]-[sae]-[kasaya-i]-Ø
 3SG.S-throw-raise-put.up-in.vain-APPL-3SG.O
 ‘He threw it up in vain’ (Margetts 2005: 67)

Jointly, these positional slots give rise to a kind of event-structural complementarity, whereby verbs take on certain "templatic" meaning components within the overall macro-event as a function of their disposition to their co-verbs.

6.4 Semantic transitivity matching

Saliba C-SVCs are subject to *semantic transitivity matching*, henceforth (*semantic*) *TM*. All constituent verb stems in C-SVCs must be either intransitive or transitive; any pairings of disparate transitivity values are ungrammatical. That is, Saliba C-SVCs allow only *intr + intr* and *trans + trans* pairings, but no *intr + trans* and *trans + intr* pairings.

Consider (6), where the V_1 ‘hit’ is transitive, but the root underlying V_2 , ‘fall’, is intransitive. Pairing these as they are is ruled out by TM, and consequently the transitivity of either verb stem must be adjusted. Because valence-decreasing operations are unproductive in Saliba, the only option is for the root of V_2 ‘fall’ to transitivize, in this case via the causative *he-*. The derived transitive V_2 ‘make fall’

can now be paired with the transitive V₁ 'hit' into a C-SVC (6):¹⁰¹

(6) *trans* + *trans*

Ye-[koi]-[*(**he**)-beku]-Ø
3SG.S-hit-CAUS-fall-3SG.O

'He made it fall down' (Margetts 1999: 99)

Similarly in (7), the intransitive root underlying V₂ 'go down' cannot combine with the transitive V₁ 'throw'. In order to do so, it must first undergo a valence increase, this time via the applicative *-(e)i*:

(7) *trans* + *trans*

Se-[gabae]-[dobi-*(**ei**)]-Ø
3PL.S-throw-go.down-APPL-3SG.O

'They threw it down' (Margetts 2005: 79f)

(8) is a minimal pair built on the same pair of roots, but differing in transitivity at the stem-level. In the intransitive C-SVC in (8a), 'sink' and 'return' can combine due to both being intransitive. By contrast, when either verb has been transitivized, so too must the other one (8b). Not transitivizing both verbs is ungrammatical (9):

(8) a. *intr* + *intr*

Ye-[yoli]-[uyo]
3SG.S-sink-return
'It sank again'

b. *trans* + *trans*

Ta-[**he**-yoli]-[uyo-**i**]-Ø
1SG.INC.S-CAUS-sink-return-APPL-3SG.O
'We sank it again'

(Margetts 1999: 105)

(9) a. *trans* + *intr*

*Ye-[**he**-yoli]-[uyo]
3SG.S-CAUS-sink-return

b. *intr* + *trans*

*Ye-[yoli]-[uyo-**i**]-Ø
3SG.S-sink-return-APPL-3SG.O

(Margetts 1999: 105)

(10) shows that [*intr* + *trans*] pairings are impossible: ¹⁰²

¹⁰¹As a language without resultatives, Saliba must resort to the means construction.

¹⁰²The intended meaning of (10) could be used in a situation where, say, someone dances on a table

(10) *intr + trans*

*Ye-[sobu]-[kesi]-Ø
3SG.S-dance-break-3SG.O

intended: 'She danced and broke it' (Anna Margetts p.c.)

Finally, when more than two verbs make up a C-SVC, TM must occur across the board on all verbs¹. This was evident in (4d) and (5) above, of which the latter is repeated here:

(11) *trans_A + trans_B + ambi-trans_C + trans_D*

Ye-[tu]-[isini]-[sae]-[kasaya-i]-Ø
3SG.S-throw-raise-put.up-in.vain-APPL-3SG.O

'He threw it up in vain' (Margetts 2005: 67)

The rarity of combinations in which all slots are filled is presumably due to strictness of TM, which heavily reduces the space of plausible complex meanings (Margetts 1999).

6.4.1 Transitivity matching has a semantic effect

One crucial property of TM as found in Saliba is that it always has a semantic import on both verbs (with one caveat in the case of a particular class of intransitive verbs, discussed in the next section). In other words, it is not a restriction that requires verbs to co-vary only formally; rather, the formal reflexes due to TM always also come with a predictable semantic effect. This is important, as there are other kinds of TM which lack semantic effects on both verbs.

First, consider the following C-SVCs involving the causative *he-*. In (12a), the presence of *he-* on the root underlying V₂, *mwaloi* 'dead', is what allows it to obtain the causative meaning 'kill', and in (12b), the presence of *he-* on V₂ entails that the action of drowning has exhaustively affected the set of object referents (Margetts 2005: 76):

to the effect of breaking it. This would have to be expressed using a bi-clausal construction with coordination:

- (i) a. Ye-sobu ede ginauli ye-ta-kesi
3SG.S-dance and.so thing 3SG.S-RES-break
'She danced and so the thing broke'
b. Ye-sobu ede ye-koi-kesi-Ø
3SG.S-dance and.so 3SG.S-hit-break-3SG.O
'She danced and so she hit-broke the thing' (Anna Margetts p.c.)

As we will see in the next chapter, *intr + trans* pairings like the one intended in (10) are well-formed in C-SVCs in other languages like Daakaka, which lack semantic TM.

- (12) C-SVC with causative *he-*
- a. Ye-[koi]-[he-mwaloi]- $\emptyset$
3SG.S-hit-CAUS-dead-3SG.O
'He hit it dead'
 - b. Ya-tupa-[he-yoli]-[he-gehe]-di
1SG.S-IMPACT-CAUS-sink-CAUS-be.finished-3PL.O
'I will drown all of them' (Margetts 2005: 70, 76)

The causative in serialization (12) thus behaves semantically exactly as it does outside serialization:

- (13)
- a. Se-lao-ko
3PL.S-go-PERF
'They left already'
 - b. Manuwa ya-he-lao- $\emptyset$
bird 1SG.S-CAUS-go-3SG.O
'I let the bird go' (Margetts 1999: 13, 170)

Second, the occurrences of valence-changing devices correlate with the presence of certain roles. We already know that the causative marker in (12) introduces a higher, causer role; regarding cases like (14), while the marker glossed as the applicative will be argued to not instantiate a canonical applicative, its presence nevertheless always correlates with an object that corresponds thematically to a lower, non-core role, which in (14) is a kind of concomitant object:

- (14) C-SVC with applicative *-(e)i*
- Se-[gabae]-[dobi-ei]- $\emptyset$
3PL.S-throw-go.down-APPL-3SG.O
'They threw it down' (Margetts 2005: 79)

Finally, if Saliba TM were a purely formal restriction, we should expect ambitransitive verbs to allow either a semantically intransitive or transitive interpretation regardless of the transitivity of their co-verb. This is not the case, as evinced by the behavior of the verb *sae*, which means either 'rise' / 'go up' or 'raise' / 'make go up' depending on which transitivity it occurs in. Once it occurs in a C-SVC, *sae* always takes on that meaning which accords with the transitivity of its co-verb:

- (15) C-SVC with *ambi-transitive* V_2
- a. *Intransitive* $V_1 \rightarrow$ *sae* is *semantically intransitive*
 Ye-[kita]-[sae]
 3SG.S-look-rise
 ‘He looked up’
- b. *Transitive* $V_1 \rightarrow$ *sae* is *semantically transitive*
 Ye-[helele]-[sae]-di
 3SG.S-reach-raise-3PL.O
 ‘She stretched them (her arms) up’ (Margetts 2005: 73)

6.4.2 Contextual bivalence

A handful of verbs occurring in Slot D cannot be transitive in isolation, but must exceptionally transitivize within C-SVCs in the context of a transitive co-verb. The verb *uyo* ‘return’, for example, has an intransitive use in isolation, with its subject referring to the person returning (16). But *uyo* ‘return’ on its own disallows any valence-increasing derivation, such as via the applicative, which would license an object referring to e.g. a concomitant object or the location returned to (16b).^{103,104} The same holds for *namwa* ‘be good’ in isolation: it occurs as an intransitive (17a), but cannot take objects (17b):

- (16) a. Se-[uyo]-ma
 3PL.S-return-hither
 ‘They came back’ (Margetts 2005: 75)
- b. *Se-[uyo-i]- $\emptyset$
 3PL.S-return-APPL-3SG.O
 intended: ‘They came back with it / to it’ (constructed)
- (17) a. Ka-na heyatu-ne ye-[namwa]
 CL1-3SG.POSS tattoo-DET 3SG.S-be.good
 ‘His tattoo is good’ (Margetts 2013: 13)
- b. *Ye-[namwa-i]- $\emptyset$
 3SG-be.good-APPL-3SG.O
 intended: ‘She is good because of it’ (constructed)

As intransitive verbs, *uyo* ‘return’ and *namwa* ‘be good’ readily occur in C-SVCs with intransitive co-verbs:

¹⁰³Either role may be introduced by the Saliba applicative *-(e)i*, depending on the base verb (Margetts 1999).

¹⁰⁴(16b) and (17b) are constructed examples. While Margetts (1999) and Margetts (2005) do explicitly list those verbs which cannot be transitive in isolation, but can exceptionally transitivize within C-SVCs, no specific examples showing their ill-formed transitive uses outside of serialization are given.

- (18) *intr + intr*
 Ye-[yoli]-[uyo]
 3SG.S-sink-return
 ‘It sank again’ (Margetts 1999: 105)

- (19) *intr + intr*
 Ye-[pitali]-[namwa-namwa]
 3SG.S-dry-RED-good
 ‘It is properly dry’ (Margetts 2005: 69)

However, *uyo* ‘return’ and *namwa* ‘be good’ can also occur in C-SVCs with transitive co-verbs. Strangely, it is only in this context that they can – and in fact must – become transitive:¹⁰⁵

- (20) Ta-[he-yoli]-[uyo-*(i)]-∅
 1INCL.S-CAUS-sink-return-APPL-3SG.O
 ‘We make it sink again’ (Margetts 2005: 80)
- (21) Ye-[he-kata]-[namwa-namwa-*(i)]-gai
 3SG.S-CAUS-learn-RED-be.good-APPL-1EXCL.O
 ‘She taught us properly’ (Margetts 2005: 66)

In (20), combining the transitive verb *he-yoli* ‘make sink’ with the intransitive verb *uyo* ‘return’ is only possible if the latter transitivizes via the applicative – despite the inability of *uyo* ‘return’ to independently function as a transitive (16b). Similarly, once the intransitive verb *namwa* ‘be good’ occurs in a C-SVC with a transitive co-verb like *he-kata* ‘make learn’, it must exceptionally transitivize despite its inability to do so as an independent verb (17b).

I term this property, i.e. the ability of a verb which is obligatorily intransitive and monovalent in isolation to exceptionally project objects within C-SVCs, *contextual bivalence*, and the verbs and C-SVCs giving rise to this kind of behavior *contextually bivalent verbs* and C-SVCs, respectively. We will see that in some languages, exceptional objects are realized obliquely, rather than structurally as in Saliba, which is why I have chosen this term over *contextual transitivity*; so verbs which are generally intransitive exceptionally undergo a valence increase in order to project an object as either a structural or an oblique dependent.

The generalization is that verbs like *uyo* ‘return’ and *namwa* ‘be good’ must occur in a specific context in order for this behavior to unfold. This licensing context corresponds to C-SVC configurations with a transitive co-verb, whereas the elsewhere, non-licensing context corresponds to any configuration outside of C-SVCs. Other contextually bivalent verbs are *nogowai* ‘be slow’ and *mwamwayau* ‘be quick’. See

¹⁰⁵It is unclear whether the meaning of ‘again’ in (20) constitutes a repetitive or restitutive reading.

Table 6.2:

Root	As independent verb	As verb within C-SVC	Contextual bivalence
<i>namwa</i>	'be good'	'properly'	via APPL, only as verb in Slot D
<i>nogowai</i>	'be slow'	'slowly'	
<i>mɔwamwayau</i>	'be quick'	'quickly'	
<i>uyo</i>	'return'	'back / again' REFL / RECIP	

Table 6.2: Contextually bivalent verbs in Saliba (Margetts 2005)

Contextually bivalent verbs systematically express meanings which would be expressed as manner adverbials in a language like English. As is common for Oceanic languages, Saliba lacks a dedicated, morpho-syntactically distinguished class of manner adverbs; instead, these meanings are expressed using verbs (Margetts 1999), which I will call *manner verbs*.

This raises the question of how contextually bivalent verbs within C-SVCs relate form and meaning. Specifically, how does the manner contribution of the contextually bivalent verb relate to the fact that it is the verb which realizes the object (we will see shortly that Saliba C-SVCs are headed syntactically by the string-final verb, meaning that objects are dependents of this verb)? That is, how does the meaning of (21), 'She teaches us properly' - where 'properly' modifies the teaching event, but stands in no thematic relationship to the 1PL thematic object - relate to the structure of (21), where the 1PL object suffix appears on the contextually bivalent verb *namwa-namwa* 'be good', rather than the verb *he-kata* 'teach', which it is thematically selected by?

As recognized by Bril (2004: 176), contextually bivalent C-SVCs do not involve proper semantic argument sharing, but rather what we can for now informally characterize as a kind of argument inheritance, because contextually bivalent verbs do not thematically select the arguments they realize. Similarly, Næss (2004: 236f) notes that contextually bivalent verbs do not express semantic transitivity, but a kind of formal "transitivity agreement". In 'We make it sink' (20), the 3SG object does not correspond to any participant of the returning event denoted by V_2 , but only to the thematic object of the sinking event. Likewise in 'She teaches us properly' (21), the 1PL object is thematically part of the teaching event, but not the doing-well event.

In other words, contextually bivalent verbs *morpho-syntactically* realize arguments which are *thematically* selected by another verb, in this case their transitive co-verb; this characterization suggests a similarity to raising. Indeed, I will argue below that – although technically distinct from classical syntactic raising – contextual bivalence is a kind of thematic raising, and that the two phenomena are conceptually analogous.

Contextual bivalence is not idiosyncratic to Saliba, but a recurring property of

C-SVCs in Oceanic. It is also found, for example, in Daakaka (Vanuatu). In the C-SVC in (22), the transitive V_1 *ling* ‘put’ combines with the V_2 *bup* ‘tumble’, which is obligatorily intransitive outside of serialization (23) (Kilu von Prince p.c.). In spite of its general resistance to transitivization, *bup* ‘tumble’ must exceptionally transitivize within the context of the transitive co-verb *ling* ‘put’:

(22) **Daakaka:** *trans* + *trans*

Ma [ling] [bup-ane] ding
REAL put tumble-TRANS mat

‘(S)he put the mat the wrong way up / with the wrong side facing up’

(von Prince 2015: 315)

(23) **Daakaka**

a. Ekuokuo mu bup
car REAL tumble
‘The car tumbled over’

b. *Ma bup-ane ding
REAL tumble-TRANS mat

(von Prince 2015: 316)

Within C-SVCs, *bup* ‘tumble’ provides manner a manner component, specifying that the action described by its co-verb is carried out in a weird, clumsy and inappropriate manner (von Prince 2015); specifically in (22), *bup* ‘tumble’ expresses that that object is caused to tumble or face the wrong way.

Other Oceanic languages with similar patterns are Äiwoo (Næss & Boerger 2008), Nêlêmwa (Bril 2004), Paamese (Crowley 2002), Pileni (Næss 2004) and Toqabaqita (Lichtenberk 2006). Note that unlike Saliba, these languages do not show semantic TM in C-SVCs, with the only kind of matching effect being limited to C-SVCs comprising a transitive V_1 and a contextually bivalent V_2 .¹⁰⁶

Morpho-syntactically, contextually bivalent verbs show an augmented transitivity potential in their serializing use relative to their use in isolation. Yet this augmentation does not come with a parallel augmentation in the semantics, given that contextually bivalent verbs express manner meanings and do not thematically select their objects. In other words, because contextually bivalent verbs are morpho-syntactically transitive but semantically intransitive, they exhibit a *form-meaning mismatch*. Directly connected to this property is the fact that contextual bivalence is always carried out

¹⁰⁶E.g. in Daakaka, an intransitive V_1 is free to combine with a transitive V_2 :

(i) **Daakaka:** *intr* + *trans*

Barar mon mwe vyan te vyan [mea] [save] apyang en=te vyan tevesye
pig also REAL go CONJ go jump pass fire DEF=MED go side

‘The pig also went and jumped over the fire to the other side’

(von Prince 2015: 314)

by a generic transitive morpho-syntax in a given language. In Saliba, for example, this is the marker *-(e)i*, which is glossed as an applicative, and the causative is never used for this purpose; cf. Table 6.2. This choice will be shown to follow from the fact that this suffix expresses Saliba’s generic strategy for deriving transitives from intransitives and for object-licensing.

Before moving on, we need to answer how contextual bivalence can be reconciled with my claim from the previous section that TM in Saliba is semantic. Recall that in that section, I hinted at a particular class of intransitive verbs for which the reflex to satisfy TM exceptionally does not come with a semantic effect – these verbs are the contextually bivalent verbs. The final generalization, then, is that TM in Saliba always has a semantic import on both verbs, except in the case of contextually bivalent verbs.

6.4.3 Interim summary

In sum: Saliba C-SVCs show semantic transitivity matching, which requires all verb stems to be either intransitive or transitive. Matching always has a semantic on both verbs, except in the case of contextually bivalent verbs; these verbs express manner meanings, and within C-SVCs in the context of a transitive co-verb, they show morpho-syntactic but not semantic transitivity.

6.5 The syntax and semantics of Saliba C-SVCs

We will now see that C-SVCs syntactically involve V-level adjunction configurations, and semantically, they involve our familiar rule of *Event Cumulation*, introduced in Section 5.3.1 of the previous chapter.¹⁰⁷

Note that I will occasionally refer to the verbal complexes underlying Saliba C-SVCs and resulting from V-level adjunction as *terminal nodes*, rather than descriptions based on the term *head*, even though the verbal complexes we will be concerned with consist of only non-phrasal, i.e. head material. However, because one of these verbs uniquely determines the projection properties of the overall complex, i.e. is the *syntactic head* (with other elements being adjuncts), terms based on *head* may make it confusing to talk about the *syntactic headedness* of such complexes.

6.5.1 Semantics

Semantic TM makes reference to the simplex argument structures of the individual verbs making up C-SVCs, and regulates the complex argument structures resulting from their combination. Because argument structure interacts closely with event structure, and because internal argument sharing and contextual bivalence will play

¹⁰⁷I am thankful to Malte Zimmermann for discussing with me in detail various formal-semantic issues in this section.

out at largely at the level of argument structure, I will present the minimally necessary event-structural semantics of C-SVCs that will allow us to model these effects.

As we know, SVCs cross-linguistically come with various semantic interpretations (Stewart 1998, Zimmermann & Amaechi 2018, Zimmermann & Amaechi 2020). I propose that Saliba C-SVCs are semantically composed via *Event Cumulation*, repeated in (24):

$$(24) \quad \textit{Event Cumulation:} \quad (\text{Zimmermann \& Amaechi 2018: 6}) \\ \exists E \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \& P(e_1) \& Q(e_2)]$$

For two-verb C-SVCs, Event Cumulation expresses a macro-event E consisting of two, symmetrically related sub-events e_1 and e_2 . The cumulation, i.e. unification, of the sub-events e_1 and e_2 into a macro-event E derives the fact that, for the purposes of inflection and operators like TAM and negation, Saliba SVCs behave as a single verb, regardless of how many verbs they contain. That is, any operators will necessarily scope over the complex macro-event E , but not be able to scope into E and modify any of its sub-events e_1 or e_2 individually. The symmetry of their cumulation is meant to reflect the fact that the semantic head verb of Saliba C-SVCs, i.e. what might plausibly be considered as expressing the main event, is variably construable and at times not easily identifiable. Event Cumulation applies to a heterogeneous set of atomic events that do not have constraints on causal, temporal or spatial cohesion. The sum operator $\oplus$ is Boolean in the sense that event composition yields a complex macro-event consisting of independent sub-events.

A comment is in order. Note that at a more abstract level, the difference between Event Cumulation and Predicate Modification is minimal: at their cores, both construals express a conjunction of events. The major difference between the two lies in whether or not the sub-event predicates end up predicating over the same event argument. This is not the case in Predicate Modification, which fuses them together without introducing a new event argument. This neutralizes the identities of the individual sub-events, making it impossible to distinguish sortally distinct internal arguments. By contrast, Event Cumulation brings the individual sub-events together under a new, higher event argument, the macro-event E . Because the individual sub-events retain their identities, sortally distinct internal arguments are automatically distinguished as a function of distinct sub-events.

Conjunction and the ability to distinguish internal arguments in terms of sub-events are the minimally necessary ingredients we need to understand semantic TM. See Section B.1 of the Appendix on why the event-structural semantics of Saliba C-SVCs is likely to be more complex.

6.5.2 Syntax

Saliba C-SVCs are adjunction configurations built at the V-level, and they are headed by the string-final verb, i.e. $V_{F=H}$. The diagnostics for syntactic headedness are (i) the argument linking profiles of transitive C-SVCs without object sharing; and (ii) certain morphological properties of transitive verbs as a function of their position within C-SVCs, in turn a reflection of their (non-)headedness.

6.5.2.1 The level of structure-building and headedness

Given that Saliba C-SVCs are strictly contiguous, I assume that their syntax involves structure-building at the V-level. In all of the Oceanic C-SVCs discussed here, in fact, no material whatsoever is ever permitted to surface inter-verbally within the verbal complex. This is noted explicitly by the authors, and nowhere in the relevant descriptive works of the Oceanic C-SVCs discussed here are there cases showing the presence of inter-verbal material.¹⁰⁸ Note that contiguity on the surface does not entail underlying contiguity, as there are various ways of deriving verb clusters from underlyingly phrasal verbal structures (Wurmbrand 2017). But Saliba C-SVCs show the hallmarks of true contiguity, as discussed in Section 2.3 of Chapter 2 and in the following. Later, we will come across a language with C-SVCs which I will argue show derived contiguity due to verb movement.

Determining syntactic headedness is less straightforward. One reason is that agreement in C-SVCs (25a) parallels that found on independent verbs (25b):

- (25) a. **Ye-[koi]-[kesi]-di**
 3SG.S-hit-break-3PL.O
 ‘He broke them’
 literally: ‘He hit-break them’
 b. **Ye-bahe-i-di**
 3SG.S-carry-APPL-3PL.O
 ‘He carried them’ (Margetts 1999: 99ff)

A third reason is that semantic and syntactic headedness need not coincide, that semantic headedness itself is a vague concept, and that translations may be misleading.¹⁰⁹ Consider the C-SVC in (26), for which Margetts (2005) gives two similar but distinct translations, each of which privileges a different verb in English:

¹⁰⁸The only exception to this which I have come across so far is one kind of construction found in Toqabaqita, though even here, the material that is allowed to occur inter-verbally is a clitic, i.e. is non-phrasal.

¹⁰⁹One clear case of semantic headedness is found in endocentric compounds like *blueberry*, which denotes a kind of berry with the property of blueness; such compounds thus involve a relationship of hyperonymy. By contrast, there is no hyperonymy holding in cases like (25a) between the verbs ‘hit’ and ‘break’; see Nóbrega & Panagiotidis (2020).

- (26) Ye-[sikwa]-[he-beku]-[dobi-ei]-Ø
 3SG.S-poke-CAUS-fall-go.down-APPL-3SG.O
 1. 'He poked it down'
 2. 'He made it fall down by poking it' (Margetts 2005: 67)

Luckily, we can tell based on the linking profiles of transitive C-SVCs without object sharing that C-SVCs are syntactically headed by the string-final verb. Consider the verb *ini* 'pour', which – when occurring on its own – has an object that denotes the thing poured (27). But when 'pour' occurs in a C-SVC as V_1 with *he-mwayau* 'CAUS-full' as $V_{2=H}$, the object of the complex verb denotes the thing filled, not the thing poured (27b). That is, the unique object of the two-verb C-SVC realizes the internal argument of the $V_{2=H}$ *he-mwayau* 'CAUS-full', while the sortally distinct internal argument of the V_1 *ini* 'pour' is blocked from realization:¹¹⁰

- (27) a. Ka-m **ti** ya-ini-Ø?
 CLF2-2SG.POSS **tea** 1SG-pour-3SG
 'Shall I pour you some tea?' (Margetts 2005: 81)
 b. **Kaputi** ku-[ini]-[he-mwayau]-Ø
cup 2SG.S-pour-CAUS-full-3SG
 'Fill the cup'
 (literally: 'Pour the cup full') (Margetts 2005: 81)

The internal argument of the V_1 'pour' remains implicit and cannot optionally be realized as an oblique (Anna Margetts p.c.). Similarly, when functioning as an independent verb, *usa* 'put in' takes an object referring to the thing undergoing a change of location (28a), while a C-SVC with *usa* 'put in' as V_1 and *he-mwayau* 'CAUS-full' as $V_{2=H}$ projects an object referring to the container that is filled, which realizes the internal argument of the $V_{2=H}$ 'CAUS-full' (28b).¹¹¹

- (28) a. **Apolo** ye-usa
apple 3SG.S-put.in
 'He put apples in'
 b. **Bosa wa labui** ye-[usa]-[he-mwayau]-di
basket GIVEN two 3SG.S-put.in-CAUS-full-3PL.O
 'He filled the two baskets' (Margetts 2005: 81f)

Based on the discussion in the previous chapter (Section 5.4.1), we know that this pattern of argument linking is systematic, as it is found in e.g. Mwotlap (François 2004) and Daaakaka (Hopperdietzel 2020).

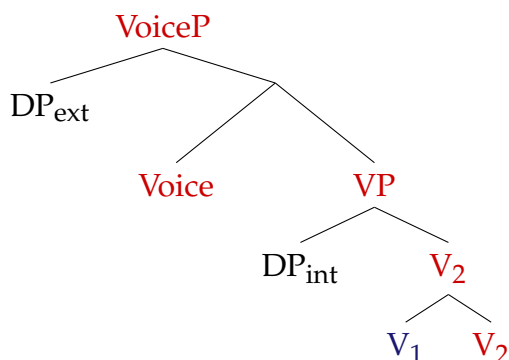
I take this pattern of argument linking, wherein the internal argument of the string-final transitive verb is privileged over that of the non-final transitive verb in

¹¹⁰The linking difference between (27) and (27b) corresponds to that of the locative alternation, as in *Alma sprayed paint (on the wall)* vs *Alma sprayed the wall (with paint)*.

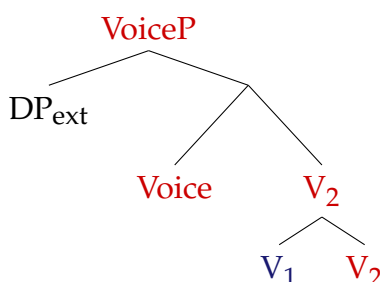
¹¹¹*usa* 'put in' in (28a) lacks an object suffix because it belongs to the class of so-called discord verbs, a subset of semantically transitive verbs that allow structurally reduced objects without an object suffix. Discord verbs are discussed in the next chapter.

terms of syntactic realization, to be a prime diagnostic for the syntactic headedness of the string-final verb in C-SVCs, and conversely, for the syntactic non-headedness of any non-final verb. Because $V_{F=H}$ uniquely projects given the MNC, only it can project its internal argument as object into the syntax. Any non-head verb cannot give rise to a projection line beyond the stem level, and so its internal argument is blocked from realization.

The basic syntax of a two-verb transitive C-SVC is given in Tree 6.1, and that of a two-verb intransitive (unergative) C-SVC in Tree 6.2:



Tree 6.1: Basic syntax of transitive C-SVC



Tree 6.2: Basic syntax of unergative C-SVC

6.5.2.2 The mode of structure-building

This brings us to the mode of structure-building, which I argue is adjunction.

First, Saliba C-SVCs are highly varied in the kinds of verbs they can consist of and meanings they can express, a great number of which would be highly unusual or even absurd as the result of complementation, which is known to yield a more restricted set of meanings (we would also need to assume rampant double lexical entries – one for a verb’s independent use, and another for its complementing, serializing use). This is not to say that all of the relevant verbs and meanings are implausible as involving complementation; rather, it is this very diversity that points to adjunction rather than complementation.

A second, directly related piece of evidence for adjunction is the fact that valence-changing morphology does not scope over the entire verbal complex, but only individual verbs. In transitive (29a), for instance, the causative *he-* occurs on, and

is interpreted on, each verb individually, and it entails exhaustivity with respect to the object. Compare intransitive (29b), where the causative is absent. If (29a) involved complementation, i.e. a configuration in which one verb were contained in the projection of the other, the causative on the matrix verb would necessarily scope over both the matrix verb and its complement, yielding a mereological meaning relationship between V_1 and V_2 , as in (30a). Yet this is not the case in (29a), where each causative occurrence is interpreted independently on each verb – i.e. the overall meaning is conjunctive, as in (30b):

- (29) a. *trans* + *trans*
 Ya-tupa-[**he**-yoli]-[**he**-gehe]-di
 1SG.S-IMPACT-CAUS-sink-CAUS-finished-3PL.O
 ‘I will drown all of them’
- b. *intr* + *intr*
 Se-[paisowa]-[gehe]
 3PL.S-work-finished
 ‘They finish working’ (Margetts 2005: 67)
- (30) *Varying construals of (29a)*
- a. *Complementation*
 (i) CAUSE(*sink*, CAUSE(*finish*))
 (ii) CAUSE(*finish*, CAUSE(*sink*))
- b. *Adjunction*
 CAUSE(*sink*) & CAUSE(*finish*)

For basically the same reason, if C-SVCs like (29a) involved complementation with V_2 as the syntactic head (in consistency with Saliba’s head-finality), we would expect the causative to occur only once on V_1 , not also on V_2 .

See Baker & Harvey (2010) and Svenonius (2016) on general differences between the kinds of meanings yielded by complementation vs adjunction in complex predication.

In fact, the presence of TM itself indicates adjunction rather than complementation, as a complementation analysis would fail to explain why changing the transitivity value of the complement verb must entail a semantically interpreted change in the transitivity of the matrix verb. The reason is that under complementation, the complement verb – be it intransitive or transitive – is always an argument of the matrix verb, and hence the matrix verb should invariantly be intransitive or transitive (depending on whether complement-taking predicates in a given language pattern (formally) as intransitives or transitives).

Also, since we know that objects are realized by $V_{F=H}$, wh-extracting an object out of a transitive C-SVC should be possible due to its complement status. This is

correct.¹¹²

- (31) a. Ye-[koi]-[he-mwaloi]-∅
 3SG.S-hit-CAUS-die-3SG.O
 ‘He hit it dead’
 b. **Saha_i** ye-[koi]-[he-mwaloi]-_i
 what 3SG.S-hit-CAUS-die
 ‘What did he hit dead?’ (Anna Margetts p.c.)

In sum, Saliba C-SVCs are composed syntactically via V-level adjunction, where the non-final verbs V_1 - V_{F-1} are verbal adjuncts to the final, head verb $V_{F=H}$. This analysis is consistent with the general head-finality of Saliba, and also with Margetts’ (1999) observation that there is hardly any indication of verbal complementation in the language, with ‘want’, ‘like’ and ‘help’ appearing to be the only verbs taking bona fide complements that are structurally smaller than full clauses (Margetts 1999: 17f).

6.5.2.3 Formal reflexes of syntactic headedness in C-SVCs

Only if a verb is the syntactic head of a C-SVC will it be able to project higher functional material, including Voice; any non-head verb within a C-SVC will consequently lack Voice. In principle, we should therefore expect to find formal reflexes in connection to Voice – which we do. We will see that non-head transitive verbs in C-SVCs may show a kind of form-meaning dissociation, in that – despite being just as semantically transitive as when occurring outside serialization as independent verbs – their stem morphology within C-SVCs is intransitive; by the same token, only head transitive verbs within C-SVCs can induce fully transitive stem morphology, in full reflection of their semantic transitivity. Verbal morphology thus diagnoses headedness asymmetries within C-SVCs.¹¹³

In Saliba, a first signature of (non-)headedness comes from the distribution of the suffix *-(e)i*, glossed as the applicative. As soon as a verb carries *-(e)i*, it is restricted to occurring as $V_{F=H}$ (Margetts 2005: 79):

- (32) a. Se-[gabae]-[dobi-**ei**]-∅
 3PL.S-throw-go.down-APPL-3SG.O
 ‘They threw it down’
 b. Ya-[deuli]-[kasaya-**i**]-di
 1SG.S-wash-in.vain-APPL-3PL.O
 ‘I washed them in vain’ (Margetts 2005: 79)

The behavior of *-(e)i* contrasts with that of the causative prefix *he-*, which is able to

¹¹²Note that this behavior is consistent with either a complementation or an adjunction account of Saliba C-SVCs, provided that the adjunction account treats the final verb as the head.

¹¹³Recall that in all the C-SVCs which are underlyingly truly contiguous discussed in this thesis, string-finality and syntactic headedness always coincide.

occur on both non-final / non-head (33a) and final / head verbs (33b):

- (33) a. *Causative he- on V_1*
 Ta-[**he**-yoli]-[uyo-i]- $\emptyset$
 1PL.INCL.S-CAUS-sink-return-APPL-3SG.O
 ‘We make it sing again’
- b. *Causative he- on $V_{2=H}$*
 Ye-[koi]-[**he**-beku]- $\emptyset$
 3SG.S-hit-CAUS-fall-3SG.O
 ‘He made it fall down’ (Margetts 2005: 80)

This restriction connects to an independent morpho-syntactic requirement on transitive verbs which are derived from intransitive verbs by a suffix that is generally glossed as an applicative or transitive marker, found not only in Saliba, but also other Oceanic languages. For Saliba, it is noted in Margetts (1999: 44) and Margetts (2005: 79) that the suffix *-(e)i* is obligatorily followed by an object suffix, and also that verbs allowing the applicative suffix cannot take an object suffix without also taking the applicative suffix (Margetts 2008: 35). More generally, Pawley & Reid (1980: 105) note that across Oceanic, nearly all transitive verbs carry both a transitivity-indicating suffix (glossed as APPL or TRANS), and – if present in the language – a φ -marking object suffix. I propose that the distribution of *-(e)i* reflects (non-)headedness: it can only surface on $V_{F=H}$, but not on any non-heads, because only $V_{F=H}$ can give rise to fully transitive stem morphology, in turn due to projecting Voice.

In examples, I have chosen to always retain the respective glossing by an author for this suffix – i.e. APPL or TRANS – though it should be kept in mind that whenever it is glossed as APPL, it often does not reflect the addition of an applied argument with one of the roles typically associated with applicative morphology, such as an Instrument or a Benefactor; as we will now see, this suffix also plays a role in nominal licensing, with its presence reflecting a fully transitive morpho-syntax. For convenience, I will call this suffix a *transitive suffix*.

In Oceanic, semantically transitive verb stems in certain configurations surface as morphologically intransitive forms, provided that the relevant verb visibly gives rise to such an alternation. While some irregularities exist, the generalization is that these verb forms (i) do not index the φ -features of objects, (ii) lack a transitive suffix, and (iii) correspond to the verb’s intransitive allomorph, if there is one (Lichtenberk 2008: 563ff.). In case I wish to call on such verbs in terms of the dissociation between their semantic transitivity and their formal intransitivity, I will refer to them as *combining forms*, following an established tradition in the Micronesian literature (Rehg 1981, Lichtenberk 2008); when making reference to them based purely on their intransitive morphology, I will refer to them as *bare forms*. This latter term will come in hand when we consider morphologically identical verb forms differing in their semantic transitivity.

It is useful at this point to venture into another Oceanic language with analogous C-SVCs (albeit without TM). In Toqabaqita (Solomon Islands), most transitive verbs occurring in isolation obligatorily bear a transitive suffix (Lichtenberk 2008). While its exact form depends on various factors, the generalization is that this suffix occurs in the presence of a full DP object denoting a Patient or some other non-core role¹¹⁴

When *thau* ‘hit’ occurs outside of C-SVCs with a full DP object (34), it obligatorily carries a transitive suffix and, depending on whether the object is lexical or pronominal, also a ϕ -marking object suffix:¹¹⁵

(34) **Toqabaqita**

- a. Keka thatham-i-a bo=naqa keki **thau-ngi-a** qaburu qeri
 3PL.SEQ want-3.O ASSRT=INTS 3PL.FUT **hit-TRANS-3.O** ogre that
 ‘They wanted very much to kill the ogre’ (Lichtenberk 2008: 1057)
- b. ... ada keka **thau-ngi** qoe
 TIMIT 3PL.SEQ **hit-TRANS** 2SG
 ‘...they might beat you up / kill you’ (Lichtenberk 2006: 267)

When *thau* ‘hit’ occurs with a structurally reduced object (35a) or without an object altogether (35b), it occurs with neither the transitive nor the object suffix, i.e. in its bare form:

(35) **Toqabaqita**

- a. Nau ku rongo-qi teqe wane qe **thau** wane
 1SG 1SG.NFUT hear-DETR one man 3SG.NFUT **hit** person
 ‘I heard that a man had killed someone’
- b. ... ma sui si doo neqe ka **thau** laqu qi ei ...
 and but PRTT thing this 3SG.SEQ **hit** ADD LOC LOC PRO
 ‘...but then this thing hit [me] there...’ (Lichtenberk 2006: 137, 1326)

Now, once *mau* ‘hit’ enters into a C-SVC as a V_1 with a transitive $V_{2=H}$, it must occur in its bare form (36); this is in spite of the fact that *mau* ‘hit’ in (36) is semantically transitive, as can be seen from the translation, in which the 2SG object is interpreted as undergoing both the beating and the killing. Only the $V_{2=H}$ *mae-li* ‘kill’ bears a transitive suffix:¹¹⁶

¹¹⁴Productive causativization in Toqabaqita is carried out by the causative prefix *faqa-* (Lichtenberk 2008: 107).

¹¹⁵Note that the verb *thau* is variously translated as ‘hit’, ‘beat’, ‘strike’ or ‘kill’, depending on context; it is simply glossed as ‘hit’ in the examples as this seems to be its most basic meaning.

¹¹⁶Note that in rare cases, TRANS also appears on transitive verbs with a causative meaning, e.g. in the form of *-li* on *mae-li* ‘kill’ in (36), based on the root *mae* ‘be dead / die’. Because this behavior is limited to a few intransitive roots denoting states or changes of states (Lichtenberk 2008: 100ff), and because I will take transitive suffixes to spell-out Voice heads which have agreed with both an internal and an external argument, I assume that such cases involve a morphologically null causative-inchoative / causative-stative alternation, akin to English *The pot broke* vs *Alma broke the pot*. So TRANS does not mark the alternation per se (regardless of whether this alternation follows from distinctly flavored *v* heads (e.g. v_{be} vs v_{become} vs v_{cause}), or automatically from the presence of both an internal and an external argument), but rather the presence of a full DP internal argument is afforded by the alternation. That *-li* in *mae-li* ‘kill’ does not directly reflect a specific thematic role, but more generally

(36) **Toqabaqita**

Ada keka [thau] [mae-li] qoe
 TIMIT 3PL.SEQ **hit**.CF die-TRANS 2SG

‘(Be careful,) they might kill you by beating you up / they might beat you to death’
 (Lichtenberk 2006: 267)

thau ‘hit’ occurs with fully transitive morphology only if it occurs with a full DP object, and in its bare form in all other cases: as an independent transitive verb with a structurally reduced object (35a), as an independent unergative (35b), and as a non-head, semantically transitive V_1 in a C-SVC (36). In the latter case (36), it is also a combining form because it remains semantically transitive. The reason is that *thau* heads its own projection as an independent transitive, but not as a non-head V_1 within C-SVCs.

Formal reflexes of headedness may also be suppletive. In isolation, the verb ‘eat’ surfaces as *fanga* when unergative (37a), and as suppletive *qani* when transitive (37b), without a transitive suffix:

(37) **Toqabaqita:**a. *Independent unergative*

Nau ku **fanga** sui naqa
 1SG 1SG.NFUT **eat** COMPL PERF
 ‘I’ve finished eating’

b. *Independent transitive*

Qani-a alo manga kai qaaqako
eat-3.O taro time 3SG.FUT be.hot
 ‘Eat the taro while it’s hot’

When ‘eat’ occurs as a non-head V_1 in a C-SVC (38), it is formally identical to its independent unergative variant *fanga* in (37a) despite being semantically transitive. Its semantic transitivity can be seen from the fact that the object of (38) originates thematically from the V_1 ‘eat’ even though it is realized as an oblique dependent of the contextually bivalent $V_{2=H}$ *baqita* ‘be big’:

the morpho-syntactic addition of a lower argument, is further supported by other cases where *mae-li* does not mean ‘kill’, but rather ‘die for’, i.e. its object refers not to a Patient of killing but to a Reason or Stimulus of dying:

- (i) Nau ku thaofa, kwa mae-li kuqu laqu boqo
 1SG 1SG.NFUT be.hungry 1SG.SEQ die-TRANS drink ADD ASSRT
 ‘I’m hungry and thirsty as well’
 literally: ‘I’m hungry and dying for drinking as well’

(Lichtenberk 2008: 972)

(38) **Toqabaqita:**

Nau ku [fanga] [baqita] qana alo
 1SG 1SG.NF **eat.CF** be.big OBL taro

‘I ate a lot of taro’

(Lichtenberk 2006: 266)

So ‘eat’ occurs as a bare form in both (37a) and (38), but it is also a combining form in (38). As above for the applicative suffix *-(e)i* in Saliba, I take the obligatory requirement on non-final transitive verbs in C-SVCs to occur in their combining forms in Toqabaqita to be a direct reflection of their non-head status within C-SVCs.¹¹⁷ This pattern is systematic: other Oceanic languages in which non-final transitive verbs in C-SVCs must occur as combining forms include Äiwoo (Næss & Boerger 2008), Daakaka (von Prince 2015, Hopperdietzel 2020), Fijian (Crowley 2002), Hoava (Davis 2003), Kokota (Palmer 2009), Paamese (Crowley 2002), Tawala (Ezard 1997) and Ughele (Frostad 2012). I take this to mean that in all these cases, non-final transitive verbs are non-heads, and conversely final transitive verbs are heads.

Following recent work on transitivity-conditioned patterns in Daakaka (Hopperdietzel 2020) and Fijian (van Urk 2020a), I propose that these formal reflexes of $V_{F=H}$ are triggered by properties of the head Voice, which projects uniquely from $V_{F=H}$. Because Voice introduces the external argument, assigns objective case and gives rise to object agreement (if present in the language), $V_{F=H}$ is the only verb that can induce a fully transitive morphology (i.e. the combination of what is in the literature glossed as either an applicative or a transitive suffix and – if present – object agreement). These formal reflexes can only arise if Voice has two sets of uninterpretable φ -features that have both been checked by an internal and an external argument in need of licensing, as schematized in Tree 6.3:

¹¹⁷The combining forms of some transitive verbs are found only in C-SVCs, i.e. they lack an independent unergative use that would correspond to a bare form. The verb *faa* ‘weave’, for example, is attested as an independent transitive (i), as an incorporating verb (ib) and as a V_1 in C-SVCs (ic), but not as an independent unergative:

(i) **Toqabaqita:**a. *Independent transitive*

faa-li-a waqi
 weave-TRANS-3.O basket
 ‘weave a/the basket’

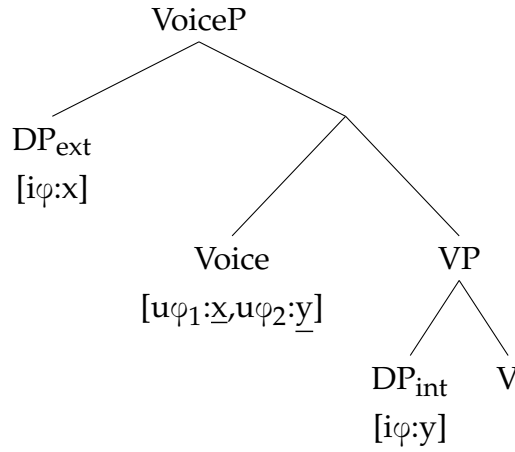
b. *Incorporation*

faa waqi
 weave.DF basket
 ‘weave a basket/baskets’

c. *V_1 in C-SVC*

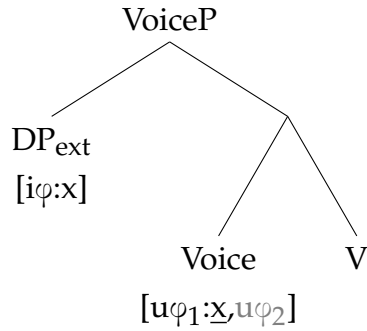
Kini qe [faa] [qaliali] qana teeteru
 woman 3SG.NFUT **weave.CF** be.quick PREP teeteru
 ‘The woman quickly wove a fan’

(Lichtenberk 2008: 265)



Tree 6.3: Voice in transitives

In the context of unergative verbs (Tree 6.4), Voice fails to find a φ -bearing goal in its c-command domain, such that its second licensing feature remains unchecked and no transitive morphology is spelled out:



Tree 6.4: Voice in unergatives

Voice also fails to find a φ -bearing internal goal in the context of structurally reduced objects, which do not require Case-licensing; in unaccusatives, Voice is either absent entirely (Hopperdietzel 2020, van Urk 2020a), or semantically expletive (Alexiadou et al. 2015), in which case it finds only an internal, but no external φ -bearing internal goal. Either way, transitive morphology is only spelled out if Voice contains two sets of checked uninterpretable φ -features.

The behavior of a verb like *thau* ‘hit’, which alternates transparently between a form without (35),(36) and with a transitive suffix (34) depending on the morpho-syntactic context, is summarized in Table 6.3:

	Transitive V₁	Independent unergative	Independent transitive
Status	non-head	head	head
Voice	–	$[u\phi_1:X, u\phi_2]$	$[u\phi_1:X, u\phi_2:Y]$
Property set	$\sqrt{hit}$	$\sqrt{hit} / \text{--- Voice}_{[u\phi_1:X, u\phi_2]}$	$\sqrt{hit} / \text{--- Voice}_{[u\phi_1:X, u\phi_2:Y]}$
Spell-out	<i>thau</i> combining form	<i>thau</i> bare form	<i>thau-ngi</i> fully transitive form

Table 6.3: Spell-outs of $\sqrt{hit}$ depending on context

In some cases, Voice may give rise to allomorphy, as with the verb *fanga* / *qani* ‘eat’; cf. (37) and (38). This behavior is summarized in Table 6.4:¹¹⁸

¹¹⁸Both as an independent unergative (37a) and as a semantically transitive V₁ in a C-SVC (38), ‘eat’ takes the form of *fanga*, while its independent transitive use outside C-SVCs requires the form *qani* (37b). The allomorphy of *fanga/qani* ‘eat’ cannot be conditioned by the semantic presence of an Agent, because all three forms denote an agentive activity of eating, and it cannot be conditioned by the semantic presence of a Patient, because both the V₁ use of ‘eat’ (*fanga*, i.e. its combining form) and its independent transitive use (*qani*) are semantically transitive. Allomorphy cannot be governed by the absolute presence of Voice alone, as this would predict the independent unergative and transitive uses of ‘eat’ to be formally identical, contrary to fact (cf. *fanga* vs *qani*). For this reason, allomorphy must be able to see whether Voice has entered into ϕ -agreement with both an internal and external argument.

	Transitive V₁	Independent unergative	Independent transitive
Status	non-head	head	head
Voice	–	$[u\phi_1:\underline{x}]$	$[u\phi_1:\underline{x}, u\phi_2:\underline{y}]$
Property set	$\sqrt{eat}$	$\sqrt{eat} / \text{— Voice}_{[u\phi_1:\underline{x}]}$	$\sqrt{eat} / \text{— Voice}_{[u\phi_1:\underline{x}, u\phi_2:\underline{y}]}$
Spell-out	<i>fanga</i> combining form	<i>fanga</i> bare form	<i>qani</i> fully transitive form

Table 6.4: Spell-outs of $\sqrt{eat}$ depending on context

Coming back to Saliba, the suffix $-(e)i$ then simply corresponds to what is glossed as TRANS in languages like Toqabaqita; $-(e)i$, together with the object suffix, reflects a Voice head with two checked sets of uninterpretable ϕ -features.

We thus derive the occurrence of fully transitive morphology in all and only those contexts with full DP objects, in a way that may dissociate from semantic transitivity. Next to diagnosing (non-)headedness within C-SVCs, these formal asymmetries also provide independent support for my claims that (i) structure-building proceeds at the V-level, and (ii) objects of transitive C-SVCs are dependents of $V_{F=H}$.¹¹⁹ Moreover, that allomorphy within C-SVCs is restricted to head verbs is consistent with the fact that adjunct-triggered allomorphy remains unattested (Weisser 2018, Weisser 2019, Hopperdietzel 2020).¹²⁰

6.6 Summary

This chapter presented the phenomenon of semantic TM as found in Saliba C-SVCs. In semantic TM, every element involved is a verb, and the matching reflex has a semantic effect on every verb. We saw that Saliba C-SVCs showing semantic TM involve Event Cumulation and adjunction at the V-level, and that they are head-final.

¹¹⁹Pileni is an exception to the generalization that transitive suffixes can only appear on $V_{F=H}$ (Næss 2004: 236). It is the only exception I am aware of, and I am presently unable to explain it.

¹²⁰Under this analysis, the Oceanic suffixes glossed as TRANS / APPL have a largely morpho-syntactic purpose, with any apparent meaning contribution they may have being (largely or entirely) predictable from the verbal root it attaches to – this is a welcome result. As shown in the next chapter, Saliba verbs which allow the suffix $-(e)i$ are a proper subset of so-called discord verbs, which may optionally occur with structurally reduced objects without the need for either $-(e)i$ or an object suffix, yet crucially without there being a change in meaning with respect to the thematic role of the object.

Chapter 7

The grammar of semantic transitivity matching

7.1 Introduction

This section provides a detailed analysis of the grammar of semantic TM in Saliba. Beginning in Section 7.2, we will see that semantic TM is restricted to V-level adjunction configurations, and that it results from the grammar forcing the semantic types of verb stems to match. Section 7.3 presents the rule that will be used to achieve this. Section 7.5 is dedicated to contextual bivalence, which I will argue instantiates a kind of proxy argument realization. Section 7.5 closes this chapter by considering a class of verbs which appear to be problematic for the present analysis, showing that they are in fact consistent with it.

7.2 Transitivity matching involves type-matching

Transitivity matching is a restriction that cannot be traced back to a property of any of the individual verbs making up C-SVCs. There is nothing about these verbs in and of themselves that is able to force a co-verb to match their transitivity. This, in conjunction with the fact that Saliba C-SVCs involve adjunction, means that the matching restriction enforced by semantic TM cannot have a selectional source.

Those grammatical sources of Saliba which come to mind as being most intimately connected to semantic TM are rigid transitivity on the one hand, and the strict contiguity of C-SVCs on other. Rigid transitivity offers an explanation for why transitive verbs may in certain contexts need to undergo a valence reduction, given the general need that their IAR be satisfied. One such context is found in contiguous V-level configurations, with an adjoined transitive V_1 and a head transitive $V_{2=H}$; because (i) V_1 cannot project its internal argument, because (ii) $V_{2=H}$ is intransitive, and if (iii) existential binding of V_1 's internal argument is unavailable, there is no

way of satisfying the adjoined V_1 's IAR, and so the derivation crashes.

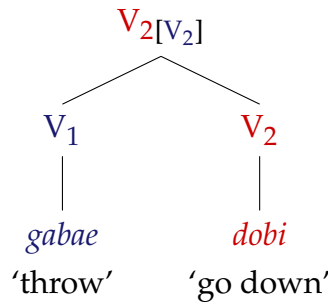
The transitive verb *gabae* 'throw', for instance, is unable to enter C-SVC formation as a V_1 with the unergative verb *dobi* 'go down' because *gabae* 'throw' cannot project its internal argument due to the MNC, and because the unergative $V_{2=H}$ *dobi* 'go down' is unable to take on the task of satisfying V_1 's IAR indirectly:

(1) *trans + intr*

*Se-[*gabae*]-[*dobi*]
3PL.S-throw-go.down

intended: 'They threw (something) down'

(Margetts 2005: 80)



Tree 7.1: [*trans + intr*] pairing

- (2) a. $\llbracket V_1 \rrbracket = \llbracket gabae \rrbracket = \lambda x \lambda e_1. [\text{throw}(e_1, x)]$ $\langle e, vt \rangle$
 b. $\llbracket V_2 \rrbracket = \llbracket dobi \rrbracket = \lambda e_2. [\text{go down}(e_2)]$ $\langle v, t \rangle$
 c. $\llbracket V_2[V_2] \rrbracket = \lambda E. \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ \text{throw}(e_1, \boxed{x}) \ \& \ \text{go down}(e_2)]$ $\langle v, t \rangle$
 $\downarrow$

Nothing more needs to be said on pairings consisting of two (unergative) intransitives, and pairings of two transitives are similarly predicted to be well-formed under rigid transitivity, provided that V_1 's IAR can be satisfied either by (i) co-indexation with $V_{2=H}$'s syntactically realized internal argument, or (ii) existential binding.

Under rigid transitivity alone, we thus predict a 3/4 pattern with respect to licit transitivity pairings in head-final C-SVCs: [*intr + intr_H*], [*trans + trans_H*], [*intr + trans_H*] pairings are well-formed, while [*trans + intr_H*] pairings are ruled out. [*intr + trans_H*] pairings thus escape the net, for which rigid transitivity offers no explanation. It remains unclear why in Saliba, an unergative V_1 and a transitive $V_{2=H}$ cannot be paired, as in (3):

(3) *intr + trans*

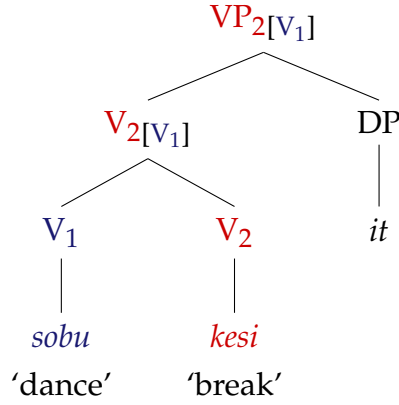
*Ye-[*sobu*]-[*kesi*]- $\emptyset$
3SG.S-dance-break-3SG.O

intended: 'She danced and broke it' / 'She broke it by dancing (on it)'

(Anna Margetts p.c.)

Given the denotations in (21), the intended meaning of (3) would instantiate a

means construction that is event-structurally well-formed, with ‘dance’ expressing the means component and ‘break’ entailing the result state. Because the projection-blocking effect of the MNC has no effect on the unergative V_1 ‘dance’, *sobu* ‘dance’ should be able to give rise to the C-SVC in Tree 7.2 with the final meaning in (5), where no IAR is left unsatisfied:



Tree 7.2: [*intr* + *trans_H*] pairing

- (4) a. $\llbracket V_1 \rrbracket = \llbracket \textit{sobu} \rrbracket = \lambda e_1. [\textit{dance}(e_1)]$ $\langle v, t \rangle$
 b. $\llbracket V_2 \rrbracket = \llbracket \textit{kesi} \rrbracket = \lambda y \lambda e_2. [\textit{break}(e_2, y)]$ $\langle e, vt \rangle$
- (5) $\llbracket VP_2[V_2] \rrbracket = \lambda E. \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ \textit{dance}(e_1) \ \& \ \textit{break}(e_2, it)]$ $\langle v, t \rangle$

Yet (3) is ungrammatical in Saliba. Compare Paamese and Daakaka, languages without semantic TM, where the intended meaning of (3) is expressible as a well-formed C-SVC comprising an unergative V_1 and a causative $V_{2=H}$:

- (6) **Paamese:** *intr* + *trans*

Kaie $\emptyset$ -[*matilu*]-[*vaasi*] aai
 3SG 3SG.REAL-sleep-split plank

‘(S)he slept on the plank, thereby splitting it’ (Crowley 2002: 95)

- (7) **Daakaka:** *intr* + *trans*

Bong ma [*tas*] [*tiwiye*] etastas
 B. REAL sit break bench

‘Bong broke the bench by sitting (down)’ (Hopperdietzel 2020: 302)

It follows that the presence of rigid transitivity alone cannot explain semantic TM. We therefore need to parametrize TM as an ad hoc grammatical requirement of Saliba

on C-SVC formation.^{121,122}

7.2.1 Semantic TM is restricted to V-level configurations

To see that semantic TM is specifically tied to C-SVC formation, as opposed to complex clause formation in general, let us consider three other types of complex clauses where it is systematically absent. First, there is coordination, which – next to allowing pairings of like transitivities, also allows disparate ones:

(8) *Coordination*

a. *intr + trans*

[Ye-sobu] ede [ye-koi-kesi-Ø]
 3SG.S-dance and.so 3SG.S-hit-break-3SG.O
 ‘She danced and so she hit-broke the thing’ (Anna Margetts p.c.)

b. *trans + intr*

Hewa-hewali-o-wa [se-kai-kaikewa-i-Ø] na kabo
 RED-young.man-PL-PM 3PL-RED-look-APPL-3SG.O CONJ TAM
 [ye-maliwai]
 3SG.S-vomit
 ‘The young men were watching her and then she vomitted’
 (Margetts 1999: 17)

Second, semantic TM is also absent in complementation configurations. Complement clauses in Saliba involve either modal, perception or desiderative matrix verbs, and are sentence-like in the sense of Noonan (1985) in generally allowing the same material as root clauses (Margetts 1999, Margetts 2004). Complement clauses can freely be headed by both intransitive (9a) and transitive verbs (9b) without the matrix verb co-varying in transitivity:

¹²¹Note that Hopperdietzel (2020: 322ff) analyzes Daakaka C-SVCs like (7) not as involving V-level adjunction configurations in which the object of the head verb is added after complex verb formation, but rather as involving V-to-VP adjunction whereby a means-denoting V_1 is adjoined to a causative VP_2 . Crucially, though, all verbs occurring as V_1 s in C-SVCs like (7) are argued to not project beyond the stem level, i.e. are terminal nodes in the present sense, to the effect that if V_1 were transitive, it would be just as unable to project its internal argument as transitive V_1 s in V-level configurations. In order to derive the fact that a non-head transitive V_1 can never project its internal argument, Hopperdietzel (2020) argues that it must be existentially bound because Voice is missing, i.e. the head that assigns objective case. As far as I can see, however, this is a stipulation. That is, why must a manner component adjoined to a maximal projection expressing a causative component be a terminal node, as opposed to also a maximal projection? Hopperdietzel (2020: 324) does note that existential binding of internal arguments is a freely available option in Daakaka, but this does not explain why it becomes obligatory with manner V_1 s in the context of SVCs. Therefore, I assume instead that (i) they involve V-to-V adjunction and not V-to-VP adjunction, and (ii) their objects are introduced as dependents of V_1 after complex predicate formation and not before; this derives the non-projection of non-head transitive V_1 s for free by simply appealing to the MNC.

¹²²Some degree of ad hoc parametrization is needed independently, e.g. in order to model the fact that some languages have contiguous but not non-contiguous C-SVCs, or vice versa. See Aikhenvald & Dixon (2006) and Aikhenvald (2018).

(9) *Complement clauses*a. *Intransitive complement clause*

Ya-henuwa [bena ku-lao-ma]
 1SG.S-like COMP.OBLIG 2SG.S-go-HITHER
 'I want you to come'

b. *Transitive complement clause*

Ye-henuwa [bena keke ye-gabu-Ø]
 3SG.S-like COMP.OBLIG cake 3SG.S-bake-3SG.O
 'She wants to bake cake'

(Margetts 2004: 74)

Third, besides C-SVCs, Saliba also has NC-SVCs expressing motion cum purpose (Margetts 2004). Even though these restrict the V_1 to an intransitive motion verb, they place no transitivity restriction on the purpose V_2 :

(10) *NC-SVCs*a. *intr + intr*

[Ya-lao] [ya-keno]
 1SG.S-go 1SG.S-sleep
 'I go and sleep'

b. *intr + trans*

[Ya-dobi] [Sioni ya-kita-Ø]
 1SG.S-go.down S. 1SG.S-see-3SG.O
 'I go down and see Sioni'

(Margetts 2004: 75)

Semantic TM is hence restricted to V-level adjunction configurations, and absent in larger, phrasal configurations.

7.2.2 TM tracks semantic types at the stem level

I propose that semantic TM is the result of the grammar tracking the semantic types of verb stems in the context of C-SVC-formation, prior to the introduction of any arguments, and requiring these to match. These stems can be simple or complex, i.e. they can contain derivational morphology such as causatives or incorporated nouns; but they must crucially lack any kind of structural argument. Because only predicates of type $\langle v, t \rangle$ or $\langle e, vt \rangle$ can respectively be paired with one another, all verb stems in Saliba C-SVCs must be either intransitive or transitive. Because TM in Saliba directly tracks semantic transitivity in terms of the types of verb stems without objects, it follows that the reflexes occurring in order to satisfy TM always have a semantic effect on all co-varying verbs.

I choose to model this sub-type of TM as an effect that is tied to the semantics of C-SVC formation, even though this could surely also be achieved syntactically, e.g. via syntactic features. My reason for going the semantic route is that – as we have

seen above and in the previous chapter – languages like Daakaka and Mwotlap have C-SVCs with a syntax identical to their Saliba cousins, except that only the C-SVCs in Saliba show matching. So while semantic TM is clearly conditioned by the syntactic configuration, having the grammar ultimately enforce it via a matching restriction on semantic types seems to me to be the most direct way to do it.

If TM indeed tracks the semantic types at the stem-level, we should expect incorporation to have an effect on a verb's potential for C-SVC formation.¹²³

This is borne out, as shown by the intransitive C-SVC in (11). In terms of stems, (11) consists of the transitive verb *kabi* 'make / touch', and the intransitive verb *gehe* 'be finished'. Due to appearing with an incorporated noun *kabole* 'sago', *kabi* 'make / touch' is rendered intransitive, and therefore able to combine with intransitive *gehe* 'be finished'. In terms of transitivity, (11) is therefore parallel to (12), which consists of two simplex intransitives:^{124,125}

¹²³Incorporated nouns in Saliba are either pre-verbal (ia) or post-verbal (ib), with pre-verbal incorporation being the dominant type (Margetts 1999: 203). The incorporate always surfaces as part of the complex verb stem within the domain of inflection. In non-incorporating clauses, the object is invariantly pre-verbal and outside the domain of inflection (ii):

(i) *Incorporation outside C-SVCs*

a. *Pre-verbal type*

Se-**koya**-tudai
3PL.S-**garden**-dig
'They garden-dig'

b. *Post-verbal type*

Se-kabi-**numa**
3PL.S-make-**house**
'They house-built'

(ii) *No incorporation*

a. Koya se-tudai-Ø
garden 3PL.S-dig-3SG.O
'They dig a garden'

b. Numa se-kabi-Ø
house 3PL.S-touch-3SG.O
'They build a house'

(Margetts 1999: 203)

As is generally the case cross-linguistically, the incorporated noun lacks semantic and syntactic independence, displaying low individuation, generic specification of the activity denoted by the verb, and strict adjacency to the verb (Margetts 1999: 206f).

¹²⁴If incorporation occurs syntactically, it must precede complex verb formation, i.e. *kabi* 'make' must incorporate *kabole* 'sago' before being adjoined to the head *gehe* '(be) finished'. Alternatively, incorporation is lexical, and the V+N complex *kabi-kabole* 'sago-make' enters the syntax as a prefabricated unit, which is subsequently adjoined to *gehe* '(be) finished'. Either way, the detransitivizing effect of incorporation on V_1 must be present in the derivation before V_1 is adjoined to $V_{2=H}$.

¹²⁵Recall how examples like the following, where the $V_{2=H}$ is based on the same root as in (11), *gehe* 'be finished', but in its causative variant, show that adjunction is involved, not complementation. Even though is expected given the causative meaning of $V_{2=H}$, the fact that it must also become transitive when its co-verb V_1 becomes transitive shows that the V_1 is not part of a projection contained within $V_{2=H}$. This co-variance follows, however, if the two are combined via adjunction, which yields a conjunctive rather than a mereological relationship:

(i) *trans + trans*

(11) *intr + intr*

Se-[kabi-kabole]-[gehe]
 3PL.S-make-sago-be.finished

‘They finished making sago’

(Margetts 2005: 69)

(12) *intr + intr*

Se-[paisowa]-[gehe]
 3PL.S-work-be.finished

‘They finish working’

(Margetts 2005: 67)

Because incorporation has a detransitivizing effect on the transitive verb *kabi* ‘make’, i.e. reduces its type from $\langle e, vt \rangle$ to $\langle v, t \rangle$, TM must be sensitive to the semantic types of verb stems. This is in line with observations in the literature that incorporated nouns (i) form part of the verbal projection, and (ii) tend to give rise to a morpho-syntax that is intransitive, or at least in some way reduced in transitivity relative to regular transitive clauses in a given language; see e.g. Mithun & Corbett (1999). While not constituting a full-fledged argument referentially and morpho-syntactically, incorporated nouns generally have a saturating effect on the verb’s IAR nonetheless, to the effect that the verb cannot take an internal argument in addition to the one it has incorporated.¹²⁶

7.3 Type-symmetric Event Cumulation

I propose that semantic TM follows from the rule of **Type-symmetric Event Cumulation**, which is simply a more restrictive version of our familiar Event Cumulation rule. This rule will be abbreviated as EC_{TS} .

EC_{TS} takes a predicate of a certain type as its first input, and then requires another predicate of that same type as its second input. It ends up yielding a complex predicate with the same type as that of its two inputs: if the first input is an intransitive predicate of type $\langle v, t \rangle$, so is the second one, such that EC_{TS} yields a complex predicate of type $\langle v, t \rangle$; if the first input is a transitive predicate of type $\langle e, vt \rangle$, so is the second one, which results in a complex predicate of type $\langle e, vt \rangle$. Note that for now, discussion will be limited to unergatives ($\langle v, t \rangle$) and transitives ($\langle e, vt \rangle$), though the same logic applies to unaccusatives; the unaccusative-unergative distinction in Saliba is taken up in the next chapter.

Ya-tupa-[he-yoli]-[he-gehe]-di
 1SG.S-IMPACT-CAUS-sink-CAUS-be.finished-3PL.O

‘I will drown all of them’

(Margetts 2005: 76)

¹²⁶This effect is notably absent in *classificatory noun incorporation*, where incorporated objects can be doubled by independent nominals serving a classifying function (Mithun 1984).

Intransitive EC_{TS} is given in (13). Transitive EC_{TS} (13b) characterizes two subtypes, depending on whether the objects of the constituent transitive verbs are shared or not. For two predicates P and Q , where P belongs to the syntactic non-head verb and Q to the syntactic head, in case of object sharing, EC_{TS} co-indexes the internal arguments of P and Q , and yields a complex predicate requiring only the internal argument of Q as its next input (13b-i). In case of non-object sharing, EC_{TS} also yields a complex predicate requiring the internal argument of Q as its next input, but instead of co-indexing the internal argument of P with that of Q , it existentially binds P 's internal argument (13b-ii):

- (13) a. *Intransitive EC_{TS} :*

$$\lambda Q_{\langle v,t \rangle} \lambda P_{\langle v,t \rangle} \lambda E. \underbrace{\exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ P(e_1) \ \& \ Q(e_2)]}_{\text{Complex Predicate: } \langle v,t \rangle}$$

$$\langle vt, \langle vt, vt \rangle \rangle$$
- b. *Transitive EC_{TS} :*
- (i) *Object sharing*

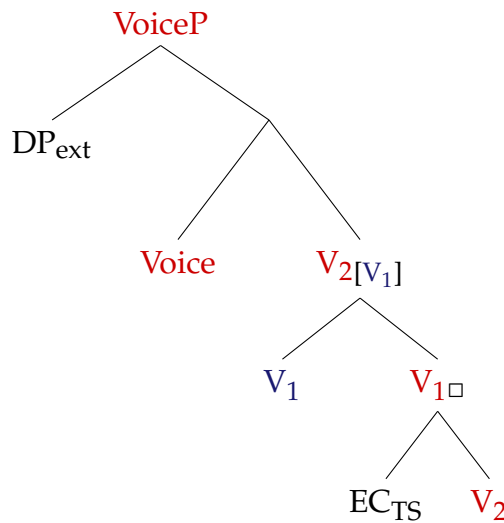
$$\lambda Q_{\langle e,vt \rangle} \lambda P_{\langle e,vt \rangle} \lambda y \lambda E. \underbrace{\exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ P(e_1, x_i) \ \& \ Q(e_2, y_i)]}_{\text{Complex Predicate: } \langle e,vt \rangle}$$

$$\langle evt, \langle evt, evt \rangle \rangle$$
- (ii) *No object sharing*

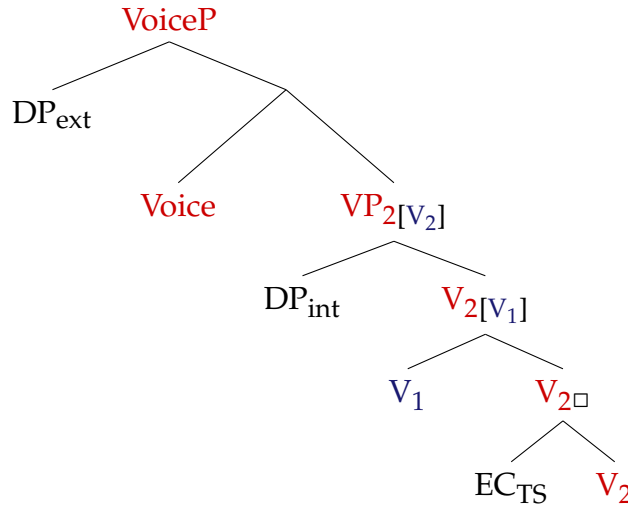
$$\lambda Q_{\langle e,vt \rangle} \lambda P_{\langle e,vt \rangle} \lambda y \lambda E. \underbrace{\exists x \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ P(e_1, x_k) \ \& \ Q(e_2, y_i)]}_{\text{Complex Predicate: } \langle e,vt \rangle}$$

$$\langle evt, \langle evt, evt \rangle \rangle$$

The basic tree structures for the derivations of intransitive C-SVCs and transitive C-SVCs via EC_{TS} are given in Tree 7.3 and 7.4, respectively:



Tree 7.3: Unergative EC_{TS}

Tree 7.4: Transitive EC_{TS}

Note that even though EC_{TS} is a rule, like Predicate Modification, and not a syntactic head, it will be included in syntactic trees from here on for compositional clarity.

Next to ensuring that only like-typed predicates can enter into C-SVC formation, EC_{TS} needs to achieve two other things: (i) yield a complex predicate that is mapped onto the format of a single verb, so that the cluster can morpho-syntactically be dealt with like a single verb; and (ii) ensure that in the resulting complex argument structure, transitive verbs end up sharing their internal arguments or – in the case of non-object sharing – that the internal argument of any non-head transitive is existentially bound. EC_{TS} also needs to be consistent with the head-finality of Saliba C-SVCs. Let us go through each of these requirements in detail.

First, complex predicate formation follows from the cumulation of the sub-events e_1 and e_2 , denoted respectively by P and Q . This is achieved by first symmetrically relating them via the sum operator $\oplus$, then equating their cumulation with the macro-event E – itself introduced by EC_{TS} – and lastly existentially quantifying over them.¹²⁷ The symmetry of their cumulation reflects the fact that the basic complex event structure of C-SVCs is conjunctive, in consistency with their syntactic lives as adjunction configurations. The cumulation of the sub-events e_1 and e_2 into a macro-event E and their subsequent existential binding ensures that only the highest λ -bound eventuality argument – i.e. the macro-event E – will be accessible to operators such as TAM and negation, which in turn derives the single-verb inflectional behavior of C-SVCs. The resulting complex predicate can either feed into another cycle of EC_{TS}, in which case its λ -bound macro-event E will likewise be the only eventuality argument accessible to EC_{TS}, or undergo event closure.

Second, in complex argument structures consisting of two or more transitives, only the internal argument of the head verb V_H remains open to syntactic projection

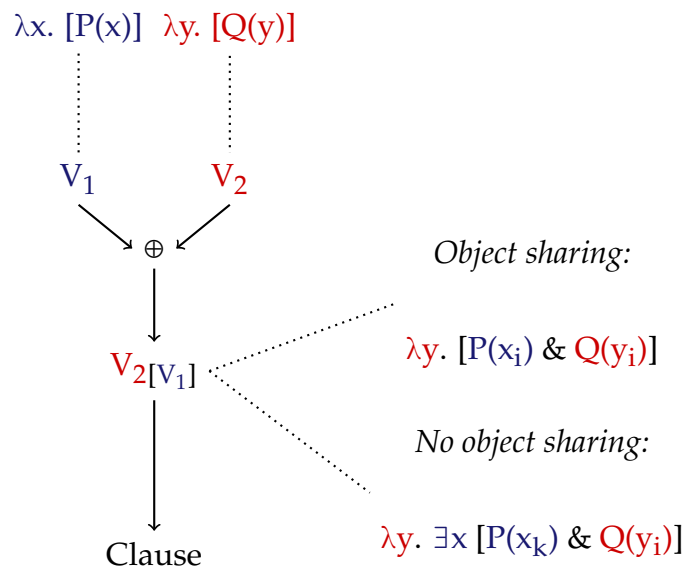
¹²⁷While these steps are left implicit in (13) and the following, I do take them to actually occur in this particular order. The sub-events e_1 and e_2 must remain open until after they have been cumulated; if they were to be closed earlier, they would be inaccessible to cumulation.

as a λ -bound argument, being required by EC_{TS} as its next input. The internal argument of the non-head verb, by contrast, is blocked from projection, due to the MNC, so that its IAR must be satisfied by some other means: any non-head internal argument is either co-indexed with that of the head verb, or existentially bound in the case of sortal distinctness. I assume that the need for co-indexation or existential binding to apply results automatically from the IAR of a verb, and not from any construction-specific property.

In the next step, if EC_{TS} does not iterate, but a DP is fed into the derivation, this DP will saturate V_H 's internal argument and, by virtue of co-indexation, also come to be interpreted as the internal argument of the non-head verb and satisfy its IAR. If EC_{TS} does iterate (adding up to two more predicates), V_H 's internal argument will likewise remain as a λ -abstract, given that EC_{TS} results in head extension, meaning that whichever verb EC_{TS} applies to is automatically the head. In other words, C-SVCs yielded by EC_{TS} behave syntactically just like their head verb would on its own, with any adjoined verb feeding into the overall event and argument structure. Recall that this kind of complex-predicate behavior otherwise associated with complementation can arise if adjoined and head material is jointly housed under the same the macro-event (Truswell 2007); cf. Section 5.8.2 of Chapter 5.

The difference between object-sharing and non-object-sharing cases is schematized in (14), which abstracts away from eventuality arguments for clarity:

(14) *Complex predicate formation via EC_{TS}*



Concerning headedness, nothing more needs to be said on why it is the string-final verb that is the head. This property does not follow from EC_{TS} , but from the simple facts that (i) Saliba is head-final, and that (ii) EC_{TS} results in head extension, such that whichever verb it applies to is trivially the syntactic head. EC_{TS} hence recapitulates syntactic headedness. Any non-first verb fed into EC_{TS} consequently becomes a non-head, in consistency with the standard assumption that adjuncts are added to main projections after the latter have entered the derivation. This gives us exactly

the configuration we need, wherein any non-final verbs preceding the head V_H are adjuncts to V_H . If we choose to iterate EC_{TS} by applying it to its own output, V_H hence remains as the syntactic head.

Also, note that the head V_H does not inherently require merger with another verb, which reflects the fact that the category an adjunct merges with does not select it. The necessity of the head V_H to merge with another verb upon application of EC_{TS} is a derived property that follows from the very application EC_{TS} to V_H . Because this requirement is introduced by EC_{TS} , we correctly capture the fact that verbs making up C-SVCs can equally occur as independent verbs.

Despite having more than one type depending on the argument it applies to, EC_{TS} behaves similarly at each of its types, and is therefore what is known as a *parametrically polymorphic function*¹²⁸. This behavior characterizes the generalized rule in (15), which takes two predicates of like types $\langle\alpha\rangle$ and yields a complex predicate of that same type $\langle\alpha\rangle$. The input required by the resulting complex predicate will depend on the predicates fed into EC_{TS} : if these are intransitive, the complex predicate will require only an event argument and so be of type $\langle v, t \rangle$; if these are transitive, the complex predicate will require both an individual and an eventuality argument, and so be of type $\langle e, vt \rangle$. This polymorphism is represented in (15) as (i) curled braces around those components which are only present if EC_{TS} applies to a transitive predicate of type $\langle e, vt \rangle$, and (ii) with transitive predicates, a variably disjoint or co-referent index on the internal argument of the non-head predicate P , depending on whether objects are shared or not:¹²⁹

$$(15) \quad \text{Generalized } EC_{TS}: \\ \lambda Q_{\langle\alpha\rangle} \lambda P_{\langle\alpha\rangle} \underbrace{\{\lambda y \lambda E. \{\exists x \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ P(e_1, \{x_i/k\}) \ \& \ Q(e_2, \{y_i\})]\}_{\langle\alpha\rangle}}_{\text{Complex Predicate: } \langle\alpha\rangle} \\ \langle\alpha, \langle\alpha, \alpha\rangle\rangle$$

In other words, the generalized rule in (15) has the three different spell-outs in (13b-i), (13b-ii) and (13), repeated here:

$$(16) \quad \text{Spell-outs of Generalized } EC_{TS}: \\ \text{a. Intransitive } EC_{TS}: \\ \lambda Q_{\langle v, t \rangle} \lambda P_{\langle v, t \rangle} \underbrace{\lambda E. \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ P(e_1) \ \& \ Q(e_2)]}_{\text{Complex Predicate: } \langle v, t \rangle} \\ \langle vt, \langle vt, vt \rangle \rangle \\ \text{b. Transitive } EC_{TS}: \\ \text{(i) Object sharing:}$$

¹²⁸Thanks to Gregory Kobele for pointing this out.

¹²⁹We can think of the type-matching effect of EC_{TS} as the Law of Coordination of Likes, known from conjunction, applied to semantic types. Note that conjunction by itself does not guarantee type symmetry; see e.g. Partee & Rooth (1983).

$$\lambda Q_{\langle e,vt \rangle} \lambda P_{\langle e,vt \rangle} \underbrace{\lambda y \lambda E. \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ P(e_1, x_i) \ \& \ Q(e_2, y_i)]}_{\text{Complex Predicate: } \langle e,vt \rangle} \langle evt, \langle evt, evt \rangle \rangle$$

(ii) *No object sharing:*

$$\lambda Q_{\langle e,vt \rangle} \lambda P_{\langle e,vt \rangle} \underbrace{\lambda y \lambda E. \exists x \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ P(e_1, x_k) \ \& \ Q(e_2, y_i)]}_{\text{Complex Predicate: } \langle e,vt \rangle} \langle evt, \langle evt, evt \rangle \rangle$$

7.4 Contextual bivalence

Recall that in contextual bivalence, verbs which are obligatorily monovalent in isolation can and must exceptionally become bivalent when occurring with a transitive co-verb in a C-SVC (17); in Saliba, contextual bivalence is always carried out morpho-syntactically via transitivization with the suffix *-(e)i*. The relevant examples are repeated here:

(17) *Contextual bivalence*

- a. Ta-[he-yoli]-[uyo-i]- $\emptyset$
1INCL.S-CAUS-sink-return-APPL-3SG.O
'We again make it sink'
- b. Ye-[he-kata]-[namwa-namwa-i]-gai
3SG.S-CAUS-learn-RED-be.good-APPL-1EXCL.O
'She teaches us properly' (Margetts 2005: 80, 66)

(18) a. Se-[uyo]-ma
3PL.S-return-hither
'They came back' (Margetts 2005: 75)

- b. *Se-[uyo-i]- $\emptyset$
3PL.S-return-APPL-3SG.O
intended: 'They came back with it / to it'

(19) a. Ka-na heyatu-ne ye-[namwa]
CL1-3SG.POSS tattoo-DET 3SG.S-be.good
'His tattoo is good' (Margetts 2013: 13)

- b. *Ye-[namwa-i]- $\emptyset$
3SG-be.good-APPL-3SG.O
intended: 'She is good because of it'

I argued that contextually bivalent verbs syntactically realize arguments which are thematically selected by their transitive co-verb. Because Saliba C-SVCs involve V-level adjunction, and since V-level adjunction takes place between (simplex or complex) terminal nodes, there is – intuitively speaking – simply no syntactic room for the internal argument of any non-head transitive verb to be realized in situ. For a non-head transitive verb within a C-SVC, the only way for its internal argument to be realized in situ would be for it to in fact not be a phrasal argument, like when it is

incorporated (cf. example (11)). This is a direct consequence of the MNC.

In terms of how adjoined transitives in C-SVCs have their IAR satisfied, then, C-SVCs with contextually bivalent verbs are no different from C-SVCs with regular object sharing. The difference, I argue, lies in the thematic nature of the syntactic object of the head V_H that ends up being co-indexed with the internal argument of the non-head V_1 : while it is thematic in regular object sharing, it is athematic in contextual bivalence, and only comes to obtain its thematic reference by virtue of being co-indexed with the thematic internal argument position of the non-head V_1 .

To see this, consider the C-SVC in (20a), in which the $V_{2=H}$ ‘return’ exceptionally becomes transitive and projects an object which is associated thematically only with the V_1 ‘CAUS-sink’; for (20a), I propose the denotation in (20b):¹³⁰

- (20) a. Ta-[he-yoli]-[uyo-i]- $\emptyset$
 1INCL.S-CAUS-sink-return-APPL-3SG.O
 ‘We again make it sink’
 b. $\llbracket(20a)\rrbracket =$
 $\exists E \exists e_1 \exists e_1' \exists s_2 [E = e_1 \oplus s_2 \ \& \ \text{CAUSE}(e_1, e_1') \ \& \ \text{sink}(e_1', x_i) \ \& \ \text{return}(s_2) \ \& \ \text{EXPL}(s_2) = it_i \ \& \ \text{AGENT}(E) = we]$

According to (20b), (20a) expresses a sinking-and-returning macro-event E , with a single syntactic object that realizes an *expletive internal argument role* of the returning eventuality s_2 (see below for arguments that ‘return’ characterizes a state eventuality); despite being realized syntactically, this expletive internal argument only comes to be interpreted thematically due to being co-indexed with the internal argument of the sinking event e_1 , i.e. that expressed by the non-head transitive V_1 .

Similarly, in the C-SVC in (21a), the $V_{2=H}$ ‘be good’ exceptionally realizes an object referring to the teachee, which is thematically selected by the non-head transitive V_1 ‘teach’; cf. the denotation in (21b):

- (21) a. Ye-[he-kata]-[namwa-namwa-i]-gai
 3SG.S-CAUS-learn-be.good-APPL-1EXCL.O
 ‘She teaches us properly’
 b. $\llbracket(21a)\rrbracket =$
 $\exists E \exists e_1 \exists s_2 [E = e_1 \oplus s_2 \ \& \ \text{CAUSE}(e_1, e_1') \ \& \ \text{learn}(e_1', x_i) \ \& \ \text{good}(s_2) \ \& \ \text{EXPL}(s_2) = we_i \ \& \ \text{AGENT}(E) = she]$

We can thus generalize the denotations of contextually bivalent C-SVCs as per (22):

- (22) *Complex event structure in contextual bivalence:*
 $\exists E \exists e_1 \exists s_2 [E = e_1 \oplus e_2 \ \& \ P(e_1, x_i) \ \& \ Q(s_2) \ \& \ \text{EXPL}(s_2) = \llbracket DP_{int} \rrbracket_i \ \& \ \text{AGENT}(E) =$

¹³⁰Following Pylkkänen (2008: 84), I assume that causatives have the denotation in (i):

(i) $\llbracket \text{CAUSE} \rrbracket = \lambda P \lambda e. \exists e' [\text{CAUSE}(e, e') \ \& \ P(e')]$

$[[DP_{ext}]]$

This means that contextually bivalent C-SVCs are not only fully parallel to C-SVCs with regular object sharing in terms of the MNC, but also in terms of their complex event and argument structure, and – as we will see – also in their syntax. The only difference lies in the fact that the internal argument of the contextually bivalent $V_{2=H}$ is athematic, or expletive.

Conceptually, we can think of EXPL as a kind of thematic proxy element or anaphor, which – just like a reflexive – does not by itself carry any thematic reference, but must obtain this reference from somewhere else in the overall argument structure. I suggest that by analogy to the syntactic well-formedness condition that reflexives must have a c-commanding antecedent within the relevant binding domain, EXPL must have a *thematic antecedent* belonging to a predicate whose event is part of the same complex event as EXPL, i.e. the licensing domain of EXPL is delimited by the highest common macro-event E .¹³¹

I propose that the relation EXPL may under certain conditions be added to verbs like *uyo* ‘return’ and *namwa* ‘be good’, thereby licensing their contextual bivalence within C-SVCs. The addition of EXPL to the argument structure of a verb will be carried out by a novel operator, which I call ADD: ADD takes a predicate of type $\langle v, t \rangle$ and yields that same predicate, yet whose argument structure has been extended by a λ -bound individual argument that stands in an expletive relation to that predicate. The original predicate has now been type-raised to $\langle e, vt \rangle$, i.e. it has an IAR. See (23):

(23) *Predicate extension via ADD:*

$$\begin{aligned} [[ADD]] &:= \\ \lambda P_{\langle v, t \rangle} \lambda x \lambda s. [P(s) \ \& \ EXPL(s) = x] &\quad \langle vt, evt \rangle \end{aligned}$$

I assume that ADD can only attach to verbs with a particular kind of event structure, which is the focus of the next section.

7.4.1 The class of contextually bivalent verbs

Table 7.1 gives a non-exhaustive list of contextually bivalent verbs from a handful of Oceanic languages. Note that I have only included verbs regarding which I know with certainty that they are able to become bivalent and express syntactic dependents only in the context of C-SVCs, based on either explicit mentioning in the respective works, or personal communication with the authors. Verbs which appear to fulfill the criteria of contextual bivalence, but for which the respective works do not explicitly show that they are obligatorily monovalent outside of C-SVCs, have been excluded. It can therefore be assumed that in most or even all of the languages below, the list

¹³¹EXPL is thus distinct from pseudo-reflexives, which are generally not taken to function as thematic proxies.

of contextually bivalent verbs is actually longer (except Saliba, for which the list is reported to be exhaustive).

Language	Contextually bivalent verbs	Morpho-syntax of valence increase
Saliba	<i>namwa</i> ‘be good’, <i>nogowai</i> ‘be slow’ <i>mwamwayau</i> ‘be quick’, <i>uyo</i> ‘return’	transitive
Daakaka	<i>bup</i> ‘tumble’	transitive
Toqabaqita	<i>baqita</i> ‘be big’, <i>qaliqali</i> ‘be quick’	oblique
Pileni	<i>themu</i> ‘be quiet’, <i>lavoï</i> ‘be good’	transitive
Paamese	<i>ramoboŋo</i> ‘do accidentally’, <i>pilu</i> ‘stick together’	transitive / oblique
Nêlêmwa	<i>sho</i> ‘be good’, <i>mwang</i> ‘be bad’, <i>khare</i> ‘be different’ <i>fwaat</i> ‘be clear’, <i>bööm</i> ‘be soft’ <i>hââhuuk</i> ‘be recent’, <i>shumwêlî</i> ‘be thus’	transitive

Table 7.1: Contextually bivalent verbs in Oceanic (non-exhaustive)

Within C-SVCs, contextually bivalent verbs function as manner verbs, i.e. they express what can by and large be characterized as manner-adverbial modification. This raises the question of whether they have an inherent disposition towards event modification. In other words, do contextually bivalent verbs in isolation require intransitive subjects that denote events?

To see that this is not the case, consider the behavior of the Saliba verb *namwa* ‘be good’ on its own outside the context of C-SVCs. In (24a), the sole argument of *namwa* refers to the smell of an object. This alone seems consistent with *namwa* restricting its sole argument to denoting an event, independently of how this event is lexicalized, viz. as a noun or as a verb, in which case (24a) would more accurately translate to ‘Its smelling is good’ (compare the adverbial use of *good* in *She speaks well* with its adjectival use in *Her speaking is good*). Other uses, however, clearly show that the sole argument of *namwa* need not be event-denoting: the intransitive subject refers to an ornament in (24b), a tattoo in (24c), and a child in (24d). This means that *namwa* ‘be good’ does not place a sortal restriction of eventhood on its sole argument: ¹³²

¹³²Note that the forms *pane*, *didiga* and *heyatu* are categorially ambiguous between being a verb and a noun: *pane* is either a labile verb, meaning ‘to smell of something’ (intr) / ‘to smell something’ (trans), or a noun, meaning ‘smell’ (Margetts 2013: 26); *didiga* is either an intransitive verb, meaning ‘to be richly ornamented’, or a noun, meaning ‘ornament’ (Margetts 2013: 7); and *heyatu* is either a transitive verb, meaning ‘to tattoo’, or a noun, meaning ‘tattoo’ (Margetts 2013: 13). This does not conflict with the above conclusion that contextually bivalent verbs do not involve inherent event modification, because even though such an analysis might be tenable for ‘smell’ (24a) - *pane* (v.) would denote a smelling event, and *pane* (n.) its object / result - and also ‘tattoo’ (24c) (a tattooing event (v.) vs its

- (24) a. Pane-na ye-namwa
smell-3SG.POSS 3SG.S-good
'Its smell is good'
- b. Didiga-na-ne i wane ye-namwa
ornament-3SG.POSS-DET COP? EMPH 3SG.S-good
'Its ornaments are really nice'
- c. Ka-na heyatu-ne ye-namwa
CL1-3SG.POSS tattoo-DET 3SG.S-be.good
'His tattoo is good'
- d. Ka-nuwatu-i natum ye-namwa-namwa-ko
1PLEXCL.S-think-APPL? child 3SG.S-INCH-be.good-PERF
'We hope your child has gotten better already'
- (Margetts 2013: 26, 7, 13, 25)

Similarly, in Nêlêmwa, the sole subject of the contextually bivalent verb *hââhuuk* 'be recent' when used in isolation need not refer to an event, but may refer to e.g. a story, as in (25a); as a $V_{2=H}$, *hââhuuk* expresses that the event denoted by its transitive co-verb V_1 is carried out recently (25b):¹³³

- (25) **Nêlêmwa**
- a. Hââhuuk fwâhuux eli
be.recent story that
'This story is a recent one' (Bril 2000: 157)
- b. *trans* + *trans*
Hla [diya] [hââhuux-e] mwa eli
3PL do be.recent-TRANS house that
'They built this house recently' (Bril 2004: 177)

These facts disconfirm an analysis that ties contextual bivalence to an inherent disposition of certain verbs towards event predication.

Another generalization is that – with the exception of the three red-colored verbs (to be discussed shortly) – all the contextually bivalent verbs in Table 7.1 classify as stative outside of C-SVCs; this is either stated explicitly by the authors, or inferrable from translation and examples. While a full treatment of the semantics of the stativity of contextually bivalent verbs is beyond the present scope, most of them qualify as expressing Kimian states, as opposed to Davidsonian states.

Recent work on stative predicates has given rise to a distinction between two classes of statives (Maienborn 2003, Maienborn 2005, Rothmayr 2009). The first class, Davidsonian statives (or *D-states*), mostly comprises verbs of position such as *lie* and *sit*, and the second class, Kimian statives (or *K-states*), largely corresponds to copular predicates. Davidsonian and Kimian statives differ with respect to the nature

object / result (*n.*)), it might not work for 'ornament' (24b), and would clearly not work for 'child' (24d).

¹³³There are no examples showing that one can say 'This house is recent', though I assume that this is possible.

of the Davidsonian argument. While Davidsonian statives contain an eventuality argument – in this case a stative one – Kimian statives contain an ontologically different argument, a Kimian state: abstract, spatio-temporally bound objects that are objects of a property instantiated by a holder at a particular time. Maienborn (2005) proposes the following lexical entry for copular forms, which I will build on:

- (26) *Copular forms*
 $\llbracket be / sein / ser... \rrbracket = \lambda P \lambda x \lambda s. [s \approx P(x)]$ (Maienborn 2005: 307)¹³⁴

According to (26), the contribution of a copula consists in introducing a referential argument s of type K-state that is characterized by the predicate P applying to an individual x . The operator " $\approx$ " relates a Kimian state s to a property P applied to a holder x , i.e. the state s instantiates the property P at the holder x and at the time t . An intuitive way of thinking about Kimian statives is that – unlike Davidsonian statives – their subjects are generally not actively and physically engaged in the state denoted by the predicate (either volitionally or non-volitionally).¹³⁵

We now need to bring this denotation together with our assumptions about verbal architecture. I assume that statives are structurally similar to unergatives: the verb expressing a state is a complement to Voice, which introduces the state-holder as an external argument in Spec,VoiceP. This analysis is in line with most existing constructionist work on stative predicates. While approaches differ in various specifics and there is yet to emerge a fully conclusive picture regarding the proper analysis of statives, there has been convergence on the following properties of stative predicates: (i) they are structurally smaller and less complex than eventive predicates; (ii) they are in many ways similar to copular constructions; and (iii) they contain only a single (structural) argument corresponding to the holder of the state denoted by the predicate.¹³⁶ Moreover, most approaches treat statives as structurally dissimilar to unaccusatives due to their lack of an aspectual or transitional component, and more similar to unergatives, in that their state-holder is merged as an external argument. This is not to say that state-holders are also like unergative subjects in their interpretation: while the latter are agentive in exerting control over the action denoted by the verb, the former generally exert a lesser degree of control over the state denoted by the verb, and often exert no such control at all. Let us briefly go through relevant works in this area to see that this analysis is warranted.

First, in Hale & Keyser (2002), which builds on their seminal work Hale & Keyser (1993), predicative adjective sentences such as *The sky is clear* are assigned a structure

¹³⁴In the original denotation in Maienborn (2005: 307), the referential argument is notated as z , not s . I have chosen to notate it as s rather than z for clarity due to its denoting a state.

¹³⁵Note that even though Kimian states are taken to be ontologically distinct from Davidsonian eventualities, we still need to treat them as the same for the purposes of C-SVC formation via EC_{TS}, which is why I will continue to refer to the non-individual argument of contextually bivalent verbs as *eventualities*.

¹³⁶See Rothmayr (2009) for cases in which these generalizations prove insufficient.

in which δ , an extended projection of A, requires a complement and a specifier, with the state expressed by the adjective in the former, and the state-holder in the latter.

Next, in Ramchand (2008)'s post-Davidsonian system, state-holders are in complementary distribution with causers and initiators, all of which are taken to be merged in Spec,InitP, while the Rheme (a cover term for non-aspectual arguments, including states) occupies Comp,Init. The analyses of stative predicates by Hale & Keyser (2002) and Ramchand (2008) are given in Table 7.2:

$ \begin{array}{c} \delta \\ \swarrow \quad \searrow \\ \text{DP} \quad \delta \quad \text{A} \\ \text{Holder} \quad \quad \text{State} \end{array} $	$ \begin{array}{c} \text{InitP} \\ \swarrow \quad \searrow \\ \text{DP} \quad \text{Init} \quad \text{DP/AP/PP} \\ \text{Holder} \quad \quad \text{Rheme} \end{array} $
Hale & Keyser (2002)	Ramchand (2008)

Table 7.2: Stative predicates

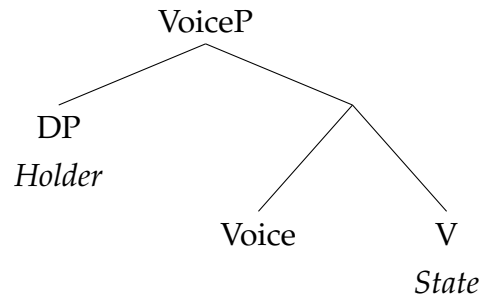
Finally, Kratzer (1996) famously proposes to sever external arguments from their verbs and have them be introduced by Voice. Her main argument for this move is that external arguments do not add to any particular interpretation of a predicate, unlike internal arguments, which may give rise to idiomatic interpretations; cf. Marantz (1984). Crucially, Kratzer argues that state-holders are a kind of external argument: depending on the type of predicate expressed by its complement, Voice introduces either an Agent or a state-holder; what matters is that Voice and the predicate of its complement have compatible Aktionsarten, so that Event Identification is defined and may take place (Kratzer 1996: 123).^{137,138} I thus assume the unergative-like structure in Tree 7.5, at least for the class verbs discussed here:¹³⁹

¹³⁷In the present context, it helps to think of external arguments as being negatively defined with respect to internal arguments in terms of their associated interpretations, given that state-holders are non-agentive, despite patterning like Agents in being expressed as external arguments under present assumptions. If we take the sole, internal arguments of unaccusatives to be (prototypically) characterized by undergoing some form of change of state, for example, we can regard external arguments as lacking this property, thereby reconciling the non-agentivity of state-holders with the agentivity of Agents via their shared status as non-Patients.

¹³⁸Event Identification is a kind of conjunction operation that makes it possible to chain together various conditions on eventualities.

(i) *Event Identification*
 $f_{\langle e,vt \rangle} + g_{\langle v,t \rangle} \rightarrow h_{\langle e,vt \rangle}$ (Kratzer 1996: 122)

¹³⁹Other works that treat state-holders as external arguments are Hallman (2004) and Rothmayr (2009). Work that does not share this assumption includes Harley (1995) *et seq.*, where only v_{CAUSE} selects an external argument, while the head assumed to underlie stative verbs, v_{BE} , does not.



Tree 7.5: Structure of contextually bivalent verbs

In Saliba, a piece of evidence supporting the view that stative verbs pattern like unergatives comes from their ability to combine as $V_{2=H}S$ with transitive verbs that have detransitivized upon incorporating a nominal. In (27a), the root underlying the complex V_1 , *kabi* ‘make’, is transitive and of type $\langle e, vt \rangle$, but becomes intransitive and of type $\langle v, t \rangle$ as a result of incorporating the noun *kabole* ‘sago’; this complex V_1 thus patterns as an unergative in now only being able to take an external argument. Because the complex V_1 is paired with the stative $V_{2=H}$ *gehe* ‘be finished’, and given the type symmetry imposed by TM, *gehe* ‘be finished’ must hence also be of type $\langle v, t \rangle$:

- (27) a. *intr + intr*
 Se-[kabi-kabole]-[gehe]
 3PL.S-make-sago-finished
 ‘They finished making sago’
- b. *intr + intr*
 Se-[niu-tutu]-[uyo]
 3PL.S-coconut-pound-return
 ‘They pound coconut again’
- (Margetts 2005: 69)

Having established that state-holders are introduced by Voice, we now have to modify Maienborn (2005)’s denotation of copular forms, repeated here:

- (28) *Copular forms*
 $\llbracket be / sein / ser... \rrbracket = \lambda P \lambda x \lambda s. [s \approx P(x)]$ (Maienborn 2005: 307)

We need to do two things: (i) because contextually bivalent verbs function as independent verbs without copulas, but (28) only expresses an abstract copular meaning, we need to substitute the λ -bound predicate argument in (28) with whatever lexical stative meaning is expressed by a given stative verb; (ii) we need to sever the state-holder from the denotation in (28), given that we are having it introduced by Voice.¹⁴⁰ For a contextually bivalent verb like *namwa* ‘be good’ in its independent use (29), we thus arrive at the denotation in (30):

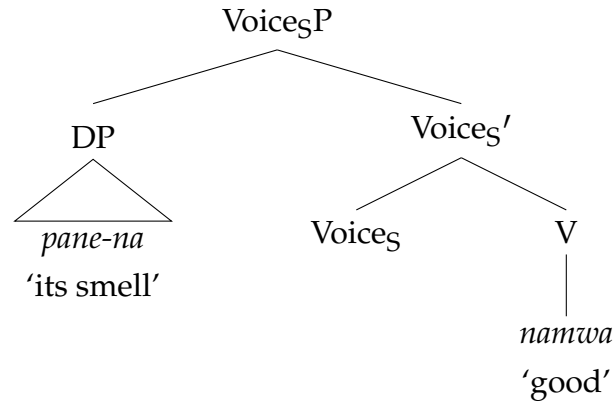
¹⁴⁰We can think of contextually bivalent verbs as constructions headed by a null copula that has incorporated its lexical, state-denoting (N or A) complement.

- (29) Pane-na ye-namwa
 smell-3SG.POSS 3SG.S-good
 'Its smell is good'

- (30) $\llbracket namwa \rrbracket = \lambda s. [s \approx \text{good}]$

Tree 7.6 gives the derivation of (31): after the stative *namwa* 'be good' enters the derivation (32a), the Voice head it combines with must have an Aktionsart that is compatible with stativity, i.e. Voice must denote a set of HOLDER-relations between a state and an individual (32b), as opposed to a set of AGENT-relations between an event and an individual. Voice then introduces the external argument corresponding to the state-holder, yielding (32c), and ultimately giving rise to (33):

- (31) Pane-na ye-namwa
 smell-3SG.POSS 3SG.S-good
 'Its smell is good'



Tree 7.6

- (32) a. $\llbracket namwa \rrbracket = \lambda s. [s \approx \text{good}]$
 b. $\llbracket \text{Voices} \rrbracket = \lambda z \lambda s. [\text{HOLDER}(s) = z]$
 c. $\llbracket \text{VoicesP} \rrbracket = \lambda s. [s \approx \text{good} \ \& \ \text{HOLDER}(s) = \text{its smell}]$
- (33) $\exists s. [s \approx \text{good} \ \& \ \text{HOLDER}(s) = \text{its smell}]$
 = 'There is a state that instantiates goodness at its smell'
 $\approx$ 'Its smell is good'
 = $\llbracket (31) \rrbracket$

Now, given what we have said about the relationship between Voice and its complement, i.e. that the two must have compatible Aktionsarten, how is Voice able to take as its complement a complex verb consisting of an active V_1 and a stative $V_{2=H}$? That is, even though Voice continues the projection of only the stative head $V_{2=H}$, how can it semantically distribute over both the event e_1 of the non-stative non-head V_1 , and the state s_2 of the stative head $V_{2=H}$? I will not attempt to fully resolve this issue, suggesting for now that while the Aktionsarten of the verbs involved may

be distinct – viz. events vs states – they are compatible under the same macro-event E because conjoining their Aktionsarten does not give rise to an event-structural contradiction.¹⁴¹ Relatedly, note that even though subjects of contextually bivalent verbs are interpreted as state-holders in isolation, they are interpreted as Agents of the macro-event E within C-SVCs, in which the predicate expressed by the contextually bivalent verb serves as a kind of modifier. This argument-structural issue is taken up in detail in Section B.1 of the Appendix.

In sum, contextually bivalent verbs are Kimian stative verbs of type $\langle v, t \rangle$ with a single external argument merged in Spec, VoiceP, thus patterning structurally like unergatives. I propose that these properties – i.e. their Kimian-stative semantics and their structural unergativity – constitute the (minimally) necessary conditions for their contextual bivalence. Regarding the operator ADD, I thus propose that it can only attach to verbs expressing Kimian stative verbs of type $\langle v, t \rangle$. It is repeated here:

(34) *Predicate extension via ADD:*

$\llbracket \text{ADD} \rrbracket :=$

$\lambda P_{\langle v, t \rangle} \lambda x \lambda s. [P(s) \ \& \ \text{EXPL}(s) = x]$ $\langle vt, evt \rangle$

I speculate that the fact that contextual bivalence should be found with structurally unergative-like verbs is not an accident: because contextual bivalence is here carried out by an operator ADD that adds an expletive internal argument to a verb's argument structure, we can intuitively think of contextually bivalent verbs as providing the necessary argument-structural "space" allowing the exceptional integration of a lower argument, in turn by virtue of their comparatively light stative semantics: ADD exploits this extra space, and hence its application to a Kimian stative does not constitute an extension proper that is anchored within that verb's argument structure, but rather augments it with an argument which simply does not contradict that argument structure. I will argue below that contextual bivalence thus construed is conceptually analogous to raising.

One last issue concerns the three red-colored contextually bivalent verbs from Table 7.1, repeated here in Table 7.3; unlike the other contextually bivalent verbs discussed here, the translations of these verbs suggest that they are non-stative:

Language	Contextually bivalent verb	Valence increase
Saliba	<i>uyo</i> 'return'	transitive
Daakaka	<i>bup</i> 'tumble'	transitive
Paamese	<i>ramobono</i> 'do accidentally'	transitive / oblique

Table 7.3: Contextually bivalent verbs which are not clearly stative

¹⁴¹This suggests that an underspecification view of Voice may provide the answer.

First, there is the Saliba verb *uyo* ‘return’, which seems to be an active unergative when occurring as an independent verb (35):¹⁴²

(35) **Saliba**

Se-uyo-ma
3PL.S-return-HITHER

‘They came back’

(Margetts 2005: 75)

Second, there is the Paamese verb *ramobono* ‘do accidentally’ (Crowley 1992, Crowley 2002), the translation of which suggests that it is agentive; unfortunately, there are no examples of this verb outside serialization.

Finally, the contextually bivalent verb *bup* ‘tumble’ from Daakaka expresses an event of tumbling or toppling over in its independent use, which suggests that it is a change-of-state verb, i.e. an unaccusative:

(36) **Daakaka**

Ekuokuo mu bup
car REAL tumble

‘The car tumbled over’

(von Prince 2015: 351)

Within C-SVCs, *bup* expresses that the action described by its co-verb is carried out in a weird, clumsy or inappropriate manner (von Prince 2015: 351):

¹⁴² Another property suggesting that *uyo* ‘return’ is active is its ability to head purposive control constructions like (i) (the complementizer *bena* expresses that the complement clause it heads stands in a relation of obligation, intention or purpose to the matrix clause (Margetts 1999: 18)):

(i) Ye-uyo-ma [bena ka-na kulutubu
3SG.S-return-HITHER COMP.OBLIG CLF-3SG.POSS ancestor
ye-dede-i-Ø]
3SG.S-talk.about.family-APPL-3SG.O
‘He returned to track down his ancestors’

(Margetts 2013: 6)

If *uyo* ‘return’ is stative, (i) is unexpected, given the general requirement of complement clauses in control constructions to require a volitional, agentive controller; compare the oddity of #*She is happy [in order to enjoy her day]*. Then again, it seems that such cases can be rescued if a plausible volitional context is provided, e.g. *I will be happy today [so that I can enjoy the party]*.

(37) **Daakaka**a. *intr + intr*

[sóró] [bup]
 speak tumble
 'to misspeak'

b. *intr + intr*

[oko] [bup]
 walk tumble
 'to walk around / stroll'

c. *trans + trans*

Ma [ling] [bup-ane] ding
 REAL put tumble-TRANS mat

'(S)he put the mat the wrong way up / with the wrong side facing up'

(von Prince 2015: 315, 351)

I assume that these verbs do not pose a problem for the present analysis of contextual bivalence, either because (i) they are actually stative, with their apparently non-stative meanings being a matter of translation (e.g. Saliba *uyo* expresses an abstract state of "being again", with 'return' as one of various possible interpretations; and Daakaka *bup* means something like 'to be in a tumbled state'); (ii) they are indeed non-stative, but stativity is brought about by ADD and / or their functioning as manner verbs within C-SVCs; or (iii) they are event-structurally ambiguous or underspecified with respect to the eventive-stative distinction. For convenience, I adopt assumption (i), i.e. that all these verbs are actually stative, both in their independent and serializing uses.

7.5 Discord verbs

There are numerous cases which look like pairings of an intransitive and a transitive verb, in apparent violation of semantic TM and my claim that TM involves type-matching. In (38), for example, *lao* 'go' forms a C-SVC as a non-head V_1 with the $V_{2=H}$ *gabae* 'throw'. That *gabae* 'throw' is transitive is clear from its C-SVC use in (38) and its independent use in (39), where it takes a ϕ -marked direct object:

(38) Kabo ya-[lao]-[gabae]-go
 TAM 1SG.S-go-throw-2SG.O
 'I will leave you'

(Margetts 2005: 83)

(39) Se-gabae- $\emptyset$
 3PL.S-throw-3SG.O
 'They throw it away'

(Margetts 2013: 8)

Considering the independent use of the verb *lao* 'go', however, we see that it can

stand alone (40a), or subcategorize for a goal PP whose nominal complement is not also φ -marked on the verb (40b):

- (40) *lao* ‘go’ in isolation
- a. Se-lao-ko
3PL.S-go-PERF
‘They left already’ (Margetts 1999: 13)
- b. Ye-lao bili wa unai
3SG.S-go room GIVEN POSTP
‘She went into the room’ (Margetts 2013: 3)

Moreover, once *lao* ‘go’ occurs in a C-SVC as the syntactic head, the C-SVC shows the same oblique linking behavior with respect to the Goal PP that is found with *lao* ‘go’ in isolation (cf. (40b)):

- (41) Ye-[kai-kaikewa]-[lao] ka-na kaha ne unai
3SG.S-red-look-go CLF2-3SG.POSS friend DET POSTP
‘He is looking over to his friend’ (Margetts 2005: 73)

A related puzzle is raised by verbs like *usa* ‘put in’, which are unable to occur with full DP objects or object suffixes in isolation without the applicative suffix *-(e)i* and object agreement (42); yet in C-SVCs, *usa* ‘put in’ is able to occur as a V_1 together with the transitive $V_{2=H}$ *he-kawa* ‘CAUS-enter’ without the applicative *-(e)i* (43):

- (42) Ye-usa-*(i)-di yo-na toba-ne unai
3SG.S-put.in-APPL-3PL.O CL-3SG.POSS bag-DET POSTP
‘He put them in his bag’ (Margetts 2005: 81)
- (43) Ye-[usa]-[he-kawa]- $\emptyset$ yo-na bosa unai
3SG.S-put.in-CAUS-enter-3SG.O CL-3SG.POSS basket POSTP
‘He put it into his basket’ (Margetts 1999: 120)

lao ‘go’ thus patterns as an intransitive or an extended intransitive in isolation (40) and when it heads a C-SVC (41), but as a transitive when it occurs as a non-head V_1 with a transitive $V_{2=H}$ (39).¹⁴³ Similarly, *usa* ‘put in’ on its own requires an applicative derivation in order to take a full DP object, thereby patterning as an intransitive (42), yet it is able to form a C-SVC with a transitive $V_{2=H}$, thus patterning as a transitive (43). In terms of types, it looks as if *lao* ‘go’ and *usa* ‘put in’ are variably of type $\langle v, t \rangle$ or $\langle e, vt \rangle$, depending on context. What do we make of this?

Fortunately, there is something special about verbs like *lao* ‘go’ and *usa* ‘put in’ from which we can derive this seemingly discrepant behavior. What is special about these verbs is that they come from the class of so-called *discord* verbs, a term which goes back to Margetts (1999).¹⁴⁴ Across Oceanic, discord is the ability of a subset of semantically dyadic verbs to alternate between fully transitive forms on the one

¹⁴³Following Dixon (1979), *extended intransitives* are intransitives with oblique dependents.

¹⁴⁴For a broader discussion of discord in Oceanic, see Margetts (2008).

hand, and forms exhibiting a kind of mixed or intermediate transitivity on the other: these verbs can appear with structurally reduced, but non-incorporated objects, yet without exhibiting the full range of morpho-syntactic properties otherwise found in regular transitive clauses.¹⁴⁵

For *lao* ‘go’, discord is reflected in its ability to also permit the expression of goals as non-oblique objects (in addition to PP goals) (Margetts 2008). This option is available for any path-encoding motion verb in Saliba, and holds for a broad variety of Goals, not just conventionalized place names. The Goal in (44), for instance, refers to a bush:

- (44) Natu-di-yao se-bahe se-lu se-lao nukula ne
 child-3PL.POSS-PL 3PL.S-carry 3PL.S-go.in 3PL.S-go bush DET
 ‘They would carry their children and go into the bush’ (Margetts 2008: 38)

As for *usa* ‘put in’, its discord variant (45a) differs from its regular variant (45b) in not being able to φ -mark its object without prior derivation by the applicative *-(e)i*:

- (45) a. Apolo ye-usa
 apple 3SG.S-put.in
 ‘He put apples in’ (Margetts 2008: 83)
 b. Apolo wa ye-usa-i-di
 apple GIVEN 3SG.S-put.in-APPL-3PL.O
 ‘He put the apples in’ (Margetts 2008: 81)

Discord objects can only occur with a subset of modifiers that are available to their non-discord counterparts: while the Goal in (44) occurs with the specificity marker *ne* (see below), it would not be able to occur with modifiers promoting high individuation, such as a singular-marked modifier or a numeral (Margetts 2008: 39).¹⁴⁶ In terms of their degree of individuation, discord objects can therefore be situated somewhere in between incorporated and full DP objects, though at times the meaning differences between discord and incorporation clauses are marginal.

Margetts (2008) shows discord constructions to be distinct from incorporation because discord objects occur outside of the inflected verb, can be phrasal and may under certain conditions be non-adjacent to the verb.¹⁴⁷ Compare the following

¹⁴⁵Discord is available with any of the following verb classes (Margetts 2008: 35): (i) root intransitive verbs that have transitive counterparts derived by the applicative *-(e)i*; (ii) the intransitive variants of ambi-transitive verbs; (iii) intransitive verbs derived from transitive verbs via the unproductive anti-passive-like prefix *kai-*; and (v) path-encoding motion verbs. In other words, discord does not arise with just any verb, but only verbs which license patientive roles by virtue of their lexical semantics; certain other intransitive verbs, for instance, never give rise to discord. What sets apart discord verbs from regular transitives is that the semantic presence of a patientive role need not be reflected in a fully transitive morpho-syntax. Also, note that because discord is not available with just any transitive verb, the discord variant of a transitive does not simply correspond to its combining form (Section 6.5.2.3 of the last chapter).

¹⁴⁶In Margetts (1999), *ne* is analyzed as a specificity marker, while in Cleary-Kemp (2006), it is analyzed as a definiteness marker.

¹⁴⁷In (i), the discord object consists of two covertly coordinated possessive phrases:

minimal triplet based on the same verb and thematic object, showing an incorporation (46a), a discord (46b) and a regular transitive clause (46c):

- (46) a. *Incorporation*
 Ye-[sada-wase]
 3SG.S-betelnut-search
 ‘She searched for betelnut’
- b. *Discord*
 Sada ye-wase
 betelnut 3SG.S-search
 ‘She searched for betelnut’
- c. *Regular transitive*
 Sada ye-wase-nei-di
 betelnut 3SG.S-search-APPL-3PL.O
 ‘She searched for the betelnuts’ (Margetts 2008: 35, 41)

The discord variants of semantically dyadic verbs have in common that their structurally lower argument, which generally co-occurs with dedicated markers such as an adposition, an applicative and / or a φ -marking object suffix, is able to occur without these, albeit at the cost of a reduced potential for modification.

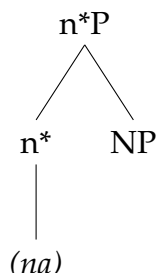
Following van Urk (2020b)’s and Hopperdietzel (2020)’s approaches to object licensing in the related and structurally similar languages Fijian and Daakaka, I propose that discord objects instantiate structurally reduced nominals that do not require Case. In Fijian, common noun objects lack number marking and cannot combine with numerals, but are able to combine with the marker *na*, which typically marks definiteness, but is crucially not a definite article (van Urk 2020b: 346f). Fijian common noun objects thus parallel Saliba discord objects, which also cannot combine with any non-lexical modifiers apart from the markers *ne* and *wa*, which mark definiteness and givenness, respectively (Margetts 2008: 38).¹⁴⁸ Building on

- (i) Yo-di puwaka yo-di gogo wa se-bahe
 CLF-3PL.POSS pig CLF-3PL.POSS things GIVEN 3PL.S-carry
 ‘They carry their pigs and their things’ (Margetts 2008: 38)

¹⁴⁸Both Fijian and Saliba lack proper indefinite and definite articles. In Fijian, the marker *na*, despite often being used in contexts in which a definite interpretation is most natural, also appears with the numeral *dua* ‘one’ (ia), and is compatible with indefinite readings (ib); a further piece of evidence against the idea that *na* is a definite article is that it surfaces with *wh*-phrases like *cava* ‘what’. Similarly in Saliba, the marker *ne* may appear with demonstratives (ii). See Margetts (1999), Schütz (2014) and van Urk (2020b).

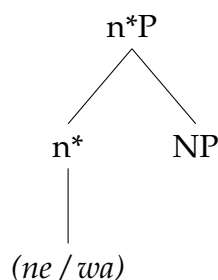
- (i) **Fijian**
- a. Au vaqa-ra tiko e dua na gone
 1SG look.for-TRANS PROG 3SG one ART child
 ‘I am looking for a child (specific / non-specific)’
- b. E levu na koli
 3SG many ART dog
 ‘There are many dogs’ (van Urk 2020b: 346f)

works like Massam (2001) and Collins (2017), van Urk (2020b) proposes that only DPs require Case, but that Fijian common noun objects only project up to n^* , an intermediate functional head between N and D, and which instantiates the marker *na*:^{149,150}



Tree 7.7: Structure of Fijian common nouns

The idea is that, like D, n^* forms part of the extended nominal projection; yet n^* hosts a different kind of functional material than D (e.g. it cannot express number), and unlike D, it does not require Case.¹⁵¹ By full analogy, Saliba discord objects thus have the structure in Tree 7.8, with the markers *ne* and *wa* (optionally) instantiating n^* :



Tree 7.8: Structure of Saliba discord objects

Full DP objects, then, which must be Case-licensed, have the structure in Tree 7.9, with DP abstracting over a range of functional projections, such as NumP:

(ii) **Saliba**

tem pilipilidai-ne
DEM story-DET

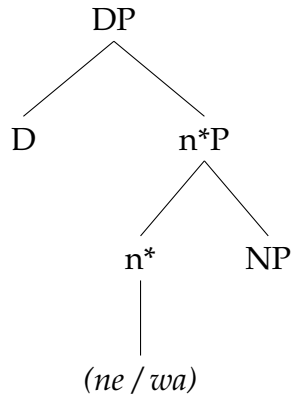
‘that story’

(Margetts 1999: 20)

¹⁴⁹This head is labeled *n* in van Urk (2020b), but I have changed its label to n^* in order to distinguish it from the nominalizing head *n*.

¹⁵⁰That n^* P arguments do not require Case-licensing in Oceanic follows from the assumption that these structurally reduced nominals lack φ -features. Scholars have argued that φ -features reside in designated functional heads articulated within DP, with person introduced by D, number introduced by Num, and gender features introduced by n^* (Ritter 1991, Danon 2011, Kramer 2016, Kalin 2018). While this implies that even n^* P s require Case-licensing, Austronesian languages have been argued to lack the grammatical category of gender altogether (Blust 2013, Corbett 2013, Potsdam & Polinsky pear).

¹⁵¹In van Urk (2020b: 342), the need for DPs to be Case-licensed is represented explicitly by a KP on top of DP, with K bearing an uninterpretable feature [uCase].



Tree 7.9: Structure of full DP objects

Because only DPs require Case, and because both object suffixes and adpositions entail Case-assigning relationships, it follows that any object nominal that projects lower than DP will be able to occur with a semantically dyadic verb without the aid of either an object suffix or an adposition.¹⁵² This means that we can treat discord verbs just like regular transitive verbs in terms of types, since their morpho-syntactically intransitive properties – viz. the absence of a transitive / applicative suffix in conjunction with a ϕ -marking object suffix or an adposition – are directly connected to the no-Case requirement of discord objects. The reduced individuation and modification potential of discord objects is hence also tied to Case, but indirectly, in that the relevant material expressing such specification is hosted in the DP shell, which in turn requires Case.

Coming back to our puzzle, we can thus assign the V_1 use of *lao* ‘go’ in the C-SVC in (47a) the type $\langle e, vt \rangle$, i.e. the same type as that of the transitive $V_{2=H}$ *gabae* ‘throw’; cf. (47b) and (47c). Because both verbs are of the same type, they satisfy the TM requirement of EC_{TS} and can jointly yield a C-SVC as co-verbs (47d):

- (47) a. Kabo ya-[*lao*]-[*gabae*]-go
 TAM 1SG.S-go-throw-2SG.O
 ‘I will leave you’
 b. $[[lao]] = \lambda x \lambda e_1. [go(e_1) \ \& \ GOAL(e_1) = x]$ $\langle e, vt \rangle$
 c. $[[gabae]] = \lambda y \lambda e_2. [throw(e_2, y)]$ $\langle e, vt \rangle$
 d. $\exists E \exists x \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ go(e_1) \ \& \ GOAL(e_1) = x_k \ \& \ throw(e_2, y_{oi}) \ \& \ AGENT(E) = I]$

Likewise, the independent use of *lao* ‘go’ in its discord variant – cf. (44), repeated here – has the type $\langle e, vt \rangle$ (cf. (47b)):

- (48) Natu-di-yao se-bahe se-lu se-lao nukula ne
 child-3PL.POSS-PL 3PL.S-carry 3PL.S-go.in 3PL.S-go bush DET
 ‘They would carry their children and go into the bush’

¹⁵²The appearance of such nP objects must, however, additionally be sanctioned by selectional and thematic considerations, since we need to rule out nP objects from freely occurring with one-place predicates.

As for when *lao* ‘go’ occurs as a $V_{2=H}$ (49) or as an independent verb in non-discord use (50) – both cases in which it takes a PP goal – we can also simply assign it the type $\langle e, vt \rangle$, and have the presence of the adposition *unai* follow from the Case-licensing needs of the goal-denoting nominal’s full DP structure:¹⁵³

- (49) Ye-[kai-kaikewa]-[lao] ka-na kaha ne unai
 3SG.S-red-look-go CLF2-3SG.POSS friend DET POSTP
 ‘He is looking over to his friend’
- (50) Ye-lao bili wa unai
 3SG.S-go room GIVEN POSTP
 ‘She went into the room’

This leaves us with the bare use of *lao* ‘go’, without a Goal (51). Here, we can either assume that *lao* ‘go’ has a second lexical entry as a one-place predicate of type $\langle v, t \rangle$ that lacks a Goal, as in (51b), or retain a single, two-place lexical entry, yet whose Goal argument may remain implicit due to being existentially bound (51c):¹⁵⁴

- (51) a. Se-lao-ko
 3PL.S-go-PERF
 ‘They left already’
- b. $\llbracket \text{lao}_1 \rrbracket = \lambda e. [\text{go}(e)]$ $\langle v, t \rangle$
- c. $\llbracket \text{lao}_2 \rrbracket = \lambda e. \exists x [\text{go}(e) \ \& \ \text{GOAL}(e) = x]$ $\langle v, t \rangle$

Turning to *usa* ‘put in’, we can also simply assign it the type $\langle e, vt \rangle$, regardless of

¹⁵³This predicts that the adposition *unai* should not contribute a directional component, and conversely, that *lao* ‘go’ should independently express a directional component without *unai*; this is correct. First, notice that in both (48) and (50), a directional meaning component expressing movement into or towards something is present regardless of the presence of the adposition *unai*. This means that the direction of movement is not contributed by *unai*, at least not exclusively, and that a basic directional component must be contained in *lao* itself. We can therefore treat *lao* as containing a semantically incorporated adposition, similar to ‘enter’.

Second, *unai* (and its plural variant *udiedi*) is a general adposition in Saliba, occurring with location, goal, source and instrument complements:

- (i) a. *Location complement*
 Iya numa-ne unai
 3SG.EMPH house-DET POSTP
 ‘She is in the house’
- b. *Source complement*
 Teina leta-ta sina-gu unai ye-lao-ma
 DEM letter-DET mother-1SG.POSS POSTP 3SG.S-go-hither
 ‘This letter comes from my mother’
- c. *Instrument complement*
 Ya-soke nigwa-wa unai
 1SG.S-open knife-GIVEN POSTP
 ‘I opened it with a knife’ (Margetts 1999: 29f)

These properties are consistent with the present analysis of *unai* as first and foremost a Case-marker.

¹⁵⁴Against the background of Saliba’s rigid transitivity, which requires transitive verbs to always surface with their objects (in the form of either a lexical DP or an object suffix), we might assume that the grammar more liberally permits Goals to be omitted than other thematic roles like Patients.

whether it occurs on its own in discord use (52a), on its own in non-discord use, or as a V_1 with a transitive $V_{2=H}$. Just like with the postposition *unai* in the case of *lao* ‘go’, we can have the presence of the applicative *-(e)i* (and the object suffix) follow obligatorily whenever the object of *usa* ‘put in’ projects up to DP. Due to being of type $\langle e, vt \rangle$, *usa* ‘put in’ can be paired with a regular transitive verb like *he-kawa* ‘CAUS-enter’ to form a C-SVC (53d):

- (52) a. Apolo ye-usa
apple 3SG.S-put.in
‘He put apples in’
b. Apolo wa ye-usa-*(i)-di
apple GIVEN 3SG.S-put.in-APPL-3PL.O
‘He put the apples in’
- (53) a. Ye-[usa]-[he-kawa]- $\emptyset$ yo-na bosa unai
3SG.S-put.in-CAUS-enter-3SG.O CL-3SG.POSS basket POSTP
‘He put it into his basket’
b. $\llbracket usa \rrbracket = \lambda x \lambda e_1. [\text{put in}(e_1, x)]$
c. $\llbracket he-kawa \rrbracket = \lambda y \lambda e_2. \exists e_2' [\text{CAUSE}(e_2, e_2') \ \& \ \text{enter}(e_2', y)]$
d. $\exists E \exists x \exists e_1 \exists e_2 \exists e_2' [E = e_1 \oplus e_2 \ \& \ \text{put in}(e_1, x_i) \ \& \ \text{CAUSE}(e_2, e_2') \ \& \ \text{enter}(e_2', it_i) \ \& \ \text{AGENT}(E) = \text{he}]$

In sum, the apparent violations of TM given rise to by verbs like *lao* ‘go’ and *usa* ‘put in’ can be traced back to the ability of these verbs to take structurally reduced nP objects that do not require Case. Type-wise, then, discord verbs pattern just like regular transitives in being of type $\langle e, vt \rangle$. We can therefore maintain the present analysis of TM as involving stem-level type-tracking, thereby ensuring pairings of like-typed predicates only. We also predict that the only cases of such apparent matching violations should be restricted to discord verbs. This is correct: Margetts (2005: 81ff) notes explicitly that such cases are found with all and only discord verbs.

The key to accommodating discord lies in recognizing that it represents a mixed or intermediate kind of transitivity, or *semi-transitivity*. The term *transitivity*, while typically used to refer to VPs consisting of a verb and a direct object, actually subsumes a whole range of properties relating to event structure and participants: high-transitive VPs denote telic events with a volitional Agent and a highly individuated and affected Patient, while low-transitive VPs denote atelic events with a non-volitional Agent and a Patient lacking individuation and affectedness (Hopper & Thompson 1980: 252). These differences may carry over into the morpho-syntax in the form of differential argument marking, structural vs oblique realization, agreement and so on, depending on the language.

The specifics of Saliba characterize a three-step cline in terms of semantically transitive verb roots of type $\langle e, vt \rangle$, depending on how their internal argument is projected and linked: (i) fully transitive VPs that consist of a verb with a structural DP object that is ϕ -marked on the verb; (ii) semi-transitive discord VPs that consist of

a verb with a structural nP object that is not ϕ -marked; and (iii) derived intransitives as the result of noun incorporation. Unergatives and statives pattern like verbs with noun incorporation in being of type $\langle v, t \rangle$. The picture on extended intransitives, i.e. morphologically intransitive verbs with oblique objects, is not conclusive, and hence they can a priori be of type $\langle v, t \rangle$ or $\langle e, vt \rangle$. The relationships among Saliba verbs between valence, morpho-syntactic transitivity and semantic types are represented in Figure 7.10 (with dotted lines indicating potential associations):

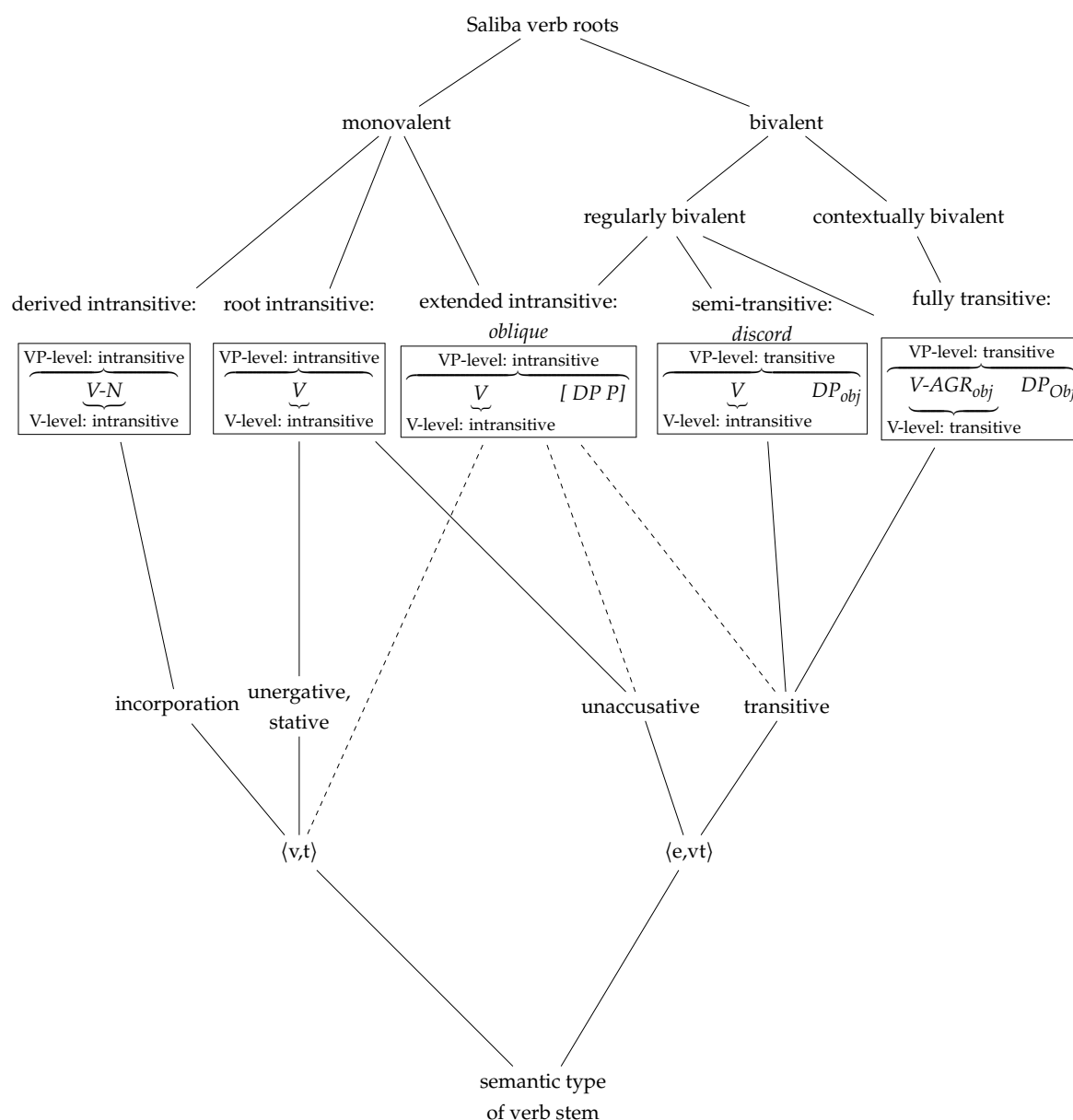


Figure 7.10: Saliba verbs

7.6 Summary

In this chapter, we saw see that semantic TM is a non-selectional, ad hoc restriction that can arise only in V-level adjunction configurations. I have chosen to model this effect as a semantic effect tied specifically to C-SVC formation, whereby the grammar

forces the semantic types of verb stems to match. Type-matching follows from the rule of *Type-symmetric Event Cumulation*, which allows only like-typed predicates to enter into C-SVC formation. In contextual bivalence, stative verbs realize expletive internal arguments that are thematically selected by their co-verb in order to satisfy the latter's IAR. Finally, we saw that discord constructions are fully compatible with the present analysis because discord verbs are only morpho-syntactically intransitive due to projecting structurally reduced objects.

Chapter 8

Derivations and discussion of semantic transitivity matching

8.1 Introduction

With all these pieces in place, we are now ready to run through the derivations of semantic TM in Saliba C-SVCs. These are presented in Section 8.2. In Section 8.3, I take up the unaccusative-unergative distinction in Saliba, arguing that it is consistent with the present analysis. Section 8.4 offers a more in-depth look at contextual bivalence, both specifically within Saliba and more generally in Oceanic. This chapter ends with a discussion of various grammar-theoretical issues in Section 8.5.

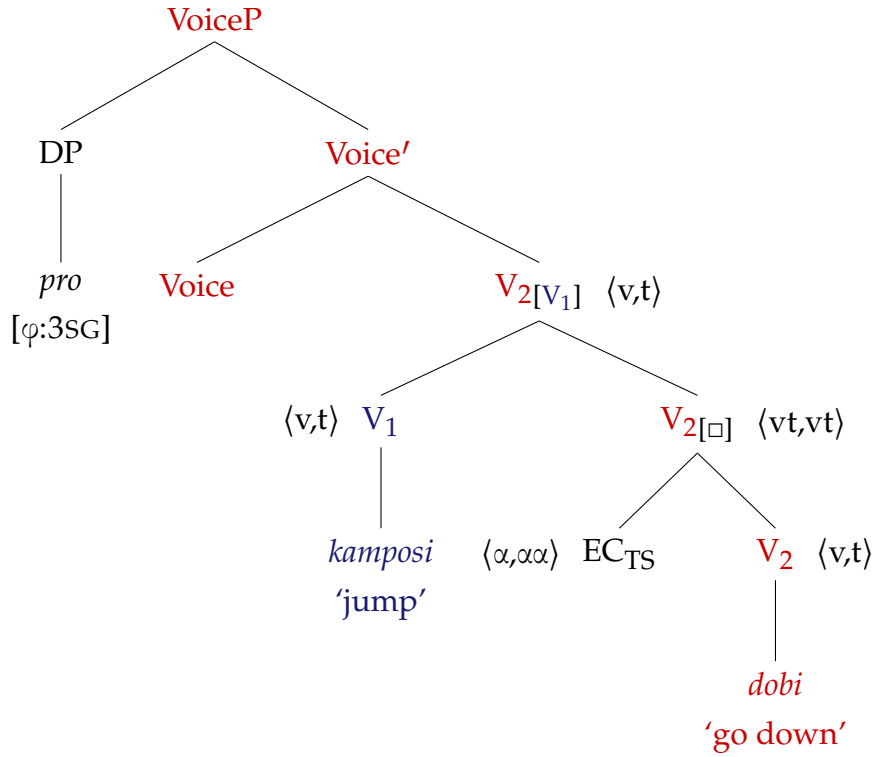
8.2 Derivations

8.2.1 Intransitive C-SVCs

The intransitive C-SVCs in (1) consist of two unergatives of type $\langle v, t \rangle$, ‘jump’ and ‘go down’, with the respective denotations in (2a) and (2b). The derivation of (1) is given in Tree 1. We begin by introducing ‘go down’ into the derivation, which is thus established as the syntactic head V_H . Next, EC_{TS} – given in (3) – applies to ‘go down’, thereafter requiring a second predicate that is also of type $\langle v, t \rangle$ (4a). After applying to its second input ‘jump’, EC_{TS} yields a complex predicate of type $\langle v, t \rangle$ which denotes a set of macro-events E consisting of a jumping sub-event e_1 and a going-down sub-event e_2 (4b). Next, the external argument is introduced by Voice via Event Identification (4c) and subsequently saturated (4d). Finally, the macro-event E undergoes closure, yielding (5), which expresses a complex jumping-and-going-down event with a sole participant:

- (1) Ye-[kamposi]-[dobi]
3SG.S-jump-go.down

‘He jumped down’



Tree 8.1: Intransitive SVC

- (2) a. $\llbracket V_1 \rrbracket = \llbracket kamposi \rrbracket = \lambda e_1. [\text{jump}(e_1)]$ $\langle v, t \rangle$
 b. $\llbracket V_2 \rrbracket = \llbracket dobi \rrbracket = \lambda e_2. [\text{go down}(e_2)]$ $\langle v, t \rangle$
- (3) **EC_{TS}**:
 $\lambda Q_{\langle \alpha \rangle} \lambda P_{\langle \alpha \rangle} \{ \lambda y \} \lambda E. \{ \exists x \} \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ P(e_1, \{x_{i/k}\}) \ \& \ Q(e_2, \{y_i\})]$
 $\langle \alpha, \langle \alpha, \alpha \rangle \rangle$
- (4) a. $\llbracket V_{2[\square]} \rrbracket = \llbracket EC_{TS}(\text{'go down'}) \rrbracket =$
 $\lambda P_{\langle v, t \rangle} \lambda E. \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ P(e_1) \ \& \ \text{go down}(e_2)]$
 b. $\llbracket V_{2[V_1]} \rrbracket = \llbracket EC_{TS}(\text{'go down'})('jump') \rrbracket =$
 $\lambda E. \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ \text{jump}(e_1) \ \& \ \text{go down}(e_2)]$
 FA('jump')
- c. $\llbracket \text{Voice}' \rrbracket =$
 $\lambda z \lambda E. \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ \text{jump}(e_1) \ \& \ \text{go down}(e_2) \ \& \ \text{AGENT}(E) = z]$
 agent
- d. $\llbracket \text{VoiceP} \rrbracket =$
 $\lambda E. \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ \text{jump}(e_1) \ \& \ \text{go down}(e_2) \ \& \ \text{AGENT}(E) = \text{he}]$
 FA('he')
- (5) $\exists E \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ \text{jump}(e_1) \ \& \ \text{go down}(e_2) \ \& \ \text{AGENT}(E) = \text{he}]$
 event closure
 = ‘He jumped and went down’
 $\approx$ ‘He jumped down’

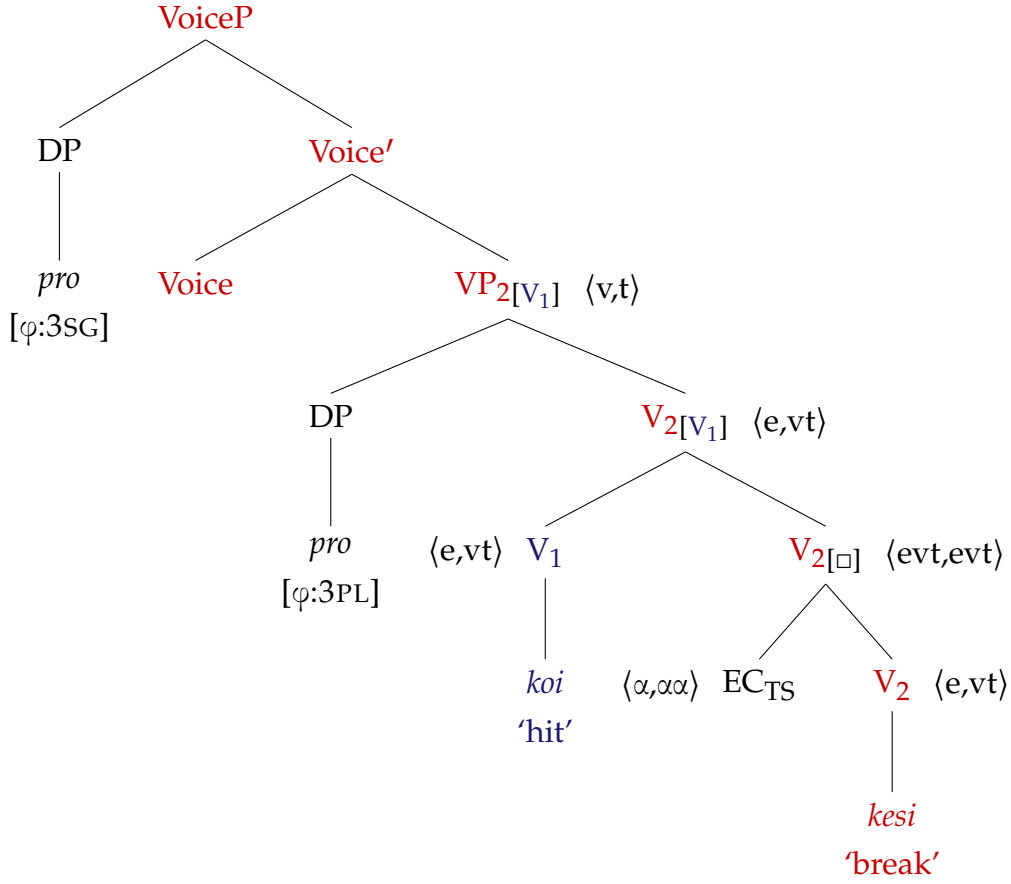
= $\llbracket(1)\rrbracket$

8.2.2 Transitive C-SVCs

8.2.2.1 Transitive C-SVCs with object sharing

- (6) Ye-[koi]-[kesi]-di
 3SG.S-hit-break-3PL.O
 ‘He hit-broke them’

The derivation of a transitive C-SVC with object sharing like (6) proceeds similarly, the difference being the presence of internal arguments. The denotations of the constituent verbs ‘hit’ and ‘break’ are given in (7a) and (7b), respectively; both are transitive, and so of type $\langle e, vt \rangle$. After EC_{TS} extends the predicate structure of ‘break’ (8a), the type of its second input predicate is fixed to also being of type $\langle e, vt \rangle$. It then combines with ‘hit’ to yield a complex predicate, also of type $\langle e, vt \rangle$, which denotes a set of macro-events E consisting of a hitting sub-event e_1 and a breaking sub-event e_2 (8b). Because the participant undergoing the hitting is identical to the one undergoing the breaking, sortal identity is fulfilled, such that the internal argument positions are co-indexed (8b). In the next step, the internal argument of the head transitive verb ‘break’ is saturated via Function Application and – by way of co-indexation – also comes to be interpreted as the internal argument of the non-head ‘hit’. After the agent is added (8d), saturated (8e) and E undergoes event closure, we end up with the final result in (9):



Tree 8.2: Transitive C-SVC (object sharing)

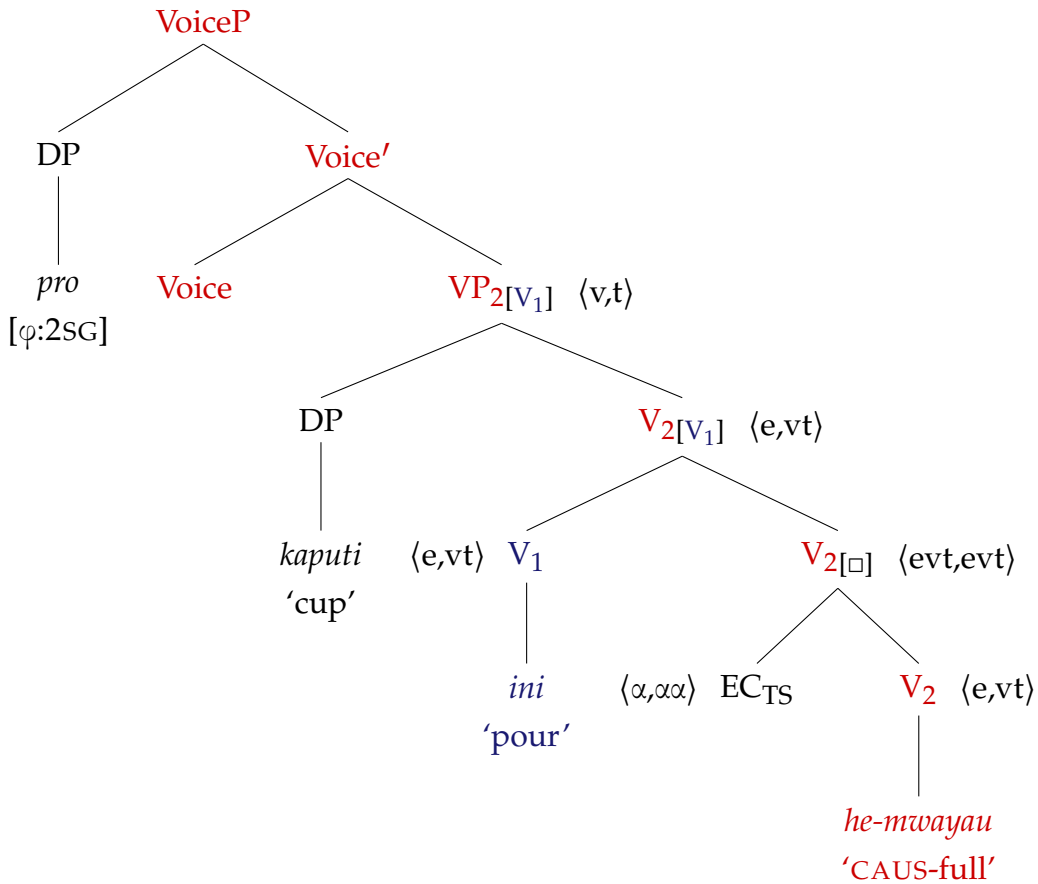
- (7) a. $\llbracket V_1 \rrbracket = \llbracket koi \rrbracket = \lambda x \lambda e_1. [\text{hit}(e_1, x)]$ $\langle e, vt \rangle$
 b. $\llbracket V_2 \rrbracket = \llbracket kesi \rrbracket = \lambda y \lambda e_2. [\text{break}(e_2, y)]$ $\langle e, vt \rangle$
- (8) a. $\llbracket V_{2[\square]} \rrbracket = \llbracket EC_{TS}(\text{'break'}) \rrbracket =$
 $\lambda P_{\langle e, vt \rangle} \lambda y \lambda E. \{ \exists x \} \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ P(e_1, x_{i/k}) \ \& \ \text{break}(e_2, y_i)]$
 b. $\llbracket V_{2[V_1]} \rrbracket = \llbracket EC_{TS}(\text{'break'})('hit') \rrbracket =$
 $\lambda y \lambda E. \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ \text{hit}(e_1, x_i) \ \& \ \text{break}(e_2, y_i)]$
 FA('hit')
- c. $\llbracket VP_{2[V_1]} \rrbracket =$
 $\lambda E. \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ \text{hit}(e_1, x_i) \ \& \ \text{break}(e_2, \text{they}_i)]$
 FA('they')
- d. $\llbracket \text{Voice}' \rrbracket =$
 $\lambda z \lambda E. \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ \text{hit}(e_1, x_i) \ \& \ \text{break}(e_2, \text{they}_i) \ \& \ \text{AGENT}(E) = z]$
 agent
- e. $\llbracket \text{VoiceP} \rrbracket =$
 $\lambda E. \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ \text{hit}(e_1, x_i) \ \& \ \text{break}(e_2, \text{they}_i) \ \& \ \text{AGENT}(E) = \text{he}]$
 FA('he')
- (9) $\exists E \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ \text{hit}(e_1, x_i) \ \& \ \text{break}(e_2, \text{they}_i) \ \& \ \text{AGENT}(E) = \text{he}]$
 event closure
 = 'He hit them and broke them'
 $\approx$ 'He hit-broke them'

= $\llbracket (6) \rrbracket$

8.2.2.2 Transitive C-SVCs without object sharing

In transitive C-SVCs without object sharing like (10), the derivation proceeds identically up to the point where the V_1 is fed as a second input into EC_{TS} . Due to the internal arguments of ‘pour’ and ‘fill’ being sortally distinct, they cannot be co-indexed, and hence the internal argument of the non-head V_1 ‘pour’ is existentially bound (12b). The DP that is then merged via Function Application will hence only specify the internal argument of the head $V_{2=H}$ ‘fill’ (12c). The subsequent derivational steps are again identical to those of transitive C-SVCs with object sharing: the external argument is added by Voice (12d) and saturated (12e), and finally event closure yields the end result in (13):

- (10) Kaputi ku-[ini]-[he-mwayau]- $\emptyset$
 cup 2SG.S-pour-CAUS-full-3SG.O
 ‘Fill the cup’
 (literally: ‘Pour the cup full’)



Tree 8.3: Transitive C-SVC (no object sharing)

- (11) a. $\llbracket V_1 \rrbracket = \llbracket ini \rrbracket = \lambda x \lambda e_1. [\text{pour}(e_1, x)]$ $\langle e, vt \rangle$
 b. $\llbracket V_2 \rrbracket = \llbracket he-mwayau \rrbracket = \lambda y \lambda e_2. \exists e_2' [\text{CAUSE}(e_2, e_2') \ \& \ \text{full}(e_2', y)]$ $\langle e, vt \rangle$

- (12) a. $\llbracket V_{2[\square]} \rrbracket = \llbracket EC_{TS}('CAUS-full') \rrbracket =$
 $\lambda P_{\langle e,vt \rangle} \lambda y \lambda E. \{ \exists x \} \exists e_1 \exists e_2 \exists e_2' [E = e_1 \oplus e_2 \ \& \ P(e_1, x_{i/k}) \ \& \ CAUSE(e_2, e_2') \ \& \ full(e_2', y_i)]$
- b. $\llbracket V_{2[V_1]} \rrbracket = \llbracket EC_{TS}('CAUS-full')('pour') \rrbracket =$
 $\lambda y \lambda E. \exists x \exists e_1 \exists e_2 \exists e_2' [E = e_1 \oplus e_2 \ \& \ pour(e_1, x_k) \ \& \ CAUSE(e_2, e_2') \ \& \ full(e_2', y_i)]$
 FA('pour')
- c. $\llbracket VP_{2[V_1]} \rrbracket =$
 $\lambda E. \exists x \exists e_1 \exists e_2 \exists e_2' [E = e_1 \oplus e_2 \ \& \ pour(e_1, x_k) \ \& \ CAUSE(e_2, e_2') \ \& \ full(e_2', cup_i)]$
 FA('cup')
- d. $\llbracket Voice' \rrbracket =$
 $\lambda z \lambda E. \exists x \exists e_1 \exists e_2 \exists e_2' [E = e_1 \oplus e_2 \ \& \ pour(e_1, x_k) \ \& \ CAUSE(e_2, e_2') \ \& \ full(e_2', cup_i) \ \& \ AGENT(E) = z]$
 agent
- e. $\llbracket VoiceP \rrbracket =$
 $\lambda E. \exists x \exists e_1 \exists e_2 \exists e_2' [E = e_1 \oplus e_2 \ \& \ pour(e_1, x_k) \ \& \ CAUSE(e_2, e_2') \ \& \ full(e_2', cup_i) \ \& \ AGENT(E) = you]$
 FA('you')
- (13) $\exists E \exists x_k \exists e_1 \exists e_2 \exists e_2' [E = e_1 \oplus e_2 \ \& \ pour(e_1, x_k) \ \& \ CAUSE(e_2, e_2') \ \& \ full(e_2', cup_i) \ \& \ AGENT(E) = you]$
 event closure
- = '(You) pour something and fill the cup'
- $\approx$ 'Fill the cup'
- = $\llbracket (10) \rrbracket$

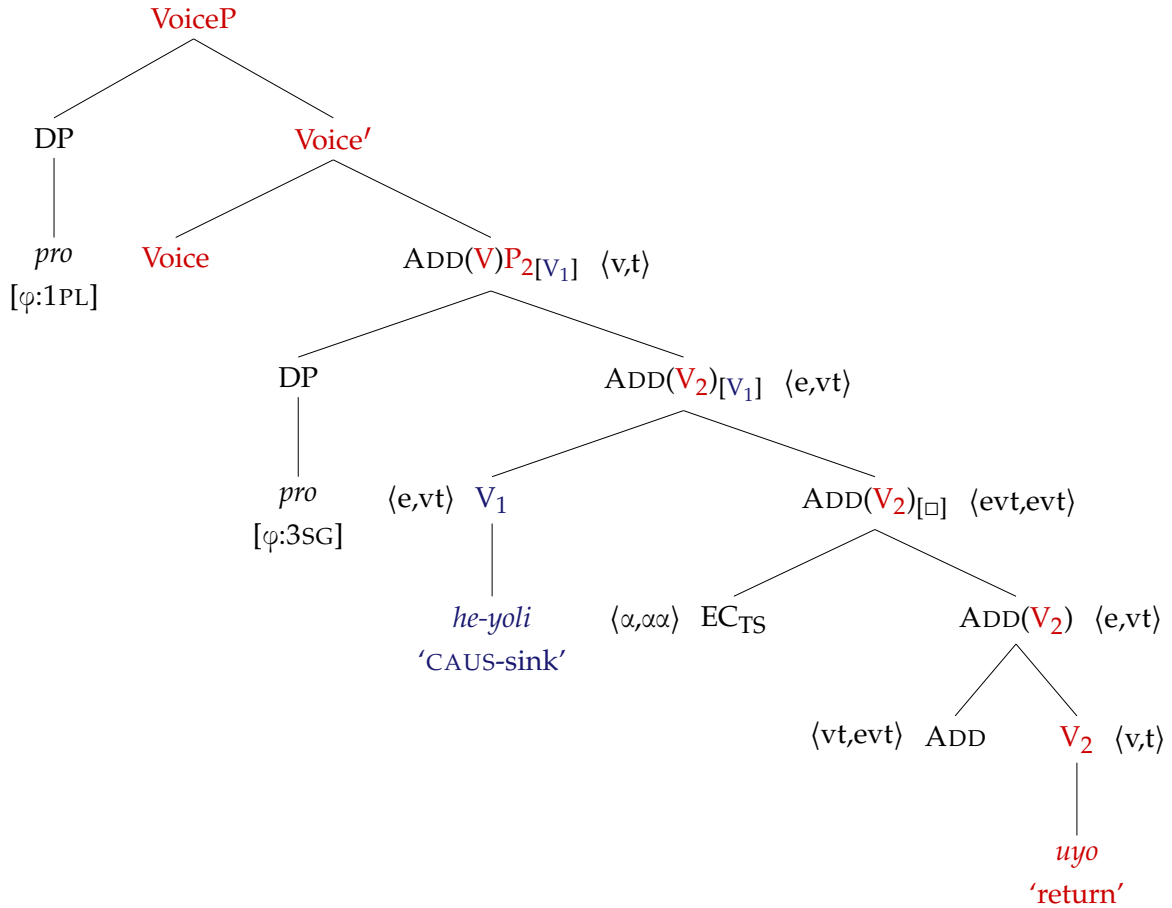
8.2.3 Contextual bivalence

Let us run through the derivation of the C-SVC in (14), where $V_{2=H}$ shows contextual bivalence. The denotations of the constituent verbs of (14), 'make sink' and 'return', are given in (15a) and (15b), respectively. Because 'make sink' is of type $\langle e,vt \rangle$ but 'return' of type $\langle v,t \rangle$, the two cannot be paired into a C-SVC as they are. But after the operator ADD (16) applies to *uyo* 'return', it acquires an expletive internal argument and is hence type-raised to $\langle e,vt \rangle$ (17). Therefore, it can now combine with 'make sink'. As will be argued in Section 8.5.1, while the application of ADD itself comes for free, its application forces verbs like 'return' to subsequently enter into C-SVC formation; this ensures that outside of C-SVCs, these verbs can never be transitive.

Now that 'return' is of the right type, it can be extended via EC_{TS} (18a) and then combine with 'make sink' (18b). The rest of the derivation proceeds just as with transitive C-SVCs without contextual bivalence (18c)-(18d), with the end result in (19). The only difference is that the projected internal argument of the $V_{2=H}$ 'return' is thematically expletive, only obtaining a thematic reference by virtue of being co-indexed with the internal argument of the non-head V_1 'make sink'. So (19) expresses

a complex sinking-and-returning event where the 3SG object is interpreted only as a thematic participant of the sinking, but not the returning eventuality:

- (14) Ta-[*he-yoli*]-[*uyo-i*]- $\emptyset$
 1INC.S-CAUS-sink-return-APPL-3SG.O
 ‘We again make it sink’



Tree 8.4: Contextually bivalent SVC

- (15) a. $\llbracket V_1 \rrbracket = \llbracket he-yoli \rrbracket = \lambda x \lambda e_1. \exists e_1' [CAUSE(e_1, e_1') \& sink(e_1', x)]$ $\langle e, vt \rangle$
 b. $\llbracket V_2 \rrbracket = \llbracket uyo \rrbracket = \lambda s_2. [return(s_2)]$ $\langle v, t \rangle$
- (16) $\llbracket ADD \rrbracket =$
 $\lambda P_{\langle v, t \rangle} \lambda x \lambda s. [P(s) \& EXPL(s) = x]$ $\langle vt, evt \rangle$
- (17) $\llbracket ADD('return') \rrbracket =$
 $\lambda y \lambda s_2. [return(s_2) \& EXPL(s_2) = y]$ $\langle e, vt \rangle$
- (18) a. $\llbracket V_2[\square] \rrbracket = \llbracket ECTS(ADD('return')) \rrbracket =$
 $\lambda P_{\langle e, vt \rangle} \lambda y \lambda E. \{ \exists x \} \exists e_1 \exists s_2 [E = e_1 \oplus s_2 \& P(e_1, x_{i/k}) \& return(s_2) \& EXPL(s_2) = y_i]$
 b. $\llbracket V_2[V_1] \rrbracket = \llbracket ECTS(ADD('return'))('CAUS-sink') \rrbracket =$
 $\lambda y \lambda E. \exists e_1 \exists e_1' \exists s_2 [E = e_1 \oplus s_2 \& CAUSE(e_1, e_1') \& sink(e_1', x_i) \& return(s_2)]$

- FA('CAUS-sink')
- c. $\llbracket \text{ADD}(\mathbf{V})\mathbf{P}_2[\mathbf{V}_1] \rrbracket =$
 $\lambda E. \exists e_1 \exists e_1' \exists s_2 [E = e_1 \oplus s_2 \ \& \ \text{CAUSE}(e_1, e_1') \ \& \ \text{sink}(e_1', x_i) \ \& \ \text{return}(s_2) \ \& \ \text{EXPL}(s_2) = it_i]$
 FA('it')
- d. $\llbracket \text{Voice}' \rrbracket =$
 $\lambda z \lambda E. \exists e_1 \exists e_1' \exists s_2 [E = e_1 \oplus s_2 \ \& \ \text{CAUSE}(e_1, e_1') \ \& \ \text{sink}(e_1', x_i) \ \& \ \text{return}(s_2) \ \& \ \text{EXPL}(s_2) = it_i \ \& \ \text{AGENT}(E) = z]$
 agent
- e. $\llbracket \text{VoiceP} \rrbracket =$
 $\lambda E. \exists e_1 \exists e_1' \exists s_2 [E = e_1 \oplus s_2 \ \& \ \text{CAUSE}(e_1, e_1') \ \& \ \text{sink}(e_1', x_i) \ \& \ \text{return}(s_2) \ \& \ \text{EXPL}(s_2) = it_i \ \& \ \text{AGENT}(E) = we]$
 FA('we')
- (19) $\exists E \exists e_1 \exists e_1' \exists s_2 [E = e_1 \oplus s_2 \ \& \ \text{CAUSE}(e_1, e_1') \ \& \ \text{sink}(e_1', x_i) \ \& \ \text{return}(s_2) \ \& \ \text{EXPL}(s_2) = it_i \ \& \ \text{AGENT}(E) = we]$
 event closure
- = 'We make it sink and return'
 $\approx$ 'We again make it sink'
 = $\llbracket (14) \rrbracket$

8.2.4 Iterating EC_{TS}

The present account correctly captures the fact that semantic TM occurs across the board when more than two verbs are combined, as in the three-verb C-SVC in (20):

- (20) *trans* + *trans* + *trans*

Kabo ya-[tupa]-[he-yoli]-[he-gehe]-di
 TAM 1SG.S-bump-CAUS-sink-CAUS-be.finished-3PL.O

'I'll drown all of them'

literally: 'I'll bump-sink-finish them'

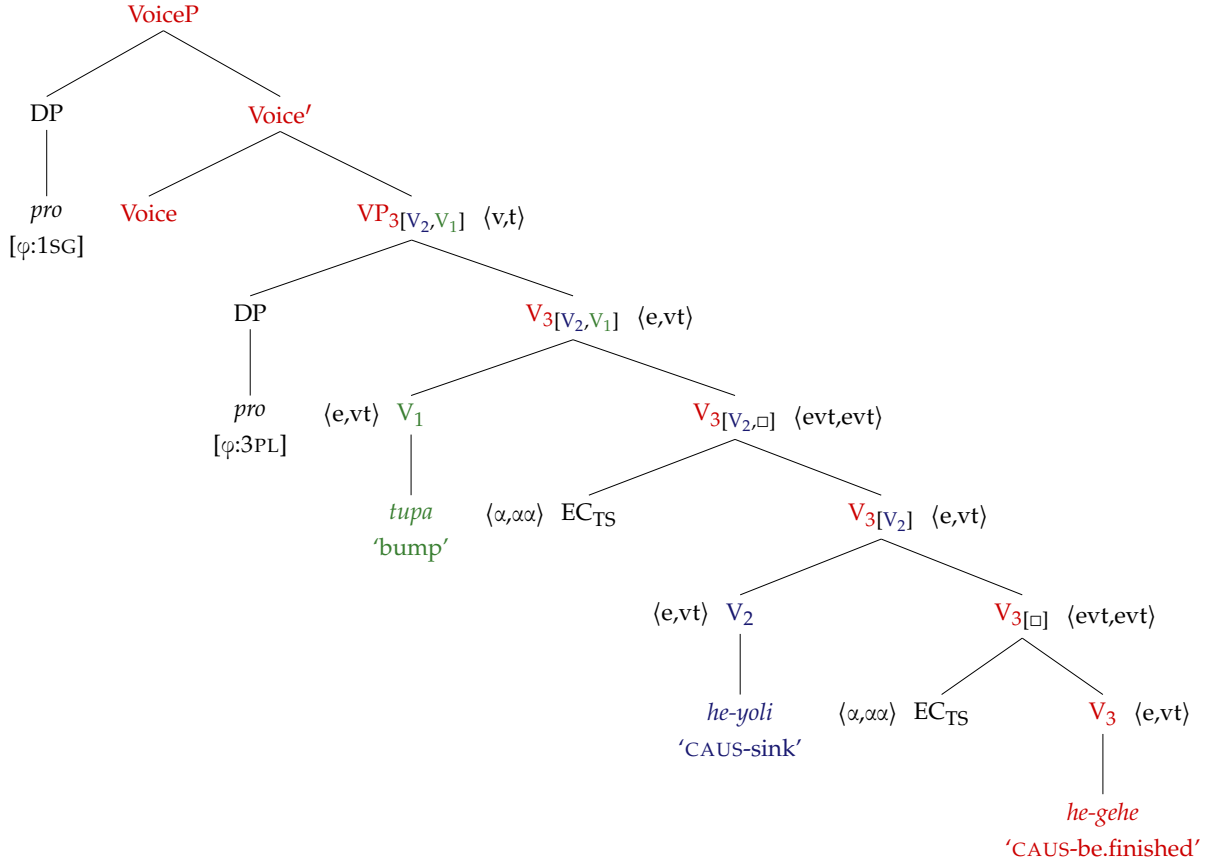
(Margetts 2005: 67)

This follows because EC_{TS} is a local binary rule that can apply to its own output.¹⁵⁵ To see this, consider the following derivation of (20), the constituent verbs of which have the denotations in (21). For clarity, the derivation is broken up into two parts, one for each cycle of EC_{TS}.

After EC_{TS} has combined the V_{3=H} 'make finished' and the V₂ 'make sink', thereby ending the first cycle, the resulting complex predicate in (22c) does not

¹⁵⁵Note that in order for the iterative application of EC_{TS} to go through, sub-events and macro-events must be ontologically identical for the purposes of C-SVC formation, differing only in their complexity; they are represented throughout with different symbols (*e* vs *E*) for clarity.

undergo event closure, but instead feeds itself as an input into EC_{TS} , thereby initiating the second cycle. Because the $V_{3=H}$ ‘make finished’ has been established as the syntactic head in the first cycle, it remains the syntactic head in. In the second cycle, after taking the complex predicate resulting from the first cycle in (22c) as its first input, EC_{TS} takes the V_1 ‘bump’ as its second input, yielding the three-verb complex predicate in (23c). The rest of the derivation proceeds normally, terminating in (24):



Tree 8.5: Transitive three-verb C-SVC

- (21) a. $\llbracket V_1 \rrbracket = \llbracket tupa \rrbracket = \lambda x \lambda e_1. [bump(e_1, x)]$ $\langle e, vt \rangle$
 b. $\llbracket V_2 \rrbracket = \llbracket he-yoli \rrbracket = \lambda y \lambda e_2. \exists e_2' [CAUSE(e_2, e_2') \& sink(e_2', y)]$ $\langle e, vt \rangle$
 c. $\llbracket V_3 \rrbracket = \llbracket he-gehe \rrbracket = \lambda z \lambda e_3. \exists e_3' [CAUSE(e_3, e_3') \& be\ finished(e_3', z)]$ $\langle e, vt \rangle$

(22) **First application**

a. *Inputs:*

- (i) $\llbracket V_2 \rrbracket = \llbracket he-yoli \rrbracket = \lambda x \lambda e_2. \exists e_2' [CAUSE(e_2, e_2') \& sink(e_2', y)]$ $\langle e, vt \rangle$
 (ii) $\llbracket V_3 \rrbracket = \llbracket he-gehe \rrbracket = \lambda z \lambda e_3. \exists e_3' [CAUSE(e_3, e_3') \& be\ finished(e_3', z)]$ $\langle e, vt \rangle$

- b. $\llbracket V_{3[\square]} \rrbracket = \llbracket EC_{TS}('CAUS-be.finished') \rrbracket =$
 $\lambda P_{\langle e, vt \rangle} \lambda z \lambda E_1. \{ \exists x \} \exists e_2 \exists e_2' \exists e_3 [E_1 = e_2 \oplus e_3 \& P(e_2, x_{i/k}) \& CAUSE(e_3, e_3') \& be\ finished(e_3', z_i)]$
 c. $\llbracket V_{3[V_2]} \rrbracket = \llbracket EC_{TS}('CAUS-be.finished')('CAUS-sink') \rrbracket =$
 $\lambda z \lambda E_1. \exists e_2 \exists e_2' \exists e_3 \exists e_3' [E_1 = e_2 \oplus e_3 \& CAUSE(e_2, e_2') \& sink(e_2', y_i) \& CAUSE(e_3, e_3') \& be\ finished(e_3', z_i)]$

FA(CAUS-sink)

(23) Second application

a. Inputs:

$$(i) \quad \llbracket V_1 \rrbracket = \llbracket tupa \rrbracket = \lambda z \lambda e_1. [\text{bump}(e_1, z)] \quad \langle e, vt \rangle$$

$$(ii) \quad \llbracket V_3[V_2] \rrbracket = \llbracket EC_{TS}('CAUS-be.finished')('CAUS-sink') \rrbracket = \\ \lambda z \lambda E_1. \exists e_2 \exists e_2' \exists e_3 \exists e_3' [E_1 = e_2 \oplus e_3 \ \& \ CAUSE(e_2, e_2') \ \& \ sink(e_2', y_i) \\ \& \ CAUSE(e_3, e_3') \ \& \ be \ finished(e_3', z_i)] \quad \langle e, vt \rangle$$

$$b. \quad \llbracket V_3[V_2, \square] \rrbracket = \llbracket EC_{TS}(EC_{TS}('CAUS-be.finished')('CAUS-sink')) \rrbracket = \\ \lambda R_{\langle e, vt \rangle} \lambda y \lambda E_2 \exists E_1 \{ \exists x \} \exists e_1 \exists e_2 \exists e_2' \exists e_3 \exists e_3' [E_2 = e_1 \oplus E_1 \ \& \ E_1 = e_2 \oplus e_3 \ \& \\ P(e_1, x_i/k) \ \& \ CAUSE(e_2, e_2') \ \& \ sink(e_2', y_i) \ \& \ CAUSE(e_3, e_3') \ \& \ be \ finished(e_3', z_i)]$$

$$c. \quad \llbracket V_3[V_2, V_1] \rrbracket = \llbracket EC_{TS}(EC_{TS}('CAUS-be.finished')('CAUS-sink'))('bump') \rrbracket = \\ \lambda z \lambda E_2. \exists E_1 \exists e_1 \exists e_2 \exists e_2' \exists e_3 \exists e_3' [E_2 = e_1 \oplus E_1 \ \& \ E_1 = e_2 \oplus e_3 \ \& \ bump(e_1, x_i) \\ \& \ CAUSE(e_2, e_2') \ \& \ sink(e_2', y_i) \ \& \ CAUSE(e_3, e_3') \ \& \ be \ finished(e_3', z_i)] \\ FA('bump')$$

$$d. \quad \llbracket VP_3[V_2, V_1] \rrbracket = \\ \lambda E_2. \exists E_1 \exists e_1 \exists e_2 \exists e_2' \exists e_3 \exists e_3' [E_2 = e_1 \oplus E_1 \ \& \ E_1 = e_2 \oplus e_3 \ \& \ bump(e_1, x_i) \ \& \\ CAUSE(e_2, e_2') \ \& \ sink(e_2', y_i) \ \& \ CAUSE(e_3, e_3') \ \& \ be \ finished(e_3', z_i)] \\ FA('they')$$

$$e. \quad \llbracket Voice' \rrbracket = \\ \lambda w \lambda E_2. \exists E_1 \exists e_1 \exists e_2 \exists e_2' \exists e_3 \exists e_3' [E_2 = e_1 \oplus E_1 \ \& \ E_1 = e_2 \oplus e_3 \ \& \ bump(e_1, x_i) \\ \& \ CAUSE(e_2, e_2') \ \& \ sink(e_2', y_i) \ \& \ CAUSE(e_3, e_3') \ \& \ be \ finished(e_3', z_i) \ \& \\ AGENT(E_2) = w] \\ agent$$

$$f. \quad \llbracket VoiceP \rrbracket = \\ \lambda E_2. \exists E_1 \exists e_1 \exists e_2 \exists e_2' \exists e_3 \exists e_3' [E_2 = e_1 \oplus E_1 \ \& \ E_1 = e_2 \oplus e_3 \ \& \ bump(e_1, x_i) \\ \& \ CAUSE(e_2, e_2') \ \& \ sink(e_2', y_i) \ \& \ CAUSE(e_3, e_3') \ \& \ be \ finished(e_3', z_i) \ \& \\ AGENT(E_2) = I] \\ FA('I')$$

$$(24) \quad \exists E_2 \exists E_1 \exists e_1 \exists e_2 \exists e_2' \exists e_3 \exists e_3' [E_2 = e_1 \oplus E_1 \ \& \ E_1 = e_2 \oplus e_3 \ \& \ bump(e_1, x_i) \ \& \ CAUSE(e_2, e_2') \\ \& \ sink(e_2', y_i) \ \& \ CAUSE(e_3, e_3') \ \& \ be \ finished(e_3', z_i) \ \& \ AGENT(E_2) = I] \\ event \ closure$$

= 'I bump them and sink them and finish them'
 ≈ 'I'll drown all of them'
 = $\llbracket (20) \rrbracket$

8.3 Unaccusativity and unergativity in Saliba

If semantic TM is the result of the grammar tracking the semantic types of verb stems prior to the introduction of any arguments, this heavily constrains the space of possi-

ble verb combinations. Assuming that verb stems characterize only the two types $\langle v, t \rangle$ and $\langle e, vt \rangle$, and given the view adopted here that unaccusatives and transitives lexically select internal arguments while unergatives have external arguments introduced by Voice, we predict that (i) unaccusatives should be combinable with other unaccusatives, that (ii) unergatives should be combinable with other unergatives, and that (iii) unaccusatives should be able to combine with transitives; conversely, we predict that (iv) unergatives should not be combinable with unaccusatives, and finally that (v) unergatives should not be combinable with transitives. Based purely on type-matching, we hence predict a syntactic argument-structural parallelism in C-SVCs. The predicted typology of verb pairings is given in Table 8.1:

	Unaccusative: $\langle e, vt \rangle$	Unergative: $\langle v, t \rangle$	Transitive: $\langle e, vt \rangle$
Unaccusative: $\langle e, vt \rangle$	✓ ?: status unclear	*	✓ ✗: thematic contradiction
Unergative: $\langle v, t \rangle$	*	✓	*
Transitive: $\langle e, vt \rangle$	✓ ✗: thematic contradiction	*	✓

Table 8.1: Typology of pairings based on type-matching

Assuming that the unaccusative-unergative distinction holds syntactically in Saliba, we will see that the predicted syntactic argument-structural parallelism predicted by type-matching is borne out. Additionally, we need to rule out pairings of unaccusatives and transitives by appealing to the condition against thematic contradictions, given that these are predicted by type-matching alone yet unattested. As for double unaccusative pairings, their status remains unclear (at least in Saliba).

According to Margetts (1999), there are no detectable phrase-structural differences between intransitive verbs that would be consistent with an unaccusative-unergative distinction. She argues that Saliba intransitive verbs are better characterized as patterning roughly along an active-stative dimension. Among the relevant criteria for this distinction is the ability of a root to occur as a nominal modifier expressing an attributive state (Margetts 1999: 81).

In Saliba, attributive modification of a noun by a verb involves a reduplicated form of the verb which cross-references the modified noun via possessive morphology. Verbs that can enter into such modification express either states (25) or changes of state (26):

(25) *Attributive nominal modification: stative verb*

- a. Leiyaha ye-pitali
pandanus 3SG.S-dry

- 'The pandanus is dry'
- b. leiyaha pita-pitali-na
pandanus RED-dry-3SG.POSS
'dry pandanus' (Margetts 1999: 81)
- (26) *Attributive nominal modification: change-of-state verb*
- a. Waga ye-yoli
boat 3SG.S-sink
'The boat sank'
- b. waga yoli-yoli-na
boat RED-sink-3SG.POSS
'sunken boat' (Margetts 1999: 82)

By contrast, active verbs like *lao* 'go' are unable to function as nominal modifiers:

- (27) *Attributive nominal modification: active verb*
- a. Waga-wa ye-lao-ko
boat-PM 3SG.S-go-PERF
'The boat went already'
- b. *waga lao-lao-na
boat RED-go-3SG.POSS
intended: 'the gone boat' (Margetts 1999: 84)

The generalization is that attributive nominal modification is restricted to verbs that entail a state, either because this state is directly expressed in the case of stative verbs (25), or because there is an entailed result state in the case of change-of-state verbs (26). Active verbs, which only express an activity but do not entail a state, are therefore precluded from attributive nominal modification (27).

Margetts (1999: 81) classifies verbs allowing attributive nominal modification as mostly stative, and verbs disallowing it as mostly active. I propose to refine this classification as follows in terms of the unaccusative-unergative distinction: first, Margetts' "stative" intransitive verbs branch out into simple statives (e.g. *pitali* 'dry'), and those expressing a change of state entailing a result state (e.g. *yoli* 'sink'); the latter I will take to be unaccusatives. Second, what Margetts labels as "active" intransitive verbs are unergatives (e.g. *lao* 'go').

Now, there is one intransitive verb which appears at first glance to falsify the prediction that unaccusatives should not be combinable with unergatives, namely the verb *lage* 'arrive'. An example of its use in isolation is given in (28). In (29), *lage* figures as a $V_{2=H}$ with the verbs *sobu* 'dance' (29a) and *nene* 'crawl' (29b):

- (28) Ya-lage
1SG.S-arrive
'I arrive' (Margetts 2013: 19)
- (29) *lage* 'arrive' in C-SVCs

- a. Ye-[sobu]-[lage]
3SG.S-dance-arrive
'He came dancing'
 - b. Ye-[nene]-[lage]
3SG.S-crawl-arrive
'He came crawling'
- (Margetts 2005: 75)

What is potentially problematic about the cases in (29) is that, cross-linguistically, 'arrive' is often unaccusative. Its co-verbs in (29), however, i.e. 'dance' and 'crawl', are clearly unergative because the referents of their intransitive subjects exert control over the actions of dancing and crawling, respectively. If *lage* 'arrive' is indeed unaccusative, the cases in (29) thus pose a genuine problem for the present analysis of TM. Type-theoretically, *lage* 'arrive' as an unaccusative of type ⟨e,vt⟩ should be able to combine with neither *sobu* 'dance' and *nene* 'crawl', both unergatives of type ⟨v,t⟩.

There is, fortunately, support for the idea that *lage* 'arrive' is in fact not unaccusative in Saliba. The first piece of support comes from attributive nominal modification. As shown above, this kind of modification is available for stative verbs like *pitali* 'dry' (30a) and change-of-state verbs like *yoli* 'sink' (30b):

(30) *Nominal modifier expressing result state*

- a. *pitali* 'dry'
leiyaha pita-pitali-na
pandanus RED-dry-3SG.POSS
'dry pandanus'
 - b. *yoli* 'sink'
waga yoli-yoli-na
boat RED-sink-3SG.POSS
'sunken boat'
- (Margetts 1999: 81f)

(31) shows that *lage* 'arrive' cannot enter into such modification:

(31) *Nominal modifier expressing result state: lage* 'arrive'

- *tamowai lage-lage-di
people RED-arrive-3PL.POSS
intended: 'arrived people'
- (Margetts 1999: 82)

The second argument favoring a treatment of *lage* as unergative is informed by the kinds of verbs which are able to occur in its place in positional slot B, i.e. those verbs paradigmatically opposed to *lage* within C-SVCs. Recall that verbs in Slot B express either a causative event entailing a result, or some other, non-causative activity; these verbs are either intransitive or transitive. Transitive verbs in Slot B are

either morphologically derived causatives based on stative or change-of-state roots, lexical causatives, or other underived transitives; these are given in Tables 8.2 and 8.3:¹⁵⁶

Verb	Meaning	Verb	Meaning
<i>he-beku</i>	'CAUS-fall'	<i>he-mwaloi</i>	'CAUS-dead'
<i>he-bida</i>	'CAUS-dirty'	<i>he-pesa</i>	'CAUS-exit'
<i>he-dudulai</i>	'CAUS-straight'	<i>he-pitali</i>	'CAUS-dry'
<i>he-dui</i>	'CAUS-bathe'	<i>he-posi</i>	'CAUS-white'
<i>he-gagili</i>	'CAUS-small'	<i>he-sigi</i>	'CAUS-move'
<i>he-kabasi</i>	'CAUS-hang'	<i>he-tolo</i>	'CAUS-stand'
<i>he-keno</i>	'CAUS-lie'	<i>he-tuli</i>	'CAUS-sit'
<i>he-laki</i>	'CAUS-big'	<i>he-yoli</i>	'CAUS-sink'
<i>he-masahala</i>	'CAUS-clear'	<i>he-you</i>	'CAUS-bend'
<i>he-mwayau</i>	'CAUS-full'		
<i>godu</i>	'break'	<i>pulisi</i>	'tear'
<i>kesi</i>	'break'	<i>utusi</i>	'snap'
<i>lapai</i>	'pierce'	<i>yawasi</i>	'clean'

Table 8.2: Transitive verbs in Slot B

¹⁵⁶Note that because some verbs are ambi-transitive, they are listed in both Table 8.3 and Table 8.4 based on their respective uses. Moreover, some verbs occur both as simplex intransitives in Table 8.4 and as causativized transitives in Table 8.2.

Verb	Meaning
<i>isini</i>	'raise'
<i>nonohai</i>	'prepare'
<i>piloi</i>	'turn'
<i>wadam</i>	'hide'
<i>basiyei</i>	'put aside'
<i>gagayo</i>	'advise / warn'
<i>kalatei</i>	'hold down'
<i>liga</i>	'cook'
<i>lobai</i>	'find'
<i>sipwa</i>	'tip' / 'trap'
<i>tonogi</i>	'try'

Table 8.3: Transitive verbs in Slot B (continued)

Given that all transitive verbs in Tables 8.2 and 8.3 are of type $\langle e, vt \rangle$, and assuming that they are all agentive, I conclude that *lage* 'arrive' is also agentive, i.e. an unergative of type $\langle v, t \rangle$.

What we also need to consider are the kinds of intransitive verbs in paradigmatic opposition to *lage* 'arrive' in Slot B, which are given in Table 8.4:

Verb	Meaning	Transitivity	Argument structure
<i>isini</i>	'rise / go up'?	intransitive	unergative?
<i>lage</i>	'arrive'	intransitive	unergative
<i>liga</i>	'cook'	intransitive	unergative
<i>pesa</i>	'exit'	intransitive	unergative
<i>tolo</i>	'get / stand up'	intransitive	unergative
<i>wadam</i>	'hide'	intransitive	unergative

Table 8.4: Intransitive verbs in Slot B

Disregarding *isini* 'rise / go up' for the moment, I propose that all of the intransitive verbs that are able to occur in Slot B in place of *lage* 'arrive' are unergative. This is

assumed based on translation and their syntagmatic opposition, or inferred from the test of attributive nominal modification.

The verb *liga* ‘cook’, for example, has an intransitive subject referring to the cooker, not the thing cooked, and occurs in C-SVCs with *lao* ‘go’, which we know is unergative (32):¹⁵⁷

- (32) Ya-[lao]-[liga]
1SG.S-go-cook
‘I go and cook’ (Margetts 2013: 20)

That e.g. *pesa* ‘exit’ is also unergative is evinced by its inability to occur as a nominal modifier expressing a state (33):

- (33) a. Se-pesa
3PL.S-exit
‘They went out’
b. *tau pesa-pesa-na
man RED-exist-3SG.POSS
intended: ‘gone out man’ (Margetts 1999: 84)

This leaves us with *isini*, an ambi-transitive verb meaning ‘rise / go up’ when intransitive, and ‘raise’ when transitive. While I have not found data showing that *isini* is clearly unergative, all examples involving this verb are consistent with an unergative analysis.¹⁵⁸ In the absence of evidence to the contrary, I will assume that the intransitive variant of *isini* is also unergative.

We thus have indirect, yet strong support for treating *lage* ‘arrive’ as unergative, meaning that the syntactic argument-structural parallelism predicted by the type-matching mechanism underlying semantic TM is borne out.

That said, even if there was no syntactic unaccusative-unergative distinction in Saliba, with, say, all intransitive arguments merged as internal arguments, to the

¹⁵⁷Similarly, consider the example in (i) involving the intransitive verb *tolo* ‘get / stand up’, which clearly shows its meaning to be unergative:

- (i) Ye-keno-keno, ye-siliyata, ye-tolo
3SG.S-RED-lie 3SG.S-become.conscious 3SG.S-stand.up
‘She was sleeping, she became conscious and stood up’ (Margetts 2013: 29)

¹⁵⁸(ia) and (ib) show *isini* within C-SVCs in its transitive use meaning ‘raise’:

- (i) a. *trans + ambi-trans*
Ya-[duku]-[isini]-Ø
1SG.S-level-raise-3SG.O
‘I level it up / make it level’ (Margetts 2013: 7)
b. *trans + ambi-trans + ambi-trans + trans*
Ye-[tu]-[isini]-[sae]-[kasaya-i]-Ø
3SG.S-throw-raise-put.up-in.vain-APPL-3SG.O
‘He threw it up in vain’ (Margetts 2005: 67)

effect that all intransitive verbs were of type $\langle e, vt \rangle$, we would still rule out pairings of unaccusatives on the one hand, and unergatives or transitives on the other, given that such pairings would assign both an agent and a patient role to one and the same argument, i.e. give rise to a thematic contradiction.^{159,160,161}

Lastly, even though non-resultative double unaccusative pairings should be perfectly well-formed under type-matching, I have not yet come across any such C-SVCs in Saliba. I suspect that this is either simply an accidental gap in the data, or that these pairings are ruled out by independent properties specific to Saliba. Support comes from the fact that non-resultative C-SVCs consisting of two verbs consistent with an unaccusative analysis are attested in other languages, e.g. in Eastern Kayah Li (Solnit 2006):

(34) **Eastern Kayah Li:** *intr + intr*

¹⁵⁹There remains the case of (i), where the apparently unaccusative V_1 ‘fall’ is paired with the unergative $V_{2=H}$ ‘go down’:

(i) *intr + intr*

Ye-[beku]-[dobi]
3SG.S-fall-go.down

‘He fell down’

(Margetts 2005: 73)

If *beku* ‘fall’ is indeed unaccusative, (i) would similarly be a counter-example to the claim that unaccusatives are unpairable with unergatives. The inability of *beku* ‘fall’ to function as a nominal modifier, however, suggests that it is not unaccusative, at least not canonically so in terms of its syntax and semantic type (iia); this contrasts with the semantically related verb *guli* ‘fall over’ (iib), and also *yoli* ‘sink’ (iic), which can indeed occur in attributive nominal modification, thus clearly patterning as unaccusative:

- (ii) a. *buka beku-beku-na
book RED-fall-3SG.POSS
‘fallen book’
b. kaiwa guli-guli-na
tree RED-fall.over-3SG.POSS
‘fallen over tree’
c. waga yoli-yoli-na
boat RED-sink-3SG.POSS
‘sunken boat’

(Margetts 1999: 82)

While explicit negative evidence is lacking, there are no examples of C-SVCs in which the clearly unaccusative verbs *guli* ‘fall over’ and *yoli* ‘sink’ are paired with unergative verbs in Margetts (1999), Margetts (2005) and Margetts (2013).

¹⁶⁰Potential counter-examples to the event-structural restriction against thematic contradictions are cases like *angetanzt kommen* ‘to arrive dancing’ (Barbara Stiebels p.c.). While the unergative status of *tanzen* ‘dance’ is uncontroversial, it seems to me that *kommen* ‘come’ may actually not be unaccusative here, at least not canonically so, but rather a kind of semi-functional element contributing a purely directional function. The reason is that switching out *kommen* ‘come’ with a clearly unaccusative verb like *fallen* ‘fall’ results in sharp ungrammaticality in my judgment, cf. **angetanzt fallen*, with the intended meaning ‘to fall dancing’. Alternatively, it is the unergative verb in this case which becomes telic.

¹⁶¹Also, recall that Saliba does not have proper resultatives, only MEANS-constructions, which means that type-symmetric pairings with intransitive V_2 s expressing result components are independently unavailable (e.g. a transitive V_1 of type $\langle e, vt \rangle$ and an unaccusative V_2 of type $\langle e, vt \rangle$).

ʔa dīpɔ [tā] [klɔ] títí
 3 rice.pot fall spill constantly

‘His pot kept falling over and spilling’ (Solnit 2006: 149)

Should double unaccusative turn out to be well-formed in Saliba C-SVCs, their derivation would be similar to that of double unergatives – cf. Section 8.2.1 above – except that their sole argument would be internal, not external.

8.4 More on contextual bivalence

This section discusses various properties connected to contextual bivalence, both within and outside Saliba. First, as we know, contextual bivalence may be carried out via transitive or oblique licensing, i.e. it has *different shapes*. Second, in a given language, this objective morpho-syntax is *comparatively unmarked*, i.e. the respective strategy for realizing objects as structural or oblique dependents is generic. Third, there are cases of *failed contextual bivalence*, i.e. configurations where the IAR of a non-head transitive verb cannot be satisfied by V_H . In these cases, contextual bivalence fails because V_H is not of the appropriate type. Finally, we also find configurations that are *functional alternatives* to contextually bivalent ones, where the need for a transitive V_H to satisfy its co-verb’s IAR does not arise. After considering contextual bivalence outside Oceanic, this section closes with a discussion of the relationship between contextual bivalence and raising.

8.4.1 Contextual bivalence across Oceanic

8.4.1.1 Oblique object realization

8.4.1.1.1 Toqabaqita Especially informative is Toqabaqita, an Oceanic language of the Solomon Islands which, like Saliba, possesses C-SVCs. In one type of N-SVC, when both V_1 and $V_{2=H}$ are transitive, the object of the complex is thematically shared with that of V_1 and realized as a structural dependent of $V_{2=H}$, clearly indicated by its φ -marking on $V_{2=H}$.¹⁶²

(35) **Toqabaqita:** *trans + trans*

- a. Nau ku [qala]-[muu-si]-a si qoko
 1SG 1SG.NFUT bite.CF-sever-TR-3.O CLF vine
 ‘I broke / severed the vine by biting it’
- b. ...kwa [riki] [thaitoqoma-na] maka nau...
 1SG.SEQ see know-3.O father 1SG
 ‘...I recognized my father by sight...’
 (literally: ‘...I saw and recognized my father...’) (Lichtenberk 2006: 267)

¹⁶²As Lichtenberk (2006: 267) himself notes, “It is the second verb that is fully transitive and sanctions the direct object”.

But when V_1 is transitive and $V_{2=H}$ comes from the class of manner intransitive verbs, the thematic object of V_1 is realized as an oblique dependent of $V_{2=H}$ via the preposition *qana*, i.e. V_2 becomes a derived extended intransitive:

(36) **Toqabaqita:** *trans* + *intr_{ext}*

- a. Nau ku [fanga] [baqita] **qana** alo
1SG 1SG.NF eat.CF be.big PREP taro
'I ate a lot of taro'
- b. Kini qe [faa] [qaliqali] **qana** teeteru
woman 3SG.NFUT weave.CF be.quick PREP teeteru
'The woman quickly wove a fan' (Lichtenberk 2006: 266, 269)

Crucially, this is in spite of the fact that outside of C-SVCs, neither verb may occur with oblique dependents (Lichtenberk 2006, Lichtenberk 2008). The verb 'eat', for instance, has two transitivity-conditioned allomorphs: transitive *qani*, and intransitive *fanga*. On its own, the transitive form *qani* can only take a direct object (37a), while the intransitive variant of *fanga* cannot occur with an oblique object (not even on a partitive reading; (37b)), and in fact must lack any non-subject dependent altogether (37c):

(37) **Toqabaqita**

- a. Wela qe qani-a alo
child 3SG.NFUT eat-3.O taro
'The child ate the taro'
- b. *Wela qe fanga qana alo
child 3SG.NFUT eat PREP taro
intended: 'The child ate (some of) the taro'
- c. Nau ku fanga sui naqa
1SG 1SG.NFUT eat COMPL PERF
'I've finished eating' (Lichtenberk 2006: 582, 266)

The preposition *qana* is the most generic preposition in Toqabaqita: in both the verbal and the nominal domain, it collapses a wide range of functions that are roughly spread across the prepositions *of*, *to*, *at*, *on*, *for* and *with* in English (Lichtenberk 2006: 488ff.). Complements to *qana* include locations (38a), goals and recipients (38b), instruments (38c) and purposes (38d):

(38) **Toqabaqita**

- a. *Location*
Keka too naqa **qana** ua qeri
3PL.SEQ stay PERF PREP hill that
'They stayed on that hill'
- b. *Goal*

According to Crowley (2002), there is no difference in meaning between these two variants. This strongly suggests that while the object of a transitive V_1 surfaces *morpho-syntactically* as a dependent of a V_H which, on its own, does not license internal dependents, the *way* in which this "inherited" object is realized is ultimately a PF matter. If ex-situ object realization were determined in the narrow syntax, we might expect the choice between the structural and the oblique variant to correlate with certain interpretive differences, such as event telicity or object individuation.¹⁶³

8.4.1.2 Unmarked objective morpho-syntax

8.4.1.2.1 Saliba In Saliba, contextually bivalent verbs always involve the suffix *-(e)i*, glossed as an applicative; they never undergo transitivization via the causative *he-* (the only other productively available valence-increasing operation). Next to occurring with DP objects referring to Patients (40), *-(e)i* occurs when other non-core roles that would otherwise be expressed as obliques project structurally: these "applied" arguments denote stimuli with experience verbs (40b), addressees with verbs of communication (40c), locations with verbs involving posture and other bodily functions (40d), and concomitant objects with motion verbs (40e) (Margetts 1999: 145ff.):

(40) **Saliba**

- a. *Patient*
 - (i) Se-lulu
3PL.S-fight
'They fought'
 - (ii) Se-lulu-**i**-gau
3PL.S-fight-APPL-1SG.O
'They attacked me'
- b. *Stimulus*
 - (i) Ye-koipili
3SG.S-angry
'She's angry'
 - (ii) Ye-koipili-**ei**-gau
3SG.S-angry-APPL-1SG.O
'She's angry with me'
- c. *Addressee*
 - (i) Ya-henamai
1SG.S-ask
'I asked'
 - (ii) Ya-henamai-**ei**-go
1SG.S-ask-APPL-2SG.O
'I asked you (a question)'

¹⁶³Note that this is part of the reason why ADD must actually be an operator, rather than a rule, given that it has effects on both form and meaning.

- d. *Location*
- (i) Teina numa unai ya-bawa
 PROX.DEM house POSTP 1SG.S-stay
 'I stay at this house' (e.g. as a guest)
 - (ii) Teina numa ya-bawa-i- $\emptyset$
 PROX.DEM house 1SG.S-stay-APPL.3SG.O
 'I live in / occupy this house' (e.g. it is my home)
- e. *Concomitant object*
- (i) Ye-heloi
 3SG.S-run
 'He ran'
 - (ii) Ye-heloi-ei- $\emptyset$
 3SG.S-run-APPL-3SG.O
 'He ran with it' (e.g. with something tied to his foot)
- (Margetts 1999: 152, 153, 154, 156, 158)

As outlined earlier, the suffix *-(e)i* does not instantiate a canonical applicative. Rather, its distribution finds a natural explanation if it is instead treated as serving the morpho-syntactic role of licensing full DP objects, with a semantics that is highly unmarked or even empty, such that the thematic interpretation of the respective DP object is determined (largely or entirely) by the base verb. Because the causative is the only other option available, yet comparatively marked due to actually making a semantic contribution by introducing a causer, the marker *-(e)i* surfaces instead.

8.4.1.2.2 Daakaka As mentioned, contextual bivalence is attested for at least one verb in Daakaka, spoken in Vanuatu. (41) shows that the verb *bup* 'tumble' in isolation can only be intransitive. Yet once it enters into a C-SVC, it undergoes exceptional transitivization via the suffix *-ane* (41b):

- (41) **Daakaka**
- a. (i) Ekuokuo mu bup
 car REAL tumble
 'The car tumbled over'
 - (ii) *ma bup-ane ding
 REAL tumble-TRANS mat
 intended: 'She tumbled the mat'
 - b. *trans* + *trans*
- Ma [ling]-[bup-ane] ding
 REAL put-tumble-TRANS mat
 '(S)he put the mat the wrong way up' (von Prince 2015: 315f)

Like the Saliba suffix *-(e)i*, the suffix *-ane* is the most common and only synchronically productive transitive suffix, occurring in morpho-syntactically transitive contexts with full DP objects. The specific thematic role of the argument flagged by its presence

is not fixed, but largely determined by the meaning of the verb it combines with, as well as the meaning of the introduced argument itself and contextual factors (von Prince 2015: 59ff). Next to patterning like a causative with stative verbs (42a), *-ane* plays an applicative-like role in introducing Patients (42b), stimuli with experiencer verbs (42c), and also goals, paths and locations with motion verbs (42d):¹⁶⁴

(42) **Daakaka:** *Various functions of -ane*

- a. Mwe ngapngap-**ane** kolir-an
REAL rest-TRANS sing-NMLZ
'He stopped singing'
- b. Mwe yas-**ane** webir
REAL steal-TRANS breadfruit
'(S)he stole the breadfruit'
- c. Na=m yungpan-**ane** wye ten
1SG-REAL thirsty-TRANS water very
'I am very thirsty'
- d. Ø-an livis mwe vyan-**ane** misy-an nyosi
CL2-3SG.POSS banana.plant REAL go-TRANS uncle-3SG.POSS 3PC
'Her banana plant goes to her uncles' (von Prince 2015: 60f.)

Again, while it is unclear to what extent causative-like or applicative-like semantics are (always) introduced by *-ane* itself, its distribution follows if its main purpose is to Case-license full DP objects (Hopperdietzel 2020), thereby reflecting a generic transitive morpho-syntax.

8.4.1.2.3 Nêlêmwa Numerous contextually bivalent verbs are also found in Nêlêmwa (New Caledonia). The verbs *mwang* 'be bad' and *hââhuux* 'be recent', for instance,

¹⁶⁴The function of *-ane* extends beyond the verbal domain to to the prepositional and nominal domains. In (ib), *-ane* occurs suffixed to the preposition *pesili* 'near' when it takes a ground-denoting complement; in (iib), *-ane* occurs on an inalienable possessum:

(i) **Daakaka:** *-ane on preposition*

- a. Na-m ling-i dal-uk nyoo pesili
1SG.S-REAL put-TRANS egg-1SG.POSS 3PL near
'I laid my eggs nearby'
- b. Ko-m tinyo pesili-**ne** lee swa
2SG.S-REAL stand near-TRANS tree one
'You stand close to a tree' (von Prince 2015: 61)

(ii) **Daakaka:** *-ane on inalienable possessor*

- a. bura ente
blood DEM
'this blood'
- b. bura-**ne** vyantem ente
blood-TRANS person DEM
'this person's (own) blood' (von Prince 2015: 70)

Based on this distribution, (Hopperdietzel 2020: 277ff) argues that *-ane* is the spell-out of a secondary licensing head that case-licenses DP-sized nominals that would otherwise remain unlicensed.

head C-SVCs with transitive co-verbs:

(43) **Nêlêmwa**

a. *trans* + *trans*

Hla [diya] [hââhuux-e] mwa eli
 3PL do be.recent-TRANS house that.ANAPH
 ‘They built this house recently’ (Bril 2004: 177)

b. *trans* + *trans*

I [hnawe] [mwang-ilî]
 3SG drop be.bad-TRANS
 ‘He dropped it awkwardly’ (Bril 2004: 176)

These verbs (and others affording this behavior) can never be transitive in isolation, only within C-SVCs in the context of a transitive co-verb (Isabelle Bril p.c.). As expected, the suffix glossed as TRANS seems to reflect a generic transitive morpho-syntax involving full DP objects (Bril 2002).

8.4.1.2.4 Pileni Pileni is spoken on the Solomon Islands. Even though it stands out from the languages discussed here in terms of contextual bivalence, it turns out to especially informative. Like in the other cases discussed here, intransitive $V_{2=HS}$ expressing manner meanings must transitivize in the context of a transitive co-verb. In (44), for example, the intransitive verb *themu* ‘be quiet’ must transitivize when combined with the transitive co-verb *kip* ‘keep’. Yet unlike in the other cases discussed so far, *themu* ‘be quiet’ is perfectly able to transitivize independently (44b):¹⁶⁵

(44) **Pileni**

a. *trans* + *trans*

Kolu-no maoli la khoulua [kip-ina]-[themu-ina]
 2.DU-TAM true DEM 2.DU keep-TRANS-quiet-TRANS
 ‘If you are telling the truth, keep it quiet’

b. *ha-themu-akina*

CAUS-quiet-APPL
 ‘to silence someone’ (Næss 2004: 236)

Yet there is a significant difference between the transitive use of *themu* ‘be quiet’ within C-SVCs and in isolation: while transitivizing *themu* in isolation always occurs via the suffix *-akina* (44b), its transitivization in C-SVCs can only occur via the suffix *-ina* (44) (Næss 2004: 236).¹⁶⁶

¹⁶⁵Recall that Oceanic transitive suffixes are sometimes glossed as APPL despite not always having an applicative function; this is the case in (44b), where the suffix only reflects a transitive morpho-syntax. If (44b) were a true applicative, the meaning of (44b) should be closer to ‘to be silent for someone’.

¹⁶⁶In the Pileni grammar, the suffix *-akina* is classified as an applicative (Næss & Hovdhaugen 2011: 211), on grounds that it generally promotes non-core roles to object. However, as noted by the authors of the grammar themselves, the functions of this suffix is at times indistinguishable from

I submit that this behavior is not at odds with my analysis of contextual bivalence, but consistent with it: just like in the other cases of contextual bivalence discussed here, the V_H realizes an internal argument which draws its thematic interpretation only from its transitive co-verb. This is remarked by the author herself, who notes that "there is no reasonable way of interpreting the V_2 as semantically transitive" (Næss 2004: 236).¹⁶⁷

As for why transitivization within C-SVCs employs *-ina* rather than *-akina*, I propose that transitivization of verbs like *themu* 'be quiet' inside the C-SVC in (44) is of a different nature than outside (44b). Specifically, the presence of the suffix *-akina* – used in the transitive variant of *themu* 'be quiet' in isolation, i.e. *ha-themu-akina* 'to silence' in (44b) – correlates with the presence of two proper thematic roles – an agent / causer silencer and a Patient silencee.

In the C-SVC in (44), by contrast, the presence of the suffix *-ina* correlates with the presence of an expletive thematic role with respect to the $V_{2=H}$ *themu* 'be quiet'. Put differently: the choice of suffix reflects a difference in the underlying thematic source of an argument, with *-akina* reflecting the absence of an expletive internal argument, while *-ina* reflects its presence.

From this, I conclude that we can maintain the core generalization about contextual bivalence, i.e. that it carries out thematic proxy realization. In Pileni, this behavior is slightly masked by the fact that *themu* happens to permit transitivization in isolation, yet in a way that is crucially distinct. Unsurprisingly, the suffix *-ina* is the most productive strategy for transitivization available in Pileni, also occurring on loanwords (Næss & Hovdhaugen 2011: 209).¹⁶⁸

8.4.1.3 Failed contextual bivalence

Consider again Toqabaqita, which we know has contextual bivalence, as in (45a). When the same $V_{2=H}$ heading (45a) forms a C-SVC with an intransitive co-verb, contextual bivalence is not carried out because there is no object to realize (45a):

(45) **Toqabaqita:**

- a. *trans* + *intr_{ext}*

those of other transitivizing suffixes available in the language, and a subset of intransitive verbs show complementary distribution with respect to whether they must be transitivized by *-akina* or some other suffix. For this reason, a characterization of the choice between *-akina* and some other suffix in terms of root-conditioned allomorphy might equally make sense, at least for some verbs. There exist verbs, however, which permit either *-akina* or some other suffix to yield a clear difference in meaning, such as *mena-kina* 'gossip about' vs *mena-ina* 'tell/inform' (Næss & Hovdhaugen 2011: 212).

¹⁶⁷Regarding analogous constructions in Nêlêmwa, Bril (2004: 176) notes similarly that "the V_1 and the V_2 evidence argument fusion not argument sharing: they have different underlying subjects and [...] the stative verb [...] inherits the Patient of V_1 ".

¹⁶⁸Unlike the other cases of contextual bivalence discussed here, then, the Pileni case is better characterized as *contextual pseudo-bivalence*: the bivalence of certain verbs (in the form of transitivity) is not generally impossible, yet their specific kind of bivalent behavior is nonetheless exceptional within the context of C-SVCs.

Nau ku [fanga] [baqita] qana alo
 1SG 1SG.NF eat.CF be.big OBL taro
 ‘I ate a lot of taro’

b. *intr + intr*

Meka [lae] [ofu] nabaa
 1DU.EXCL.SEQ go be.together there.then
 ‘We went together’

(Lichtenberk 2006: 266)

In other words, the manner $V_{2=H}$ can remain while V_1 can be either intransitive (45b) or transitive (45a) (Lichtenberk 2006: 258). As soon as $V_{2=H}$ is a different kind of intransitive verb, however, i.e. does not come from the closed class of manner verbs, V_1 cannot be transitive; in these cases, $V_{2=H}$ can only be paired with an intransitive V_1 (Lichtenberk 2006: 258). Unfortunately, while it is explicitly stated in Lichtenberk (2006) and Lichtenberk (2008) that such pairings are ill-formed, the only examples given are those illustrating well-formed pairings; the example in (47) is constructed based on the transitive V_1 *riki* ‘see’ from (46c) and the intransitive $V_{2=H}$ *karangi* ‘be close’ from (46a), meant to illustrate what an ill-formed, failed case of contextual bivalence might look like:

(46) **Toqabaqita**

a. *intr + intr*

[Qidu] [karangi] mai!
 move be.close VENT
 ‘Move close(r)!’

b. *intr + trans*

Dani qe [qaru] [labi-taqi] nau qasia naqa
 rain 3SG.NFUT fall affect.negatively-TRANS 1SG INTENS INTENS
 ‘I got rained on very badly’

c. *trans + trans*

Kwa [riki] [thaitoqoma-na] maka nau...
 1SG.SEQ see know-3.O father 1SG.POSS
 ‘I recognized my father by sight...’

(Lichtenberk 2006: 258)

(47) **Toqabaqita: Failed contextual bivalence**

*Kwa [riki] [karangi] qana maka nau
 1SG.SEQ see be.close PREP father 1SG.POSS

intended: ‘I looked closer at my father’

(constructed)

The impossibility of [*trans + intr*] pairings when $V_{2=H}$ is not the proper kind of verb showcases a failed attempt on part of the grammar to have the internal argument of the non-head V_1 be realized indirectly by the head $V_{2=H}$. I propose that contextual bivalence fails in these configurations because $V_{2=H}$ lacks whatever crucial property

would allow it to become contextually bivalent, such as some relevant Kimian-stative or manner-related semantic quality.

8.4.1.4 Functional alternatives to contextual bivalence

8.4.1.4.1 Ambient serialization If contextual bivalence occurs due to the MNC, it should be absent in NC-SVCs with a transitive V_1 and a manner $V_{2=H}$. The reason is that, by configuration, NC-SVCs do not give rise to the MNC, meaning that the arguments of any verb can project; consequently, there is no need for the operator ADD to apply, and hence no contextual bivalence occurs.

This prediction is borne out for Oceanic, where contextual bivalence is restricted to C-SVCs. This is best illustrated by Paamese, which next to C-SVCs (39) also has C-SVCs (48a). Note that, just like in the C-SVC in (39), the V_2 s in the NC-SVCs in (48) are manner verbs modifying the event denoted by V_1 . Unlike in (39), however, the manner V_2 s in (48) do not occur strictly adjacent to V_1 , but to the right of the object of V_1 - i.e. the VP_2 headed by the manner verb is adjoined to VP_1 . Because V_1 can project its internal argument in situ, the manner V_2 does not undergo argument extension via ADD:¹⁶⁹

(48) **Paamese**

- a. [A-ro-saani-tei] **seluusiene** [vo-tahosi]
 3SG-NEG1-send-NEG2 message 3SG.FUT-good
 ‘They didn’t send the message properly’ (Crowley 2002: 87)
- b. [Ki-huli-n] **ato** **kail** [he-mal]
 2SG-count-OBJ chicken PL 3SG-be.correct
 ‘You count the chickens correctly’ (Crowley 1987: 40)

8.4.1.4.2 Apparent licensing at a distance There are certain configurations which appear to contradict my analysis of contextual bivalence. In cases like (49), from Toqabaqita, V_1 is transitive and $V_{2=H}$ intransitive, so that, all else being equal, the object of the C-SVC should be realized as an oblique dependent of $V_{2=H}$. But it is not, instead surfacing as a structural object that is next to $V_{2=H}$, but φ -marked on V_1 :

(49) **Toqabaqita: trans + intr**

- a. Kulu [ilitoqo-na] [ofu] **qidu-la-na** fau naqi
 PL.INCL.NFUT try-3.O be.together move-NMLZ-3.PERS rock this
 ‘Let’s try together to move this rock’
 (literally: ‘Let’s try together the moving of this rock’)

¹⁶⁹In the literature, this kind of serialization is known as *ambient* or *event-argument* serialization (Aikhenvald 2006b: 18) because it does not involve sharing of any individual arguments; instead, one of the constituent verbs (in this case V_2) serves a modifying function with respect to the event(s) of the other verb(s). The eventive nature of the modifying verb’s sole argument is in Paamese reflected by its default-like, 3SG subject prefix, regardless of the φ -features of the modified VP’s individual arguments. Outside Oceanic, ambient serialization is also found in Tariana (Arawak) (Aikhenvald 2006a) and Eastern Kayah Li (Sino-Tibetan) (Solnit 2006), among others.

- b. Kawi [soetoqo-**na**] [kokoto] fasi maka nau
 1SG.FUT ask-**3.O** be.correct PREC father 1SG
 ‘I will first ask my father to make sure it’s correct’
 (literally: ‘I will first ask correctly my father’)

(Lichtenberk 2006: 264, 269, 270)

The answer to this puzzle lies in the fact that, in configurations like (49), all else is actually not equal. Toqabaqita has two classes of transitive verbs, called *Class 1* and *Class 2* (Lichtenberk 2006, Lichtenberk 2008). Those C-SVCs above which realize their object as an oblique dependent of the intransitive $V_{2=H}$ have a transitive V_1 from Class 1 (36). By contrast, those in which the object is not realized as an oblique dependent of the intransitive V_H , but seemingly as a structural dependent of V_1 , have a V_1 from Class 2 (49). See Table 8.5:

C-SVC Type	Verbal complex	Internal argument
A: $V_1 \in \text{Class 1}$	V_1 + $V_{2=H}$ <i>transitive</i> <i>intransitive</i>	[_{PP} <i>qana</i> DP]
	V_1 + $V_{2=H}\text{-AGR}_i$ <i>transitive</i> <i>transitive</i>	DP _i
B: $V_2 \in \text{Class 2}$	$V_1\text{-AGR}_i$ + $V_{2=H}$ <i>transitive</i> <i>intransitive</i>	DP _i

Table 8.5: Object licensing in Toqabaqita C-SVCs

Class 1 and Class 2 transitive verbs differ in a number of respects, two of which help us understand this behavior. First, while Class 1 transitives only show object suffixes with 3rd-person lexical objects (50a-ii), Class 2 transitive verbs always obligatorily have object suffixes:¹⁷⁰

(50) **Toqabaqita**

- a. *Class 1 transitive verbs*
- (i) Nau ku rongo-**a** kini qeri
 1SG 1SG.NFUT hear-**3.O** woman that
 ‘I heard the woman’
- (ii) Nau ku rongo qoe
 1SG 1SG.NFUT hear 2SG
 ‘I heard you’

¹⁷⁰More specifically, class 1 transitive verbs only index 3rd-person objects (50a-i), while local-person objects must be encoded as free-standing pronouns (50a-ii). Class 2 transitives index objects of any person. 3rd-person objects can be lexical (50b-i), while local-person objects can only be encoded by suffixes (50b-ii).

b. *Class 2 transitive verbs*

- (i) Keka qanitoqo-**na qofa qeri...**
 3PL.SEQ taste-**3.O** betel.pepper that
 'They tasted the betel pepper...'
- (ii) Toqa neqe kera thaitoqoma-**ku**
 PEOPLE this 3PL.NFUT know-**1SG.O**
 'These people know me' (Lichtenberk 2006: 256)

Second, Class 2 transitives are morphologically more complex than Class 1 transitives. They are either (i) V-N compounds, with the verb stemming from Class 1; or (ii) transparently complex verbs consisting of a Class 1 transitive and the suffix *-toqo*, which expresses a kind of investigative action (Lichtenberk 2008: 117ff). So Class 2 transitives are derived from Class 1 transitives.¹⁷¹

These differences carry over into C-SVCs. When a Class 1 transitive appears as a non-head in a C-SVC, it appears in a morphologically intransitive form despite being semantically transitive, i.e. in its combining form (cf. Section 6.5.2.3 of Chapter 6). While the combining forms of Class 1 transitives never show any kind of transitive morphology including an object suffix, Class 2 transitives always retain their object suffix as non-heads in C-SVCs.

In other words, objects in C-SVCs with Class 2 transitives are not the same kinds of dependents with respect to $V_{2=H}$ in C-SVCs with Class 1 transitives, despite both surfacing adjacent to $V_{2=H}$ but not V_1 . I propose that in cases like (49), the internal argument of the non-head transitive V_1 is realized in situ as a clitic that is borne by the Class 2 transitive verb as a kind of lexically incorporated obligatory argument (for whatever reason). I take this clitic to be a D element, which – as a terminal node – can occur within the syntactic space of the contiguous verbal complex. As for the lexical object (if present), I assume that it is a kind of adjunct that serves a purely specificational function with respect to the clitic, which corresponds to the actual argument. So because (i) the internal argument of V_1 is a clitic that can be licensed in situ within the projection of V_1 , and (ii) the lexical object is not a proper argument but a double, $V_{2=H}$ has no need for ADD and hence remains intransitive. I will not spell out the derivational details here, referring the reader to works like Torrego (1988), van de Visser (2003), Matushansky (2006), Preminger (2009), Kramer (2014) and Harizanov (2014).

8.4.2 Contextual bivalence outside Oceanic

8.4.2.1 Tlachihilco Tepehua

Outside Oceanic, there are at least two languages with SVCs appearing to show contextual bivalence. The first one is Tlachihilco Tepehua, a Totonacan language

¹⁷¹There is a small subset of Class 2 transitives that are neither V-N compounds nor suffixed with *-toqo*, yet most of which also show transparent internal complexity (Lichtenberk 2008: 121ff).

which we have encountered before.

As we know, the only illicit pairing in C-SVCs involves a transitive V_1 and an intransitive $V_{2=H}$. The most common verbs serving as $V_{2=H}$ in C-SVCs are the motion verbs *ʔan* ‘go’ and *min* ‘come’, which, when combined with the so-glossed applicative prefix *ʔi:-*, usually translate to ‘bring’ and ‘take’, respectively.¹⁷² However, when they occur as $V_{2=H}$ s with a transitive co-verb V_1 , the addition of *ʔi:-* on $V_{2=H}$ seems to lack a semantic reflex:

(51) **Tlachichilko Tepehua:** *trans* + *trans*

- a. [maqat'aʔ]-[ʔi:-min]-ta wa:kaf
 throw.at-APPL-come-PERF cow
 ‘(S)he comes throwing (something) at cows’
- b. [laqts'in]-[ʔa:-ʔan]-ta ts'o'o-n
 see-APPL-go-PERF bird-PL
 ‘(S)he goes looking at birds’ (Watters 2017: 293)

Just like *ʔi:-min* does not yield ‘take’ in (51a), *ʔa:-ʔan* does not yield ‘bring’ in (51b); instead, these formally transitive verbs retain their intransitive meanings of ‘come’ and ‘go’, respectively. As made explicit by the author, “[t]he applicative prefix *ʔi:-* contributes nothing to the semantics but is required by the syntax” (Watters 2017: 293). An analogous case is found in the Pisaflores Tepehua dialect (Watters 2017: 284). Because the verbs *ʔan* ‘go’ and *min* ‘come’ can be transitive outside C-SVCs, the Tepehua cases qualify as contextual pseudo-bivalence, thus patterning like Pileni (cf. Section 8.4.1.2.4).

8.4.2.2 Dyirbal

The other non-Oceanic candidate for contextual bivalence is Dyirbal, a Pama-Nyungan language of Australia. Among Dyirbal SVCs, there are a few involving verbs with a transitive morphology that does not appear to be contributing anything to the semantics. In (52), for example, the root-intransitive verb *ɲurba* ‘go’ has been transitivized via the applicative *-m(b)a(l)* in order to be able combine with the root-transitive verb *wuga* ‘give’:

(52) **Dyirbal:** *trans* + *trans*

- ɲinda bayi ɲaygu aña [ɲurba-yarray-ma] [wuga]
 2SG.NOM he.ABS 1SG.POSS new go.back-START-APPL.IMP give.IMP
 ‘You give back (my son)!’ (Dixon 2011: 197)

¹⁷²In Watters (2017), *ʔi:-* is glossed as DIR(ection) because it also has a basic spatial use that marks the ground or direction a figure is oriented towards, next to its applicative function of introducing comitatives; in the context of C-SVCs, however, the author explicitly refers to *ʔi:-* as an applicative (Watters 2017: 293).

The resulting SVC denotes a complex giving-and-returning event, i.e. an event of giving something back (in this case, the speaker's son). Given the translation of (52), however, it is not clear that there is a meaning component involved which we would expect from applicativizing *ɲurba* 'go back'. According to Dixon (2011: 197), *-m(b)a(l)* in (53a) "serves just to ensure transitivity agreement with *wuga*". This contrasts with other uses of *-ma* which have a clear semantic reflex, as in (53a) and (53b), where *-ma* introduces a comitative meaning (in (53b), this is brought out explicitly in the translation):¹⁷³

(53) **Dyirbal**

- a. *trans* + *trans* + *trans*

baŋgul [bulayi-mba-n] [budi-n] [yanu-**ma**-n]
he.ERG two-CAUS-PST carry-PST go-APPL-PST

'Then he carried the two (girls) off' (Dixon 2011: 198)

- b. *trans* + *trans*

[banagay-**mba**-li] yali-gala [yambil-**ma**-li]
go.back-APPL-PURP HERE.TO.PLACE-DOWN fly-APPL-PURP

'(The flying star spirit flew down south to get a painted basket) in order to fly back up here with it' (Dixon 2011: 199)

8.4.3 The locus of contextual bivalence

This section briefly clarifies on the issue of where contextual bivalence takes place. For a two-verb C-SVC, I have argued that a generally intransitive $V_{2=H}$ undergoes an operation of argument extension via ADD *before* combining with its transitive co-verb V_1 . An alternative, it seems, would be to assume that ADD applies *after* C-SVC-formation, i.e. to the $[V_1-V_{2=H}]$ complex as a whole, rather than to one of its parts.¹⁷⁴

First of all, we know that there is a headedness asymmetry within the $[V_1-V_{2=H}]$ complex that goes back to one of its parts, with the non-final V_1 as the non-head and the final $V_{2=H}$ as the head.¹⁷⁵ This, in turn, we know based on numerous asymmetries within in these complexes, such as the inability of non-final transitives to project, and the fact that non-final transitives may be formally intransitive despite being semantically transitive. So because $[V_1-V_{2=H}]$ is headed by V_2 , applying ADD to $[V_1-V_{2=H}]$ amounts to applying ADD to V_2 .

¹⁷³Admittedly, we cannot rule out that *ɲurba-yarray-ma* in (52) expresses a comitative-like meaning, perhaps in a more abstract sense due to serialization, and which is difficult to render in translation (e.g. 'go back with him and give him').

¹⁷⁴I thank Jens Hopperdietzel and the audience at AFLA 27 for raising this issue.

¹⁷⁵Earlier work on SVCs with object sharing has argued that their constituent verbs are co-heads of a single VP (Baker 1989: 145, Baker 1999: 656, Baker 1998: 590). However, as noted in Baker & Stewart (2002: 12), the notion that a single phrase can have two heads is now generally considered theoretically problematic, and most theories of phrase structure rule this possibility out (Kayne 1995: 262, Chomsky 1995: 465).

Second, if the $[V_1-V_{2=H}]$ complex constituted a projection that does not stem from either V_1 or V_2 , this would leave unexplained the fact that the presence of the (transitive or oblique) marker which spells out contextual bivalence shows a systematic sensitivity to properties that are specific to $V_{2=H}$, rather than either V_1 or properties determined jointly by V_1 and $V_{2=H}$. In Saliba, for example, the suffix *-(e)i* occurs only on intransitive verbs, never on transitive verbs (Margetts 1999: 145f). Similarly, the general preposition *qana* in Toqabaqita only ever occurs with intransitive verbs (Lichtenberk 2008: 488ff). The same is true of the transitivizers *-ane* in Daakaka (von Prince 2015: 59ff), *-ilî* in Nêlêmwa (Bril 2000: 227), and *-ina* in Pileni (Næss & Hovdhaugen 2011: 209): all of these are sensitive to transitivity in only being able to occur on intransitive verbs, and all serve to Case-license full DP objects.¹⁷⁶ Given that it is V_2 which is intransitive, and not V_1 , it follows that ADD must enter into an immediately local relationship with V_2 .

If ADD were to apply to a $[V_1-V_{2=H}]$ complex headed by neither V_1 nor V_2 , but, say, a coordinator-like functional head F mediating the combination of V_1 and V_2 , we would not expect it to selectively be sensitive to properties of one of its lexical parts, viz. the intransitivity of V_2 , and conversely, to selectively be blind to properties of another one of its lexical parts, viz. the transitivity of V_1 . What we find, however, is that these properties are consistently determined by V_2 alone.

This sensitivity to $V_{2=H}$ is especially clear in Toqabaqita, where we saw that – as soon as a V_1 is to be combined with an intransitive $V_{2=H}$ that does not allow contextual bivalence – V_1 cannot be transitive; cf. Section 8.4.1.3.

8.4.4 Contextual bivalence and raising

The fact that V_H projects an argument that has its thematic source elsewhere raises the question of whether contextual bivalence is a kind of raising. In classical raising, the subject of an embedded infinitival clause is not realized as a dependent of the non-finite verb it is thematically associated with, but rather as a dependent of an athematic yet finite verb in the matrix clause. This dependency is standardly derived by case-driven A-movement: because non-finite T is unable to assign case, the embedded subject must move into the finite matrix clause in order to receive case from finite T (perhaps also in order to satisfy an EPP feature on T).

Cross-linguistically, typical subject-raising verbs express temporal, aspectual or modal meanings (54a), while object-raising verbs express meanings relating to causation or perception (54b) (for an overview of raising and its relationship to control, see e.g. Polinsky (2013)):¹⁷⁷

¹⁷⁶In Toqabaqita, *qana* also occurs with transitive verbs like ‘give’ and ‘hit’ in order to mark recipients and instruments, respectively (Lichtenberk 2008: 490f).

¹⁷⁷Here, I am ignoring the potential distinction between raising to object and ECM.

- (54) a. *Raising to subject*
 Alma_i seems [_{ec_i} to be reading *Dune*].
 b. *Raising to object*
 Alma considers *Dune*_i [_{ec_i} to be her favorite book].

There are a few arguments which favor treating contextual bivalence as a kind of raising. First, contextually bivalent verbs in C-SVCs realize morpho-syntactic dependents which are not also their thematic dependents, because the corresponding underlying arguments are "inherited" from their transitive co-verb; this dissociation between structural and thematic dependence lies at the core of what it means to be a raising verb.

Second, raising verbs are free to take any type of (non-finite) complement clause given their unaccusative-like status, and contextually bivalent verbs are likewise unconstrained with respect to the kind of transitive verb they may combine with.

Third, just like raising is cross-linguistically limited by the lexical idiosyncrasies of a given language (some languages have raising verbs, while others don't, and semantically similar verbs are raising verbs in some, but not other languages), so does the distribution of contextual bivalence seem lexically idiosyncratic in nature, in that some intransitive verbs (with the appropriate semantics) are contextually bivalent, while others simply aren't.

A final argument for the raising analysis comes from constructions which are functionally equivalent to contextual bivalence, but derived by distinct morpo-syntactic means. Recall ambient serialization in languages like Paamese, where an intransitive manner verb may attach to a transitive VP rather than directly to the transitive verb (55); at least in Paamese, ambient serialization is found alongside contextual bivalence (55b):¹⁷⁸

- (55) **Paamese**
- a. *trans + intr*
- (i) [A-ro-saani-tei] seluusiene [vo-tahosi]
 3SG-NEG1-send-NEG2 message 3SG.FUT-good
 'They didn't send the message properly' (Crowley 2002: 87)
- (ii) [Ki-huli-n] ato kail [he-mal]
 2SG-count-OBJ chicken PL 3SG-be.correct
 'You count the chickens correctly' (Crowley 1987: 40)
- b. *trans + trans*
- Kai [Ø-mutau]-[ramoboŋo-ni] tirausise one-ne
 3SG 3SG-REAL-defecate-do.accidentally-TRANS shorts POSS-3SG
 '(S)he accidentally shat his / her shorts' (Crowley 2002: 61)

¹⁷⁸Unfortunately, it is unclear whether contextual bivalence (V-level adjunction) and ambient serialization (VP-level adjunction) are equally available options for any intransitive verb with the appropriate manner-adverbial meaning, or whether some verbs strictly require one but disallow the other; see Crowley (1987) and Crowley (2002).

Given that the V_2 s in (55) and (55b) are both manner verbs, ambient serialization and contextual bivalence are functional equivalents derived by distinct levels of structure-building: the former involve VP-level adjunction, the latter V-level adjunction.

This difference seems analogous to the alternation found with raising verbs in a language like English, where the complement clause may be non-finite (56a) or finite (56b):

- (56) a. *Non-finite raising complement*
 Alma seems [to be reading *Dune*]
 b. *Finite raising complement*
 It seems [that Alma is reading *Dune*]

This relationship is analogous because in both contextual bivalence and non-finite raising with a non-finite complement, the ex-situ object and the raised subject are realized as syntactic dependents of a verb they are not also thematic dependents of, whereas in both ambient serialization and finite raising, the in-situ object and the non-raised subject are realized as syntactic dependents of a verb that also thematically selects them. We thus arrive at the analogy in (57) and Table 8.6:

- (57) *Contextual bivalence : non-finite raising complement*
 ::
 ambient serialization : finite raising complement

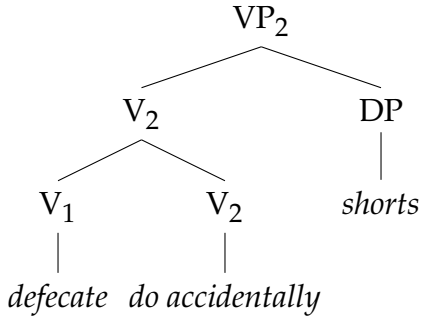
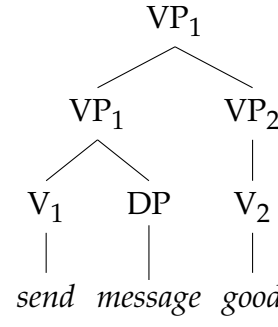
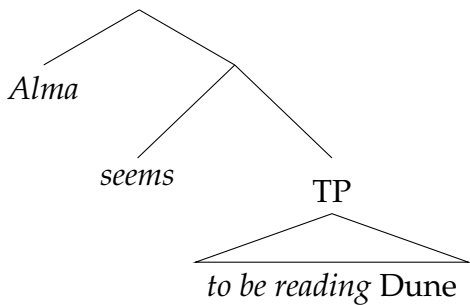
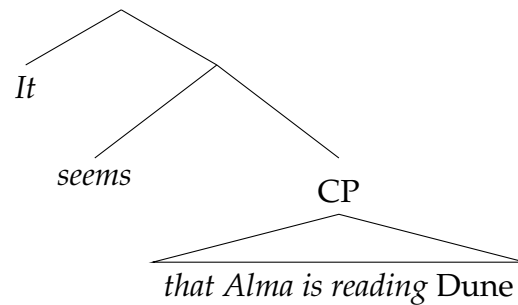
 Contextual bivalence	 Ambient serialization
 Non-finite raising complement	 Finite raising complement

Table 8.6: Raising, contextual bivalence and ambient serialization

Other considerations lessen the credibility of a raising analysis. For one, given that contextually bivalent verbs occur either as independent intransitives or within C-SVCs, whereas raising verbs are ill-formed as standalone elements (which we can attribute to their need for a complement), a raising analysis would need to posit double lexical entries for contextually bivalent verbs, along the lines of (58): a raising variant that takes an event-denoting complement, and a non-raising one that doesn't:

- (58) a. *Double lexical entries for uyo 'return'*
- (i) $\llbracket uyo_1 \rrbracket = \lambda P \lambda e. [P(e) \ \& \ \text{return}(e)]$ / within C-SVCs (raising)
 - (ii) $\llbracket uyo_2 \rrbracket = \lambda e. [\text{return}(e)]$ / elsewhere (non-raising)
- b. *Double lexical entries for namwa 'be good'*
- (i) $\llbracket namwa_1 \rrbracket = \lambda P \lambda s. [P(e) \ \& \ \text{be good}(s)]$ / within C-SVCs (raising)
 - (ii) $\llbracket namwa_2 \rrbracket = \lambda s. [\text{be good}(s)]$ / elsewhere (non-raising)

Yet such multiple listing of variants misses important generalizations, i.e. the fact that – at least in Oceanic – contextually bivalent verbs express Kimian-stative manner meanings, that their bivalence arises predictably as a function of context, and finally that they are a recurring trait within Oceanic. By analogy to other patterns of argument-structural variability – such as the dative or locative alternation – this pattern should instead be captured at the level of (lexical or syntactic) rule; see

Ramchand (2013) for discussion of such alternations, both generally and within generative grammar.

Second, we know that the internal arguments of non-final transitive verbs are realized by V_H – not only in contextual bivalence, but in all transitive C-SVCs. Prior to complex verb formation, there is no syntactic argument present in the structure for the contextually bivalent verb to attract via raising, because it is the contextually bivalent verb itself that introduces and realizes this argument after complex verb formation.

Just to drive home the point, even if non-final transitive verbs did introduce an internal argument in situ, which were subsequently to raise (such that verb-to-verb constituency were merely a surface result), there is a strong counter-argument to be found in transitive C-SVCs with non-object-sharing: assuming that raising occurs for V_1 's internal argument to get case, this argument would have nowhere to raise to because the syntactic object position of V_H would be occupied by V_H 's internal argument (which is sortally distinct); alternatively, assuming that raising additionally occurs to satisfy a requirement of V_H , V_1 's internal argument would equally be unable to raise to V_H because the latter's object position would also be filled by V_H 's internal argument, which would in turn also satisfy V_H 's raising requirement. All else being equal, V_1 's internal argument would remain low either way, and the derivation would crash. Furthermore, we will see in Section 8.5.1 that projecting the non-head V_1 's internal argument prior to complex verb formation makes it impossible for the grammar to enforce TM via type-matching.

A final property of contextually bivalent verbs that sets them apart from raising is that, syntactically, raising verbs target subjects of complement clauses, and semantically, they canonically elaborate on propositions (though not always, cf. *The rain continued*; Barbara Stiebels p.c.); see e.g. Grimm (2010).¹⁷⁹ The disposition of raising verbs towards modifying propositions is reflected in the fact that cross-linguistically, typical raising verbs are mental, perception, modal, phasal, speech, emotive and aspectual verbs (Serdobolskaya 2009: 259). Contextually bivalent verbs, on the other hand, are not merged with their transitive co-verbs via complementation, but adjunction, and as verbs serving a manner-adverbial purpose, their meanings elaborate on events, not propositions; of course, this difference falls out from the fact that contextual bivalence is carried out much lower in the structure than classical raising,

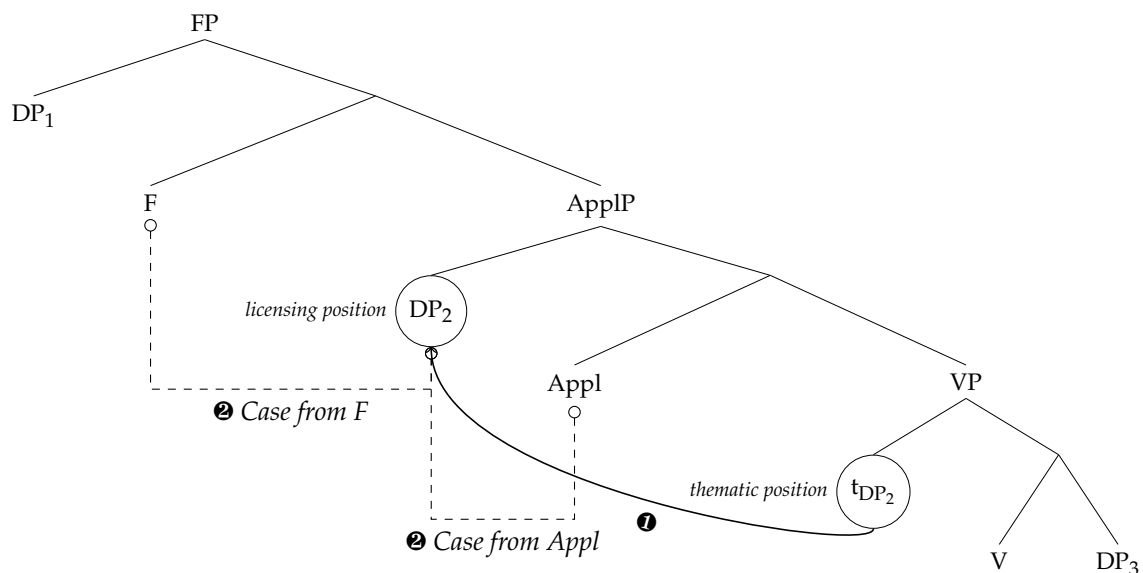
¹⁷⁹That raising predicates necessarily target subjects of complement clauses has been disputed; see Serdobolskaya (2009) and references therein for what appears to be non-subject raising in Kipsigis, and raising to oblique from adverbial clauses in Mongolian. It seems to me, however, that these are not true counter-examples. First, the data show that Kipsigis has a voice system similar to that found in Austronesian languages, which promotes one of various thematic roles to pivot, a subject-like grammatical function that roughly collapses subject and topic properties; in Kipsigis, it is always only this pivot, flagged by dedicated verbal morphology, that can be the target of raising. Second, while adverbial clauses out of which subjects may raise in Mongolian are not themselves complement clauses of the matrix verb, the adposition to which the infinitival subject raises and from which it receives oblique case is itself the head of the adverbial clause. The generalization can thus be maintained that raising always targets subjects of complement clauses.

i.e. within VP rather than between clauses.

My conclusion is that – while contextual bivalence does not qualify as an instance of classical raising – there is an analogous, abstract mechanism at work at the level of argument structure. The main property that makes contextual bivalence distinct from raising is that the argument in question is not transferred syntactically in complementation, but rather via the operator ADD in adjunction. If we wish to emphasize contextual bivalence as a kind of raising, then, an appropriate term would be something like *VP-internal thematic raising*.

8.4.5 Raising applicatives

Contextually bivalent verbs have more in common with so-called *raising applicatives*. Starting with Georgala et al. (2008) and Georgala (2012), a growing body of work has argued that non-core arguments may raise to a derived applicative position in order to be licensed, despite being introduced in a lower thematic position.¹⁸⁰ While differing in the specifics, all these accounts build on the idea that certain applicatives may be co-opted as elements without a thematic import, instead serving the purely morpho-syntactic purpose of attracting some DP to feeds its Case-licensing. After this DP raises to Spec,ApplP, it receives Case either from a higher functional head like Voice, or from Appl itself, as schematized in Tree 8.6:



Tree 8.6: Raising applicative configuration

¹⁸⁰In Georgala et al. (2008), markers passing low-applicative diagnostics in Mandarin and Chi-Mwi:ni are argued to also function as Case-assigners in the presence of VP-internal non-core arguments; for Niuean, Massam (2015) shows that the marker *aki* - next to applying instrument roles - doubles as a case marker for causees in causativized transitives; and in Nie (2019), Tagalog external possession is taken to involve raising of the possessor to Spec,ApplP in order to become licensed for Case assignment. The idea that applicative morphology may contribute to syntactic licensing goes back to at least the account in Baker (1988), which treats applicatives as incorporated prepositions.

8.5 Discussion

This section turns to various grammar-theoretical issues. I discuss timing issues in semantic TM, both generally and specifically in the context of contextual bivalence. I also discuss an alternative to semantic co-indexation, arguing that the co-indexation approach should be maintained. Lastly, I show why semantic TM via type-matching is necessarily restricted to V-level adjunction configurations, and briefly discuss existing approaches to semantic TM.

8.5.1 On the timing of type-matching

This section provides additional support for my claim that semantic TM involves type-matching prior to object introduction. By considering what happens in configurations involving VP-level adjunction, we will see that semantic TM is necessarily restricted to V-level configurations.

Recall that one of the arguments presented for this analysis was the fact that incorporation of a nominal into a transitive verb renders this verb intransitive, such that it can only be paired with another intransitive verb:

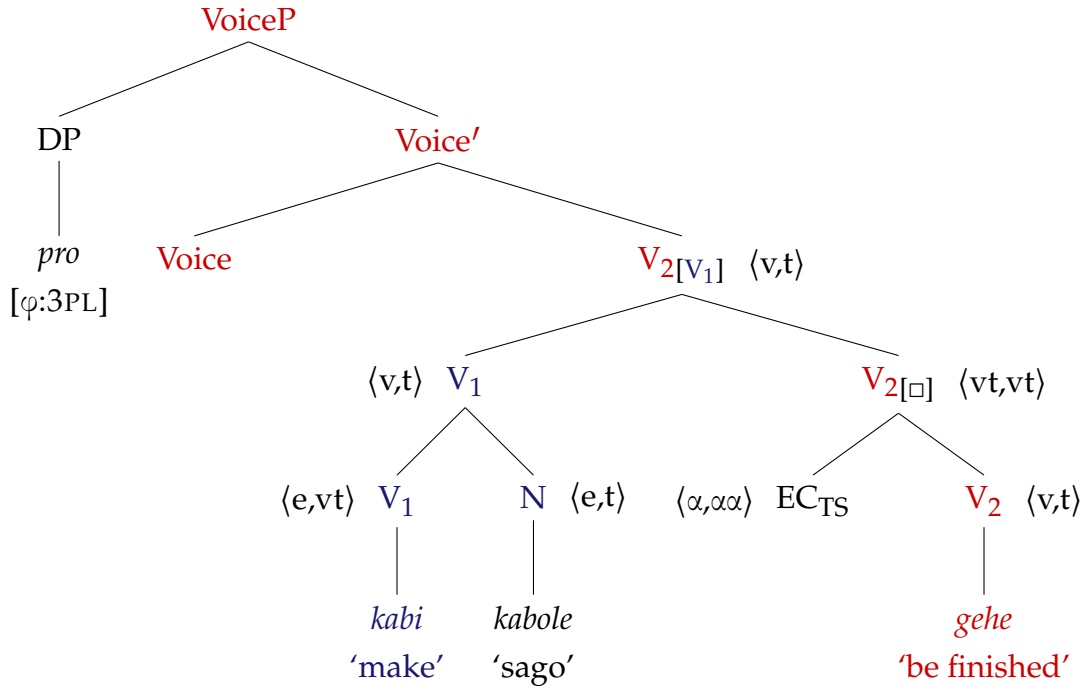
(59) *intr + intr*

Se-[*kabi-kabole*]-[*gehe*]
 3PL.S-make-sago-be.finished
 ‘They finished making sago’

(59) shows that the property tracked by semantic TM is not a property of VPs, but one of (simplex or complex) verb stems without any arguments. Moreover, it shows that this property is mutable, in that incorporation alters the specific value taken on by this property, viz. intransitive as opposed to transitive, or $\langle v, t \rangle$ as opposed to $\langle e, vt \rangle$.

For illustration, I adopt the approach to incorporation by Chung & Ladusaw (2004), according to which the operation RESTRICT composes nouns of type $\langle e, t \rangle$ with transitive verbs of type $\langle e, vt \rangle$, thereby restricting the reference set of the verb’s internal argument to a proper subset characterized by the property denoted by the noun. After RESTRICT applies, existential closure closes off the now referentially restricted internal argument slot. Note that any approach to incorporation will do, so long as incorporation has a saturating effect.

Let us quickly run through the derivation of (59): as an intransitive verb of type $\langle v, t \rangle$, *gehe* ‘be finished’ must combine with another verb of type $\langle v, t \rangle$. The lexical transitive *kabi* ‘make’ is of type $\langle e, vt \rangle$, but incorporation of the noun *kabole* ‘sago’ renders it an intransitive of type $\langle v, t \rangle$ (61). After EC_{TS} applies to *gehe* ‘be finished’ and extends its predicate structure, *kabi-kabole* ‘sago-make’ can now function as the second input into EC_{TS} , yielding (62). The structure of (59) is given in Tree A.1:



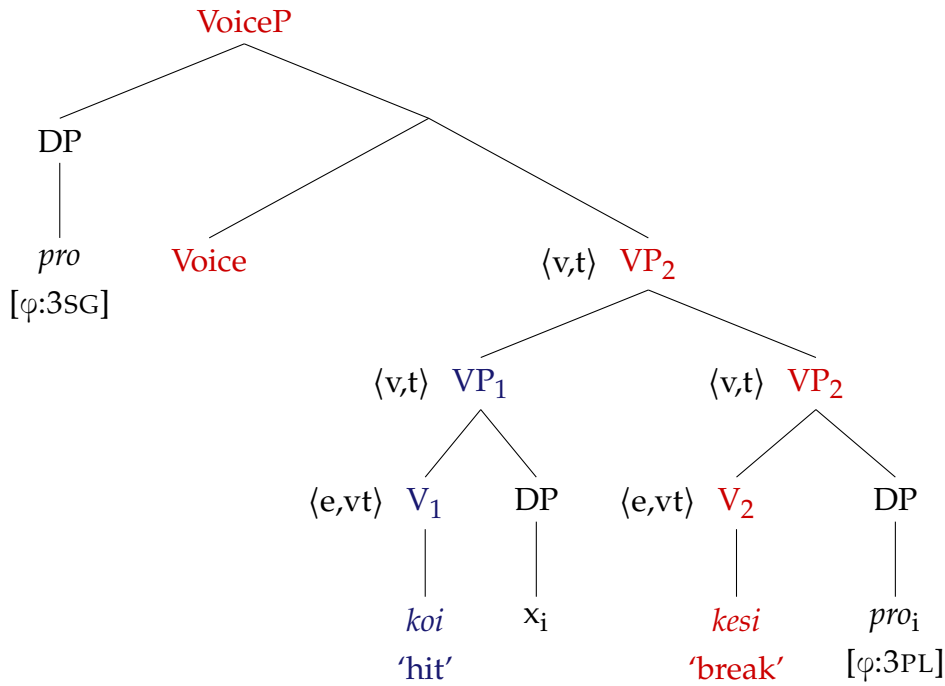
- (60) a. $\llbracket V_1 \rrbracket = \llbracket kabi \rrbracket = \lambda x \lambda e_1. [\text{make}(e_1, x)]$ $\langle e, vt \rangle$
 b. $\llbracket N \rrbracket = \llbracket kabole \rrbracket = \lambda y. [\text{sago}(y)]$ $\langle e, t \rangle$
 c. $\llbracket V_2 \rrbracket = \llbracket gehe \rrbracket = \lambda e_2. [\text{be finished}(e_2)]$ $\langle v, t \rangle$
- (61) a. $\lambda x \lambda e_1. [\text{make}(e_1, x) \ \& \ \text{sago}(x)]$ Restrict
 b. $\lambda e_1. \exists x [\text{make}(e_1, x) \ \& \ \text{sago}(x)]$ Existential closure
- (62) $\exists E \exists x \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ \text{make}(e_1, x) \ \& \ \text{sago}(x) \ \& \ \text{be finished}(e_2) \ \& \ \text{AGENT}(E) = \text{they}]$
 = 'They sago-make and are finished'
 $\approx$ 'They finished making sago'
 = $\llbracket (1) \rrbracket$

A similar argument can be mounted against an alternative analysis according to which Saliba C-SVCs involve VP-level adjunction, i.e. the same kind of analysis we have employed for NC-SVCs in languages like Edo, based on Baker & Stewart (2002). If we apply VP-level adjunction to the Saliba C-SVC in (63), we get Tree 8.8 (which abstracts away from argument movement for illustration):

- (63) *trans* + *trans*

Ye-[koi]-[kesi]-di
 3SG.S-hit-break-3PL.O

'He hit-broke them'



Tree 8.8: Saliba SVC with VP-level adjunction

This structure, however, poses a problem for type-matching: because VP₁ is necessarily adjoined to VP₂ after V₂ has merged with its object, VP₂ will no longer count as transitive when VP₁ is to be adjoined to VP₂. Instead, it will count as intransitive because it no longer has an open internal argument slot. Both VP₁ and VP₂, in fact, will count as intransitive at the point of VP-level adjunction given that both have had their IARs satisfied. The property of transitivity is thus neutralized in terms of types at the VP-level because regardless of whether or not a verb combines with an internal argument, the projected VP will always be of type $\langle v, t \rangle$. The stem-level distinction between type $\langle e, vt \rangle$ and type $\langle v, t \rangle$ breaks down once an object is introduced and the phrasal level is reached. In other words, type-matching comes in too late for transitivity to be distinguished in terms of semantic types, i.e. object introduction bleeds semantic TM.¹⁸¹

In sum, if type-tracking were to come in above the level of the verb stem, it would be unable to backtrack transitivity in terms of semantic types. Contiguous serialization via V-level adjunction feeds type-matching, while serialization at any higher structural level bleeds type-matching because the information required for verbs to be so distinguished is neutralized at the VP-level.

8.5.1.1 Timing in contextual bivalence

Let us now turn to ordering in contextual bivalence. As we know, whichever verb enters C-SVC formation first is the syntactic head V_H. How does this assumption

¹⁸¹Note how this problem arises independently of (i) which VP is the main projection, and (ii) which VP is introduced first into the derivation – it is tied specifically to the structural level at which matching operates.

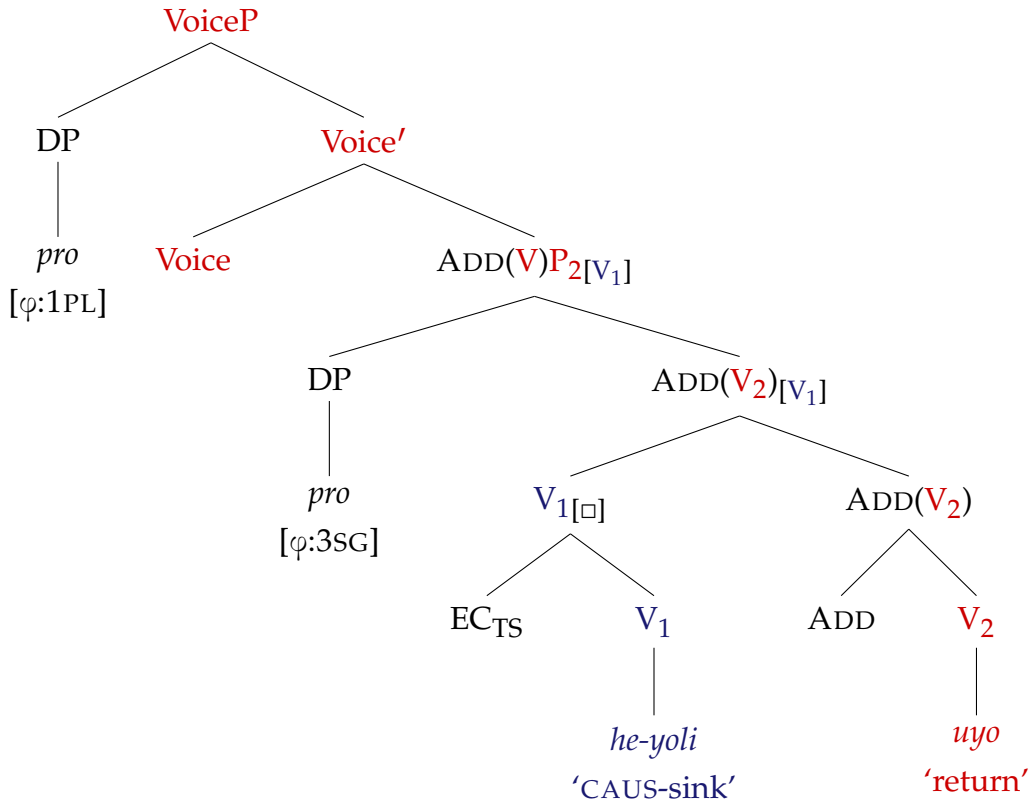
interact with the need to ensure that contextually bivalent verbs unfold their exceptional bivalence only within C-SVCs in the context of a transitive co-verb, but never in isolation? In a two-verb C-SVC, if we think of the transitive V_1 as the "trigger" and the contextually bivalent $V_{2=H}$ as the "target" – in that the former's transitivity and IAR represent the context that is required to feed and force the contextual bivalence of $V_{2=H}$ – it would seem to make more sense for V_1 to be present in the derivation before $V_{2=H}$, rather than vice versa. In the theory of grammar assumed here, where derivations proceed incrementally and bottom-up, nothing rules out this possibility.

Under this alternative approach, the derivation runs the other way around, with EC_{TS} now constituting a kind of *adjunct extension*, whereby the adjunct accommodates to the head. Aside from this difference, the derivation proceeds normally:

(64) *trans* + *trans*

Ta-[he-yoli]-[uyo-i]- $\emptyset$
1INCL.S-CAUS-sink-return-APPL-3SG.O

'We again make it sink'



Tree 8.9: Contextual bivalence (adjunct extension)

- (65) a. $\llbracket V_1 \rrbracket = \llbracket he-yoli \rrbracket = \lambda x \lambda e_1. \exists e_1' [CAUSE(e_1, e_1') \ \& \ sink(e_1', x)]$ $\langle e, vt \rangle$
 b. $\llbracket V_2 \rrbracket = \llbracket uyo \rrbracket = \lambda s_2. [return(s_2)]$ $\langle v, t \rangle$
- (66) $ADD(\lambda s_2. [return(s_2)])$
 $= \lambda y \lambda s_2. [return(s_2) \ \& \ EXPL(s_2) = y]$ $\langle e, vt \rangle$

- (67) a. $\llbracket V_{1[\square]} \rrbracket = \llbracket EC_{TS}(CAUS-sink) \rrbracket =$
 $\lambda Q_{\langle e,vt \rangle} \lambda y \lambda E. \{ \exists x \} \exists e_1 \exists e_1' \exists e_2 [E = e_1 \oplus e_2 \ \& \ CAUSE(e_1, e_1') \ \& \ sink(e_1', x_{i/k})$
 $\& \ Q(e_2, \{y_i\})]$
- b. $\llbracket ADD(V)_{2[V_1]} \rrbracket = \llbracket EC_{TS}(CAUS-sink)(ADD(return)) \rrbracket =$
 $\lambda y \lambda E. \exists e_1 \exists e_1' \exists s_2 [E = e_1 \oplus s_2 \ \& \ CAUSE(e_1, e_1') \ \& \ sink(e_1', x_i) \ \& \ return(s_2)$
 $\& \ EXPL(s_2) = y_i]$
- c. $\llbracket ADD(V)P_{2[V_1]} \rrbracket =$
 $\lambda E. \exists e_1 \exists e_1' \exists s_2 [E = e_1 \oplus s_2 \ \& \ CAUSE(e_1, e_1') \ \& \ sink(e_1', x_{i/k}) \ \& \ return(s_2)$
 $\& \ EXPL(s_2) = it_i]$
- d. $\llbracket Voice' \rrbracket =$
 $\lambda z \lambda E. \exists e_1 \exists e_1' \exists s_2 [E = e_1 \oplus s_2 \ \& \ CAUSE(e_1, e_1') \ \& \ sink(e_1', x_{i/k}) \ \& \ return(s_2)$
 $\& \ EXPL(e_2) = it_i \ \& \ AGENT(E) = z]$ agent
- e. $\llbracket VoiceP \rrbracket =$
 $\lambda E. \exists e_1 \exists e_1' \exists s_2 [E = e_1 \oplus s_2 \ \& \ CAUSE(e_1, e_1') \ \& \ sink(e_1', x_{i/k}) \ \& \ return(s_2)$
 $\& \ EXPL(s_2) = it_i \ \& \ AGENT(E) = we]$ FA(*we*)
- (68) $\exists E \exists e_1 \exists e_1' \exists s_2 [E = e_1 \oplus s_2 \ \& \ CAUSE(e_1, e_1') \ \& \ sink(e_1', x_i) \ \& \ return(s_2) \ \& \ EXPL(s_2)$
 $= it_i \ \& \ AGENT(E) = we]$
 $= \text{'We make it sink and return'}$
 $\approx \text{'We again make it sink'}$
 $= \llbracket (64) \rrbracket$

Extensionally, we end up with the exact same result as the head-extension approach. Just like under the head-extension approach, we can derive independent monovalence of contextually bivalent verbs by assuming that the expletive thematic role introduced by ADD must receive a thematic interpretation; consequently, in the absence of a transitive co-verb that can provide this information, applying ADD in isolation is impossible. For C-SVCs, we can assume that (i) the IAR of the co-verb V_1 must be satisfied, and that (ii) it is satisfied by co-indexing V_1 's internal argument with the expletive internal argument of the contextually bivalent $V_{2=H}$, which ends up projecting.

Arguably though, what also drives argument extension via ADD under the adjunct-extension approach is the type mismatch that arises at the point in the derivation after EC_{TS} has applied to its first input V_1 , but before it applies to its second input $V_{2=H}$; this step corresponds to (67a) above, and is repeated below in (69). Because (69) requires a predicate of type $\langle e,vt \rangle$, but the prospective $V_{2=H}$ *uyo* 'return' has not yet been extended by ADD and hence remains of type $\langle v,t \rangle$, *uyo* 'return' cannot function as the second input to EC_{TS} due to a type mismatch:

- (69) $\lambda Q_{\langle e,vt \rangle} \lambda y \lambda E. \{ \exists x \} \exists e_1 \exists e_1' \exists e_2 [E = e_1 \oplus e_2 \ \& \ CAUSE(e_1, e_1') \ \& \ sink(e_1', x_{i/k}) \ \& \ Q(e_2, \{y_i\})]$
- $\langle evt, evt \rangle$

$$(70) \quad \llbracket uyo \rrbracket = \lambda s_2. [\text{return}(s_2)] \quad \langle v, t \rangle$$

The only way for *uyo* ‘return’ (70) to become a well-formed input to (69) is by having this type mismatch resolved via ADD. Assuming that extension via ADD comes in later, after *uyo* ‘return’ (70) is composed with (69), is not an option since this would require pairing a type $\langle e, vt \rangle$ with $\langle v, t \rangle$, in violation of TM.

In conclusion, the adjunct-extension approach comes with the trade-off that it must additionally take into account a type mismatch. For this reason, I contend that the head-extension approach is more adequate because it can make do without this extra consideration. As outlined above, I propose that ADD can freely apply to any predicate with the relevant properties licensing its application, but that its application must subsequently be followed by the application of EC_{TS} because it is only way that the dummy argument EXPL can end up with a thematic interpretation. This way, we force any predicate extended via ADD to enter into C-SVC formation, conversely deriving the impossibility of contextual bivalence outside C-SVCs.

A similar logic is employed in Müller (2014)’s approach to resumption in German relative clauses: because the information that forces the presence of the resumptive pronoun in the relative clause (i.e. an island) is only added later in the derivation after the relative clause is built, how can the grammar enforce resumption? From the perspective of the relative clause, this constitutes a lookahead problem, while from the perspective of the island, this constitutes a backtracking problem. Simplifying somewhat, Müller (2014) resolves this dilemma by proposing that resumption of an XP, taken to involve a copying operation applied to XP, is a freely available option, but that the information that XP is now associated with a copy is registered on the original, moved XP as a kind of self-serving edge feature. Now endowed with this information, the moved XP can and in fact must cross what would otherwise be a barrier.

This logic parallels that of the head-extension approach to contextual bivalence, in that the result of applying ADD to a verb constitutes the information that not only permits, but in fact forces the introduction of a transitive co-verb with a thematic internal argument. In short, the very introduction of the “target” forces the creation of its “trigger”.

8.5.2 Argument pooling

In C-SVCs with object sharing, it seems that an elegant alternative to specifying any non-head internal argument by way of co-indexation with the head internal argument or existential binding would be to fuse both internal argument slots into a single, λ -bound argument that ends up saturated via Function Application. This mechanism of complex argument structure formation is known as *argument pooling* (Haider 2001, Haider 2010); it will be compared to the present approach, which I call the *indexing approach*.

In thematic terms, argument pooling combines the roles of multiple theta-providing verbs into a single thematic grid; in lambda terms, pooling abstracts over the individual arguments of both of its operands, such that the internal arguments of both P and Q end up being bound by a single λ -operator. Pooling instantiates a kind of argument fusion, as opposed to co-indexation (Baker & Harvey 2010), and is employed by Haider (2001) and Haider (2010) to derive argument sharing in restructuring configurations. It also figures in approaches to other types of complex predicates: Williams (2015), for instance, proposes an analogous mechanism in his analysis of resultatives.

The argument-pooling variant of transitive EC_{TS} is given in (71); observe that the internal arguments of P and Q have now been reduced to a single, λ -bound argument:

(71) *Transitive EC_{TS} with argument pooling:*

$$\lambda Q_{\langle e,vt \rangle} \lambda P_{\langle e,vt \rangle} [\lambda z \lambda E. \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ P(e_1, [z]) \ \& \ Q(e_2, [z])]] \\ \langle \text{evt}, \langle \text{evt}, \text{evt} \rangle \rangle$$

Similar to the difference between Predicate Modification and Event Cumulation, argument pooling neutralizes the syntactic headedness asymmetry of constituent verbs at the level of argument structure, given that the internal argument slot open to Function Application now belongs to both predicates, not just the head verb.

If we apply argument pooling to the constituent verbs of (72), we end up with the complex predicate in (73):

(72) Ye-[koi]-[kesi]-di
 3SG.S-hit-break-3PL.O
 ‘He hit-broke them’

(73) $\lambda z \lambda E. \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ \text{hit}(e_1, z) \ \& \ \text{break}(e_2, z)]$

Just like in the indexing approach, (73) requires an individual argument as its next input if EC_{TS} does not iterate; but unlike in the indexing approach, merging a DP object saturates both internal argument slots simultaneously. Object sharing thus follows from identity, not co-indexation.

The pooling approach is clearly more elegant than the indexing approach for C-SVCs with object sharing, but it comes with a trade-off once we turn to C-SVCs without object sharing. Because the internal arguments in cases like (74) refer to sortally distinct entities, pooling would yield the wrong result in (75), according to which the object of pouring identifies the object of filling:

(74) Kaputi ku-[ini]-[he-mwayau]- $\emptyset$
 cup 2SG.S-pour-CAUS-full-3SG.O
 ‘Fill the cup’
 (literally: ‘Pour the cup full’)

(75) $\exists E \exists e_1 \exists e_2 \exists e_2' [E = e_1 \oplus e_2 \ \& \ \text{pour}(e_1, \text{cup}) \ \& \ \text{CAUSE}(e_2, e_2') \ \& \ \text{full}(e_2', \text{cup}) \ \& \]$

AGENT(E) = you]
 = 'You pour the cup and fill the cup'
 ≠ [(74)]

Again, this reproduces the undesired neutralization effect that arises with Predicate Modification as opposed to Event Cumulation.

In order to accommodate C-SVCs without object sharing alongside those with object sharing, we would thus need to ensure that sortally distinct internal arguments cannot be pooled into a single λ -bound argument, and that the argument belonging to the non-head verb must instead undergo existential binding. This by itself is unproblematic, given that under the indexing approach, we must also call on and regulate existential binding. What is potentially problematic, however, is that the pooling approach actually requires two distinct operations to work: pooling on the one hand (object sharing), and non-pooling plus existential binding on the other (no object sharing). By contrast, the indexing approach allows for a uniform treatment of the two sub-types of transitive C-SVCs, with the choice between co-indexation and existential binding following systematically from the sortal properties of internal arguments.

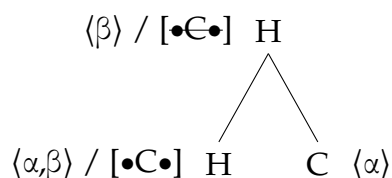
8.5.3 Transitivity matching and modes of structure-building

We will now see that TM with semantic effects on all co-varying elements can only arise in V-level adjunction configurations. The reason is that only V-level adjunction configurations show two crucial properties which are necessary for the grammar to be able to distinguish verb stems in terms of their types.

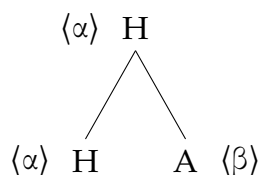
First, by definition, an adjunct is not intrinsically selected by a head element, i.e. merging an adjunct A to a head H does not check off any intrinsic selectional features on H. As a consequence, merging A with H does not only leave the syntactic featural properties of H unaltered, but also its semantic type. By contrast, merging a complement C with a head H does satisfy some intrinsic selectional requirement of H due to C being an argument of H, thereby altering both its syntactic featural properties and its semantic type.

Second, as argued in the preceding section, the point in the derivation prior to object introduction is crucial for allowing the grammar to distinguish verb stems based on their types, because its type and selectional properties are reduced as a result of merging an argument.

Complementation is represented in Tree 8.10, and adjunction in Tree 8.11:



Tree 8.10: Complementation



Tree 8.11: Adjunction

Even though a complement C to a head H is itself free to vary in its transitivity, because the very act of merging C with H reduces the syntactic selectional feature on H and also its semantic type, the grammar is unable to bring C and H together under the condition that both must match in their semantic types. Configurations other than V-level adjunction can only give rise to syntactic or morphological TM, discussed in the next chapters.

Coming from a different perspective, Nishiyama & Ogawa (2014) arrive at the same conclusion that certain kinds of matching effects are necessarily restricted to adjunction and absent in complementation. In an attempt to derive Kageyama (1993)'s *Principle of transitivity harmony*, which states that any two verbs are serializable if and only if they both share or lack an external argument position, Nishiyama & Ogawa propose that Voice may bear an uninterpretable feature $[\pm uF]$, which must agree with all and only same-valued corresponding interpretable features within its complement domain. If its verbal complement is a complex V or VP built via adjunction, the non-head and head verbs are both equidistant from Voice, such that Voice will encounter $[F]$ on both and undergo multiple Agree. But if its verbal complement is a VP that is itself built via complementation, Voice will only find $[F]$ on the higher verb because the lower verb is too deeply embedded. While Nishiyama & Ogawa's insight is correct, it crucially also relies on diagnostics tied to Agree, and is therefore only contingently tied to the distinction between complementation and adjunction.

8.5.4 Existing approaches to semantic transitivity matching

As of now, I am not aware of any existing analyses dedicated to semantic TM. That said, authors have come up with mechanisms to derive other kinds of matching effects in non-resultative SVCs, mainly for Korean and Japanese, which I will briefly go over in terms of their extendability to semantic TM.¹⁸²

¹⁸²Japanese SVCs were once discussed in connection with semantic TM (Jacobsen 1982, Jacobsen 1992), but this is actually only a strong tendency; see e.g. Gamerschlag (2005) and Nishiyama &

First, Chung (1993) proposes for Korean that two verbs can only be serialized if the highest argument of the non-head verb does not outrank the highest argument of the head verb on the thematic hierarchy. The net effect of Chung's condition is similar to that of our event-structural condition against thematic contradictions, which we know is needed independently in order to rule out cases which would incorrectly be ruled in by type-matching alone. By itself, however, this condition is insufficient to derive semantic TM because of the existence of e.g. [*unerg* + *trans*] pairs, which are thematically well-formed but lack TM.

Second, Kang (1997) argues that only unaccusatives, unergatives or transitives can respectively be paired with one another in Korean.¹⁸³ In other words, the syntactic type of a verb determines whether it can enter into serialization. Conceptually, this approach comes closer to being extendable to semantic TM because serialization is sensitive to some transitivity-related property, in this case syntactic argument structure. However, this sensitivity to a verb's syntactic type is essentially stipulated as a global construction-specific requirement, rather than being made to follow in any principled manner from its parts. This is exacerbated by the fact that the Korean SVCs discussed by Kang are non-contiguous, so that it is unclear how [*unerg*₁ + *trans*₂] or [*trans*₁ + *unerg*₂] can be ruled out as there is no way for the grammar to distinguish the syntactic type of the two VoicePs at the point of serialization (moreover, we know that these are actually grammatical).

Third, Lee (2003) proposes – again for Korean – that two verbs must match in their semantic type in order to be serializable. In Lee's system, this means that only verbs of type ⟨e,vt⟩ or ⟨e,evt⟩ can be paired, which gives rise to double unaccusative, unergative or transitive pairings. Conceptually, this is exactly how I have proposed to derive semantic TM, namely in terms of type-matching. However, the approach faces some shortcomings, such as wrongly predicting pairings of unaccusatives and unergatives to be well-formed, which are type-identical in their system. Also, as with Kang's syntactic-type approach, it is unclear how Lee's type-matching condition – which must make reference to the semantic types of verbs prior to argument introduction – can be reconciled with the non-contiguity of Korean SVCs, i.e. the semantic types Lee's system needs to refer to would have to be backtracked down to the V-level from the point of serialization at VoiceP.¹⁸⁴

For a more detailed critique of these approaches, see Ko & Sohn (2015).

Ogawa (2014).

¹⁸³Similarly, Kageyama (1993) argues for Japanese that two verbs are serializable if and only if their highest arguments are both internal or both external, respectively.

¹⁸⁴Ko & Sohn (2015) show that Lee's approach comes with a few other unwelcome results, such as ruling out grammatical [*unerg* + *trans*] pairings.

8.5.5 Transitivity restrictions

There are several transitivity restrictions at play in Saliba semantic TM. First, there is the configurational restriction of the MNC, which blocks any non-head verb from projecting. Second, there is semantic TM itself, which is ad hoc due to not following naturally from what we know about the structural context. Crucially, even though semantic TM via type-matching itself is an arbitrary restriction stipulated by the grammar on C-SVC formation, its availability is necessarily conditioned by V-level adjunction configurations. Third, there are the IARs of transitive verbs, which must generally be satisfied in some way. Given V-level adjunction and the MNC, the IARs of transitive verbs result in the need for the grammar to co-index their internal argument positions with that of the head verb or – if this fails due to sortal distinctness – existentially bind these. By extension, IARs give rise to contextual bivalence. Lastly, we invoked the restriction against thematic contradiction in order to rule out certain verb pairings that are predicted to be well-formed by type-matching alone. We thus have a fully derivational account of semantic TM, which captures all of its properties by appealing solely to the properties of its parts.

8.6 Summary

This chapter presented the derivations of Saliba C-SVCs with semantic TM. We also saw how the proposed analysis is consistent with the unaccusative-unergative distinction in Saliba, and discussed various properties connected to contextual bivalence in depth. Specifically, I showed that (i) the particular morpho-syntactic strategies employed to spell out contextual bivalence – while variant across languages – are systematically comparatively unmarked; (ii) contextual bivalence may fail either if the syntactic head verb is not of the appropriate type, or if a structural alternative to the same end is available; and (iii) contextual bivalence can be viewed thematic raising. We also discussed various grammar-theoretical issues, concluding most importantly that type-matching can only arise in V-level adjunction configurations.

Chapter 9

Semantic transitivity matching in Tariana

9.1 Introduction

In this chapter, we will go over C-SVCs in Tariana, a subset of which shows the same kind of semantic TM found in Saliba. We can therefore apply the exact same analysis developed for semantic Saliba to Tariana, with one caveat, which is that we must also account for why all C-SVCs in Tariana are contiguous, despite only some of them showing TM. I will argue that C-SVCs with TM are truly contiguous, while those without semantic TM show derived contiguity. The data and generalizations are presented in SVCs and semantic TM are presented in Section 9.2, and the analysis in 9.3.

9.2 Serial verb constructions in Tariana

Tariana is a North Arawakan language spoken in Brazil and Colombia. It is a polysynthetic agglutinating language with some fusion, with a relatively free constituent order. Subjects are cross-referenced via prefixes, roughly on an active-stative basis (Aikhenvald 2003).

On the surface, all SVCs in Tariana are strictly contiguous (Aikhenvald 2006a). What is noteworthy about Tariana C-SVCs is that semantic TM is not found across the board – as in Saliba – but restricted to a subset of C-SVCs. Following Aikhenvald’s symmetry-based distinction for the moment, among asymmetrical C-SVCs, TM is only found in the directional sub-type, whereas it is found across the board in symmetrical C-SVCs.

Consider the C-SVCs in (1), both of which have the ambi-transitive V_1 *ka* ‘see’. When V_2 is the lexical intransitive *ruku* ‘go down’, V_1 ‘see’ must also be interpreted intransitively (1a); and once the same root underlying V_2 is causativized, V_1 ‘see’

must also have a transitive interpretation (1b):

(1) *Asymmetrical directional C-SVCs*

a. *ambi-trans + intr*

[di-ka] [di-ruku]
3SG.NF-see 3SG.NF-go.down

1. 'He looked going down'

2. 'He looked down'

b. *ambi-trans + trans*

[di-ka] [di-ruku-ita]
3SG.NF-see 3SG.NF-go.down-CAUS

1. 'He looked (at something) making (something) go down'

2. 'He looked down (at something)' (Aikhenvald 2006a: 186)

Note that these C-SVCs allow two readings with respect to the directional verb: they can either entail movement on part of the subject, or only specify the direction along which the subject's action is carried out.

Cases like (1), in which an ambi-transitive verb must have an intransitive or transitive interpretation in strict dependence on whether its co-verb is morphologically intransitive or transitive, yet without itself also visibly co-varying, already show that TM in Tariana comes with semantic effects on all co-varying elements.

In other sub-types of asymmetrical C-SVCs, TM is absent, as in the motion sub-type in (2):

(2) *Asymmetrical motion C-SVC: intr + trans*

ñaama-ita [nu-eku] [nu-pinita-ka-na]
two-NUM.CL 1SG-run 1SG-pursue-DEC-PST

'I pursued two (pigs) by running' (Aikhenvald 2006a: 190)

Once we turn to symmetrical C-SVCs, we again find TM, yet this time across the board. (3a) shows an intransitive symmetrical C-SVC, (3b) a transitive one:

(3) *Symmetrical C-SVC*

a. *intr + intr*

Ãpia kesani-wani du-thaku-se [di-wha] [de:ru-pidana]
pigs smell-CL:COLL 3SG.F-nose-LOC 3SG.NF-fall 3SG.NF.get.stuck-PST

'She felt the smell of wild pigs'

literally: 'The smell of wild pigs fell-got stuck in her nose'

(Aikhenvald 2006a: 193)

b. *trans + trans*

[Nhalitu] [na-inu-pidana] ñaña
3PL.fish 3PL-kill-REM.P.REP madi.fish

‘They caught (fish-kill) *madi-fish*’

(Aikhenvald 2003: 440)

To summarize: symmetrical C-SVCs show semantic TM across the board, but semantic TM is absent in asymmetrical C-SVCs, except in the directional sub-type.

9.3 Analysis

9.3.1 C-SVCs with semantic TM involve adjunction

Beginning with what Aikhenvald classifies as symmetrical C-SVCs, which show TM across the board, notice that TM in these comes with an interpretable effect on both verbs, as made evident in the translation of (1) and (3); they hence show semantic TM. For this reason, they are straightforwardly captured by the analysis developed for Saliba C-SVCs: they involve V-level via adjunction, with type-matching enforced by EC_{TS}. Consistent with this is the fact that the meanings expressed in symmetrical C-SVCs correspond to simple conjunctions of the meanings of their constituent verbs (Aikhenvald 2003: 440).

9.3.2 C-SVCs without semantic TM involve complementation

Next, we need to restrict semantic TM among what are classified by Aikhenvald as asymmetrical C-SVCs to only the directional sub-type. What we clearly want to avoid is to stipulate this difference into EC_{TS} with brute force, i.e. by making it sensitive to whether one of its inputs is a directional verb. This move would be highly dubious, for two reasons: first, there is nothing about the lexical semantics of specifically directional verbs that would plausibly be visible to the grammar so as to be able to enforce type-matching – if anything, we would expect such a selective effect to arise with meanings like causation and resultativity, which we know can give rise to restrictions on argument structure; second, it ignores the possibility of any structural differences that may arise in connection with the presence and absence of TM, respectively.

What about drawing on the notion of symmetry? Recall that the symmetry distinction in the SVC literature is based on the relative paradigmatic productivities and semantic contributions of constituent verbs: in symmetrical SVCs, each verb comes from an open class and contributes more or less equally to interpretation; in asymmetrical SVCs, at least one verb comes from a restricted class, and the semantic contributions of verbs are disparate (Aikhenvald & Dixon 2006).

From a grammar-theoretical perspective, however, classifying SVCs on the basis of symmetry should be handled with care, and treated more as a hint at certain grammatical differences, the main reason being that SVCs and their sub-types do not constitute single grammatical categories. Moreover, as criticized by Enfield (2009), relative openness is less a grammatical or lexical property than it is a property of

constructional slots, and diagnostics for "openness" vs "restrictedness" are not only set up based on language-specific factors, but at times also authors' subjective judgments. This is evident in the collection in Aikhenvald & Dixon (2006), which – despite being masterful in its coverage – shows that there is not only significant range in what is taken to fall into one type of SVC or the other, but also that the same diagnostic used by one author for regarding a certain slot as open may be used by another for regarding a certain slot as restricted.

This means that we cannot tie the distribution of TM in Tariana C-SVCs directly to the property of symmetry in a way that is particularly informative grammatically. Just like we cannot plausibly make EC_{TS} sensitive to whether one of its inputs is a directional verb, we can even less so make EC_{TS} sensitive to whether one of its input comes from what is in some undefined sense regarded as an open or restricted verbal class.

Fortunately, Aikhenvald (2003) provides detailed coverage of Tariana C-SVCs which I believe warrants the following classification, based on clear grammatical principles, and which nevertheless aligns roughly with her symmetrical-asymmetrical distinction: her asymmetrical C-SVCs involve VP-level complementation, while her symmetrical C-SVCs involve V-level adjunction; as for directional C-SVCs, which she classifies as asymmetrical, I propose that these, too, involve V-level adjunction. Under this analysis, because directional and symmetrical C-SVCs involve V-level adjunction, only these can give rise to semantic TM, and conversely, because asymmetrical C-SVCs (now excluding the directional sub-type) involve VP-level complementation, they cannot give rise to semantic TM.

Let us go through why Tariana asymmetrical C-SVCs (without the directional sub-type) should be analyzed as complementation configurations. First, they express modal, aspectual, desiderative, benefactive, causative and reflexive / reciprocal meanings (Aikhenvald 2003: 430), all of which involve relatively functional meanings components characteristic of complementation. Periphrastic causation with the V₂ *pala* 'put' is shown in (4a), and in (4b) the V₂ *yeka* 'be able to' expresses a root-modal meaning as V₂:

(4) *Asymmetrical C-SVC*

a. *intr + 'put': causation*

Phia nu-na [karu] [pi-pala-ka]
you 1SG-OBJ be.afraid 2SG-put-REC.P.VIS
'You made me scared'

b. *trans + 'be able to': root modality*

Wa-dalipaki-pe [ma-de-kade-mha] [wa-yeka]
1PL-blood.relative-PL NEG-have-NEG-PRES 1PL-be.able.to
'We cannot have our blood relatives (as marriage partners)'

(Aikhenvald 2003: 437, 439)

This is not to say that each of these meanings per se indicates complementation (some of them may also found in adjunction); rather, it is the fact that specifically these meanings should cluster collectively under the label of asymmetrical SVCs which points to complementation.

Second, many of these verbs stand in an opaque meaning relationship to their independent uses (Aikhenvald 2003: 432). To give just two examples, the verb *a* means ‘go’ in isolation, but means something like ‘become’ when serialized (5a), and similarly the verb *pe* means ‘throw’ in isolation, but expresses a kind of telicity when serialized (5b):

(5) *Asymmetrical C-SVC*

- a. *intr* + ‘go’: ‘become’

Iya [sereni] [di-a-pidana]
rain drizzle 3SG.NF-go-PST
‘The rain became a drizzle’

- b. *trans* + ‘throw’: *telic aspect*

[Di-ka] [di-pe]
3SG.NF-see 3SG.NF-throw
‘He looks and sees it all’

(Aikhenvald 2003: 430, 431)

This shows that verbs like *a* ‘go’ and *pe* ‘throw’ have split off onto a path of grammaticalization into more functional elements, which calls for a heterosemy account with multiple lexical entries (cf. Enfield (2006)), one of which takes a verbal complement.

Third, there are asymmetrical C-SVCs in which the proposed relationship of complementation is brought out especially clearly, such as (6), where the V_1 *hpani* ‘work’ must be a complement to the V_2 *adeta* ‘prevent’, with a control dependency holding between the grammatical object of the V_2 *adeta* ‘prevent’ and the understood subject of the V_1 *hpani* ‘work’:

(6) *Asymmetrical CS-VC: object control*

Emite-tiki nu-na [di-hpani] [di-adeta-naka]
child-DIM 1SG-OBJ 3SG.NF-work 3SG.NF-prevent-PRS

‘The little boy is preventing me from working’ (Aikhenvald 2003: 439)

This leaves us with directional C-SVCs, classified as symmetrical. A first indication that this sub-type differs from the other asymmetrical sub-types is that – unlike the functional verbs just discussed above for asymmetrical C-SVCs, which show some degree of abstraction and grammaticalization – the independent and serializing uses of directional verbs are always transparently related, with any differences being systematically predictable. In fact, the relatedness we find is just like the one found in Saliba, which is that a directional verb within a C-SVC may lack a motion entailment

with respect to the subject, in turn is reminiscent of zero-marked causative-inchoative alternations. On a purely directional reading of (1a), repeated here, there is no entailment that the subject moves downwards, only that the direction of seeing is downwards:

(7) *Asymmetrical directional C-SVC*

[di-ka] [di-ruku]
3SG.NF-see 3SG.NF-go.down

1. 'He looked going down'

2. 'He looked down'

(Aikhenvald 2006a: 186)

Without running into circularity, if – as argued here – semantic TM can only arise in V-level adjunction, the very fact that directional C-SVCs exhibit semantic TM is a hint that they involve V-level adjunction (even if their descriptive classification says otherwise), and conversely, that all sub-types lacking TM do not.

Having separated the directional sub-type from the other asymmetrical C-SVCs, we should now check whether the latter, argued to involve complementation, map consistently onto properties associated with complementation in Tariana; for the most part, this is correct.

First, a large number of the closed-class verbs from all sub-types of asymmetrical C-SVCs are unable to occur on their own, i.e. they are obligatorily serializing (Aikhenvald 2003: 244).¹⁸⁵ We can thus assign two lexical entries to the former, one of which requires a verbal complement and one which doesn't, and a single entry that requires a verbal complement to the latter. The obligatory need of some verbs to enter into serialization follows naturally under a complementation analysis, according to which these verbs have an inherent requirement for a verbal argument. Symmetrical C-SVCs, on the other hand, never contain verbs which are forced to enter serialization and likewise, the directional verbs occurring in directional SVCs are always free to occur as standalone verbs. This follows if all and only this class of C-SVCs (symmetrical + directional) are built via adjunction.

Second, syntactically larger complement clauses in other, more obvious cases of complementation always precede the matrix verb (Aikhenvald 2003: 457ff), meaning that we should expect the closed-class verb to uniformly correspond to the string-final verb in asymmetrical C-SVCs.¹⁸⁶ While this generalization is mostly true, it

¹⁸⁵These include *a* 'be going to / want / intend', *keña* 'begin', *sita* 'finish / complete / succeed / manage', *wa* 'try', *na* 'want', *mayē* 'pretend / do wrongly', *adeta* 'prevent', *a* 'cause / make / let / give', *matara* 'let', *isa* 'order', *pala* 'put' and *wake(ta)* 'help / put together / join' (Aikhenvald 2003: 244ff).

¹⁸⁶E.g. complement clauses to perception verbs always precede the matrix verb:

(i) [Diha nahipa dira-ka] di-ka
he big.CLF 3SG.NF.drink-SUB 3SG.NF-see
'He (the fox) saw that he (the jaguar) had drunk a lot'

(Aikhenvald 2003: 548)

does have exceptions: while the closed-class verb corresponds to the string-final verb V_F in most sub-types, as in the aspectual and object-control C-SVCs above in (5) and (6), there are a few in which it corresponds to V_1 . In (8), for example, the verb *na* ‘want’ precedes its complement (which in this case is itself a C-SVC):

- (8) Pathina [du-na] [[du-eku] [du-a-pidana]] panisi-se
suddenly 3SG.F-want 3SG.F-run 3SG.F-go-PST house-LOC
‘All of a sudden she wanted to run out of the house’

(Aikhenvald 2003: 436)

Such variation in the placement of a head relative to its complement is unexpected from a first-Merge perspective, leading me to assume that cases like (8) involve movement.

9.3.3 Deriving strict contiguity in asymmetrical C-SVCs

We still have one problem to solve, namely the strict contiguity of C-SVCs. As stated explicitly in Aikhenvald (2006a: 181f), no material whatsoever may ever occur between constituent verbs across all C-SVC sub-types (apart from the occasional clitic). Strict contiguity comes for free in symmetrical and directional C-SVCs, which involve V-level structure-building; more, though, needs to be said on asymmetrical C-SVCs.^{187,188}

Recall that in the domain of SVCs, the idea that C-SVCs are actually NC-SVCs to which movement has applied is an old one (Collins 2002, Déchaine 1993). More generally, a surface-contiguous verb cluster does not entail that it was base-generated as a cluster – as shown by the in-depth discussion in Wurmbrand (2017), there are various ways of deriving surface-contiguous clusters from underlyingly non-

¹⁸⁷There is another property distinguishing asymmetrical C-SVCs from symmetrical ones, which points to complementation in the former and adjunction in the latter: verb order in symmetrical C-SVCs is always iconic, whereas it is sometimes non-iconic in asymmetrical ones. This suggests a disparity between verbs in asymmetrical C-SVCs, with one of the verbs being more functional and hence syntactically more rigid than the other, and conversely, with all the verbs being equally lexical in symmetrical C-SVCs.

¹⁸⁸Only symmetrical C-SVCs may give rise to idiomatic meanings (Aikhenvald 2003: 424). Two examples are *kalite-pala* ‘tell-put’, which means ‘boast, complain’ and *yena-ema* ‘pass-stand’, which means ‘be arrogant’ (Aikhenvald 2003: 428). The fact that non-compositional meanings are found only in symmetrical C-SVCs suggests that symmetrical C-SVCs form more cohesive units than asymmetrical ones. Arad (2003) proposes that idiomatic meanings are subject to a strict locality condition, which is that the first nominal or verbal head that merges with a root serves as the immediate environment for determining its interpretation. Consequently, any further derivation takes as its input not the root itself, but an element whose semantic (and phonological) properties have been cashed out. In this way, idiomatic meanings are restricted to syntactic units at the “derivational cradle”, so to say. Though more needs to be said on the timing of adjunction and categorization, I take the distribution of idiomatic meanings in Tariana C-SVCs to nevertheless be consistent with Arad’s more abstract claim that special meanings are restricted to smaller syntactic domains. Applied to the present analysis, this would mean that idiomatic meanings may arise in V-level adjunction, but not VP-level complementation, because serialization in the latter applies to units whose meanings have been cashed out. That said, it would seem that it should be complementation that gives rise to idiomatic meanings, rather than adjunction (Barbara Stiebels p.c.). I leave this issue open.

contiguous structures. Here specifically, I propose to make use of tools that have been proposed for German restructuring configurations in Keine & Bhatt (2016).

For German long passives (9), wherein under a certain class of restructuring verbs like *versuchen* ‘try’, the thematic object of the embedded verb moves into the matrix subject position upon passivization of the matrix verb (Wurmbrand 2001), Keine & Bhatt (2016) propose that the matrix and the embedded verb are strictly contiguous due to forming a verb cluster, as schematized in Tree 9.1:¹⁸⁹

(9) **German:** *Long passive restructuring*

Gestern wurde [der Traktor]_i [t_i zu reparieren] versucht
yesterday was the tractor.NOM to repair tried

‘Yesterday it was tried to repair the tractor’ (Keine & Bhatt 2016: 2)

¹⁸⁹ Among the diagnostics for the strict contiguity of the matrix and the embedded verb are the impossibility of (i) inter-verbal negation (ia), and (ii) an inter-verbal, extra-posed relative clause (iia); in the local-passive variants, these elements are free to occur inter-verbally (ib)-(iib):

(i) *Placement of negation*

a. *Long passive*

Gestern wurde der Traktor zu reparieren (***nicht**) versucht
yesterday was the tractor.NOM to repair not tried
‘Yesterday it was (*not) tried to repair the tractor’

b. *Local passive*

Gestern wurde den Traktor zu reparieren (**nicht**) versucht
yesterday was the tractor.ACC to repair not tried
‘Yesterday it was (not) tried to repair the tractor’ (Keine & Bhatt 2016: 5)

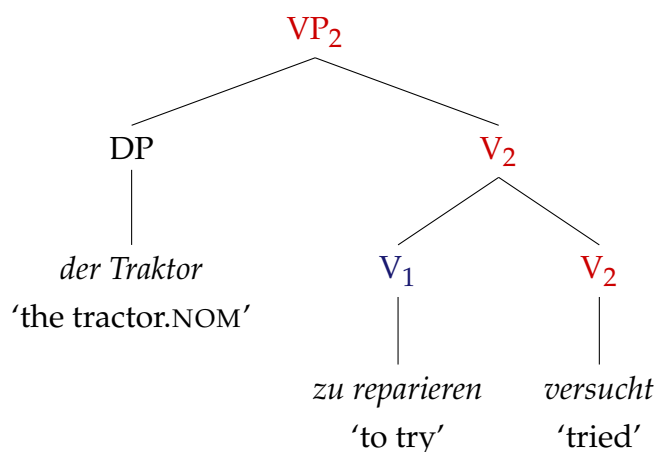
(ii) *Placement of extra-posed relative clause*

a. *Long passive*

weil der Brief einzuwerfen, (*[**der hier liegt**]), vergessen wurde
since the letter.NOM in-to-put that here lies forgotten was
‘since it was forgotten to post the letter (*that is lying here)’

b. *Local passive*

weil den Brief einzuwerfen, ([**der hier liegt**]), vergessen wurde
since the letter.ACC in-to-put that here lies forgotten was
‘since it was forgotten to post the letter (that is lying here)’ (Haider 2010: 334)



Tree 9.1: Verb clusters in long passives

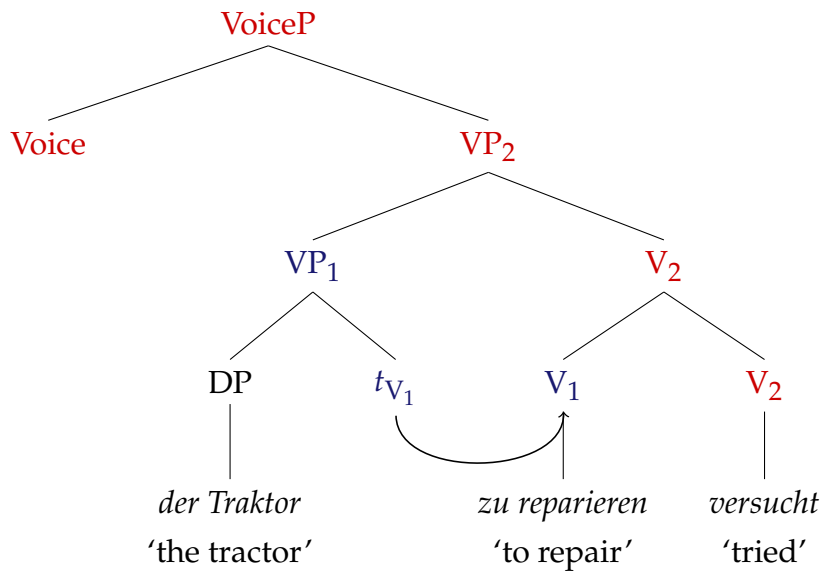
In order to derive verb cluster formation, Keine & Bhatt invoke the notion of *Distinctness*:

- (10) DISTINCTNESS (Richards 2010: 5)
 If a linearization statement $\langle \alpha, \alpha \rangle$ is generated, the derivation crashes.

Distinctness is a pre-condition on linearization which requires that elements within a phasal Spell-Out domain be categorially distinct from one another, such that if a Spell-Out domain contains elements which are non-distinct, some operation must be employed to remedy this violation. Keine & Bhatt extend Distinctness to instances of head movement, thus allowing verb incorporation as one of various available options for satisfying Distinctness.¹⁹⁰

Verb cluster formation in the long passive in (9) proceeds as follows: after the matrix verb *versucht* 'tried' takes the bare VP *der Traktor zu reparieren* 'repair the tractor' as its complement, because the two lexical V heads *reparieren* 'to repair' and *versucht* 'tried' co-exist within the same Spell-Out domain (demarcated by matrix Voice), they give rise to the linearization statement $\langle V^\circ, V^\circ \rangle$, which incurs a Distinctness violation; consequently, incorporation of *zu reparieren* 'to repair' into *versucht* 'tried' must occur in order to remedy this violation and rescue the derivation, as in Tree 9.2:

¹⁹⁰These fall into three basic types: structure-adding, structure-deleting and structure-moving.



Tree 9.2: Verb clusters in long passives

Because Distinctness is not evaluated within complex syntactic heads, but only at the super-head level, Tree 9.2 is now a well-formed input to the linearization algorithm due to no longer characterizing any offending statements, and so the derivation converges. Conceptualizing verb incorporation in this way offers an account of why it obligatorily applies if (i) the complement is a bare VP, but not a VoiceP (as is the case in local passives, see footnote 189); and (ii) no other operation in place of verb incorporation applies that similarly derives a well-formed input to linearization, such as movement of the complement VP to a position in a higher Spell-Out domain, as in (11) (Keine & Bhatt 2016: 33):

(11) **German:**

[Zu reparieren]_i wurden [nur blaue Autos t_i] vergessen
 to repair were only blue cars.NOM forgotten

'Only blue cars were forgotten to be repaired' (Keine & Bhatt 2016: 6)

If the surface-contiguity of Tariana asymmetrical C-SVCs is truly the result of verb incorporation of the embedded into the matrix verb, this predicts – as with German long passives – that strict contiguity should be lifted under two conditions, both to the effect of the embedded and the matrix verb no longer inhabiting the same Spell-Out domain: (i) the complement clause constitutes its own Spell-Out domain; and (ii) the embedded VP vacates the Spell-Out domain of Voice by moving across it into the matrix clause (before linearization statements are calculated, and Distinctness evaluated).

Unfortunately, the descriptive work on Tariana lacks clear examples involving VP-movement, leaving the status of prediction (ii) unclear, but prediction (i) is borne out: interrogative complement clauses, for example, which we can safely assume project

up to CP, allow the occurrence of material between the matrix and the embedded verb, and they can either precede (12a) or follow the matrix verb (12b):

- (12) a. [Diha kani-mhade di-ayha diha] nu-ka-sida
 he where-FUT 3SG.NF-swim he 1SG-see-PROX
 'I will see straight away where he will swim'
- b. Nhua-de nu-ka [kani-pu-mha nha na-rapa
 I-FUT.CERT 1SG-see where-AUG-PRES.NONVIS they 3PL-dance
 nemhani]
 3PL.walk
 'I will see where they really go and dance' (Aikhenvald 2003: 556f)

9.3.4 Interim summary

So symmetrical and directional C-SVCs show semantic TM and involve V-level adjunction, while asymmetrical C-SVCs without the directional sub-type lack TM and involve VP-level complementation. The surface contiguities of Tariana C-SVCs do not reflect a uniform structure: symmetrical and directional C-SVCs are underlyingly contiguous, but asymmetrical C-SVCs show a derived surface contiguity as a result of verb incorporation. These results are summarized below:

	Symmetrical + directional	Asymmetrical – directional
Semantic TM	✓	*
Level of structure-building	V	VP
Mode of structure-building	adjunction	complementation
Contiguity	underlying	derived via V-to-V incorporation

Table 9.1: Tariana C-SVCs

9.3.5 Derivations

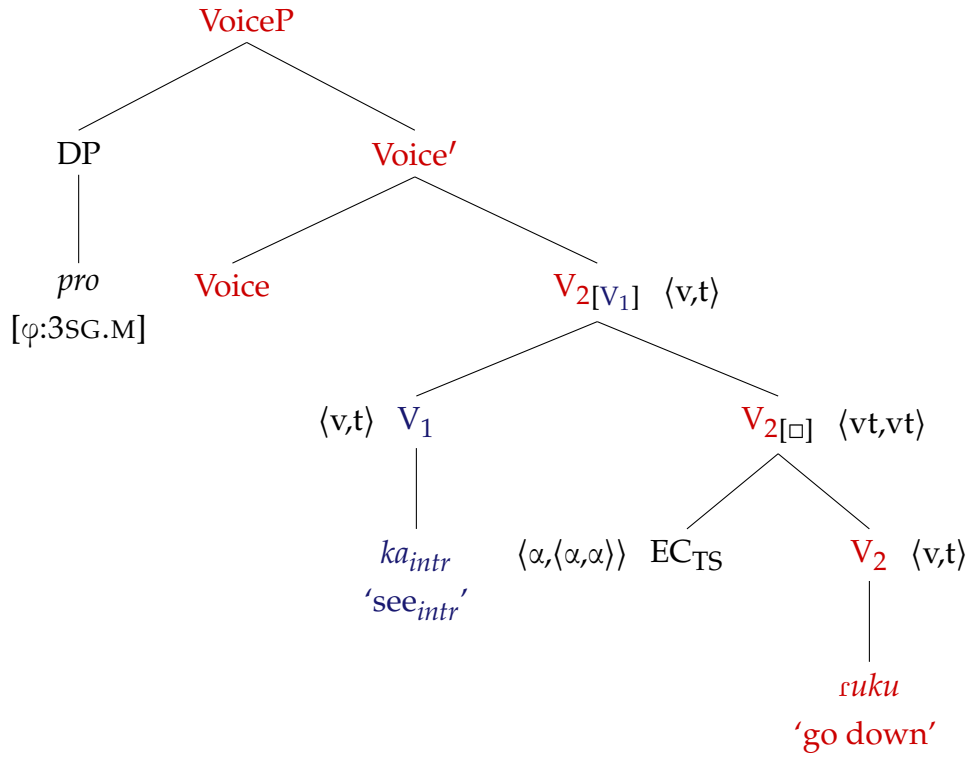
This section presents the derivations of Tariana C-SVCs. I begin with true C-SVCs with semantic TM, before moving on to derived C-SVCs without semantic TM.

9.3.5.0.1 True C-SVCs with semantic transitivity matching The derivation of the non-directional version of (13), with an intransitive $V_{2=H}$ and an ambi-transitive V_1 in its intransitive use, is given in Tree 9.3:

- (13) *ambi-trans + intr*

[di-ka] [di-ruku]
 3SG.NF-see 3SG.NF-go.down

'He looked going down'



Tree 9.3: Tariana 1

- (14) a. $\llbracket V_1 \rrbracket = \llbracket ka_{intr} \rrbracket = \lambda e_1. [see(e_1)]$ $\langle v, t \rangle$
 b. $\llbracket V_2 \rrbracket = \llbracket ruku \rrbracket = \lambda e_2. [go\ down(e_2)]$ $\langle v, t \rangle$

Given the denotations of the constituent verbs in (14), the derivation proceeds just like with intransitive C-SVCs in Saliba: first EC_{TS} applies to unergative V_2 (15a), then it takes a second input verb that is also of type $\langle v, t \rangle$ (15b), after which the agent slot is introduced (15c) and saturated with the subject DP (15d); this gives us (16):

- (15) a. $\llbracket V_{2[\square]} \rrbracket = \llbracket EC_{TS}('go\ down') \rrbracket =$
 $\lambda P_{\langle v, t \rangle} \lambda E. \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ P(e_1) \ \& \ go\ down(e_2)]$
 b. $\llbracket V_{2[V_1]} \rrbracket = \llbracket EC_{TS}('go\ down')('see_{intr}') \rrbracket =$
 $\lambda E. \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ see(e_1) \ \& \ go\ down(e_2)]$
 c. $\llbracket Voice' \rrbracket =$
 $\lambda z \lambda E. \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ see(e_1) \ \& \ go\ down(e_2) \ \& \ AGENT(E) = z]$
agent
 d. $\llbracket VoiceP \rrbracket =$
 $\lambda E. \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ see(e_1) \ \& \ go\ down(e_2) \ \& \ AGENT(E) = he]$
FA(pro)

- (16) $\exists E \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ see(e_1) \ \& \ go\ down(e_2) \ \& \ AGENT(E) = he]$

event closure

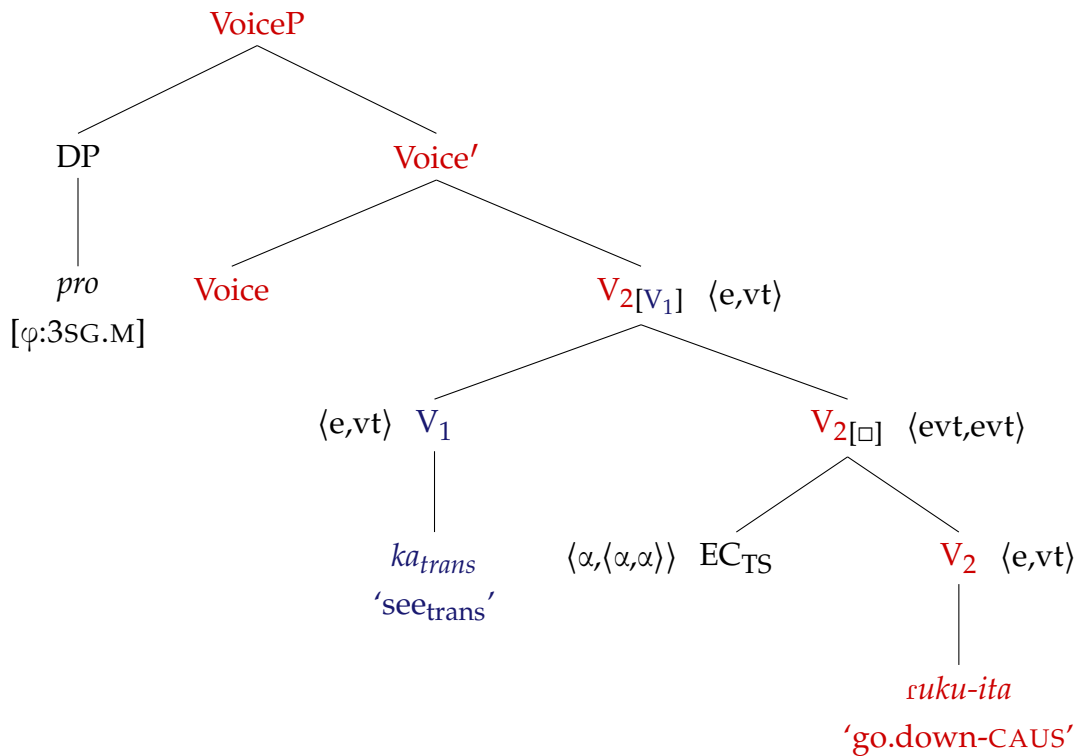
= 'He saw and went down'
 ≈ 'He looked going down'
 = $\llbracket (13) \rrbracket$

The derivation of the non-directional version of the transitive C-SVC in (17) again fully parallels its Saliba counterpart, shown in Tree 9.4.

(17) *ambi-trans* + *trans*

[di-ka] [di-ruku-ita]
 3SG.NF-see 3SG.NF-go.down-CAUS

'He looked (at something) making (something) go down'



Tree 9.4: Tariana 2

(17) is the transitive version of (13), with V_1 now occurring in its transitive use and V_2 with the causative, cf. (18). EC_{TS} ensures that both verbs must have the type $\langle e, vt \rangle$ and co-indexes their internal argument positions (19a)-(19b); because the translation of (17) suggests an existential reading of the object, I assume that the complex predicate's internal argument slot is closed via existential closure (19c). After the agent is introduced (19d) and saturated (19e), we arrive at (20):

- (18) a. $\llbracket V_1 \rrbracket = \llbracket ka_{trans} \rrbracket = \lambda x \lambda e_1. [see(e_1, x)]$ $\langle e, vt \rangle$
 b. $\llbracket V_2 \rrbracket = \llbracket ruku-ita \rrbracket = \lambda y \lambda e_2. \exists e_2' [CAUSE(e_2, e_2') \ \& \ go.down(e_2', y)]$ $\langle e, vt \rangle$

- (19) a. $\llbracket V_{2[\square]} \rrbracket = \llbracket EC_{TS}('go.down-CAUS') \rrbracket =$
 $\lambda P_{\langle e,vt \rangle} \lambda y \lambda E. \{ \exists x \} \exists e_1 \exists e_2 \exists e_2' [E = e_1 \oplus e_2 \ \& \ P(e_1, \{x_i/k\}) \ \& \ CAUSE(e_2, e_2') \ \& \ go \ down(e_2', y_i)]$
- b. $\llbracket V_{2[V_1]} \rrbracket = \llbracket EC_{TS}('go.down-CAUS')('see_{trans}') \rrbracket =$
 $\lambda y \lambda E. \exists e_1 \exists e_2 \exists e_2' [E = e_1 \oplus e_2 \ \& \ see(e_1, x_i) \ \& \ CAUSE(e_2, e_2') \ \& \ go \ down(e_2', y_i)]$
- c. $\lambda E. \exists e_1 \exists y \exists e_2 \exists e_2' [E = e_1 \oplus e_2 \ \& \ see(e_1, x_i) \ \& \ CAUSE(e_2, e_2') \ \& \ go \ down(e_2', y_i)]$
 existential closure(*y*)
- d. $\llbracket Voice' \rrbracket =$
 $\lambda z \lambda E. \exists e_1 \exists y \exists e_2 \exists e_2' [E = e_1 \oplus e_2 \ \& \ see(e_1, x_i) \ \& \ CAUSE(e_2, e_2') \ \& \ go \ down(e_2', y_i) \ \& \ AGENT(E) = z]$
 agent
- e. $\llbracket VoiceP \rrbracket =$
 $\lambda E. \exists e_1 \exists y \exists e_2 \exists e_2' [E = e_1 \oplus e_2 \ \& \ see(e_1, x_i) \ \& \ CAUSE(e_2, e_2') \ \& \ go \ down(e_2', y_i) \ \& \ AGENT(E) = he]$
 FA(*pro*)
- (20) $\exists E \exists e_1 \exists y \exists e_2 \exists e_2' [E = e_1 \oplus e_2 \ \& \ see(e_1, x_i) \ \& \ CAUSE(e_2, e_2') \ \& \ go \ down(e_2', y_i) \ \& \ AGENT(E) = he]$
 event closure
- = 'He saw something and made it go down'
 ≈ 'He looked (at something) making (it) go down'
 = $\llbracket (17) \rrbracket$

Given that Tariana symmetrical C-SVCs show semantic TM, we should ask whether they also give rise to contextual bivalence; this appears to be the case. First, Aikhenvald (2006a: 268) notes that certain motion verbs, e.g. *se* 'go up', only appear with transitivity matching causative morphology as $V_{2=HS}$ in the context of C-SVCs; and second, the causative *-i* / *-ita* is Tariana's only productive morphological strategy for deriving transitives from intransitives (Aikhenvald 2006a: 268) (i.e. there is not also an applicative like in Saliba). These facts support a contextual-bivalence analysis in which the causative *-i* / *-ita* may morpho-syntactically license the projection an expletive internal argument which, upon Function Application of the object DP, is co-indexed with the thematically contentful internal argument of V_1 .

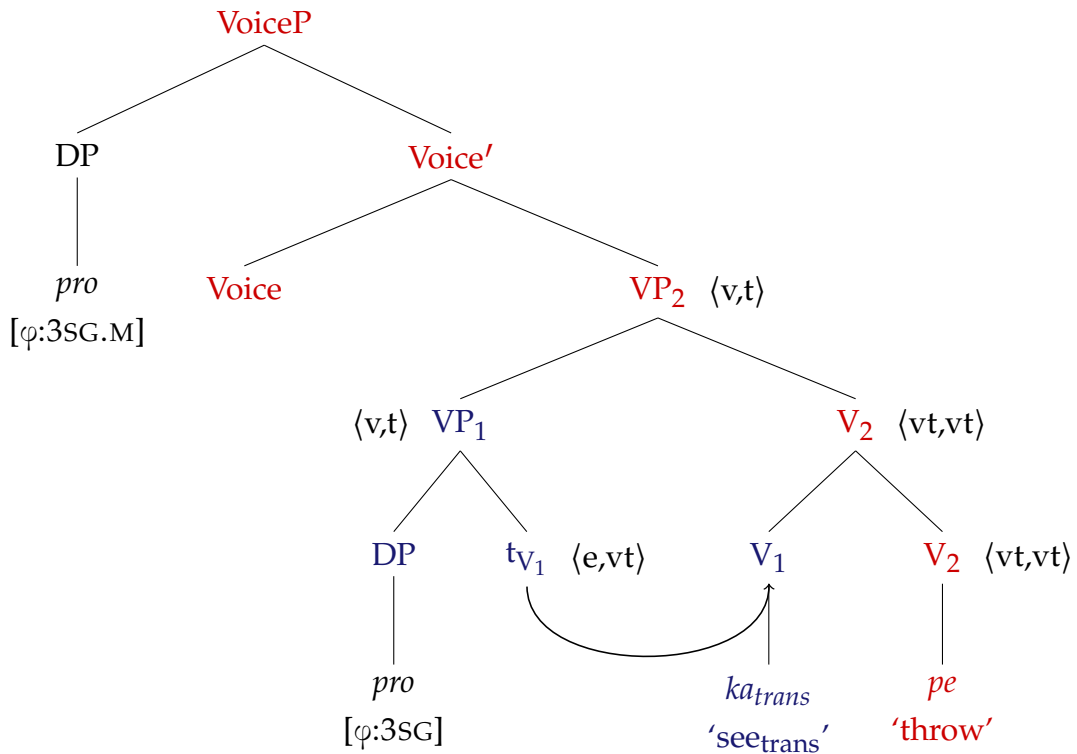
9.3.5.0.2 Derived C-SVCs without transitivity matching The derivation of C-SVCs like (21), which lack TM and whose contiguity is the result of verb incorporation, is given in Tree 9.5.

- (21) *Asymmetrical C-SVC: trans + 'throw': telic aspect*

[Di-ka] [di-pe]
 3SG.NF-see 3SG.NF-throw

'He looks and sees it all'

(Aikhenvald 2003: 431)



Tree 9.5: Tariana 1

The constituent verbs of (21) have the denotations in (22). After the embedded V_1 takes its object (23), the resulting VP_1 is merged as a complement to the matrix aspectual verb (23b). The agent is introduced and saturated (23c)-(23d), and we end up with (24):

- (22) a. $\llbracket V_1 \rrbracket = \llbracket ka_{trans} \rrbracket = \lambda x \lambda e_1. [see(e_1, x)]$ <e,vt>
 b. $\llbracket V_2 \rrbracket = \llbracket pe \rrbracket = \lambda P \lambda e_2. [throw(e_2, P)]$ <vt,vt>
- (23) a. $\llbracket VP_1 \rrbracket = \lambda e_1. [see(e_1, it)]$
 FA(*pro*)
 b. $\llbracket VP_2 \rrbracket = \lambda e_2. [throw(e_2, \lambda e_1. [see(e_1, it)])]$
 FA(VP_1)
 c. $\llbracket Voice' \rrbracket = \lambda z \lambda e_2. [throw(e_2, \lambda e_1. [see(e_1, it)]) \& AGENT(e_2) = z]$
 agent
 d. $\llbracket VoiceP \rrbracket = \lambda e_2. [throw(e_2, \lambda e_1. [see(e_1, it)]) \& AGENT(e_2) = he]$
 FA(*pro*)
- (24) $\exists e_2. [throw(e_2, \lambda e_1. [see(e_1, it)]) \& AGENT(e_2) = he]$
 = $\llbracket (21) \rrbracket$

Assuming that Voice is a phase head whose Spell-Out domain is shipped off upon completion of VoiceP, i.e. after the external argument is merged (23d) (rather than, say, upon merger of the next higher phase head), and because the presence of two verbs

within the same Spell-Out domain gives rise to an unlinearizable $\langle V^\circ, V^\circ \rangle$ statement, V_1 must incorporate into V_2 in order for this Distinctness violation to be remedied and the derivation rescued. Verb incorporation does not occur when Distinctness goes unviolated, like when the complement of V_2 gives rise to its own Spell-Out domain, cf. (12a).

As for the absence of TM in asymmetrical C-SVCs, given that they do not involve base-generated V-level adjunction prior to object introduction, but instead complementation of a transitive VP that already includes an object, the grammar is unable to enforce TM via type-matching; consequently, the embedded verb can be transitive, as in (21), or intransitive, as in (5a), repeated here in (25):

(25) *Asymmetrical C-SVC: intr + 'go': inchoative aspect*

Iya [sereni] [di-a-pidana]
rain drizzle 3SG.NF-go-REM.P.REP

'The rain became a drizzle'

9.4 Summary

This chapter extended the analysis proposed for semantic TM in Saliba to Tariana C-SVCs. Among these, semantic TM was shown to arise in adjunction configurations, and conversely to be absent systematically in C-SVCs built via complementation. This confirms my proposal that semantic TM via type-matching is restricted to adjunction configurations. I proposed to derive the surface contiguity of Tariana C-SVCs lacking semantic TM via Distinctness, which in this case was remedied by incorporation of the embedded verb into the matrix verb.

Chapter 10

Syntactic and morphological transitivity matching

10.1 Introduction

This chapter turns to two other kinds of TM: syntactic and morphological. Syntactic TM involves two or more verbs, one of which has a more functional meaning, and on which the matching reflex does not come with a semantic effect. Morphological TM is similar to syntactic TM, except that the functional element need not be a verb, and that the matching reflex on the higher element is non-transparent. Syntactic TM is presented and analyzed in Section 10.2. Section 10.3 does the same for morphological TM. All our findings regarding TM and its sub-types are brought together in Section 10.4, which also concludes our second case study.

10.2 Syntactic transitivity matching in Panoan

This section looks into two Panoan languages, Shipibo (Peru, Brazil) (Valenzuela 2003, Valenzuela 2011) and Kashibo (Peru) (Biondi 2011); see also Fleck (2013). In these, we can distinguish two phenomena which involve the co-variance of one verbal element with that of another verb with respect to some transitivity-related property. First, there are *switch-reference* clauses, which do not only encode whether their subject is identical to the subject of a main clause, but in which the switch-reference marker also appears to co-vary with the transitivity of the verb in the main clause.

Second, there are TM constructions where, like in Saliba and Tariana, the transitivity of two verbs must co-vary with one another. Unlike in Saliba and Tariana, however, Panoan has what I call *syntactic TM*, based on the fact that matching comes with a semantic effect on only the more lexical verb, whereas the reflex on the other, more functional verb is purely formal. In syntactic TM, the grammar does not track the semantic types of verbal stems, but instead expresses an agreement relationship

between the matrix and the embedded Voice in terms of a feature that encodes whether the embedded VoiceP induces only one argument, or both an internal and an external argument. Correlated with this is the difference that structure-building proceeds via complementation, not adjunction.

In line with Valenzuela (2003: 587), Camacho (2010) and Baker & Souza (2020), I will keep the phenomenon of Panoan switch-reference distinct from TM, on grounds that switch-reference does not track transitivity per se. Panoan switch-reference is outlined here to show that it is distinct from TM.

10.2.1 Switch-reference clauses

Shipibo has a switch-reference system in which verbs in dependent clauses carry a morpheme that expresses whether their subject is identical to or distinct from the matrix subject (Valenzuela 2003: 413ff). In what follows, I will refer to this morpheme as the *switch-reference marker*, the clause it is a part of as the *switch-reference clause*, the non-finite verb that the switch-reference marker is attached to as the *dependent verb*, and the finite verb and clause that switch-reference clauses are dependent on as the *matrix verb* and *matrix clause*, respectively. *Switch-reference* will be abbreviated as *SR*.

SR clauses cannot stand alone, and their temporal interpretation is dependent on that of a matrix clause. The SR marker bundles three different kinds of information: (i) information about co-reference between the subjects of the SR and the matrix clause; (ii) whether the event time of the SR clause precedes or co-occurs with that of the matrix clause; and (iii) in some cases, whether the verb in the matrix clause is intransitive or transitive (Camacho 2010: 245). This section will focus on property (iii), i.e. the apparent ability of the SR marker to track the transitivity of a verb in the matrix clause.

In (1a), the matrix verb is intransitive, and this is reflected in the SR marker on the dependent verb, which additionally encodes that (i) the dependent subject is co-referent with the matrix subject, and that (ii) the dependent verb's event time coincides with that of the matrix verb; in (1b), by contrast, the SR marker encodes that the matrix verb is transitive (next to expressing subject co-reference and event precedence):

(1) **Shipibo:**

- a. *Matrix verb: intr* $\Rightarrow$ *SR marker: intr*

Noa-ra- ka-tan-iba-ke [jono bena-i]
 1.PL go-*DIR-RCNT.PST-PERF* peccary look.for-*SS.INTR*
 ‘We went to look for a peccary’ (Faust 1973: 107)

- b. *Matrix verb: trans* $\Rightarrow$ *SR marker: trans*

[Nima oin-**son**]-ra Jose-kan ken-ai
N. see-SS.TRANS-EVID J.-ERG call-IMPERF
'When he (Jose) saw Nima, Jose called (him)' (Loriot et al. 1993: 55f)

The verbs in the SR clause in (1) are both transitive. To further see that the SR marker consistently co-varies with the transitivity of the matrix verb, regardless of the transitivity of its host dependent verb, consider (2), where the matrix and the dependent verb differ in transitivity: in (2a), the matrix verb is intransitive and the dependent verb is transitive, but the SR marker reflects the transitivity of the matrix verb, while the situation is reversed in (2b):¹⁹¹

(2) **Shipibo:**

- a. *Matrix verb: intr; dependent verb: trans* $\Rightarrow$ *SR marker: intr*

[En_i-ra atsa pi-i] *pro*_i ka-ke
 1.ERG-EVID yucca eat-SS.INTR go-PERF
 ‘When I was eating yucca, I left’

- b. *Matrix verb: trans; dependent verb: intr* $\Rightarrow$ *SR marker: trans*

[Ea_i-ra ka-**kin**] *pro*_i mananşawe b(i)-iba-ke
 1.ABS-EVID go-SS.TRANS turtle find-RCNT.PST-PERF
 ‘When I was going away, I found a turtle’ (Camacho 2010: 261)

Finally, notice that (i) the matrix and the dependent verb are both equally lexical, and that (ii) the dependent and the matrix verb need not be adjacent to one another, cf. (1) and (2b). The relevant properties of SR markers are summarized in Table 10.1:

Argument co-reference	Intransitive matrix verb	Transitive matrix verb
Same subject	-aş / -i	-şon / -kin
Different subject	-n / -tian	
Dependent object = matrix subject	-a	

Table 10.1: Shipibo switch-reference marking (Camacho 2010: 246)

A crucial generalization emerges: only those SR markers encoding a same-subject relation may co-vary with the transitivity of the matrix verb; with markers expressing all other argument co-reference relations, this distinction is neutralized (Camacho 2010: 258).¹⁹² For this reason, Camacho (2010) argues that SR markers do not track transitivity per se, but rather the case of the matrix subject. In Shipibo, transitive subjects are marked ergative, and objects and intransitive subjects are marked absolutive. So because the SR marker co-varies with the case of the matrix subject, and

¹⁹¹SR constructions in Shipibo are subject to the requirement that, for every pair of SR and matrix clause, at least one subject position must be overt, whereas the other may host an empty category that is argued to be *pro* (Camacho 2010: 256). In (2), it is the SR clause whose subject is overt, while the matrix subject is *pro*.

¹⁹²In addition, same-subject relations may also be expressed by *-taanan* and *-anan*. While the former is used for event precedence and the latter for event simultaneity, these markers also do not co-vary with the transitivity of the matrix verb, just like in the different-subject and object=subject cases; cf. Camacho (2010: 246).

given ergativity, it looks as if the property it is sensitive to were transitivity. This is implemented as Agree-driven case concord between the SR marker and the matrix verb.

Using the idea that Agree decomposes into Agree-Link and Agree-Copy, Baker & Souza (2020) dissociate SR even from Case, arguing that SR morphology spells out different referential dependencies via index-like pointers; these are established by Agree-Link between the embedded subject and embedded T on the one hand, and the matrix subject and embedded C on the other. Because Agree-Copy does not apply to transfer the ϕ -features from subjects to T or C, LF interprets these pointers as expressing a dependency of referential identity or distinctness between the embedded and the matrix subject.

SR clauses are CPs adjoined to the main projection at TP or CP, with the SR marker itself occupying C (Camacho 2010, Baker & Souza 2020). Even though Camacho's and Baker & Souza's analyses differ in their implementations, they converge towards the notion that Agree is at work in SR. Importantly for us, these approaches entail that the apparent co-variance of the SR marker with the matrix verb's transitivity does not involve type-tracking, because there is no point in the derivation at which the grammar refers to the semantic type of the dependent verb stem. This supports my claim that any kind of transitivity-related information transferred along dependencies which go beyond the V-level cannot refer to the semantic types of verb stems.

10.2.2 Constructions with syntactic transitivity matching

10.2.2.1 Shipibo

In Shipibo, constructions involving the phasal verbs *peo* 'begin' and *keyo* 'finish' (and also *jene* 'stop / leave') give rise to syntactic TM. These verbs are lexical transitives which may take objects as independent verbs. Their independent simple-transitive uses are exemplified in (3), and their independent derived intransitive uses in (4); in the latter, the phasal verbs are detransitivized via the middle, which yields a reflexive, anti-causative or inchoative meaning (the middle suffix has various allomorphs) (Valenzuela 2011: 189):¹⁹³

(3) Root-transitive phasal verbs in isolation

- a. Nokon xobo-ra e-n moa peo-ke
1.GEN house.ABS-EV 1-ERG already begin-COMPL
'I began (to build) my house'
- b. Koka-n yawa rete-ibat-a-ra moa
maternal.uncle-ERG white-lipped.peccary.ABS kill-PST2.ABS-EV already
no-n keyo-ke
1PL-ERG finish-COMPL

¹⁹³The derived intransitive forms in (4) seem a certain degree of lexicalization, with meanings not entirely predictable from the root and the reflexive.

'We already finished (eating) the white-lipped peccary that maternal uncle killed'

(4) *Derived intransitive phasal verbs in isolation*

- a. Ja yosi-nko-ni-a-x peo-koot-a-ronki iki ayawaska
that old.man-LOC-LIG-ABS-SO begin-MID-PP2-HSY AUX ayahuasca
'And so the old man turned into ayahuasca'
literally: 'From the old man originated the ayahuasca'
- b. Tita-ra keyo-t-ai, ja-tian no-a
mother.ABS-EVID finish-MID-INC that-TEMP 1PL-ABS
banet-i ka-[a]i ja-n no-a axe-a-a
stay-SIM.EVENT.SS.SO go-INC 3-ERG 1PL-ABS get.used.to-do.TR-PP2
jawéki-bo-ya
thing-PL-PROPR
'Our mother dies, and then we stay with the things that she taught us'
(Valenzuela 2011: 200)

These phasal verbs also have a complementing use, in which they combine with a lexical verb to give rise to a construction in which the phasal verb is finite and the lexical verb carries SR marking. I will keep the terminology established in connection with SR in the previous section, referring to the finite verb as the *matrix verb*, and the non-finite verb with SR marking as the *dependent verb*. In the following, it is always the phasal verb which is the matrix verb, and the lexical verb which is dependent; within TM constructions, the matrix verb takes on a more functional role relative to the dependent verb.

The SR marker on the dependent verb (i) co-varies with the transitivity of the phasal verb and (ii) always expresses a same-subject relation; yet unlike in the SR constructions discussed above, the stem-level transitivity of the verb itself also co-varies with the transitivity of the phasal verb: in the (a) examples, both verbs are transitive, while both verbs have been made intransitive in the (b) examples via the middle:

(5) *Phasal constructions*

- a. *Inchoative*
- (i) $trans_{dep} + trans_{mat}$
E-n-ra (xeki) [bana-kin] [peo-ke]
1-ERG-EVID corn.ABS sow-SS.TRANS begin-COMPL
'I began to sow (it / the corn)'
- (ii) $intr_{dep} + intr_{mat}$
E-a-ra [ransa-i] [peo-koo-ke]
1-ABS-EVID dance-SS.INTR begin-MID-COMPL
'I began to dance (e.g. at a party)'
- b. *Completive*
- (i) $trans_{dep} + trans_{mat}$

- E-n-ra (nami) [pi-kin] [keyo-ke]
 1-ERG-EVID meat.ABS eat-SS.TRANS finish-COMPL
 ‘I finished eating (the meat) / I ate it up’
- (ii) *intr_{dep}* + *intr_{mat}*
 E-a-ra [ransa-i] [keyó-ke]
 1-ABS-EVID dance-SS.INTR finish.MID-COMPL
 ‘I finished dancing’ (Valenzuela 2011: 201, 202)

10.2.2.2 Kashibo

Like Shipibo, Kashibo has various constructions showing syntactic TM. First, there is the frustrative construction, which expresses the non-accomplishment of an event due to reasons beyond the subject’s control (Biondi 2011: 596). In (6), an auxiliary-like matrix verb occurs with a dependent verb marked for SR; depending on whether the latter is intransitive or transitive, the matrix verb occurs in its intransitive (6a) or transitive (6b) form:^{194,195}

(6) *Frustrative construction*

- a. *intr_{dep}* + *intr_{mat}*

Paën-i-bi kana ‘ë=x [ië-a-n]
 get.drunk-SS.INTR-SAME NAR.1SG 1SG=ABS FRUST.INTR-PERF-1 / 2P
 ‘I almost got drunk’

- b. *trans_{dep}* + *trans_{mat}*

Chupa [maru-kin-bi] kana ‘ë=n
 clothes.ABS buy-SS.TRANS-SAME NAR.1SG 1SG=ERG
 [iëtan-a-n]
 FRUST.TRANS-PERF-1 / 2P
 ‘I almost bought some clothes’ (Biondi 2011: 597)

Next, there are phasal constructions similar to those in Shipibo:

(7) *Inchoative phasal construction*

- a. *trans_{dep}* + *trans_{mat}*

[pi-kin] kana [pëu-i-n]
 eat-SS.TRANS NAR.1SG begin-IMP-1 / 2P
 ‘I begin to eat’

- b. *intr_{dep}* + *intr_{mat}*

¹⁹⁴Even though the transitive form of the frustrative matrix verb in (6b) appears to be morphologically more complex than its intransitive variant in (6a), it is not clear whether one form is derived from the other (Biondi 2011: 596).

¹⁹⁵Every Kashibo sentence requires a second-position clitic which, among other things, indicates the register used. *kana*, for example, decomposes into *ka*, which indicates narrative register, and *na*, which expresses 1SG agreement (Biondi 2011: 480ff).

[‘ux-i] kana pëu-**ukut**-i-n
 sleep-SS.INTR NAR.1SG begin-REFL-IMP-1 / 2P
 ‘I begin to sleep’ (Biondi 2011: 597)

(8) *Completive phasal construction*

a. *intr_{dep} + intr_{mat}*

[‘ux-i] kana [sënë-a-n]
 sleep-SS.INTR NAR.1SG finish-PERF-1 / 2P
 ‘I finished sleeping’

b. *trans_{dep} + trans_{mat}*

[pi-kin] kana [sënë-**o**-a-n]
 eat-SS.TRANS NAR.1SG finish-FACT-PERF-1 / 2P
 ‘I finished eating a long time ago’ (Biondi 2011: 598)

Finally, there is what Biondi (2011) labels the "should-not-have" construction, which expresses regret on part of the subject over some action (syntactically, this construction requires the dependent verb to double). Again, depending on the dependent verb’s transitivity, the matrix verb is intransitive (9a) or transitive (9b):

(9) *"Should-not-have" construction*

a. *[intr_{dep} + intr_{mat}] + intr_{mat}*

[‘ux-ax-ma-ri] [‘i-ti-bi] kana [‘ux-a-n]
 sleep-SS.INTR-NEG-COUN **be**-NOM-SAME NAR.1SG sleep-PERF-1 / 2P
 ‘I should not have slept, but I did’

b. *[trans_{dep} + trans_{mat}] + trans_{mat}*

A piti [pi-xun-ma-ri] [‘a-ti-bi] kana
 that food eat-SS.TRANS-NEG-COUN **do**-NOM-SAME NAR.1SG
 [pi-a-n]
 eat-PERF-1 / 2P
 ‘I should not have eaten that food, but I did’ (Biondi 2011: 599)

10.2.3 Analysis

In a nutshell, I propose that syntactic TM as found in Panoan is the result of agreement between Voice heads with respect to a feature that reflects whether the embedded verb characterizes one or two arguments.

10.2.3.1 Panoan transitivity matching lacks semantic effects

Panoan TM lacks a semantic import on both verbs: while the meaning of the dependent, more lexical verb changes in consistency with its transitive morphology, the co-variance of the matrix, more functional verb’s transitivity via some valence-changing marker is purely formal, i.e semantically vacuous. Note that in cases like (9),

where neither form of the matrix verb seems to be derived from the other, I still take the alternation to be transparent despite the absence of a visible valence-changing device, in the sense that each form corresponds to a clearly identifiable, independent verb; it just happens to be that in pairs like ‘be’ vs ‘do’, the alternation is not formally derivational, like in ‘die’ vs ‘kill’.

First, consider a rather clear case of a semantically vacuous valence change, involving the phasal verb *sënë* ‘finish’, which is a root intransitive.¹⁹⁶ When it is combined with a transitive lexical verb, it undergoes transitivization via the factitive *-o* (10b):

(10) **Kashibo:** *Completive phasal construction*

a. *intr_{dep} + intr_{mat}*

[‘ux-i] kana [sënë-a-n]
sleep-SS.INTR NAR.1SG finish-PERF-1/2P
‘I finished sleeping’

b. *trans_{dep} + trans_{mat}*

[pi-kin] kana [sënë-**o**-a-n]
eat-SS.TRANS NAR.1SG finish-FACT-PERF-1/2P
‘I finished eating a long time ago’

(Biondi 2011: 598)

According to Biondi (2011), the factitive *-o -a* is a generic transitivizer. However, he explicitly mentions that its use in other contexts is restricted to deriving transitive verbs from nouns and adjectives, never verbs (Biondi 2011: 390ff). The nouns *naë* ‘garden’ and *xubu* ‘house’, for example, can in this way be derived into *naë-o* ‘make a garden’ and *xubu-a* ‘make a house’ Biondi (2011: 391). Cases like (10b) appear to be the only ones in which the factitive suffix takes a verbal base, and its presence does not add to the meaning in any way.

Second, consider the frustrative (6) and “should-not-have” construction (9) from above. In both, the choice of one form of the matrix verb over the other as a function of the dependent verb’s transitivity does not effect any semantic change in the matrix verb.

This brings us to the reflexive, employed in Kashibo on the root-transitive phasal verb *pëu* ‘begin’.¹⁹⁷ With an intransitive dependent verb, it must bear the reflexive (11b):

(11) **Kashibo:** *Inchoative phasal construction*

a. *trans_{dep} + trans_{mat}*

[pi-kin] kana [pëu-i-n]
eat-SS.TRANS NAR.1SG begin-IMPF-1/2P

¹⁹⁶Unfortunately, Biondi (2011) gives no examples of *sënë* ‘finish’ in isolation.

¹⁹⁷Likewise, Biondi (2011) gives no examples of *pëu* ‘begin’ in isolation.

- 'I begin to eat'
- b. *intr_{dep}* + *intr_{mat}*
- ['ux-i] kana [pëu-**ukut**-i-n]
 sleep-SS.INTR NAR.1SG begin-REFL-IMP-1 / 2P
 'I begin to sleep' (Biondi 2011: 597)

Outside phasal constructions, the reflexive suffix – which has various allomorphs – has the predictable semantic effect of deriving an action that a subject performs on themselves:

(12) **Kashibo:**

- ʼĒ=x kana policia=nu ʼinan-**mēt**-i-n
 1SG=ABS NAR.1SG police=LOC give-REFL-IMP-1 / 2P
 'I will give myself to the police' (Biondi 2011: 673)

Turning to (11b), however, its use in satisfying TM does not impart any meaning at all. Also, the fact that the cognate phasal verb *peo* 'begin' in Shipibo is detransitivized not via the reflexive, but rather the middle (13b) raises the question of what determines the choice of one valence-decreasing operation over the other in each language:

(13) **Shipibo:** *Inchoative phasal construction*

- a. *trans_{dep}* + *trans_{mat}*
- E-n-ra (xeki) [bana-kin] [peo-ke]
 1-ERG-EVID corn.ABS sow-SS.TRANS begin-COMPL
 'I began to sow (it / the corn)'
- b. *intr_{dep}* + *intr_{mat}*
- E-a-ra [ransa-i] [peo-**koo**-ke]
 1-ABS-EVID dance-SS.INTR begin-MID-COMPL
 'I began to dance (e.g. at a party)' (Valenzuela 2011: 201)

Just like the Kashibo reflexive, the Shipibo middle shows semantically vacuous behavior specifically within the context of satisfying syntactic TM. Its use on *peo* 'begin' in the phasal constructions in (13) contrasts sharply with its use on *peo* 'begin' as an independent verb in (14) as it is only in isolation where it has a clear semantic effect:

(14) **Shipibo:** 'begin' *in isolation*

- a. *peo* 'begin'
- Nokon xobo-ra e-n moa peo-ke
 1.GEN house.ABS-EV 1-ERG already begin-COMPL
 'I began (to build) my house'

b. *peo-koot* ‘begin-MID’

Ja yosi-nko-ni-a-x peo-koot-a-ronki iki ayawaska
 that old.man-LOC-LIG-ABS-SO begin-MID-PP2-HSY AUX ayahwasca
 ‘And so the old man turned into ayahwasca’
 literally: ‘From the old man originated the ayahwasca’

(Valenzuela 2011: 200)

Finally, Panoan TM constructions often express meanings relating to speaker evaluation, mood, modality and aspect. The fact that there exist constructions with similar meanings yet without TM is compatible with the analysis that Panoan TM lacks semantic effects on all verbs. Consider (15), which like the frustrative construction in (6) expresses regret on part of the subject, in this case over an action that was not carried out; unlike the frustrative construction, however, the dependent verb in (15) lacks SR marking, the dependent clause is nominalized, and the matrix verb is invariantly intransitive ‘be’:

(15) **Kashibo:** *Counterfactual construction*a. *intr_{dep} + intr_{mat}*

...[u-ti=bi] ka Juan [‘i-a-x-a]
 come-NMLZ=same NAR.3P J. be-PERF-3P.NON.PROX
 ‘...Juan should have come (but he didn’t)’

b. *trans_{dep} + intr_{mat}*

‘ë [ñuka-ti=bi=ri] ka [‘i-a-x-a]
 1SG.O ask-NMLZ=same=COUN NAR.3P be-PERF-3P.NON.PROX
 ‘He should have asked me about it (but he didn’t)’ (Biondi 2011: 603)

In sum: Panoan TM lacks semantic effects, and the semantic vacuousness of voices such the Kashibo reflexive and the Shipibo middle arises specifically within the context of TM.

10.2.3.2 **The level and mode of structure-building**

This brings us to the level and mode of structure-building. As we know, semantic TM via type-tracking can only arise in V-level adjunction configurations, because it is only in pair-Merge relationships prior to object introduction that the grammar can force the semantic types of two verb stems to match. Given that our Panoan constructions involve syntactic TM, this predicts that they cannot involve V-level adjunction – which is correct.

In the Kashibo cases, this is evident because all TM constructions allow the appearance of phrasal material between verbs; the dependent clause must therefore be at least VP. While analogous data are missing for Shipibo, the possibility of objects in the dependent clause likewise shows that the dependent clause must be at least

VP. We know, in fact, that dependent clauses project up to CP because they contain SR markers, which are located in the C-space (Camacho 2010, Baker & Souza 2020).

What about the mode of structure-building? As outlined above, Shipibo SR clauses are adjuncts to TP or CP (Camacho 2010); in more recent work, Baker & Souza (2020: 1083) show that SR is also found in complement clauses to certain auxiliary-like verbs. I propose that Panoan TM constructions involve complementation, not adjunction.¹⁹⁸ Analyzing these as complementation configurations explains the often auxiliary-like and light-verb-like nature of the matrix verb, brought out especially clearly in the frustrative (6), counterfactual (15) and "should-not-have" constructions (9), and is consistent with and expected of the semantics of these verbs.^{199,200}

10.2.3.3 Semantically vacuous voice

In order to shed light on semantically vacuous voice in Panoan TM, which instantiates a kind of contextual deponency, it is useful to consider the repertoire of valence-decreasing operations in each language. In Kashibo, the middle suffix *-t* derives intransitives from transitives, yielding a reflexive, anti-causative or stative meaning depending on the base verb (Biondi 2011: 394f). In Shipibo, the middle suffix *-t* (and its allomorphs) attaches to transitive verbs and yields intransitive verbs with a reflexive, anti-causative or inchoative meaning. Unlike Kashibo, Shipibo lacks a dedicated reflexive function, which is subsumed under the middle suffix (Valenzuela 2003: 263ff). Both languages have a verbal reciprocal, yet both lack passive, anti-passive and anti-causative operations (Valenzuela (2003: 265,407), Biondi (2011: 391)).

I propose that in both languages, the particular operation that is employed on the more functional, matrix verb in order to satisfy syntactic TM corresponds to whichever one is least marked in that language for achieving the relevant valence change. In Kashibo, this is the reflexive for detransitivization and the factitive for transitivity; in Shipibo, this is the middle for detransitivization (which subsumes the reflexive function).²⁰¹ The use of more marked valence-decreasing devices is system-

¹⁹⁸As a side note, the absence of examples in Biondi (2011) showing the phasal verbs *pëu* 'begin' and *sënë* 'finish' as independent verbs outside complex verbs may be an indication that they are actually unable to occur on their own. This would support a complementation analysis, but note that this is mere speculation absent explicit negative evidence, given that the same verbs lead independent lives in Shipibo (see Section 10.2.2).

¹⁹⁹Following Anderson (2006), I take auxiliary verbs to be verbs that are at least somewhat semantically bleached, and grammaticalized to express one or more of a range of functional categories relating to tense, aspect, mood and modality. Unlike Anderson, who allows auxiliaries to sometimes express meanings pertaining to argument structure and voice, I take these kinds of meanings to fall within the domain of what are classified as light verbs in by Butt (2010). From a thematic perspective under this distinction, auxiliaries are roughly similar to raising verbs, while light verbs similar to control verbs.

²⁰⁰In Valenzuela (2003), Valenzuela (2011) and Biondi (2011), many of these verbs are explicitly referred to as auxiliaries.

²⁰¹We can thus take the Shipibo middle suffix to be functionally underspecified, with its specific contribution determined by the base verb and/or the syntactic context. Alternatively, the Shipibo reflexive is homophonous with the middle.

atically avoided: Kashibo, for instance, has a range of productive valence-decreasing operations in addition to the factitive, including a causative and a benefactive applicative (Biondi 2011: 387ff), all of which are more marked due to imparting more meaning than the factitive; yet none of these are ever used for the purposes of TM. This behavior is parallels that found in contextual bivalence in Oceanic, which is also always carried out via a comparatively unmarked morpho-syntax.

As for why the grammar should specifically employ the reflexive – as opposed to the reciprocal, which both languages also have – first, recall that these languages lack certain other valence-decreasing operations such as the anti-passive, i.e. part of the reason is a paradigmatic gap.

Second, reflexives are semantically less complex than reciprocals: while reflexive relations comprise a single participant acting on themselves, reciprocal relations consist of at least two distinct participants acting on each other, with variation as to the particular kind of reciprocity involved, e.g. each participant may be acted upon by every other participant (*Strong Reciprocity*), or each participant may be acted upon by at least one other participant (*Weak Reciprocity*) (Langendoen 1978, Dalrymple et al. 1998, Bruening 2006). Reciprocals are hence more complex than reflexives both in terms of the number of participants involved, and also the kinds of relations holding between them. They are also more complex than middles, whose subjects carry out an action they are also affected by.

Third, reflexive markers across languages tend to have a variety of non-reflexive uses besides reflexivity, appearing in various non-active intransitive contexts such as passives, anti-causatives and middles (with some verbs lexically bearing reflexive morphology). In this connection, I draw on work by Alexiadou et al. (2015), who argue that certain voice elements may have underlying syntactic correspondents, yet remain devoid of any semantic import (Alexiadou et al. 2015: 108). One instance of this kind of vacuous voice are anti-causative constructions in Romance languages, which productively employ reflexivation to derive anti-causatives from transitive inputs:

(16) **Italian:** *Anti-causative via reflexive*

a. *Transitive*

Gianni ha chiuso la finestra in un secondo
J. has closed the window in one second
'John closed the window in one second'

b. *Anti-causative*

La finestra **si** è chiusa in un secondo
the window REFL is closed in one second
'The window closed in one second' (Alexiadou et al. 2015: 83)

Building on Schäfer (2008), Alexiadou et al. (2015: 110) argue that reflexives like in (16b) can give rise to vacuous voice because only reflexives may under certain condi-

tions remain without any denotation, thereby being allowed not to be semantically integrated into the verbal event (i.e. they need neither a variable slot nor a thematic role). Consequently, the presence of the reflexive pronoun *si* in (16b) has syntactic and morphological, but no semantic effects.

Semantically vacuous voice in Panoan is reminiscent of the behavior of Formosan voice marking in restructuring configurations, where certain instances of voice morphology lack the syntactic effects they are otherwise associated with. This is illustrated by Mayrinax Atayal (Austronesian, Taiwan), where extraction of a certain grammatical function must always be accompanied by dedicated verbal morphology which flags that function's thematic role. Object extraction in simple transitive contexts is thus blocked when the verb is marked for Agent Voice (17a), and only licit under Patient Voice marking (17b):

(17) **Mayrinax Atayal**

- a. *Agent Voice blocks object extraction*

*Nanuan_i ku —_i t<um>aluk i Yumin?
 what KU AV.cook KU Y.
 intended: 'What is Yumin cooking?'

- b. *Patient Voice licenses object extraction*

Nanuan_i ku —_i tuting-un i Yumin?
 what KU beat-PV CS Y.
 'What is Yumin beating?'

(Chen 2010: 8)

When objects are extracted out of complements to restructuring verbs, this restriction disappears, i.e. the verb from which they are extracted can and in fact must carry Agent Voice marking (note that the matrix verb must carry Patient Voice marking):

(18) **Mayrinax Atayal:** *Agent Voice does not block object extraction in restructuring*

Nanuan_i ku naqaru-un i [—_i t<um>uting] ni Yumin?
 what KU finish-PV LNK AV.beat CS Y.

'What did Yumin finish beating?'

(Chen 2010: 8)

So even though the Agent Voice markers in (17a) and (18) are morphologically identical, they are syntactically distinct (Chen 2010, Levin 2016, Wurmbrand 2016). Just like the Panoan middle and reflexive voices, the vacuous behavior of the Agent Voice marker arises only in certain contexts.

Following Levin (2016) and Wurmbrand (2016), I take the fact that Agent Voice in (18) exceptionally does not block object extraction to mean that it is a kind of *default voice* that lacks the syntactic effects it is generally associated with. Levin (2016) argues that at PF, an uninterpretable Case-agreement feature [uCAGR] is inserted on the embedded verb for morphological well-formedness, which subsequently gets

spelled out as Agent Voice due to failing to find an appropriate goal. In Pietraszko & Wurmbrand (2021), in turn a continuation of Wurmbrand & Shimamura (2017), there is an Agree-based voice dependency between the matrix and the embedded verb in the narrow syntax, with vacuous Agent Voice arising if only a subset of voice features successfully get transferred to the embedded verb. Even though these approaches differ in certain regards, both agree that what appears as Agent Voice in certain contexts is devoid of the syntactic effects it is otherwise accompanied by.

Another way in which the Panoan and the Formosan data are similar concerns so-called *voice concord* languages, where – unlike in languages like Mayrinax Atayal (18), where the restructuring complement always carries Agent Voice marking – voice marking in restructuring complements always reflects the voice marking on the matrix verb:²⁰²

(19) **Isbukun Bunun**

Iliskin-**un**-ku bunbun-a [tu baliv-**un**]
want-PV-1SG.ACC banana-that.NOM PRT buy-PV

‘I wanted to buy the bananas’

literally: ‘The bananas are wanted to be bought by me’ (Wu 2013: 40)

Again following Levin (2016) and Wurmbrand (2016), I assume that voice concord instantiates a kind of vacuous voice: the morphology either responds to the insertion of the [uCAGR] feature by filling it with the voice features of the matrix verb (Levin 2016), or syntactic conditions permit the spreading of the full set of morpho-syntactic features from the matrix to the embedded Voice head (Pietraszko & Wurmbrand 2021). The syntactico-semantic vacuousness of voice is thus expressed grammatically either as a kind of anaphoric Voice (voice concord), or by falling back to a default (default voice).

10.2.3.4 Syntactic transitivity matching via Voice

I adopt Pietraszko & Wurmbrand (2021)’s Agree-based mechanism to model the formal co-variance of the matrix verb with the dependent verb’s transitivity in Panoan syntactic TM.²⁰³ Important in this approach are the ideas that (i) Voice heads can enter into morpho-syntactic feature-sharing dependencies via Agree; and (ii) the morphological properties of Voice may dissociate from its syntactic and semantic properties, as similarly argued in Alexiadou et al. (2015). That the matrix and the dependent verb should communicate via Voice is expected given that Voice is the clausal locus of transitivity alternations.

²⁰²Agent Voice in restructuring complements also does not interact with cliticization, unlike Agent Voice in regular transitive contexts and full CP complements (Chen 2010).

²⁰³Note that these works comes from the phenomenon of restructuring, where complements are taken to only project up to VP or VoiceP, not CP.

I propose that matrix Voice reflects whether the embedded clause has only one argument (which can be internal or external), or whether it has both an internal and an external argument. The grammar thus makes reference to the embedded verb's transitivity in term of the number of its arguments. What this predicts is that embedded unaccusatives and unergatives should behave the same in triggering intransitive morphology on the matrix verb.

The analysis of ergative case in Shipibo by Baker (2014) entails that unaccusatives and unergatives are phrase-structurally distinct in the language, yet I have been unable to find TM data from Shipibo or Kashibo which clearly involve embedded unaccusatives; all cases with embedded intransitives that I am aware of are more consistent with an unergative analysis. Nevertheless, I believe we can leverage data from Dulong – a language which has syntactic TM just like in Panoan – for an indirect argument that Panoan unaccusatives and unergatives pattern alike for the purposes of TM.

In Dulong, the prefix *v-* yields intransitive verbs with unintentional, anti-causative or reciprocal meanings (LaPolla 2000: 288f). I assume that the embedded verb in (20), prefixed with *v-*, is a derived unaccusative, while in (20b), it is unergative; crucially, the lexically transitive aspectual auxiliary is made intransitive via the reflexive / middle in both (20a) and (20b):

- (20) **Dulong:** *intr_{dep}* + *intr_{mat}*
- a. [v-ngaq-lým] [dýn-shì-ě]
 DETR-push-PURP about.to-REFL/MID-NPST
 'It seems like it is about to fall down'
- b. *intr_{dep}* + *intr_{mat}*
 àng-í [ým]-[dýng-shì] bǒ-ì
 3SG eat-finish-REFL/MID PFV-3.INTR.PST
 'He finished eating' (LaPolla 2000: 294)

Given that unaccusatives and unergatives pattern together for the purposes of syntactic TM in Dulong, I conclude that this must also true in Panoan.

I propose that in syntactic TM, matrix Voice reflects whether the embedded clause has one or two arguments by way of the structure-building features of embedded Voice. These structure-building features correspond to a feature I descriptively call [$\pm$ TRANS], as follows: unaccusative Voice only requires the merger of a verbal complement, while unergative Voice also triggers the merger of an external argument – since both of these induce only a single argument, they characterize the feature [-TRANS]; by contrast, transitive Voice has both a merge feature for an external argument and a Case feature for an internal argument, which characterizes the feature [+TRANS]. This is summarized in 10.2:

Structure-building features	Argument structure	Transitivity	[±TRANS]
[•V•<*CASE*•D•]	internal + external argument	transitive	[+TRANS]
[•V•<•D•]	external argument	unergative	[-TRANS]
[•V•]	internal argument	unaccusative	[-TRANS]

Table 10.2: Voice and [±TRANS]

This way, [±TRANS] connects directly to features on Voice and the syntactic argument structure they induce. Unlike SR marking, which only indirectly reflects transitivity by tracking subjective Case, syntactic TM thus directly tracks a verb's transitivity in terms of the number of its arguments. Note that [±TRANS] must be a feature borne by Voice in addition to its structure-building features, not simply a shorthand for these feature sets, since structure-building features are checked in the course of the derivation, and can therefore not be backtracked by the probe on matrix Voice; [±TRANS], in contrast, remains even after these features are checked.

I suggest that those verbs in Panoan which undergo syntactic TM lexically select a Voice head bearing a feature [uTRANS], which probes downwards for a [±TRANS] feature. I leave open why specifically these verbs should exhibit this behavior, merely adding that it should be tied to their verb-taking variants, but not their simple object-taking ones, given that in the latter, voice morphology is not semantically vacuous (cf. (4)).

With respect to the particular voice marker chosen to spell out [uTRANS] and satisfy TM, recall that it is either the factitive, reflexive or middle, all of which instantiate comparatively unmarked valence-changing operations. Based on the idea in Alexiadou et al. (2015) that reflexive morphology is compatible with various non-active / intransitive contexts, I claim that this is no accident, i.e. that the grammar systematically resorts to whichever strategy is least specific within a language's repertoire in order to satisfy syntactic TM.

[uTRANS] is spelled out according to rules in (21) and (22): in Shipibo, [uTRANS: -] on the Voice head of the lexically transitive verb *peo* 'begin' is spelled out as *-koo*, which corresponds formally to the reflexive, and as $\emptyset$ elsewhere; in Kashibo, this pattern is reversed with respect to the verb *sēnē* 'finish', which is lexically intransitive and therefore spelled out with the suffix *-o*, formally the factitive, when Voice bears [uTRANS: +] (and as $\emptyset$ elsewhere):

(21) **Shipibo:** *peo* 'begin' (lexical transitive)

$$[\text{uTRANS}] \Leftrightarrow \begin{cases} \emptyset & [\text{uTRANS} : +] \\ -koo & [\text{uTRANS} : -] \end{cases}$$

(22) **Kashibo:** *sĕnĕ* ‘finish’ (lexical intransitive)

$$[\text{uTRANS}] \Leftrightarrow \begin{cases} -o & [\text{uTRANS} : +] \\ \emptyset & [\text{uTRANS} : -] \end{cases}$$

Additionally, we have to model the fact that Spell-Out is sensitive to the lexical transitivity of *peo* ‘begin’ and *sĕnĕ* ‘finish’, i.e. we have to ensure that the reflexive on transitive *peo* ‘begin’ is only spelled out when [uTRANS] is positive, and conversely that the factitive on intransitive *sĕnĕ* ‘finish’ is only spelled out when [uTRANS] is negative. Assuming that the verb-taking variants of *peo* ‘begin’ and *sĕnĕ* ‘finish’ also bear [$\pm$ TRANS], the generalization is that an overt valence-changing marker is only spelled out if the value of [uTRANS] is opposite to the verb’s lexical [$\pm$ TRANS] value: $\{[+\text{TRANS}], [\text{uTRANS}:-]\}$ and $\{[-\text{TRANS}], [\text{uTRANS}:+]\}$ are thus spelled out as an overt valence-changing marker, while $\{[+\text{TRANS}], [\text{uTRANS}:+]\}$ and $\{[-\text{TRANS}], [\text{uTRANS}:-]\}$ are not. See Table 10.3:

	<i>peo</i> ‘begin’ ([+TRANS])	<i>sĕnĕ</i> ‘finish’ ([-TRANS])
[uTRANS: +]	-	$\Rightarrow$ transitivization via factitive
[uTRANS: -]	$\Rightarrow$ detransitivization via reflexive	-

Table 10.3: Variable spell-outs of [uTRANS]

Consistent with the proposal that syntactic TM involves complementation is that while certain grammatical processes can take place out of adjuncts, probing into adjuncts is generally impossible (McFadden et al. 2019). I also assume that Voice can probe downward until the next Voice head, based on Keine (2020).

10.2.3.5 Interim summary

Shipibo and Kashibo show syntactic TM, whereby certain matrix, more functional verbs co-vary formally with the transitivity of a dependent, more lexical verb in terms of the feature [$\pm$ TRANS], which encodes whether embedded VoiceP contains only one argument, i.e. is intransitive ([-TRANS]), or two arguments, i.e. is transitive ([+TRANS]). Agree regulates the spreading of [$\pm$ TRANS] from the embedded to the matrix Voice, while a language’s voice repertoire, the markedness of a particular voice and property sets determine how [$\pm$ TRANS] is spelled out.

10.2.3.6 Derivations

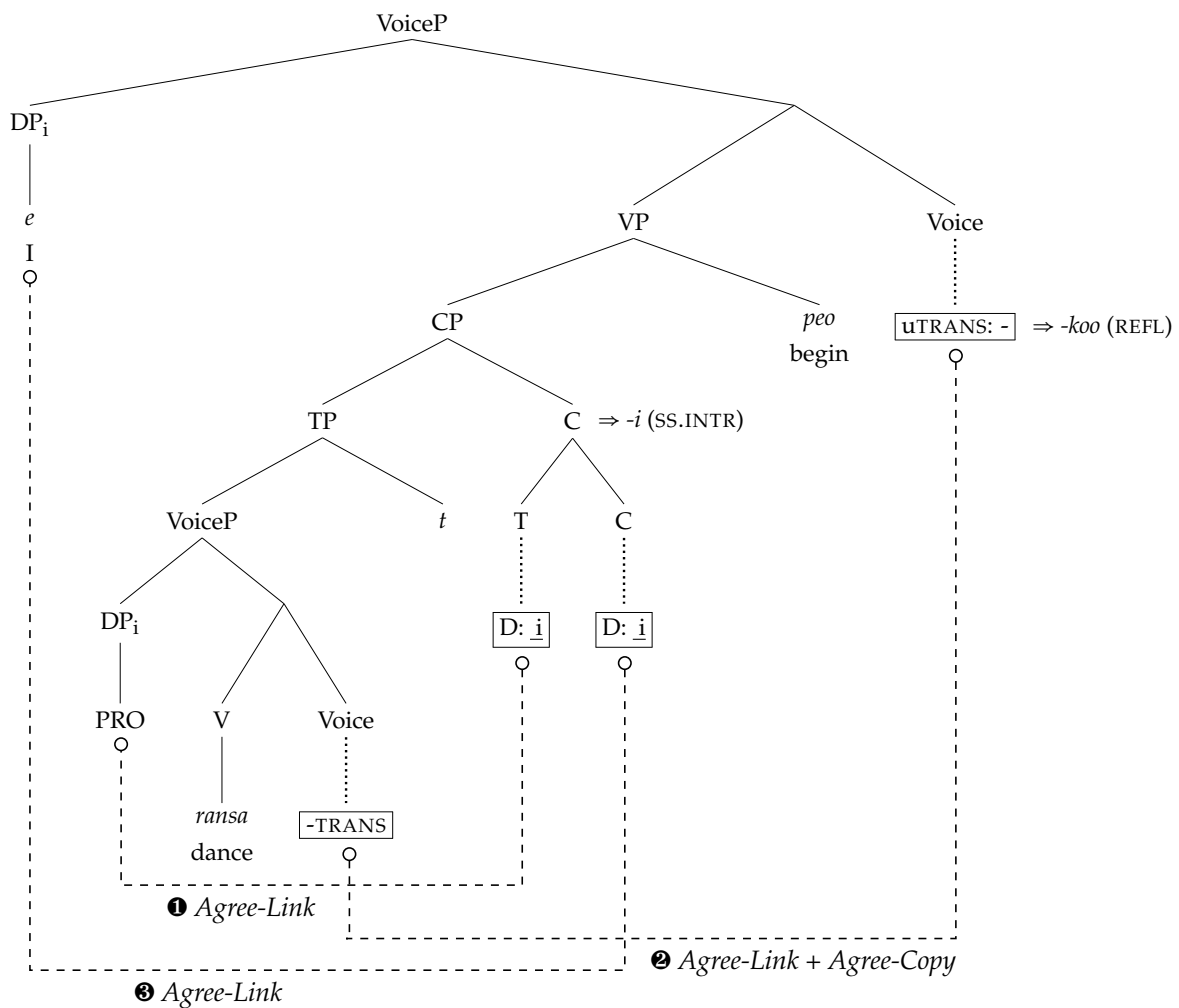
We can now assemble all the pieces into derivations. I begin with the intransitive TM phasal construction in (23) from Shipibo, where the matrix verb undergoes semantically vacuous detransitivization. (23) is derived in Tree 10.1:²⁰⁴

(23) **Shipibo:** *intr_{dep}* + *intr_{mat}*

E-a-ra [ransa-i] [peo-koo-ke]
1-ABS-EVID dance-SS.INTR begin-MID-COMPL

‘I began to dance (e.g. at a party)’

(Valenzuela 2011: 201)



Tree 10.1: Inchoative phasal construction + intransitive lexical verb

The dependent clause in (23) projects up to CP, with the unergative verb ‘dance’ projecting a PRO external argument in Spec, VoiceP. Following Baker & Souza (2020)’s analysis of Panoan SR, embedded T enters into Agree-Link with the embedded subject, which creates an index-like pointer to the embedded subject on T, yet without there also being a transfer of its φ -features to T. T then undergoes head movement to

²⁰⁴Note that Trees 10.1 and 10.2 both leave out head movement of V to C, through Voice and T.

C. Because embedded VoiceP has only an external argument, it counts as intransitive and hence Voice bears [+TRANS]. After the matrix Voice is merged, it enters into both Agree-Link and Agree-Copy with the embedded Voice in terms of the feature [+TRANS]. As per the rule in (21), [+TRANS] on matrix Voice is spelled out as the reflexive marker *-koo*.

Embedded C subsequently probes upwards and enters into Agree-Link with the matrix subject, again resulting in a pointer. Because T and C in the complex T+C head point to two distinct, yet co-referent DPs, LF interprets these pointers as expressing a semantic binding relation (Baker & Souza 2020: 5). The T+C head is therefore spelled-out as the same-subject SR marker. Although the derivation of SR by itself is irrelevant to our purposes, I have included it here in order to show that it is not a necessary condition for Panoan TM: in Tree 10.1, feature-sharing along the Voice channel occurs independently of the SR mechanism.²⁰⁵ Syntactic TM constructions with dependent unaccusatives are derived just like in Tree 10.1, except that the sole argument in the embedded clause is internal.

Next, consider the transitive phasal construction in (24), in which the matrix verb retains its lexical transitivity, in reflection of the transitive dependent verb. Tree 10.2 shows the derivation of (24):

(24) **Shipibo:** *trans_{dep} + trans_{mat}*

E-n-ra (xeki) [bana-kin] [peo-ke]
1-ERG-EVID corn.ABS sow-SS.TRANS begin-COMPL

‘I began to sow (it / the corn)’ (Valenzuela 2011: 201)

²⁰⁵This is not to say that SR and TM cannot interact. Consider (i), a uniquely peculiar case restricted to the phasal verb *sënë* ‘finish’, in which *sënë* ‘finish’ can be either intransitive or transitive despite the dependent verb remaining invariantly transitive. The choice of transitivity does, however, come with a temporal-aspectual difference: while (ia) has a perfective interpretation, (ib) has a past interpretation:

(i) **Kashibo:** *Completive phasal construction*

a. ‘finish’ is intransitive → *Perfective reading*

[pi-i] kana [sënë-a-n]
eat-SS.INTR NAR.1SG finish-PERF-1/2P

‘I just finished eating’

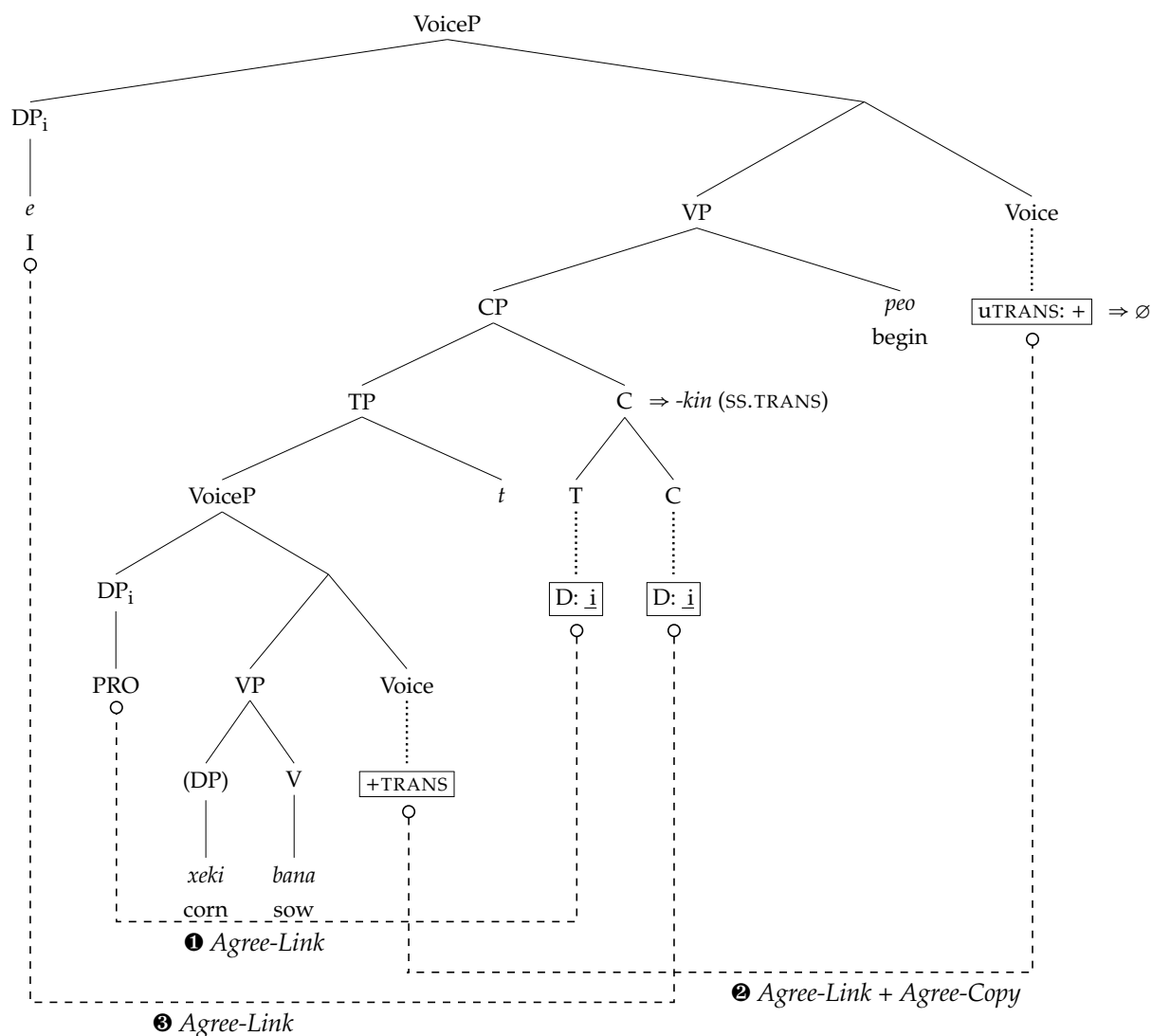
b. ‘finish’ is transitive → *Past reading*

[pi-kin] kana [sënë-o-a-n]
eat-SS.TRANS NAR.1SG finish-FACT-PERF-1/2P

‘I finished eating a long time ago’

(Biondi 2011: 598)

Moreover, notice that the dependent verb carries an intransitive same-subject SR marker in (ia), but a transitive same-subject SR marker in (ib), in co-variance with the matrix verb’s transitivity. This shows us that there is a third Agree-Link dependency involved, not shown in Tree 10.1, namely between the embedded SR marker on the one hand, and matrix Voice on the other. So even though TM is independent of SR in that neither entails or requires the other, it is possible for SR to piggyback on TM. I do not know why only *sënë* ‘finish’ behaves this way, except that it is consistent with the present analysis, which treats the feature [uTRANS] as a lexically arbitrary property of certain verbs.



Tree 10.2: Inchoative phasal construction + transitive lexical verb

The derivation proceeds just as before, the only difference being that because embedded Voice now requires the merger of both an internal and an external argument, it bears the feature $[+TRANS]$.²⁰⁶ This feature gets transferred to the matrix Voice and receives a null spell-out, i.e. the phasal matrix verb retains its lexically transitive form.²⁰⁷

The derivations of the Kashibo pair in (25), where the matrix verb is lexically intransitive and undergoes pseudo-transitivization via the factitive if the dependent verb

²⁰⁶As the brackets in the translation of (24) show, while the object is optional, the form of the phasal verb is invariantly transitive. This follows naturally from the present analysis, which ties syntactic TM to transitivity in terms of features on Voice heads, rather than directly to syntactic arguments present in the structure. I assume that the while the feature $[*CASE*]$ obligatorily enters the derivation, it need not be discharged, following ideas in Hopperdietzel (2020) and Nie (2020); cf. also Preminger (2012).

²⁰⁷The present account does not explain why in the intransitive phasal construction in (23), the matrix subject is absolutive, whereas in the transitive phasal construction in (24), it is ergative. All else being equal, the case of the matrix subject is predicted to remain invariant. What seems to be going on here is that the matrix subject reflects the case of the embedded subject. One way to derive this is to assume that (i) in their complementing uses, these phasal verbs involve raising; and (ii) the matrix subject does not receive case in the matrix clause, instead inheriting it from below.

is transitive, are mirror images of Trees 10.1 and 10.2 with respect to how Voice on the matrix verb ends up being spelled out: when the dependent verb is intransitive (25a), [-TRANS] on matrix Voice is spelled out as $\emptyset$, and when the dependent verb is transitive (25b), [+TRANS] on matrix Voice is spelled out as the factitive *-o*, as per the rule in (22):

(25) **Kashibo:** *Completive phasal construction*

a. *intr_{dep} + intr_{mat}*

[‘ux-i] kana [sënë-a-n]
sleep-SS.INTR NAR.1SG finish-PERF-1 / 2P
‘I finished sleeping’

b. *trans_{dep} + trans_{mat}*

[pi-kin] kana [sënë-o-a-n]
eat-SS.TRANS NAR.1SG finish-FACT-PERF-1 / 2P
‘I finished eating a long time ago’

(Biondi 2011: 598)

10.2.4 Other languages with syntactic transitivity matching

Another case in point is Dulong, a Tibeto-Burman language spoken in Myanmar and China. It is rigidly transitive, and has at its disposal a reflexive/middle and causative operation, but no passive or anti-passive LaPolla (2000: 284, 287ff). There is a second valence-decreasing operation, the prefix *v-*, which takes transitive inputs and yields intransitive verbs expressing either unintentional, anti-causative or reciprocal action (LaPolla 2000: 288f).

Like in Panoan syntactic TM, Dulong has a small number of verbs which are able to occur on their own, and which may combine with another, more lexical verb and co-vary with its transitivity in a kind of complex verb construction (LaPolla 2008: 3). LaPolla calls these *auxiliaries*, but I will continue the distinction between *matrix* and *dependent* for the more functional and lexical verbs, respectively, in keeping with the practice established above.

The verb *mōnòè* ‘to continue’, for example, is lexically transitive and therefore able to appear with a transitive dependent verb without further ado (26a), but must bear the reflexive / middle marker *-shì* when occurring with an intransitive dependent verb (26b):²⁰⁸

²⁰⁸All transitive verbs with third-person objects are obligatorily suffixed with either *-ò* or *-à* in non-past and past tenses, respectively (LaPolla 2008).

(26) **Dulong**

- a.
- trans_{dep} + trans_{mat}*

àng-í shǒng [rí]-[mǔn-ò]-ē
 3SG-AGT tree/wood carry-continue-3.TRANS-NPST
 ‘He is continuing to carry the wood’

- b.
- intr_{dep} + intr_{mat}*

àng [yǒp]-[mǔn-shì]-ē
 3SG sleep-continue-REFL/MID-NPST
 ‘He is continuing to sleep’

(LaPolla 2000: 293f)

When the dependent verb is ambi-transitive, the matrix verb systematically co-varies with either transitive (27a) or intransitive morphology via the reflexive / middle marker (27b), depending on whether the verb is used as a transitive (27a) or an unergative (27b):

(27) **Dulong**

- a.
- ambi-trans_{dep} + trans_{mat}*

àng ǔmpà-lòng [ǔm]-[dǔng] bó-à
 3SG-INST food-CLF eat-finish PFV-3.TRANS.PST
 ‘He has finished eating the food’

- b.
- ambi-trans_{dep} + intr_{mat}*

àng-í [ǔm]-[dǔng-shì] bó-ì
 3SG eat-finish-REFL/MID PFV-3.INTR.PST
 ‘He finished eating’

(LaPolla 2000: 294)

This pattern also holds under non-adjacency of the two verbs. In (28), the auxiliary reflects the transitivity of the lexical verb just as above, despite the two being separated by the purposive suffix *-lóm*:

(28) **Dulong**

- a.
- trans_{dep} + trans_{mat}*

àng-í lègā [vrū-lǔm] [dǔn-ò-ē]
 3SG-AGT letter write-PURP about.to-TRANS-NPST
 ‘He is about to write a letter’

(LaPolla 2008: 58)

- b.
- intr_{dep} + intr_{mat}*

[v-nqāq-lǔm] [dǔn-shì-ē]
 DETR-push-PURP about.to-REFL/MID-NPST
 ‘It seems like it is about to fall down’

(LaPolla 2000: 294)

Other verbs following this pattern include *dǔndē* ‘be about to’, *mūdē* ‘be used to’, *dǔngdē* ‘be finished’, *pǔngdē* ‘begin to’ and *nēdē* ‘be willing to’ (LaPolla 2000: 294), and

also *daqòē* ‘be able to’ (LaPolla 2008: 3).

Just like in Panoan, the detransitivization reflex these verbs exhibit in order to satisfy TM is achieved via a device which – despite generally yielding a reflexive / middle meaning – lacks any such meaning in the context of TM. When the matrix verb needs to satisfy syntactic TM, it always undergoes semantically expletive detransitivization via the reflexive / middle suffix *-shì*, never the prefix *v-*.²⁰⁹

This means that *-shì* should be less marked than *v-*, which finds support in the fact that the functions of *v-* include the expression of unintentional and reciprocal actions, which are semantically more loaded than reflexive and middle actions. I therefore conclude that exactly the same analysis for Panoan should be applied to Dulong, i.e. Dulong has syntactic TM.²¹⁰ The Dulong pattern also shows – as stated above – that TM is independent of SR because Dulong has TM but lacks SR.

Other languages that have syntactic TM with analogous patterns are Araona (Tacanan, Bolivia) (Emkow 2006), Gta? (South Munda, India) (Mahapatra, Pujari & Panda 1989) and Dumi (Tibeto-Burman, Nepal) (van Driem 1993).

10.3 Morphological transitivity matching in Kiowa

We have seen that TM may have semantic effects, as in Saliba and Tariana, and that it may remain without semantic effects yet have a transparent reflex in the form of an overt valence-changing device, as in Panoan and Dulong. In semantic and syntactic TM, all of the elements involved are verbs. This section will present a third kind of TM which lacks these properties, *morphological TM*, which is basically transitivity-conditioned allomorphy.

In Kiowa (Tanoan, Oklahoma), for example, there is a modal suffix that co-varies with the transitivity of the lexical verb: when the lexical verb is intransitive, this suffix surfaces as *t!ɔɔ*, and when it is transitive, as *tɔɔ* (29):

(29) **Kiowa**

- a. *intr_{lex} + intr_{mod}*

[héíb-é]-[t!ɔɔ]

enter-INTR-MOD_(INTR)

‘will come in’

- b. *trans_{lex} + trans_{mod}*

[héíb-e]-[tɔɔ]

enter-TRANS-MOD_(TRANS)

‘will bring in’

(Bonet & Harbour 2012: 231)

Co-variance does not require adjacency:

²⁰⁹*v-* may, however, occur on the lexical verb, cf. (28b).

²¹⁰It is unclear whether all complementation configurations without TM form a natural class.

(30) **Kiowa**

- a.
- $trans_{lex} + trans_{mod}$

[héɪb-é]-gʷu-mɔɔ-[t!ɔɔ]

enter-INTR-DISTR-NEG-MOD_(INTR)

‘will not come in at different times / locations’

- b.
- $intr_{lex} + intr_{mod}$

[héɪb-e]-gʷu-mɔɔ-[tɔɔ]

enter-TRANS-DISTR-NEG-MOD_(TRANS)

‘will not bring in at different times / locations’

(Bonet & Harbour 2012: 231)

I adopt Bonet & Harbour’s analysis that this is an instance of allomorphy which (i) is sensitive to the transitivity of the lexical verb, and which (ii) is not restricted to structurally adjacent elements, as per the lexical insertion rule in (31):

$$(31) \quad [\text{MOD}] \Leftrightarrow \begin{cases} t!ɔɔ & \text{INTR} \\ tɔɔ & \text{TRANS} \end{cases} \quad (\text{Bonet \& Harbour 2012: 231})$$

For conceptual and empirical reasons, such as the fact that there is little basis both within Kiowa and across languages for positing an agreement relation between transitivity and modality, the authors reject the notion that this kind of co-variance involves Agree. This is consistent with the generalizations in Weisser (2019) on differences between agreement and allomorphy, according to which the types of features governing allomorphy are often arbitrary, in the sense that it is not always clear why one element (e.g. a modal) should track some property (e.g. transitivity) of another element. Concerning agreement, by contrast, we have a relatively good understanding of the kinds of relations involved (e.g. φ -agreement is connected to nominal licensing).²¹¹

Bonet & Harbour do not go into the particular transitivity-related property governing the choice of the form of the modal. I propose that as allomorphy, it may be governed by various properties, e.g. structure-building features on Voice, a feature jointly determined by the verb and the (particular kind of) internal argument (i.e. a property of VP), some lexical feature on V (which may correlate with any of these), etc. What matters is that (i) the relevant feature somehow align with transitivity classes in verbal typology (i.e. it must induce a partition between intransitive and transitive verbs); and that (ii) this feature be morpho-syntactic in nature, given that – despite being distinct from Agree – allomorphy is still subject to syntactic conditions, just like Agree.²¹² For Kiowa, I assume that the relevant features are connected to

²¹¹See Bobaljik & Harley (2017) for similar points regarding Hiaki, where verbal root suppletion governed by the plural features of one of the verb’s arguments is argued to instantiate allomorphy, rather than agreement.

²¹²Works discussing locality constraints on contextual allomorphy include David (2010), Bobaljik &

the presence of the suffixes glossed as INTR and TRANS in (29) and (30), which I take to instantiate Voice or v.

Concerning the elements involved, I define morphological TM as cases in which a transitivity-related property on some lexical or functional verbal element induces an allomorphic change on another, more functional element that is not itself a verb, but part of the extended verbal projection.

In terms of the inventory of alternations, allomorphy usually shows around two alternating forms, whereas agreement often gives rise to full paradigms; moreover, within a language, agreement behaves highly regularly (Weisser 2019). The inventory of alternations is not helpful in telling apart syntactic from morphological TM because in all the cases discussed here, the inventory of alternations is restricted to two, which we can trace to the fact that the general opposition involved is binary, i.e. between the values *intransitive* and *transitive*.²¹³ I nevertheless take the fact that syntactic TM involves morphologically transparent reflexes while morphological TM does not (at least synchronically) to be further support for the view that the former involves agreement, and the latter allomorphy.

Finally, I assume that in terms of the mode of structure-building, morphological TM is restricted to complementation, in light of the fact that adjunct-triggered allomorphy remains unattested (Weisser 2018, Weisser 2019); see also Hopperdietzel (2020: 238).

10.4 Towards a typology of transitivity matching

The descriptive works on transitivity matching generally treat it as a uniform phenomenon, without considering whether a given formal change on a verb also entails a change in that verb's semantics. In LaPolla (2010), for example, the kind of TM found in Dulong is discussed on a par with the kind of TM found in Saliba and Panoan; we know, though, that TM in Saliba is semantic, while TM in Dulong and Panoan is syntactic.

Biondi (2015) is first to recognize at least a two-way distinction between what he calls *transitivity harmony* and *allomorphic alternation based on transitivity*. For Biondi, transitivity harmony refers to situations in which the transitivity-related co-variance of a given element occurs transparently via adjustment by some valence-changing morpheme; allomorphic alternation, on the other hand, refers to situations where there is no such explicit change to an element, but where an entirely different, non-transparently related element is selected instead. In our terms, the former correspond to syntactic TM, the latter to morphological TM. Somewhat ironically, even though Biondi (2015) explicitly mentions the language, the kind of TM that is missing from

Harley (2017), Arregi & Nevins (2012), Merchant (2015) and Moskal (2015), among many others.

²¹³So far, I have not come across a TM construction which exhibits a tripartite distinction between intransitives, transitives and ditransitives.

his typology is the kind instantiated by Saliba, i.e. semantic TM.

The different kinds of TM we have uncovered here, along with their similarities and differences, are summarized in Table 10.4:

Transitivity matching	Elements involved		Mode of SB	Mechanism
Morphological	V ₁ / VP ₁ / Voice(P) ₁ : -lexical -reflex has semantic effect	X ₂ (X may be non-verbal): -functional -reflex has no semantic effect	complementation	allomorphy wrt transitivity-related property
Syntactic	Voice ₁ : -V is lexical -reflex has semantic effect	Voice ₂ : -V is more functional -reflex has no semantic effect	complementation	agreement wrt structure-building features
Semantic	V ₁ : -lexical -semantic effect	V ₂ : -lexical -semantic effect	adjunction	matching of semantic types

Table 10.4: Typology of transitivity matching

How do these different sub-types of TM relate to one another? I propose that the different kinds of TM arise as a function of the respective kinds of elements that are involved, i.e. whether both elements are lexical verbs (semantic TM), the formally co-varying element is a more functional verb (syntactic TM), or the formally co-varying element is a functional non-verb (morphological TM). This synchronic characterization, in turn, recapitulates a certain diachronic trajectory. It is well-known, for example, that auxiliaries typically develop out of lexical sources, which is reflected in the synchronic grammar of a language by the fact that auxiliaries tend to also exist as independent verbs, or at least have obvious lexical correspondents (Heine 1993, Kuteva 2004). Syntactically, the most common diachronic path towards auxiliation departs from SVCs and verb-complement configurations (Anderson 2006: 302) – i.e. it proceeds from more cohesive to less cohesive domains.

Based on this generalization, I suggest the following diachronic path: TM starts out as the semantic sub-type, before evolving into the syntactic sub-type and finally ending as the morphological sub-type. In other words, semantic TM is initially enforced via a rule like EC_{TS}, which forces type-matching with semantic effects between two adjoined verb stems. Over time, this maximally cohesive domain develops into a less cohesive one by virtue of one of the verbs acquiring a more abstract meaning; this now semi-functional verb has been freed from the semantic restrictions once entailed by type-matching, but retains as a lexical rudiment the requirement to formally reflect this erstwhile restriction in a transparent way. Finally, this semi-functional verb abstracts further into a fully functional non-verbal element that is nevertheless part of the verbal extended projection, similarly retaining a

rudimentary reflex of semantic TM, the difference being that this reflex is now no longer transparent.²¹⁴

10.5 Transitivity restrictions

With respect to the transitivity restrictions involved, both syntactic and morphological have a configurational basis, namely complementation, which is necessary for them to arise. The sufficient ingredient that ultimately gives rise to these kinds of TM, however, is a purely formal ad hoc property in each case: [uTRANS] on a Voice head, and the allomorphy of functional elements like a modal. Just like with semantic TM, then, we have a fully a derivational account of syntactic and morphological TM.

10.6 Summary

This chapter presented and analyzed syntactic and morphological TM. The former is the result of agreement between Voice heads with respect to the feature [$\pm$]TRANS, which reflects whether a verb characterizes a single argument – i.e. is intransitive – or two arguments, i.e. is transitive. This was achieved indirectly through structure-building features on Voice. Depending on its lexical transitivity, the matrix verb is spelled out with either detransitivizing or intransitivizing morphology – in each case, the particular strategy employed is comparatively unmarked. Morphological TM is simply an instance of allomorphy, conditioned by some component of transitivity in the verbal projection.

This concludes our second case study. I have shown that TM is the co-variance of two or more elements with respect to some transitivity-related property. Even though each sub-type involves some kind of ad hoc component constituting the sufficient condition for matching, these components interact with clear grammatical principles – semantic types that must match, agreement and allomorphy – to give us a fully derivational account of TM and its sub-types.

²¹⁴I leave open the possibility that syntactic and morphological TM may reduce to a single analysis.

Part III

Non-finite clauses in Mayan

Chapter 11

The restriction against non-finite transitivity

11.1 Introduction

Mayan languages often restrict transitivity in non-finite contexts. While some languages in the family readily allow the expression of transitive VPs in non-finite contexts, such as Ch'ol (1), most do not. In Popti', for example, transitive verbs can only enter into non-finite contexts if they undergo some valence-decreasing operation like anti-passivization (38):

(1) **Ch'ol**

K-om [j-kãñ-ety]
A.1-want A.1-know-B.2

'I want to know you' (Vázquez Álvarez 2011: 99)

(2) **Popti'**

Ch-ach to [col-wa-l y-iñ naj]
ASP-B.2SG go help-ANTIP-NMLZ A.3SG-RN.OF 3SG

'You are going (there) to help him' (Craig 1979: 5)

I will call this the *Restriction Against Non-finite Transitivity*, or *RANT* for short. The RANT is defined in (3):

(3) ***Restriction against non-finite transitivity (RANT)***

A language exhibits the RANT if and only if fully transitive VPs cannot occur in non-finite contexts. A VP is fully transitive if it consists of (i) a transitive verb stem and (ii) a structural, full DP object.

So Ch'ol lacks the RANT because the non-finite clause in (1) consists of a transitive

verb stem and a structurally projected, full DP object (which I take to be *pro*). Popti', by contrast, does have the RANT because the non-finite clause in (38) comprises a verb stem that has been detransitivized – here via the anti-passive – and an oblique object.

We will see that non-finite VPs in languages with the RANT may also involve transitive verb stems but oblique objects, or vice versa; in other words, a VP must lack both a transitive verb stem and a structural DP object in order to not count as fully transitive.

This chapter sets the stage for investigation of the RANT. Section 11.2 outlines Mayan morpho-syntax, and Section 11.3 presents the basic properties of Mayan non-finite clauses. The RANT is taken up in Section 11.4, along with the various strategies used in languages in order to work around it.

11.2 Mayan morpho-syntax

This section lays out Mayan morpho-syntax, based mainly on Coon (2016) and Aissen et al. (2017). All Mayan languages are characterized by agglutination, head-marking and ergative-absolutive alignment, and most show verb-initial word order in discourse-neutral contexts (England 1991, Aissen 1992). In the Mayanist literature, ergative morphology – which also marks possession – is known as *Set A*, and absolutive as *Set B*, a convention which I will adopt here.

The verbal root is typically preceded by one of a series of TAM markers. While *Set A* precedes the root throughout the Mayan family, the location of *Set B* varies depending on the language: in so-called HIGH-ABS languages, *Set B* occurs pre-verbally, whereas in LOW-ABS languages *Set B* is marked post-verbally at the end of the verbal complex (Bricker 1977, Tada 1993). The root may be followed by one or more voice suffixes, and – under the right conditions – also by a status suffix, which encodes whether a verb is morpho-syntactically intransitive or transitive, and at times also whether its transitivity is a root property, or the result of derivation. A simplified general template of the Mayan verbal complex is given in (4):²¹⁵

(4) *The Mayan verbal complex:*

$$\text{TAM} - \underbrace{\{\text{SET B}\}}_{\text{HIGH-ABS}} - \underbrace{\text{SET A}}_{\text{ERG}} - \boxed{\text{ROOT}} - \text{VOICE} - \text{STATUS.SUFFIX} - \underbrace{\{\text{SET B}\}}_{\text{LOW-ABS}}$$

²¹⁵See Kaufman & Norman (1984), Grinevald & Peake (2012) and Coon (2016).

The following examples illustrate these components:

(5) **Q'anjob'al**

X-in ha-mitx'-a'
ASP-B.1SG A.2SG-catch-STAT.TRANS

'You caught me' (Mateo Toledo 2008: 49)

(6) **Tz'utujil**²¹⁶

X-oq-kee-ch'ey
ASP-B.1PL-A.3PL-hit

'They hit us' (Dayley 1985: 104)

(7) **Tsotsil**

Ch-a-mil-at
ASP-B.2-kill-PASS

'You're going to be killed' (Aissen 1987: 62)

(8) **Ch'ol**

Tyi k-wäy-is-añ-ety
ASP A.1-sleep-CAUS-STAT.DTRANS-B.2

'I made you sleep' (Coon 2016: 515)

The basic order of the components in (4) is fairly consistent across the family. Next to the locus of Set B, languages vary in the inventory and distribution of status suffixes, as well as the location of morpho-phonological boundaries within the verbal complex. Note that the predicate may also be more complex, additionally comprising e.g. directional suffixes.

Core arguments are cross-referenced on the verb stem by Sets A and B, and they are often *pro*-dropped. The following transitive examples show the expression of two lexical and two pronominal arguments:

(9) **Tz'utujil de Santiago**

a. Ja siaw x-Ø-u-k'ay-e-ej jaw ak'
the cat COMPL-B.3SG-A.3SG-bite-V-STAT.DTRANS the chicken
'The cat bit the chicken'

b. Atet x-in-a-ch'ey nen
2SG COMPL-B.1SG-A.2SG-hit 1SG
'You hit me' (own notes, RP)

Verbal predicates generally appear with a pre-verbal TAM marker, whereas non-

²¹⁶The Tz'utujil data from Dayley's (1985) grammar are based primarily on the San Juan la Laguna variety, while the variety spoken by my consultant comes from Santiago de Atitlán.

verbal predicates obligatorily lack such a marker:²¹⁷

(10) **Q'anjob'al**

a. *Verbal predicate*

Max hach way-i
ASP B.2SG sleep-STAT.INTR
'You slept'

b. *Adjectival predicate*

Yel wal yob' hach
INTS INTS bad B.2SG
'You are really mean'

(Mateo Toledo 2011)

The homophony of ergative and possessive morphology is illustrated by the following sentence from Tz'utujil, where the same Set A marker occurs once on the verb to index the transitive subject, and once again on the object to index the possessor:

(11) **Tz'utujil de Santiago**

X-Ø-a-b'an a-chooch
COMPL-B.3SG-A.2SG-make A.2SG-house

'You built your house'

(own notes, RP)

Possessive Set A morphology is also found on so-called *relational nouns*, i.e. obligatorily possessed nouns which serve to introduce oblique elements and denote grammatical or thematic relations (Aissen 1987, Larsen 1988). Relational nouns case-mark DP complements which are formally expressed as grammatical possessors, and hence indexed via Set A on the relational noun. We can think of relational nouns as φ -marked adpositions, and indeed they generally correspond functionally to adpositions in other languages.²¹⁸ In the following example from K'ichee', the relational noun *-uk'* 'with' reflects the φ -features of its DP complement 'the women':

(12) **K'ichee'**

k-uk' ri ixoq-iib'
A.3PL-with the woman-PL

'with the women'

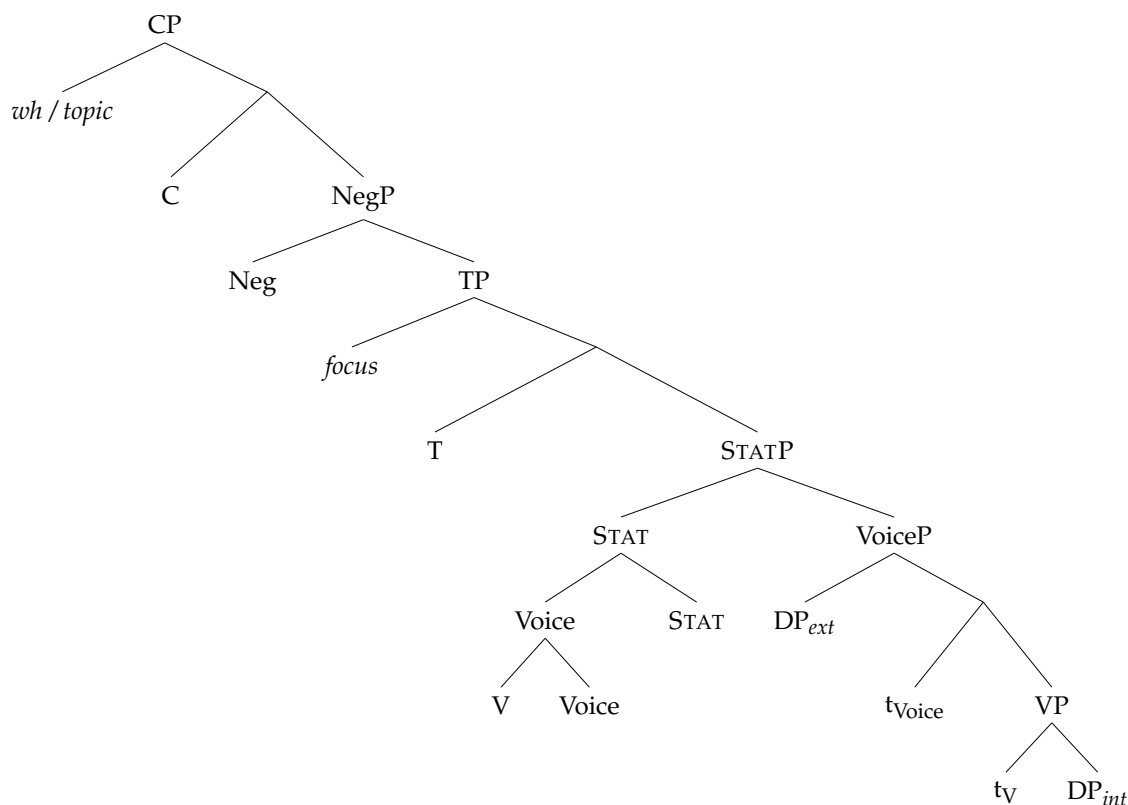
(Larsen 1988: 128)

²¹⁷In some cases, including (10a), the TAM marker is actually optional, but even then (10a) can only have a single TAM interpretation, in this case perfective / completive. By contrast, (10b) is open to more than one TAM interpretation.

²¹⁸Mayan languages may employ adpositions in conjunction with relational nouns. K'ichee', for example, has the prepositions *chi* and *pa*, which express generalized locative meanings and may combine with a relational noun to yield more specific meanings (Can Pixabaj 2009: 23).

11.2.1 Clause structure and the Mayan absolutive parameter

Following Aissen (1992), Aissen (1996) and much subsequent work, I adopt a tripartite clause structure for Mayan with three main levels of verbal projection: VP/VoiceP, TP and CP. Aspect marking is located in T, such that the presence of aspect marking in a clause entails that the structure projects at least up to TP, and for our purposes the converse is also true, i.e. the absence of aspect marking means that the structure does not project up to TP.²¹⁹²²⁰ A projection hosting the status suffix may be sandwiched between VoiceP and TP, and depending on the language and structural context the status suffix may be phonologically overt or null; see e.g. Henderson (2012). Negation is above TP but below CP, and there are pre-verbal operator positions for *wh*-elements / topics and foci, corresponding to Spec,TP and Spec,CP, respectively (in Mayan, arguments appear obligatorily pre-verbally for topic, focus, *wh*-questions and relativization). This structure is schematized in Tree 11.1:²²¹



Tree 11.1: Mayan clause structure (Aissen 1992, 1996)

Following Clemens & Coon (2018), I take the verb stem to be formed by head

²¹⁹Note that Mayan languages actually lack tense, and are instead aspect-based. For this reason, some generative works label the projection that is taken to host aspect marking *IP* rather than *TP*. For consistency with tradition, I remain with the latter over the former.

²²⁰The absence of a visible aspect marker does not necessarily mean there is no aspectual specification in the underlying structure. In some languages, certain verbal aspects are marked by the absence of an overt morpheme, a fact which is not to be confused with non-verbal predicates, which obligatorily lack aspect marking (Coon 2016: 530). See footnote 217.

²²¹Another feature of Aissen's clausal syntax is that the specifiers of lexical projections are to the right, while functional projections have left-branching specifiers. Here, I have abstracted away from this property since it is not crucial to our purposes.

movement of the root up through verbal functional projections related to transitivity and argument structure, including the projection hosting status suffixes. This head movement will be left implicit in derivations throughout, unless when necessary for argumentation. Finally, I follow Coon (2019) in assuming that Mayan languages are voice-bundling languages, where *v* and Voice are bundled into a single fused head (Harley 2017), represented here as *Voice*.²²²

Mayan languages fall into two broad types depending on the locus of Set B, i.e. absolutive morphology. On the one hand, there are HIGH-ABS languages, where Set B occurs pre-verbally (13); on the other, there are LOW-ABS languages, which mark Set B post-verbally. Tz'utujil is an instance of a HIGH-ABS language (13), Ch'ol an instance of a LOW-ABS language (14):

(13) **Tz'utujil de Santiago:** HIGH-ABS

X-at-nu-q'et-eej
COMPL-B.2SG-A.1SG-hug-STAT.DTRANS

'I hugged you' (own notes, RP)

(14) **Ch'ol:** LOW-ABS

Tyi k-mek'-e-yety
PRFV A.1-hug-STAT.TRANS-B.2

'I hugged you' (Coon 2010: 33)

Regarding this variation, I follow Coon et al. (2014) *et seq.* in assuming that this difference in the linear occurrence of Set B stems from an underlying syntactic difference: while transitive subjects are uniformly assigned inherent ergative by transitive Voice, variation in how transitive objects (absolutives) are licensed boils down to distinct inventories of functional verbal heads. The idea is that unlike nominative, accusative and ergative, which can straightforwardly be characterized as cases that are invariably assigned by the functional heads T and transitive Voice, respectively, absolutive can come about in at least two different ways (Legate 2008).

In what Legate (2008) refers to as ABS=NOM languages, both transitive objects and intransitive subjects receive what is actually nominative case from T. In so-called ABS=DEF languages, transitive objects and intransitive subjects are licensed by distinct functional heads: transitive objects are licensed by Voice, thus receiving accusative, while intransitive subjects are licensed by T, thus receiving nominative. Following Coon et al. (2014), I take HIGH-ABS languages to correspond to Legate's ABS=NOM languages, and LOW-ABS languages to her ABS=DEF languages. The differences

²²²See Harley (2017: 5) for diagnostics specific to Ch'ol, showing that accusative assignment, external argument introduction and verbalization are all accomplished by a single head.

between HIGH-ABS and LOW-ABS languages are schematized in Table 11.1:^{223,224}

Absolutive parameter	Intransitive clause	Transitive clause
HIGH-ABS		
LOW-ABS		

Table 11.1: HIGH-ABS vs LOW-ABS languages

Following Coon et al. (2014), I further assume that (i) Set A morphemes are the spell-out of a low, in-situ agreement relationship between the external argument and Voice; but (ii) that Set B morphemes do not simply spell-out an analogous agreement relationship between the absolutive argument and T. Rather, Set B morphemes are the result of clitic doubling which arises due to an Agree relationship between the absolutive argument and the licensing head.²²⁵ In finite contexts, the head which licenses absolutive arguments is T/Voice; accordingly, after T/Voice licenses the

²²³Note that Table 11.1 abstracts away from the difference between unergative and unaccusative predicates. The HIGH vs LOW-ABS parameter has no direct bearing on this distinction.

²²⁴Note that even though Tseltal and Tsotsil are basically also LOW-ABS languages, a subset of constructions appears to show a HIGH-ABS profile, thus falling into murky waters in terms of the HIGH-ABS vs LOW-ABS parameter. See Aissen (1987) on Tsotsil and Polian (2013) on Tseltal.

²²⁵One major piece of evidence supporting the view that Set B morphemes are clitics comes from the fact that throughout Mayan, absolutive morphemes appear to be reduced versions of full emphatic pronouns (Coon et al. 2014: 11); conversely, the fact that ergative morphemes lack this property supports a treatment of Set A morphemes as bona fide agreement markers. Another piece of evidence comes from morpho-phonologically conditioned placement: in e.g. Q'anjobal, Set B morphemes attach to the sentence-initial TAM marker in clauses with overt aspect marking, but appear low when there is no such marking (Coon et al. 2021: 36), (Mateo Toledo 2008: 54,69). On the clitic status of Set B morphemes, see also Woolford (2000) for Jakalteq and Popti', Mateo Toledo (2008) and Grinevald & Peake (2012) for Q'anjob'al, Coon (2010) and Coon (2013) for Ch'ol, Preminger (2014) for Kaqchikel and Coon & Carolan (2017) for Chuj. For historical evidence to the same effect, see Kaufman (1990) and Robertson (1992). For general discussion and diagnostics for telling clitic doubling from agreement,

absolutive argument proper, it triggers encliticization of the Set B clitic double to the TAM particle (in T) or the verb stem (in Voice).

11.2.2 Transitivity and voice in Mayan

Based on stem-forming morphology, Mayan languages generally characterize three main classes of verbal roots: intransitive, transitive and positional. Intransitives and transitives are distinguished by whether they are (natively) able to take one (Set B) or two person markers (Set A + Set B); positional roots refer to configurational and physical properties, and they enter verbal or adjectival frames depending on the stem-forming suffix they combine with. These three main classes are illustrated by the following examples from Ch'ol:

(15) **Ch'ol**

a. *Intransitive*

tyi wäy-i-x-oñ
 PFV sleep-STAT.INTR-EP-B.1
 'I slept' (Coon 2013)

b. *Transitive*

tyi k-mek'-e-y-ety
 PFV A.1-hug-STAT.TRANS-EP-B.2
 'I hugged you' (Coon 2013)

c. *Positional*

Koty-ol jiñi wakax tyi cholel
 standing.on.4.legs-POS DET cow PREP field
 'the cow is standing in the field' (Coon 2016: 537)

Positional roots will not play any role in this work. Among transitives, we can further distinguish root from derived transitives (e.g. Dayley (1985)). Root transitives are morphologically simple CVC roots, while derived transitives are morphologically complex and may or may not contain overt derivational morphology. This distinction is generally reflected in the choice of different voice allomorphs, and also in the kind of status suffix that attaches to a verb stem.

In (16a) from Q'anjob'al, a root transitive bears the corresponding transitive status suffix; (16b) shows a transitive verb that has been derived from a noun by the verbalizer *-le*, and the transitive in (16c) is a causative verb derived from an intransitive verb. Accordingly, the derivational history of (16b) and (16c) is reflected in their containing a derived transitive suffix:

see Anagnostopoulou (2003), Harizanov (2014) and Kramer (2014). Ixil is a potential exception to this generalization, where absolutive markers are fully identical to full pronominal forms and behave as separate words (Ayres 1991: 134).

1988). Similarly, next to demoting anti-passives like (17b), which demote the Patient to an optional oblique DP, we find anti-passives which require the Patient to appear as bare NPs, and also those which prohibit expression of the Patient altogether.²²⁷

Transitivity plays a significant role in Mayan overall, as reflected in how it is tracked by the grammar in various domains. First and most obviously, morpho-syntactic transitivity is overtly indicated on the verb stem by the head-marking of grammatical relations in ergative-absolutive alignment, which distinguishes transitive and intransitive subjects.

Second, Mayan languages make use of status suffixes to indicate various kinds of transitivity-related information. By themselves semantically vacuous given that they may be absent under certain syntactic and prosodic conditions, these suffixes encode whether a verb is morpho-syntactically intransitive or transitive, and also whether the verb is a root or derived transitive. Consider the following triplet from K'ichee':²²⁸

(18) **K'ichee'**: *Status suffixes*

a. *Root intransitive*

X-in-war-**ik**
COMPL-B.1SG-sleep-STAT.INTR
'I slept'

b. *Root transitive*

X-Ø-in-chap-**o**
COMPL-B.3SG-A.1SG-grab-STAT.TRANS
'I grabbed it' (Henderson 2012: 744)

c. *Derived transitive*

X-Ø-u-kam-isa-**j** ri tz'i' l a
COMPL-B.3SG-A.3SG-die-CAUS-STAT.DTRANS DEF dog DET MASC
Xwan
J.
'John killed the dog' (Gluckman 2016: 44)

See Henderson (2012) for a recent analysis of status suffixes in K'ichee'.

Third, while languages vary in whether or not they allow or *pro*-drop (most do, including subjects, objects and possessors), they generally show rigid transitivity, meaning that any omission or acquisition of an argument must be accompanied

²²⁷In addition, some languages have what Ayres (1983) calls *verbal indexing*. On the one hand, verbal indexing constructions pattern like applicatives in promoting peripheral arguments such as Instruments to core status; on the other, they seem similar to Philippine-type voice systems found in Austronesian in having a rearranging effect on argument structure, privileging a single argument to subject-like function while leaving unaffected all other grammatical relations. See Ayres (1983) on Ixil, Dayley (1981) on Tz'utujil and Henderson (2007) on Kaqchikel.

²²⁸Semantic and morpho-syntactic transitivity need not coincide. As shown by the Chuj example in (35), agent focus forms require intransitive status suffixes due to being morpho-syntactically intransitive despite being semantically transitive.

Finally, the area in which Mayan grammar most famously brings out transitivity are agent focus contexts (Kaufman 1990, Quesada 1997, Stiebels 2006a, Pascual 2007, Aissen 2017b). As is well known, most Mayan languages restrict $\bar{A}$ -extraction of ergative subjects from regular transitive verb forms. In order for ergatives to be focused, questioned or relativized, a special verb form must be used that carries an agent focus marker. (19) illustrates the basic paradigm on the basis of *wh*-questions in Q'eqchi:

a. *Transitive Baseline*

X-at-in-sak'
COMPL-B.2SG-A.1SG-hit
'I hit you'

Ani x-sak'-o-k aw-e?
who COMPL-hit-**AF**-NFUT.INTR A.2SG-RN.OF
'Who hit you?'

Ani x-a-sak’?
 who COMPL-A.2SG-hit
 ‘Who did you hit?’

(Dayley 1981: 27ff)

I follow Aissen (2017b) in labeling this restriction the *Ergative Extraction Constraint*, itself part of the more general phenomenon of *syntactic ergativity*, whereby certain grammatical processes are sensitive to the distinction between ergatives and absolutes.²²⁹ For discussions of and approaches to agent focus, see Smith-Stark (1978), Ordóñez (1995), Stiebels (2006a), Pascual (2007), Coon et al. (2014), Erlewine (2016), Aissen (2017b) and Coon et al. (2021); for a detailed analysis that takes into account its non-homogeneous range of contexts across Mayan, see specifically Stiebels (2006a).

11.2.3 Complement clauses in Mayan

In general terms, the choice of a particular type of complement clause depends on both the matrix predicate and the semantic relation between the matrix and the embedded predicate; while some predicates only allow one type, others may

²²⁹See (Deal 2016) for a recent overview.

allow more than one type, with potential semantic differences correlating with the choice of one over another. Overall, the distribution of complement clauses across Mayan aligns with typological patterns documented in works like Givón (2001), Noonan (2007) and Wurmbrand & Lohninger (2022), according to which the more the embedded event is semantically dependent on the matrix event, the more likely it is to be encoded as a structurally reduced, dependent clause.

On one end of the continuum, we find clauses which most strongly resemble independent root clauses, corresponding roughly to CPs and typically found under verbs of communication, propositional attitude and knowledge; on the other end lie clauses which least resemble independent root clauses, corresponding roughly to Vs, VPs or VoicePs, typically occurring as complements to desiderative and manipulative verbs. Depending on the precise semantic and syntactic properties involved, complements to modal, phasal and perception verbs may fall anywhere in between.²³⁰ A useful distinction in this regard is how sentence-like, or *s-like*, a complement clause is: a property in a particular kind of complement clause is *s-like* if it behaves exactly or mostly like it would in an independent root clause in that language, and *non-s-like* if its complement-clause behavior deviates from its root-clause behavior (Noonan 2007).

Specifically in Mayan, complement clauses fall into two broad types: *finite* and *non-finite*. Finite clauses contain verbs which inflect for aspect, and non-finite clauses contain verbs which do not; in the following, a *non-finite clause* will hence refer to a clause lacking aspect. Unless noted, the following overview of Mayan complement clauses is based on Aissen (2017a).

Among finite clauses, we can further distinguish between those headed by a complementizer, and those which lack a complementizer, with the choice of one over the other generally determined by the matrix predicate.²³¹ With respect to non-finite clauses, there exists a further subdivision made by scholars like Mateo Toledo (2013) and Aissen (2017a) into *aspectless* and *infinitival* clauses, where aspectless clauses index the φ -features of (one or more) arguments, and infinitives show no φ -marking at all.

Some languages, like Tsotsil, distinguish all four types of complement clauses: in (20a), the complement clause is fully finite by virtue of being inflected for aspect and φ -features, it is headed by a complementizer, and the embedded subject is

²³⁰These generalizations reflect non-absolute tendencies which may be overridden: if a language lacks a certain complement type, for instance, obviously no head can select it, and relatedly, a predicate that imposes temporal constraints on its complement event may allow (or require) a complement type whose syntax is comparatively unreduced (e.g. ‘want’).

²³¹Additionally, the presence vs absence of a complementizer correlates with a low vs high integration of the embedded event into the matrix event. As a rule of thumb, co-reference between a matrix argument and the embedded subject is possible only in the absence of a complementizer, and conversely the presence of a complementizer generally implies disjoint reference between a matrix argument and the embedded subject. Some complement-taking predicates allow complement clauses with and without complementizers, in which case the choice usually reflects different senses of the predicate and its relation to the complement.

referentially disjoint from the matrix subject; (20b) is just like (20a), except that it lacks a complementizer and that the matrix and embedded subjects are co-referent; the aspectless non-finite clause in (20c) lacks aspect marking, but φ -marks the subject via Set A (the object, being 3SG, is not overtly marked via Set B); finally, infinitives like (20d) lack any φ -marking whatsoever:

(20) **Tsotsil**

a. *Finite + complementizer*

I-y-il ti s-me' un-e [ti muk'=bu ta
COMPL-A.3-see DET A.3-mother PAR-ENC COMP never INCOMPL
s-sa' y-ajnil ti s-krem un-e]
A.3-see A.3-wife DET A.3-son PAR-ENC
'His mother saw that her son was never going to find a wife'

b. *Simple finite*

Mu s-k'an [ch-k-uch'be-tik] li
NEG A.3-want INCOMPL-A.1-drink-APPL-1PL.INCL DET
yajval=balamil-e
lord=earth-ENC
'The Earth lord didn't want us to drink it'

c. *Aspectless*

K'u=yu'un mu x-av-ak' [k-uch' vo']-e?
why NEG NT-A.2-let A.1-drink water-ENC
'Why don't you let me drink water?'

d. *Infinitive*

Vo'on-e mu j-k'an [mil-el]
1SG-ENC NEG A.1-want kill-INF.PASS
'Me, I didn't want to be killed'

(Laughlin 1977: 55, 155, 45, 139)

The aspectless vs infinitive sub-division, which is not found across the family, will not figure prominently here. Other Mayan languages may distinguish less than four sub-types, but all show a minimal distinction between finite and non-finite clauses. Finite complement clauses with a complementizer show the strongest similarity to independent, finite root clauses: next to marking aspect and φ -features as in root clauses, they admit sentential negation (21):

(21) **Tsel'tal:** *Finite complement with complementizer + negation*

Ya j-k'an [te ma x-ch'ay ta aw-ot'an-ik]
INCOMPL A.1-want COMP NEG INCOMPL-lose PREP A.2-heart-PL

'I want you all not to forget'

(Polian 2013: 821)

Complement clauses headed by a complementizer also always allow interrogative

material and focus fronting (Laughlin 1977: 139). Simple finite clauses without complementizers are like their complementizer-headed counterparts, except that their aspect marking usually matches that of the matrix clause in what can be taken to be a kind of concord, and they may sometimes restrict internal negation or focus fronting.

Non-finite clauses, on the other hand, lack aspect marking altogether:

(22) **Chuj**

Lan [hach=ko-chel-an-i]
 PROG B.2=A.1PL-hug-AF-STAT.INTR

‘We’re hugging you’ (Coon & Carolan 2017: 2)

Moreover, they allow neither internal negation (23) nor focus fronting (24):²³²

(23) **Jakaltek**

*K’ul [mach/mat j-uk’-i]
 good NEG A.1PL-drink-STAT

intended: ‘It is good not to drink’ (Craig 1977: 243)

(24) **Tseltal**

*Ma la y-ak’ [joon j-k’o-pon-at]
 NEG COMPL A.3-let FOC.B.1SG A.1-speak.to-B.2SG

intended: ‘He didn’t let me speak to you’ (Polian 2013: 823)

Contrast this with the finite complement clause in (21) above, and the following simple finite complement clause from Tseltal:

(25) **Tseltal:** *Simple finite complement + focus fronting*

Ja’ xan=ix j-k’an [jo’on xan=ix x-jajch’-on bel
 FOC more=now A.1-want FOC.1SG more=now INCOMPL-begin-B.1 DIR
 at’el]
 work

‘I just wanted to begin working myself’ (Polian 2013: 821)

²³²In order to express the intended meaning of (23), a finite clause with a complementizer must be resorted to:

(i) **Jakaltek**

K’ul [ta(to) mach ch-onh uk’-i]
 good COMP NEG INCOMPL-B.1PL drink-STAT

‘It is good that we do not drink’ (Craig 1977: 243)

Unlike finite complement clauses, where φ -marking is fully s-like, φ -marking in non-finite clauses is non-s-like because it is governed not simply by grammatical functions, but also factors like transitivity and voice, the structural size of the thematic object (NP vs DP), how this object is linked (structural vs oblique), whether or not the matrix predicate introduces its own subject, and subject obviation (i.e. whether or not the subject of the complement clause co-refers with a matrix argument).

I follow Aissen (2017a) in assuming that complement clauses map onto the tripartite clause structure presented in Section 11.2.1 as follows (cf. Tree 11.1): finite clauses with complementizers are roughly CPs, simple finite clauses without complementizers are roughly TPs, and non-finite clauses are smaller than TP, i.e. V, VP or VoiceP. The main differences between these clause types are summarized in Table 11.2:

Property	Finite		Non-finite
	Finite + COMP	Simple finite	
Aspect	✓	✓ (often concord)	*
φ -marking	s-like	mostly s-like	non-s-like
Internal negation	✓	may be restricted	*
Internal focus	✓	may be restricted	*
Size	CP	TP	$\leq$ VoiceP

Table 11.2: Complement clauses in Mayan (adapted from Aissen 2017: 277)

11.3 Non-finite clauses in Mayan

In Mayan, non-finite clauses – from here on abbreviated as *NFCs* – have a verbal internal syntax and a nominal external syntax (Norman & Campbell 1978, Larsen & Norman 1979, Robertson 1980, Bricker 1981, Dayley 1981, Law et al. 2006, Can Pixabaj 2009, Mateo Pedro 2010, Henderson 2012, Coon 2013, Mateo Toledo 2013, Can Pixabaj 2015, Can Pixabaj & Aissen 2021). In other words, they are deverbal nominalizations with mixed categorial properties. As projections from a verbal core, NFCs may contain functional material such as voice and status suffixes, and allow adverbial modification. Recall that all Mayan NFCs lack aspect, which means that their point of transition from the verbal to the nominal domain lies below TP. As projections with a nominal shell, they afford most of the properties otherwise found with regular nouns. Mayan NFCs will now be illustrated on the basis of various languages, though most (if not all) of the properties discussed apply to NFCs across Mayan in general.

11.3.1 NFCs have a verbal internal syntax

Mayan NFCs retain numerous verbal properties. First, they can manifest voice operations such as causative (26a) and passive (26b):

(26) *NFCs may contain voice*

a. **Ch'ol:** *Causative*

Mi [i-tsäñ-s-añ jiñi wakax]

PROG A.3-die-CAUS-NMLZ DET cow

'He kills the cow'

(Coon 2010: 123)

b. **Q'anjobal:** *Passive*

Max-Ø xew [ha-maq'-lay y-uj]

COMPL-B.3SG finish A.2SG-hit-PASS A.3SG-RN

'You finished being hit by him/her'

(Mateo Pedro 2010: 53)

Second, in some languages, NFCs retain the status suffixes found in finite clauses. This is the case in Jakaltek:²³³

(27) **Jakaltek:** *NFCs may contain status suffix*

K'ul [ta(to) mach ch-onh uk'-i]

good COMP neg INCOMPL-B.1PL drink-STAT.INTR

'It is good that we do not drink'

(Craig 1977: 243)

Third, NFCs behave like verbs with respect to adverbial modification. In (28) from Kaqchikel, the NFC admits the adverb *aninäq* 'quickly':²³⁴

²³³(i) shows that the same status suffix *-i* found in (27) also occurs on finite intransitives:

(i) **Jakaltek:**

Xc-ach il-lax-i

COMPL-B.2SG see-PASS-STAT.INTR

'You were seen'

(Craig 1979: 3)

²³⁴(ia) shows the modification of a finite verb via *aninäq* 'quickly'; *aninäq* 'quickly' cannot modify nouns (ib):

(i) **Kaqchikel**

a. Ri a Juan **aninäq** x-Ø-u-pax-iiij

ri läq

DET CL J. **quickly** COMPL-B.3SG-A.3SG-break-STAT.DTRANS DET plate

'Juan broke the plate quickly'

b. ***aninäq** tzij

quickly word

intended: 'quick word'

(Imanishi & Mateo Pedro 2013: 57)

- (28) **Kaqchikel:** *NFCs allow adverbial modification*

[Ri choy-oj che' **aninäq**] k'ayew
DET cut-NMLZ tree **quickly** difficult

'Cutting trees quickly is difficult'

(Imanishi 2020: 11)

11.3.2 NFCs have a nominal external syntax

In this section, we will see that non-finite clauses across Mayan have a nominal external syntax (Norman & Campbell 1978, Larsen & Norman 1979, Robertson 1980, Bricker 1981, Dayley 1981, Law et al. 2006, Can Pixabaj 2009, Mateo Pedro 2010, Henderson 2012, Coon 2013, Mateo Toledo 2013, Can Pixabaj 2015, Can Pixabaj & Aissen 2021).

First, NFCs are free to enter any argument position that is open to regular nominals; next to objects, NFCs thus also occur as subjects (29a), on a par with regular nominals (29b):²³⁵

- (29) **Chuj**

- a. *NFC subject*

Man te wach'-ok laj [ko-xik-an te' k'atzitz]
NEG very good-IRR NEG A.1PL-chop-AF CLF wood
'Our chopping wood isn't very good'

- b. *Regular nominal subject*

Man te wach'-ok laj [ko-kape]
NEG very good-IRR NEG A.1PL-coffee
'Our coffee isn't very good'

(Coon & Carolan 2017: 15)

Second, non-finite clauses are (sometimes) able to take determiners (30a), and – when functioning as complements to intransitive matrix predicates – are linked as obliques to the same relational nouns / adpositions (30b) which also take simple nominal

²³⁵Note that the subject of the NFC in (29a) must be interpreted as a thematic agent, as made clearer by the following example, based on the exact same NFC:

- (i) **Chuj**

Lan [ko-xik-an te' k'atzitz]
PROG A.1PL-chop-AF CLF wood

'We're chopping wood'

(Coon & Carolan 2017: 15)

complements (30c):^{236,237}

(30) **Kaqchikel**

a. *NFC headed by determiner*

X-Ø-qa-chäp [(ri) choy-ik che']

COMPL-B.3SG-A.1PL-start DET cut-NMLZ tree

'We started to cut trees'

(Imanishi & Mateo Pedro 2013: 56)

b. *NFC complement to preposition*

Y-in-ajin [che wa'-in]

INCOMPL-B.1SG-PROG PREP eat-NMLZ

'I am eating'

(Imanishi & Mateo Pedro 2013: 56)

c. *Simple nominal complement to preposition*

Y-in-ajin che jun b'ix

INCOMPL-B.1SG-PROG PREP INDEF song

'I am singing a song'

(Imanishi 2020: 24)

²³⁶Imanishi & Mateo Pedro (2013) report no difference in meaning between the presence vs absence of the determiner in (30a). But NFCs are not always free to combine with determiners across the board. Coon (2010: 117) reports that determiners degrade certain kinds of NFCs, Can Pixabaj (2015: 115) notes a complication with respect to determiners in connection with NFCs containing oblique agent *by*-phrases, and Mateo Toledo (2013: 274) shows them to be possible with some, but not all kinds of NFCs; similarly, my Tz'utujil consultant would not always accept them. So far, I have been unable to make out a clear picture of the general conditions governing NFCs headed by determiners, which may turn out to vary from language to language in the details (see Coon (2010: 119)) for a proposal specific to Ch'ol).

²³⁷Imanishi (2020: 8) notes that finite clauses cannot be headed by determiners in Kaqchikel (unless they function as complementizers or relativizers).

Third, NFCs can be modified by quantifiers and adjectives.²³⁸

(31) **Kaqchikel**

a. *NFC modified by quantifier*

[K'iy ru-choy-ik che'] y-a-r-kosir-saj
lots.of A.3SG-cut-NMLZ tree INCOMPL-B.2SG-A.3SG-tire-CAUS
'Lots of cutting trees makes you tired'

b. *NFC modified by adjective*

Ri y-a-tiker n-Ø-a-b'än peligroso
DET INCOMPL-B.2SG-can INCOMPL-B.3SG-A.2SG-do dangerous
[choy-oj che'] k'ayew
cut-NMLZ tree difficult
'That you can do dangerous cutting trees is difficult' (Imanishi 2020: 9)

Fourth, NFCs condition case and agreement just like regular nominals. As elements without ϕ -features, they always condition default 3SG agreement. While 3SG Set B is virtually always zero-marked, the presence of 3SG agreement is clearly brought out via the non-null Set A marker, such as when a NFC itself recurs within a NFC; in (32), the more embedded NFC occurs as the grammatical possessor of the higher NFC, a relationship that is marked via 3SG Set A morphology on the higher NFC:

(32) **Ch'ol:** *NFC triggers 3SG agreement*

Choñkol [y_i-ujty-el [k-wuts' jiñi pisil]_i]
PROG A.3-finish-NMLZ A.1-wash DET clothes
'I'm finishing washing the clothes' (Coon 2010: 117)

Because NFCs are themselves case-marked, they condition Set A agreement with respect to matrix subjects when functioning as complements to transitive predicates:

(33) **Tz'utujil**

X-Ø-qaa-maj [wa'-iim]
COMPL-B.3SG-A.1PL-begin eat-INF
'We began to eat' (Dayley 1985: 394)

²³⁸That the quantifier *k'iy* and adjectives like *peligroso* modify nouns elsewhere is shown by the following:

(i) **Kaqchikel**

- a. K'iy che' x-e-tzaq
lots.of tree COMPL-B.3PL-fall
'Lots of trees fell'
b. peligroso samaj
dangerous work
'dangerous work'

(Imanishi 2020: 9)

Accordingly, when the matrix predicate is intransitive, as in (30b) above, the NFC must be introduced by a preposition because the single argument slot of the matrix predicate has been saturated by the matrix subject.

Fifth, in many languages of the family, NFCs give rise to so-called *extended ergativity*, a form of split ergativity wherein Set A morphology has been extended to also mark subjects of intransitive predicates. This is schematized in Table 11.3:^{239,240}

	Intransitive	Transitive
Subject	A	A
Object	-	B

Table 11.3: Extended ergativity

Consider the following NFCs from Mopan, in which Set A occurs on both intransitive and transitive verbs:

(34) **Mopan**

- a. *Intransitive NFC: Set A indexes intransitive patient subject*

Tan [a-lub'-ul]
 PROG A.2SG-fall-NMLZ
 'You are falling'

- b. *Transitive NFC: Set A indexes transitive agent subject*

Tan [in-lox-ik-ech]
 PROG A.1SG-hit-NMLZ-B.2SG
 'I am hitting you'

(Larsen 1990)

While its precise distribution depends on the language and construction – notice, for instance, that all individual arguments in (34) are expressed downstairs within the NFC – Set A in NFCs is not restricted to a unique thematic role. In (34a), Set A indexes an intransitive patient subject, while in (34b), it indexes a transitive agent subject; in (35) from Jakaltek, the argument indexed via Set A is an intransitive agent subject:

(35) **Jakaltek:** *Intransitive NFC: Set A indexes intransitive agent subject*

K'ul [j-uk'-i]
 good A.1PL-drink-STAT

'It is good to drink'

(Craig 1977: 243)

²³⁹This term goes back to Dixon (1979).

²⁴⁰Because Set B now only marks transitive objects, and intransitive subjects and transitive objects thus no longer pattern alike, this kind of alignment has been likened to a nominative-accusative pattern. Unlike in a nominative-accusative pattern, however, markedness is reversed because all subjects are formally more marked than thematic objects.

As argued in Coon (2010) *et seq.*, the fact that both intransitive and transitive subjects are indexed via Set A rather than Set B in NFCs falls out from the following: first, the fact that ergative and genitive are systematically homophonous across Mayan; second, NFCs are nominalizations; and third, intransitive and transitive subjects are expressed as grammatical possessors of these nominalizations.

Finally, at least in Tz'utujil de Santiago, NFCs in object position pattern like regular transitive objects in conditioning agent focus upon $\bar{A}$ -extraction of the transitive subject of the matrix verb.^{241,242,243}

(36) **Tz'utujil de Santiago**

- a. A Tru' x-Ø-u-moj [r-mich-x-ik jaw
CLF P. COMPL-B.3SG-A.3SG-start A.3SG-pluck-PASS-NMLZ the
ak']
chicken
'Pedro started plucking the chicken'
- b. Naqi —i x-e-mj-ow-a [r-mich-x-ik
WH COMPL-B.3SG-start-AF-STAT.INTR A.3SG-pluck-PASS-NMLZ
jaw ak']?
the chicken
'Who started plucking the chicken?' (own notes, RP)

Note that nominalizing suffixes may be phonologically overt, as in (30), (34) and (30), or null, as in (29) and (35).²⁴⁴

In sum, NFCs have a nominal external syntax. Numerous scholars have therefore proposed nominalization analyses for these clauses, including Bricker (1981), Mateo Pedro (2010), Coon (2010), Coon (2013), Mateo Toledo (2013), Coon et al. (2014) and Can Pixabaj (2015). Throughout, I will take as a given that Mayan NFCs are deverbal nominalizations. The NFCs we are mainly interested in here pattern largely

²⁴¹Note that by "conditioning", I do not mean that agent focus is triggered directly by objects (be they NFCs or regular nominals). As we will see, agent focus arises specifically due to properties of transitive subjects, not objects. Objects nevertheless factor into triggering agent focus indirectly in that transitive subjects entail objects, to the effect that some work – e.g. Coon et al. (2014), Coon et al. (2021) – has argued that transitive subjects require agent focus upon extraction because of their syntactic disposition towards objects.

²⁴²In Tz'utujil de Santiago, the 3SG Set B marker is generally null, though under certain phonological conditions, like in (36b), it surfaces as *-e-*. Alternatively, the appearance of what appears to be a non-null 3SG Set B marker in (36b) does not instantiate phonologically conditioned allomorphy, but rather a phonotactically conditioned epenthetic vowel.

²⁴³Agent Focus does not appear on the matrix verb when the complement clause is finite:

(i) **Tz'utujil**

- Naqi —i x-Ø-u-moj [ec_i n-Ø-u-mich' jaw ak']?
WH COMPL-B.3SG-A.3SG-start INCOMPL-B.3SG-A.3SG-pluck the chicken
'Who started plucking the chicken?' (own notes, RP)

²⁴⁴The overtness of the nominalizer may also depend on the construction and / or verb class, as in (32) from Ch'ol (Coon 2010).

like *Complex Event Nominals* in the sense of (Grimshaw 1990), which have a fixed internal argument structure (i.e. arguments within the NFC have fixed thematic interpretations).

11.3.3 Distribution of NFCs

As mentioned, Mayan NFCs are typically found in contexts where they stand in a relation of relatively high semantic dependence on the matrix clause. These include complements to thematic predicates such as desiderative, phasal and manipulative verbs (37); athematic phasal predicates (38); and purpose clauses under motion verbs (39):²⁴⁵

(37) *Complements to thematic predicates*

a. **Ch'ol:** *Desiderative verb*

K-om [j-kãñ-ety]
A.1-want A.1-know-B.2

'I want to know you'

(Vázquez Álvarez 2011: 99)

b. **Tz'utujil de San Juan:** *Phasal verb*

X-Ø-qaa-maj [wa'-iim]
COMPL-B.3SG-A.1PL-begin eat-INF

'We began to eat'

(Dayley 1985)

c. **Tsotsil:** *Manipulative verb*

L-i-y-ak' [ak'otaj-ik-on]
COMPL-B.1SG-A.3SG-CAUS dance-STAT-B.1SG

'He let me dance'

(Aissen 1987: 215)

(38) **Chuj:** *Athematic phasal predicate*

Lan [hach=ko-chel-an-i]
PROG B.2=A.1PL-hug-AF-STAT.INTR

'We're hugging you'

(Coon & Carolan 2017: 2)

(39) **Popti':** *Motion cum purpose clause*

Ch-ach to [col-wa-l y-iñ naj]
ASP-B.2SG go help-ANTIP-NMLZ A.3SG-RN.OF 3SG

'You are going (there) to help him'

(Craig 1979: 5)

²⁴⁵Note that by *thematic predicate*, I mean predicates that introduce their own external argument in addition to taking an internal argument as complement; *athematic predicates*, then, lack this property.

In addition, NFCs are also found under perception verbs (40) and non-verbal predicates such as evaluative adjectives (41):

(40) **Popti'**: *Perception verb*

X-Ø-(y)-il ix [hin ha-mak-ni]
COMPL-B.3SG-A.3SG-see CLF B.1SG A.2SG-hit-AF

'She saw you hit me' (Craig 1977: 237)

(41) **Chuj**: *Evaluative adjective*

Wach' [hin-chel-an-i]
good A.1SG-hug-AF-STAT.INTR

'I hug well / My hugging is good' (own notes: MT)

Across the family, there are robust tendencies as to which kinds of predicates can or must take NFC complements, my impression being that the most common of these are phasal, desiderative and evaluative predicates. Still, there exists considerable variation in this regard, both across languages with respect to predicates with similar meanings and also between dialects of the same language.²⁴⁶

In terms of grammatical function, NFCs occur as objects of transitive predicates (e.g. (37a)), as well as subjects of both intransitive (e.g. (41)) and transitive predicates.²⁴⁷ In terms of linking, they may occur structurally (e.g. (37a)) or as obliques (e.g. (30b)), depending on whether the matrix predicate is intransitive or transitive.

Based on generalizations in the literature and my elicitations, the specific embedding context of a given NFC is independent of its internal properties, i.e. the internal behavior of a given NFC does not in any way co-vary with its embedding context (subject to syntactic and semantic well-formedness). For this reason, we can establish generalizations about the morpho-syntax of Mayan NFCs without having to worry about confounding factors resulting from variation in matrix environments.²⁴⁸ One

²⁴⁶For instance, the verb 'learn' takes NFC complements in K'ichee' (Can Pixabaj 2009: 42) but not Tz'utujil de Santiago (own notes), and while the San Juan dialect of Tz'utujil allows NFC complements under 'want', my consultant of the Santiago dialect systematically rejected them (allowing only finite complements).

²⁴⁷Consider (i), where the NFC occurs in transitive subject function:

(i) **K'ichee'**

X-at-u-yak ub'i [ri k'ayi-n tra's]
COMPL-B.2SG-A.3SG-wake.up DIR DET sell-ANTIP peach

'Selling peaches (woke you up), motivated you to wake up' (Can Pixabaj 2009: 43)

²⁴⁸With my Kaqchikel, Tz'utujil, Chuj and Q'anjob'al consultants, I have been unable to detect any NFC-internal differences as a function of the choice of the matrix predicate. Works like Verhoeven (2007), Can Pixabaj (2009), Coon (2010), Can Pixabaj (2015) and Aissen (2017a) similarly lack reports on any such differences. Additionally, Verhoeven (2007: 291ff) reports for Yucatec that the mode of syntactic integration – i.e. complementation vs adjunction – does not affect the internal properties of

of the only exceptions to this is constituted by binding scenarios, where we will see the presence of a matrix binder become relevant.

11.4 The restriction against non-finite transitivity

We now come back to the RANT, repeated here:

(42) *Restriction against non-finite transitivity (RANT)*

A language exhibits the RANT if and only if fully transitive VPs cannot occur in non-finite contexts. A VP is fully transitive if it consists of (i) a transitive verb stem and (ii) a structural, full DP object.

Languages lacking the RANT allow the occurrence of transitive VPs in NFCs. Throughout (43), Sets A and B consistently index the thematic subject and object, respectively, just as in finite contexts (in (43c), 3SG Set B is zero-marked):

(43) *NFCs with transitive VPs*

a. **Ch'ol**

K-om [j-käñ-ety]
A.1-want A.1-know-B.2

'I want to know you'

(Vázquez Álvarez 2011: 99)

b. **Yucatec**

In-k'áat [inw-il-ech]
A.1SG-want A.1SG-see-B.2SG

'I want to see you'

(Bricker 1981: 96)

c. **Itzaj**

K-u-jo'm-ol [ki-b'et-ik]
ASP-A.3SG-end-STAT.INTR A.1PL-make-STAT.TRANS
kiw-uk'-ul-ej]
A.1PL-drink-NMLZ-TOP

'... after we make our drink'

(Hofling 2000: 486)

As we know, however, most Mayan languages actually restrict transitivity in NFCs, to the effect that transitive VPs in NFCs obligatorily require some kind of workaround. Workaround strategies fall into three types: those which target the verb stem, those which target the object, and those which target both the verb and the object.

11.4.1 Verb-based workarounds to the RANT

By far, the majority of workarounds to the RANT are verb-based. These manifest in the form of changes to a verb's stem transitivity via some valence-decreasing

NFCs.

operation. The resulting verb is morphologically intransitive, with the object's linking form following from the particular voice operation employed on the verb. In Popti', for instance, transitive VPs in NFCs undergo anti-passivization:

(44) **Popti'**: *Anti-passive NFC*

Ch-ach to [col-**wa**-l y-iñ naj]
 ASP-B.2SG go help-**ANTIP**-NMLZ A.3SG-RN.OF 3SG

'You are going (there) to help him' (Craig 1979: 5)

The thematic object in (44) is not linked structurally and φ -marked on the verb, but expressed as an oblique via a relational nou; in some cases, anti-passive NFCs involve bare NP objects in what is known as the *incorporation anti-passive* (Pascual 2007, Coon 2019). The fact that the object is oblique in (38) follows from anti-passivization itself, which licenses a relational noun (alternatively, the object is existentially bound). Notice anti-passive NFCs lack Set B morphology. The anti-passive strategy is also found in Kaqchikel (García Matzar & Rodríguez Guaján 1997), K'ichee' (Can Pixabaj 2015) and Sakapultek (DuBois 1981).

Next, there are RANT languages where NFCs based on transitive VPs are *pseudo-passive*, i.e. they show passive morphology while retaining an active interpretation:

(45) *Pseudo-passive NFCs*

a. **Tz'utujil**

X-Ø-qaa-maj [ki-ch'e<j>y-iik]
 COMPL-B-3SG-A.1PL-begin A.3PL-hit.**PASS**-NMLZ

'We began to hit them' (Dayley 1985: 393)

b. **Kaqchikel**

Röj y-oj-ajin [che ki-q'ete-x-ik ri ak'wal-a]
 we INCOMPL-B.1PL-PROG PREP A.3PL-hug-**PASS**-NMLZ DET child-PL

'We are hugging the children' (Imanishi 2020: 12)

Unlike the anti-passive NFCs in (44), pseudo-passive NFCs consistently use Set A to φ -mark the thematic object, which is realized structurally; like in anti-passive NFCs, Set B marking is absent. The pseudo-passive strategy is also found in Achi (Sis Iboy 2007), K'ichee' (Can Pixabaj 2015), Sakapultek (DuBois 1981) and Sipakapense (Barrett 1999), and is especially favored within the K'ichean branch.

The astute reader may have noticed a peculiarity, namely that the matrix subject in these cases identifies the agent external argument of the embedded passive verb, even though passivization should render this position inaccessible to such identification. We will see that formally passive NFCs like (45) are in fact syntactically active and contain external arguments, a result which confirms the intuition voiced in Can Pixabaj (2015).

Third, there are languages which require transitive verbs to bear the agent focus marker in NFCs, i.e. the same verbal suffix that also appears on finite transitive verbs under $\bar{A}$ -extraction of the ergative subject:^{249,250}

(46) *Agent focus NFC*

a. **Chuj**

Lan [hach=ko-chel-**an-i**]
 PROG B.2=A.1PL-hug-AF-STAT.INTR
 ‘We’re hugging you’ (Coon & Carolan 2017: 2)

b. **Popti’**

X- $\emptyset$ -(y)-il ix [hin ha-mak-**ni**]
 COMPL-B.3SG-A.3SG-see CLF B.1SG A.2SG-hit-AF
 ‘She saw you hit me’ (Craig 1977: 237)

In agent focus NFCs, the thematic object is indexed via Set B. Among transitivity-restricted NFCs, those employing agent focus are the only ones in which Set B morphology is present; in anti-passive NFCs (44) and pseudo-passive NFCs (45), Set B is notably absent. This property will turn out to be directly connected to the presence of the agent focus marker, and also the fact that Set A indexes the thematic subject of the embedded clause. Importantly, one of the central properties that come with the agent focus marker is that semantically transitive verbs are rendered morpho-syntactically intransitive, as evinced by e.g. the presence of the intransitive status suffix (35a).²⁵¹ In this regard, then, agent focus is on a par with anti-passivization and pseudo-passivization. Next to Chuj and Popti’, the agent focus strategy is also found in Q’anjob’al (Mateo Toledo 2003).

²⁴⁹(i) shows the use of the agent focus marker under finite ergative extraction:

(i) *Agent Focus in finite ergative extraction*

a. **Chuj**

Mach ix-ach-chel-an-i?
 who PFV-B.2SG-hug-AF-SS
 ‘Who hugged you?’ (Coon & Royer 2021: 5)

b. **Popti’**

Mak k=ach ?il-ni?
 who ASP=B.2SG see-AF
 ‘Who saw you?’ (Dayley 1981: 38)

²⁵⁰Note that even though this subset of languages makes use of the same marker in finite transitive contexts and NFCs, the respective constructions are crucially not the same in all respects, as will be discussed. For example, while the NFC in (35a) has both Set A and Set B marking, agent focus in regular transitive contexts only ever has Set B marking, a fact which traces back to the nominal external syntax of NFCs.

²⁵¹Despite remaining semantically transitive, agent focus verb forms are always morphologically intransitive. In finite contexts, they always have only a single Set B agreement slot which agrees either consistently with the object, or with whichever argument ranks highest on a salience hierarchy (Stiebels 2006a, Coon et al. 2021). That argument which goes unagreed with on the verb is either not ϕ -marked at all, or may be ϕ -marked obliquely via a relational noun, as in Q’eqchi’ (19b).

A language may have more than one strategy to work around the RANT at its disposal, though there appears to be a tendency to opt for a single one (at least if the internal argument is to be referential and realized as a full DP). Popti', for example, has both the anti-passive (38) and the agent focus (35b), but not the pseudo-passive strategy (Craig 1977: 311ff); K'ichee' has the passive and the anti-passive, but not the agent focus strategy (Can Pixabaj 2015: 104ff); in contrast, Q'anjob'al and Chuj have only the agent focus strategy (Mateo Toledo 2013, Coon & Carolan 2017), and Tz'utujil only the pseudo-passive strategy (Dayley 1985: 394ff), just like the Patzún variety of Kaqchikel (García Matzar & Rodríguez Guaján 1997), even though other varieties additionally allow anti-passivization (García Matzar & Rodríguez Guaján 1997: 457).

11.4.2 Object-based workarounds to the RANT

Next to verb-based workarounds, we find at least one instance of a purely object-based workaround to the RANT. So far, I have only come across the following strategy in Mam, where objects in NFCs based on transitive VPs are linked via relational nouns, yet without the verb stem showing any sign of a valence reduction:

(47) **Mam:** *NFC with oblique object*

N=ch=ul aaj=x=kya xjaal tzluu' [b'inchaa-l
INCOMPL=B.3PL=DIR return=always=3PL.FOC people here do-INF
t-ee]
A.3SG-RN.PAT

'People come here to do it' (England 2013: 282)

(47) patterns like an anti-passive NFC in that the object is linked via a relational noun, on which it is also ϕ -marked, cf. (44); unlike in an anti-passive, however, the verb stem in (47) lacks any (visible) change to its transitivity. Another way in which the workaround in (47) deviates from regular anti-passivization concerns the choice of relational noun: in Mam, the relational noun that introduces Patients in finite anti-passive contexts is not *-e(e)* – which is used in (47) – but rather *-i'j* (i):

(48) **Mam:** *Finite anti-passive*

Ma Ø-tzyuu-n Cheep ky-i'j xiinaq
PROX B.3SG-grab-ANTIP J. A.3PL-RN.OF man

'José grabbed the men' (England 2013: 5)

This behavior contrasts sharply with Popti', where the relational nouns used in non-finite (49a) (=38)) and finite anti-passives (49b) are identical:

(49) **Popti'**a. *Anti-passive NFC*

Ch-ach to [col-wa-l y-iñ naj]
 ASP-B.2SG go help-ANTIP-NMLZ A.3SG-RN.OF 3SG
 'You are going (there) to help him'

b. *Finite anti-passive*

X-ach-col-wa y-iñ naj
 COMPL-B.2SG-help-ANTIP A.3SG-RN.OF 3SG
 'You helped him'

(Craig 1979: 5, 2)

A different object-based workaround in Mam involves projecting the object as a structurally reduced N or NP, rather than a full DP. Unlike DPs, these structurally reduced nominals must appear strictly adjacent to the verb stem, are unable to host functional material (i.e. classifiers, possessors, numerals or determiners), and are interpreted non-specifically:

(50) **Mam:** *NFC with bare object*

Ma tz'=ok n-q'o-'n=a [tx'eema-l sii']
 PROX B.3SG=DIR A.1SG-give-DEP=1SG cut-INF **firewood**

'I had him cut firewood' (England 2013: 283)

Again, the verb shows no sign of a valence reduction.²⁵²

Note that Mam is an outlier in terms of how it handles the RANT because its object-based strategy is not at all typical of the family as a whole.

11.4.3 Mixed workarounds to the RANT

Finally, there is what I will consider a mixed workaround to the RANT, where both the verb and the object are targeted by some kind of adjustment.²⁵³ These are similar to the NFCs with bare objects in Mam (i), except that the verb additionally carries a non-voice marker that is tied specifically to the presence of this bare object. Because

²⁵²The Mam suffix *-l*, which I take to be a nominalizer, occurs on both transitives – regardless of whether the object is bare (50) or oblique (i) – and intransitives:

(i) **Mam**

Ma chin-x aaj-a [b'eeta-l]
 PROX A.1SG-DIR RETURN-1SG walk-INF

'I went to walk' (England 1983: 299)

²⁵³Mayan NFCs show complementary distribution with respect to how objects are linked as a function of their degree of individuation: objects with high individuation, i.e. DP objects, must be linked structurally or obliquely and end up φ -marked, while objects with low individuation, i.e. N or NP objects, remain bare and unindexed.

this marker has the form *-oj* in various languages of the family, I refer to this particular workaround strategy as the *-oj strategy*.

The *-oj* strategy is found in e.g. Chuj, where the suffix *-oj* entails the presence of an obligatorily bare object that is non-referential:

- (51) **Chuj:** *-oj* NFC + bare object

Ix-a-yamoch [mol-**oj** kape]
PFV-A.2SG-begin gather-NMLZ coffee

‘I began coffee-gathering’

(Coon & Royer 2021: 11)

In (51), the verb is suffixed with *-oj*, a nominalizing suffix (Buenrostro 2013, Coon & Carolan 2017), and the object cannot bear D-level material (52a).²⁵⁴ The only way for a full DP object – be it a lexical noun, an overt pronoun or *pro* – to be licensed in NFCs is via the agent focus marker, repeated here in (52b) (=35a)):

- (52) **Chuj**

- a. *-oj* NFC + DP object

*Ol-ach-b’at [mol-**oj** te’ kape]
PROSP-B2-go gather-NMLZ DET coffee

intended: ‘You will gather the coffee’

(Coon & Carolan 2017: 28)

- b. *Agent Focus* NFC + DP object

Lan [hach=ko-chel-**an-i**]
PROG B.2=A.1PL-hug-AF-STAT.INTR

‘We’re hugging you’

(Coon & Carolan 2017: 2)

Crucially, transitive verbs suffixed with *-oj* must appear with a bare object, i.e. the object cannot be omitted (Coon & Royer 2021). Whether it is the bare object that subsequently requires the introduction of *-oj* or vice versa, what matters for now is that *-oj* is a non-voice marker whose presence goes hand in hand with a bare object. Bare objects are never indexed via Set A or B, which is why they have been analyzed as incorporated (Baker 1988) or pseudo-incorporated nominals (Massam 2001) that

²⁵⁴ As Maxwell (1976) describes for Chuj, the object is not restricted to a bare N, but may appear with certain modifiers like adjectives, and thus be phrasal:

- (i) **Chuj**

Ix-in-man-w-i niwak kaxlan
PFV-B.1SG-buy-AG-STAT.INTR fat chicken

‘I fat chicken-bought’

(Maxwell 1976)

See also Coon (2019: 52), and Craig (1979: 8) for Popti’.

do not require Case.²⁵⁵

Even though structurally reduced objects are a generally available option that is found across the family in various constructions independently of the RANT, the *-oj* strategy in particular – which involves (i) a designated verbal suffix that does not directly alter valence, and (ii) a bare object – is primarily found in RANT languages, and may turn out to be restricted to these. Unlike Mam’s object-based workaround strategy, which lies outside the family’s typical behavior, the *-oj* strategy is found in numerous languages.

11.4.4 Non-finite intransitives

We lastly turn to the behavior of intransitive verbs in NFCs. We already know that in both RANT and non-RANT languages, derived intransitive verbs formed via anti-passive, agent focus and pseudo-passive operations are well-formed. Unsurprisingly, non-finite root intransitives are always fine, occurring in both non-RANT (53) and RANT languages (54):

(53) **Ch’ol:** *Root intransitive NFC (non-RANT)*

Aj-Juan y-om [wäy-el]
DET-J. want sleep-NMLZ

‘John wants to sleep’ (Coon 2010: 114)

²⁵⁵(i) shows for Kaqchikel that NFCs which φ -mark bare NP objects are ungrammatical:

(i) **Kaqchikel:** *No agreement with bare NP objects*

*X-Ø-qa-chäp [ru/ki-choy-øj che’]
COMPL-B.3SG-A.1PL-begin A.3SG/A.3PL-cut-NMLZ tree

‘We began to cut tree(s)’ (Imanishi 2020: 29)

But this is not a necessary property of NFCs with bare objects, as shown by the following example from Uspantek:

(ii) **Uspantek**

X-Ø-w-eta’ma-j [j-b’an-ik poop]
COMPL-B.3SG-A.1SG-know-STAT.DTRANS A.3SG-make-NMLZ petate

‘I learned to make petates’ (Can Pixabaj 2007: 425)

For such cases, I assume either that what appears as a bare NP is actually a covert DP, as suggested in Coon et al. (2021: 26) for superficially NP nominals which pattern syntactically like DPs in K’ichee’ (Aissen 2011), or – following Lyskawa & Ranero (2021a) for Tz’utujil – that 3SG Set A marking is a default resulting from a failed attempt to Agree with a DP.

(54) **Tz'utujil:** *Root intransitive NFC (RANT)*

[Paj-ik] m'=utz ta
 fall-NMLZ NEG=good IRR

'Falling is not good'

(own notes: RP)

11.5 Summary

Mayan languages often restrict the occurrence of transitive VPs in NFCs, a property we have termed the *RANT*. Languages with the RANT show verb-based, object-based and mixed workaround strategies. In verb-based workarounds, the verb undergoes a valence-decreasing operation like anti-passivization, agent focus or pseudo-passivization. In the single-attested object-based workaround, the object is linked as either an oblique DP or in structurally reduced fashion, yet without the appearance of verbal morphology from which this linking form obviously follows. In mixed workarounds, which recur as the *-oj* strategy, we find co-occurring changes to both elements: the object is bare, and the verb bears a suffix tied specifically to this linking form. An initial overview of the RANT languages discussed in this section in terms of their workaround strategies is given in Table 11.4, which we will elaborate on in the next chapter (empty cells indicate either missing or insufficient data):

Language	Workaround strategy				
	Anti-passive	Verb-based agent focus	Pseudo-passive	Object-based Oblique object	Mixed <i>-oj</i> + bare object
Achi			✓		
Chuj		✓	*		✓
Kaqchikel		*			✓
K'ichee'	✓		✓		✓
Mam				✓	
Popti'	✓	✓			✓
Q'anjob'al		✓	*		✓
Sipakapense			✓		
Sakapultek	✓		✓		✓?
Tz'utujil		*	✓		✓

Table 11.4: Workaround strategies to the RANT

This gives us two perspectives from which to characterize the RANT: from the perspective of the verb, the RANT is satisfied if the verb is detransitivized; from

the perspective of the object, the RANT is satisfied only if the object is licensed via a reduction (bare linking) or an augmentation of its structure (oblique linking). Regardless of whether the specific workaround employed targets the verb, the object or both, the overall generalization that emerges is that the RANT always entails the manipulation of some component of transitivity, either at the level of the verb stem or at the VP-level.

In this dissertation, I will focus on the differences between RANT languages that employ verb-based workarounds on the one hand, and non-RANT languages on the other, placing less emphasis on object-based and mixed workarounds.

Chapter 12

Existing approaches to the RANT

12.1 Introduction

This chapter examines the RANT more closely with the aim of isolating its source. We begin by discussing existing approaches to the RANT in Section 12.2, concluding in Section 12.3 that the RANT is intimately connected to nominalization.

12.2 Existing approaches to the RANT

I discuss three existing approaches to the RANT: the first one ties the RANT to syntactic ergativity, the second one to Case, and the third one to nominalization. I will show that – despite providing insights that will continue to be important here – both the syntactic ergativity and the Case approach provide inadequate explanations for the RANT, and that the nominalization approach – despite some shortcomings – is on the right track.

12.2.1 The syntactic ergativity approach

According to a first potential explanation of the RANT, which I call the *syntactic ergativity approach*, transitive verbs are banned from NFCs because certain grammatical processes cannot target ergative subjects; consequently, some workaround must be employed in order to render the subject absolutive and thus accessible to these processes. To my knowledge, the syntactic ergativity approach has not figured prominently in discussions of the RANT.

Polinsky (2017: 15) explicitly cites the RANT in Popti' as an instance of this effect. The general idea is that just like detransitivized verb forms function as way-stations in order for subjects to be able to participate in $\bar{A}$ -extraction, control, anaphor binding

and other processes, so too do detransitivized verb forms function as way-stations that feed said processes for NFCs (Polinsky 1994, Polinsky 2017).

In Chukchi, for instance, subjects cannot be relativized with ergative positions (1a), such that transitive verbs must be anti-passive; this renders the subject position absolutive, and hence relativizable (1b):²⁵⁶

(1) **Chukchi**

a. *Relativization of ergative subject*

*[_{ERG} _i ŋinqey rəyagtala-lʔ-ən] tumgətum_i
 boy.ABS save-PART-ABS friend.ABS

intended: ‘the friend that saved the boy’

b. *Relativization of absolutive subject via anti-passive*

[_{ABS} _i ŋinqey-ək ine-nyegtelewə-lʔ-ən] tumgətum_i
 boy-LOC ANTIP-save-PART-ABS friend

‘the friend that saved the boy’

(Polinsky 2017: 15)

When applied specifically to Popti’ (Polinsky 2017: 15), and more generally to Mayan languages with the RANT, the syntactic ergativity approach makes the basic prediction that transitive verbs in (a subset of) NFCs must undergo some kind of detransitivization that renders their null subject absolutive, which is then accessible to control. Now, given that this restriction is construed not as an absolute general ban against ergative null subjects, but rather specifically against relations such as control between matrix arguments and ergative controllees (i.e. PRO), this view predicts that if there is no controller in the matrix clause, the NFC subject should be licit as an ergative by virtue of no longer being part of a control relation.

This prediction is incorrect: numerous counterexamples show that the RANT remains in the absence of a matrix controller. One such configuration arises in embeddings where the matrix predicate does not introduce an argument in its specifier, i.e. behaves syntactically like an unaccusative. In Chuj, for instance, the positional predicate *lan*, which expresses progressive aspect, takes as its sole argument a NFC containing the principal meaning (Buenrostro 2013, Coon & Carolan 2017). Regardless, transitive verbs in NFC complements to *lan* must have the Agent Focus workaround (2a); the same holds for NFCs based on transitive verbs functioning as complements to evaluative adjectives like ‘good’ (2b):

(2) **Chuj**

a. *Matrix predicate is progressive predicate*

Lan [hach=in-chel-an-i]
 PROG A.2-walk-STAT.INTR

²⁵⁶Specifically for control complements, a similar restriction appears in e.g. Chukchi (Skorik 1977) and some Australian languages (Dixon 1979, Dixon 1994). Aldridge (2008) and Aldridge (2011) extend this proposal to a number of Austronesian languages, suggesting it may be tied to ergativity.

‘You’re walking’ (Coon & Carolan 2017: 7)

- b. *Matrix predicate is evaluative adjective*

Wach’ [hin-chel-an-i]

good A.1SG-hug-AF-STAT.INTR

‘I hug well / My hugging is good’ (own notes: MT)

The same is true in Q’anjob’al and Tz’utujil: ²⁵⁷

- (3) *Matrix predicate is evaluative adjective*

- a. **Q’anjob’al**

[Ø h-il-on] kawal watx’

B.3SG A.2SG-see-AF very good

‘Your seeing him / her / it is very good’

(Pedro Mateo Pedro p.c., in Imanishi (2020: 19))

- b. **Tz’utujil de Santiago**

[R-ch’iy-ik ja siaw] m’=utz ta

A.3SG-hit.PASS-NMLZ DET cat NEG=good IRR

‘Hitting the cat is not good’ (own notes: RP)

I assume these cases involve an arbitrarily controlled PRO.

A second problem faced by the syntactic ergativity approach also has to do with the EEC. In recent years, a growing body of work has converged on the insight that the Mayan EEC is not about properties of ergative subjects per se, but rather about properties of objects and / or their relationship to ergative subjects (Stiebels 2006a, Coon et al. 2014, Coon et al. 2021). Among those languages with the EEC, this is most clearly brought out in contexts where certain changes to the object obviate the need for Agent Focus – crucially in spite of the fact that the subject remains ergative and $\bar{A}$ -extracted. In one such context, structurally reduced objects with non-specific

²⁵⁷Relatedly, under the assumption that (i) finite complement clauses with null subjects that are obligatorily identified with matrix subjects involve finite control, and that (ii) the syntactic ergativity approach extends to finite control, the ban against ergative controllees is falsified given that control targets an ergative controllee. In the following examples from Poqomam and K’ichee’, the null subject of the finite complement clause must identify the matrix subject; as made evident by the presence of Set A morphology on the embedded verb, the controllee bears ergative:

- (i) *Finite clauses with ergative controllees*

- a. **Poqomam**

X-u-reej [ec x-u-tik ab’iix] [la sa imaas]_i
COMPL-A.3SG-want_{ERG} COMPL-A.3SG-plant cornfield DET DIM gentleman

‘The gentleman wanted to seed his cornfield’

(Santos Nicolás & Benito Pérez 1998: 436)

- b. **K’ichee’**

X-oj-peet-ik [ec x-Ø-ol-qa-k’am-a’]
COMPL-B.1PL-come-STAT.INTR_{ERG} COMPL-B.3SG-MOV-A.1PL-receive-DEP

‘We came to take her’

(Can Pixabaj 2015: 27)

reference exceptionally permit $\bar{A}$ -movement of the ergative subject without Agent Focus, as found in K'ichee' (López Ixcoy 1997, Aissen 2011, Coon et al. 2021):²⁵⁸

- (4) **K'ichee'**: *Bare NP object → no Agent Focus*

Ma jun achi taj k-e'-u-b'oq alaj taq chee'
NEG INDEF man IRR INC-B.3PL-A.3SG-uproot DIM PL tree

'It's not A MAN that is uprooting little trees'

(López Ixcoy (1997), in Aissen (2011: 12))

Another environment that lifts the EEC is when an $\bar{A}$ -moved subject binds a reflexive or extended reflexive object, i.e. an object properly containing a reflexive. An example is provided by Q'anjob'al:

- (5) **Q'anjob'al**: *Extended reflexive object → no Agent Focus*

Maktxel max s-bon s-a?
who PFV A.3-paint A.3-house

'Who_i painted his_{i/*k} house?'

(Coon et al. 2014: 226)

In both (4) and (5), the verb exceptionally lacks Agent Focus marking even though the ergative subject has been $\bar{A}$ -extracted. The subject is φ -marked on the verb via Set A, meaning that it remains ergative, and hence the verb morpho-syntactically transitive. Regardless of their correct analysis, these environments show clearly that it is specifically some change to the object – viz. a structural reduction to NP, or the addition of a reflexive – which obviates the EEC.

Given the persistence of the RANT in the absence of a matrix controller, the availability of ergative controllees, and the fact that the RANT is not entailed by the EEC, I conclude that the syntactic ergativity approach cannot explain the RANT. These facts are a crucial first indication that the RANT is actually connected specifically to non-finiteness.

12.2.2 The Case approach

According to perhaps the most well-known line of research in this regard (Coon et al. 2014, Coon et al. 2021), the RANT is inextricably linked to Case and the absolutive parameter:²⁵⁹ in NFCs, HIGH-ABS languages lack the regular source of Set B that is found in finite root contexts, namely finite T, which makes it impossible for transitive verbs to project their internal arguments as full DPs; consequently, some kind of

²⁵⁸In (4), the object does not project up to DP, but at least up to NumP because it can bear plural marking (Coon et al. 2021).

²⁵⁹This work builds on insights in Coon et al. (2014), where NFCs do not yet play as large a role as in Coon et al. (2021).

workaround is required to bring about an alternative Case source.²⁶⁰ Conversely, because LOW-ABS languages do provide a low source of Set B, transitive verbs can enter into NFCs and project their internal arguments as DPs without further ado. In this way, the RANT is also indirectly tied to the EEC, given that the vast majority of HIGH-ABS languages require Agent Focus under ergative $\bar{A}$ -extraction.

Table 12.1 schematizes this effect according to the Case approach for transitive verbs in finite and non-finite clauses depending on whether the locus of absolutive assignment is high or low (note that in the HIGH-ABS language, the transitive VoiceP is embedded in a NFC without any workaround for the sake of illustration, even though we will see that such configurations are actually ruled out):

ABS	Transitive finite clause	Transitive non-finite clause
HIGH		
LOW		

Table 12.1: Transitive clauses in HIGH-ABS vs LOW-ABS languages

Consider first the LOW-ABS languages Ch'ol and Yucatec in (6), which – due to the presence of a low, transitive Voice that can assign Set B – require no workaround for transitive VPs in NFCs:

²⁶⁰Internal arguments cannot receive Set B from matrix T because this Case goes the NFC itself, which is nominal and hence requires Case.

(6) LOW-ABS languages: Set B from transitive Voice

a. **Ch'ol**

K-om [j-käñ-ety]
 A.1-want A.1-know-B.2

'I want to know you'

(Vázquez Álvarez 2011: 99)

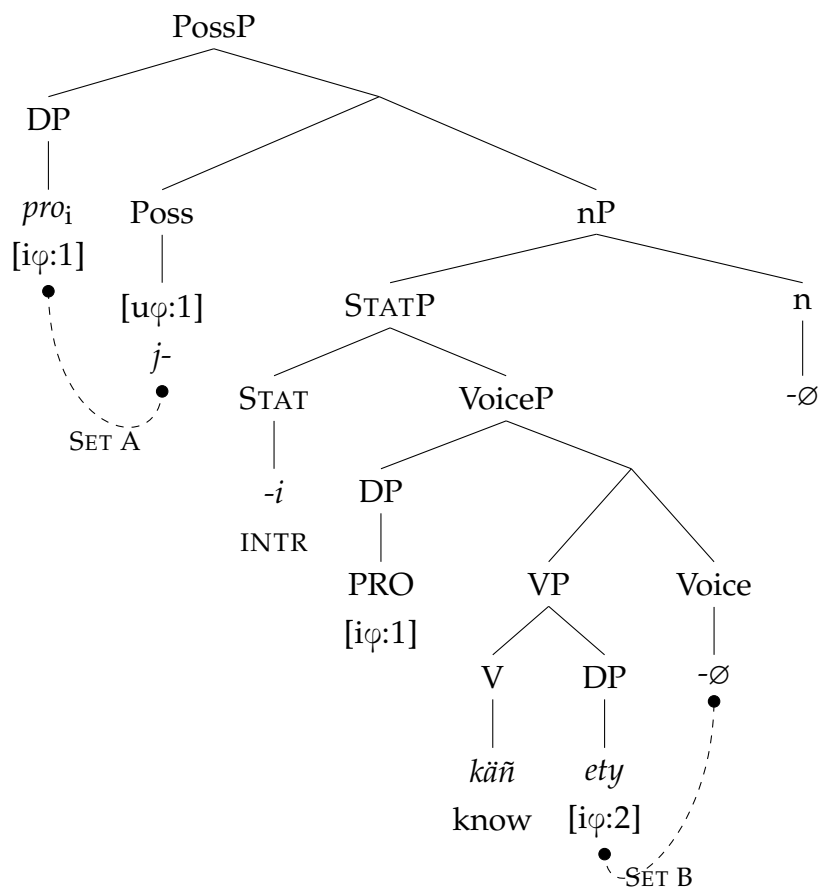
b. **Yucatec**

In-k'áat [inw-il-ech]
 A.1SG-want A.1SG-see-B.2SG

'I want to see you'

(Bricker 1981: 96)

(6a) has the structure in Tree 12.1, following Coon (2010):



Tree 12.1: Ch'ol: LOW-ABS

Now, compare the NFC in (7) from the HIGH-ABS language Chuj, where the verb bears Agent Focus marking: Coon et al. (2021) extend their analysis of Agent Focus in finite contexts to non-finite contexts like (7), arguing that the Agent Focus marker provides an alternative, low source of Set B for the object in the absence of finite T (7):

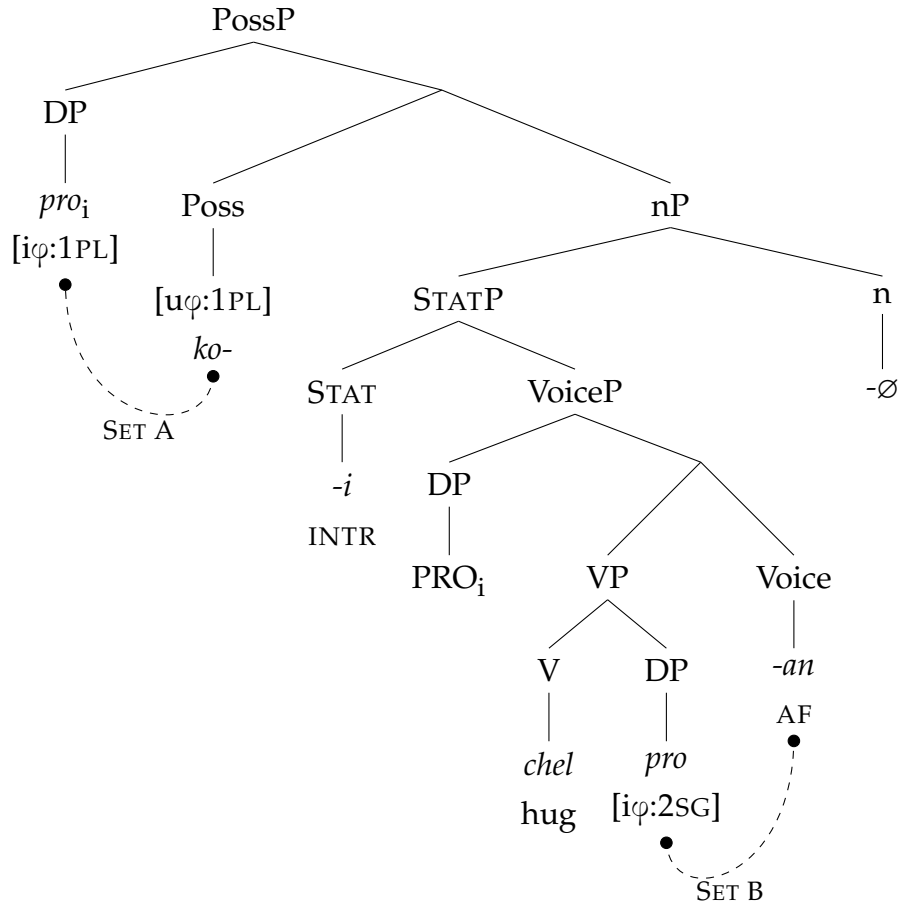
(7) **Chuj:** *Set B from Agent Focus marker*

Lan [hach=ko-chel-an-i]
 PROG B.2=A.1PL-hug-AF-STAT.INTR

‘We’re hugging you’

(Coon & Carolan 2017: 2)

(7) has the structure in Tree (7), following Coon & Carolan (2017):²⁶¹



Tree 12.2: Chuj: Agent Focus

Only when an object is a bare N(P) can it forgo Case, and instead be licensed via incorporation (Baker 1988) or pseudo-incorporation (Massam 2001):²⁶²

²⁶¹With respect to the Set B marker in the Chuj example in (7), recall that Set B markers are the result of a clitic doubling operation triggered by an Agree relationship between the absolutive argument and its licensing head. Because the clitic double cannot cliticize to T in (7) given that it is absent, it instead cliticizes to the entire embedded stem, i.e. the minimal word containing the head which triggers the doubled clitic (Coon et al. 2014, Coon & Carolan 2017).

²⁶²See Coon (2019) and Coon & Royer (2021) for an explicit application of this analysis to Ch’ol and Chuj. The authors assume that DP objects are integrated semantically via Function Application, while bare objects compose with their verbs via Restrict (Chung & Ladusaw 2004).

(8) *NFC with bare object*a. **Chuj:** *-oj verb*

Ix-a-yamoch [mol-oj kape]
 PFV-A.2SG-begin gather-NMLZ coffee

'I began coffee-gathering'

(Coon & Royer 2021: 11)

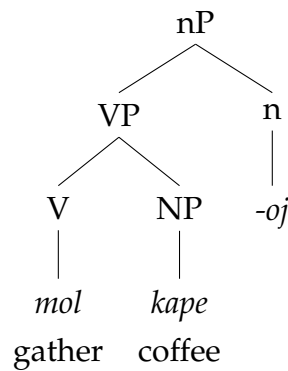
b. **Q'eqchi':** *Anti-passive*

Laa'in t-inw-aj [lo'-o-k tul]
 1SG ASP-A.1SG-want eat-ANTIP-INF banana

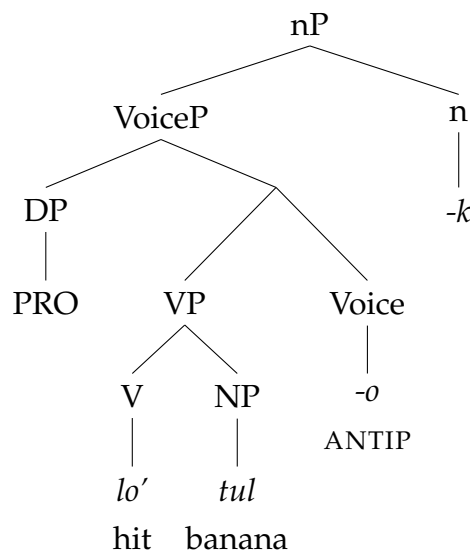
'I want to eat bananas'

(Berinstein 1985: 265ff)

(8a) and (8b) have the structures in Trees 12.3 (Coon & Royer 2021) and 12.4, respectively:

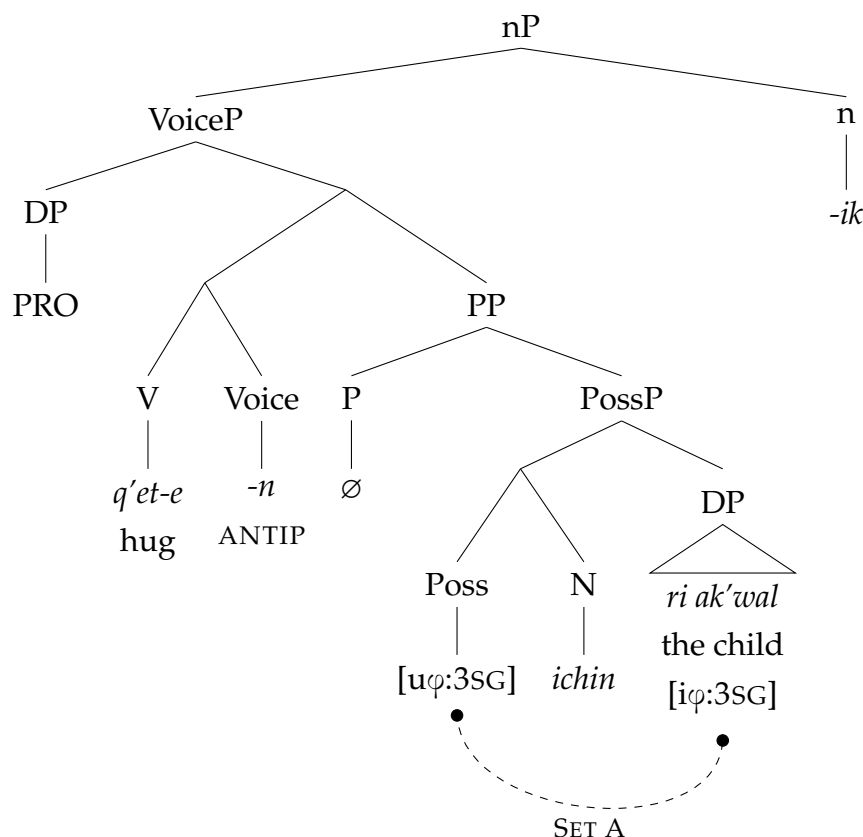


Tree 12.3: Chuj: bare verb + bare NP object



Tree 12.4: Q'eqchi': anti-passive verb + bare NP object

According to the Case approach, the RANT arises due to a property of the transitive verb's internal argument (its need for Case), which goes unsatisfied within certain structural contexts (NFCs); consequently, some workaround is required that can alternatively Case-license the internal argument (e.g. the Agent Focus marker). On



Tree 12.5: Kaqchikel: anti-passivization

So for anti-passive NFCs, the Case approach seems to adequately capture how the relational noun afforded by anti-passivization provides a low source of Case – in alternation to, but on a par with Agent Focus. This variation in the specific strategy employed comes with variation in the how the object is cross-referenced: while it is marked on the verb via Set B in Agent Focus NFCs, it is marked on the relational noun via Set A in anti-passive NFCs.²⁶⁵

Turning to pseudo-passive NFCs, however, we find that the Case approach is insufficient. In (10), the verb is formally passive and the thematic object is ϕ -marked on the verb via Set A:

- (10) **Tz'utujil:** *Pseudo-passive NFC → Set A via grammatical possession*

X-Ø-qaa-maj [ki-ch'e<j>y-iik]
 COMPL-B-3SG-A.1PL-begin A.3PL-hit.PASS-NMLZ

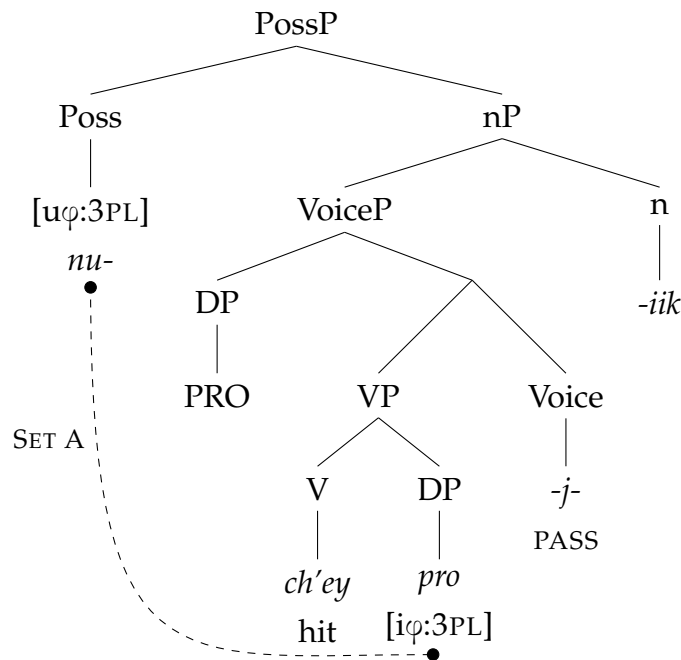
'We began to hit them'

(Dayley 1985: 393)

Given the absence of both an Agent Focus marker and a relational noun licensed by anti-passivization in (10) (again for whatever reason), the Case approach would explain the internal argument's reception of Case in pseudo-passive NFCs by means of the categorial switch from the verbal to the nominal domain, brought about by

²⁶⁵In finite agent focus contexts, there is significant variation in terms of which argument is indexed via Set B; see e.g. Stiebels (2007).

nominalization, and its subsequent entering into grammatical possession with the resulting nominal (recall that in Mayan, Set A marks ergatives and possessors):



Tree 12.6: Tz'utujil de San Juan: pseudo-passivization

However, this would leave unexplained the fact that the verb in (10) must undergo what appears to be passivization. Leaving aside for now the pseudo-passive nature of these NFCs, if Set A becomes available to the internal argument due to Agree with the nominalization, there is actually no need for the verbal complement to *n* to detransitivize – the assignment of Set A via grammatical possession is available independently without passivization. This essentially amounts to the prediction that the RANT in the form of a restriction on the stem-level transitivity of a verb should go away once the object's need for Case is taken care of – contrary to fact: in all languages that (i) employ the pseudo-passive strategy in NFCs, and (ii) in which the thematic object is formally expressed as a possessor via Set A, passive verbal morphology remains obligatory.

It is useful in this regard to briefly reconsider the Agent Focus and anti-passive strategies by thinking backwards from their afforded Case-assigning elements: in Agent Focus NFCs, Case comes directly from the Agent Focus marker itself, while in anti-passive NFCs, it comes not from anti-passivization per se, but an element that is licensed by specifically this operation (i.e. the relational noun requires anti-passivization).²⁶⁶ Passivization, by contrast, does not come with any such Case-assigning element, neither directly (as in the Agent Focus strategy) or indirectly (as in the anti-passive strategy). Put differently: if the object gets Set A via grammatical possession anyway, why must the verb also be (formally) passive?

²⁶⁶In finite contexts, my Tz'utujil and Kaqchikel consultants systematically rejected oblique objects without anti-passive morphology on the verb.

A further shortcoming of the Case approach is revealed by a related prediction that it makes about HIGH-ABS languages: all else being equal, if a transitive verb in a NFC does not project an internal argument at all, its need for a workaround should likewise disappear because there would no longer be any argument in need of Case.

This prediction is also incorrect as there are numerous examples from HIGH-ABS languages showing that transitive verbs must detransitivize in NFCs even in the absence of an object. This is illustrated by the following pseudo-passive and anti-passive NFCs, all of which occur without objects:²⁶⁷

(11) **Tz'utujil de Santiago:** *Pseudo-passive NFC*

X-Ø-a-moj [ch'iy-ik]
COMPL-B.3SG-A.2SG-begin hit.PASS-NMLZ

'You started hitting' (own notes: RP)

(12) **K'ichee':** *Anti-passive NFC*

X-Ø-r-eta'ma-j [kuna-n-ik]
COMPL-B.3SG-A.3SG-know-ACT cure-ANTIP-NMLZ

'(S)he learned to cure' (Can Pixabaj 2015: 107)

(13) **Kaqchikel**

a. *Pseudo-passive NFC*

Ri ixoq n-Ø-ajin chi [war-sa-x-ik]
the woman INCOMPL-B.3SG-PROG PREP sleep-CAUS-PASS-NMLZ

'The woman is causing to sleep' (own notes: SR)

b. *Anti-passive NFC*

X-Ø-u-chäp [tz'ib'-a-n-ik] ri ak'wal
COMPL-B.3SG-A.3SG-grab write-SUF-ANTIP-NMLZ DET child

'The boy started to write' (Ajsivinac Sián 2007)

²⁶⁷I assume that in cases like (11)-(15), the NFC lacks an object entirely. I leave open how exactly this reconciles with the rigid transitivity of Mayan languages mentioned in Chapter 1. One option is to assume that Mayan languages show rigid transitivity to varying degrees of strictness, e.g. because of the availability of an operation of (non-specific) object deletion without the aid of an operation like anti-passivization in some, but not other languages.

(14) **Sakapulteko**a. *Pseudo-passive NFC*

X-in-ak [cha tiix-iik]
 COMPL-B.1SG-begin PREP eat.PASS-NMLZ
 'I began eating'

b. *Anti-passive NFC*

Ø-Ø-tiikir [cha ch'iy-in-eem]
 COMPL-B.3SG-begin PREP hit-ANTIP-NMLZ
 'He began to hit'

(DuBois 1981: 254)

The same is true for Agent Focus NFCs:

(15) **Chuj**

Ix-in-ya-moch [hin-chel-an-i]
 COMPL-A.1-begin A.1-hug-AF-STAT.INTR

'I began to hug'

(own notes MT)

Popti' NFCs in which the complement to the embedded non-finite verb is itself a finite clause point to the same conclusion. In (16), the most deeply embedded clause is headed by the complementizer *ta*, which only heads fully finite complement clauses with aspect marking (Craig 1977: 232f). Assuming that this clause is a CP and that CPs are not Case competitors in Popti', it is unexpected that the NFC one level higher up must show the Agent Focus workaround regardless:²⁶⁸

(16) **Popti'**

Lañan [Ø w-abe-**ni** [ta x-Ø-ul naj]]
 PROG B.3SG A.1SG-hear-AF that COMPL-B.3SG-come 3SG

'I (am) hear(ing) that he came'

(Craig 1977: 64)

Based on cross-linguistic linking types in nominalization (Koptjevskaja-Tamm 1993, Stiebels 2006b), we can establish a further variant of the Case approach that ties the RANT specifically to the number of arguments that must be Case-licensed within the NFC. The Salishan language Shuswap, for example, makes use of a mixed linking profile in nominalizations: in nominalizations of intransitives, the sole argument undergoes nominal linking via possessor agreement (17a); but nominalizations of transitives retain a fully verbal linking pattern (17b):

²⁶⁸Though see Henderson & Coon (2018) and Coon, Baier & Levin (2021) for Mayan languages in which CPs do compete for Case.

(17) **Shuswap**

- a.
- Nominalization of intransitive → nominal linking*

ʔ-n-s-ʔiʔiʔlx

DET-POSS.1SG-NMLZ-jump

'my jumping'

- b.
- Nominalization of transitive → verbal linking*

ʔ-s-qʰl-n-cécm-s

DET-NMLZ-summon-TRANS-1SG.ABS-3SG.ERG

'his summoning me'

(Kuipers 1974: 41ff)

Just like the Case approach, this idea is resource-based: nominal frames have a general linking potential for arguments, but this linking potential may be exhausted and therefore insufficient to license two arguments, such that the verbal frame must be resorted to instead (Stiebels 2006b: 208). Applied to Mayan, this view states that detransitivization must occur in order for the number of arguments that must be (canonically) Case-licensed to be reduced by one. This explanation does not work either since we can mount the same argument against it as we did above for the purely object-based Case approach: we know from examples (11)-(15) (and (16)) that detransitivization in RANT languages must occur even if NFCs lack objects. In other words, the RANT is blind to the number of arguments in need of Case-licensing within NFCs because the RANT is blind to Case-licensing.

The persistence of the RANT in the absence of an object in need of Case (either because Case can be provided by grammatical possession, or there is no object to begin with), and its absence in at least one HIGH-ABS language show us that the RANT is independent of Case per se and at least partially dissociated from the absolutive parameter (albeit tightly correlated with the HIGH-ABS property). We thus have conclusive evidence that the RANT is tied to non-finiteness, which we know in Mayan is defined by nominalization.

12.2.3 The nominalization approach

This brings us to the *nominalization* approach, which ties the RANT directly to nominalization. Departing from a point of variation in alignment found in NFC constructions, Imanishi (2020) argues that some nominalizing heads – instances of *n* – may restrict the transitivity of their verbal complement. Basically, he argues for Kaqchikel that (i) because unaccusatives (18a) and pseudo-passives (18b) can occur in NFCs, but not unergatives (18c) (or transitive verbs without workarounds), and (ii) because unaccusatives and passives pattern alike in lacking external arguments, the Kaqchikel nominalizer *-ik* must require its verbal complement to lack an external argument:²⁶⁹

²⁶⁹See Burukina (2021a) for independent morpho-syntactic diagnostics for the unaccusative-unergative distinction in Kaqchikel, based on those developed for related Tz'utujil in Levin et al.

(18) **Kaqchikel NFCs**a. *Unaccusative* + n

[Ri ru-tzaq-ik ri a Juan] üt
 DET A.3SG-fall-NMLZ DET CLF J. good
 'Juan's falling is good'

b. *Pseudo-passive* + n

Y-in-ajin [che ki-k'ul-ik ak'wl-a']
 INCOMPL-B.1SG-PROG PREP A.3PL-meet.PASS-NMLZ child-PL
 'I am meeting children'

c. *Unergative* + n

*ru-tzopin-ik ri xta Maria
 A.3SG-jump-NMLZ DET CLF M.
 intended: 'Maria's jumping'

(Imanishi 2020: 2, 19)

This is formulated as the *Restriction on Nominalization*:²⁷⁰

(19) *Restriction on Nominalization*

(Imanishi 2020: 18)

Nominalized verbs must lack a syntactically projected external argument.

(19) is argued to be parametrized. In languages like Ch'ol, which allows both transitive verbs and unergatives to occur freely in NFCs – i.e. as complements to nominalization – the *Restriction on Nominalization* is inactive:

(20) **Ch'ol NFCs**a. *Transitive* + n

Choñkol [i-jats'-oñ]
 PROG A.3SG-hit-B.1SG
 'She's hitting me'

(Coon 2013: 11)

b. *Unergative* + n

Choñkol [k-majl-el]
 PROG A.1-go-NMLZ
 'I'm going'

(Coon 2010: 136)

As it stands, Imanishi's nominalization approach comes with a few issues. The most pressing of these is the fact that pseudo-passive NFCs actually contain syntactically projected external arguments. Consider the following pseudo-passive NFCs from K'ichee' and Tz'utujil, which contain reflexive objects:²⁷¹

(2020) and Lyskawa & Ranero (2021a). She also provides data showing that Imanishi's generalizations are partially incorrect, though this is orthogonal to Imanishi's general claims regarding the role played by nominalization.

²⁷⁰Imanishi (2020) bases (19) on Alexiadou (2001) and Imanishi (2014).

²⁷¹The earliest mention of reflexive objects in pseudo-passive NFCs that I am aware of is Can Pixabaj (2015).

(21) *Pseudo-passive NFC + reflexive*a. **Tz'utujil de Santiago**

Anen n-Ø-e-mjon [r-tz'et-ik
 1SG INCOMPL-B.3SG-A.1SG-PROG A.3SG-see.PASS-NMLZ
w-ii' chpaam tzetb'al]
 A.1SG-RN.REFL in mirror
 'I am seeing myself in the mirror' (own notes: RP)

b. **K'ichee'**

X-Ø-inw-eta'ma-j [r-iil-ik
 COMPL-B.3SG-A.1SG-learn-ACT A.3SG-see.PASS-NMLZ
w-iib']
 A.1SG-RN.REFL
 'I learned to take care of myself' (Can Pixabaj 2015: 119)

These reflexives remain well-formed even in the absence of a matrix binder, like when the matrix predicate is an evaluative adjective (28); but they are impossible in finite passives (22b):

(22) **Tz'utujil de Santiago**a. *Pseudo-passive NFC + reflexive*

[R-tz'et-ik w-ii' chpaam tzetb'al] kungan
 A.3SG-see.PASS-NMLZ A.1SG-RN.REFL in mirror much
 raro
 strange
 'Seeing myself in the mirror is very strange'

b. *Finite passive + reflexive*

*X-Ø-tza'at-a w-ii'
 COMPL-B.3SG-see.PASS-STAT.INTR A.1SG-RN.REFL
 intended: 'Myself was seen / I saw myself' (own notes: RP)

It follows that non-finite pseudo-passives (but not finite passives) actually contain external arguments that can bind reflexives, i.e. they are syntactically active despite their passive morphology. This means that the specific way in which the RANT restricts transitivity cannot be reduced to the selection of projections without external arguments, which runs directly counter to the Restriction on Nominalization (19) – especially in view of the fact that (19) is predicated on the assumption that pseudo-passive NFCs lack external arguments. This issue will be re-addressed in the next chapter.

Despite its shortcomings, Imanishi's nominalization approach is on the right track because it correctly recognizes that the source of the RANT lies specifically in nominalization – not Case or syntactic ergativity.²⁷² To my knowledge, his approach is

²⁷²Note that Imanishi (2020) himself does not arrive at his conclusions regarding the role of nominalization in the RANT from the perspective of either ergativity or Case, considering mainly alignment

also the first attempt at formalizing generalizations presented in Mateo Pedro (2010), a survey showing that the grammars of numerous Mayan languages systematically distinguish transitive from intransitive verbs under nominalization (though exactly how this occurs is left open).

12.3 Nominalization and the RANT

Having established that nominalization plays a crucial role in the RANT, we will see that the RANT follows neither from nominalization by itself, nor from the structural level at which it takes place. I suggest that the major way the RANT manifests is in the form of a selectional restriction imposed by certain kinds of *n* on Voice, in the spirit of work like Alexiadou (2001). Crucially, even though nominalization and its structural level of occurrence *per se* are not enough to explain the RANT, they do constitute conditions which are necessary for the RANT to arise.

12.3.1 The RANT does not follow from nominalization *per se*

As far as inquiries have been able to reveal, there does not appear to be anything intrinsic to the mere categorial switch from the verbal to the nominal domain that should induce the RANT. In her seminal typology of nominalizations, Koptjevskaja-Tamm (1993) lists numerous examples of nominalizations containing fully transitive VPs; likewise, the in-depth juxtaposition of how arguments are linked depending on whether the linking domain is purely verbal or deverbally nominal in Stiebels (2006b) does not uncover any property associated with nominalization (or nominality) that would force verbs to undergo a valence change before they can serve as inputs to nominalization. The same generalization emerges in various anthologies on nominalizations (Alexiadou & Rathert 2010, Rathert & Alexiadou 2010, Ha Yap et al. 2011, Alexiadou & Borer 2021). From a Mayan-internal perspective, the very fact that NFCs based on transitive VPs are restricted in some but not other languages shows that the RANT must not only be connected to nominalization, but also properties specific to Mayan.

That said, the literature has brought forward several cases in which the grammar is clearly sensitive to event- and / or argument-structural properties in the context of nominalization, with potential indirect effects on verbal transitivity. Grimshaw (1990) famously distinguishes *Result Nominals*, *Simple Event Nominals* and *Complex Event Nominals*, a tripartition which continues to inform much generative work on nominalization. Roughly, Result Nominals lack argument structure and have a result reading; Simple Event Nominals are like Result Nominals but with an event reading; and Complex Event Nominals, by contrast, have argument structure and require an obligatory event reading. Consider the following triplet:

patterns and the distribution of the nominalizer *-ik*.

(23) a. Result Nominal:
the examination was on the table

b. Simple Event Nominal:
the examination took a long time

c. Complex Event Nominal
the examination of the patients took a long time

(based on Grimshaw 1990: 49ff)

verbs implying external causation (24) and those implying internal causation (25).^{274,275} These two accounts, which share the intuition that the restriction on nominalization correlates directly with causative verbal typology, is disputed in Harley & Noyer (2000), who source the restriction in thematic properties of external arguments. In an attempt to bring these insights together, Sichel (2010) concludes that the restriction is actually due to conditions on the kinds of events that can be expressed in nominalization, rather than any properties associated with lexical meanings or external arguments; she proposes that this event-structural constraint be encoded as a selectional restriction on inputs to nominalization.

In his investigation of apparently equivalent suffixes competing over verbal bases in Spanish, Fabregas (2010) shows that different nominalizers yield outputs with distinct clusters of semantic and syntactic properties, to the extent that if a single verbal base allows more than one nominalizer, it will differ predictably in those properties. He relates the distribution of affixes again not to verb's stem-level transitivity, but rather event-structural constraints centered around properties of internal arguments; like Sichel, he proposes to encode it as a selectional restriction on verbal inputs.

The picture that emerges is that nominalization can make demands on event- and / or argument-structural components, with potential effects on transitivity if these happen to align with a certain verbal transitivity value. In fact, the demands imposed by nominalizers on their verbal complements may extend beyond the core-eventive domain into higher verbal functional structure: as argued by Borsley & Kornfilt (2000) and Kornfilt & Whitman (2011) for Turkish, TP-level nominalizations involve a *n* that selects a TP headed by a *T* which does not show the full range of tense specifications found in purely verbal TPs.²⁷⁶ So there are clearly recurring constraints associated with nominalization, although the extent to which these are necessary remains to be uncovered. What is clear is that nominalization per se does not intrinsically require a verb's stem-level transitivity to reduce.

12.3.2 The RANT does not follow from the level of nominalization per se

The view on nominalization which I adopt here assumes that verbal projections become nominal by virtue of a functional nominalizing head. This is the so-called *Functional Nominalization Thesis* (FNT) (Borsley & Kornfilt 2000, Kornfilt & Whitman 2011), and it is espoused in much recent work on various kinds of nominalizations

²⁷⁴Smith (1972) also gives a lexical explanation, but unlike Marantz (1997), she draws the division in terms of whether the affix is drawn from the Latin or the Anglo-Saxon vocabulary of English.

²⁷⁵This contrast is seemingly absent in German: *die Eiswürfel schmelzen* 'the ice cubes melt' vs *das Schmelzen der Eiswürfel* 'the melting of the ice cubes' vs *Marias Schmelzen der Eiswürfel* 'Maria's melting of the ice cubes' (Barbara Stiebels p.c.). I suspect this has to do with differences between the English suffix *-ing* on the one hand, and the German suffix *-en* on the other.

²⁷⁶See also Demirok (2019)

across languages. The FNT operates from the premise that each particular category is either fully nominal or fully verbal – i.e. there are no hybrid verbal-nominal categories – but there is a range of choices as to which particular categories are nominal or verbal. After a verbal projection merges with a nominalizing functional head at some point in the derivation, the projection becomes nominal. This yields a tree with an internally homogeneous verbal syntax, and an externally homogeneous nominal syntax.²⁷⁷

Given that the FNT permits a range of levels for (different kinds of) *n* to merge at, different kinds of nominalizations show varying degrees of nominal behavior, while at the same time retaining characteristics of the input verbal projection to varying degrees.²⁷⁸ Kornfilt & Whitman (2011) identify four basic levels of nominalization (with leeway for more cut-off points within finer clausal articulations): VP, VoiceP, TP and CP.²⁷⁹ The question arises whether the RANT can be explained by appealing to the particular structural level at which nominalization occurs.

What we find, however, is that the level of nominalization per se generally affects transitivity only in terms of the object's linking form. That is, when (i) Voice assigns objective case, (ii) *n* merges above Voice, and (iii) Voice is non-defective – i.e. retains all the features it has in regular finite clauses – the object retains the same linking form as with normal verbs. This scenario is illustrated by the Turkish suffix *-mA*, which takes a TP complement in which the object retains accusative case (26) (Borsley & Kornfilt 2000, Kornfilt & Whitman 2011). But when *n* merges below Voice, objective case from Voice becomes unavailable and so the object is linked non-canonically, as in Chuj, where the suffix *-oj* attaches directly to verbal root projections, and objects are linked in structurally reduced fashion (27):

(26) **Turkish:** *TP-nominalization* → *object is accusative*

Hasan [uşağ-in oda-yı temizle-me-sin]-i isti-yor
H. servant-GEN room-ACC clean-NMLZ-ACC want-PST

'Hasan wants that the servant should clean the room'

(Kornfilt & Whitman 2011: 1301)

²⁷⁷The FNT is meant to predictably derive a typology of nominalizations in which various morpho-syntactic property clusters arise as functions of the specific structural level at which *n* is merged, which most notably include argument linking and inflection, but also syntactic properties like extraction and binding. See Ehrich (2002), Alexiadou (2001), Markova (2010), Sleeman & Brito (2010), Baker (2011), Carnie (2011) Cole & Hermon (2011), among numerous others. As remarked in Baker (2011), this approach reconciles the generative theory of discrete lexical categories with the cline between nominal and verbal properties observed in nominalizations.

²⁷⁸Discussions of degrees of nominalization are equally found in functionalist-typological work, such as Comrie (1976), Croft (1991), Koptjevskaja-Tamm (1993) and Malchukov (2006).

²⁷⁹See e.g. Sleeman & Brito (2010) for arguments that various fine-grained readings in nominalizations are accounted for under the split vP hypothesis (Ramchand 2008).

(27) **Chuj:** $\sqrt{\text{ }}$ -nominalization $\rightarrow$ object is bare

Ix-a-yamoch [mol-oj kape]
PFV-A.2SG-begin gather-NMLZ coffee

‘I began coffee-gathering’

(Coon & Royer 2021: 11)

As is well-known, English shows a similar split between so-called *poss-ing* gerunds on the one hand (28a), with accusative objects, and derived nominals on the other (28b), with oblique objects:^{280,281,282}

(28) a. Poss-ing gerund: VoiceP-nominalization $\rightarrow$ object is accusative

[Alma’s criticizing the book thoroughly] was helpful.

b. Derived nominal: $\sqrt{\text{ }}$ -nominalization $\rightarrow$ object is oblique

[Alma’s thorough criticism of the book] was helpful.

(based on examples from Yoon (1996) and Borsley & Kornfilt (2000))

Now, scholars assuming the FNT have explicitly drawn connections to intransitivity regarding VoiceP-nominalizations based on transitive verbs, though here again we find that the observed effects differ substantially from those of the RANT. Alignment-wise, these nominalizations have been argued to follow an ergative pattern in that transitive external arguments are marked differently from intransitive arguments (Alexiadou 2001, Alexiadou 2017, Salanova 2007, Imanishi 2014). To the extent that transitive external arguments appear (if not expressed as possessors), they must do so in the form of a *by*-phrase. Consider the following English triplet:²⁸³

²⁸⁰*poss-ing* constructions are also known as *verbal gerunds*.

²⁸¹I assume that derived nominals like *criticism* are derived directly at the verbal root - relevant here is that *n* is merged below where Voice would be merged, thereby bleeding its introduction. On objection that nouns like *criticism* are not derived syntactically, the fact that sub-VoiceP-nominalizations induce nominal object linking is evident in examples involving productive nominal gerunds like *Your giving of the book to Bill*, from Yoon (1996).

²⁸²For explicit criticism of the FNT, see e.g. Kiparsky (2017), who argues that contrary to what the FNT seems to promise, the morpho-syntactic properties of a nominalization cannot be fixed just by locating its nominal head in a universal hierarchy of verbal functional categories. Kiparsky concludes that all nominalization proceeds in the lexicon. Šereikaitė (2022) is similarly critical of the FNT as it stands, though maintains a syntactic approach to nominalization by showing how the system can be adapted to account for seemingly problematic patterns, as found in e.g. Lithuanian, if one extends the parameter of Voice-bundling vs Voice-splitting (Pylkkänen 2008, Harley 2017) from the verbal to the nominal domain.

²⁸³Stiebels (2006b) argues that this allegedly ergative linking pattern is epiphenomenal, and derivable from the fact that it is always the lowest argument which is privileged in terms of structural nominal linking.

- (29) a. *Transitive*
 the destruction of the city by the barbarians
- b. *Unergative*
 the jumping of the cow
- c. *Unaccusative*
 the arrival of the policemen (Alexiadou 2019: 357)

In terms of argument structure, VoiceP-nominalizations based on transitive verbs have been likened to passive structures (Grimshaw 1990, Alexiadou 2001, Bruening 2013, Imanishi 2014, Alexiadou 2019, Borer 2021). According to Alexiadou (2001), the particular kind of Voice found in nominalizations neither introduces an external argument, nor assigns accusative case (a point which will prove important); Grimshaw (1990) notes that external arguments are demoted to adjunct, cf. (1a); Bruening (2013) claims that *n* is like passivization in selecting VoiceP complements lacking an external argument in their specifier; and Borer (2021) argues that internal arguments pass subjecthood diagnostics just like in verbal passives.

It should be emphasized that these approaches mostly do not take the verbal input to *n* to undergo the same process of passivization found in regular verbal contexts; if this were the case, VoiceP-nominalizations should always host passive morphology – contrary to fact.²⁸⁴ Rather, the idea is that they resemble verbal passives in certain regards, with consensus that Voice under nominalization may in some sense be defective compared to regular verbal contexts. Most relevant for us is the generalization that – while the level at which *n* merges may predictably affect linking – it cannot predictably restrict a verb’s stem-level transitivity, and hence does not suffice to explain the Mayan RANT.

12.3.3 The RANT is due to a selectional requirement of *n*

So neither nominalization, nor the specific level at which nominalization occurs per se can explain the RANT. I propose that a major part of the solution comes from a selectional requirement imposed by certain nominalizers on their verbal complement.

The idea that nominalization may impose restrictions on its complement is familiar from the previous section. Recall that *n* tends to restrict the temporal (Kornfilt & Whitman 2011) or eventive structure of its complement (Fabregas 2010, Sichel 2010) to a proper subset of the range of values it can take in regular verbal contexts, and authors have proposed various ways in which complements to *n* may be restricted syntactically, e.g. in terms of the obligatory non-projection of external arguments (Alexiadou 2001, Bruening 2013, Borer 2021). These restrictions all have in common that the potential of the verbal complement is somehow defective, and they are consistent with broader generalizations in the literature that nominalizations support

²⁸⁴See Šereikaitė (2022) for other arguments that VoiceP-nominalizations do not involve passivization, and that they may license a full-fledged transitive syntax.

less functional verbal structure than their purely verbal counterparts; see e.g. Comrie (1976), Croft (1991), Koptjevskaja-Tamm (1993), Malchukov (2006), Sichel (2010), and references therein.

We next need to identify what exactly in the verbal complement it is that *n* restricts. To do this, we will consider the distribution of various kinds of nominalizers as a function of their verbal complement, both within and across languages. Insofar as it is made available in the relevant sources, this information is compiled in Table 12.2. Table 12.2 shows for a given language (i) whether its locus of Set B morphology is high or low, (ii) whether or not it exhibits the RANT, and (iii) the distribution of nominalizers depending on the type of the verbal complement. Table 12.2 also indicates for transitive stems whether the resulting nominal takes a full DP object, or an obligatorily bare object (via subscripts). Pseudo-passive forms are also indicated via subscripts. In cases where the verb lacks an overt nominalizing suffix, whichever overt suffix productively appears is additionally referenced.²⁸⁵ For transitive verbal complements without valence-changing morphology, subscripts indicate whether the object can be a full DP or must be bare N(P). Note that (i) asterisks indicate ungrammaticality, (ii) question marks indicate slight uncertainty, (iii) empty cells indicate that the relevant data were missing or unclear, and (iv) only productive nominalizers were considered.²⁸⁶

The entry on Kaqchikel, for example, reads as follows: Kaqchikel is a HIGH-ABS RANT language; transitive verbs can only occur without a valence change in the form of *-oj* nominals, which comprise underived transitive roots and bare objects; unaccusative, unergative, pseudo-passive and anti-passive stems all share the suffix *-ik*; Agent Focus is unavailable as a workaround to the RANT. Chontal, on the other hand, is a LOW-ABS non-RANT language; transitive VPs (derived or not) with full DP objects can hence occur in NFCs without any kind of workaround; even though non-finite intransitive stems lack overt nominalizing suffixes, they always bear an overt imperfective suffix regardless of matrix aspect; the question of Agent Focus as a workaround in NFCs does not apply because Chontal lacks Agent Focus.

²⁸⁵There is a strong tendency for verb stems in Mayan NFCs to bear either an overt nominalizing suffix but not an (overt) status suffix, or vice versa, but not both; cf. footnote 288.

²⁸⁶The following works on individual languages were used for Table 12.2: Kaqchikel: García Matzar & Rodríguez Guaján (1997), Ajsivinac Sián (2007), Imanishi & Mateo Pedro (2013), Imanishi (2020) and my own notes (SR); K'ichee': Can Pixabaj (2009), Henderson (2012), Can Pixabaj (2015) and Can Pixabaj & Aissen (2021); Sakapulteko: DuBois (1981); Sipakapense: Barrett (1999); Tz'utujil: Dayley (1985), Levin et al. (2020), Lyskawa & Ranero (2021a) and my own notes (RP); Mam: England (1983), England (1988) and England (2013); Ch'ol: Gutiérrez Sánchez (2004), Coon (2010), Coon (2013), Vázquez Álvarez (2011); Chontal: Knowles (1984); Chuj: Buenrostro (2004), Buenrostro (2007), Buenrostro (2013), Coon & Carolan (2017), Coon (2019) and my own notes (MT); Popti' (also known as Jakaltek): Craig (1977) and Craig (1979); Q'anjob'al: Mateo Toledo (2003), Mateo Pedro (2010), Mateo Toledo (2011), Mateo Toledo (2013), Preminger (2012) and my own notes (PMP); Itzaj: Hofling (2000); Yucatec: Bricker (1978), Bricker (1981), Bohmeyer (2002), Gutiérrez Bravo (2002) and Verhoeven (2007). Additionally, the following works on generalizations across multiple languages were used: Coon et al. (2014), Aissen (2017a), Coon et al. (2021) and Coon & Royer (2021). Note that Achi (Sis Iboy 2007) and Uspantek (Can Pixabaj 2009) show profiles which are consistent with those of the other RANT languages in Table 12.2, but have been excluded due to numerous gaps in the data.

Branch	Language	ABS	RANT	Nominalizer as a function of verbal complement					
				transitive	unaccusative	unergative	passive	anti-passive	Agent Focus
K'ichean	Kaqchikel	HIGH	✓	$-oj\sqrt{+NP}$	$-ik^a$	$-ik^a$	$-ik_{pseudo}$	$-ik$	*
	K'ichee'	HIGH	✓	$-oj\sqrt{+NP} / -ow\sqrt{+NP}$	$-iik^{b,d}$	$-iik / -VVm^{b,c}$	$-iik_{pseudo}$	$-iik / -VVm$	*
	Sakapulteko	HIGH	✓	$-ow_{NP?}$	$-ik^c$	$-ik / -VVm^c$	$-ik_{pseudo}$	$-ik / -VVm$	$-ik?$
	Sipakapense	HIGH	✓		$-ik$	$-ik$	$-ik_{pseudo}$	$-ik?$	
	Tz'utujil	HIGH	✓	$-n\sqrt{+NP} / -ooj\sqrt{+NP}$	$-ik^a$	$-VVm^e$	$-ik_{pseudo}$	$-VVm^e$	*
Mamean	Mam	HIGH	✓	$-l_{DP_{obl}}$	$-l$				
Tzeltalan	Ch'ol	LOW	*	$-\emptyset_{DP}$	$-el$				n.a.
	Chontal	LOW	*	$-\emptyset_{DP}$	$-\emptyset$				n.a.
					(cf. IMPF)				
Q'anjobalan	Chuj	HIGH	✓	$-oj\sqrt{+NP}$	$-\emptyset$				
					(cf. STAT.INTR) ^d				
	Popti'	HIGH	✓	$-o\sqrt{+NP} / -a\sqrt{+NP}$	$-oj$			$-l$	
	Q'anjob'al	HIGH	✓	$-oj\sqrt{+NP}$	$-\emptyset$				
				$-oj\sqrt{+NP}$	(cf. STAT.INTR) ^d				
Yucatecan	Itzaj	LOW	*	$-VI_{DP}$	$-VI / -ik?$				n.a.
	Yucatec	LOW	*	$-\emptyset?_{DP}$	$-\emptyset?$				n.a. ^f

a. See Burukina (2021a) for agreement data showing that Kaqchikel syntactically distinguishes unergatives from unaccusatives. Her diagnostics are based on Levin et al. (2020) and Lyskawa & Ranero (2021a) for Tz'utujil de Santiago, which shows an analogous distinction.

b. Explicit phrase-structural evidence for the unergative-unaccusative distinction in K'ichee' is unavailable, but verbs that are consistent with an unergative and unaccusative analysis, respectively, systematically either allow only *-iik*, or allow both *-iik* and *-VVm* (Can Pixabaj 2009: 25ff).

c. With respect to verbs that are consistent with an unergative and unaccusative analysis, respectively.

d. Subject to language-specific syntactic and prosodic constraints on the distribution of status suffixes. In K'ichee', the nominalizer *-iik* is partially homophonous with the intransitive status suffix, but these are actually distinct as only the latter may delete phrase-finally (Henderson 2012, Can Pixabaj 2015, Can Pixabaj & Aissen 2021).

e. My Tz'utujil consultant, who speaks the Santiago dialect, only sometimes accepted non-finite unergative and anti-passives stem with the suffix *-VVm*, preferring so-called unergative nouns.

f. In Yucatec, the so-called Agent Focus construction differs considerably in both form and function from other Mayan languages; consistent with this is the fact that it is LOW-ABS. See e.g. Bricker (1978), Tonhauser (2007) and Norcliffe (2009).

Table 12.2: Mayan nominalizers as a function of verbal complement

Keep in mind that the following generalizations are all established with respect to only those languages sampled in Table 12.2. Also, a few languages absent from Table 12.2 will be discussed separately due to being outliers.

First, RANT languages lack nominalizers that can take transitive VPs with full DP objects (the entire K'ichean branch, Mam, and the entire Q'anjobalan branch). Instead, there is a class of nominalizers that attaches exclusively to intransitives, i.e. unaccusative, unergative, (pseudo)-passive, anti-passive and Agent Focus stems.

Among these languages, we find two sub-patterns: in the first sub-pattern – found in Kaqchikel, Q’eqchi’, Sipakapense, Chuj and Q’anjob’al – there is a single nominalizer for all intransitive stems, which I label $n_{\text{VOICE.INTR}}$:

(30) **Kaqchikel:** $n_{\text{VOICE.INTR}}$ takes intransitive stems

a. *Unaccusative*

[Ri ru-tzaq-**ïk** ri a Juan] üt
 DET A.3SG-fall-NMLZ DET CLF J. good
 ‘Juan’s falling is good’

b. *Unergative*

Rat x-Ø-a-chäp [atin-**ïk**]
 2SG COMPL-B.3SG-A.2SG-begin bathe-NMLZ
 ‘You began to bathe’

c. *Pseudo-passive*

Röj y-øj-ajin [che ki-q’ete-x-**ïk** ri ak’wal-a]
 we INCOMPL-B.1PL-PROG PREP A.3PL-hug-PASS-NMLZ DET child-PL
 ‘We are hugging the children’ (Imanishi 2020: 19, 21, 12)

d. *Anti-passive*

X-Ø-u-chäp [q’et-e-n-**ïk** r-ichin ri
 COMPL-B.3SG-A.3SG-begin hug-SUF-ANTIP-NMLZ A.3SG-RN.OF DET
 ak’wal]
 child
 ‘He began to hug the child’

(García Matzar & Rodríguez Guaján 1997: 457)

The second sub-pattern – found in K’ichee’, Sakapulteko and Tz’utujil – involves a partition along what can be characterized as an unaccusative-unergative dimension: unergatives and anti-passive stems take one nominalizer, which I label $n_{\text{VOICE.INTR.1}}$; unaccusatives and pseudo-passives take another one, which I label $n_{\text{VOICE.INTR.2}}$.²⁸⁷

²⁸⁷Of these languages, only Tz’utujil and Kaqchikel provide first-hand evidence for a robust phrase-structural distinction between unaccusatives and unergatives, as diagnosed by agreement data for Tz’utujil (Levin et al. 2020, Lyskawa & Ranero 2021a) and Kaqchikel (Burukina 2021a). For K’ichee’ and Sakapulteko, this analysis is based indirectly on the documented verbs’ consistency with an unaccusative and unergative analysis, respectively. Yet the given (i) the fact that the distinction exists in Tz’utujil and Kaqchikel, and (ii) the obviously cognate status of the suffix -VVm among these languages, we have clear support for this distinction.

- (31) **Tz'utujil:** $n_{\text{VOICE.INTR.1}}$ takes unergative or anti-passive stems

a. *Unergative*

X-Ø-qaa-maj [wa'-iim]
COMPL-B.3SG-A.1PL-begin eat-NMLZ

'We began to eat' (Dayley 1985: 393)

b. *Anti-passive*

[Ch'i-on-eem] m=utz ta
hit-ANTIP-NMLZ NEG=good NEG

'Hitting is not good' (Santiago dialect; own notes: RP)

- (32) **Tz'utujil:** $n_{\text{VOICE.INTR.2}}$ takes unaccusative or pseudo-passive stems

a. *Unaccusative*

[R-yar-ik a-welado] kungan b'iis
A.3SG-melt-NMLZ A.1SG-ice.cream much sadness

'My ice cream's melting is very sad' (Santiago dialect; own notes, RP)

b. *Pseudo-passive*

X-Ø-qaa-maj [ki-ch'e<j>y-iik]
COMPL-B.3SG-A.1PL-begin A.3PL-hit<PASS>-NMLZ

'We began to hit them' (Dayley 1985: 393)

Note that $n_{\text{VOICE.INTR.1}}$ occurs either in addition to also allowing $n_{\text{VOICE.INTR.2}}$ (K'ichee', Sakapulteko) or in complementary distribution with $n_{\text{VOICE.INTR.2}}$ (Tz'utujil).

Second, all RANT languages have at their disposal the familiar *-oj* nominals as a mixed workaround, which always involve an underived transitive root and a bare object. I call this particular nominalizer n_{ROOT} :

- (33) **Chuj:** n_{ROOT} takes transitive root with bare object

Ix-a-yamoch [mol-øj kape]
PFV-A.2SG-begin gather-NMLZ coffee

'You began coffee-gathering' (Coon & Royer 2021: 11)

In non-RANT languages, *-oj* nominals appear to be absent entirely, i.e. they seem to lack n_{ROOT} .

We can thus characterize the nominalizing inventories of RANT languages as follows: first, most RANT languages show a broad partition between n_{ROOT} (for underived transitive roots) and n_{INTR} (for intransitive stems). Additionally, RANT languages may induce a sub-partition on n_{INTR} between $n_{\text{VOICE.INTR.2}}$ (for unaccusatives and pseudo-passives) and $n_{\text{VOICE.INTR.1}}$ (for unergatives and anti-passives).

Turning to non-RANT languages (Ch'ol, Chontal, Itzaj and Yucatec), these unsurprisingly all have a nominalizer which readily takes transitive VPs with full DP objects, and which I label n_{TRANS} :

(34) **Ch'ol:** $n_{\text{VOICE.TR}}$ *takes transitive stems*

K-om [j-kãñ-ø-ety]
 A.1-want A.1-know-NMLZ-B.2

'I want to speak to you'

(Vázquez Álvarez 2011: 99)

What we also find is that in no non-RANT language sampled here, nominalizers for intransitive inputs shows a sub-partition along an unaccusative-unergative (or some other) dimension. Rather, there is what appears to be a single $n_{\text{VOICE.INTR}}$ for intransitives alike:

(35) **Ch'ol:** $n_{\text{VOICE.INTR}}$ *takes intransitive stems*a. *Unaccusative base*

Choñkol [k-yajl-**el**]
 PROG A.1-fall-NMLZ

'I'm falling'

(Coon 2010: 136)

b. *Passive base*

Choñkol [k-mejk'-**el**]
 PROG A.1-hug.PASS-NMLZ

'I'm being hugged'

(Coon 2010: 136)

c. *Anti-passive base*

Ma'añ mi i-mul-añ [mãñ-oñ-**el**]
 NEG.EXT IMPF A.3-like-D.NML buy-ANTIP-NMLZ

'She doesn't like buying'

(Coon 2010: 130)

In Ch'ol and also Q'eqchi', $n_{\text{VOICE.TRANS}}$ and $n_{\text{VOICE.INTR}}$ are formally distinct, while in the other non-RANT languages, both are null. I assume that two distinct suffixes may be involved, with potential homophony.²⁸⁸

The final, perhaps most important generalization about Table 12.2 is that all RANT languages are HIGH-ABS languages.

These results are summarized for RANT languages in Table 14.4, and for non-RANT languages in Table 14.5, in which adjacent cells with the same shading indicate the same nominalizer:

²⁸⁸In Chuj and Q'anjob'al, n_{INTR} is null, while the phrase-final intransitive status suffix is overt (Coon & Carolan 2017). The situation is reversed in Ch'ol, i.e. Ch'ol NFCs show an overt n_{INTR} but no intransitive status suffix (cf. (35)). As noted in Coon & Carolan (2017: 24), the fact that we do not find two overt morphemes is probably a morphological accident, given (i) the tendency for status suffixes to delete in non-phrase-final position, and (ii) the fact that vowel hiatus is frequently resolved by syncope.

Verbal complement	n _{ROOT}	n _{VOICE}			
		n _{VOICE.TR}	n _{VOICE.INTR}	n _{VOICE.INTR.1}	n _{VOICE.INTR.2}
Underived transitive root + bare NP object					
Transitive (no workaround) + DP object					
Unergative					
Anti-passive					
Unaccusative					
(Pseudo-)passive					
Agent focus					

Table 12.3: Mayan nominalizers in RANT languages

Verbal complement	n _{ROOT}	n _{VOICE}			
		n _{VOICE.TR}	n _{VOICE.INTR}	n _{VOICE.INTR.1}	n _{VOICE.INTR.2}
Underived transitive root + bare object					
Transitive stem + DP object					
Unergative					
Anti-passive					
Unaccusative					
(Pseudo-)passive					
Agent focus	n.a.	n.a.	n.a.	n.a.	n.a.

Table 12.4: Mayan nominalizers in non-RANT languages

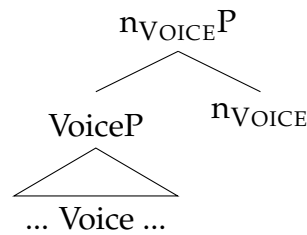
The picture that emerges is that in the context of nominalization, Mayan grammars are systematically sensitive to transitivity.²⁸⁹ This sensitivity manifests prominently as the RANT, which requires languages to resort to a detransitivizing workaround

²⁸⁹The same general conclusion is reached in Mateo Pedro (2010), although this work glosses over several fine-grained details observed here, such as the connection between *-oj* nominals and obligatorily bare objects, or the sensitivity of nominalizers to the unergative-unaccusative distinction.

in NFCs; but it is also present in non-RANT languages, which robustly distinguish nominalizers for transitive and intransitive stems, respectively.

12.3.3.1 Selection between *n* and Voice

I propose that all nominalizers other than n_{ROOT} – i.e. $n_{\text{VOICE.TR}}$, $n_{\text{VOICE.INTR}}$, $n_{\text{VOICE.INTR.1}}$ and $n_{\text{VOICE.INTR.2}}$ – select VoicePs. For this reason, they will be grouped under the label n_{VOICE} . From here on, we will be interested in the class of n_{VOICE} as a whole, only returning to potential further sub-divisions briefly at the end. There is hence a selectional relationship between n_{VOICE} and Voice:



Tree 12.7: n_{VOICE} selects Voice

Let us review the evidence for this. First, there is the transparent fact that complements to n_{VOICE} productively contain overt anti-passive, (pseudo-)passive and Agent Focus derivations, which I assume spell out distinct kinds of Voice.

Second, that it is specifically Voice which can be selected for is supported by the fact that there exist nominalizers which specifically do not select Voice, i.e. n_{ROOT} . We know that n_{ROOT} selects underived transitive roots, which – given their maximally reduced structure – necessarily lack Voice (Coon & Royer 2021).²⁹⁰

Third, a restrictive selectional relationship between *n* and Voice helps explain why in the absence of an object, a transitive verb in a RANT language must employ a workaround regardless. Assuming that (i) the formal absence of an object entails its syntactic absence (in turn supported by the absence of Set A/B agreement), and that (ii) some Mayan languages have at their disposal an operation of unspecified object deletion in spite of their rigid transitivity, this restriction is unexpected if stated in purely Case-theoretic terms, but ceases to be so if restated in terms of selection between *n* and Voice. Put differently, the workaround no longer occurs in order to bring about an alternative Case source for the internal argument, but rather because it is mandated by *n*, independently of the internal argument. The object's need for Case is in this way only indirectly tied to non-finiteness, being mediated by Voice.

Finally, from a conceptual standpoint, it is unsurprising that the RANT should manifest as restrictions on Voice given that (i) Voice introduces external arguments, (ii) Voice may Case-license up to two arguments (depending on the language and

²⁹⁰I leave open the issue of nominalizers selecting specifically VP complements in Mayan.

construction), and (iii) VoiceP constitutes the clausal locus of transitivity alternations; see e.g. Nie (2020) for recent work in this regard, among many others.

12.3.3.2 Selection between *n* and unaccusatives

Before moving on, a few words are in order about the presence of Voice in unaccusatives. While it is uncontroversial that unergatives, transitives, passives and anti-passives contain Voice, unaccusatives are generally assumed to lack Voice. The reasoning is that because Voice introduces agentive external arguments, but unaccusatives are non-agentive, they must lack Voice. Specifically in our case, however, endowing unaccusatives with Voice seems like a desirable move because we would very easily be able to model the fact that they pattern together either with pseudo-passives (when selected by $n_{\text{VOICE.INTR.2}}$), or with all other intransitives (when selected by n_{INTR}). So having unaccusatives contain Voice in light of my assumption that selection makes reference to Voice would make possible a uniform and parsimonious treatment of the elements falling within n_{VOICE} 's selection range.

Fortunately, scholars have come some way to slightly refine this assumption. Based on the existence of a certain class of unaccusatives, specifically marked anti-causatives – i.e. semantically causative intransitives lacking agentivity, but which are nevertheless marked morphologically – work such as Alexiadou et al. (2015) and (Oikonomou & Alexiadou 2022) argues for the dissociation of the semantic and the morpho-syntactic import of Voice, to the effect that Voice may be present syntactically and expounded morphologically, yet remain devoid of a semantic contribution.²⁹¹ In this system, regular Voice has the denotation in (36a), and expletive Voice that in (36b) – the former returns a one-place predicate requiring an agentive external argument, while the latter returns the same predicate that was passed into it, in need of no argument:

- (36) a. $\llbracket \text{VOICE}_{\text{Agt}} \rrbracket = \lambda P_{\langle v,t \rangle} \lambda x \lambda e. [P(e) \ \& \ \text{AGENT} = x] \quad \langle vt, evt \rangle$
 b. $\llbracket \text{VOICE}_{\text{Expl}} \rrbracket = \lambda P_{\langle v,t \rangle} \lambda e. [P(e)] \quad \langle vt, vt \rangle$
- (Oikonomou & Alexiadou 2022: 22)

This view predicts that we should in principle find unaccusative verbs which are morphologically marked; this is borne out.²⁹² In Tz'utujil, for example, numerous unaccusatives are derived from nouns or adjectives via the suffix $-(V)r$:²⁹³

²⁹¹This proposal builds on Alexiadou et al. (2006), Schäfer (2008) and Alexiadou (2010); see also Oikonomou & Alexiadou (2022), and references in these works.

²⁹²Note that this prediction would equally confirm an analysis which sources the morphology found in unaccusatives in *v*, rather than Voice. See Coon (2019) for arguments that Mayan languages bundle Voice and *v* into a single head.

²⁹³Elicitation with my consultant showed that the verbs in (37) pass the anti-causative diagnostics laid out in Alexiadou et al. (2015): they support agentivity-denying modifiers like *by itself*, allow Causer *from*-phrases, and reject agentivity-entailing modifiers like Agent or Instrument *by*-phrases.

(37) **Tz'utujil**

- a. X-Ø-ya'-r-i
COMPL-B.3SG-water-V-STAT.INTR
'It melted'
- b. X-Ø-kaq-ar-i
COMPL-B.3SG-red-V-STAT.INTR
'It reddened'

(Dayley 1985: 122)

Dayley (1985: 122) notes that the suffix $-(V)r$ also derives passives from a handful of transitive verbs. This can be seen as further support for the view that (a subset of) unaccusatives contain Voice, and that the presence of Voice need not coincide with the expression of agentivity, just like anti-causatives derived by reflexive or passive morphology do not have reflexive and passive meanings, respectively.^{294,295}

As for those unaccusatives lacking overt derivational morphology, we can assume that expletive Voice is in these cases null. This assumption is actually widespread in work that assumes Voice, usually implicitly, given that many languages do not overtly spell out (active and transitive) Voice. The overtness of expletive Voice as a designated morpheme that is separate from the verb may further be obscured by the bundling of Voice and the verbalizer *v* into a single head (Pylkkänen 2008), a property which Mayan languages have independently been argued to instantiate (Harley 2017, Coon 2019).

An alternative to assuming expletive Voice in unaccusatives would be to assume that – in keeping with the standard view – unaccusatives lack Voice but project up to *vP*, and that the particular kind of *v* found in unaccusatives has whichever property necessary for it to pattern with complements which unquestionably contain Voice.

I leave this issue open, but choose the expletive Voice option here for convenience.

12.3.3.3 Selection between *n* and verbal roots

Lastly, I follow Coon & Carolan (2017), Burukina (2021b) and Coon & Royer (2021) in assuming that in *-oj* nominals like (38), *n*_{ROOT} merges with smaller root projections that preclude higher functional material, including Voice.

²⁹⁴Relatedly, it is argued in Embick (1997) and Embick (1998) that what is classified as passive morphology may realize all structures lacking external arguments, including anti-causatives, irrespective of whether it also ends up receiving a passive interpretation.

²⁹⁵Cross-linguistically, there are no principled and consistent semantic and thematic differences between marked and unmarked anti-causatives (Alexiadou et al. 2015), although there is an overall tendency for anti-causative morphology to occur on those verbs which express changes that are construed as less likely to occur spontaneously, compared to unmarked anti-causatives (Haspelmath 1993).

(38) **Chuj**

Ix-a-yamoch [mol-**oj** kape]
PFV-A.2SG-begin gather-NMLZ coffee

‘I began coffee-gathering’

(Coon & Royer 2021: 11)

12.4 Summary

This chapter discussed existing approaches to the RANT, concluding that neither the syntactic ergativity approach, the Case approach nor Imanishi’s nominalization approach as it stands can explain it. Imanishi’s account is nonetheless on the right track because it recognizes that nominalization plays a crucial role in giving rise to the RANT. We then saw that the RANT does not follow from nominalization or the level of nominalization per se, but rather from a selectional requirement holding between *n* and its verbal complement, with *n*_{VOICE} selecting VoicePs and *n*_{ROOT} smaller root projections.

Chapter 13

The internal syntax of transitivity-restricted NFCs

13.1 Introduction

This chapter delves into the internal syntax of complements to n_{VOICE} and n_{ROOT} in RANT languages. With n_{VOICE} , the VoiceP-level NFCs considered show anti-passive, agent focus or pseudo-passive workarounds. We will see that they share a PRO external argument in Spec,VoiceP, whereas the internal argument projects depending on the Case potential associated with each particular strategy. This is expected with anti-passive and agent focus NFCs, but less so with pseudo-passive NFCs in light of their passive veneer.

We begin with pseudo-passive NFCs in Section 13.2, then move on to agent focus and anti-passive NFCs in Sections 13.3 and 13.4, respectively. Next to these verb-based workarounds, RANT languages also have *-oj* nominals, which constitute a mixed workaround strategy; these are the topic of Section 13.6.

13.2 Pseudo-passive NFCs

We begin with pseudo-passive NFCs because much of what these reveal extends directly to agent focus and anti-passive NFCs.

The pseudo-passive strategy is attested in at least Kaqchikel, K'ichee', Sakapulteko, Sipakapense, Tz'utujil and Tojol'ab'al. In this strategy, favored especially within the K'ichean branch, the verb carries passive morphology, and the internal argument is realized structurally and φ -marked via Set A on the verb:

(1) *Pseudo-passive NFCs*a. **Tz'utujil de San Juan**

X-Ø-qaa-maj [ki-ch'e<j>y-iik]
 COMPL-B-3SG-A.1PL-begin A.3PL-hit.PASS-NMLZ
 'We began to hit them' (Dayley 1985: 393)

b. **Kaqchikel**

Röj y-oj-ajin [che ki-q'ete-x-ik ri ak'wal-a]
 we INCOMPL-B.1PL-PROG PREP A.3PL-hug-PASS-NMLZ DET child-PL
 'We are hugging the children' (Imanishi 2020: 12)

As we know, pseudo-passive NFCs have active meanings. In this section, we will see that even though pseudo-passive NFCs are morphologically passive, their syntax is indeed active, as surmised in (Can Pixabaj 2015). Evidence for this comes from (i) their robustly active argument structure, (ii) the fact that their pseudo-passive property is immanent within the pseudo-passive NFC itself, and (iii) the presence of a PRO external argument. Pseudo-passive NFCs thus instantiate a kind of contextual deponency, contrasting with other formally passive NFCs showing truly passive behavior.

13.2.1 Two types of formally passive non-finite clauses

Pseudo-passive NFCs contrast sharply with other, likewise formally passive NFCs found in languages like Q'eqchi', Chuj and Q'anjob'al. The former have an active meaning (2), whereas the latter retain an interpretation consistent with their passive morphology (3). In order to distinguish them from pseudo-passive NFCs like in (2), I will refer to formally passive NFCs like those in (3), with predictably passive meanings given their morphology, as *true passive NFCs*:

(2) *Pseudo-passive NFCs*a. **Kaqchikel**

Röj x-Ø-qa-chäp [ki-q'ete-x-ik ri ak'wal-a']
 we COMPL-B.3SG-A.1PL-begin A.3PL-hug-PASS-NMLZ DET child-PL
 'We began to hug the children' (Imanishi 2020: 22)

b. **Sakapulteko**

K-ee-tin cha [tiix-iik]
 INCOMPL-B.3PL PREP eat.PASS-NMLZ
 'They are eating' (DuBois 1981: 254)

c. **Tz'utujil de Santiago**

Atet x-Ø-a-moj [n-ch'iy-ik nen]
 2SG COMPL-B.3SG-A.2SG-start A.1SG-hit.PASS-NMLZ 1SG
 'You started hitting me' (own notes: RP)

(3) *True passive NFCs*a. **Q'eqchi'**

T-in-xic chi [ban-e'-c]
 INCOMPL-B.1SG-go PREP cure-PASS-INF

'I will go to be cured'

(Berinstein 1985: 262)

b. **Chuj**

Ix-in-yamoch [hin-chel-chaj-i]
 COMPL-A.1SG-begin A.1SG-hug-PASS-ITV

'I began to be hugged'

(Coon & Carolan 2017: 27)

c. **Q'anjob'al**

Max-Ø xew [ha-maq'-lay-i]
 COMPL-B.3SG finish A.2SG-hit-PASS-SS.INTR

'You finished being hit'

(own notes: PMP)

I term the puzzling interpretive dependency which pseudo-passive NFCs give rise to – i.e. the fact that the agent position of the formally passive verb remains accessible to control – as *passive agent control*.

13.2.2 Pseudo-passive NFCs are morphologically passive

Pseudo-passive NFCs involve a marker that is identical to the passive marker found in finite contexts. This is made evident by stem-conditioned allomorphy, illustrated here for Tz'utujil.

In Tz'utujil, a transitive verb requires a distinct passive allomorph depending on whether it is a root transitive or a derived transitive. The root transitive *ch'ey* 'hit', for instance, takes the vocalic infix -V- (4a) (whose precise form seems to vary lexically), but the derived transitive *k'ayeej* 'bite' takes the segmental suffix -x (4b):²⁹⁶

(4) **Tz'utujil de Santiago: Finite passives**a. *Root transitive* → -V- allomorph

Anen x-in-cha'ay-a (aw-maaq tet)
 1SG COMPL-B.1SG-hit.PASS-STAT.INTR A.2SG-RN.BY 2SG
 'I was hit (by you)'

b. *Derived transitive* → -x allomorph

Jaw ak' x-Ø-k'ay-e-x-a (r-maaq
 the chicken COMPL-B.3SG-bite-SUF-PASS-STAT.INTR A.3SG-RN.BY
 siaw)
 cat

'The chicken was bitten (by the cat)'

(own notes: RP)

NFCs preserve this allomorphic behavior: the non-finite form of the root transitive

²⁹⁶In (4a), the surface form *cha'ay-a* is the result of what appears to be regressive vowel harmony, conditioned by the intransitive status suffix -a. See Lyskawa & Ranero (2021c).

‘hit’ takes the vocalic infix -V- (5a), while the non-finite form of the derived transitive ‘bite’ takes the segmental suffix -x (5b):²⁹⁷

(5) **Tz’utujil de Santiago: Non-finite passives**

a. *Root transitive* → -V- allomorph

Atet x-Ø-a-moj [n-ch’iy-ik nen]
 2SG COMPL-B.3SG-A.2SG-start A.1SG-hit.PASS-NMLZ 1SG
 ‘You started hitting me’

b. *Derived transitive* → -x allomorph

Ja siaw x-Ø-u-moj [r-k’ay-[e]-x-ik jaw
 the cat COMPL-B.3SG-A.3SG-start A.3SG-bite-V-PASS-NMLZ the
 ak’]
 chicken
 ‘The cat started biting the chicken’ (own notes: RP)

In just the same way, stem-conditioned allomorphy in closely related Kaqchikel reveals that pseudo-passive NFCs involve a marker that is identical to the passive marker in finite contexts (root transitives with lax vowels undergo vowel tensing, while derived transitive take the suffix -x) (Imanishi & Mateo Pedro 2013). The same is true in other languages with pseudo-passive NFCs, i.e. K’ichee’ (Can Pixabaj 2015), Sakapulteko (DuBois 1981), Sipakapense (Barrett 1999) and Tojol’ab’al (Curiel Ramírez del Prado 2017).

13.2.3 Pseudo-passive NFCs have an active argument structure

In this section, we will see that pseudo-passive NFCs have a robustly active argument structure, with the embedded DP systematically interpreted as patient and the matrix DP – if present – systematically as agent. Moreover, if the pseudo-passive NFC occurs without either a clause-internal object or a matrix argument, its active meaning remains. Next to showing us that – unsurprisingly – the embedded DP is an object of the pseudo-passive verb, these facts confirm that the meaning of pseudo-passive NFCs is truly active, and not just a matter of diathetical framing, and that this active meaning is immanent within the pseudo-passive NFC itself.²⁹⁸ The data in this section are drawn in large part from Tz’utujil.

In the sentence involving a phasal matrix verb and a pseudo-passive NFC in (6), the only interpretation available is that the cat started biting the chicken. This thematic relation is fixed, i.e. (6) cannot mean that the chicken started biting the cat, or – equivalently, yet expressed in a passive frame – that the cat started being bitten by the chicken:

²⁹⁷In (5a), the presence of the vocalic infix -V- is slightly obscured by the application of both regressive vowel harmony and subsequent vowel shortening. See Dayley (1985: 45ff).

²⁹⁸Cf. *Alma read the book* vs *The book was read by Alma*, which are truth-conditionally equivalent yet different in how they are framed via diathesis.

(6) **Tz'utujil de Santiago**

Ja siaw x-Ø-u-moj [r-k'ay-[e]-x-ik jaw ak']
 the cat COMPL-B.3SG-A.3SG-start A.3SG-bite-V-PASS-NMLZ the chicken

'The cat started biting the chicken'

*'The chicken started biting the cat'

*'The cat started being bitten by the chicken' (own notes: RP)

This interpretation is independent of the presence of an embedded thematic object, and also Set A marking: if both of these are removed, the matrix DP retains an active interpretation with respect to the hitting (and starting), with the understanding that there is someone or something that the referent of the matrix DP is hitting:²⁹⁹

(7) **Tz'utujil de Santiago**

Atet x-Ø-a-moj [ch'iy-ik]
 2SG COMPL-B.3SG-A.2SG-start hit.PASS-NMLZ

'You started hitting (someone / something)'

*'You started being hit'

*'Someone started to hit you' (own notes: RP)

To see that this property is not specific to Tz'utujil, consider the pair in (8) from Sakapulteko. Regardless of whether the formally passive verb has an object to index (8a) or not (8b), the interpretation of the embedded clause remains active with respect to the matrix DP, with the understanding that this DP's referent is acting on someone or something:

(8) **Sakapulteko**

a. X-im-pee cha [aa-ch'iyy-iik]
 COMPL-B.1SG-come PREP A.2SG-hit.PASS-NMLZ
 'I came to hit you'

b. K-ee-tin cha [tiix-iik]
 INCOMPL-B.3PL PREP eat.PASS-NMLZ
 'They are eating'

(DuBois 1981: 254)

²⁹⁹This argument is based on minimal pairs like the following, which show that the interpretation of the embedded subject position correlates with the presence vs absence of a matrix object (I take (ib) to involve Exceptional Case Marking):

- (i) a. *I want to be loved*
 b. *I want him to be loved*

The same holds for Kaqchikel (9):

(9) **Kaqchikel**

Ri ixoq n-Ø-ajin chi [war-sa-x-ik]
the woman INCOMPL-B.3SG-PROG PREP sleep-CAUS-PASS-NMLZ

‘The woman is causing to sleep’

*‘The woman is being caused to sleep’ (own notes: SR)

So far, the matrix predicates involved have been thematic verbs, i.e. verbs which introduce their own subjects, meaning that there is a matrix argument with potential influence on the meaning of the embedded clause in terms of argument identification. In order to control for this potential confound, we can replace thematic matrix predicates with athematic predicates such as evaluative adjectives, in which case the formally passive verb retains an active meaning (regardless of the presence of Set A marking):^{300,301}

(10) **Tz’utujil de Santiago**

- a. [Ch’iy-ik] m=utz ta
hit.PASS-NMLZ NEG=good IRR
‘Hitting is bad’
*‘Being hit is bad’
- b. [Mich’-x-ik] kungan peen
pluck-PASS-NMLZ much hard
‘Plucking is hard’
*‘Being plucked is hard’
- c. [R-kum-s-s-ik] m=utz ta
A.3SG-die-CAUS-PASS-NMLZ NEG=good IRR
‘Killing is bad’
*‘Being killed is bad’

(own notes: RP)

These facts show us that the pseudo-passive property is immanent in the pseudo-passive clause itself, and not conditioned by the matrix environment. They also show that the embedded DP has a consistent patient interpretation, and the matrix DP – if present – a consistent agent interpretation. Our expectation regarding the matrix DP, however, is that it should actually have a patient interpretation, brought about either via control of or raising from the embedded null subject position, which should be patientive given downstairs passive morphology.

(6) and (7) actually already show clearly that such a passive interpretation with respect to the matrix DP is unavailable. But let us bolster this generalization, and

³⁰⁰It seems that a generic, impersonal interpretation with respect to thematic objects is available in Tz’utujil both without Set A marking ((10a),(10b)), and with what I take to be default 3SG Set A marking (10c).

³⁰¹Next to vowel syncope, (10c) also involves regressive sibilant harmony between the causative suffix *-sa* and the segmental passive suffix *-x* (Lyskawa & Ranero 2021b).

try to force it by (i) using a scenario which strongly favors a passive construal, (ii) removing both Set A marking and an overt thematic object, and (iii) adding an oblique phrase intended to express the agent of the event denoted by the formally passive verb. (11) has all of these properties, yet the matrix DP *ixiim* ‘corn’ must refer to the eater, while the oblique DP *ak’* ‘chicken’ can at best be coerced into an indirect causer.³⁰²

(11) **Tz’utujil de Santiago**

- a. Jaw *ixiim ru-mjon* [ti’ij-ik r-maaq ak’]
 the corn A.3SG-PROG eat.PASS-NMLZ A.3SG-RN.BY chicken
 ‘The corn is eating (something) because of the chicken’ (dispreferred)
 *‘The chicken is eating the corn’
 *‘The corn is being eaten by the chicken’
- b. Jaw *ixiim x-Ø-u-moj* [ti’ij-ik r-maaq
 the corn COMPL-B.3SG-A.3SG-start eat.PASS-NMLZ A.3SG-RN.BY
 ak’]
 chicken
 ‘The corn started eating (something) because of the chicken’
 *‘The chicken started eating the corn’
 *‘The corn started being eaten by the chicken’ (own notes: RP)

We thus have conclusive evidence that if present, matrix DPs in constructions with embedded pseudo-passive NFCs have a consistent agent interpretation, that the embedded DP has a consistent patient interpretation, and that pseudo-passive NFCs have active meanings. This raises the question of the internal syntax of pseudo-passive NFCs, to which we turn next.

13.2.4 Pseudo-passive NFCs according to Imanishi

As of yet, there is no analysis of the internal syntax of pseudo-passive NFCs that adequately takes into account their pseudo-passive property. Coon (2010, 2013) on the one hand, and Imanishi (2020) on the other propose structures which, despite sharing empirical coverage, turn out to be inadequate as they are.

I begin with the *Imanishi variant*, based on his work on Kaqchikel. For pseudo-passive NFCs like in (12), Imanishi (2020) proposes the structure in Tree 13.1: a transitive verb is merged with a DP internal argument, passivized, and then nominalized, after which the internal argument receives Set A from the D head heading the nominalized clause.³⁰³

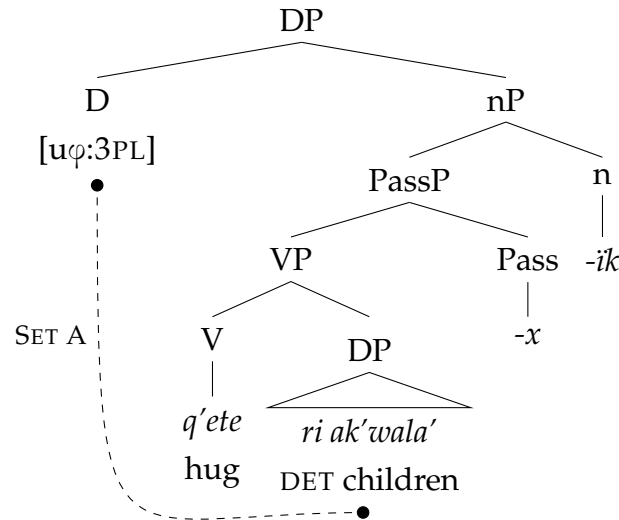
³⁰²In (11), the relational noun *-maa*q ‘by’ is the same one that is used to mark agents in finite passives:

- (i) Jaw *ixiim x-Ø-ti’ij-a* r-maaq ak’
 the corn COMPL-B.3SG-eat.PASS-STAT.INTR A.3SG-RN.BY chicken
 ‘The corn was eaten by the chicken’

³⁰³In Imanishi’s system, some instances of Set A are assigned by a phase head to the highest case-less

(12) **Kaqchikel:** *Pseudo-passive NFC*

- a. Røj x-ø-qa-chäp [ki-q'ete-x-ik ri ak'wal-a']
 we COMPL-B.3SG-A.1PL-begin A.3PL-hug-PASS-NMLZ DET child-PL
 'We began to hug the children' (Imanishi 2020: 22)

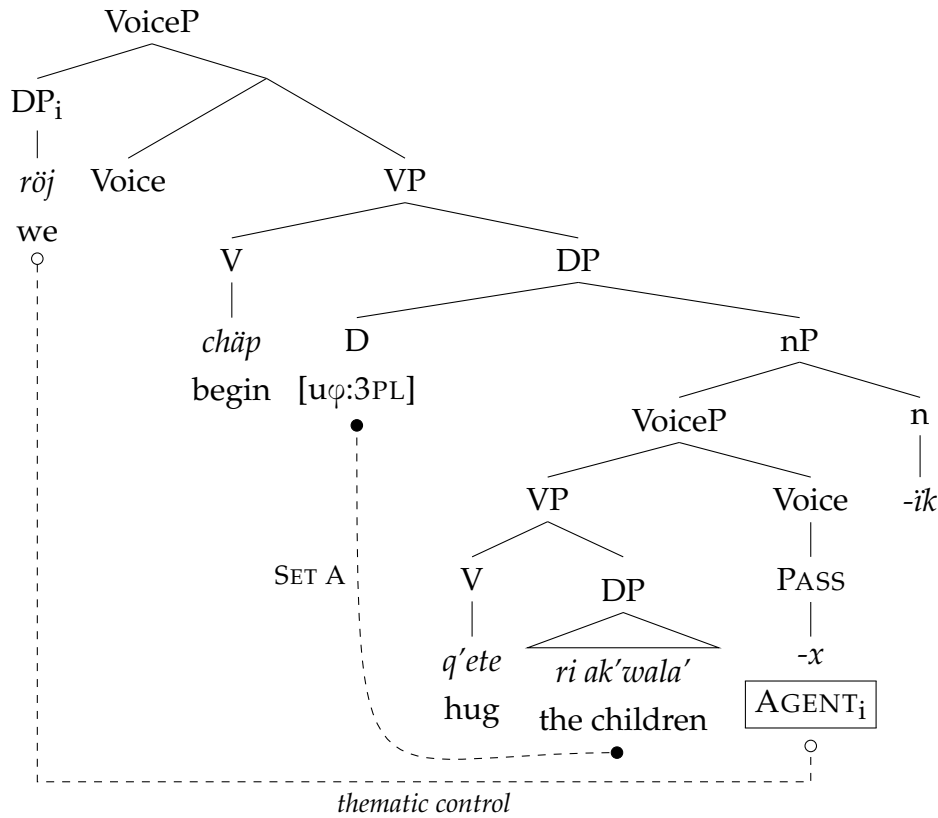


Tree 13.1: Pseudo-passive NFC following Imanishi (2020)

When a pseudo-passive NFC is embedded under a phasal verb like *chäp* 'begin', the subject of the phasal verb (introduced in its specifier) identifies the understood agent of the embedded passive verb.³⁰⁴ Now, because passivization has removed the agent of the transitive verb from the structure, but this argument remains accessible to identification, Imanishi (2020) suggests that the agent remains accessible thematically despite being absent syntactically, and is thus not identified via syntactic, but rather thematic control. This mechanism is invoked from Herd et al. (2011), who argue for the necessity of thematic control based on Polynesian relative constructions in which arguments do not c-command the implicit agent positions of passive-like verbs which they are understood to control. This is schematized in Tree 13.2:

nominal within its Spell-Out domain, which – in the case of Tree 13.1 – is D. See Imanishi (2014).

³⁰⁴Tree 13.2 omits the issue of how the matrix subject and the NFC receive Case. The former receives Set A in Spec,VoiceP from the transitive phasal verb, and the latter Set B from finite T due to itself being nominal (recall that Kaqchikel is a HIGH-ABS language).



Tree 13.2: Phasal verb + pseudo-passive NFC following Imanishi (2020)

In sum, the Imanishi variant assumes that the passive morphology of pseudo-passive NFCs reflects an actual process of passivization, to the syntactic effect of removing the external argument from the structure. Passive agent control is achieved thematically. In terms of transitivity, pseudo-passive NFCs according to Imanishi are both syntactically (the internal argument is the sole argument) and morphologically intransitive (this argument receives sole Set A marking).

13.2.5 Pseudo-passive NFCs according to Coon

I now turn to the *Coon variant*. Regarding true passive NFCs like in (13), Coon (2010, 2013) proposes that the transitive verb first merges with a PRO internal argument, after which it undergoes passivization and then nominalization. This PRO internal argument receives its interpretation via clause-internal syntactic control by a *pro* that is expressed as a grammatical possessor in the extended nominal projection, Spec,PossP; *pro* receives Set A from Poss. See Tree 13.3:³⁰⁵

³⁰⁵Note that *pro* serves a purely specificational function with respect to PRO, since Spec,PossP – unlike Comp,V – is itself athematic.

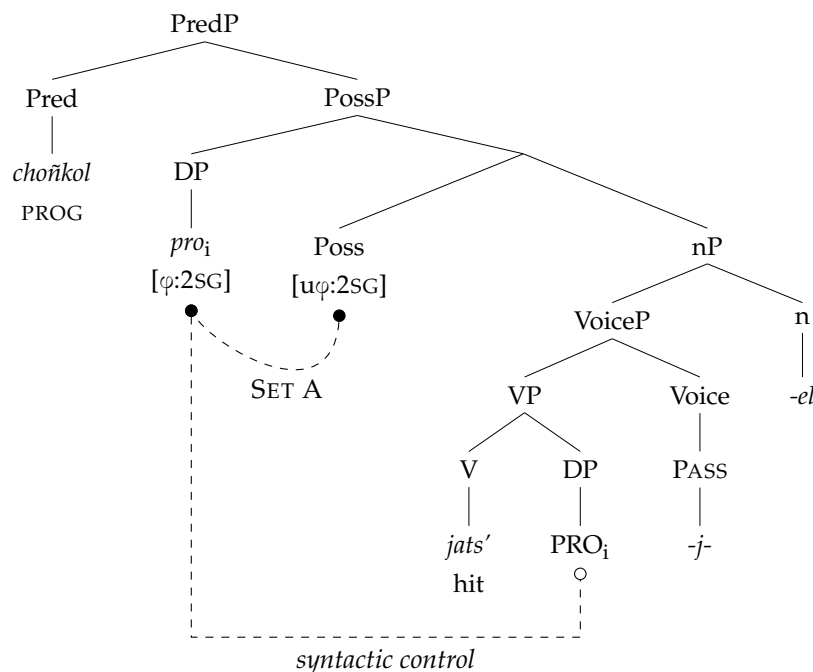
(13) **Ch'ol:** *True passive NFC*

Choñkol [a-ja<j>ts'-el]

PROG A.2SG-hit<PASS>-NMLZ

'You're being hit'

(Coon 2013: 133)



Tree 13.3: True passive NFC following Coon (2010, 2013)

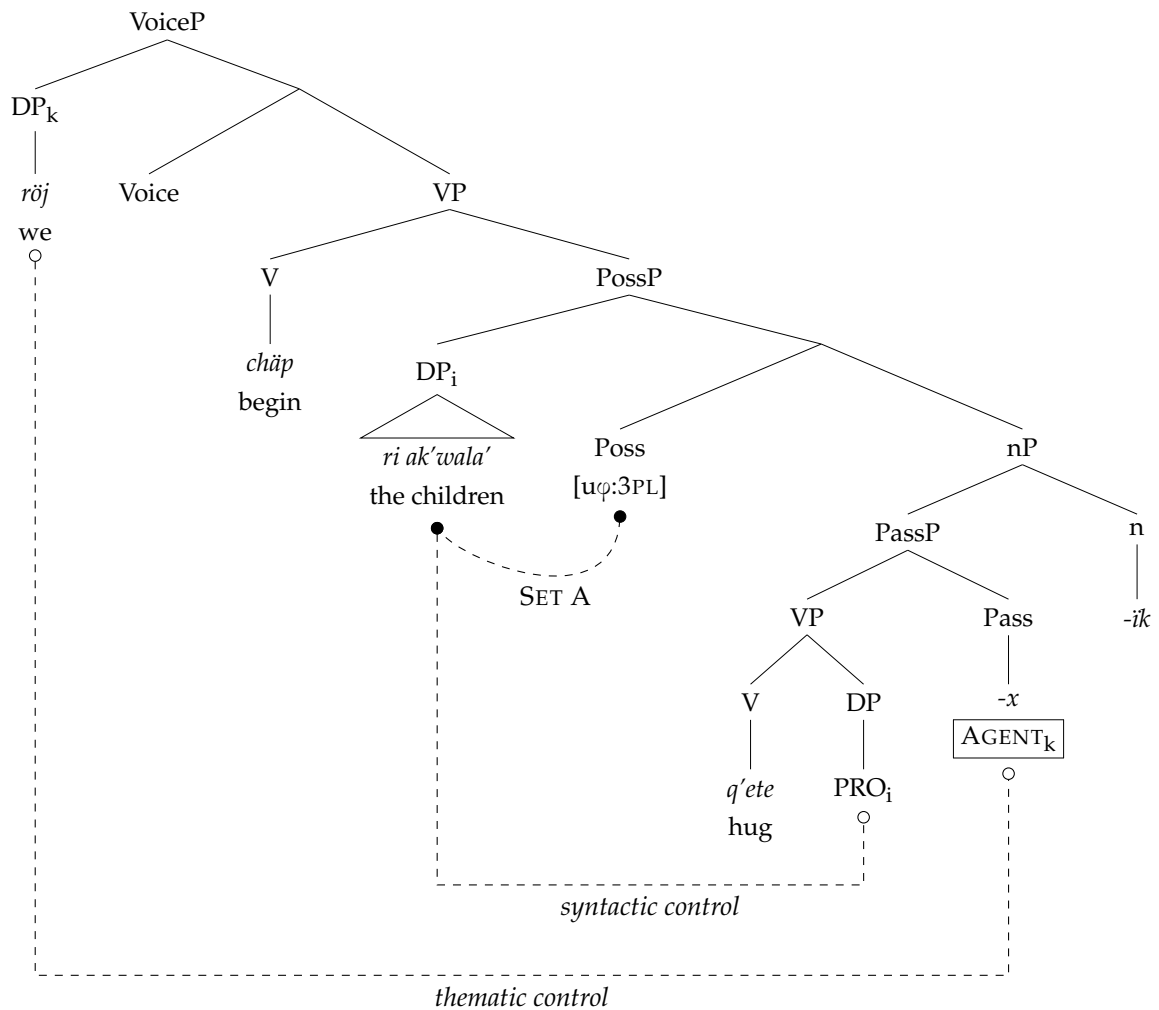
Neither Coon (2010) nor Coon (2013) directly consider pseudo-passive NFCs, but we can straightforwardly apply the logic of the proposed true passive structure in Tree 13.3 to our Kaqchikel case, repeated here in (14). This gives us Tree 13.4:^{306,307}

(14) **Kaqchikel:** *Pseudo-passive NFC*

- a. Røj x-Ø-qa-chäp [ki-q'ete-x-ik ri ak'wal-a']
 we COMPL-B.3SG-A.1PL-begin A.3PL-hug-PASS-NMLZ DET child-PL
 'We began to hug the children' (Imanishi 2020: 22)

³⁰⁶That the structure in Tree 13.4 is indeed how pseudo-passive NFCs would be handled in her approach was confirmed with Jessica Coon (p.c.).

³⁰⁷Note that Tree 13.4 abstracts away from the linear order of the possessor with respect to nominalized verb.



Tree 13.4: Phasal verb + pseudo-passive NFC following Coon (2010, 2013)

Although Coon glosses over the issue of passive agent control, it is clear the structure in Tree 13.4 must make use of a mechanism that is analogous to Imanishi's thematic control because just like Imanishi, she takes pseudo-passive NFCs to involve an actual operation of passivization; accordingly, I have included this relation in Tree 13.4.

Transitivity-wise, pseudo-passive NFCs according to Coon are also syntactically and morphologically intransitive.³⁰⁸

13.2.6 Imanishi vs Coon

The Imanishi and the Coon variant have in common that the operation of passivization is taken to apply just as in regular verbal contexts, most importantly in that the external argument is removed. Also, both variants require the non-standard mechanism of thematic control for implementing passive agent control.³⁰⁹

³⁰⁸Note that purely in terms of how many nominals it contains, the Coon variant is dyadic because in addition to the *PRO* internal argument, it houses the *pro* grammatical possessor. But only one of these arguments is thematic (*PRO* in *Comp,V*), while the other one is athematic (*pro* in *Spec,PossP*).

³⁰⁹Moreover, both variants are able to capture the fact that the thematic object patterns formally as a possessor of the nominalized verb, because in both variants the former and the latter enter into an

The main difference between the two variants concerns the relations in which the arguments within the pseudo-passive NFC are involved. In the Imanishi variant, there is a single syntactic argument inside the NFC, and only one instance of control, namely that of passive agent control; this relation is achieved thematically and holds cross-clausally. The Coon variant, on the other hand, is characterized by two arguments within the NFC, and hence two control relations: a syntactic one within the NFC between the possessor (*pro*) and the internal argument (PRO), and a cross-clausal thematic one instantiating passive agent control. These differences are summarized in Table 13.1:

	Imanishi	Coon
Spec,PossP	-	<i>pro</i>
Comp,V	DP	PRO
Control relations	1 thematic + cross-clausal: $[DP_i \dots [AGENT_i]]$	1 syntactic + intra-clausal: $[[pro_i \dots PRO_i]]$
		2 thematic + cross-clausal: $[DP_k \dots [AGENT_k]]$

Table 13.1: Pseudo-passive NFCs: Imanishi vs Coon

The Coon variant makes a major prediction with respect to control: if *pro* in Spec,PossP is omitted, there will cease to be a clause-internal controller of the PRO internal argument, leading to PRO being instead controlled by the matrix subject. Assuming that the omission of *pro* leaves passive agent control unchanged, we should end up with a reflexive interpretation, in which the matrix subject is understood as both the agent of the matrix and the embedded event, and also the patient of the embedded event,

agreement relationship that induces Set A on the nominalized verb.

as schematized here:³¹⁰

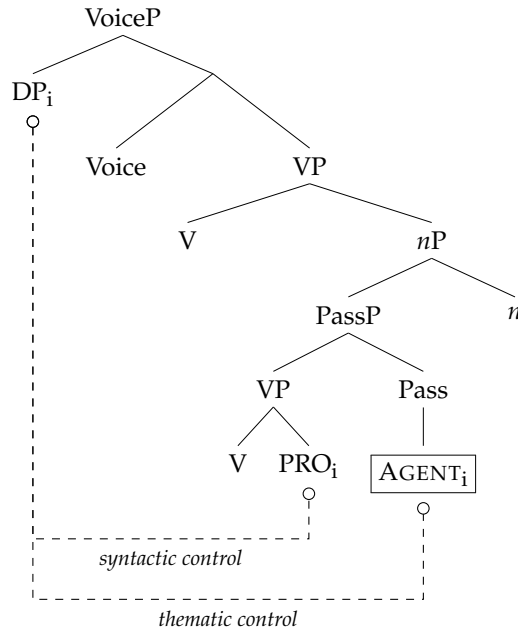


Figure 13.5: The Coon variant without *pro* in Spec,PossP

This prediction is incorrect. As we know for numerous languages, if the pseudo-passive verb lacks Set A marking, which entails the absence of a *pro* in Spec,PossP under Coon's analysis, the only possible interpretation of the embedded clause is active and non-reflexive. The following sentences can only mean that the subject acts on someone or something else:

(15) a. **Kaqchikel**

Ri ixoq n-Ø-ajin chi [war-sa-x-ik]
the woman INCOMPL-B.3SG-PROG PREP sleep-CAUS-PASS-NMLZ

'The woman is causing to sleep'

*'The woman is causing herself to sleep' (own notes: SR)

b. **Sakapulteko**

X-in-ak [cha tiix-iik]
COMPL-B.1SG-begin PREP eat.PASS-NMLZ

'I began eating'

(DuBois 1981: 254)

c. **Tz'utujil de Santiago**

Atet x-Ø-a-moj [ch'iy-ik]
2SG COMPL-B.3SG-A.2SG-start hit.PASS-NMLZ

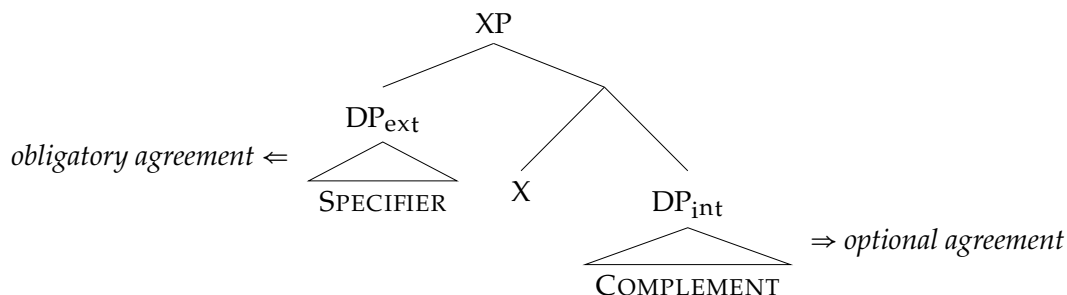
'You started hitting (something / someone else)'

*'You started hitting yourself' (own notes: RP)

There is further evidence showing that the Coon variant is not on the right track, which is specific to Tz'utujil. In Tz'utujil, the φ -features of inanimate 3PL arguments

³¹⁰In other words, *pro* in Spec,PossP acts as a kind of structural intervener for control between the matrix subject and PRO.

in Tz'utujil are expounded obligatorily in some configurations, but optionally in others (Levin et al. 2020, Lyskawa & Ranero 2021a). The availability of optional agreement thus serves as a diagnostic for underlying syntactic structure, the generalization being that all arguments which are base-merged as external arguments in specifier position must always control agreement, while inanimate arguments base-merged as internal arguments in complement position optionally control agreement:³¹¹



Tree 13.6: Optional agreement as a function of an argument's base-merge position

In the Coon variant, the fact that *pro* is a grammatical possessor in Spec,PossP means that it should disallow optional agreement. What we find, however, is that pseudo-passive NFCs allow optional agreement with 3PL inanimate objects, in which case Set A on the verb falls back to default 3SG marking:

(16) **Tz'utujil de Santiago**

Anen nu-mjuon [{**ki/r**}-kon-x-ik] i-k'e' nuu-mees
 1SG A.1SG-PROG {**A.3PL/A.3SG**}-search-PASS-NMLZ PL-two A.1SG-table

'I am looking for my two tables' (Lyskawa & Ranero 2021a: 33)

Compare regular possessors of simple nominals, which are incompatible with optional agreement:

(17) **Tz'utujil de Santiago**

[{**Ki/*ru**}-skil uxi' k'um] x-Ø-q'oy-a
A.3PL/*A.3SG-seed three squash COMPL-B.3SG-rot-STAT.INTR

'The seeds of three squashes rotted' (Lyskawa & Ranero 2021a: 16)

So in terms of agreement, objects of pseudo-passive NFCs pattern like internal, not external arguments. ³¹²

³¹¹The following arguments, merged externally, always show obligatory agreements: transitive subjects, agent focus subjects, unergative subjects, subjects of phasal verbs, subjects of nominal predicates, and possessors; the following arguments, merged internally, allow optional agreement with 3PL inanimates: transitive objects, unaccusative subjects, passive subjects and agent focus objects. See Levin et al. (2020) and Lyskawa & Ranero (2021a).

³¹²As Lyskawa & Ranero (2021a) note in passing, the fact that an argument is base-merged in a complement position does not preclude it from later moving to a specifier position. The details would

Extraction data from Tz'utujil point to the same conclusion. In regular possession as in (18a), possessors can only extract via *naq xu xiin* ($\approx$ 'of who / what'), which pied-pipes the possessum under inversion (18b); it is impossible for the possessor to extract via the simple interrogative *naq* 'who / what' (18c):

(18) **Tz'utujil de Santiago**

- a. Ya Mriiy x- $\emptyset$ -[u]-pax-i-ij [ru-xa'ar a
CLF M. COMPL-B.3SG-A.3SG-break-CAUS-SS.DTJ A.3SG-glass CLF
Tru'
P.
'Maria broke Pedro's glass'
- b. [**Naq xu xiin ru-xa'ar**]_i ya Mriiy
wh XU RN.OF A.3SG-glass CLF M.
x- $\emptyset$ -[u]-pax-i-ij [_i]?
COMPL-B.3SG-A.3SG-break-CAUS-SS.DTJ
'Whose glass did Maria break?'
- c. ***Naq_i** ya Mriiy x- $\emptyset$ -[u]-pax-i-ij
wh CLF M. COMPL-B.3SG-A.3SG-break-CAUS-SS.DTJ
[ru-xa'ar _i]?
A.3SG-glass
intended: 'Whose glass did Maria break?' (own notes: RP)

The simple strategy with *naq* 'who / what' also occurs under object extraction out of simple finite transitives:

(19) **Tz'utujil de Santiago**

- a. X- $\emptyset$ -a-mich' ak'
COMPL-B.3SG-A.2SG-pluck chicken
'You plucked a chicken'
- b. **Naq_i** x- $\emptyset$ -a-mich' _i?
wh COMPL-B.3SG-A.2SG-pluck
'What did you pluck?' (own notes: RP)

take us too far afield, but my notes show that objects of pseudo-passive NFCs in Tz'utujil de Santiago actually have mixed properties with respect to possessorhood. The data are compatible with an analysis according to which even though all objects are base-merged in Comp,V, only those referring to local-person participants must later also move to Spec,PossP, for what appear to be licensing reasons, thereby becoming derived possessors..

Turning to pseudo-passive NFCs, we find that their objects can only extract via *naq*, not *naq xu xiin*:

(20) **Tz'utujil de Santiago**

- a. Atet a-mjon [r-mich-x-ik jun ak']
2SG A.3SG-PROG A.3SG-pluck-PASS-NMLZ a chicken
'You are plucking a chicken'
- b. **Naq_i** a-mjon [r-mich-x-ik —_i]?
wh A.2SG-PROG A.3SG-pluck-PASS-NMLZ
'What are you plucking?'
- c. ***[Naq xu xiin r-mich-x-ik]_i** a-mjon —_i?
wh XU RN.OF A.3SG-pluck-PASS-NMLZ A.2SG-PROG
intended: 'What are you plucking?' (own notes: RP)

This is unexpected if pseudo-passive NFC objects are external possessors, but follows if they are internal arguments.³¹³

I am going to make the assumption that Mayan pseudo-passive NFCs all share the same internal syntax. In light of this assumption, we have seen that data from control, agreement and extraction show that objects of pseudo-passive NFCs pattern like internal arguments. This behavior hence disfavors the Coon variant and favors the Imanishi variant, because only the latter takes objects to be merged as lexical internal arguments in Comp,V.³¹⁴ I will therefore adopt the Imanishi variant, with some modification.

13.2.7 Pseudo-passive NFCs have external arguments

The previous section provided support for Imanishi's analysis of pseudo-passive NFCs based on how it treated the internal argument. There remains, however, the thorny issue of passive agent control, the puzzling nature of which is twofold: first, that passivization does not alter the control relation by shifting the thematic status of the controllee, cf. (6)-(11); and second, that the agent remains accessible for identification with a matrix argument, despite the fact that passive agents generally

³¹³In English, the $\bar{A}$ -extraction profiles of formal possessors are identical, regardless of whether their possessum is a simple noun (i) or a deverbal nominalization (ii). In either case, the possessor must pied-pipe its possessum to clause-initial position - stranding the possessum is impossible:

- (i) a. *I broke [Alma's glass].*
b. *[Whose glass]_i did you break —_i?*
c. **Whose_i did you break [—_i glass]?*
- (ii) a. *I observed [Alma's examination].*
b. *[Whose examination]_i did you observe —_i?*
c. **Whose_i did you observe [—_i examination]?*

³¹⁴It seems that the Coon variant can be sanctioned if we assume that *pro* is not base-merged in Spec,PossP, but moves there from Comp,V, with PRO actually corresponding to its trace (cf. footnote 312). However, this would leave unexplained both the control (15) and the agreement facts (16).

cannot be controlled or bound (Williams 1987, Partee 1989, Bruening 2013). In English, for example, passive agents cannot be controlled by a matrix argument:

- (21) *Every journalist wants the president to be interviewed*
 (cannot mean ‘Every journalist wants to interview the president’)
 (Bruening 2013: 19)

It is therefore standard practice to assume that agents in passives are existentially quantified over (Bach 1980, Keenan 1985, Williams 1987).

Passive agent control in the context of Mayan NFCs has not gone unnoticed by scholars. Stiebels (2015) surmises that passive agent control instantiates a typologically marked kind of control relation which targets thematic, instead of structural prominence.³¹⁵ This idea is taken up in Imanishi (2020) for Kaqchikel, and Can Pixabaj & Aissen (2021) for K’ichee’, who invoke an approach that renders the passive agent accessible thematically despite being suppressed syntactically. Unfortunately, these works do not spell out how exactly this mechanism is supposed to work. Imanishi’s proposal that the passive agent is targeted via thematic control is basically a stipulation because it serves only to explain the phenomenon itself, and does not follow from any principles which take into account properties of NFCs, or any other specifically Mayan properties, for that matter.

Based on the domain in which passive agent control arises, the availability of reflexives, and the distribution of agent *by*-phrases, I offer a much simpler solution: pseudo-passive NFCs actually contain external arguments, and they lack passivization proper. In other words, unlike Imanishi (and Coon), who take passive morphology to faithfully reflect a passive syntax, I claim that pseudo-passive NFCs involve a mismatch between an underlying active syntax and a superficial passive morphology. This result confirms Can Pixabaj (2015), who questions the passive status of pseudo-passive NFCs, suspecting that they are actually active.

13.2.7.1 Domain of passive agent control

Passive agent control is a property that arises specifically within Mayan pseudo-passive NFCs. Neither non-finite passivization, nor passivization under nominalization, nor finite passivization under complementation per se can explain its availability.

First, in English and other languages, passivization on an embedded non-finite verb shifts the controllee from agent to patient, such that only the derived patient subject is controllable:

³¹⁵See e.g. Kroeger (1991) and Kroeger (1993) for the same idea applied to control in languages with Austronesian voice systems.

- (22) a. *The cat started / wants [to bite]*
 ⇒ The cat does the biting
 cannot mean: The cat undergoes the biting
- b. *The cat started / wants [to be bitten]*
 ⇒ The cat undergoes the biting (by something else)
 cannot mean: The cat does the biting

This shows that passive agent control cannot be due to passivization in non-finite contexts alone, under the assumption that it might be afforded by some kind of transparency effect of non-finiteness.

Second, to see that passive agent control does not follow from passivization in the context of non-finiteness qua nominalization, consider Turkish, which permits nominalization over passivization. When a nominalized passive verb is embedded under a matrix verb like ‘want’, which allows disjoint reference between the wanter and the understood subject of the embedded event, it is impossible for the wanter to control the agent position of the embedded passive verb. In (23), Hande can only be understood as the wanter, and the finisher only as someone else:³¹⁶

- (23) **Turkish:** *Nominalized passive verb*

Hande [iʃ-in bit-ir-il-me-si-ni] iste-di
 H. task-GEN finish-CAUS-PASS-NMLZ.MA-3SG.POSS-ACC want-PST
 ‘Hande wanted the task to be finished (by someone else)’
 literally: ‘Hande wants the being finished of the task’
 *‘Hande wanted to finish the task’ (Furkan Atmaca p.c.)

Third, passive agent control is systematically absent in finite contexts. Next to NFCs, Tz’utujil also has finite complement clauses with aspect and φ -marking for both subjects and objects, as in (24):

- (24) **Tz’utujil de Santiago:** *Finite complement clause*

Atet_i x-Ø-a-moj [—_{i/*k}
 2SG COMPL-B.3SG-A.2SG-start
 n-Ø-a-josq-iiij ksiiin]
 INCOMPL-B.3SG-A.2SG-clean-STAT.DTJ kitchen
 ‘You started cleaning the kitchen’
 (literally: ‘You started (it), you clean the kitchen’) (own notes: RP)

If we keep constant the matrix verb and a formally passive embedded verb, but vary the finiteness of the complement clause, passive agent control only arises in

³¹⁶Concerning (23), note that the verb *bitmek* ‘to be done / run out’ is a root intransitive, e.g. *Iş bitti* ‘The task is done’.

non-finite complement clauses (25a); in finite complement clauses, the agent of the formally passive verb is inaccessible to control (25b):

(25) **Tz'utujil de Santiago**

- a. *Non-finite complement clause → passive agent control*

Atet x-Ø-a-moj [r-josq-s-ik ksíin]
 2SG COMPL-B.3SG-A.2SG-start A.3SG-clean-PASS-NMLZ kitchen
 'You started cleaning the kitchen'

- b. *Finite complement clause → no passive agent control*

?Atet x-Ø-a-moj
 2SG COMPL-B.3SG-A.2SG-start
 [n-Ø-josq-e-s-a ksíin]
 INCOMPL-B.3SG-clean-V-PASS-STAT.INTR kitchen
 *'You started cleaning the kitchen'

'You started (it), (such that) the kitchen is cleaned (by someone else)'

(own notes: RP)

13.2.7.2 Reflexives

In Mayan, reflexive meanings are expressed with reflexive relational nouns, which index the object via Set A; on the verb, the subject is marked via Set A, while the relational noun itself triggers null 3SG Set B marking. (26) is a finite reflexive clause:

(26) **Tz'utujil de Santiago**

Anen x-Ø-in-tz'at w-ii' chpaam tzetb'al
 1SG COMPL-B.3SG-A.1SG-see A.1SG-RN.REFL in mirror

'I saw myself in the mirror'

(own notes: RP)

Now, a peculiar property of pseudo-passive NFCs, noticed and discussed in Larsen (1988) and Can Pixabaj (2015) for K'ichee', is their ability to host reflexives, a fact replicated in Tz'utujil:

(27) *Pseudo-passive NFCs allow reflexives*a. **K'ichee'**

La x-at-ok k'u chi [r-atinsa-x-ik
 Q COMPL-B.2SG-start PART PREP A.3SG-bathe-PASS-NMLZ
 aw-iib']?
 A.2SG-RN.REFL

'Did you start bathing yourself?' (Can Pixabaj 2015: 119)

b. **Tz'utujil de Santiago**

Anen ne-mjon [r-tz'et-ik **w-ii'** chpaam
 1SG INCOMPL-PROG A.3SG-see.PASS-NMLZ **A.1SG-RN.REFL** in
 tzetb'al]
 mirror

'I am seeing myself in the mirror' (own notes: RP)

Pseudo-passive NFCs with reflexive objects remain well-formed in the absence of a matrix binder:

(28) **Tz'utujil de Santiago**

a. [R-tz'et-ik w-ii' chpaam tzetb'al] kungan
 A.3SG-see.PASS-NMLZ A.1SG-RN.REFL in mirror much
 raro
 strange
 'Seeing myself in the mirror is very strange'

b. [R-q'ete-x-ik r-ii'] kungan b'iis
 A.3SG-hug-PASS-NMLZ A.3SG-RN.REFL much sadness
 'Hugging oneself is very sad' (own notes: RP)

The availability of reflexives in pseudo-passive NFCs contrasts sharply with two other contexts in which they are unavailable. First, in languages with pseudo-passive NFCs, reflexives are never possible in finite passives:

(29) **Tz'utujil de Santiago**

a. *X-Ø-cha'ay-a w-ii'
 COMPL-B.3SG-hit.PASS-SS.INTR A.1SG-RN.REFL
 intended: 'I hit myself' / 'I was hit by myself'

b. *X-Ø-q'et-{e}-x-a r-ii'
 COMPL-B.3SG-hug-V-PASS-SS.INTR A.3SG-RN.REFL
 intended: 'She hugged herself' / 'She was hugged by herself'

(own notes: RP)

Second, nor are they possible in true passive NFCs:

(30) *True passive NFCs → no reflexives*

a. **Chuj**

*Lan [hin-chel-chaj hin-b'a]
 PROG A.1SG-hug-PASS A.1SG-RN.REFL
 intended: 'I am hugging myself' (own notes: MT)

b. **Q'anjob'al**

*Lanan [∅ w-ante-lay hin-b'a]
 PROG B.3SG A.1SG-cure-PASS A.1SG-RN.REFL
 intended: 'I am curing myself' (own notes: PMP)

In order to express reflexive meanings in NFCs, Chuj and Q'anjob'al must resort to the agent focus strategy:

(31) *Agent Focus NFCs → reflexives*

a. **Chuj**

Lan [hin-chel-an hin-b'a]
 PROG A.1SG-hug-AF A.1SG-RN.REFL
 'I am hugging myself' (own notes: MT)

b. **Q'anjob'al**

Lanan [∅ w-ante-n hin-b'a]
 PROG B.3SG A.1SG-cure-AF A.1SG-RN.REFL
 'I am curing myself' (own notes: PMP)

Reflexive relational nouns are anaphors which require a c-commanding antecedent in order to be licensed. We thus have particularly strong evidence from binding that – while there is no c-commanding external argument in either finite passives or truly passive NFCs in addition to the internal argument – pseudo-passive NFCs do host external arguments.

13.2.7.3 Agent *by*-phrases

A final diagnostic for the presence of external arguments in pseudo-passive NFCs is the distribution of agent *by*-phrases, which in Mayan are headed by relational nouns. Can Pixabaj (2015) notes that in K'ichee', the same *by*-phrase that is licensed in finite passives in order to express the agent (32a) becomes illicit in pseudo-passive NFCs (32b):

(32) **K'ichee'**a. *Finite passive*

X-Ø-looq' ri wuuj **r-umal** ri ali
 COMPL-B.3SG-buy.PASS DET book A.3SG-RN.BY DET girl
 'The book was bought by the girl'

b. *Pseudo-passive NFC*

*X-Ø-inw-il [u-kuna-x-iik le achi
 COMPL-B.3SG-A.1SG-see A.3SG-cure-PASS-NMLZ DET man
r-umal le ixoq]
 A.3SG-RN.BY DET woman
 intended: 'I saw the curing of the man by the woman'

(Can Pixabaj 2015: 69, 115)

Likewise, *by*-phrases in pseudo-passive NFCs are heavily degraded in Kaqchikel:

(33) **Kaqchikel**

?*[Ri ru-k'at-ik ri tinamit **r-oma** ri a Juan]
 DET A.3SG-burn.PASS-NMLZ DET city A.3SG-RN.BY DET CL J.
 x-Ø-xib'i-n
 COMPL-B.3SG-scare-ANTIP

intended: 'The burning of the city by Juan was scary' (Imanishi 2020: 18)

In Tz'utujil pseudo-passive NFCs, nominals in *by*-phrases reject agent readings, but may allow indirect causer readings (cf. (11)).

A comparison with true passive NFCs again reveals a crucial minimal difference: both Chuj and Q'anjob'al readily allow agent *by*-phrases:

(34) a. **Chuj**

Lan [ha-chel-chaj **w-uj]**
 PROG A.2SG-hug-PASS **A.1SG-RN.BY**

'You are being hugged by me' (own notes: MT)

b. **Q'anjob'al**

Max-Ø xew [ha-maq'-lay-i **y-uj]**
 COMPL-B.3SG finish A.2SG-hit-PASS-SS.INTR **A.3SG-RN.BY**

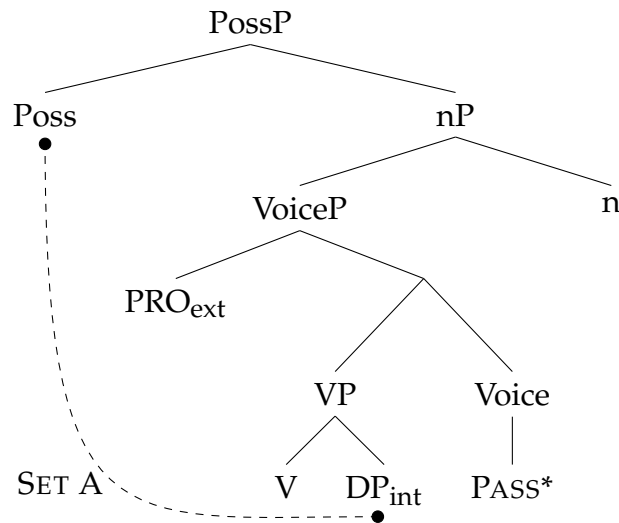
'You finished being hit by him / her' (own notes: PMP)

Under the premise that agent external arguments, agent *by*-phrases, and existentially bound agents are in complementary distribution (i.e. assuming that the specification of the agent argument must be unique), this restriction against agent *by*-phrases in pseudo-passive NFCs follows if pseudo-passive verbs actually retain their external arguments.

13.2.8 Interim summary

In sum, pseudo-passive NFCs contain external arguments, i.e. they are syntactically active. It follows that they involve a mismatch because this active syntax is exponed with passive morphology. In terms of its syntactic effects, then, the presence of the passive marker in pseudo-passive NFCs does not come with the same effects as in finite clauses, given that the external argument is retained. In fact, the passive marker in pseudo-passive NFCs comes with no directly identifiable syntactic effect at all.

I assume the following structure for pseudo-passive NFCs, in which the internal argument is a DP that receives Set B from Poss, the external argument is PRO and the pseudo-passive nature of the Voice head is now indicated with an asterisk:³¹⁷



Tree 13.7: General structure of pseudo-passive NFCs

Recognizing the mismatched nature of pseudo-passive NFCs does away with the need for thematic control, especially in the absence of clear principles from which to derive it. In other words: we no longer need to model passive agent control into pseudo-passive NFCs because there simply is no passive agent control to begin with.³¹⁸ What we do want to account for, of course, is why pseudo-passive NFCs are

³¹⁷I leave open how, under standard assumptions, the head Poss can assign Set A across PRO in Spec,VoiceP. One possibility is to assume that the silent nature of PRO does not follow from its lack of Case, but rather that the lack of Case follows from the nature of PRO and its structural context; see e.g. Satk (2022).

³¹⁸The notion of thematic control into pseudo-passives is predicated on the assumption that the pseudo-passive clause has two semantic arguments but only one syntactic argument – which we know to be wrong. An account that would actually be able to handle precisely this kind of configuration is the argument sharing mechanism developed for Voice restructuring in Bryant et al. (2023). Voice restructuring (Wurmbrand & Shimamura 2017, Pietraszko & Wurmbrand 2021) involves a control-like relationship from a matrix agent to the agent of a VoiceP complement, with either the matrix or the embedded Voice entering the derivation as a special restructuring variant lacking valued features. Argument sharing holds between two arguments even though only one of these is present syntactically (either in the matrix or the embedded clause). This mechanism of argument identification piggybacks on a very particular syntactic configuration that is afforded specifically by Voice restructuring, which is not given in our Mayan NFCs (as diagnosed by e.g. the availability of pseudo-passive NFCs under athematic predicates, cf. (10)). This leads me to conclude that the two constructions are distinct, which

formally passive – an explanation that will follow in the next chapter. For now, we should take away that pseudo-passive NFCs involve VoicePs with a PRO external argument and a DP internal argument.

13.3 Agent focus NFCs

The agent focus strategy is attested in at least Sakapulteko, Chuj, Popti' and Q'anjob'al. In this strategy, the verb carries the agent focus suffix, and the internal argument is realized structurally and indexed via Set B on the verb:

(35) *Agent Focus NFC*

a. **Chuj**

Lan [hach=ko-chel-**an**-i]
 PROG B.2=A.1PL-hug-AF-SS.INTR
 'We're hugging you' (Coon & Carolan 2017: 2)

b. **Popti'**

X-Ø-(y)-il ix [hin ha-mak-**ni**]
 COMPL-B.3SG-A.3SG-see CLF B.1SG A.2SG-hit-AF
 'She saw you hit me' (Craig 1977: 237)

Unlike anti-passive and pseudo-passive complements, which never index the internal argument, agent focus NFCs index both the external and the internal argument: the former is indexed via a Set A prefix, and the latter via a Set B suffix.

Morphologically, agent focus NFCs involve the same marker as in finite contexts, although its syntactic distribution in finite contexts may suggest otherwise. In finite contexts in Chuj, for example, the agent focus marker *-an* only occurs when both arguments are 3rd-person, but not when both are local-person (Buenrostro 2013: 248ff). Consider the contrast between (36a) and (36b):

(36) **Chuj:** *Agent Focus in finite contexts*

a. *3/3 → agent focus*

A waj Xun' ix-Ø-il-**an** ix Malin
 FOC CL J. COMPL-B.3-see-AF CL M.
 'It was Juan who saw Maria' 263

b. *1/2 → no agent focus*

A in=tik ix-ach-w-il-a'
 FOC B.1=DEM COMPL-B.2-A.1-see-SS.TRANS
 'It was me that saw you' (Buenrostro 2013: 263, 266)

in turn further brings out the inadequacy of the thematic control approach.

Yet in NFCs, *-an* surfaces regardless of the person features of either argument:³¹⁹

(37) **Chuj:** *Agent Focus in non-finite contexts*

- a. 3/3 → *agent focus*

Lan [Ø-y-il-**an** heb']

PROG B.3-A.3-see-AF PL

'They are seeing it'

(Buenrostro 2004: 262)

- b. 1/2 → *agent focus*

Ix-in-yamoch [hach=in-chel-**an**-i]

COMPL-A.1-begin B.2=A.1-hug-AF-SS.INTR

'I began to hug you'

(Coon & Carolan 2017: 13)

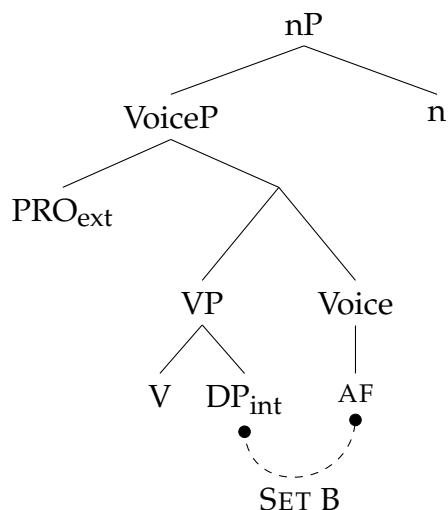
Even though this suggests that the marker found in NFCs is syntactically distinct from the one in finite contexts, I contend that we are dealing with one and the same syntactic *element* despite dealing with distinct *constructions* overall, the reason being that the respective kinds of arguments involved are not the same: while both arguments are lexical nominals in finite contexts (36), non-finite contexts involve PRO external arguments – which I assume does not require Case (see below). In light of the recent work mentioned here, which takes agent focus constructions to be the grammar's response to not only an intervention problem but also a nominal-licensing problem, this means that only one argument, the internal argument, needs to be Case-licensed in non-finite contexts. Because (i) two arguments require Case in finite contexts, and (ii) the person features of both nominals are jointly tracked in their function as co-arguments, but (iii) only the internal argument requires Case in non-finite contexts, the person restriction that arises in finite contexts simply does not arise in non-finite contexts, from which I conclude that the exact same element is involved in both finite and non-finite contexts.³²⁰ That said, the *Agent Focus* label is admittedly rather misleading, especially in a language like Chuj, where it appears in a range of contexts, only one of which constitutes ergative extraction (Buenrostro 2004).

I follow Coon et al. (2014), Coon & Carolan (2017) and Coon et al. (2021) in assuming that agent focus instantiates a special type of Voice head which provides a low source of Set B for the internal argument. Regarding the unexpected presence of Set A for the external argument, in light of the fact that finite agent focus contexts

³¹⁹Syntactic ergativity is also voided in Q'anjob'al when the ergative subject is 1st- or 2nd-person (Pascual 2007, Coon et al. 2014), and Craig (1979) reports a similar pattern for Popti'; see also Robertson (1980) and Stiebels (2006a).

³²⁰To varying degrees, Mayan languages allow ergative extraction with 1st- or 2nd-person arguments. That the Ergative Extraction Constraint should be relaxed in local rather than non-local settings is unexpected from a licensing perspective, given that cross-linguistically, local-person arguments are generally subject to stricter conditions than 3rd-person arguments. Tolan & Clemens (2021) argue for a processing-based account of this discrepancy, drawing on evidence that long-distance dependencies are processed more easily with local-person arguments than 3rd-person arguments because local-person arguments do not introduce new discourse referents. Of course, this by itself leaves open why this property should grammaticalize to varying extents across languages.

never show Set A, I adopt the proposal in Coon & Carolan (2017) that there is a higher *pro* merged as a grammatical possessor controlling the thematic PRO subject in Spec,VoiceP, while leaving to future work the question of why specifically agent focus complements φ -mark the external argument on top of the internal argument.^{321,322} So Set A is not assigned by transitive Voice to its specifier like in finite contexts, instead arising from possession. Agent focus NFCs are structured as follows:



Tree 13.8: General structure of agent focus NFCs

In terms of syntactic effects, the low source of Set B in agent focus NFCs allows internal arguments to be linked structurally. Syntactically, agent focus NFCs are transitive due to housing two structural arguments; morphologically, they are transitive with respect to the internal argument, but classify as intransitive once we also take into account the external argument given that only the internal argument is indexed verbally (the external argument, recall, gets Set A via possession).

13.4 Anti-passive NFCs

The anti-passive strategy is attested in at least Kaqchikel, K'ichee', Q'eqchi', Chuj, Popti' and Q'anjob'al. In this strategy, the verb is marked anti-passive, and the internal argument is typically demoted to an oblique, in which case it is φ -marked on the relational noun via Set A (38):

(38) **Popti'**: *Anti-passive NFC*

Ch-ach to [col-wa-l y-iñ naj]
 ASP-B.2SG go help-ANTIP-NMLZ A.3SG-RN.TO 3SG

'You are going (there) to help him'

(Craig 1979: 5)

³²¹Regarding the presence of PRO in Spec,VoiceP, recall from (31) that agent focus NFCs allow reflexives, a fact which entails the presence of a c-commanding antecedent.

³²²My Chuj consultant robustly rejected agent focus NFCs which lacked Set A marking.

The following finite counterpart of (38) shows that the same construction is involved in both finite and non-finite contexts:

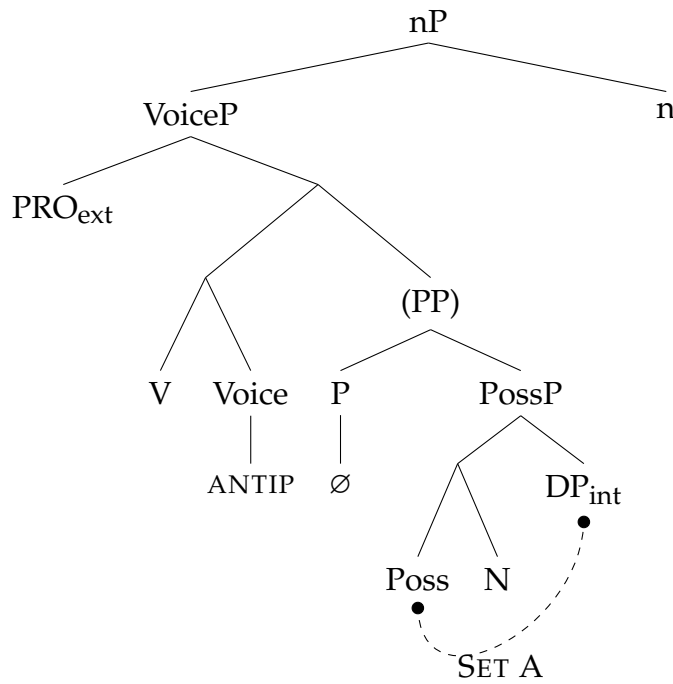
(39) **Popti'**: *Finite anti-passive*

X-Ø-col-wa naj t-aw-iñ
 ASP-B.3SG-help-ANTIP he AUG-A.2SG-RN.TO

'He helped you'

(Craig 1979: 2)

I take anti-passive NFCs to have the general structure in Tree 13.9:³²³



Tree 13.9: General structure of anti-passive NFCs

In terms of its syntactic effects, the anti-passive demotes the internal argument, thereby ensuring that it is realized as an oblique or bare object. The anti-passive marker does not itself provide Case for the internal argument, but licenses an alternative way for the internal argument to get Case, viz. a relational noun or (pseudo-)incorporation. Syntactically, anti-passive NFCs are intransitive given that they contain a sole, external argument; morphologically, they technically speaking show a lower transitivity than regular intransitive contexts (i.e. they are "atransitive" in the context of Mayan) due to not indexing any argument on the verb, only ever showing Set A on the relational noun if the object is realized obliquely. For our purposes, it suffices to

³²³The binding tests used for diagnosing PRO external arguments in pseudo-passive and agent focus NFCs are unavailable for anti-passive NFCs because reflexive relational nouns are ungrammatical with anti-passives, at least in Kaqchikel and Tz'utujil (own notes: SR, RP); see Burukina (2019) for an explanation. I nonetheless assume that anti-passive NFCs are just like their pseudo-passive and agent focus counterparts in this regard, differing only with respect to properties associated with the internal argument. See Burukina (2021a) and Burukina (2021b) for independent arguments for the presence of PRO in semantically transitive NFCs, on the basis of Kaqchikel.

classify them as morphologically intransitive.

13.5 Interim summary

Let us take stock of these VoiceP-level NFCs based on transitive VPs, all of which show a verb-based workaround to the RANT. Crucially, anti-passive, agent focus and pseudo-passive NFCs all retain their semantic transitivity, all have a PRO external argument in Spec,VoiceP, and all are morphologically intransitive.^{324,325} Where they differ is in syntactic properties relating to the internal argument: in terms of the number of structural arguments, anti-passive NFCs are syntactically intransitive, while agent focus and pseudo-passive NFCs are syntactically transitive; in terms of syntactic effects, anti-passive and agent focus NFCs directly affect the Case of the internal argument, while pseudo-passives lack any such effect. See Table 13.2:

³²⁴From a VoiceP-internal perspective, agent focus NFCs can be regarded as morphologically transitive because the internal argument receives structural objective case (Set B); a similar reasoning applies to internal arguments of pseudo-passive NFCs, which receive structural nominal case. This trickiness in classifying the morphological transitivity of finite vs non-finite contexts shows how the notion may be confounded by an argument's linking form and finiteness, in particular the fact that the NFC-internal subject is PRO, whose Case-theoretic properties and effects on its co-arguments differ from those of regular nominals (even more so in an ergative languages, where dependent case is marked on subjects). For our purposes, I am considering all three types of NFCs as morphologically intransitive given how their agreement profile relates to finite contexts.

³²⁵See e.g. Coon et al. (2014), Coon & Carolan (2017), Burukina (2019), Burukina (2021a), Burukina (2021b) and Coon et al. (2021) for the presence of PRO in VoiceP-level NFCs.

Property			
	Anti-passive	Agent Focus	Pseudo-passive
External argument	PRO	PRO	PRO
Internal argument	[_{PP} P DP] / NP	DP	DP
Object linking form	oblique Set A	structural (verbal) Set B	structural (nominal) Set A
Semantic transitivity	transitive	transitive	transitive
Syntactic properties			
Transitivity (# of struct. args.)	intransitive	transitive	transitive
Effect of Voice	demotes int. arg.	Set B for int. arg.	–
Morphological transitivity	intransitive	intransitive	intransitive

Table 13.2: VoiceP NFC types in RANT languages

Because these VoiceP complements to n_{Voice} have an argument structure with fixed thematic interpretations, they qualify as Complex Event Nominals in the sense of Grimshaw (1990); cf. Coon (2010), Coon & Carolan (2017) and Burukina (2021a). On the other hand, a subset of them pattern like English gerunds in terms of their Case potential for internal arguments: compare *the criticism* *(of) *the book* vs *her criticizing* *(of) *the book*, where only the latter (the gerund) assigns accusative case.

I will refer to this general species of Mayan nominalizations as *VoiceP-level NFC complements*, or *VoiceP NFCs* for short, leaving their precise classification within the broader typology of nominalizations to future work.³²⁶

13.6 Root-level complements

Regarding *-oj* nominals, I follow Coon & Carolan (2017), Burukina (2021b) and Coon & Royer (2021) in assuming that n_{ROOT} merges with root projections. An *-oj* nominal like in (40) thus has the structure in Tree 13.10:

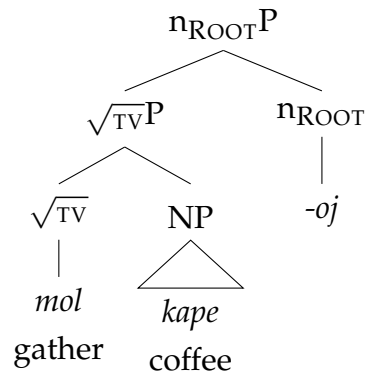
³²⁶For an important work leading the way in this regard, see Burukina (2021b).

(40) Chuj

Ix-a-yamoch [mol-**oj** kape]
 PFV-A.2SG-begin gather-NMLZ coffee

‘I began coffee-gathering’

(Coon & Royer 2021: 11)



Tree 13.10: Chuj: n_{ROOT} takes transitive root with bare object

Due to n_{ROOT} directly taking root projections, *-oj* nominals do not support any kind of voice or valence morphology, nor may they carry D-level material including possession, or be modified by adverbs (Gutiérrez Sánchez 2004, Vázquez Álvarez 2011, Coon 2013, Coon & Carolan 2017, Coon & Royer 2021). Regarding the obligatory presence of the internal argument, I follow Coon (2019) and Coon & Royer (2021) in assuming that Mayan transitive roots are generally subject to a strict IA requirement, which mandates that they must either explicitly appear with an internal argument in the form of a DP or NP, or have their internal argument slot closed off by some other means such as anti-passivization. As for the internal argument's reduced structure, I likewise adopt the proposal in Coon (2019) that in the absence of functional nominal structure to Case-license it – i.e. transitive Voice assigning Set B – the internal argument must be a structurally reduced N or NP of type $\langle e, t \rangle$ that is (pseudo-)incorporated and interpreted semantically via Restrict and Existential Closure (Chung & Ladusaw 2004).^{327,328} From the perspective of the verbal projection, merging the *-oj* nominalizer bleeds the structure that would Case-license a DP object; from the perspective of the object, merging a full DP structure precludes the merger of the *-oj* nominalizer because it would remain unlicensed.

³²⁷Consistent with the lowest-possible attachment site of n_{ROOT} is the fact that *-oj* nominals sometimes give rise to unpredictable meanings, such as *k'an-oy ix* (ask.for-NMLZ woman) from Chuj, which refers to a particular form of marriage (Hopkins 2012, Coon & Royer 2021); Can Pixabaj (2009) notes the same property in connection with *-oj* nominals in K'ichee'. As argued in e.g. Arad (2003), the domain of specialized meaning is low in the structure.

³²⁸The fact that *-oj* nominals cannot be possessed offers an explanation for why the internal argument cannot be a DP licensed via Set A, by virtue of grammatical possession.

13.7 Summary

In this chapter, pseudo-passive, agent focus and anti-passive NFCs were all shown to host external arguments in Spec,VoiceP. This is particularly noteworthy for pseudo-passive NFCs in light of their passive morphology, but welcome given that they pattern like transitives in other respects; there is hence no need for us to make special assumptions about the accessibility of agents in passives because pseudo-passives are only morphologically passive. Next to PRO external arguments, these NFCs are all characterized by a retention of their semantic transitivity, and an intransitive morphology; where they differ is with respect to the particular kind of Voice head involved. We also saw that mixed workarounds to the RANT in the form of *-oj* involve smaller, root projections below Voice, such that the internal argument is precluded from projection up to DP.

Chapter 14

Proposal and derivations

14.1 Introduction

This chapter presents the analysis of the RANT and its derivations. The basic proposal is given in Section 14.2, with derivations following in Section 14.3. I discuss various open issues in Section 14.4, and end both this chapter and our third case study in Section 14.5.

14.2 Proposal

We now have everything we need in order to be able to derive the RANT: we know that (i) in RANT languages, n_{VOICE} imposes a selectional restriction on VoiceP complements based on transitive VPs; (ii) this restriction brings about a workaround in the form of an anti-passive, agent focus or pseudo-passive strategy; (iii) these NFCs based on transitive VPs all have PRO external arguments.

14.2.1 n_{VOICE} selects against ergative case

Based purely on these generalizations, which relate to properties that are internal to VoiceP-level NFCs, we can surely come up with more than one way of ensuring that VoiceP complements to n_{VOICE} based on transitive VPs must show some valence-decreasing operation. Yet if nothing more is taken into account, any such analysis would miss the bigger picture, which is that the RANT and the absolutive parameter are tightly correlated. Recall that despite the inadequacy of a purely Case-based approach to the RANT, there remains a one-to-one correspondence between the RANT and the HIGH-ABS property among the languages sampled here, and conversely no non-RANT language sampled here exhibits the RANT (apparent exceptions are discussed later). It should therefore be our null hypothesis that these two properties are connected.

I argue that this is indeed true: the RANT arises due to how selectional properties

of n_{VOICE} interact with the Mayan absolutive parameter. Specifically, I propose that the kind of Voice head selected by n_{VOICE} cannot assign ergative case, which in a LOW-ABS language gives rise to VoiceP NFCs based on transitive VPs which necessarily show some kind of workaround.

Recall that in both HIGH-ABS and LOW-ABS languages, finite transitive Voice assigns Set A to the DP in its specifier. In RANT languages, because these are HIGH-ABS, there is independently no Voice head that can assign Set B, and so selection by n_{VOICE} of a Voice head without a Set A feature means that all Voice heads in NFCs will never assign Set A or Set B. Given that Voice under n_{VOICE} will simply have no Case to assign, the internal argument must be Case-licensed in some other way if it is to project as a full DP. Depending on the particular Voice head employed in a given VoiceP under n_{VOICE} , the internal argument gets either Set A from a relational noun (anti-passive strategy), Set B from Voice (Agent Focus strategy) or Set A via grammatical possession (pseudo-passive strategy).

In contrast, because non-RANT languages are LOW-ABS, they do have a Voice head that can provide a low source of structural Set B, and so selection by n_{VOICE} of a Voice head that does not assign Set A means that Voice heads in NFCs will still be able to assign Set B. Internal arguments can therefore be Case-licensed via Set B from Voice and project as full DPs.

This analysis adequately brings together the absolutive parameter and the RANT in a one-to-one fashion, deriving the latter from the value to which the former is set: even though n_{VOICE} 's selection of Voice heads that do not assign ergative case occurs independently in both RANT and non-RANT languages, once this restriction is overlaid on top of a given language's global HIGH-ABS property, the two jointly yield the RANT. Put differently, the selectional restrictions imposed by n_{VOICE} on its VoiceP complement conspire with the paradigmatic gap in the functional inventory of a HIGH-ABS language, namely the utter absence of a Set B-generating Voice head, to give rise to the RANT.

14.2.2 Defective Voice under nominalization

The idea that the kind of Voice head under n_{VOICE} does not assign ergative case is based on Alexiadou (2001). She argues that cross-linguistically, nominalizations based on transitive verbs which yield Complex Event interpretations contain a Voice head that is deficient in two ways: it does not project an external argument in its specifier, and does not check accusative case; if an agent projects, it must do so as a *by*-phrase:³²⁹

³²⁹She hence takes nominalization to be a reflex of Burzio's Generalization.

- (1) a. **English**
 the destruction of the city by the enemy
- b. **Greek**
 I katastrofi tis polis apo tus Italus
 the destruction the.GEN city by the Italians
 ‘the destruction of the city by the Italians’ (Alexiadou 2001: 12, 40)

In our case, we know that anti-passive, agent focus and pseudo-passive NFCs actually have external arguments in the form of PRO.³³⁰ As for why Mayan n_{VOICE} should restrict ergative case rather than absolutive, which would pattern like accusative case in being objective case, I submit that the kind of n standing in a local selectional relationship with Voice systematically restricts the *dependent case*: accusative in a nominative-accusative system, and ergative in an ergative-absolutive system.

At least for regular transitive Voice, then, this analysis entails that it must come in at least two sub-types within a given language: one that occurs in regular finite contexts, and another one in non-finite contexts. While this is left somewhat implicit in Alexiadou (2001), it is made explicit in Coon (2010: 115), who argues that in the LOW-ABS language Ch’ol, transitive Voice is spelled out $-V$ in finite contexts (2a), but $-e'$ under n_{VOICE} (2b):³³¹

- (2) **Ch’ol**
- a. Tyi k-ch’äx-ä jiñi ja’
 COMPL A.1-boil-TV DET water
 ‘I boiled the water’
- b. Mi [k-ch’äx-e’ jiñi ja’]
 INCOMPL A.1-boil-DEP DET water
 ‘I boil the water’ (Coon 2010: 111)

I take the restriction against dependent case under n_{VOICE} to be purely syntactic, with no semantic effects.

14.2.3 A sub-typology of Mayan Voice

Let us generalize the Case-theoretic behavior of transitive Voice, $Voice_{TRANS}$, for external and internal arguments depending on the language type (LOW-ABS vs HIGH-ABS) and structural context (finite vs non-finite).

In LOW-ABS languages, $Voice_{TRANS}$ bears both $[*ERG*]$ and $[*ABS*]$ in finite contexts. In non-finite contexts, which involve n_{VOICE} , the sub-type of $Voice_{TRANS}$ that is

³³⁰Note that the status of external arguments in Complex Event Nominals is still very much debated. In my view, some sub-types contain external arguments, while others simply don’t.

³³¹Coon does not frame her proposal in terms of ergative case assignment as a function of structural context, but in terms of whether the external subject demanded by the respective Voice head is lexical or PRO (Coon 2010: 115). Since – all else being equal – her and my proposal induce the same partitioning of Voice heads, we should want to reduce the occurrence of PRO in Spec,VoiceP to the absence of ergative, or vice versa. I leave this issue open.

selected lacks [**ERG**], but retains [**ABS**] – hence the RANT does not arise.

In HIGH-ABS languages, Voice_{TRANS} in finite contexts bears only [**ERG**], meaning that Voice_{TRANS} under n_{VOICE} ends up completely stripped of any Case features. This is summarized in Table 14.1, where (i) each shaded area represents a distinct Voice_{TRANS} sub-type; (ii) "–" indicates an argument that needs Case, yet which the respective Voice head does not provide; (iii) "PRO" means that the argument is PRO, which I assume does not need Case (see below); and (iii) each sub-type is indicated with a shorthand reflecting its absolutive parameter and structural context (e.g. "Voice_{TRANS.L.NF}" is the sub-type of Voice found in non-finite contexts in a LOW-ABS language):

Argument	LOW-ABS / non-RANT		HIGH-ABS / RANT	
	Finite	Non-finite	Finite	Non-finite
external	[<i>*ERG*</i>]	PRO	[<i>*ERG*</i>]	PRO
internal	[<i>*ABS*</i>]	[<i>*ABS*</i>]	– ^a	– ^b
	TRANS.L.F	TRANS.L.NF	TRANS.H.F	TRANS.H.NF

a. This argument gets Set B from finite T.

b. The internal argument cannot get Set B from finite T because it goes to the NFC, which is itself nominal.

Table 14.1: Voice_{TRANS} across language types and structural contexts

Now, the crucial sub-type of Voice_{TRANS} – i.e. Voice_{TRANS.H.NF} – which occurs in HIGH-ABS languages in non-finite contexts and does not assign ergative – is shaded pink. I claim that the RANT reflects different workaround strategies to one and the same problem, which is that Voice_{TRANS} under n_{VOICE} cannot bear [**ERG**]: in some RANT languages, either because Voice_{TRANS.H.NF} cannot be spelled out or it simply does not exist, the grammar resorts to an entirely different Voice head such as Voice_{ANTIP} or Voice_{AF} in order to express transitive meanings; in others, Voice_{TRANS.H.NF} exists, but it is spelled out with passive morphology. The latter have the pseudo-passive strategy, while the former do not. In fact, the present sample turns out to show a complementary distribution with respect to pseudo-passive and agent focus NFCs, with pseudo-passives occurring in the K'ichean branch and agent focus in the Q'anjob'alan branch.³³²

Let us first consider languages without pseudo-passive NFCs, like Chuj and Q'anjob'al. In these, formally passive NFCs always have true passive interpretations (3a), and non-finite transitive meanings must be expressed using the anti-passive or

³³²Sakapulteko is a potential exception to this as it may allow agent focus NFCs. The data on this are unclear, cf. (DuBois 1981).

agent focus strategy (3b):³³³

(3) **Chuj**

a. *True passive NFC*

Ix-in-yamoch [hin-chel-chaj-i]
COMPL-A.1SG-begin A.1SG-hug-PASS-ITV
'I began to be hugged'

b. *Agent focus NFC*

Lan [hach=ko-chel-an-i]
PROG B.2=A.1PL-hug-AF-STAT.INTR
'We're hugging you'

(Coon & Carolan 2017: 27, 2)

The idea is that in Chuj-type languages, the inability of Voice_{TRANS} to bear [*ERG*] under n_{Voice} results in a kind of ineffability: either in terms of a paradigmatic syntactic gap because the appropriate Voice_{TRANS} sub-type without [*ERG*] – i.e. Voice_{TRANS.H.NF} – is missing entirely from the grammatical inventory; or morphologically in that Voice_{TRANS.H.NF} exists, yet without the grammar providing any way of spelling it out. The only remaining option, then, is to employ an alternative such as Voice_{ANTIP} or Voice_{AF}, each of which comes with its own Case potential for the internal argument. Passive NFCs are available, but not to express transitive meanings.

This is made clear in Table 14.2: because Chuj-type languages lack or cannot pronounce Voice_{TRANS.H.NF}, they must instead use Voice_{ANTIP.H.NF} or Voice_{AF.H.NF} in NFCs; Voice_{ANTIP.H.NF} either provides ergative for the internal argument via the relational noun or obviates its need for Case via (pseudo-)incorporation, while Voice_{AF.H.NF} assigns absolutive to it (note that Voice_{AF} is absent in LOW-ABS languages, indicated by "n.a.").

³³³Formally passive NFCs in Chuj can never have active meanings:

(i) **Chuj**

Lan [ha-chel-chaj]
PROG A.2SG-hug-PASS

'You are being hugged'
* 'You are hugging'

(own notes: MT)

Voice	Arg.	LOW-ABS / non-RANT		HIGH-ABS / RANT	
		Finite	Non-finite	Finite	Non-finite
Voice _{TRANS}	ext.	[*ERG*]	PRO	[*ERG*]	PRO
	int.	[*ABS*]	[*ABS*]	_ a	_ b
		TRANS.L.F	TRANS.L.NF	TRANS.H.F	TRANS.H.NF ❶ ineffable
Voice _{ANTIP}	ext.	_ a	PRO	_ a	PRO
	int.	_ c	_ b,c	_ c	_ b,c
		ANTIP.L.F	ANTIP.L.NF	ANTIP.H.F	ANTIP.H.NF ❶ instead of TRANS.H.NF
Voice _{AF}	ext.	n.a.	n.a.	_ d	PRO
	int.	n.a.	n.a.	[*ABS*] ^d	[*ABS*]
		AF.L.F	AF.L.NF	AF.H.F	AF.H.NF ❶ instead of TRANS.H.NF

a. This argument gets absolutive from finite T.

b. The internal argument cannot get Set B from finite T because it goes to the NFC, which is itself nominal.

c. The internal argument in anti-passives is either existentially bound, assigned ergative from the relational noun, or projected as a bare NP.

d. In finite agent focus contexts, agreement is either object-based or hierarchy-based. Following Coon et al. (2021), in object-based languages, the internal argument remains low and receives Set B from the agent focus marker ([*ABS*]) while the external subject remains Case-less (-). In hierarchy-based languages, the agent focus marker assigns no Case but attracts the internal argument to its specifier thereby making it equidistant from T alongside the subject, a configuration which allows T to agree with whichever argument ranks higher on the hierarchy (cf. also Oxford (2019) for Algonquian). Table 14.2 only includes the object-based agent focus marker, given that the agent focus strategy is unavailable as a workaround to the RANT in hierarchy-based languages.

Table 14.2: Voice_{TRANS} / ANTIP / AF across language types and structural contexts

Turning to languages like Tz’utujil and Kaqchikel, which we know have pseudo-passive NFCs (4a), we find that agent focus NFCs are unavailable (4b):³³⁴

(4) **Tz’utujil de Santiago:**

a. *Pseudo-passive NFC*

Atet x-Ø-a-moj [n-ch’iy-ik nen]
 2SG COMPL-B.3SG-A.2SG-start A.1SG-hit.PASS-NMLZ 1SG
 ‘You started hitting me

b. *Agent focus NFC*

Atet Ø-a-mjon [ch’y-ow-eem]
 2SG B.3SG-A.2SG-PROG hit-AF-NMLZ
 ‘You are hitting’

(own notes: RP)

In Tz’utujil-type languages, Voice_{TRANS.H.NF} without [*ERG*] exists and is spelled out with passive morphology because the syntactic context in which the passive marker generally occurs (both finite and non-finite) overlaps significantly with that of Voice_{TRANS.H.NF}: all have in common that the internal argument remains caseless, and the external argument is non-canonical in either being absent entirely or present as PRO. This overlap is indicated with pink shading in Table 14.3, which expands on Table 14.2 by including Voice_{PASS}. Because Voice_{TRANS.H.NF} is Case-neutral, the internal argument can be linked by entering into grammatical possession, thereby receiving Set A. Recall that internal arguments cannot get Set B from matrix finite T because this Case is stolen by the NFC as it is itself nominal.

Conversely, spelling out Voice_{TRANS.H.NF} with either the anti-passive or the agent focus marker is avoided because these are too specific: the anti-passive marker entails absolute absorption, and – depending on the particular sub-type – either existentially binds the internal argument, links it obliquely via a relational noun or forces its expression as a bare NP; the agent focus marker entails absolute for the internal argument. Voice_{ANTIP.H.NF} and Voice_{AF.H.NF} are shaded light gray in Table 14.2.

³³⁴I have also confirmed this for Kaqchikel (own notes: SR).

Voice	Arg.	LOW-ABS / non-RANT		HIGH-ABS / RANT	
		Finite	Non-finite	Finite	Non-finite
Voice _{TRANS}	ext.	[*ERG*]	PRO	[*ERG*]	PRO
	int.	[*ABS*] [TRANS.L.F]	[*ABS*] [TRANS.L.NF]	– ^a [TRANS.H.F]	– ^b [TRANS.H.NF] ① ineffable ② spelled out as PASS
Voice _{PASS}	ext.	∅	∅	∅	∅
	int.	– ^a [PASS.L.F]	PRO [PASS.L.NF]	– ^a [PASS.H.F]	PRO [PASS.H.NF] ② spells out [TRANS.H.NF]
Voice _{ANTIP}	ext.	– ^a	PRO	– ^a	PRO
	int.	– ^c [ANTIP.L.F]	– ^{b,c} [ANTIP.L.NF]	– ^c [ANTIP.H.F]	– ^{b,c} [ANTIP.H.NF] ① instead of [TRANS.H.NF]
Voice _{AF}	ext.	n.a.	n.a.	–	PRO
	int.	n.a. [AF.L.F]	n.a. [AF.L.NF]	[*ABS*] [AF.H.F]	[*ABS*] [AF.H.NF] ① instead of [TRANS.H.NF]

a. This argument gets absolutive from finite T.

b. The internal argument cannot get Set B from finite T because it goes to the NFC, which is itself nominal.

c. The internal argument in anti-passives is either existentially bound, assigned ergative from the relational noun, or projected as a bare NP.

Table 14.3: Voice_{TRANS} / PASS across language types and structural contexts

It is useful to think about this in Optimality-Theoretic terms: compared to the anti-passive and agent focus markers, the passive marker induces the fewest violations resulting from differences between its morphological property set on the one hand, and the syntactic property set of Voice_{TRANS.H.NF} on the other (i.e. it is least incompatible). I do not pursue an OT approach here, referring the reader to the discussion in the next chapter.³³⁵

³³⁵Tz'utujil also has a so-called completive passive, which is syntactically just like the regular passive, but additionally expresses that the patient is now in an affected state resulting from the action denoted by the verb. Morpho-syntactically, it patterns just like the regular passive, but completive passive

Because pseudo-passive NFCs are a response by the grammar of a HIGH-ABS language to the impossibility of [**ERG**]-bearing Voice under n_{VOICE} , they, too, are now accommodated within the bigger picture, being connected to the Mayan-specific properties of the RANT, the absolutive parameter and finiteness. We capture the fact that pseudo-passives are only formally passive by not appealing to morphology alone, but morphology as a function of predictable syntactic context, which in turn falls out from the the absolutive parameter and the defectiveness of Voice under nominalization. Pseudo-passive NFCs thus instantiate a kind of *contextual deponency*, i.e. a form-function mismatch that arises only in a certain structural context.

In a similar vein, Coon et al. (2021) provide an explanation for why RANT languages in Q'anjob'alan do not allow agent focus in NFCs, while those of the Q'anjobalan branch do. Q'anjob'alan languages show object-based agreement in finite agent focus contexts, and K'ichean languages show hierarchy-based agreement. Coon et al. (2021) argue that this is because Q'anjob'alan Voice_{AF} assigns absolutive to the internal argument, while K'ichean Voice_{AF} assigns no absolutive case but attracts the internal argument to its specifier via EPP-driven movement.³³⁶ Consequently, agent focus NFCs are ruled out in K'ichean because the agent focus marker would not provide a low source of Set B.³³⁷

We also derive the fact that the RANT does not have a Case-theoretic basis, as reflected in the persistence of workarounds even if no object is present: the crucial ingredient of the RANT manifests as a selectional restriction between n_{VOICE} and Voice, i.e. a relationship that does not directly involve the internal argument. Therefore, the RANT is independent of its presence (provided that the language allows its omission).

Lastly, what about the Case of PRO? The present analysis entails that PRO remains Case-less because it is present but not assigned ergative. While this may seem problematic under the standard contemporary Minimalist view that PRO bears Case just like regular nominals – see e.g. Bobaljik & Landau (2009) – it is actually a welcome result in light of recent work like Satik (2022), who argues that PRO may or may not bear Case depending on the language and construction. PRO is therefore independent of Case. I adopt this assumption, meaning that the Caselessness of PRO in NFCs is unproblematic.

morphology can never give rise to pseudo-passive NFCs (own notes: RP). I take this to support my claim that the general passive marker is employed in pseudo-passives specifically because it is least marked.

³³⁶This movement to Spec,VoiceP, which makes the external and the internal argument equi-distantly accessible to finite T, is meant to derive the hierarchy effect in agreement. This builds on Oxford (2019), who uses a similar mechanism for Algonquian verb paradigms.

³³⁷Technically, the K'ichean agent focus marker should actually be able to occur in NFCs based purely on the fact that it lacks [**ERG**]. Yet it would have no Case-theoretic effect, and is arguably also ruled out because it would be unable to satisfy its EPP feature.

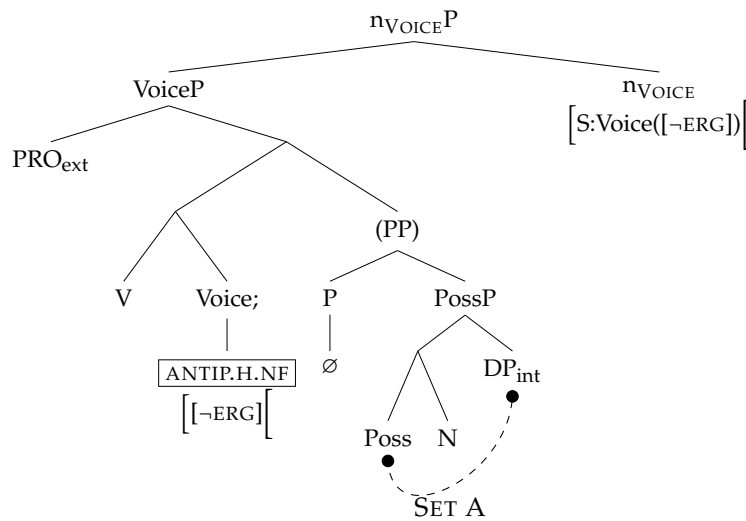
14.3 Derivations

I model the restriction against ergative assignment under nominalization as follows: n_{VOICE} selects Voice heads bearing the feature $[-\text{ERG}]$, where $[-\text{ERG}]$ characterizes any Voice head that does not assign ergative case, i.e. lacks the feature $[\ast\text{ERG}\ast]$; only if Voice bears $[-\text{ERG}]$ will it be able to merge with n_{VOICE} . Based on the notation in Bruening (2013), I represent this as $\text{S:Voice}([-\text{ERG}])$.³³⁸

Because derivations proceed bottom-up, the presence of an ergative-assigning $\text{Voice}_{\text{TRANS}}$ in the derivation bleeds the merger of n_{VOICE} , thereby forcing the addition of projections that end up yielding a finite clause.

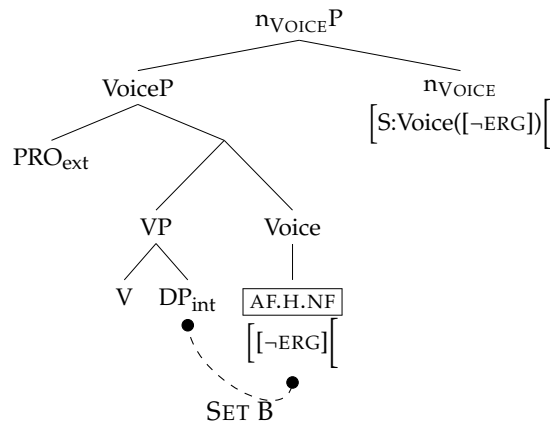
With this in place, full derivations of verb-based workarounds to the RANT are straightforward. The following trees are enriched versions of those from above, now including (i) the specific sub-type of non-finite Voice in relation to the absolutive parameter; and (ii) the selection requirement of n_{VOICE} that its complement lack $[\ast\text{ERG}\ast]$.

For RANT languages, anti-passive, agent focus and pseudo-passive NFC derivations are given in Trees 14.1, 14.2 and 14.3, respectively. Note that in Tree 14.3, the pseudo-passive nature of Voice is no longer indicated with "PASS*" given that we now know that passive morphology spells out $\text{Voice}_{\text{TRANS.H.NF}}$.

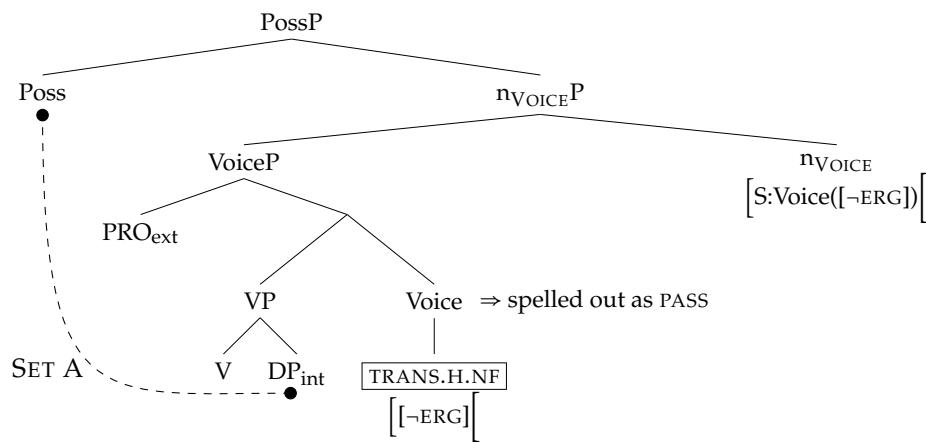


Tree 14.1: Anti-passive NFC in RANT language

³³⁸A different way of thinking about this is that n_{VOICE} undergoes an *anti-Agree* operation with respect to the feature $[\ast\text{ERG}\ast]$: only if n_{VOICE} "agrees" with the absence of $[\ast\text{ERG}\ast]$ on Voice can the two merge, and conversely Merge cannot apply whenever it encounters $[\ast\text{ERG}\ast]$ on Voice.

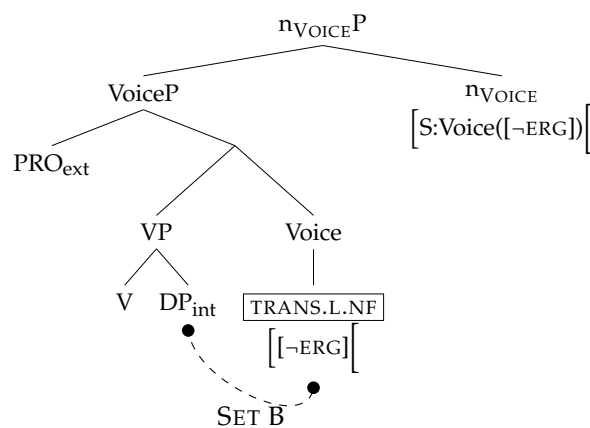


Tree 14.2: Agent focus NFC in RANT language



Tree 14.3: Pseudo-passive NFC in RANT language

Tree 14.4 shows the derivation of transitive NFC in non-RANT languages, in which $\text{Voice}_{\text{TRANS.L.NF}}$ retains the ability to assign Set B:



Tree 14.4: Transitive NFC in non-RANT language

Finally, recall that NFCs based on root intransitives are always fine in both RANT (5) and non-RANT languages (6):

(5) **Tz'utujil:** *Root intransitive NFC (RANT)*

[Paj-ik] m'=utz ta
fall-NMLZ NEG=good IRR

'Falling is not good'

(own notes: RP)

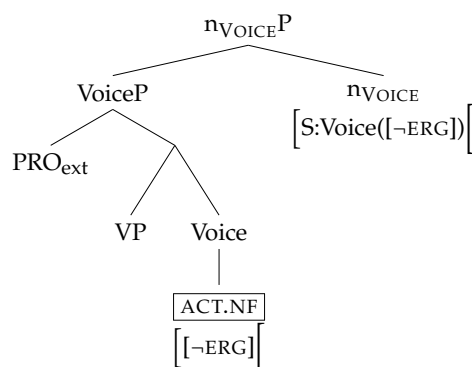
(6) **Ch'ol:** *Root intransitive NFC (non-RANT)*

Aj-Juan y-om [wäy-el]
DET-J. want sleep-NMLZ

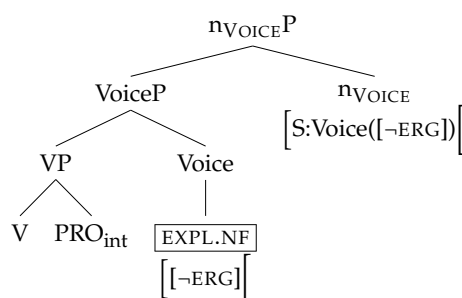
'John wants to sleep'

(Coon 2010: 114)

Given that neither unaccusatives nor unergatives assign ergative, they lack [**ERG**], i.e. bear [*−ERG*]. Merging with n_{VOICE} is therefore unproblematic, regardless of whether the intransitive is unergative – Tree 14.5 – or unaccusative – Tree 14.6:³³⁹



Tree 14.5: Unergative NFC in RANT / non-RANT language



Tree 14.6: Unaccusative NFC in RANT / non-RANT language

³³⁹Recall that I take unaccusatives to involve expletive Voice. In a system where they lack Voice altogether, we can assume that n_{VOICE} 's selection range allows for the categories *v* or *V* – in any case, these Voice-less unaccusatives must have something in common with Voice for them to pattern together with respect to selection by n_{VOICE} .

14.4.2 Mam: an object-based workaround to the RANT

Recall that the workaround strategy to the RANT found in Mam lies outside the others discussed here in being purely object-based. In (8a), the verb shows no sign of a valence decrease and the object is oblique:

(8) **Mam:**

- a. N=ch=ul aaj=x=kya xjaal tzluu'
 INCOMPL=B.3PL=DIR return=always=3PL.FOC people here
 [b'inchaa-l t-ee]
 do-INF **A.3SG-RN.PAT**
 'People come here to do it'
- b. Ma chin-x aaj-a [b'eeta-l]
 PROX A.1SG-DIR RETURN-1SG walk-INF
 'I went to walk' (England 1983: 282, 299)

I was unable to find certain data points necessary for a conclusive analysis, such as the availability of clear valence-changing morphology under the same nominalizer *-l*. That said, one possible analysis is to treat the NFC in (8a) as involving the same Voice_{TRANS.H.NF} without [**ERG**], except that (i) Mam does not choose to spell out Voice_{TRANS.H.NF} with the passive, but with a null marker; and (ii) Mam does not allow internal arguments in NFCs to enter into grammatical possession. Consequently, DP internal arguments must be linked as obliques.³⁴²

14.4.3 Q'eqchi': an apparent exception to the RANT

Another puzzle is raised by Q'eqchi', which is HIGH-ABS and exhibits the EEC:

(9) **Q'eqchi'**

- a. X-**at**-in-sak'
 COMPL-**B.2SG**-A.1SG-hit
 'I hit you'

³⁴²(8a) would on this analysis be a kind of zero-marked anti-passive. Note that Mam does have a dedicated anti-passive marker (i). So even though (8a) patterns like an anti-passive NFC in terms of the object's linking form, it does not in terms of the verb stem. Another way in which the workaround in (8a) deviates from true finite anti-passivization is the choice of relational noun, which is *-i'j* (i) rather than *-e(e)* (8a):

(i) **Mam:** *Finite anti-passive*

Ma Ø-tzyuu-n Cheep ky-i'j xiinaq
 PROX B.3SG-grab-ANTIP J. **A.3PL-RN.OF** man

'José grabbed the men' (England 2013: 5)

The exact distribution of these relational nouns is unclear. Given that anti-passivization typically has a semantic import, the choice of relational noun in (8a) vs (i) may correlate with telicity or object affectedness.

- b. Ani x-sak'-**o**-k aw-e?
who COMPL-hit-**AF**-NFUT.INTR A.2SG-RN.TO
'Who hit you?' (Dayley 1981: 29)

Yet surprisingly, Q'eqchi' seems to be free of the RANT – the verb in (10) lacks a clear valence decrease and the object is non-oblique:

- (10) **Q'eqchi'**
- T-inw-aj [aaw-il-bal]
ASP-A.1SG-want **A.2SG**-see-NMLZ
- 'I want to see you' (Berinstein 1985: 269)

The first thing to note about (10) is that the internal argument is not indexed via Set B – like in LOW-ABS languages – but via Set A, which means that it is a possessor. I propose that just like Mam, Q’eqchi’ chooses to spell out Voice_{TRANS.H.NF} with a null marker, but that unlike Mam, Q’eqchi’ does allow the internal argument to enter into grammatical possession in NFCs.

14.4.4 Non-canonical object linking

The analytic sketches I have proposed for the above patterns in K'ichee', Mam and Q'eqchi' predict that we should also find a language with pseudo-passive NFCs in which the object is not linked via possession, but rather a relational noun. This is correct. As first noticed by Paulina Lyskawa and Rodrigo Ranero (p.c.), Tz'utujil allows internal arguments in pseudo-passive NFCs to also be linked as obliques via the same relational noun used in anti-passives:^{343,344}

- (12) **Tz'utujil de Santiago**
- a. Anen n-Ø-e-mjon [a-kon-x-ik tet]
1SG INCOMPL-B.3SG-A.1SG-PROG A.2SG-look.for-PASS-NMLZ 2SG
'I am looking for you' (own notes: RP)

³⁴³In (12b), 3SG Set A marking on the verb either agrees with the relational noun, which lacks person features, or is a default. Note that the sentences in (12) were produced by the same speaker. My notes show that two variants are not optional across all contexts, depending on factors like verb-object adjacency and the person features of the object.

³⁴⁴Oblique object linking is ungrammatical in finite transitive frames. This is expected because it shows that the object's linking form via either possessive or oblique linking in NFCs truly follows from the fact that internal arguments cannot be canonically Case-licensed in the context of Voice_{TRANS.H.NF}.³⁴⁵

- (11) **Tz'utujil de Santiago**
- *Atet x-{\emptyset-/n-}a-ch'éy uw-xiin nen
2SG COMPL-{B.3SG-/B.1SG-}-A.2SG-hit A.1SG-RN.OF 1SG
- intended: ‘You hit me’
- (own notes: RP)

- b. Anen n-Ø-e-mjon [r-kon-x-ik
 1SG INCOMPL-B.3SG-A.1SG-PROG A.3SG-look.for-PASS-NMLZ
 aw-xiin tet]
 A.2SG-RN.OF 2SG
 'I am looking for you'
 (Paulina Lysaka & Rodrigo Ranero p.c.)

The inverse pattern to (12b), where an anti-passive verb occurs with a possessor object, is also attested in e.g. Q'eqchi':

(13) **Q'eqchi'**

- X-in-lub' [(chi) r-iiq-a-n-k-il]
 COMPL-B.1SG-tired COMP A.3SG-carry-SUF-ANTIP-SS-NMLZ
 'I got tired of carrying it' (Xol-Choc 2007)

This variation in object linking is expected, the reason being that because Voice_{TRANS.H.NF} is Case-neutral, the internal argument is in principle open to more than one licensing strategy.

14.4.5 Tojol'ab'al: an in-between case

Finally, there is Tojol'ab'al, which shows inconclusive properties with respect to the RANT. What is clear is that Tojol'ab'al is LOW-ABS (14a) and lacks the EEC (14b):

(14) **Tojol'ab'al**

- a. J-nuts-u-w-ex
 A.1-chase-STAT-EP-B.2PL
 'I / we chased you guys'
 b. Ja' ja=Patrisya sma s-ta'-a
 FOC DET=P. maybe A.3-reach-STAT
 'Maybe it was Patricia who reached it'
 (Curiel Ramírez del Prado 2017: 583, 593)

This leads us to expect that Tojol'ab'al should be a non-RANT language; unfortunately, the data on this are inconclusive. On the one hand, Furbee-Losee (1976) reports (15a), which suggests that Tojol'ab'al lacks the RANT, in consistency with its being LOW-ABS; according to Peake & Curiel (2007), on the other hand, NFCs based on transitive verbs must always be pseudo-passive, which implies that the RANT is present despite the LOW-ABS property:

(15) **Tojol'ab'al**

- a. Hose x-y-il-a [s-mak-e' Manwel]
 J. COMPL-A.3SG-see-STAT.TRANS A.3SG-hit-B.3PL M.
 'Jose sees Manuel hit them' (Furbee-Losee 1976: 207)

- b. Kala wab' lek [ja s-k'uts'-j-el ja si'i]
 I told you good DET A.3SG-cut-PASS-NMLZ the firewood
 'I promised you to split the firewood' (Peake & Curiel 2007)

Tojol'ab'al from deviates from other LOW-ABS languages sampled here in being the only one with *-oj* nominals:

(16) **Tojol'ab'al**

Ajyita niwan [mil-uj chan]
 exist already big.kill-NMLZ bird

'There was already a big killing of birds'

(Curiel Ramírez del Prado 2017: 583)

It is possible that Tojol'ab'al is currently undergoing a transition from one language type to another in terms of the absolutive parameter and / or the RANT; alternatively, the discrepant examples are drawn from different dialects. I leave this issue open.

14.4.6 On sub-types of n_{VOICE}

The derivations above did not refer to the sub-types of n_{VOICE} because we were interested in the syntactic effect of the absence of [**ERG**], which holds across all n_{VOICE} sub-types. But recall that next to n_{ROOT} , RANT languages either have one n_{VOICE} for all (formally) intransitive stems, $n_{VOICE.INTR}$, and they may further distinguish $n_{VOICE.INTR}$ along an unaccusative-unergative dimension, giving rise to $n_{VOICE.INTR.1}$ vs $n_{VOICE.INTR.2}$. Non-RANT languages distinguish two sub-types of n_{VOICE} , one for transitive ($n_{VOICE.TR}$) and one for intransitive stems ($n_{VOICE.INTR}$), which may be homophonous. The relevant tables are repeated here:

Verbal complement	n _{ROOT}	n _{VOICE}			
		n _{VOICE.TR}	n _{VOICE.INTR}	n _{VOICE.INTR.1}	n _{VOICE.INTR.2}
Undersived transitive root + bare NP object					
Transitive (no workaround) + DP object					
Unergative					
Anti-passive					
Unaccusative					
(Pseudo-)passive					

Table 14.4: Mayan nominalizers in RANT languages

Verbal complement	n _{ROOT}	n _{VOICE}			
		n _{VOICE.TR}	n _{VOICE.INTR}	n _{VOICE.INTR.1}	n _{VOICE.INTR.2}
Undersived transitive root + bare object					
Transitive stem + DP object					
Unergative					
Anti-passive					
Unaccusative					
(Pseudo-)passive					

Table 14.5: Mayan nominalizers in non-RANT languages

Assuming that formal distinctness entails distinct elements, I propose the following sketches. For those RANT languages with the relevant sub-division, given that n_{VOICE.INTR.1} – for unergatives and anti-passives – is more specific than n_{VOICE.INTR.2} – for unaccusatives and (pseudo-)passives – we can endow only Voice_{ACT} and Voice_{ANTIP} with a feature that reflects their agentivity. Unaccusatives and (pseudo-)passives would lack this feature due to being non-agentive.^{346,347}

³⁴⁶This is not problematic for pseudo-passives given that the passive marker is in these cases co-opted, i.e. it does not faithfully reflect the agentivity of Voice_{TRANS.H.NF}.

³⁴⁷In Uspanteko, the choice of nominalizer over another is lexically conditioned with some verbs, and may correlate with differences in Aktionsart (Can Pixabaj 2007).

For non-RANT languages, we can simply establish two n_{VOICE} sub-types, one of which selects complements bearing [$*_{\text{ABS}}$] ($n_{\text{VOICE.TRANS}}$), and one which does not ($n_{\text{VOICE.INTR}}$).

14.5 Transitivity restrictions and summary

We close this chapter and case study on Mayan NFCs by summing up the restrictions they show on transitivity.

The RANT arises due to the interaction of two crucial ingredients: first and foremost, there is the requirement imposed by n_{VOICE} on its complement that it not assign dependent case, i.e. ergative. This restriction is selectional. There is reason to believe that it is a deeper property of the grammatical principles surrounding n . We know independently that n tends to restrict certain properties in its verbal complement, though it remains to be seen whether this is universal. Yet Alexiadou herself notes languages like Hebrew, which allows internal arguments to be linked via accusative (17a) or genitive (17b):

(17) Hebrew

- a. ha-harisa ha-cava 'et ha-'ir
the-destruction the-city ACC the-city
 - b. ha-harisa ha-cava 'et ha-'ir
the-destruction the-city ACC the-city
'The army's destruction of the city'
- (Alexiadou 2001: 88)

Assuming that n merges directly with VoiceP in (17) (and that all else remains equal), we would have an instance of well-formed dependent case under n_{VOICE} . There is also a domain condition involved, as evinced by e.g. Turkish TP-level nominalizations, which allow the presence of accusative (Borsley & Kornfilt 2000); that is, in order for n to be able to restrict Voice, it must unsurprisingly be in an immediately local configuration with it. For this reason, I take it that this restriction is a robust, but non-universal selectional requirement imposed by n_{VOICE} on its complement. It therefore classifies as ad hoc.

Second, there is the absolutive parameter, which takes on the values HIGH or LOW. This property actually characterizes two kinds of restrictions, depending on how we approach it: in terms of its effects, the absolutive parameter is a Case-theoretic restriction; with respect to its source, however, it is a paradigmatic restriction stemming from distinct inventories of Voice heads. Whenever a language is HIGH-ABS, the ban against ergative under n_{VOICE} gives rise to the RANT. In this interaction, the paradigmatic side of the absolutive parameter is what gives rise to the RANT in the first place, whereas the Case-theoretic side is what ultimately determines the specific workaround strategies, i.e. the particular morpho-syntactic shapes that are employed.

In sum: we arrive at an account of the RANT that is again fully derivational, due to capturing all of its properties by appealing solely to the properties of its parts and the ways in which these are brought together.

Part IV

Concluding remarks

Chapter 15

Concluding remarks

This final chapter summarizes the results, bringing them together to an initial typology of complex transitivity restrictions that can arise in grammar; this is done in Section 15.1. I also outline the application of repairs in the sense of Optimality Theory to two kinds of exceptionally well-formed grammatical properties that we encountered along the way (Section 15.2), show how a recurring logic used in the results helps explain a certain other peculiar construction (Section 15.3), and touch on other phenomena to be explored in future work (Section 15.4).

15.1 Summary of findings

15.1.1 Co-occurrence restrictions: serial verb constructions

In our first case study, we investigated co-occurrence restrictions on the transitivity values of verbs in SVCs. The attested transitivity profiles were derived by cross-cutting the level and mode of structure-building, and also by appealing to independent principles on how elements must be integrated, projection, and the fact that transitive verbs must generally have their internal argument slot closed off.

In NC-SVCs built via adjunction, VP-level adjunction yields profile Type B ($\geq[trans + trans]$), while VoiceP-level adjunction yields profile Type A (no restrictions). Type B follows because only internal arguments are available at the VP-level, and because serialization via adjunction requires integration to proceed through predication, which entails syntactic argument sharing at the phrasal level. The same conditions apply to the VoiceP-level, with the difference that the external argument position is freed up as an additional locus for argument sharing; consequently, there are no transitivity restrictions at this level.

In C-SVCs, V-level adjunction yields profile Type C ($*[trans + intr_H]$). Because V-level adjunction configurations yield base-generated complex heads, C-SVCs sidestep the need for integration to proceed via syntactic argument sharing. Consequently, the only transitivity restriction that arises at this level reduces to the IAR of adjoined

transitive verbs and the MNC: whenever an adjoined transitive cannot have its IAR satisfied indirectly via co-indexation or existential binding through a head transitive verb, the head verb cannot itself be intransitive.

In SVCs built via adjunction, the co-occurrence restrictions on transitivity are intimately connected to the restriction that arguments in adjoined verbal elements cannot canonically project into the syntax: they must either project as null arguments, modeled here as minimal pronouns (NC-SVCs), or remain entirely implicit (C-SVCs). In NC-SVCs, this follows from our theory of serialization via adjunction, while in C-SVCs, this follows from the MNC.

This effect is absent in a subset of NC-SVCs involving systematically recurring verbs with an argument-adding function, which I analyzed as verbal periphrastic counterparts of morphological applicatives. As applicatives, these NC-SVCs involve complementation, and because integration in complementation proceeds via selection, there is no requirement that an argument in either the complement or the main projection be realized in some special way. The applicative NC-SVCs discussed here – derived via VP-level complementation – do appear to exhibit a transitivity effect due to robustly involving lexical verbs with direct objects, though this was surmised to likely be due to a coincidence and / or event-structural conditions.

Likewise, we do not expect any transitivity restrictions to arise from VoiceP-level complementation *per se*, a level which is often not regarded as falling under serialization due to the frequent obviously complementing nature of the kinds of verbs that are typically involved, e.g. causatives. Finally, the status of V-level complementation remains unclear.

The co-occurrence restrictions on transitivity in SVCs have a largely configurational source, alongside fundamental well-formedness conditions on syntactic and semantic integration. The restrictions follow naturally from what we know about the structures involved, without the need for additional, ad hoc grammatical statements.

The main contribution of this first case study was to show how complex transitivity restrictions can arise without the grammar making reference to any direct component of transitivity *per se*, but e.g. argument positions. This came with three additional, welcome results: first, where previous work has often made claims about serialization as strictly involving either adjunction or complementation, but not both, we found that adjunction and complementation must both be recognized as available serialization strategies. Second, there are genuine deep-syntactic differences between contiguous and non-contiguous serialization. This point is important in light of the fact that – just as some scholars have attempted to reduce serialization to a single mode of structure-building – there have been attempts to reduce serialization to a single level at which structure-building takes place, with structures that are contiguous on the surface argued to be derived from underlyingly non-contiguous structures. Finally, I proposed a novel definition of SVCs in terms of necessary instead of absolute conditions that makes reference to clear-defined grammatical principles, rather than

the purely formal criteria that are usually employed.

15.1.2 Matching restrictions: transitivity matching

Our second case study was dedicated to transitivity matching, where two verbal elements co-vary with respect to some component of transitivity. Here, too, both the mode and the level at which structure-building takes place played an important role.

In semantic TM, the matching reflex comes with a semantic effect on all verbs. The kind of semantic TM found in Saliba and Tariana follows from semantic type-matching at the level of the verb stem. In order for semantic TM to arise via type-matching, it must occur in V-level adjunction configurations prior to the introduction of any arguments, because it is only in these that the grammar can force semantic types to match. Matching at the phrasal level is unavailable due to the unavailability of backtracking and lookahead, while the very act of complementation alters the semantic type of the complementing head.

In syntactic TM, the matching reflex on the matrix verb is transparent but lacks a semantic effect. This sub-type was modeled as the result of an agreement relationship between a feature [uTRANS] borne by certain matrix verbs on the one hand, and structure-building features on embedded Voice on the other. Syntactic TM is limited to complementation because probing into adjuncts is unavailable.

Finally, in morphological TM the matching reflex on the second element is neither transparent nor associated with a semantic effect. Morphological TM was argued to be an instance of transitivity-conditioned allomorphy, following existing work. Given that adjunct-triggered allomorphy is unattested, this sub-type of TM must be limited to complementation, like syntactic TM.

The restrictions at play in transitivity matching are all conditioned configurationally. These configurational sources, however, only made up the necessary conditions for a particular type of matching to arise – we therefore had to additionally formulate ad hoc grammatical requirements in order to fully derive each matching sub-type. Once we allowed for these, the various sub-types of matching fell out as the result of these requirements interacting with distinct transitivity components to jointly be targeted by the grammatical processes of type-matching (semantic), agreement (syntactic) or allomorphy (morphological).

This case study constitutes the first comprehensive, generative account of transitivity matching, which crucially recognizes three distinct sub-types. Transitivity matching adds to the typology of grammatical matching effects, showcasing how distinct components of transitivity are distributed across elements within VoiceP, and how these may variously be targeted by different grammatical operations.

15.1.3 Reductive restrictions: Mayan non-finite clauses

The third case study looked into non-finite clauses in Mayan, which often restrict transitive verbs, a property that was dubbed the RANT. The RANT arises due to the interaction of two crucial ingredients: first, the selectional requirement imposed by certain nominalizers, n_{VOICE} , on their verbal complement that it must lack an ergative case feature; second, the absolutive parameter, which was classified as a kind of global configurational property that can take on the values HIGH or LOW. Whenever a language is HIGH-ABS, the ban against ergative under n_{VOICE} gives rise to the RANT.

This restriction is the only one discussed here with a selectional source. While the ability of certain nominalizers to restrict their verbal complement in this way is clearly itself conditioned configurationally – i.e. the two must be in a local selectional relationship – it does not clearly follow naturally from nominalization and selection per se, meaning that the sufficient condition on this restriction – i.e. the selection against an ergative case feature – is ad hoc.

Based on existing and novel data from elicitations, this third case study offered an original account of the Mayan RANT that is surprisingly simple. On the one hand, it makes reference to properties which are more specific to Mayan, such as ergativity, and whether absolutive is assigned high or low; on the other hand, it builds on an idea that has independently been proposed for nominalization in general, which is that nominalizers may select complements with a defective Voice head that does not assign accusative case (Alexiadou 2001).

15.1.4 An initial typology of syntactic transitivity restrictions

The main intention of this dissertation was to uncover the kinds of complex syntactic interactions that can arise in terms of transitivity. We have established the existence of three general types: co-occurrence, matching and reductive interactions – each of which involves restrictive interactions on distinct components of transitivity in various ways. This initial typology raises the question of whether it exhausts the set of possible complex transitivity restrictions, or whether further types exist.

Based on these three, we can envisage at least two other, logically possible kinds of systematically recurring transitivity restrictions, which I call *dissimilative* and *augmentative*. A dissimilative restriction is one where two elements would have to be opposite with respect to some component of transitivity when co-occurring, despite each element being free to occur with any value in isolation. A hypothetical example would be a construction similar to syntactic TM as found in Panoan, except that a transitive embedded verb would give rise to intransitive morphology on the matrix verb, and vice versa. We can also think of such a situation as instantiating an *anti-matching* restriction, reminiscent of e.g. the dissimilation phenomena that have been described in phonology.

What I am calling an augmentative restriction is meant to be the conceptual opposite of a reductive restriction: a verbal element occurring in the context of another morpheme shows a value range of some transitivity component that is larger than its range in isolation. In a construction showing this kind of restriction, for example, a verb that can only be transitive in isolation would additionally be able to be intransitive when co-occurring with another verb. In fact, we have already come across this kind of restriction in contextual bivalence, where a verb can do something transitivity-wise in a complex verb construction that it cannot do as an independent verb.

I leave these restrictions, and potentially further (aub-)types, to future work.

15.1.5 The necessity of a derivational grammar

These case studies show how transitivity-importing elements can interact and give rise to different kinds of complexity, thus supporting a notion of transitivity in which it consists of various components that are distributed across VoiceP. They also provide strong support for a derivational model of grammar, where the properties of larger structures follow from the properties of their building blocks, and the processes by which these are brought together. In general, we saw how the ability of certain restrictions to arise is systematically tied to certain necessary and sufficient conditions, and conversely how certain other structural contexts simply cannot produce a given restriction.

In SVCs, this was brought out clearly by how all and only the attested transitivity profiles (and other properties connected to these, such as how arguments must project) follow from how complementation and adjunction interact with the level of structure-building. The level of structure-building thus goes hand in hand with a timing effect: serialization at the V-level occurs prior to argument introduction, at the VP-level after internal argument introduction, and at the VoiceP-level after external argument introduction.

Similarly, the various sub-types of transitivity matching show how the availability of various components of transitivity is strictly tied to specific points in the derivation. Type-matching in semantic TM, for example, cannot occur at the VP-level because the mechanism that builds SVCs would be unable to backtrack the semantic types of verb stems. One consequence of this is that syntactic and morphological TM must necessarily involve a matching reflex that is devoid of meaning.

The necessity of a derivational grammar was also demonstrated by the Mayan RANT, where certain conditions interacting with one another needed to be overlaid on the more global property of the absolutive parameter. Not only did we need to recognize the role of nominalization and Voice in the conditioning of the RANT, but also the structural closeness between the two, i.e. two needed to be in a maximally local relationship of selection in order for the nominalizer to be able to make certain

demands on Voice.

15.2 Optimality Theory and repairs

Along the way, we encountered two properties which have in common some kind of grammatically conditioned exceptionality: contextual form-function mismatches, and the exceptional existential binding of internal arguments. This naturally begs the question of whether they can and should be handled by *Optimality Theory* (OT), where a grammar consists of a set of ranked and violable constraints (Prince & Smolensky 1993, McCarthy & Prince 1995, Prince & Smolensky 2004). Reasons of time and space keep me from working out the details here, but these effects do lend themselves to OT-based analyses, which I will briefly sketch out.

15.2.1 Contextual form-function mismatches

We encountered three cases of contextually conditioned form-function mismatches: first, semantic TM may give rise to contextual bivalence, where verbs that are strictly monovalent in isolation exceptionally become bivalent within C-SVCs in the context of a transitive co-verb; second, in syntactic TM, matrix verbs bear voice morphology that is semantically expletive only in the context of matching with the transitivity of an embedded verb; and third, numerous Mayan languages exhibiting the RANT produce syntactically transitive VoicePs with passive morphology specifically in non-finite contexts, i.e. pseudo-passive NFCs.

15.2.1.1 Contextual bivalence

In contextual bivalence (Sections 6.4.2, 7.4 and 8.2.3), there is a mismatch between a verb's semantics and its morpho-syntax: the internal argument is not thematically licensed by the contextually bivalent verb, but realized as its morpho-syntactic dependent – a situation which was likened to raising at the thematic level. Contextual bivalence itself was carried out by the operator ADD, which augments the argument structure of a verb with an expletive thematic role, provided that the verb meets certain conditions, which include being of type $\langle v, t \rangle$ and expressing a particular kind of stative eventuality. The way I have chosen to model it here, nothing about the core mechanism underlying contextual bivalence per se involves conflicting grammatical requirements. I propose that it is after the application of ADD, at Spell-Out, where OT might come into play.

Recall that the particular strategy chosen to spell out contextual bivalence varies from languages to language, with some choosing to realize the dummy argument as a direct object, and others opting for oblique licensing; moreover, the respective transitivity and oblique morpheme is always comparatively unmarked within a given language. Assuming that these morphemes carry some semantic import, as

minimal as it may be, OT can be used to resolve a conflict arising between (i) the need to spell out the expletive argument on the one hand, in turn the result of a generalized need to project arguments; and (ii) the requirement that thematic roles be maximally coherent with the semantic components of morphemes on the other, subsumed by a generalized requirement on coherence between semantics and morphemes. This could be handled by a faithfulness constraint like *REALIZEARGUMENT* for the former, and a markedness constraint like *SEMMORPHCOHERENCE* for the latter, with *REALIZEARGUMENT* outranking *SEMMORPHCOHERENCE* such that whichever morpheme induces the fewest violations of *SEMMORPHCOHERENCE* wins by virtue of being least marked.³⁴⁸ The fact that the least marked option emerges is consistent with the cross-linguistic tendency of repairs to take comparatively unmarked forms.

15.2.1.2 Contextual deponency

Contextual deponency is instantiated by voice morphology in syntactic TM (Section 10.2.3.3), and pseudo-passive NFCs in Mayan languages with the RANT (Section 13.2). The mismatches here reside between syntax and morphology, and they are crucially conditioned by specific structural contexts – hence *contextual deponency*. In syntactic TM, contextually deponent behavior is limited to a subset of matrix verbs, while in Mayan pseudo-passives, it is absolute, arising with any transitive verb that enters into a non-finite structure.

Canonical deponency is an all-or-nothing property of particular lexical items; see e.g. Embick (2000), Baerman et al. (2007) and Müller (2013). Grestenberger (2018) shows how deponency can be suspended in non-finite contexts, offering a derivational account that makes reference to the structural context; even here, however, whether or not a verb gives rise to deponent behavior boils down to a lexical matter.

We can handle the contextual deponency of both syntactic TM and Mayan pseudo-passive NFCs with the same general logic as above for contextual bivalence. In each case, the particular marker that is chosen to spell out the morpho-syntactic property set at hand falls out as comparatively unmarked because it induces the fewest violations when the need to pronounce syntactic elements – the feature [uTRANS] in syntactic TM, and the non-finite transitive Voice head in Mayan – is ranked higher than the need for coherence between a morpheme and the property set it spells out, i.e. a constraint like *PRONOUNCEX* outranks *SEMMORPHCOHERENCE*.

Under these analyses, contextual deponency – and also the choice of linking morpheme in contextual bivalence – qualify as involving morphological *repairs*, i.e. exceptionally well-formed structural changes that occur in order to avoid illicit

³⁴⁸This predicts that we should find languages where *SEMANTICCOHERENCE* outranks *REALIZEARGUMENT*, and hence contextual bivalence fails because expletive arguments cannot be pronounced. This prediction is extensionally correct for e.g. Toqabaqita (Lichtenberk 2006, Lichtenberk 2008), but not necessarily intensionally because the facts are also compatible with the grammar banning the application of *ADD* for a subset of stative verbs.

output configurations (Murphy 2019).

15.2.2 Exceptional existential binding of internal arguments

A final property that should be modeled as an OT-based repair is the existential binding of internal arguments of non-head transitive verbs, which we encountered in the context of C-SVCs (Section 5.4.1). The generalization here was that even though existential binding does not come for free in simple transitive clauses, it becomes available for internal arguments of non-head transitive verbs in C-SVCs whenever the head transitive verb characterizes a sortally distinct internal argument (subject to language-specific conditions).

In order to restrict existential binding to C-SVC contexts, we can assume (i) an undominated constraint MNC, which makes it impossible for non-head verbs to project structural arguments in V-level adjunction configurations; (ii) a constraint like NOEXISTENTIALBINDING, which incurs a violation for every argument slot that is existentially bound; and (iii) the constraint REALIZEARGUMENT from above, which requires arguments to project. Given that existential binding is generally unavailable, NOEXISTENTIALBINDING must dominate REALIZEARGUMENT, which gives us the ranking $MNC \gg NOEXISTENTIALBINDING \gg REALIZEARGUMENT$. This derives the fact that while internal arguments cannot be existentially bound and must project in the general case (simple transitive clauses), existential binding becomes an exceptionally available option in the specific case (C-SVCs).

15.3 Extension: spurious causative constructions in Formosan

A recurring notion in this thesis is that changing the mode of structure-building may be crucial in deriving the patterns we see. In SVCs, switching from adjunction to complementation helped explain why SVCs built by the latter no longer show the restriction that one of the objects must be null; in transitivity matching, semantic type-matching was shown to be restricted to V-level adjunction configurations, and conversely complementation was shown to be necessary for the syntactic and morphological sub-types to arise. I will now extend this idea to another complex verb construction, which gives rise to a peculiar interaction with effects on transitivity, suggesting that its pattern can be explained if we simply assume that rather than complementation, it actually involves adjunction or coordination.

In some Formosan languages, spoken in Taiwan, constructions involving two verbs which jointly express a kind of causative meaning show a peculiar property: one of the verbs inherently expresses a manipulative meaning, such as ‘force’, while the other obligatorily bears causative morphology. The overall meaning of this complex verb construction – which I term a *spurious causative* – is that the subject of

the manipulative verb is identified with the causer of the morphologically causative verb, with the action denoted by the former specifying the manner by which the action denoted by the latter is brought about. Consider (1) from Paiwan, where V_1 is the manipulative ‘force’ and V_2 is the morphological causative ‘make eat’:

(1) **Paiwan:** *spurious causative*

[ʔadil] ti palang a [pa-kan] tjay kalalu
force<AV> NOM.PS.SG P. LNK CAUS.AV-eat OBL.PS.SG K.

‘Palang forced Kalalu to eat’

(Chang 2006: 264)

The presence of causative morphology on V_2 in (1) is unexpected under the assumption that it involves object control and complementation: if the causative V_2 heads a control complement to the manipulative V_1 , why does the causative V_2 have to have its transitivity increased via causativization? Assuming that it is not semantically expletive, the causative adds a causer argument, which in turn should give rise to a meaning in which it is the object referent of (1) that is forced to cause someone to eat, i.e. ‘Palang forced Kalalu to cause (someone) eat’. Yet this is not the case: (1) means that Kalalu ends up eating, not that Kalalu ends up causing someone else to eat – hence the name *spurious causative*.

One proposal that has been brought forward for spurious causativization – and control in some Formosan languages more generally – is that controllers must always be agents (Chang & Tsai 2001). According to Chang and Tsai, because controllers cannot be non-agents, object control configurations as found in other languages cannot surface as such; consequently, causativization occurs on the embedded verb in order to identify the newly added causer subject with the matrix agent subject. Put differently, what corresponds to object control in languages like English must be construed as a kind of non-canonical subject control that can exceptionally shift controller status from matrix object to subject (Chang 2018).

According to this approach, spurious causativization is a restriction rooted in what I will on the one hand characterize as a thematic source, namely the constraint that controllers not be non-agentive; on the other, Chang and Tsai argue that it also has a morpho-syntactic basis in the so-called *actor sensitivity* often found in these languages, i.e. a general orientation of numerous grammatical processes to systematically privilege agentive arguments over non-agentive ones, such as verbal agreement and voice morphology in non-finite contexts, which is limited to Agent Voice.

I propose an alternative, arguing that spurious causativization ceases to be spurious if they are analyzed not via complementation, but adjunction. A first piece of evidence for this view is that spurious causative constructions allow verbs to be variably ordered: either the manipulative verb can precede the causative verb, as in (1) above, or vice versa as in (2):

(2) **Paiwan:** *causative verb < manipulative verb*

[pa-kan] a [ʔadil] ti palang tjay kalalu
 CAUS.AV-eat LNK force<AV> NOM.PS.SG P. OBL.PS.SG K.

‘Palang forced Kalalu to eat’

(Chang 2006: 264)

Elsewhere, such variable ordering is also found in motion cum purpose (3a) and (3b) constructions:

(3) **Paiwan**a. *Motion cum purpose*(i) ‘go’ < V_2

[sa-taihuku-u] a [v<en>eLi] tua djamay!
 go.to-T.-IMP LNK buy<AV> OBL.CM dishes
 ‘Go to Taipei to buy dishes!’

(ii) V_1 < ‘go’

[yeLi-u] a [sa-taihuku] tua djamay!
 buy-IMP LNK go.to<AV>-T. OBL.CM dishes
 ‘Go to Taipei to buy dishes!’

b. *Phasal constructions*(i) ‘start’ < V_2

[patagil] a [sane-cekui] ti zepul tucu
 start.AV LNK make<AV>-table NOM.PS.SG Z. now
 ‘Zepul is starting to make tables now’

(ii) V_1 < ‘continue’

[ka-kan] a [pazurung] tua ciʔaw ti zepul
 RED<AV>-eat LNK continue.AV OBL.CM fish NOM.PS.SG Z.
 ‘Zepul continues to eat fish’ (Chang 2006: 261, 260, 262)

I take these constructions to involve adjunction, and adjunct clauses in Paiwan are free to occupy various positions in the clause (Tang 1998, Chang 2006). That swapping up the order of verbs is not merely the result of movement is shown by the presence of the linker in (1) and (2), and also the fact that imperative morphology consistently surfaces on V_1 in (3a), which I take to reflect headedness; that is, a change in order correlates with a change in projection status. Crucially, except for cases involving direct speech predicates, which cross-linguistically tend to show more liberty with respect to the placement of their complement clause, neither Chang (2006)’s grammar of Paiwan nor Tang (1998)’s work on complementation in Paiwan contain any examples of a complementation configurations in which the complement precedes the head. Complements systematically follow their head:

(4) **Paiwan:** *head < complement*

Kival-amen [tu macevung a kirimu tua
 want.AV-1.NOM.PL.EXCL COMP finish.AV LNK soon OBL.CM
 nia-sengsengan]
 1PL.EXCL.GEN-work

‘We want to finish our work soon’ (Chang 2006: 364)

Second, spurious causative (1)-(2), motion cum purpose (3a) and (3b) phasal constructions all involve the linker *a*, which serves the general purpose of marking head-modifier relations (Chang 2006: 118ff), and also shows numerous conjunctive characteristics (Tsai & Wu 2012). Most complementation configurations, by contrast, involve a distinct marker, typically *tu*, cf. (4).

My proposal is that spurious causatives do not involve complementation. By extension, the manipulative verb does not induce object control with respect to the causative verb. Because neither verb is embedded under the other, both verbs end up predicating over the same subject, and hence we derive the necessary presence of causative marking on the non-manipulative verb.

Spurious causatives thus have structures similar to the ones proposed for Korean NC-SVCs built via adjunction at the VoiceP-level (Section 4.2.4 of Chapter 4), with either the manipulative verb heading the main projection and the causative verb the adjoined one (1), or vice versa (2).³⁴⁹ The meanings of spurious causatives are therefore better paraphrased as means constructions, i.e. ‘Palang caused Kalalu to eat by forcing them’, or simple conjunctions, i.e. ‘Palang forced Kalalu and caused them to eat’.³⁵⁰

Tsou and Kavalan show the same pattern (Chang & Tsai 2001, Lin 2014), for which I propose the same solution. For Kavalan, reflexive data are available which further support my proposal: once the manipulative verb is reflexive or reciprocal, a causative marker on the non-manipulative verb becomes ungrammatical:

³⁴⁹Regarding the linker *a*, I assume that it obligatorily spells out the adjunction nexus, given that it marks all kinds of head-modifier relations in Paiwan (Chang 2006: 188ff). Alternatively, spurious causatives involve coordination, with *a* spelling out the coordinator.

³⁵⁰Consistent with this analysis is the fact that some varieties of Paiwan allow the causative prefix to be omitted, in which case the linker must change to the one that is used elsewhere in complementation:

(i) **Paiwan**

- a. [ʔ-əm-adil] ti kina tua alak a/*tu(a) [pa-kan]
 force-AV NOM mother ACC child LNK CAUS.AV-eat
 ‘Mother forces her child such that she causes him/her to eat’
 b. [ʔ-əm-adil] ti kina *a/tu(a) [k-əm-an] a alak
 force-AV NOM mother LNK eat-AV NOM child
 ‘Mother forces that the child eats’

(Yeh 1997: 99)

(5) **Kavalan**

[pawRat-an-na] ni utay_i ya izip-na_i [{q<m>an} / {*pa-qan}] tu
 force-PV-3ERG ERG U. ABS boy-3GEN <AV>eat / CAUS.AV-eat OBL
 mutun
 mouse

‘Utay forced himself to eat a mouse’

(Lin 2014: 293)

This follows from the proposed adjunction analysis: because both verbs predicate over the same grammatical subject and there is no object control, the event denoted by the non-manipulative verb has a conjunctive relationship with the event denoted by the manipulative verb, rather than a mereological one; cf. Svenonius (2016). Accordingly, the actual translation of (5) is something like ‘Utay ate a mouse by forcing himself’ or ‘Utay forced himself and ate a mouse’.

I follow Chang (2018) in taking the non-object-control-inducing nature of the verbs giving rise to spurious causatives to simply be quirks of languages like Paiwan, Tsou and Kavalan.³⁵¹ We can model this in our terms as a restriction with an ad hoc selectional basis: in these languages, ‘force’-type verbs obligatorily select nominal complements (i.e. they select against verbal complements). I leave open how this is best modeled.³⁵²

Under this alternative proposal, spurious causatives pattern like the SVCs from our first case study in showing how a transitivity effect can arise without there being any grammatical demand placed on a component directly pertaining to transitivity. Just like with applicative NC-SVCs, where acknowledging complementation instead of adjunction allowed us to dispense with certain construction-specific stipulations, acknowledging adjunction instead of complementation in spurious causativization gives us a surprisingly analysis without the need for construction-specific assumptions that lack independent motivation.³⁵³

15.4 Outlook

To come to a close, I will briefly touch on two other kinds of constructions that should be investigated as future case studies from the perspective of the concepts and vocabulary established in this thesis: causatives and light verb constructions.

³⁵¹While Chang (2018)’s account is right in recognizing the inability of certain verbs to induce object control, it retains a major drawback, which is that the workaround he proposes in order to achieve the desired control interpretation uses a “shifting” mechanism that is abstractly similar to Chang & Tsai (2001)’s.

³⁵²The ad hoc nature of ‘force’-type verbs in Paiwan is clearly brought out by the fact their counterparts in a language like English allow verbal complements. Though see Chang (2018) for arguments that these verbs are not simple translations of their purported English counterparts.

³⁵³Stiebels (2015: 422f) notes that (i) the evidence for the kind of control proposed by Chang & Tsai (2001) is sparse anyway; and (ii) under a complementation analysis of spurious causativization, some other strategy should be resorted to when the presence of the causative would be inadequate, e.g. under matrix negation, or a non-implicative directive verb like ‘ask’.

While work with respect to transitivity-related issues does exist on these, especially on causatives, there is no comprehensive account that accommodates all the patterns we see (as far as I am aware). A seminal generative work in this regard is Pylkkänen (2008)'s work on causatives, although further patterns have since been uncovered and discussed which do not easily fit into her typology.

15.4.1 Causatives

Causatives are known to often impose transitivity-related restrictions on their complement. Let us quickly go over some of the interesting patterns in this regard, some of which relate directly to transitivity, while others appear to fall more on the argument-structural side of things.

In Halkomelem (Salishan), for instance, causatives are only possible with unergative verbs (6a); transitive verbs cannot be causativized as they are (6b), and must hence be anti-passivized in order to be eligible for causativization as derived unergatives (6c):

(6) Halkomelem

a. *Causative of unergative*

Ni cən ʔiməš-stəx^w k^wθə John
AUX 1SUB walk-CAUS DET J.
'I made John walk'

b. *Causative of transitive*

*Ni cən q^wəl-ət-stəx^w k^wθə səplíl ʔə tə sléniʔ
AUX 1SUB bake-TR-CAUS DET bread OBL DET woman
intended: 'I had the woman bake the bread'

c. *Causative of anti-passive*

Ni cən q^wəl-ə^m-stəx^w k^wθə səplíl ʔə tə sléniʔ
AUX 1SUB bake-ANTIP-CAUS DET bread OBL DET woman
'I had the woman bake the bread' (Gerds 1984: 194-5)

Causatives of passives (7a) and unaccusatives (7b) are also impossible:

(7) Halkomelem

a. *Causative of passive*

*Ni q^wəl-ət-ə^m-stəx^w-əs k^wθə səplíl ʔə tə sléniʔ
AUX bake-TR-PASS-CAUS-3ERG DET bread OBL DET woman
intended: 'He had the bread baked by the woman'

b. *Causative of unaccusative*

*Ni cən wəc'əš-stəx^w k^wθə sc'éšt
AUX 1SUB fall-CAUS DET stick
intended: 'I made the stick fall' (Gerds 1984: 195-6)

This reductive restriction – i.e. the ability to take only unergatives – is rooted in a selectional requirement intrinsic to the causative *-əm*, not any general properties of Halkomelem (Gerds 1984). Potential solutions include having the Halkomelem causative impose an event-structural restriction on its complement (e.g. by selecting certain only certain kinds of activities), or a morpho-syntactic one (e.g. by selecting against a structure-building feature that merges internal arguments).

This behavior is the mirror image of what we find in e.g. Mohawk (and other poly-synthetic languages), where causatives can only take unaccusative complements (Baker 1996). According to Baker, this restriction does not have a simple selectional basis, but is the result of a complex interplay of restrictions relating to argument structure, syntax and morphology.

A restriction with major effects on linking is found in Southern Tiwa (Tanoan), where – like in Halkomelem – causatives of transitive verbs come with certain restrictions. In causatives, the patient argument of a transitive verb cannot be a free-standing nominal (8a), but must incorporate (8b):

(8) **Southern Tiwa**

- a. **ʔu-de i-kur'am-ban*
baby-SUF 1S:2SO-hold-CAUS-PST
intended: 'I made you hold the baby'
- b. *L-ʔu-kur-'am-ban*
1S:2SO-baby-hold-CAUS-PST
'I made you hold the baby' (Baker 1988: 194f)

Unlike in Halkomelem, however, this restriction on argument linking is not due to the causative *-am* itself, but rather something else that is external to *-am*. While the entire range of restrictions on argument realization in Southern Tiwa is rather complex (Baker 1988, Rosen 1990, Heck & Richards 2010), the incorporation of patients is generally obligatory in the context of dative co-arguments, meaning it has a partially configurational source.³⁵⁴

This thesis cannot do justice to the other countless phenomena in this regard. There remains much to be done, and it is clear that the restrictions imposed by causatives on transitivity and argument structure deserve a thesis of their own. Where the case studies in this dissertation have been well-behaved in that most of the patterns investigated involved interactions contained within a single domain,

³⁵⁴This is illustrated by ditransitive dative-shift-type verbs:

(i) **Southern Tiwa**

- a. **Ta-wia-ban hliawra-de ʔu-de*
1S:A/A-give-PST woman-SUF baby-SUF
'I gave the woman the child'
- b. *Ta-ʔu-wia-ban hliawra-de*
1S:A/A-baby-give-PST woman-SUF
'I gave the woman the child' (Baker 1988: 194)

this is by far not the case, as argued in work like Baker (1996), which shows that the lines between argument structure, morphology and syntax are often blurred.

15.4.2 Light verb constructions

Finally, another class of constructions where there remains much to investigate are so-called *light verb constructions*. Following Butt (2010), I take light verbs to impart both a thematic and a functional component, and hence to fall between lexical verbs and auxiliaries. In light verb constructions, light verbs are combined with morpho-syntactically (and in part thematically) dependent lexical verbs under various argument-, event-structural and transitivity-based conditions; among others, see Butt (1993) on Hindi-Urdu, Bowerman (2008) and Bowerman (2014) on Australian languages, and Balusu (2014) on Telugu.

On the one hand, light verbs are interesting because they bind interactions stemming from sources in lexical semantics, event and argument structure, and morpho-syntax. On the other hand, they occasionally give rise to form-function mismatches reminiscent of the contextual deponency discussed here, where morphemes that have a consistent semantic interpretation in isolation come to be expletive within light verb constructions.

Appendix A

Serial verb constructions

A.1 True vs derived contiguity

If we generalize the derived contiguity approach and take all C-SVCs to be underlying NC-SVCs to which movement operations have applied – either of arguments, or verbs or both – we predict that the (non-)contiguity of SVCs should never correlate with any semantic or morpho-syntactic properties, except for those connected to the movement operations themselves. That is, we should never find any morpho-syntactic differences resulting from V-level structure-building because there would be no actual V-level structure-building involved – only ever VP-level structure-building, the non-contiguity of which is subsequently obscured by movement operations. Also, the distribution of superficial contiguity over the various sub-types of SVCs should in principle be random, in that we should not expect a certain sub-type of SVC to correlate with (non-)contiguity. As it turns out, this prediction is incorrect.

Recall that in instrumental SVCs, a verb taking on an applicative function in the context of serialization introduces an object with the non-core role of instrument. Instrumental SVCs do not involve syntactic argument sharing – the applied instrument is referentially distinct from the patient of the lexical verb, and integration proceeds via complementation. This lack of argument sharing is directly reflected in how arguments are linked: while SVCs whose internal arguments are shared project only one overt object, instrumental SVCs project one object per verb, i.e. an instrument and a patient.

Now, the derived contiguity approach predicts that we should in principle find instrumental serialization in both contiguous and non-contiguous form. As far as I am aware, however, all instrumental SVCs that have been described in the literature show non-contiguity on the surface (with one potential exception, discussed below). I claim that instrumental SVCs are truly NC-SVCs by necessity. Specifically, instrumental SVCs must be underlyingly non-contiguous because only non-contiguous configurations allow the overt projection of one structural argument per verb under disjoint reference, in accordance with the applicative nature of instrumental SVCs. In

other words, if instrumental SVCs were composed as contiguous V-level configurations in their underlying syntax, only either the applicative or the lexical verb would be able to realize its internal argument in the syntax, but not both, due to the MNC, which prohibits non-head verbs from projecting in V-level configurations.

This is reflected in the fact that languages with otherwise productive C-SVC formation may ban instrumental C-SVCs. Take, for example, #Hoan, which has C-SVCs with argument sharing like in (1a). An instrumental event, however, cannot be expressed as a C-SVC, as shown by the ungrammatical attempt to do so in (1b):

(1) **#Hoan**

a. *Consecutive C-SVC*

Ma qo kí [cook]-[’am] | | a’’e
 1SG FUT PL cook-eat meat
 ‘I will cook and eat meat (repeatedly)’

b. *Instrumental C-SVC*

*Ma a (kí) [!u]-[tcxa] | | a’’e ki g | | ama
 1SG PROG PL take-cut meat PART knife
 intended: ‘I will cut meat with a knife’ (Collins 2002: 4, 20)

Instrumental events can only be expressed using a non-contiguous structure, which crucially does not involve serialization because each VP has its own TAM specification (a property that is otherwise absent in #Hoan SVCs, compare the single-marked TAM specification in (1)). In other words, the construction is bi-clausal:

(2) **#Hoan: Instrumental complex clause**

Ma a [!u] g | | ama a [tcxa] | | a’’e
 1SG PROG take knife PROG cut meat
 ‘I am taking the knife and cutting the meat’ (Collins 2002: 20)

Similarly, in the Oceanic language Pileni, which has both contiguous (3a) and non-contiguous SVCs (3b), instrumental events can only be expressed using the non-contiguous type (3c):

(3) **Pileni**

a. *C-SVC: trans + trans*

Ko-i mot-ia na mara na, [mot-ia] [vakao-ina]
 TAM-3SG cut-TRANS 3SG.POSS eye DEM cut-TRANS encircle-TRANS
 ‘He cut off its (the coconut’s) end, he cut around it’

b. *NC-SVC: trans + trans*

Nekepo hua-mua mai a te tupua na e-i [pnepen-age]
 PART CAUS-front DIR PART ART creator DEM TAM-3SG prepare-DIR

a hihiga ne-i [tuhāmac]

ART RED.thin TAM-3SG distribute

‘In the beginning the creator made all things and distributed them’

c. *Instrumental NC-SVC*

Ko-i [toa] na noko ko-i [mue-ia] i te paua

TAM-3SG take 3SG.POSS bottom TAM-3SG slam-TRANS LOC ART clam

‘He slammed into the clam with his bottom’ (Næss 2004: 236, 238 243)

I claim that this is not accidental, but systematic: because instrumental serialization does not involve argument sharing, but argument addition, V-level structure-building simply does provide the necessary morpho-syntactic resources for both verbs to be able to do so because of the MNC.

The only way in which the expression of something like an instrumental event might be sanctioned in V-level configurations after all would be if either the instrument or the patient were not projected as a regular, but rather a structurally reduced argument; this brings us to a potential exception to the generalization that instrumental serialization must necessarily take place above the V-level.

Consider the instrumental SVC in (4), from Barai (Trans-New Guinea, Papua New Guinea). In (4), the applicative and the lexical verb are strictly adjacent, yet preceding the superficial verbal complex are two arguments, the patient and the instrument:

(4) **Barai:** *Instrumental SVC*

Fu {burede ije} {sime} [abe] [ufu]

he bread DEF knife take cut

‘He cut the bread with a knife’

(Foley & Olson 1985: 44)

While the strict adjacency of V_1 and V_2 indicates contiguous serialization, the presence of two objects indicates non-contiguous serialization. This example, noted in Owens (2011: 97), constitutes the only attested potential exception to my claim that instrumental SVCs cannot be built at the VP-level.³⁵⁵ As always when faced with an outlier to what is otherwise a robust pattern, we should seek to derive its apparent exceptionality before revising the generalization. I propose two possible solutions to this issue (which are not entirely mutually exclusive).

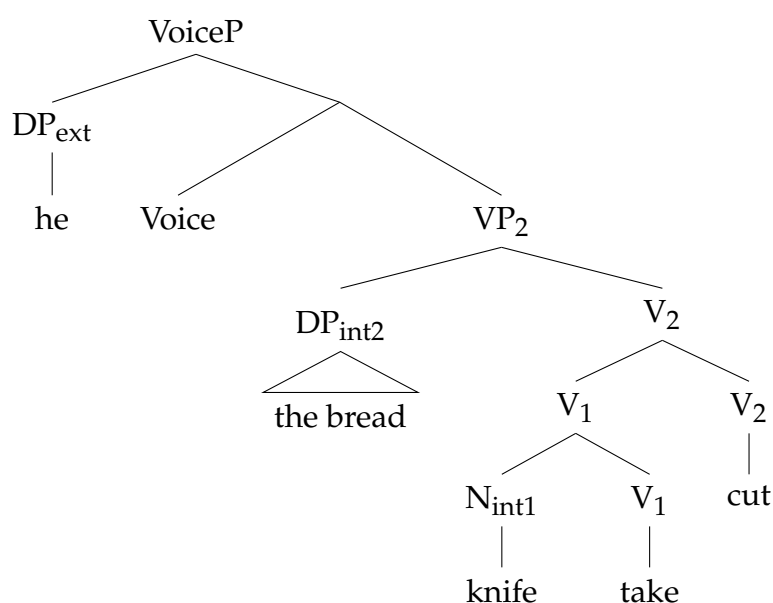
First, (4) might involve contiguous V-level structure-building, i.e. its surface syntax is a direct reflection of its underlying syntax, but with an instrument that is structurally reduced. Notice that unlike the patient *burede ije* ‘the bread’, the instrument *sime* ‘knife’ does not occur with a determiner and is translated accordingly as indefinite.³⁵⁶ As a structurally reduced argument, i.e. one that does not project

³⁵⁵For a typological overview of ‘take’ SVCs, see Lee (2019), which also lists the Barai SVC in (4) as the only instrumental ‘take’ SVC with multiple objects.

³⁵⁶Barai determiners are post-nominal (Oraro, Unini & Bradshaw 2015: 19). Unfortunately, neither Oraro et al. (2015) nor Foley & Olson (1985) provide enough information to ascertain whether the

up to DP (but only *n* / *N* or *nP* / *NP*), the instrument would not need to be licensed in the same way as a regular DP argument, yet might require strict adjacency to the applicative verb. This kind of analysis has been proposed in various guises for (pseudo-)incorporation and related phenomena; cf. Levin (2015), Hopperdietzel (2020) and van Urk (2020b), among others.

One possible structure based on this approach is given in Tree A.1, under the assumptions that (i) the *V*₁ ‘take’ is adjoined to the head *V*₂ ‘cut’, and (ii) the instrument corresponds to a bare noun that is incorporated. After the bare noun ‘knife’ incorporates into the *V*₁ ‘take’, the resulting complex *V*₁ is adjoined to the head *V*₂ ‘cut’, and given the MNC, the head *V*₂ can project its internal argument as a full DP; the linear order of (4) follows without further assumptions:



Tree A.1: *intr* + *intr*

Alternatively, (4) involves non-contiguous VP-level structure-building in its underlying syntax, but that its surface-contiguity results from verb and / or argument movement, in the spirit of Déchaine (1993) and Collins (2002). While this idea is hard to verify without fine-grained tests, there is a hint that argument movement might obscure what is actually an underlying NC-SVC configuration.

According to Olson (1978: 140f), the ordering of nominals in Barai clauses is governed not only by thematic roles and grammatical function, but also referentiality. For our purposes, the generalization is that for any two arguments differing in referentiality, the referentially more prominent one must precede the less prominent one; elsewhere, the thematically more prominent one must precede the less prominent one. To see this, consider the following simple transitive clauses: in (5a), both the subject and object are definite, and hence the subject precedes the object; in (5b), however, the subject is indefinite and the object definite, and so the object

instrument in (4) optionally or obligatorily occurs without a determiner.

must precede the subject; having an indefinite subject precede a definite object is ungrammatical (5c):

(5) **Barai:**

- a. $[3SG, +DEF]_{Subj} / [1SG]_{Obj} \Rightarrow Subject < Object$

Fu na kan-ie
3SG 1SG strike-1SG.O
'He struck me'

- b. $[3SG, -DEF]_{Subj} / [1SG]_{Obj} \Rightarrow Object < Subject$

Na e-be kan-ie
1SG person-INDEF strike-1SG.O
'Someone struck me'

- c. $[3SG, -DEF]_{Subj} / [1SG]_{Obj} \Rightarrow Subject < Object$

*E-be na kan-ie
person-INDEF 1SG strike-1SG.O
intended: 'Someone struck me'

(Olson 1978: 141f)

We can understand this correlation between syntactic position and referentiality if objects higher in referentiality may need to undergo raising (Diesing 1992, Bhatt & Anagnostopoulou 1996, van Urk 2020b). Coming back to our instrumental SVC in (4), notice that it is the patient, marked with a determiner, that precedes the instrument, which lacks a determiner, i.e. it is the referentially more prominent one that precedes the less prominent one. Assuming an underlying NC-SVC syntax, and that definite objects must indeed vacate their VP and raise to a structurally higher position, I tentatively propose the derivation in Tree A.2 for the instrumental SVC in (4): after the applicative V_1 takes the lexical VP_2 as its complement, the patient – being higher in referentiality than the instrument – must move to a position that ends up linearly preceding the instrument, which I assume for simplicity corresponds to an outer specifier of VoiceP; finally, the subject moves from Spec,VoiceP to Spec,TP, thus giving us the linear order of (4):³⁵⁷

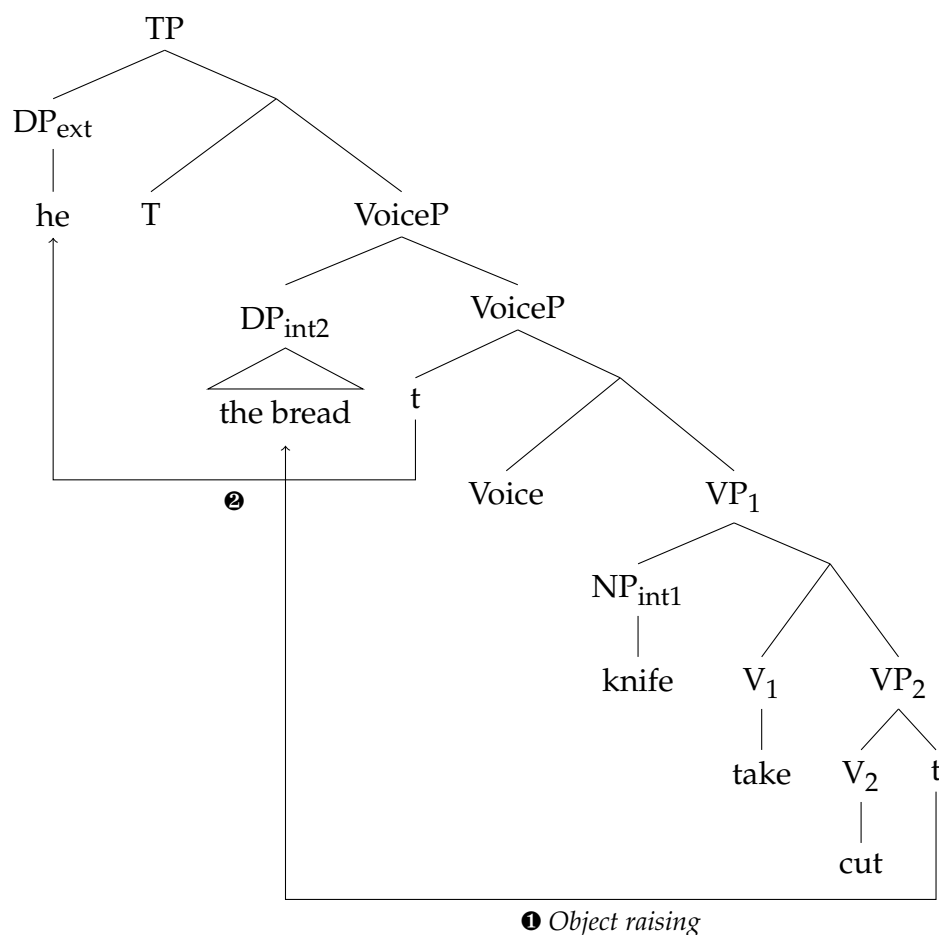
³⁵⁷This behavior carries over into ditransitive clauses with non-oblique indirect objects, where definite Recipients must precede indefinite patients:

(i) **Barai:** *Ditransitive clause*

Fu na ire vaj-ie
3SG 1SG food give-1SG.O

'She gave food to me'

(Oraro, Unini & Bradshaw 2015: 23)



Tree A.2: Barai instrumental NC-SVC + DP movement

In sum, instrumental SVCs constitute evidence for an actual, deep-syntactic difference between contiguous and non-contiguous serialization configurations. Because instrumental SVCs are always surface-contiguous, and because of their addition of an instrument role that is distinct from the patient of the lexical verb, they are necessarily composed as underlying non-contiguous configurations since only these allow the projection of both the instrument and the patient.

A.2 Argument blocking in Lexical Decomposition Grammar

I have argued that because non-head transitive verbs in C-SVCs do not project beyond the stem level, they are by extension unable to project their internal arguments. Argument blocking is thus the result of a morpho-syntactic configurational restriction. It is useful to turn to how argument blocking has been handled for strong resultatives within an alternative framework, Lexical Decompositional Grammar, and see how this approach relate to the current one.³⁵⁸

In strong resultatives based on transitive manner verbs, i.e. resultatives in which

³⁵⁸The *weak vs strong resultative* distinction goes back to Washio (1997).

the argument of the result predicate is not sub-categorized by the transitive manner verb, internal arguments are productively blocked from realization. In (6), for example, the argument that comes to be empty as a result of drinking (the wine cellar) is distinct from the argument which undergoes the drinking (a liquid):

- (6) Sie tranken den Weinkeller leer
 they drank the wine cellar empty
 ‘They drank the wine cellar empty’ (Kaufmann & Wunderlich 1998: 7)

Even though (6) involves three individual arguments, (the drinker, the liquid drunk and the cellar that becomes empty as a result of drinking) only two end up being projected: the internal argument of the manner verb, referring to the liquid that is drunk, is blocked from realization, while that of the result predicate is projected. We can hence assign to (6) the denotation in (7), where the internal argument of ‘drink’ remains implicit:

- (7) $\exists y \exists s. [drink(he, \boxed{y}) \& BECOME(empty(wine\ cellar))](s)$

Given the operation of strung resultative extension in (8), which – similar to our operation of Event Cumulation – instantiates a kind of head extension by extending the predicate structure of a manner-denoting verb by a result component that is predicated of a non-subcategorized argument, the transitive verb ‘drink’ is thus extended from its base denotation in (9a) to ‘drink empty’ in (9b), in which the internal argument referring to the liquid that is drunk is blocked from structural realization: by contrast, ‘drink’ in its independent use specifies two arguments which can both be realized structurally (9a):

- (8) *Strong resultative extension:*
 $... \lambda s. [P(...)](s) \Rightarrow \lambda Q \lambda z ... \lambda s. [P(...) \& BECOME Q(z)](s)$
 (Kaufmann & Wunderlich 1998: 10)
- (9) a. $\llbracket drink \rrbracket = \lambda y \lambda x \lambda s. [drink(x, y)](s)$
 b. $\llbracket drink\ empty \rrbracket = \lambda z \lambda x \lambda s. [drink(x, \boxed{y}) \& BECOME\ empty(z)](s)$
 (Wunderlich 1997: 36)

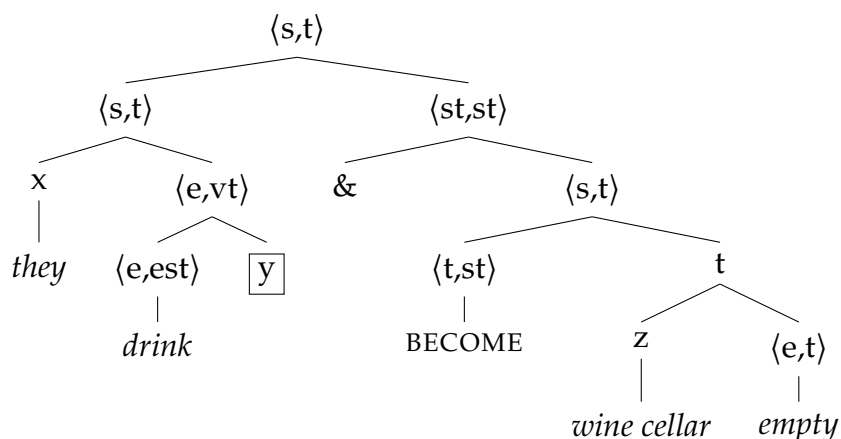
To capture this systematic pattern of argument linking, Kaufmann & Wunderlich appeal to the notion of *Lexical command*, or *L-command*, defined in (10):

- (10) *L-command*
 α L-commands β iff the node γ , which either directly dominates α or dominates α via a chain of nodes that are type-identical to γ , also dominates β .
 (Wunderlich 1997: 41)

L-command is similar to c-command, yet defined over SF structures rather than syntactic structures. After ‘drink’ undergoes extension by the result state ‘empty’, it

has the SF in (11a), represented as the tree structure in Tree A.3. From Tree A.3, we can derive the L-command relations in (11b), according to which x L-commands both y and z, but y does not L-command z:

- (11) a. SF: $[drink(x,y) \ \& \ BECOME \ empty(z)](s)$ (see Tree A.3)
 b. L-command: $\langle x,y \rangle, \langle x,z \rangle$ (but not $\langle y,z \rangle$)



Tree A.3: The tree structure of the SF of *drink empty*

Since it is y, the liquid that is drunk, that is blocked from realization, and since y (i) does not L-command z, and (ii) is not the structurally lowest argument, this gives rise to the following generalization on argument linking:

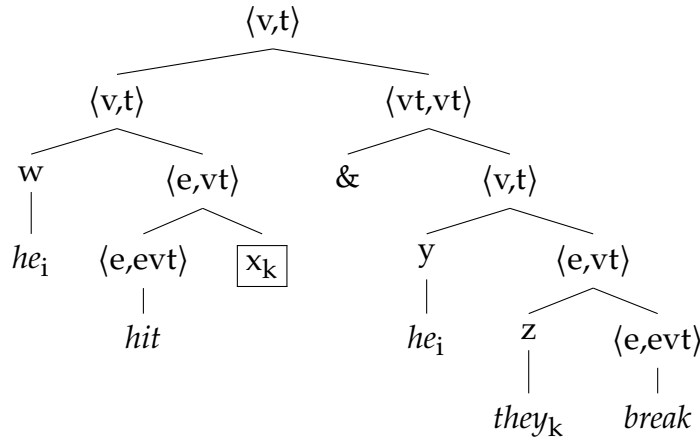
- (12) *Restriction on structural arguments*
 An argument is structural only if it is either the lowest argument or (each of its occurrences) L-commands the lowest argument.
 (Wunderlich 1997: 41)

(12) is motivated by the fact that "the lowest argument is always linked to the direct object and the highest argument always L-commands the lowest one and thus is linked to the subject, but medial arguments may be kept from being linked into syntax under certain conditions" (Wunderlich 1997: 41). Informally, y is too deeply embedded in Tree A.3 to be targeted for structural realization. Given the L-command relations in (11b) and the condition in (12), we finally arrive at the TS in (13) (=9b)), which correctly reflects in its λ -grid that y is blocked from structural realization:

- (13) TS: $\lambda z \lambda x \lambda s. [drink(x, \boxed{y}) \ \& \ BECOME \ empty(z)](s)$

Because structural arguments (i.e. arguments mapped onto the subject, direct object and indirect object functions) can in this way be defined in terms of L-command, SF determines the mapping of arguments onto syntax. It should be clear by now that there is a certain similarity between argument blocking in strong resultatives and C-SVCs, not only empirically but also in terms of theoretical handling. In fact,

the LDG approach is able to correctly derive the argument linking profiles of Saliba C-SVCs. For an object-sharing C-SVC like (14a), I assume the SF in (14b) (with the tree structure in A.4), which in turn characterizes the L-command relations in (14c) and, finally, the TS in (14d). Ignoring subject sharing, TM and V-level contiguity for the sake of illustration, we correctly derive the structural realization of $V_{2=H}$'s internal argument, and the blocking of V_1 's internal argument:



Tree A.4: The SF tree structure of *hit-break*

(14) C-SVC with object sharing

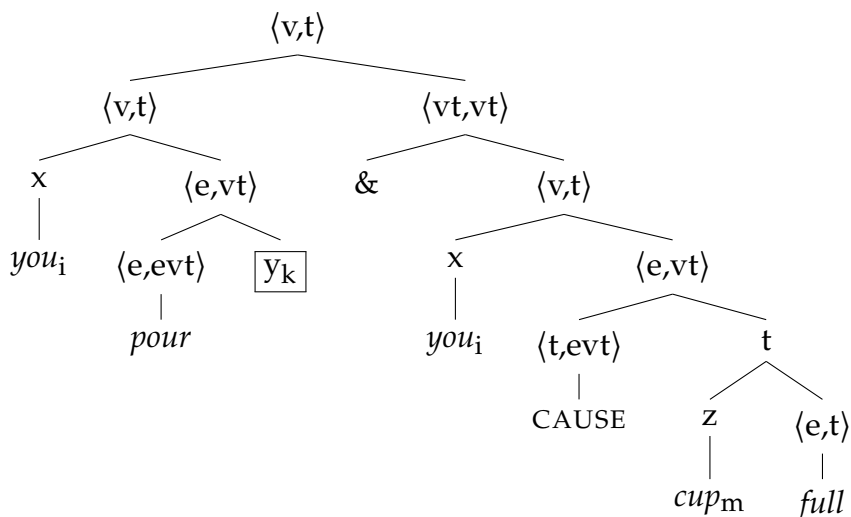
- a. Ye-[koi]-[kesi]-di
3SG.S-hit-break-3PL.O
'He broke them'
- b. SF: $[hit(w_i, x_k) \ \& \ break(y_i, z_k)](s)$
- c. L-command: $\langle w, x \rangle \ \& \ \langle w, z \rangle$ (but not $\langle x, y \rangle \ \& \ \langle x, z \rangle$)
- d. TS: $\lambda z \lambda w \lambda s. [hit(w_i, \boxed{x_k}) \ \& \ break(\boxed{y_i}, z_k)](s)$

This also applies to C-SVCs without object sharing, in which the blocking of V_1 's internal argument is more obvious; see (15) and Tree A.5:

(15) C-SVC without object sharing

- a. Kaputi ku-[ini]-[he-mwayau]- $\emptyset$
cup 2SG.S-pour-CAUS-full-3SG.O
'Fill the cup'
- b. SF: $[pour(w_i, x_k) \ \& \ CAUS-full(y_i, z_m)](s)$
- c. L-command: $\langle w, x \rangle \ \& \ \langle w, z \rangle$ (but not $\langle x, y \rangle \ \& \ \langle x, z \rangle$)
- d. TS: $\lambda z \lambda w \lambda s. [pour(w_i, \boxed{x_k}) \ \& \ CAUS-full(\boxed{y_i}, z_m)](s)$

Under the LDG account, certain internal arguments are blocked from realization if they do not fulfill the condition on structural argument linking in (12), while under the present, MNC account, they are blocked because their parent verb does not syntactically project. Note that even though L-command is a property of SFs, which are not themselves syntactic structures, it is nonetheless a property that determines

Tree A.5: The SF tree structure of *pour-fill*

which arguments can be mapped from SF onto syntactic structures, such that the net effect we get is the same as under the MNC account: under both accounts, the internal argument of what corresponds syntactically to the non-head verb is correctly blocked from realization.³⁵⁹

Even though the LDG account gives us the same argument linking profile extensionally, it is not clear in how far it is generalizable to the cases at hand in all the relevant details, an issue which I leave open. What I see to be an apparent advantage of the current approach is that – at least for C-SVCs – apart from syntactic headedness, no additional information, i.e. the level of representation (SF), along with a property of this level of representation (L-command) and the condition on argument linking (6), needs to be referred to by the grammar in order to derive their argument linking profiles. This follows naturally from what we know about projection and headedness.

I will mention just one notable difference between (strong) resultatives like (6) and C-SVCs, which is the role of syntactic headedness in argument blocking: while it is the non-head verb whose internal argument remains blocked in C-SVCs, it is the head verb whose internal argument remains blocked in (6) (assuming that the fact that finite inflection occurs only on the manner verb, rather than on the entire complex, is a direct reflection of its syntactic headedness).³⁶⁰

³⁵⁹Note that this generalizes to C-SVCs consisting of three or more verbs, since any transitive verb added on top of the Trees A.4 and A.5 will necessarily have an internal argument that does not L-command the structurally lowest argument, such that it will not satisfy the condition in (12) and be blocked from realization.

³⁶⁰Alternatively, we can assume that syntactic headedness in (6) is held not by the manner verb, but rather (i) by a silent syntactic element mediating between the manner and result components, (ii) by the result component, or (iii) by jointly the manner and the result component (i.e. the two from a complex syntactic head); additionally, one would need to assume (e.g. via morpho-syntactic property sets and DM-style contextual spell-out rules) that finite inflection occurs only on the manner verb because it is the only element in the configuration that can be inflected for finiteness. See (Williams 2015: 307ff) for recent theoretical discussion of the underlying syntax of resultatives.

Appendix B

Transitivity matching

B.1 Contextual bivalence

While the fine-grained semantics of SVCs lie outside the scope of the present work, this issue actually bears on the larger issue of morpho-syntactic constraints on argument structure and argument linking because – within contextually bivalent C-SVCs – subjects are not interpreted as state-holders like in isolation, even though the contextually bivalent $V_{2=H}$ constitutes the syntactic head; instead, they are interpreted as agentive subjects of the linearly preceding co-verbs, to which the contextually bivalent verb serves as a modifier: the subject is not interpreted as someone who returns (1a), is good (1b) or is slow (1c), but rather as an agent that carries out an action of sinking in a way that is repeated (or restituted) (1a), an action of teaching in a way that is good (1b), and an action of talking in a way that is slow (1c), respectively:

- (1) a. Ta-[he-yoli]-[uyo-i]- $\emptyset$
 1INCL.S-CAUS-sink-return-APPL-3SG.O
 ‘We again make it sink’
 b. Ye-[he-kata]-[namwa-namwa-i]-gai
 3SG.S-CAUS-learn-RED-be.good-APPL-1EXCL.O
 ‘She teaches us properly’
 c. Ku-[hedede]-[nogo-nogowai]!
 2SG.S-talk-slow
 ‘Speak slowly!’
- (Margetts 2005: 75)

We thus find a "thematic dissociation" between the subject and the contextually bivalent verb within C-SVCs, relative to its independent uses; crucially, this effect arises as a function of not only the lexical semantics of the contextually bivalent verb, but also its syntactic configuration. The first thing to note about this discrepancy is that the relation introduced by Voice, which holds between the denotation of its complement and the participant corresponding to the external argument, does not semantically associate directly with either of the individual sub-eventualities e_1 and s_2 ; this relationship is only ever directly established with the highest eventuality, i.e.

the macro-event E , meaning that the relationship between Voice on the one hand and e_1 and s_2 on the other is mediated by E and hence indirect. What, then, is the Aktionsart of the macro-event E ?

As argued in Kratzer (1996), Voice may conjoin with any predicate so long as the Aktionsarten involved are compatible. Kratzer takes Aktionsarten to originate from selectional restrictions of a predicate's eventuality argument, meaning that if a predicate's eventuality argument denotes an action, state, etc, so will the eventuality argument of Voice denote an action, state, etc. This rules out, for example, the combination of a state-holder with an action predicate, or that of an agent with a stative predicate. While she notes some complications with verbs whose subjects show both unergative and unaccusative properties (e.g. experiencer verbs), this means that the kind of external argument introduced by Voice is computed semantically, in a way that is (largely) predictable from the Aktionsart of its complement (Kratzer 1996: 123).

Since the subjects in (1a)-(1c) are clearly agentive external arguments, and because external arguments are introduced after complex verb formation, the Aktionsart of E must be agentive, i.e. an action. But how does this follow from the Aktionsarten of its sub- eventualities e_1 and s_2 ? Part of the answer, I propose, lies in the proper event-structural treatment of C-SVCs involving manner modification. We will now see that once we amend our event-structural semantics of C-SVCs with the appropriate components, we can derive the above-mentioned effect of thematic dissociation between subjects and contextually bivalent verbs within C-SVCs.

According to the present analysis, the contextually bivalent C-SVCs in (2a) and (2b) have the denotations in (3a) and (3b), respectively. Notice, however, that the manner-adverbial contributions of the contextually bivalent verbs *uyo* 'return' and *namwa* 'be good', indicated clearly in translation, are not reflected in the semantic representations since they are in no way related to the event they are supposed to modify, i.e. e_1 :

- (2) a. Ta-[he-yoli]-[uyo-i]- $\emptyset$
 1INCL.S-CAUS-sink-return-APPL-3SG.O
 'We again make it sink'
- b. Ye-[he-kata]-[namwa-namwa-i]-gai
 3SG.S-CAUS-learn-be.good-APPL-1EXCL.O
 'She teaches us properly'
- (3) a. $\llbracket(2a)\rrbracket =$
 $\exists E \exists e_1 \exists s_2 [E = e_1 \oplus s_2 \ \& \ \text{CAUS-sink}(e_1, x_i) \ \& \ \text{return}(s_2) \ \& \ \text{EXPL}(s_2) = it_i \ \& \ \text{AGENT}(E) = we]$
- b. $\llbracket(2b)\rrbracket =$
 $\exists E \exists e_1 \exists s_2 [E = e_1 \oplus e_2 \ \& \ \text{CAUS-learn}(e_1, x_i) \ \& \ s_2 \approx \text{be good} \ \& \ \text{EXPL}(s_2) = we_i \ \& \ \text{AGENT}(E) = she]$

Assuming that contextually bivalent verbs and manner adverbs differ only in category, but are otherwise parallel semantically, we ultimately want representations along the lines of the following alternatives for a sentence like *She teaches us properly*: (4a) represents a plain neo-Davidsonian treatment, according to which the manner adverbial *properly* is a further predicate to the event argument, semantically on a par with the event predicate corresponding to the verb; the manner component in (4b), by contrast, is treated as an ontologically distinct, manner coordinate that is linked to the event argument via an underspecified relation *R*:

- (4) $\llbracket \textit{She teaches us properly} \rrbracket =$
- a. *Manners as neo-Davidsonian event predicates*
 $\exists e. [\text{teach}(e, us) \ \& \ \text{AGENT}(e) = \text{she} \ \& \ \text{proper}(e)]$
based on Maienborn (2003)
 - b. *Manners as coordinates of neo-Davidsonian events*
 $\exists e. [\text{teach}(e, us) \ \& \ \text{AGENT}(e) = \text{she} \ \& \ \exists m [R(e, m) \ \& \ \text{proper}(m)]]$
based on Piñón (2007)

While each of these approaches comes with its own benefits and drawbacks, both differ from the representations in (3a) and (3b) in somehow reflecting the modificational contribution of the contextually bivalent verb. Specifically, the manner component explicitly relates to the event that it modifies, due to either being a further predicate of the singular event (4a), or a coordinate related to the event (4b). Compare this to the representations in (3), where the s_2 expressed by the contextually bivalent $V_{2=H}$ stands in no direct relation to the e_1 expressed by V_1 . Even though these are cumulated via the sum operator $\oplus$ into a macro-event E , cumulation itself does not establish any specific relation between its two operands; in fact, it does not establish any relation at all, but merely expresses that the complex macro-event E is a symmetric sum over e_1 and s_2 . As they stand, the representations in (3) are too "flat" because the manner component is not integrated into the overall event structure in a way that makes clear its modification of e_1 .

I discuss two possible solutions to this shortcoming, choosing one over the other. First, we might leave our representations unaltered, yet aided by a meaning postulate that yields the desired interpretation. One possible meaning postulate to this end is given in (5a), with the paraphrase in (5b), according to which any macro-event E "inherits" the properties of its sub-eventualities:

- (5) *Meaning postulate I: part-to-whole inheritance*
- a. $\exists E \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ P(e_1) \ \& \ Q(e_2)]$
 $\rightarrow P(E) \ \& \ Q(E)$
 - b. If two eventualities e_1 and e_2 are part of the same macro-event E , the event predicates P and Q of e_1 and e_2 , respectively, also apply to E .

As applied to (6), for example, because the macro-event E consists of a teaching sub-eventuality e_1 and a being-good sub-eventuality e_2 , it is also both a teaching and a being-good macro-event. This approach reproduces the flat neo-Davidsonian treatment of manner modification – repeated in (8) – in that applying the meaning postulate in (5) to the representation in (7a) entails that the predicates of the sub- eventualities e_1 and e_2 are fully on a par in terms of their relationship to the macro-event E :

- (6) Ye-[he-kata]-[namwa-namwa-i]-gai
 3SG.S-CAUS-learn-RED-be.good-APPL-1EXCL.O
 ‘She teaches us properly’
- (7) a. $\exists E \exists e_1 \exists s_2 [E = e_1 \oplus e_2 \ \& \ \text{CAUS-learn}(e_1, x_i) \ \& \ s_2 \approx \text{be good} \ \& \ \text{EXPL}(s_2) = w_{e_i} \ \& \ \text{AGENT}(E) = \text{she}]$
 b. $\rightarrow \text{CAUS-learn}(E) \ \& \ \text{be good}(E)$
- (8) *Manners as neo-Davidsonian event predicates*
 $\llbracket \text{She teaches us properly} \rrbracket$
 $= \exists e. [\text{teach}(e, \text{us}) \ \& \ \text{proper}(e) \ \& \ \text{AGENT}(e) = \text{she}]$

This approach, while seemingly elegant due to making do with only a minor additional assumption, nevertheless suffers from a drawback. The final meaning yielded in (7a) is that the macro-event E is characterized by teaching and goodness, which is not the meaning we are after, namely that the way in which the teaching is carried out is good. In Piñón (2007), a number of diagnostics are presented showing that the difference between manner adverbs and verbs – and by extension adjectives – is not merely syntactic, but also semantic. Specifically for our purposes, we need to establish an asymmetric event-structural relationship between the two verbs such that one is interpreted as modifying the other. This can be achieved by the alternative meaning postulate in (9):

- (9) *Meaning postulate II: MANNER entailment*
- a. $\exists E \exists e_1 \exists s_2 [E = e_1 \oplus s_2 \ \& \ e_1 \ \& \ s_2]$
 $\rightarrow \exists R [R(e_1, s_2) \ \& \ \text{MANNER}(s_2)]$
- b. If an event e and a state s are part of the same complex event E , s is interpreted as the MANNER of e .

According to (9), a stative verb is interpreted as expressing the manner of the event denoted by its co-verb whenever the two are part of the same macro-event. The manner interpretation is thus not present as a proper component at the level of complex event structure, but derived from it under certain conditions.

While this approach gives us the desired interpretation, it is essentially stipulative. Moreover, because the representation which licenses the entailment remains flat, it predicts that any state that is part of a macro-event should be interpreted as a manner

component. This is false, as shown by (10), in which both the V_1 *pitali* ‘dry’ and the $V_{2=H}$ *namwa* ‘be good’ are statives, but only $V_{2=H}$ *namwa* ‘be good’ functions as a manner modifier with respect to V_1 :

(10) *intr + intr*

Ye-[pitali]-[**namwa-namwa**]
3SG.S-dry-**be.good**

‘It is properly dry’

(Margetts 2005: 69)

Compare this to (11), which contains an C-SVC where *namwa* ‘be good’ is now the V_1 , and another verb, *kalili* ‘very’, is $V_{2=H}$; crucially, it is now *namwa* ‘be good’ that expresses an event that is modified:

(11) Ye-namwa, na nige ye-[**namwa**]-[kalili]
3SG.S-be.good CONJ NEG 3SG.S-**be.good**-very
‘It’s good, but not very good’

(Margetts 2005: 77)

This shows us that manner verbs systematically give rise to manner-modificational meanings as a function of their positional slot. Recall from Section BLA that Saliba C-SVCs obey an event-structural complementarity in terms of the distribution of verbs, between what can roughly be characterized as a means, an activity, a directionality and a manner/phasal component, which jointly give rise to the observed positional Slots A-D, repeated here in Table B.1:

	A	B	C	D
Meaning	main event means	CAUSE + result main event	directionality	manner phasal
# of stems	open	≥ 39	closed, 7	closed, 12

Table B.1: Positional slots in Saliba C-SVCs (Margetts 2005)

namwa ‘be good’ occupies Slot D in (25), and Slot A in (11), yet it is only in the latter that it expresses a manner meaning. I conclude that we need to amend our representations with an event-structural component that somehow makes reference to these slots, i.e. to the distinct event-structural functions that are induced. In place of invoking meaning postulates, I therefore propose that EC_{TS} carries out more than just the cumulation of eventualities, in that it also always comes with a certain relation holding between the two eventualities it cumulates. I leave open whether this relation is introduced directly by EC_{TS} , or whether it is forced independently by some deeper constraint on complex event structures, which may feed into the grammar from grammar-external sources; what matters for our purposes is that the relation itself be explicitly represented and visible to the grammar, at the level of

complex event structure.

Specifically, I assume that EC_{TS} comes with an event-structural relation R that is itself underspecified, yet the specific kind of which is determined by the meanings of the predicates related by EC_{TS} . For any two verbs and their associated eventualities, R specifies one of the various event-structural relationships reflecting the positional slots. In case V_1 is paired with a certain kind of stative $V_{2=H}$, for example – as in (12a) – R specifies a MANNER relation between e_1 and s_2 (12b):

- (12) a. Ye-[he-kata]-[namwa-namwa-i]-gai
 3SG.S-CAUS-learn-be.good-APPL-1EXCL.O
 ‘She teaches us properly’
 b. $\exists E \exists e_1 \exists s_2 [E = e_1 \oplus s_2 \ \& \ \boxed{R_{\text{MANNER}}(e_1, s_2)} \ \& \ \text{CAUS-learn}(e_1, x_i) \ \& \ s_2 \approx \text{be good} \ \& \ \text{EXPL}(s_2) = \text{we}_i \ \& \ \text{AGENT}(E) = \text{she}]$

Because R_{MANNER} in (12b) specifies a MANNER-relation between the teaching sub-eventuality e_1 and the being-good sub-eventuality s_2 , we correctly model that the latter is interpreted as a manner-modifier with respect to the former. This modification to EC_{TS} exploits the logic of Piñón (2007), yet without treating manners as separate ontological categories.³⁶¹ In contextually bivalent C-SVCs like (12a), the V_1 fills Slot A, and the $V_{2=H}$ Slot B. Let us now see how our modification to EC_{TS} extends to C-SVCs consisting of other slots.

In the unergative C-SVC in (13a), the V_1 ‘jump’ fills Slot A, and the $V_{2=H}$ ‘go down’ fills Slot C. The relation holding between the sub-eventualities e_1 and e_2 is thus one of directionality, as shown in (13b):

- (13) a. Ye-[kamposi]-[dobi]
 3SG.S-jump-go.down
 ‘He jumped down’
 b. $\exists E \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ \boxed{R_{\text{DIRECTION}}(e_1, e_2)} \ \& \ \text{jump}(e_1) \ \& \ \text{go.down}(e_2) \ \& \ \text{AGENT}(E) = \text{he}]$

The transitive C-SVC in (14a) is an instance of the MEANS-construction, in which the V_1 specifies the means by which the event denoted by the causative $V_{2=H}$ is carried out. Therefore, we can take R to specify a MEANS relation between e_1 and e_2 (14b):

- (14) a. Ye-[koi]-[kesi]-di
 3SG.S-hit-break-3PL.O
 ‘He hit-broke them’

³⁶¹As far as I can tell, it is orthogonal for the overall reasoning of Piñón (2007)’s approach whether or not we grant manners their own ontological status. Moreover, even in English, one would need to derive the relationship between the non-adverbial use of *good* and its adverbial use as *well*. Because *good* does not intrinsically express manner modification, its status as a manner modifier when occurring as *well* must be a property derived from adverbial modification. That is, what is taken to be an ontologically special status of manner predicates must anyway be a property that arises specifically in contexts of adverbial modification, such that it seems to me to be sufficient to model the manner component as a relation, rather than an independent semantic object (at least for present purposes).

- b. $\exists E \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ \boxed{R_{\text{MEANS}}(e_1, e_2)} \ \& \ \text{hit}(e_1, x_i) \ \& \ \text{break}(e_2, \text{they}_i) \ \& \ \text{AGENT}(E) = \text{he}]$

This way, we end up with the modified version of generalized EC_{TS} in (15), which has now been amended by the underspecified relation R_α , whose specific type is determined by the two input eventualities fed into EC_{TS} , and which establishes an asymmetric event-structural relation between these two:

- (15) *Generalized EC_{TS} (amended by R):*
 $\lambda Q_{\langle \alpha \rangle} \lambda P_{\langle \alpha \rangle} \{ \lambda y \} \lambda E. \underbrace{\{ \exists x \} \exists e_1 \exists e_2 [E = e_1 \oplus e_2 \ \& \ \boxed{R_\beta(e_1, e_2)} \ \& \ P(e_1, \{x_i/k\}) \ \& \ Q(e_2, \{y_i\})]}_{\text{Complex Predicate: } \langle \alpha \rangle} \rangle \langle \alpha, \langle \alpha, \alpha \rangle \rangle$

Notice that the specific kind of event-structural relation assumed by R is never stipulated, but always determined by the meanings of any two cumulated eventualities: if V_n is a causative verb, it will always stand in a **MEANS** relation to V_{n-1} ; if V_n is a motion or directional verb, it will always stand in a **DIRECTION** relation to V_{n-1} ; and if V_n is a manner verb, it will always stand in **MANNER** relation to V_{n-1} ; and so on.³⁶²

We are now ready to return to our argument-structural puzzle, i.e. the issue of how Voice is able to associate with a complex verb consisting of an active V_1 and a stative $V_{2=H}$ in cases like (16), especially in light of the fact that the subject of (16) is agentive, despite Voice projecting syntactically from the non-agentive $V_{2=H}$:

- (16) a. Ye-[he-kata]-[namwa-namwa-i]-gai
 3SG.S-CAUS-learn-be.good-APPL-1EXCL.O
 ‘She teaches us properly’
 b. $\exists E \exists e_1 \exists s_2 [E = e_1 \oplus s_2 \ \& \ \boxed{R_{\text{MANNER}}(e_1, s_2)} \ \& \ \text{CAUS-learn}(e_1, x_i) \ \& \ s_2 \approx \text{be good} \ \& \ \text{EXPL}(s_2) = \text{we}_i \ \& \ \text{AGENT}(E) = \text{she}]$

I propose that, given the newly introduced presence of the event-structural R , which specifies a **MANNER**-relation between e_1 and s_2 in cases like (16), the Aktionsart of the macro-event E in (16) is active, thus licensing the combination with a Voice head that introduces an agentive external argument. The idea is that because s_2 is now related to e_1 via R_{MANNER} , it is now no longer the exact same kind of eventuality as in isolation, i.e. a state characterizing a state-holder subject; instead, s_2 now functions as a manner modifier to e_1 , a property which “repeals”, as it were, the characterization of any external-argument denoting a state-holder. In place of the individual state-holder found in isolation, we can intuitively think of e_1 as now corresponding to the non-individual holder of the state of goodness.

Alternatively, and again not necessarily in mutual exclusion, we might adopt an

³⁶²More fine-grained data would be needed to determine whether the relation characterized by any V_n scopes only over the eventuality denoted by V_{n-1} , or – in the case of three- or four-verb C-SVCs – over the macro-event denoted by $[V_1 + \dots + V_{n-1}]$.

approach along the lines of Wood (2017), where the thematic properties of external arguments are underspecified and determined allosemically as a function of their structural context.

B.2 Dyirbal

Dyirbal is a Pama-Nyungan language spoken in Australia. Like in Saliba, Dyirbal SVCs show strict semantic TM. (17a) consists of two simplex intransitives, and (17b) shows the pairing of two simplex transitives:

(17) a. *intr + intr*

Jañjja [ñina-ñu] [wurba-ñu] wiyamay-gu
 now sit-PST talk-PST do.what-PURP
 '(All the men) now sat around talking about what to do'

b. *trans + trans*

Bayi gugar [bura-n] [wuba-n] banjun nayi-jarran-du
 he.ABS black.goanna look.at-PST look.up-PST she-ERG girl-DU-ERG
 'The two girls looked up at the black goanna'

In case two simplex verbs are to be paired which differ in transitivity, either the intransitive one must be made transitive, or the transitive one must be made intransitive. In (18a), the simplex transitive V_2 *ɲamba* 'listen' is detransitivized by the reflexive so it can be paired with the simplex intransitive V_1 *banaga* 'go back'; in (18b), the transitive V_1 *guñja* 'drink' can only be paired with the simplex intransitive V_2 *ɲabay* 'bathe' because the latter is transitivized by the applicative:

(18) a. *intr + intr*

Balay bayi [banaga-ñu] jañja [ɲamba-yirri-ñu] bayi
 there.LOC he.ABS go.back-PST now listen-REFL-PST he.ABS
 'Now he went back there thinking (as he went)'

b. *trans + trans*

Bana banju.mangan-du [guñja-guñja-li] [ɲaba-ɲabay-ma-li]
 water 3PL-ERG RED-drink-PURP RED-bathe-APPL-PURP
 'For them all to drink water a lot and bathe in it a lot' (Dixon 2011: 201, 206)

Compared to most Oceanic languages, which generally lack productive valence-reducing operations, Dyirbal has at its productive disposal both valence-increasing and valence-reducing operations. Next to the reflexive and the applicative, used in (18a) and (18b) above, Dyirbal also possesses an anti-passive operation, which is similarly attested as a transitivity-adjusting device in SVCs; see the V_1 in (19a), or

both V₁ and V₂ in (19b):^{363,364}

(19) a. *intr* + *intr*

gaji [bural-ay-gu] [yanu-li]
TRY look.at-ANTIP-PUR go-PURP
'(We) should try to go on looking as we go'

b. *intr* + *intr*

[nudil-ŋa-ñu] bayi [bañil-ŋa-ñyu] jambun-du
cut-ANTIP-PST he.ABS chop-ANTIP-PST grub-INSTR
'(The younger brother,) he was cutting and chopping out grubs' (Dixon 2011: 202)

Just like Saliba (and Tariana, see below), Dyirbal requires all verbs to undergo overt derivation if arguments are to be acquired or omitted (from argument structure; lexical DPs are often dropped), i.e. displays rigid transitivity. But Dyirbal SVCs differ from the SVCs found in these two languages in a number of respects. First, although they are in most cases surface-contiguous, they do allow the appearance of inter-verbal material: (Dixon 2011: 187) notes that in about three-quarters of SVCs, the verbs occur immediately next to each other, but in the remainder they may be separated by one or more words, which can be any type of constituent. In (20a), for example, V₁ and V₂ are separated by an adverb, and in (20b), they are separated by both an adverb and the subject:

(20) a. *trans* + *trans*

[banagay-mba-li] yali-gala [yambil-ma-li]
go.back-APPL-PURP HERE.TO.PLACE-DOWN fly-APPL-PURP
'(The flying star spirit flew down south to get a painted basket) in order to fly back up here with it'

³⁶³Dyirbal lacks a properly distinguished causative operation. Even though the marker *-ma-l* ~ *(m)ba-l* is sometimes glossed as a causative in Dixon (2011) and Dixon (2015), it is almost identical to the marker that is glossed as an applicative in these same works depending on context, namely *-ma-l* ~ *(m)ba-l* ~ *ba-l*. With (i) adjectives or nouns, (ii) locative and temporal adverbs, or (iii) numerals, *-ma-l* ~ *(m)ba-l* derives transitive verbs and is glossed as a causative, e.g. *mulgu-ma-l* 'noise-CAUS / make a noise' and *ŋulga-gu-mba* 'tomorrow-until-CAUS.IMP / Do (it) until tomorrow!' (Dixon 2011: 194, 211); with intransitive verbs, *-ma-l* ~ *(m)ba-l* ~ *ba-l* derives transitive verbs with comitative objects and is glossed as an applicative, e.g. *banagay-mba-l* 'go.back-APPL / go back with' (Dixon 2011: 203); finally, with transitive verbs, *-ma-l* ~ *(m)ba-l* ~ *ba-l* has a rearranging effect in advancing an Instrument to core function while demoting the old Patient to oblique, and is glossed as an applicative, e.g. *yugu balga-lma-n* 'stick.ABS hit-APPL-NFUT / hit with a stick' (Dixon 1972: 95). The generalization is that *-ma-l* ~ *(m)ba-l* (~ *ba-l*) is glossed as a causative when it derives transitive verbs from non-verbal predicates, but as an applicative when it derives transitive verbs from intransitive or transitive verbs; it thus functions as a transitive verbalizer in the former case, and as a (rearranging) transitivizer in the latter. For an overview and discussion of valence-increasing operations in Dyirbal and other Australian languages, see Austin (2005).

³⁶⁴Dyirbal also has a reciprocal, *-(n)barri-y*. In the corpus on which the data in Dixon (2011) and Dixon (2015) are based, however, there are no instances of SVCs in which a verb's transitivity has been adjusted by the reciprocal, though Dixon notes that such cases might exist in a larger corpus.

- b. [jɪŋgali-ɲu] aɲja bayi [banaga-ɲu]
 run-PST NEW he.ABS go.back-PST
 ‘He ran back’ (Dixon 2011: 187, 199)

A second point of flexibility of Dyirbal SVCs is that their verbs are not subject to rigid ordering. This is illustrated by the following pair: in (21a), V_1 is an adverbial, i.e. a verb expressing a manner-adverbial-like meaning, whereas this function is taken on by V_2 in (21b):

- (21) a. *intr + intr; V_1 is adverbial*
 yaburri bala [wiɣama-njay] [banda-njay]
 INTERJ it.ABS do.how-FUT burst-FUT
 ‘Oh dear! I wonder if (the pipe) is going to burst’
 b. *intr + intr; V_2 is adverbial*
 jaɲja [wurrba-ɲu] [yalama-ɲu]
 now speak-PST do.like.this-PST
 ‘(The man) now spoke like this’ (Dixon 2011: 195f)

Dyirbal is the only language I have come across so far in which semantic TM is available in NC-SVCs – a fact which I presently cannot derive. I conjecture that Dyirbal semantic TM results from a mechanism that is distinct from type-matching. It is possible, for instance, that Dyirbal does not distinguish internal and external arguments, as has been proposed for e.g. Ch’ol (Coon 2010), and that serialization occurs at the VP-level. An alternative approach could be based in *multi-dominance*, whereby there is a single object that is dominated by multiple VerbPs. If an obligatory one-to-one object-verb relationship can be made to conversely entail the obligatory absence of an object in each VerbP, it seems that the net effect of semantic TM can be achieved. See Hiraiwa & Bodomo (2008) for a multi-dominance analysis of SVCs in Dagaare. It might also turn out that both the non-contiguity of SVCs and the non-rigid order of their constituent verbs are connected to Dyirbal’s *non-configurationality*, a property for which it is well-known (Dixon 1972, van de Visser 2003). I leave this issue to future work.

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